BetterMath

IB Diploma Programme

Mathematics

Analysis & Approaches

Higher Level

today > yesterday

Hasan Khan · First edition, 2026

BetterMath

IB Mathematics: Analysis and Approaches, Higher Level

Hasan Khan

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Set in TeX Gyre Pagella, TeX Gyre Heros and Latin Modern, using XeLaTeX.

Corrections and unclear passages are worth reporting at ibxhasan@gmail.com

How to use this book

Read a section, work its examples, then do the **Your Turn** questions that follow it. Each badge below means the same thing every time it appears.

IN THE NOTES

DEF	A **definition**: the precise meaning of a term.
KEY	A **key result** or formula you will use again.
KEY · IB	The same, and **printed in the IB formula booklet**: you are given it in the exam. Its panel has a heavier cobalt rule.
HL	Examined **only at Higher Level**. A chapter that is entirely HL says HL only once, on its opening page.
CAUTION	A mistake made here often enough to be worth naming.
TIP	Exam craft: what earns the mark, or what loses it.

IN THE REFLECTIONS

TOK	**Theory of Knowledge**: how we know this.
IM	**International mindedness**: who built the idea, and where.
RW	**Real world**: where it is actually used.
EXT	**Extension**: where to go next, past the syllabus.

These close every chapter, on a tinted page. None of it is examined.

IN THE PRACTICE

1.	**Your Turn**: one technique at a time, straight after its section.
1.	**End-of-Chapter Problems**: exam-style, whole chapter, in section order. The running head names them.
1.	**Challenge Questions**: Paper 3 style investigations, and good starting points for the Exploration.
GDC	A calculator is needed. Everything else is answerable without one.

FINDING YOUR WAY

The cobalt bar on the outer edge steps down the book as the chapters advance, so a chapter can be found with the book shut. The running head names the chapter outside and its strand inside.

Contents

NUMBER & ALGEBRA

Chapter 1

Exponents & Logarithms

1

SYLLABUS 1.1 1.5 1.7

Japan lies in one of the world's most active seismic zones. Suppose a seismologist asks you to plot two recordings on one axis. The first has an amplitude (the height of the wave) of 10^4. The second has an amplitude of 10^9, which is $10^5 = 100{,}000$ times the first.

The task sounds simple, but an ordinary scale, with evenly spaced marks, cannot show both. If the marks are fine enough for the first recording, the second falls far beyond the page. If the scale is long enough for the second, the first sits so close to zero that you cannot see it.

Now suppose equal spaces along the axis meant multiplication by 10 rather than addition of 1. The scale would read $10^0, 10^1, 10^2$, and so on. The two recordings have exponents 4 and 9, so they would sit five equal steps apart, and both would fit on one page. That exponent has a name. It is the **logarithm** of the amplitude, and it is the idea this chapter is built on.

This is how earthquake magnitude scales work. One step up in magnitude means ten times the recorded amplitude. Writing a quantity as a power of ten turns multiplication into addition, and that is what fits an enormous range into a small space. The same idea gives us pH, decibels, and the scale for the brightness of stars. It is useful because the hard operation becomes the easy one: quantities that would have to be divided can instead be subtracted, and a range too wide to plot fits on an axis you can read.

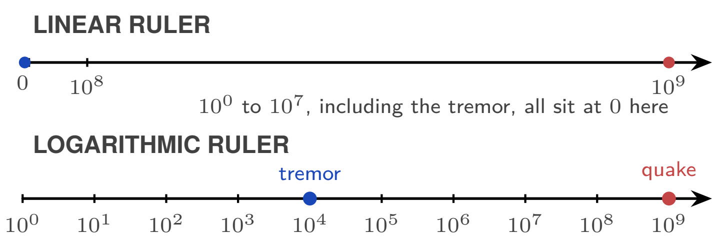


Figure 1.1 A logarithmic scale separates values that collapse near zero on a linear scale.

By the end of this chapter you will be able to use the laws of exponents, write quantities in standard form, and move between exponential and logarithmic form. You will also solve equations in which the unknown sits in an exponent. Each of these does one job: it puts a quantity into the form that makes a comparison easy. That is why a scale built on powers of ten can hold the mass of an electron and the mass of a star on one axis.

1.1 What an Exponent Counts

An exponent, also called a *power* or an *index*, is shorthand for repeated multiplication. In a^n, a is the **base** and n is the **exponent**. When n is a positive integer,

$$a^n = \underbrace{a \cdot a \cdots a}_{n \text{ factors}}.$$

For example, $a^3 = a \cdot a \cdot a$.

When you multiply powers with the same base, it is tempting to multiply the exponents. Writing out the factors shows what actually happens:

$$a^3 \cdot a^2 = (a \cdot a \cdot a)(a \cdot a) = a^5.$$

There are five factors of a, not six. Since the exponent counts how many times the base appears as a factor, the exponents are **added**:

$$a^m \cdot a^n = a^{m+n}.$$

Division cancels common factors, so the exponents are **subtracted**:

$$\frac{a^5}{a^2} = \frac{a \cdot a \cdot a \cdot a \cdot a}{a \cdot a} = a^3 = a^{5-2}.$$

Raising a power to another power repeats the whole power, so the exponents are **multiplied**:

$$(a^2)^3 = a^2 \cdot a^2 \cdot a^2 = a^6 = a^{2\times 3}.$$

A power outside a bracket applies to every factor inside:

$$(ab)^n = a^n b^n, \qquad \left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}.$$

The same laws extend naturally to zero and negative exponents. For $a \neq 0$, the subtraction rule gives $a^3/a^3 = a^{3-3} = a^0$; since any nonzero number divided by itself is 1, this forces $a^0 = 1$. Continuing the same pattern gives $a^{-n} = 1/a^n$. These values are not imposed on us; they are the only ones that keep the laws consistent. The main laws are collected below.

DEF For $a, b \neq 0$ and $m, n \in \mathbb{Z}$:

$a^m \cdot a^n = a^{m+n}$	product of powers: add the exponents
$\dfrac{a^m}{a^n} = a^{m-n}$	quotient of powers: subtract the exponents
$(a^m)^n = a^{mn}$	power of a power: multiply the exponents
$(ab)^n = a^n b^n$	power of a product: raise each factor
$\left(\dfrac{a}{b}\right)^n = \dfrac{a^n}{b^n}$	power of a quotient: raise top and bottom
$a^0 = 1$	zero exponent: gives 1
$a^{-n} = \dfrac{1}{a^n}$	negative exponent: take the reciprocal

CAUTION Do not confuse $(a^m)^n$ with a^{m^n}. In $(a^m)^n$, the power of a power, you multiply the exponents. In a^{m^n} you start at the top: first work out m^n, then raise a to that result. The two are usually not equal: $(2^3)^2 = 2^6 = 64$, but $2^{3^2} = 2^9 = 512$.

CAUTION A negative exponent means *reciprocal*, not *negative value*. $2^{-3} = \frac{1}{8}$, not -8.

Worked examples

How do these rules combine when a problem has several operations at once?

EXAMPLE 1.1

Simplify $\dfrac{6c^9}{2c^4}$.

Split the coefficient and the variable and handle them separately.

$$\frac{6c^9}{2c^4} = \frac{6}{2} \cdot c^{9-4} = 3c^5.$$

Apply one rule at a time.

EXAMPLE 1.2

$(2x^2y^{-4})^5$ contains several operations: a bracket raised to a power and a negative exponent.

Raise each part in turn: the 2 to the 5th, x^2 to the 5th, and y^{-4} to the 5th.

$$(2x^2y^{-4})^5 = 32x^{10}y^{-20}.$$

The most common error here is forgetting to raise the coefficient.

CAUTION The bracket in $(2x^2)^3$ applies to everything inside it, including the 2. $(2x^2)^3 = 8x^6$, not $2x^6$.

Beyond counting factors

The idea of a^n as repeated multiplication only makes sense when n is a positive integer. For example, what could 2^{-3} mean? You cannot multiply 2 by itself -3 times. And what is $3^{1/2}$, or 4^0? You cannot take half a factor, or no factors at all. Yet each of these still has exactly one sensible value. We choose that value so the exponent laws keep working:

$$2^{-3} = \frac{1}{2^3} = \frac{1}{8}, \qquad 3^{1/2} = \sqrt{3}, \qquad 4^0 = 1.$$

The next sections show where these values come from.

Surds and fractional exponents

What is $9^{1/2}$? Since $\frac{1}{2}$ is not an integer, repeated multiplication cannot answer this directly. But the power-of-a-power law gives

$$\left(9^{1/2}\right)^2 = 9^{(1/2)\times 2} = 9,$$

so $9^{1/2}$ must be the non-negative number whose square is 9:

$$9^{1/2} = \sqrt{9} = 3.$$

Notice the value is 3, not ± 3: the symbol $\sqrt{9}$ means the **principal** (non-negative) square root.

DEF For $a > 0$, $n \in \mathbb{Z}^+$ and $m \in \mathbb{Z}$:

$$a^{1/n} = \sqrt[n]{a}$$

unit fraction: the denominator is the root

$$a^{m/n} = \left(\sqrt[n]{a}\right)^m = \sqrt[n]{a^m}$$

in general: root then power, in either order

A useful way to remember this:

$$\boxed{\text{denominator } n \longrightarrow \text{root}, \qquad \text{numerator } m \longrightarrow \text{power}}$$

These are not new laws; they are the meanings chosen so that the familiar exponent laws keep holding.

Using fractional exponents

For $a^{m/n}$, it is usually easier to take the root first, then apply the power:

$$64^{2/3} = \left(\sqrt[3]{64}\right)^2 = 4^2 = 16.$$

Either order gives the same result, $64^{2/3} = \sqrt[3]{64^2} = 16$, but taking the root first keeps the numbers small.

Working with surds

A **surd** is an irrational root left in exact form, such as $\sqrt{2}$ or $\sqrt[3]{5}$, rather than rounded to a decimal. On a non-calculator paper, surds should be kept exact and simplified using the root laws

$$\sqrt{ab} = \sqrt{a}\,\sqrt{b}, \qquad \sqrt{\tfrac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}},$$

valid wherever the quantities are real. Two techniques cover most questions:

1. **Simplify** a surd by extracting its largest perfect-square factor.
2. **Combine** only like surds, the terms that have the same value under the root.

Rationalising the denominator

An exact answer is usually written without a surd in the denominator; removing it is called **rationalising the denominator**.

- For a single surd, multiply top and bottom by that surd.
- For a sum or difference, multiply by its **conjugate**, formed by changing the sign between the terms.

EXAMPLE 1.3

Rationalise $\dfrac{1}{\sqrt{3}}$ and $\dfrac{6}{2+\sqrt{3}}$.

For the first, multiply by $\dfrac{\sqrt{3}}{\sqrt{3}}$:

$$\frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}.$$

For the second, multiply by the conjugate $2-\sqrt{3}$:

$$\frac{6}{2+\sqrt{3}} = \frac{6(2-\sqrt{3})}{(2+\sqrt{3})(2-\sqrt{3})} = \frac{6(2-\sqrt{3})}{4-3} = 12-6\sqrt{3}.$$

The conjugate works because the product is a difference of two squares, $(2+\sqrt{3})(2-\sqrt{3}) = 2^2 - (\sqrt{3})^2 = 1$, so the surd leaves the denominator.

TIP In the real numbers, $(-4)^{1/2}$ is undefined: no real number squares to -4. In general, the radicand, the number inside the root, must be non-negative for an even root to have a real value. Square roots of negative numbers are complex numbers, which are introduced in Chapter 18.

Exponents as continuous growth

There is another way to read a non-integer exponent. Suppose 2^t means growth by a factor of 2 over a time t. Then 2^3 is three complete doublings, while

$$2^{1/2} = \sqrt{2}$$

is the growth factor halfway through one doubling period. Because the growth is exponential, "halfway" refers to the time, not to half the final increase. This reading extends to every real value of t, whether negative, fractional, or irrational:

$$2^{-3}, \qquad 2^{0}, \qquad 2^{1/2}, \qquad 2^{\sqrt{2}}.$$

Read as growth over time t, each has a plain meaning. $2^0 = 1$ is no growth at all: at time zero the population is unchanged, one hundred per cent of where it started. $2^{-3} = \frac{1}{8}$ looks backwards in time, to three periods before now (three seconds ago, if one doubling takes a second), when the population was an eighth of its present size. $2^{1/2} = \sqrt{2}$ is the factor after half a period, and $2^{\sqrt{2}}$ the factor after an irrational stretch of time. For any positive base a, the expression a^x has a meaning for every real exponent x, which is where the exponential function of Chapter 6 begins.

The number *e*

In the previous section, 2^t described growth that doubles every period. But real growth is usually smooth. A population does not wait until the end of the hour to add all its new members; it grows a little at every moment. What number describes growth that happens continuously, at every instant?

Take a quantity that would grow by 100% in one hour. If all of that growth arrives at the very end, the quantity simply doubles. But suppose the same 100% is spread over two half-hours. After the first half-hour it is $1 \times 1.5 = 1.5$ times its start, and that larger amount then grows by the next 50%, reaching $1.5 \times 1.5 = 2.25$ times. Spreading the growth into more steps lets each new part start growing sooner, so the hour's total rises a little each time. In n equal steps the quantity multiplies by $\left(1 + \frac{1}{n}\right)^n$:

steps in the hour, n	**size after one hour**
1 (once)	2.000
2 (half-hourly)	2.250
60 (every minute)	2.6960
3600 (every second)	2.7179
$n \to \infty$ (continuous)	$2.71828\ldots = e$

The totals do not grow without bound. They approach a single limit, $2.71828\ldots$, and this number is called e. Like π, it cannot be written as a finite decimal, and it appears throughout mathematics. Here it is the result of 100% growth acting continuously, at every instant instead of in fixed steps. A continuous rate r acting for time t multiplies the starting amount by e^{rt}. This is the model behind population growth, radioactive decay, and cooling. Chapter 14 gives a deeper reason for e: the function e^x grows at a rate equal to its own value.

EXT Formally, $e = \lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n$. Each n is a number of compounding steps per year, so letting n grow without bound is what "continuously" means. Limits are defined properly in Ch. 14.

YOUR TURN 1.1 What an Exponent Counts

1. Simplify each of the following.

(a) $\dfrac{6c^9}{2c^4}$ [1] (b) $(3x^2y^{-3})^4$ [2]

(c) $\dfrac{12a^5b^3}{4a^2b^{-1}}$ [2] (d) $\dfrac{(2x^3)^2 \cdot x^{-4}}{4x^2}$ [2]

2. Simplify by dividing each term of the numerator separately.

(a) $\dfrac{4x^3 - 8x^2}{2x}$ [2] (b) $\dfrac{16a^5b^2 - 12ab^4}{4ab^2}$ [2]

3. Show that $\dfrac{(a^2b^{-1})^3}{a^{-3}b^2} = \dfrac{a^9}{b^5}$. [3]

4. Evaluate each of the following.

(a) $27^{2/3}$ [1] (b) $81^{-3/4}$ [2]

(c) $\left(\dfrac{8}{27}\right)^{-2/3}$ [2] (d) $32^{-2/5}$ [2]

5. Write each of the following in the form x^n.

(a) $\sqrt[4]{x^3}$ [1] (b) $\dfrac{1}{\sqrt[3]{x^2}}$ [1]

(c) $\dfrac{\sqrt{x^5}}{x^2}$ [2] (d) $\dfrac{x^2 \cdot \sqrt{x}}{x^{3/2}}$ [2]

6. Solve $x^{3/2} = 64$. [2]

7. Simplify each of the following.

(a) $\sqrt{50} - \sqrt{8}$ [1] (b) $\sqrt{12} + \sqrt{27}$ [2]

8. Rationalise the denominator of $\dfrac{4}{3 - \sqrt{5}}$, giving your answer in the form $a + b\sqrt{5}$. [3]

1.2 Standard Form

The mass of a proton, written out in full, is

$$0.00000000000000000000000000167 \text{ kg},$$

that is, 26 zeros after the decimal point before the first significant figure. The mass of the Sun runs to the opposite extreme,

$$1{,}990{,}000{,}000{,}000{,}000{,}000{,}000{,}000{,}000{,}000 \text{ kg},$$

a 1 followed by 30 digits. Both numbers are quite difficult to write, let alone compare or calculate with.

There is a shorter way to write both. Instead of all the digits, record *the power of 10* that places the decimal point where it belongs. Mathematicians call the result **standard form** (also known as scientific notation).

DEF A number is in **standard form** when written as $a \times 10^k$, where $1 \leq a < 10$ and $k \in \mathbb{Z}$.

The proton becomes 1.67×10^{-27} kg. The Sun becomes 1.99×10^{30} kg. The exponent k tells you immediately whether you are dealing with something microscopic or cosmic.

Calculating with standard form

Multiplication and division follow straight from the exponent laws. For a product, multiply the coefficients and add the powers of 10; for a quotient, divide the coefficients and subtract the powers:

$$(a \times 10^m)(b \times 10^n) = (a \times b) \times 10^{m+n}.$$

One check matters afterwards. If the new coefficient falls outside the range $1 \leq a < 10$, shift the decimal point and adjust the power. For example, $(5 \times 10^3)(4 \times 10^2) = 20 \times 10^5 = 2 \times 10^6$: the coefficient 20 becomes 2, and the exponent rises by one to account for the extra factor of 10.

Addition and subtraction need one more move first. Just as you cannot add metres and kilometres without converting, you cannot add two numbers in standard form until they share the same power of 10. Rewrite the one with the smaller exponent so the powers match, then add the coefficients.

EXAMPLE 1.4

Calculate $(3.5 \times 10^4) \times (2 \times 10^{-7})$.
Group and multiply:

$$(3.5 \times 2) \times 10^{4+(-7)} = 7 \times 10^{-3}.$$

EXAMPLE 1.5

Calculate $(4.8 \times 10^5) + (3.2 \times 10^4)$.
Before adding, you need the same power of 10, just as you cannot add metres and kilometres without converting first.

$$4.8 \times 10^5 + 0.32 \times 10^5 = 5.12 \times 10^5.$$

CAUTION In addition and subtraction, do **not** add the exponents. Only multiplication and division combine powers of 10.

CAUTION Calculator display notation is not acceptable in written answers. If your graphic display calculator (GDC) shows 5.2E30, you must write 5.2×10^{30}.

YOUR TURN 1.2 Standard Form

9. Write 0.000047 in standard form. [1]

10. Write 3.2×10^5 as an ordinary number. [1]

11. Calculate $(2.4 \times 10^3) \times (3 \times 10^{-5})$, giving your answer in standard form. [2]

12. Calculate $\dfrac{6.6 \times 10^8}{2.2 \times 10^3}$, giving your answer in standard form. [2]

13. Calculate $(5.1 \times 10^4) + (3.9 \times 10^3)$, giving your answer in standard form. [2]

14. GDC The mass of an electron is 9.11×10^{-31} kg. The mass of a proton is 1.67×10^{-27} kg. Find how many times heavier a proton is than an electron. Give your answer in standard form to 3 significant figures. [2]

1.3 The Logarithm: an Exponent Reversed

Many familiar operations come in inverse pairs: addition with subtraction, multiplication with division. So what is the inverse of exponentiation? If $a^x = b$, and you know a and b, how do you find x? That is what a logarithm does. It is not a new idea: it is the exponential equation $a^x = b$ rearranged to make the exponent x the subject.

DEF · IB For $a > 0$, $a \neq 1$, and $b > 0$:

$$a^x = b \iff x = \log_a b$$

The **common logarithm** ($\log x$) uses base 10. The **natural logarithm** ($\ln x$) uses base $e \approx 2.718$, the continuous-growth constant met earlier in this chapter.

The word *logarithm* comes from the Greek *logos* (ratio) and *arithmos* (number).

The graph of $y = \log_a x$ is the mirror image of $y = a^x$ across the line $y = x$, with a vertical asymptote at $x = 0$. These graphs and their transformations are developed in Chapter 6 (Functions).

Because the exponential and logarithm are inverses, each undoes what the other does:

KEY For base 10 and base e:

$\log(10^x) = x$	base 10: true for every real x
$10^{\log x} = x$	the same pair reversed, but only for $x > 0$, since $\log x$ is undefined otherwise
$\ln(e^x) = x$	base e: true for every real x
$e^{\ln x} = x$	reversed, and again only for $x > 0$

Two special values follow at once from the definition, and are worth knowing on sight: since $a^1 = a$ and $a^0 = 1$,

KEY For any base $a > 0$, $a \neq 1$:

$\log_a a = 1$	since $a^1 = a$
$\log_a 1 = 0$	since $a^0 = 1$

Working with logarithms

Evaluating a logarithm by hand is always the same move in reverse. Since $\log_a b$ is defined as the exponent that turns a into b, every logarithm question is an exponent question asked backwards. Read $\log_a b$ as **"a to what power gives b?"** and in many cases the answer is immediate.

EXAMPLE 1.6

Find $\log_3 81$.

Ask the inverse question: 3 to what power gives 81? You know $3^4 = 81$, so $\log_3 81 = 4$. The logarithm is just the exponent.

EXAMPLE 1.7

Solve $\log_{10}(y + 1) = 3$.

The equation says: the exponent you need to get $(y + 1)$ from base 10 is 3. So $y + 1 = 10^3 = 1000$, giving $y = 999$.

EXAMPLE 1.8

Without a calculator, find $e^{3 \ln 2}$.

Think of $e^{\ln 2}$ first: the exponential and natural log cancel, giving 2. So $e^{3 \ln 2} = (e^{\ln 2})^3 = 2^3 =$

8.

Laws of logarithms

WHERE the laws come from

Why do the logarithm laws work? A logarithm is an exponent. When you multiply two powers with the same base, you add the exponents: $a^p \cdot a^q = a^{p+q}$. In terms of logarithms, this means that multiplication becomes addition.

The first three laws below are the three exponent laws restated for logarithms. A fourth, change of base, is a different kind of rule: it rewrites any logarithm as a ratio of two logarithms in a base of your choice.

KEY · IB For $a, b, x, y > 0$ with $a, b \neq 1$:

$\log_a(xy) = \log_a x + \log_a y$	product: a product becomes a sum
$\log_a\left(\dfrac{x}{y}\right) = \log_a x - \log_a y$	quotient: a quotient becomes a difference
$\log_a(x^m) = m \log_a x$	power: an exponent becomes a coefficient
$\log_a x = \dfrac{\log_b x}{\log_b a}$	change of base: rewrite in any base you can compute

All three proofs share one setup. Write $x = a^p$ and $y = a^q$, so that $p = \log_a x$ and $q = \log_a y$ by definition. Each law is then a one-line consequence of the matching exponent law:

- **Product.** $xy = a^p \cdot a^q = a^{p+q}$, so $\log_a(xy) = p + q = \log_a x + \log_a y$.
- **Quotient.** $\dfrac{x}{y} = a^{p-q}$, so $\log_a\left(\dfrac{x}{y}\right) = p - q = \log_a x - \log_a y$.
- **Power.** $x^m = (a^p)^m = a^{mp}$, so $\log_a(x^m) = mp = m \log_a x$.

The third law is the most useful in this course. When x appears in an exponent, the power law $\log_a(x^m) = m \log_a x$ moves the exponent into a coefficient. This often converts an exponential equation into a simpler algebraic equation.

CHANGE of base

What do you do when your GDC only has log (base 10) and ln (base e), but you need a logarithm in a different base, or you want to rewrite a base-10 logarithm in terms of base e? The change-of-base law above rewrites any logarithm as a ratio of two logarithms you can compute. In practice you take base e, which turns the law into $\log_a x = \dfrac{\ln x}{\ln a}$.

The proof is elegant: let $\log_a x = y$, so $x = a^y$. Take ln of both sides: $\ln x = y \ln a$. Solve for y.

EXAMPLE 1.9

Evaluate $\log_3 5$ on a GDC.

Convert to natural logarithms:

$$\log_3 5 = \frac{\ln 5}{\ln 3} \approx \frac{1.609}{1.099} \approx 1.46.$$

Applying the laws

These laws can be used in either direction: they split one logarithm into several, or gather several back into one. Both directions appear constantly, in simplifying expressions and in solving equations.

EXAMPLE 1.10

Express $\log_5\left(\frac{a^3}{b}\right)$ in terms of $p = \log_5 a$ and $q = \log_5 b$.

Apply the quotient law, then the power law:

$$\log_5\left(\frac{a^3}{b}\right) = \log_5 a^3 - \log_5 b = 3\log_5 a - \log_5 b = 3p - q.$$

EXAMPLE 1.11

Write $2\ln a + 6\ln b$ as a single logarithm.

Use the power law in reverse first, then the product law:

$$2\ln a + 6\ln b = \ln a^2 + \ln b^6 = \ln(a^2 b^6).$$

EXAMPLE 1.12

Solve $\log_2 x + \log_2(x-2) = 3$.

The two logs on the left can combine into one product:

$$\log_2[x(x-2)] = 3 \implies x(x-2) = 2^3 = 8.$$

$$x^2 - 2x - 8 = 0 \implies (x-4)(x+2) = 0.$$

So $x = 4$ or $x = -2$. But $\log_2(-2)$ does not exist, so reject $x = -2$. The answer is $x = 4$.

Always check domain when logarithms are involved. It is easy to generate solutions that look valid algebraically but are meaningless.

CAUTION $\log(x+y) \neq \log x + \log y$. The product law is about $\log(xy)$, not a sum inside the log.

Solving exponential equations

SAME base first

Is there ever a way to solve an exponential equation without using a logarithm at all? Before anything else, look for a common base. If both sides can be written as powers of the same number, the exponents must be equal, provided the base is not 0 or 1: $a^p = a^q$ forces $p = q$ for $a > 0,\ a \neq 1$. This is the fastest route whenever it is available, and the intended one on non-calculator papers.

EXAMPLE 1.13

Solve $\left(\frac{1}{3}\right)^x = 9^{x+1}$.

Both sides are powers of 3: $\frac{1}{3} = 3^{-1}$ and $9 = 3^2$. Rewrite and equate exponents:

$$3^{-x} = 3^{2(x+1)} \implies -x = 2x + 2 \implies x = -\tfrac{2}{3}.$$

No logarithm was needed. Check for a shared base before anything else; look for common bases such as $2, 4, 8$ and $3, 9, 27$.

THE general strategy

When no common base exists, take a logarithm of both sides, use $\log(x^m) = m \log x$ to bring the exponent down, then solve what is now a linear or quadratic equation. The choice of which logarithm to use (log or ln) does not matter: pick whichever gives cleaner arithmetic.

EXAMPLE 1.14

Solve $5^x = 30$.

Take ln of both sides:

$$x \ln 5 = \ln 30 \implies x = \frac{\ln 30}{\ln 5} \approx 2.11.$$

Or equivalently, $x = \log_5 30$.

EXAMPLE 1.15

Solve $7^{x-2} = 5^{x+3}$ exactly.

With different bases on each side, take ln of both sides and use the power law:

$$(x-2)\ln 7 = (x+3)\ln 5.$$

Collect x on one side:

$$x(\ln 7 - \ln 5) = 3\ln 5 + 2\ln 7 \implies x = \frac{\ln 5^3 + \ln 7^2}{\ln(7/5)} = \frac{\ln 6125}{\ln(7/5)}.$$

TIP Give an exact answer, like $x = \dfrac{\ln 6125}{\ln(7/5)}$, whenever a question says *exact* or on a non-calculator paper (Paper 1). When a decimal is asked for, or on a calculator paper (Paper 2), round to three significant figures unless told otherwise. Three significant figures is the IB default, and rounding too early or giving too few figures costs accuracy marks.

From words to an equation

Many growth and decay problems use one model,

$$A(t) = A_0 \times r^t,$$

where A_0 is the starting amount and r is the multiplier for each period. The wording gives you r. A rise of 6% per year means $r = 1.06$. A fall of 6% means $r = 0.94$. A quantity that doubles each period has $r = 2$. When the change is continuous rather than in steps, the model uses base e:

$$A(t) = A_0\, e^{kt}.$$

Once the model is written, the logarithm provides the final step.

EXAMPLE 1.16

Sea ice in a Finnish bay covers 1200 km^2 and shrinks by 6% each year. After how many years will it have shrunk to 800 km^2?

Each year multiplies the area by 0.94, so after t years the area is $A(t) = 1200(0.94)^t$. Set this equal to the target and solve:

$$1200(0.94)^t = 800 \implies (0.94)^t = \tfrac{2}{3} \implies t = \frac{\ln(2/3)}{\ln 0.94} \approx 6.55.$$

The ice passes 800 km^2 partway through the seventh year.

EXAMPLE 1.17

Algae in a Finnish lake grow continuously at 12% per year, starting from 200 grams. When does the mass reach 500 grams?

Continuous growth uses base e, so $A(t) = 200\, e^{0.12t}$. Set it equal to the target:

$$200\, e^{0.12t} = 500 \implies e^{0.12t} = 2.5 \implies t = \frac{\ln 2.5}{0.12} \approx 7.6 \text{ years}.$$

The 12% here is a continuous rate, so it sits in the exponent as 0.12 directly, with no conversion to a yearly factor.

EXAMPLE 1.18

A medical isotope has a half-life of 6 hours: every 6 hours, half of it decays. An 80 mg sample is

prepared. How long until only 12 mg remains?
Halving every 6 hours means $A(t) = 80\left(\frac{1}{2}\right)^{t/6}$, with t in hours. Set this equal to the target and take a logarithm:

$$80\left(\tfrac{1}{2}\right)^{t/6} = 12 \implies \left(\tfrac{1}{2}\right)^{t/6} = 0.15 \implies t = 6 \cdot \frac{\ln 0.15}{\ln 0.5} \approx 16.4 \text{ hours.}$$

Every half-life problem has this form: the base is $\frac{1}{2}$, and the exponent is the elapsed time divided by the half-life.

YOUR TURN 1.3 The Logarithm: an Exponent Reversed

15. Evaluate each of the following without a calculator.

(a) $\log_2 32$ [1]

(b) $\log_3 \dfrac{1}{9}$ [2]

(c) $e^{2\ln 3}$ [2]

(d) $\ln e^5$ [1]

16. Rewrite each statement in the other form.

(a) Write $3^4 = 81$ in logarithmic form. [1]

(b) Write $\log_5 x = 3$ in exponential form, and hence find x. [2]

17. Solve each equation.

(a) $\log_4(x+1) = 2$ [2]

(b) $\log_2 x + \log_2(x-2) = 3$ [4]

18. Write each expression as a single logarithm.

(a) $\log_3 5 + \log_3 7$ [1]

(b) $\log_2 24 - \log_2 3$, and evaluate it [2]

(c) $3\log_a x + \log_a y$ [2]

(d) $2\ln a - 3\ln b + \ln c$ [2]

19. Express each logarithm in terms of the quantities given.

(a) $\log_5\left(\dfrac{a^2}{b}\right)$, in terms of $\log_5 a$ and $\log_5 b$ [2]

(b) $\log_3 18$, given $p = \log_3 2$ [3]

(c) $\log_{10} 72$, given $\log_{10} 2 = p$ and $\log_{10} 3 = q$ [3]

(d) $\log_5 75$, given $\log_5 3 = 0.682$ [3]

(e) $\log_4 9$, in terms of natural logarithms [1]

(f) $\log_8 x$, in terms of $\log_2 x$ [2]

20. Show that each statement is true.

(a) $\log_a(x^2-1)-\log_a(x+1)=\log_a(x-1)$, for $x>1$ [2]

(b) $\log_a b = \dfrac{1}{\log_b a}$ [2]

21. GDC Evaluate $\log_3 50$ to 3 significant figures. [2]

22. Solve each equation, giving your answer as an exact value.

(a) $5^{2x}=125$ [2]

(b) $4^x=8^{x-1}$ [3]

23. GDC Solve each equation, giving your answer to 3 significant figures.

(a) $2^x=10$ [2]

(b) $3^{x+1}=20$ [2]

24. Solve each equation. *(Hint: in (a) and (b), substitute u for the base raised to x)*

(a) $e^{2x}-5e^x+6=0$ [4]

(b) $2^{2x}-6\cdot 2^x+8=0$ [4]

(c) $3\cdot 2^x=2\cdot 3^x$ [3]

Chapter 1 · Exponents & Logarithms

Reflections

TOK John Napier spent twenty years computing logarithms by hand and published his tables in 1614, not out of mathematical curiosity, but to help sailors and astronomers multiply enormous numbers. He never used the word "logarithm" to mean what we mean today. Does mathematics develop because we need it, or do we find uses for what mathematicians have already built?

TOK The exponent laws force $a^0 = 1$ for every $a \neq 0$, but 0^0 is left undefined, because approaching it along different routes gives different answers. Does leaving 0^0 undefined reveal a limitation of mathematics or the precision of mathematics?

IM Base-10 logarithms feel natural to us because we count in base 10. But the choice is cultural, not mathematical. The Babylonians used base 60 (traces of which survive in our 60-minute hour and 360-degree circle). What does it mean for a mathematical tool to be "natural"?

IM The word *algebra* comes from *al-jabr*, a technique for solving equations described by the Persian mathematician al-Khwarizmi in ninth-century Baghdad; the word *algorithm* is a Latinisation of his own name. His treatises on equations, and the work of mathematicians across the wider Islamic world who followed him, were translated into Latin centuries later and became a foundation of European algebra, including the exponent notation used throughout this chapter. The rules you are applying here developed over many centuries and across many cultures.

RW The decibel (dB) scale for sound is logarithmic: the sound level is $10\log_{10}(I/I_0)$, comparing a sound's intensity I to a fixed reference intensity I_0, so each increase of 10 dB means ten times the intensity. The stellar magnitude scale in astronomy and the octave system in music work the same way. In each case, human perception responds approximately to the logarithm of the physical quantity, not to the quantity itself.

EXT Euler's identity: $e^{i\pi} + 1 = 0$. Five fundamental constants, one equation. It is the exponential function extended into the complex plane, connecting the circular motion of $e^{i\theta} = \cos\theta + i\sin\theta$ to the real exponential growth you have been studying here.

End-of-Chapter Problems — Chapter 1: Exponents & Logarithms

1. Solve $3^{2x+1} = 27^{x-1}$. [3]

2. Show that the equation $2^{x+3} - 2^x = k$ has a real solution only when $k > 0$, and find this solution in terms of k. [5]

3.

(a) Simplify $(2x^{-2}y^3)^3 \times (4x^4y^{-1})^{-1}$, giving your answer with positive indices. [3]
(b) Solve $8^x = 2^{x+6}$. [2]
(c) Show that $\left(a^{1/2} + a^{-1/2}\right)^2 = a + 2 + a^{-1}$, for $a > 0$. [2]

4. GDC The mass of the Earth is 5.97×10^{24} kg and the mass of the Moon is 7.35×10^{22} kg.

(a) Find how many times heavier the Earth is than the Moon, to 3 significant figures. [2]
(b) Find their combined mass, giving your answer in standard form to 3 significant figures. [2]

5. Prove that $\log_a b \cdot \log_b a = 1$ for $a, b > 0$, $a, b \neq 1$. Hence simplify $(\log_2 3)(\log_3 8)$. [4]

6. Given that $\log_a x = 3$ and $\log_a y = 5$, find each of the following.

(a) $\log_a(x^2y)$ [1] (b) $\log_a \sqrt{xy}$ [2]
(c) $\log_a\left(\dfrac{x}{y^2}\right)$ [1]

7. Let $\log_a 2 = p$ and $\log_a 3 = q$.

(a) Write $\log_a 6$ in terms of p and q. [1]
(b) Write $\log_a \sqrt{12}$ in terms of p and q. [2]
(c) Write $\log_a \dfrac{4}{9}$ in terms of p and q. [2]

8.

(a) Show that $\log_a \sqrt[n]{x} = \dfrac{1}{n}\log_a x$. [2]
(b) Hence solve $\log_2 \sqrt{x} + \log_4 x = 6$. [4]

9. Prove that $\log_a b \cdot \log_b c \cdot \log_c a = 1$ for all valid bases and arguments. Hence evaluate $\log_2 3 \cdot \log_3 5 \cdot \log_5 8$ without a calculator. [6]

10. Given that $\log_2 3 = a$, express each of the following in terms of a:

(a) $\log_2 9$ [1]
(b) $\log_4 3$ [2]
(c) $\log_8 27$ [2]
(d) $\log_3 8$ [2]

11. Solve $\log_3(x+5) - \log_3(x-1) = 2$. [4]

12. Solve $\log_6(x+2) + \log_6(x-3) = 1$, justifying any rejection of solutions. [4]

13. Solve the simultaneous equations:

$$\log_2 x + \log_2 y = 5, \qquad \log_2(x-y) = 3.$$

[5]

14. Solve $\log_x 8 = \log_4 x$. [5]

15. GDC Solve $\log_2(x^2 - 2x) = \log_4(x+4)$. [5]

16. GDC

(a) Express $\log_3 x - 2 = \log_9(x+6)$ as a quadratic equation in x, using change of base. [3]
(b) Hence find the value of x. [2]

17. Solve $\log_5(x^2 - 2x + 1) = 2\log_5(x-1) + 1$, or show that no solution exists. [5]

18. Solve $7^{x-2} = 5^{x+1}$, giving your answer in exact logarithmic form. [4]

19. Find the exact solution to $e^{2x} = 3e^x + 10$. *(Hint: let $u = e^x$)* [4]

20. Solve the equation $9^x - 4 \cdot 3^{x+1} + 27 = 0$. *(Hint: let $u = 3^x$; note $9^x = u^2$)* [5]

21. The number of bacteria in a culture doubles every 3 hours. Initially there are 500 bacteria.

(a) Write an expression for the number of bacteria after t hours. [2]
(b) Find the time, to the nearest minute, when there are 50 000 bacteria. [3]

22. GDC The population of a city is modelled by $P(t) = 250\,000 \times e^{0.032t}$, where t is the number of years after 2020.

(a) Write down the population in 2020. [1]

(b) Find the population in 2030, to the nearest thousand. [2]

(c) Find the year in which the population first exceeds 400 000. [3]

23. GDC The value V of a car after t years is given by $V = 32\,000 \times (0.85)^t$.

(a) Write down the initial value of the car. [1]

(b) Find the value after 5 years, to the nearest dollar. [2]

(c) Find the number of years until the value falls below $10 000. [3]

24. GDC The pH of a solution is defined by pH $= -\log_{10}[\mathrm{H}^+]$, where $[\mathrm{H}^+]$ is the hydrogen ion concentration in mol/L.

(a) A solution has $[\mathrm{H}^+] = 3.5 \times 10^{-4}$ mol/L. Find its pH to 2 decimal places. [2]

(b) A solution has pH $= 8.3$. Find its hydrogen ion concentration in standard form. [2]

(c) Solution A has pH $= 3$ and solution B has pH $= 7$. Find how many times more acidic solution A is than solution B. [2]

25. Let $f(x) = \log_3(2x-1)$ and $g(x) = 3^x + 2$. (Inverse functions and graph transformations are developed in Ch. 6; this question looks ahead to them.)

(a) Find the domain of f. [1]

(b) Find $f^{-1}(x)$. [3]

(c) Show that $f^{-1}(x) = \frac{1}{2}g(x) - \frac{1}{2}$, and describe the transformation of g this represents. [3]

26. GDC A radioactive substance decays so that its mass M grams after t years satisfies $M = M_0 e^{-kt}$, where M_0 and k are positive constants. After 10 years the mass is halved.

(a) Show that $k = \dfrac{\ln 2}{10}$. [2]

(b) Find the percentage of the original mass remaining after 25 years. [3]

(c) Find the time for the mass to reduce to 10% of its original value. [3]

27. GDC This question is about writing very large and very small quantities in standard form.

(a) A human cell has diameter about 2×10^{-5} m, and a water molecule about 2.8×10^{-10} m. Find how many times wider the cell is, giving your answer in standard form to three significant figures. [3]

(b) The observable universe is about 8.8×10^{26} m across, and the shortest length with any physical meaning, the Planck length, is about 1.6×10^{-35} m. Find the ratio of the two, in standard form. [3]

(c) A sheet of paper is 0.1 mm thick. It is folded in half, then in half again, and so on. Find the thickness after 20 folds, in metres and in standard form. [3]

(d) Find the least number of folds needed for the thickness to exceed the distance to the

Moon, 3.84×10^8 m. [4]

(e) Your answer to part (d) is far smaller than most people expect. Explain why, referring to how the thickness grows. [2]

28. GDC An amount P is invested. Bank A pays 4% interest compounded annually. Bank B pays 3.9% compounded monthly, so its value after t years is $P\left(1+\frac{0.039}{12}\right)^{12t}$.

(a) Write down an expression for the value in Bank A after t years. [2]

(b) Show that Bank B's value can be written as Pr^t, and find r to five decimal places. [3]

(c) Determine which bank gives more after 10 years, and by how much as a percentage of P. [3]

(d) Use logarithms to find how long an investment in Bank A takes to double. [4]

(e) Bank C pays 4% compounded continuously, giving $Pe^{0.04t}$. Show that Bank C always beats Bank A, and find the difference after 10 years as a percentage of P. [3]

29.

(a) Prove the change of base rule $\log_a b = \dfrac{\log_c b}{\log_c a}$, for valid bases a and c. *(Hint: let $x = \log_a b$, so $a^x = b$, then take $\log_c$ of both sides)* [3]

(b) Hence show that $\log_4 x = \frac{1}{2}\log_2 x$. [2]

(c) Solve $\log_2 x + \log_4 x + \log_{16} x = 7$. [4]

(d) Solve $\log_x 2 + \log_{x^2} 2 = 3$, giving an exact answer. [5]

Challenge Questions

30. *Investigation.* Consider $f(x) = \log_a x$ for different values of $a > 0$, $a \neq 1$.

(a) Show that $\log_{1/a} x = -\log_a x$ for all valid x. [3]

(b) Show that $\log_{a^n} x = \dfrac{1}{n}\log_a x$. [3]

(c) Using part (b), find all values of a for which $\log_a x = \log_{a^2}(x+2)$ has a solution, and find that solution. [5]

31. The loudness L of a sound (in decibels) is given by $L = 10\log_{10}\left(\dfrac{I}{I_0}\right)$, where I is the intensity in W/m^2 and $I_0 = 10^{-12}$ W/m^2 is the threshold of hearing.

(a) A concert has loudness 110 dB. Find the intensity I. [3]

(b) A whisper has loudness 30 dB. Show that the concert is 10^8 times more intense than the whisper. [3]

(c) Two sounds with intensities I_1 and I_2 are played together. Show that the combined loudness is not simply $L_1 + L_2$. Find the combined loudness when two 80 dB sounds are played simultaneously. [4]

32. Counting digits, and the first digit. This investigation uses logarithms to count digits, and then to explain a pattern that auditors use to detect faked data.

(a) Let N be a positive integer. Show that N has exactly $\lfloor \log_{10} N \rfloor + 1$ digits, where $\lfloor t \rfloor$ is the greatest integer not exceeding t. [3]

(b) Hence find the number of digits in 2^{1000}. [2]

(c) The first digit of N is d exactly when the fractional part of $\log_{10} N$ lies in $[\log_{10} d, \ \log_{10}(d+1))$. Verify this for $N = 2^7$. [3]

(d) Assume the fractional parts of $\log_{10}(2^n)$ are spread evenly over $[0, 1)$ as n increases. Show that the proportion of powers of 2 whose first digit is d is $\log_{10}\left(1 + \dfrac{1}{d}\right)$. [4]

(e) Evaluate this proportion for $d = 1$ and for $d = 9$. Then count the first digits of $2^1, 2^2, \ldots, 2^{30}$ on your GDC and compare. [5]

(f) The pattern in part (d) is known as Benford's law. Comment on how well your counts in part (e) support it, and state one reason why a sample of 30 might mislead. [3]

NUMBER & ALGEBRA

Chapter 2

Sequences & Series

2

SYLLABUS 1.2 1.3 1.4 1.8 1.9 1.10

A quantity that keeps doubling grows slowly at first, and then very fast. A sheet of paper is about 0.1 mm thick. Each fold doubles the thickness, so 42 folds would give 0.1×2^{42} mm, which is about $440{,}000$ km. The Moon is about $384{,}000$ km away. Nobody can fold paper 42 times, but the calculation is correct.

One more fold would reach the Moon and come back. Can you see why?

This chapter is about lists of numbers built by repeatedly adding a fixed amount, and lists built by repeatedly multiplying by a fixed factor. You will learn to find any term without writing out the terms before it, to add the first n terms in a single calculation, and to see why some sums that never end still have a finite total. The same ideas are used to value a savings plan, and, with a counting argument, to expand a bracket like $(a+b)^{12}$ without multiplying out twelve brackets.

2.1 Arithmetic Sequences

A theatre has 20 seats in the front row, 23 in the second, and 26 in the third. Each row has 3 more seats than the one before it. To find how many seats are in row 50, you should not have to count up one row at a time. There is a clear pattern: each row adds 3 seats. A regular pattern like this can be written as a formula, and finding that formula is the point of this section.

A sequence where each term is obtained by adding a fixed constant to the previous one is called **arithmetic** (also called an **arithmetic progression**, or **AP**). That constant is the **common difference** d.

DEF An **arithmetic sequence** has the form $u_1,\ u_1+d,\ u_1+2d,\ \ldots$, where u_1 is the first term and d is the common difference.

For any two consecutive terms: $d = u_{n+1} - u_n$.

The general term

The n-th term is the first term plus $(n-1)$ steps of size d. Starting from u_1, you take one step to reach u_2, two steps to reach u_3, and so on.

KEY · IB

$$u_n = u_1 + (n-1)d$$

EXAMPLE 2.1

An arithmetic sequence has first term 7 and common difference -3. Find u_{20}.

You need 19 steps of size -3 from $u_1 = 7$:

$$u_{20} = 7 + (20-1)(-3) = 7 - 57 = -50.$$

A negative common difference means the sequence is decreasing. The general term can be negative even when the first term is positive.

EXAMPLE 2.2

In an arithmetic sequence, $u_5 = 11$ and $u_9 = 27$. Find u_1 and d.

From u_5 to u_9 is exactly 4 steps:

$$4d = u_9 - u_5 = 27 - 11 = 16 \implies d = 4.$$

Then $u_1 = u_5 - 4d = 11 - 16 = -5$.

EXAMPLE 2.3

Find the number of terms in the arithmetic sequence $5, 9, 13, \ldots, 101$.

Here $d = 4$. Set $u_n = 101$:

$$5 + (n-1)(4) = 101 \implies 4(n-1) = 96 \implies n = 25.$$

CAUTION The formula is $u_1 + (n-1)d$, not $u_1 + nd$. For the first term $(n = 1)$, you add zero differences, not one.

YOUR TURN 2.1 Arithmetic Sequences

1. Find the term indicated.

(a) u_7 of $3, 7, 11, \ldots$ [1]

(b) u_{15} of the AP with $u_1 = 12$, $d = -4$ [1]

(c) u_{10} of $-5, -1, 3, \ldots$ [1]

(d) u_{25} of the AP with $u_1 = 100$, $d = -7$ [2]

2. Find the quantity indicated.

(a) d, given $u_1 = 5$ and $u_9 = 37$ [2]

(b) u_1, given $u_6 = 23$ and $d = 3$ [2]

(c) d, given $u_3 = 8$ and $u_{11} = 32$ [2]

(d) u_1, given $u_{12} = 50$ and $d = 4$ [2]

3. Find the number of terms in the AP $4, 7, 10, \ldots, 100$. [2]

4. Given $u_4 = 11$ and $u_{10} = 29$, find u_1 and d. [3]

5. Show that the sequence $5, 11, 17, \ldots$ has a term equal to 101. State which term it is. [3]

2.2 Arithmetic Series

When you need the total of the first n terms of an arithmetic sequence, adding them one at a time is always possible but rarely necessary. A pattern reduces the whole sum to one short calculation.

Consider adding all integers from 1 to 100. Write the sum forwards and backwards:

$$1 + 2 + 3 + \cdots + 99 + 100$$

$$100 + 99 + 98 + \cdots + 2 + 1$$

Every vertical pair sums to 101, and there are 100 such pairs. So twice the sum equals 100×101, giving $S_{100} = 5050$.

The same pairing works for any arithmetic series: the first and last term always sum to the same value, $u_1 + u_n$. There are n such pairs when you write the series twice, so $2S_n = n(u_1 + u_n)$, and therefore:

KEY · IB

$$S_n = \frac{n}{2}(u_1 + u_n) \qquad = \qquad \frac{n}{2}(2u_1 + (n-1)d)$$

The first form is faster when you know u_n. Substituting $u_n = u_1 + (n-1)d$ gives the second, which works when only u_1 and d are known.

EXAMPLE 2.4

Find the sum of the first 30 terms of the arithmetic sequence $2, 7, 12, \dots$

Here $u_1 = 2$ and $d = 5$. The last term is not given, so use the second form:

$$S_{30} = \frac{30}{2}(2(2) + 29(5)) = 15(4 + 145) = 15 \times 149 = 2235.$$

EXAMPLE 2.5

Find the sum $4 + 7 + 10 + \cdots + 100$.

First, $d = 3$. Find the number of terms: $4 + (n-1)(3) = 100 \Rightarrow n = 33$.

Then use the first form, since both u_1 and u_{33} are known:

$$S_{33} = \frac{33}{2}(4 + 100) = \frac{33 \times 104}{2} = 1716.$$

EXAMPLE 2.6

The sum of the first n terms of an arithmetic series is given by $S_n = 3n^2 + 5n$. Find u_1, d, and the general term u_n.

$u_1 = S_1 = 3 + 5 = 8$.

$u_2 = S_2 - S_1 = (12 + 10) - 8 = 14$, so $d = u_2 - u_1 = 6$.

For the general term: $u_n = S_n - S_{n-1} = 3n^2 + 5n - 3(n-1)^2 - 5(n-1)$.

Expanding: $u_n = 3n^2 + 5n - 3n^2 + 6n - 3 - 5n + 5 = 6n + 2$.

Check: $u_1 = 8$ and $u_2 = 14$. Consistent.

CAUTION S_n is the total of the first n terms; u_n is just the n-th term. Confusing the two is the most common error in this topic. A question asking for a ``term'' wants $u_n$; a question asking for a ``sum'' wants S_n.

Application: simple interest

The standard financial application of an arithmetic sequence is **simple interest**: the same fixed amount of interest is added each year, so the balance increases by the same amount each year.

EXAMPLE 2.7

$2000 is deposited in an account paying 3% simple interest per year. Find the balance after 15 years.

The interest is $3\% \times 2000 = \$60$ every year, so the balances form an arithmetic sequence with $d = 60$. After 15 years, 15 payments of interest have been added:

$$2000 + 15 \times 60 = \$2900.$$

YOUR TURN 2.2 Arithmetic Series

6. Find the sum indicated.

(a) S_{12} for the AP $2, 5, 8, \ldots$ [2]

(b) S_{20} for the AP with $u_1 = 100$, $d = -5$ [2]

(c) S_{15} for the AP $7, 11, 15, \ldots$ [2]

(d) S_{25} for the AP with $u_1 = -12$, $d = 5$ [2]

7. Find the sum $6 + 10 + 14 + \cdots + 90$. [3]

8. Find the sum of all integers from 1 to 200 that are multiples of 6. [3]

9. Find n if $S_n = 234$ for the AP $3, 6, 9, \ldots$ [3]

10. An AP has $u_1 = 15$, $u_n = -9$ and $n = 13$. Find S_n. [2]

11. Given $S_n = 4n^2 - n$, find u_1, d and the general term u_n. [4]

2.3 Geometric Sequences

A geometric sequence multiplies by the same number at each step. Start at 1 and double: $1, 2, 4, 8, 16, \ldots$ Many real quantities change in this way. Bacteria double in every generation. A ball bounces to 60% of the height it fell from. Light loses half its intensity for each extra filter. In every case, each new value is the previous value multiplied by a fixed number.

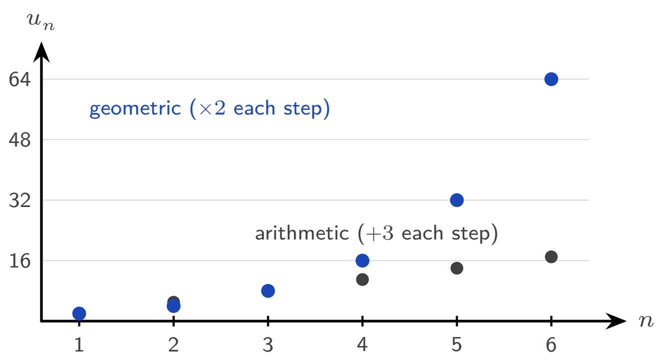


Figure 2.1 Adding a fixed amount grows in a straight line; multiplying by a fixed factor grows explosively.

A sequence where each term is obtained by multiplying the previous term by a fixed constant is called **geometric** (also called a **geometric progression**, or **GP**). That constant is the **common ratio** r.

DEF A **geometric sequence** has the form $u_1,\ u_1 r,\ u_1 r^2,\ \ldots$, where u_1 is the first term and r is the common ratio.
For any two consecutive terms: $r = \dfrac{u_{n+1}}{u_n}$.

The general term

Starting from u_1, each step multiplies by r. After $(n-1)$ multiplications you reach u_n:

KEY · IB

$$u_n = u_1 \cdot r^{n-1}$$

EXAMPLE 2.8

A geometric sequence has $u_1 = 5$ and $r = 3$. Find u_6.

$$u_6 = 5 \cdot 3^5 = 5 \times 243 = 1215.$$

EXAMPLE 2.9

A geometric sequence has $u_3 = 12$ and $u_6 = 96$. Find u_1 and r.
From u_3 to u_6 is 3 multiplicative steps: $u_6 = u_3 \cdot r^3$, so

$$r^3 = \frac{96}{12} = 8 \implies r = 2.$$

Then $u_1 = u_3/r^2 = 12/4 = 3$.

EXAMPLE 2.10

Find the number of terms in the geometric sequence $2, 6, 18, \dots, 4374$.

Here $r = 3$. Set $u_n = 4374$:

$$2 \cdot 3^{n-1} = 4374 \implies 3^{n-1} = 2187 = 3^7 \implies n = 8.$$

TIP To test whether a sequence is arithmetic, check that differences are constant. To test whether it is geometric, check that ratios are constant. Never apply both tests to the same sequence expecting both to be true.

YOUR TURN 2.3 Geometric Sequences

12. Find the term indicated.

(a) u_6 of the GP $2, 6, 18, \dots$ [1]

(b) u_7 of the GP with $u_1 = 96$, $r = \frac{1}{2}$ [2]

(c) u_5 of $\frac{1}{4}, \frac{1}{2}, 1, \dots$ [2]

(d) u_8 of the GP with $u_1 = 3$, $r = -2$ [2]

13. Find the quantity indicated.

(a) r, given $u_1 = 5$ and $u_4 = 40$ [2]

(b) u_1, given $u_4 = 24$ and $r = 2$ [2]

(c) r, given $u_1 = 81$ and $u_5 = 1$ [3]

(d) u_1, given $u_3 = 45$ and $r = 3$ [2]

14. Find the number of terms in the GP $3, 6, 12, \dots, 768$. [2]

15. Given $u_2 = 10$ and $u_5 = 1250$, find u_1 and r. [3]

16. The 3rd term of a GP is 12 and the 6th term is 96. Find the first term. [3]

2.4 Geometric Series

Adding the terms of a geometric sequence needs a different method. In an arithmetic series, the first and last terms have the same total as the second and second-to-last terms, so pairing works. In a geometric series these pair totals are all different, so pairing does not help. Instead there is an algebraic method: multiply the sum by r, line the two sums up, and subtract.

Let $S = u_1 + u_1 r + u_1 r^2 + \cdots + u_1 r^{n-1}$. Multiply by r:

$$rS = u_1 r + u_1 r^2 + \cdots + u_1 r^n.$$

Every term appears in both lines except u_1 in the first line and $u_1 r^n$ in the second. Subtract rS from S:

$$S(1-r) = u_1 - u_1 r^n = u_1(1-r^n).$$

Divide both sides by $(1-r)$ to get the formula.

KEY · IB

$$S_n = \frac{u_1(1-r^n)}{1-r} \quad = \quad \frac{u_1(r^n-1)}{r-1}, \qquad r \neq 1$$

Both forms are equivalent. When $0 < r < 1$, use the first form: both $1 - r^n$ and $1 - r$ are then positive, so no negative signs appear. When $r > 1$ the second form is tidier, for the same reason. If r is negative, either form works, but r^n changes sign with n, so take care with the arithmetic.

EXAMPLE 2.11

Find the sum of the first 10 terms of the geometric series $3 + 6 + 12 + \cdots$

Here $u_1 = 3$ and $r = 2$:

$$S_{10} = \frac{3(2^{10}-1)}{2-1} = 3 \times 1023 = 3069.$$

EXAMPLE 2.12

A geometric series has $u_1 = 80$ and $r = \frac{1}{2}$. Find the least value of n such that $S_n > 159$.

$$S_n = \frac{80\left(1 - \left(\frac{1}{2}\right)^n\right)}{1 - \frac{1}{2}} = 160\left(1 - \frac{1}{2^n}\right).$$

Require $160\left(1 - \frac{1}{2^n}\right) > 159$, so $\frac{1}{2^n} < \frac{1}{160}$, giving $2^n > 160$.

Since $2^7 = 128 < 160$ and $2^8 = 256 > 160$, the least value is $n = 8$.

CAUTION The formula requires $r \neq 1$. If $r = 1$, every term equals u_1 and $S_n = nu_1$. Substituting $r = 1$ into the formula gives $0/0$, which is undefined.

Solving for n in geometric series often requires taking logarithms of both sides. See Ch. 1, §1.3.

YOUR TURN 2.4 Geometric Series

17. Find the sum indicated.

(a) S_8 for the GP $1, 3, 9, \ldots$ [2]

(b) S_{10} for the GP with $u_1 = 6$, $r = \frac{1}{3}$ [2]

(c) S_6 for the GP $128, 64, 32, \ldots$ [2]

(d) S_6 for the GP with $u_1 = 1$, $r = -2$ [2]

18. The sum of the first three terms of a GP is 21 and the common ratio is 2. Find the first term. [2]

19. Find n such that $S_n = 63$ for the GP $1, 2, 4, \dots$ [2]

20. Find the sum $3 + 6 + 12 + \cdots + 1536$. [2]

21. Given $S_3 = 7$ and $S_6 = 63$, find u_1 and r. [4]

2.5 Sigma Notation

By now you have written several sums in the form $u_1 + u_2 + \cdots + u_n$. There is a compact notation for this that appears throughout mathematics, physics, and statistics, and it is worth learning well.

The Greek letter Σ (capital sigma) is used as shorthand for "sum". The expression

$$\sum_{k=1}^{n} u_k = u_1 + u_2 + u_3 + \cdots + u_n$$

is read "the sum of u_k from $k = 1$ to n." The variable k is the **index of summation**: it takes each integer value from the lower limit to the upper limit, the corresponding term is evaluated, and all terms are added.

DEF $\displaystyle\sum_{k=m}^{n} f(k) = f(m) + f(m+1) + f(m+2) + \cdots + f(n)$, where $m \leq n$.
The index k is a **dummy variable**: the result depends only on f, m, and n, not on the name chosen for the index.

Properties of sigma notation

These three properties are just familiar rules of arithmetic, written in sigma form. They are used constantly when simplifying or manipulating sums.

KEY For a constant c and sequences $\{a_k\}$, $\{b_k\}$:

$\sum_{k=1}^{n} c = nc$	a constant summed n times
$\sum_{k=1}^{n} c\, a_k = c \sum_{k=1}^{n} a_k$	a constant factor comes outside the sum
$\sum_{k=1}^{n} (a_k + b_k) = \sum_{k=1}^{n} a_k + \sum_{k=1}^{n} b_k$	a sum splits over addition

EXAMPLE 2.13

Write $3 + 7 + 11 + 15 + 19$ in sigma notation.

The terms form an arithmetic sequence with $u_1 = 3$, $d = 4$: the k-th term is $u_k = 4k - 1$.

$$3 + 7 + 11 + 15 + 19 = \sum_{k=1}^{5} (4k - 1).$$

EXAMPLE 2.14

Evaluate $\sum_{k=1}^{6} 5 \cdot 2^{k-1}$.

This is a geometric series with $u_1 = 5$ and $r = 2$:

$$\sum_{k=1}^{6} 5 \cdot 2^{k-1} = S_6 = \frac{5(2^6 - 1)}{2 - 1} = 5 \times 63 = 315.$$

EXAMPLE 2.15

Evaluate $\sum_{k=3}^{10} (2k + 1)$.

The lower limit is 3, not 1. Use the fact that the full sum minus the missing part gives the partial sum:

$$\sum_{k=3}^{10} (2k + 1) = \sum_{k=1}^{10} (2k + 1) - \sum_{k=1}^{2} (2k + 1).$$

The first sum is an arithmetic series: $u_1 = 3$, $u_{10} = 21$, so $S_{10} = \frac{10}{2}(3 + 21) = 120$.

The second sum is just $3 + 5 = 8$.

Result: $120 - 8 = 112$.

CAUTION $\sum_{k=1}^{n} (a_k \cdot b_k) \neq \left(\sum a_k\right)\left(\sum b_k\right)$. You can split a sum over addition but not over multiplication. This is one of the most common misapplications of sigma notation.

YOUR TURN 2.5 Sigma Notation

22. Write each sum in sigma notation.

(a) $5+10+15+20+25$ [1]
(b) $4+8+16+32+64$ [1]
(c) $3+6+9+\cdots+30$ [2]
(d) $1+4+9+16+25$ [2]

23. Evaluate $\displaystyle\sum_{k=1}^{6}(3k-2)$. [2]

24. Evaluate $\displaystyle\sum_{k=1}^{5}2\cdot 3^{k-1}$. [2]

25. Evaluate $\displaystyle\sum_{k=4}^{8}(2k+1)$. [2]

26. Find $\displaystyle\sum_{k=1}^{n}5$ and $\displaystyle\sum_{k=1}^{n}5k$ using the sigma properties. [2]

2.6 Infinite Geometric Series

Cut a pizza in half. Eat one half and keep the other. Cut what is left in half again, eat one piece, and keep the rest. If you could continue forever, how much pizza would you eat?

The pieces you eat are $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{8}$, $\frac{1}{16}$, ... There are infinitely many of them, and every piece is larger than zero. Adding infinitely many positive numbers sounds like it must give an infinite total. But you began with one pizza, so you can never eat more than one pizza. The total can only be 1.

That is the surprise of this section: *an addition that never ends can still have a finite answer.*

Add the pieces one at a time and record the running total after each step. Each running total is a **partial sum**:

$$S_1=0.5,\quad S_2=0.75,\quad S_3=0.875,\quad S_4=0.9375,\quad \ldots$$

Two things happen at once. Every partial sum is larger than the one before it, and every partial sum is still smaller than 1. Look at the distance from 1: it is 0.5, then 0.25, then 0.125. The distance halves at every step, so it never becomes zero, but it becomes as small as you wish. This value that the partial sums approach is called the **limit**, and here the limit is 1.

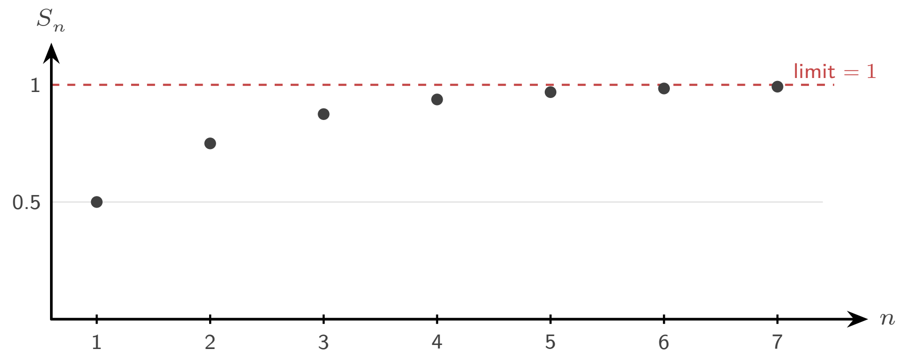


Figure 2.2 The partial sums of $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots$ climb toward 1 but never pass it.

The pizza works because each piece is smaller than the piece before it. That is exactly what $|r| < 1$ means: every term is a fraction of the previous term, so the terms shrink toward zero.

Now change one number and the behaviour changes completely. Take $1 + 2 + 4 + 8 + \cdots$, where $r = 2$. Each term is larger than the last, the partial sums are $1, 3, 7, 15, \ldots$, and they pass every value you can name. There is no finite total.

A series whose partial sums approach a finite limit is said to **converge**. A series whose partial sums grow without any bound is said to **diverge**. A geometric series converges when $|r| < 1$ and diverges when $|r| \geq 1$.

To turn this into a formula, start from the finite sum and let n grow without bound. When $|r| < 1$, the power r^n shrinks toward 0:

$$S_n = \frac{u_1(1 - r^n)}{1 - r} \xrightarrow{n \to \infty} \frac{u_1(1 - 0)}{1 - r} = \frac{u_1}{1 - r}.$$

KEY · IB For a geometric series with $|r| < 1$:

$$S_\infty = \frac{u_1}{1 - r}$$

EXAMPLE 2.16

Find the sum to infinity of $6 + 4 + \frac{8}{3} + \cdots$

The common ratio is $r = 4/6 = \frac{2}{3}$. Since $|r| < 1$ the series converges:

$$S_\infty = \frac{6}{1 - \frac{2}{3}} = \frac{6}{\frac{1}{3}} = 18.$$

EXAMPLE 2.17

Express $0.\overline{27} = 0.272727\ldots$ as an exact fraction.

Write as an infinite geometric series with $u_1 = \frac{27}{100}$ and $r = \frac{1}{100}$:

$$0.272727\ldots = \frac{27/100}{1 - 1/100} = \frac{27/100}{99/100} = \frac{27}{99} = \frac{3}{11}.$$

EXAMPLE 2.18

A geometric series has first term a and common ratio r. Given $S_\infty = 8$ and $u_2 = 2$, find a and r.

From S_∞: $\dfrac{a}{1-r} = 8$, so $a = 8(1-r)$.
From $u_2 = ar = 2$: substitute a:

$$8(1-r)r = 2 \implies 8r^2 - 8r + 2 = 0 \implies 4r^2 - 4r + 1 = 0 \implies (2r-1)^2 = 0.$$

So $r = \frac{1}{2}$, and $a = 8\left(1 - \frac{1}{2}\right) = 4$.

CAUTION Always verify $|r| < 1$ before applying the infinite sum formula. If $|r| \geq 1$, no finite sum to infinity exists.

YOUR TURN 2.6 Infinite Geometric Series

27. Find S_∞ for each series.

(a) $12 + 4 + \frac{4}{3} + \cdots$ [2]
(b) the GP with $u_1 = 10$, $r = \frac{3}{5}$ [2]
(c) $9 + 3 + 1 + \cdots$ [2]
(d) the GP with $u_1 = 8$, $r = -\frac{1}{4}$ [2]

28. Find the quantity indicated.

(a) u_1, given $S_\infty = 20$ and $r = \frac{1}{5}$ [2]
(b) r, given $u_1 = 6$ and $S_\infty = 9$ [2]
(c) u_1, given $S_\infty = 12$ and $r = \frac{1}{4}$ [2]
(d) r, given $u_1 = 15$ and $S_\infty = 12$ [2]

29. Express $0.\overline{5} = 0.5555\ldots$ as an exact fraction. [2]

30. State whether $6 + 9 + 13.5 + \cdots$ converges. Give a reason. [2]

31. A GP has $u_1 = 16$ and $u_3 = 4$. Find S_∞. [3]

2.7 Financial Applications

A bank pays you 4% per year. In the first year you earn interest on your deposit. In the second year you earn interest on the deposit *and* on the interest from the first year. The amount added is different every year, so the balances do not form an arithmetic sequence. They form a geometric one: each year the balance is multiplied by 1.04.

This is **compound interest**, and it is why money left alone for a long time grows far more than most people expect. The balance after any number of years is simply a term of a geometric sequence.

Interest is often added more than once a year. If a present value PV earns interest at a nominal annual rate of $r\%$, compounded k times per year, then after n years:

KEY · IB

$$FV = PV \times \left(1 + \frac{r}{100k}\right)^{kn}$$

The quantity $\left(1 + \frac{r}{100k}\right)$ is the common ratio. The total number of compounding periods is kn.

EXAMPLE 2.19

\$5 000 is invested at 4% per year, compounded monthly. Find the value after 3 years.

Here $PV = 5000$, $r = 4$, $k = 12$, $n = 3$:

$$FV = 5000 \times \left(1 + \frac{4}{1200}\right)^{36} \approx \$5636.36.$$

EXAMPLE 2.20

How many complete years does it take for an investment to double at 6% per year, compounded annually?

Set $FV = 2 \cdot PV$:

$$(1.06)^n = 2 \implies n = \frac{\ln 2}{\ln 1.06} \approx 11.9.$$

After 12 complete years the investment has more than doubled.

TIP The Rule of 72: dividing 72 by the annual interest rate gives an approximate doubling time in years. At 6%, that is $72/6 = 12$ years, matching the exact calculation above.

Depreciation

Some things lose value instead of gaining it. A car, a laptop, or a machine may lose a fixed *percentage* of its value each year. This is called **depreciation**.

No new formula is needed. Use the same compound-interest formula with a **negative** rate r. A loss of 12% per year means $r = -12$, so the common ratio becomes

$$1 + \frac{r}{100} = 1 - \frac{12}{100} = 0.88,$$

a number less than 1. Multiplying by a ratio less than 1 at every step makes the values shrink, so the sequence of values is still geometric.

EXAMPLE 2.21

A car bought for \$28 000 loses 12% of its value each year. Find its value after 5 years.
Here $r = -12$ and $k = 1$, so each year the value is multiplied by 0.88:

$$FV = 28\,000 \times (0.88)^5 \approx \$14\,776.$$

After five years the car has lost nearly half its value. Notice that the value never reaches zero: each year removes 12% of a smaller amount than the year before.

Inflation and real value

There is a second geometric process working against your savings. **Inflation** is the yearly rise in prices. Interest increases the number in your account, while inflation reduces what that number can buy. If your account grows by 2% in a year and prices rise by 3% in the same year, you hold more money but you can buy less with it.

The **real value** of an investment is its future value adjusted for the price rises over the same period. It measures the gain in purchasing power rather than the gain in the number.

EXT Real value is not required by the syllabus, but it uses the same geometric reasoning. With interest producing a future value FV over n years, and prices rising at $i\%$ per year:

$$\text{real value} = \frac{FV}{\left(1 + \frac{i}{100}\right)^n}$$

Divide by the inflation factor because the same goods cost $\left(1 + \frac{i}{100}\right)^n$ times as much after n years.

EXAMPLE 2.22

\$10 000 is invested for 5 years at 4.5% per year, compounded annually. Prices rise at 2.5% per year. Find the future value and the real value of the investment.

The balance grows by a factor of 1.045 each year:

$$FV = 10\,000 \times (1.045)^5 \approx \$12\,461.82.$$

Now adjust for five years of price rises:

$$\text{real value} = \frac{12\,461.82}{(1.025)^5} \approx \$11\,014.43.$$

The account shows a gain of about $2460, but in purchasing power the investor is only about $1014 better off. An interest rate means little until you compare it with inflation.

TIP Your GDC's built-in Finance (TVM) solver handles compound interest, depreciation, and unknown rates or times directly: enter the known values of N, $I\%$, PV, FV, and the compounding frequency, and solve for the missing one. In examinations, state the values you entered.

Solving for n in financial problems requires exponential equations. See Ch. 1, §1.3.

YOUR TURN 2.7 Financial Applications

32. GDC Find the future value of each investment.

(a) $3 000 at 4% p.a., compounded annually, for 5 years [2]

(b) $8 000 at 6% p.a., compounded monthly, for 3 years [2]

(c) $500 at 3.6% p.a., compounded monthly, for 10 years [2]

(d) $2 000 at 5% p.a., compounded quarterly, for 4 years [2]

33. GDC Find the annual interest rate if $5 000 grows to $5 828.84 in 3 years, compounded annually. [3]

34. GDC How many complete years does it take $2 000 to double at 9% p.a., compounded annually? [3]

35. GDC Find the present value needed to accumulate $15 000 in 8 years at 5% p.a., compounded quarterly. [3]

36. A car valued at $25 000 depreciates at 15% per year. Find its value after 4 years. [2]

2.8 Counting Principles HL

Before the chapter's final result, a short look at a question that sounds simple: *how many ways?* The answer is what makes the binomial theorem work, and it starts with one principle.

If one choice can be made in m ways and a second, independent choice in n ways, then the pair of choices can be made in $m \times n$ ways. This is the **multiplication principle**, and it extends to more choices: a menu offering 4 starters, 5 mains, and 3 desserts allows $4 \times 5 \times 3 = 60$ different meals.

Arranging n distinct objects in a row is the principle applied n times: n choices for the first position, $n-1$ for the second, and so on down to 1. The total is n **factorial**:

KEY

$$n! = n \times (n-1) \times (n-2) \times \cdots \times 2 \times 1, \qquad 0! = 1$$

Permutations and combinations

Two standard questions arise when selecting r objects from n distinct ones. If the *order* of selection matters, the count is a **permutation**. If only the final collection matters, it is a **combination**.

The two counts are closely related. Suppose you choose 3 letters from A, B, C, D, E. The selection $\{A, B, C\}$ can be written in $3! = 6$ different orders: ABC, ACB, BAC, BCA, CAB, CBA. A permutation count treats these six as six different results. A combination count treats them as one. So the combination count is the permutation count divided by $r!$:

$$\binom{n}{r} = \frac{{}^{n}P_{r}}{r!}.$$

KEY · IB HL Selecting r from n distinct objects:

$${}^{n}P_{r} = \frac{n!}{(n-r)!} \quad \text{(ordered)}, \qquad \binom{n}{r} = {}^{n}C_{r} = \frac{n!}{r!\,(n-r)!} \quad \text{(unordered)}$$

EXAMPLE 2.23

Eight runners compete. In how many ways can gold, silver, and bronze be awarded?

The three medals are distinguishable, so order matters:

$$^8P_3 = 8 \times 7 \times 6 = 336.$$

EXAMPLE 2.24

A committee of 5 is chosen from 6 boys and 5 girls. How many committees contain exactly 2 girls?

Choose the girls and the boys separately, then multiply:

$$\binom{5}{2}\binom{6}{3} = 10 \times 20 = 200.$$

A committee has no internal order, so combinations are correct throughout, and the multiplication principle combines the two independent choices.

CAUTION Ask one question before every counting problem: *does order matter?* Medals, passwords, and rankings: yes, permutations. Committees, hands of cards, and subsets: no, combinations.

TIP Your GDC computes both: nPr and nCr. Arrangements with repeated identical objects and circular arrangements are not required in this course.

YOUR TURN 2.8 Counting Principles HL

37. In how many ways can 6 different books be arranged on a shelf? [1]

38. A 4-digit security code uses digits from 1 to 9 with no digit repeated. How many such codes are there? [2]

39. Find the number of ways of choosing a committee of 3 from 12 people. [1]

40. From 7 men and 6 women, find the number of ways of choosing a group of 2 men and 2 women. [2]

41. Solve $\binom{n}{2} = 45$. [3]

2.9 The Binomial Theorem

Expanding $(a+b)^2$ is quick. Expanding $(a+b)^3$ takes longer. Expanding $(a+b)^{12}$ by multiplying out twelve brackets is not realistic. Fortunately the answer follows a pattern, so no bracket multiplication is needed at all.

Look at the small cases:

$$(a+b)^1 = a+b, \qquad (a+b)^2 = a^2+2ab+b^2, \qquad (a+b)^3 = a^3+3a^2b+3ab^2+b^3.$$

Two features are regular. The power of a falls from n to 0 while the power of b rises from 0 to n, and in every term the two powers add to n. The coefficients form a row of **Pascal's triangle**, in which each entry is the sum of the two entries above it:

$$\begin{array}{ccccccccccc} & & & & & 1 & & & & & \\ & & & & 1 & & 1 & & & & \\ & & & 1 & & 2 & & 1 & & & \\ & & 1 & & 3 & & 3 & & 1 & & \\ & 1 & & 4 & & 6 & & 4 & & 1 & \\ 1 & & 5 & & 10 & & 10 & & 5 & & 1 \end{array}$$

Counting explains why. Multiplying out $(a+b)^n$ means choosing a or b from each of the n brackets. A term $a^{n-r}b^r$ arises once for every way of choosing which r brackets contribute a b, and that count is exactly $\binom{n}{r}$. The coefficients of the expansion *are* the combinations of the previous section.

KEY · IB For $n \in \mathbb{N}$:

$$(a+b)^n = a^n + \binom{n}{1}a^{n-1}b + \cdots + \binom{n}{r}a^{n-r}b^r + \cdots + b^n,$$

$$\text{where} \quad \binom{n}{r} = \frac{n!}{r!\,(n-r)!}.$$

Pascal's triangle is fastest for small n. When n is large, or when a question asks for one coefficient only, use the single term $\binom{n}{r}a^{n-r}b^r$ shown in the middle of the expansion above and choose the r you need.

TIP The booklet shows that single term inside the expansion, but it is worth naming it. The **general term** is

$$T_{r+1} = \binom{n}{r}a^{n-r}b^r,$$

so $r = 0$ gives the first term, $r = 1$ the second, and r gives term number $r+1$. The subscript is $r+1$ because r counts how many factors of b the term contains, and the first term contains none. In an examination, write down this term, find the r that produces the power you want, then substitute.

EXAMPLE 2.25

Expand $(x+2)^4$.

Row 4 of the triangle gives coefficients $1, 4, 6, 4, 1$, with the powers of 2 increasing as the powers of x decrease:

$$(x+2)^4 = x^4 + 4x^3(2) + 6x^2(4) + 4x(8) + 16 = x^4 + 8x^3 + 24x^2 + 32x + 16.$$

EXAMPLE 2.26

Find the coefficient of x^3 in $(2x-3)^5$.

The general term is $\binom{5}{r}(2x)^{5-r}(-3)^r$. For x^3, take $5-r=3$, so $r=2$:

$$\binom{5}{2}(2x)^3(-3)^2 = 10 \times 8x^3 \times 9 = 720x^3.$$

The coefficient is 720. Keep the sign inside the bracket with $b=-3$: the $(-3)^r$ produces the alternating signs automatically.

EXAMPLE 2.27

Find the term independent of x in $\left(x^2 + \frac{2}{x}\right)^6$.

Track the power of x in the general term:

$$T_{r+1} = \binom{6}{r}(x^2)^{6-r}\left(\frac{2}{x}\right)^r = \binom{6}{r}2^r\, x^{12-3r}.$$

Independent of x means $12-3r=0$, so $r=4$:

$$\binom{6}{4}2^4 = 15 \times 16 = 240.$$

TIP $\binom{n}{r}$ can also be read from a GDC table. To solve $\binom{6}{r} = 20$, generate $\binom{6}{r}$ for $r = 0, \dots, 6$ and read off $r=3$. Symmetry, $\binom{n}{r} = \binom{n}{n-r}$, halves the search.

YOUR TURN 2.9 The Binomial Theorem

42. Expand or write down the terms requested.

(a) Expand $(x+3)^4$. [2]

(b) Write down the first three terms, in ascending powers of x, of $(1 + 2x)^7$. [2]

(c) Expand $(2x-1)^4$. [2]

(d) Write down the first three terms, in ascending powers of x, of $(1 - 3x)^6$. [2]

43. Find the coefficient of x^2 in the expansion of $(3x-2)^5$. [3]

44. Find the term independent of x in $\left(x+\dfrac{1}{x^2}\right)^9$. [3]

45. GDC Find the values of r for which $\dbinom{8}{r} = 56$. [2]

46. Find the middle term in the expansion of $(x-2y)^6$. [3]

2.10 The Binomial Series HL

So far n has been a positive whole number, because $(a+b)^n$ meant n brackets multiplied together. But expressions such as $\sqrt{1+x} = (1+x)^{1/2}$ and $\dfrac{1}{1+x} = (1+x)^{-1}$ also have exponents. Can they be expanded too?

They can, and the same coefficient pattern still works for *any* rational n. Two things change. The expansion never ends, and it is only valid when x is small enough.

KEY · IB HL For $n \in \mathbb{Q}$:

$$(a+b)^n = a^n\left(1 + n\frac{b}{a} + \frac{n(n-1)}{2!}\left(\frac{b}{a}\right)^2 + \cdots\right)$$

In practice you first factor the expression into the form $(1 + \text{something})^n$, which turns the booklet formula into the version you actually compute with:

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \cdots, \qquad |x| < 1.$$

Why does the expansion stop for whole numbers but not otherwise? Look at the coefficients $n(n-1)(n-2)\cdots$. If $n=3$, the third factor is $n-3=0$, so every later coefficient is zero and the series ends: that is the finite theorem of §2.9. If $n=\frac{1}{2}$ or $n=-1$, no factor is ever zero, so the terms continue forever.

The condition $|x| < 1$ should look familiar. Put $n = -1$:

$$(1+x)^{-1} = 1 - x + x^2 - x^3 + \cdots$$

This is a geometric series with first term 1 and common ratio $-x$, and it converges exactly when $|-x| < 1$. The two infinite series of this chapter are the same idea seen from two directions.

EXAMPLE 2.28

Expand $(1 - 3x)^{1/3}$ up to and including the term in x^2, and state the values of x for which the expansion is valid.

Apply the series with $n = \frac{1}{3}$ and x replaced by $-3x$:

$$(1-3x)^{1/3} = 1 + \tfrac{1}{3}(-3x) + \frac{\frac{1}{3}\left(-\frac{2}{3}\right)}{2}(-3x)^2 + \cdots = 1 - x - x^2 - \cdots$$

Validity requires $|-3x| < 1$, that is $|x| < \frac{1}{3}$.

When the bracket does not already start with 1, factor out the larger term first, as the booklet formula does. The expansion is then valid for $\left|\frac{b}{a}\right| < 1$.

EXAMPLE 2.29

Expand $\dfrac{1}{2+x}$ in ascending powers of x up to x^2, stating the validity.

Factor out the 2:

$$(2+x)^{-1} = \frac{1}{2}\left(1+\frac{x}{2}\right)^{-1} = \frac{1}{2}\left(1 - \frac{x}{2} + \frac{x^2}{4} - \cdots\right) = \frac{1}{2} - \frac{x}{4} + \frac{x^2}{8} - \cdots$$

valid for $\left|\frac{x}{2}\right| < 1$, that is $|x| < 2$.

EXAMPLE 2.30

By expanding $(1 - x)^{1/2}$ up to x^2 and choosing a suitable value of x, find an approximation to $\sqrt{2}$.

The expansion is

$$(1-x)^{1/2} = 1 - \frac{x}{2} - \frac{x^2}{8} - \cdots$$

Now pick x so that $1 - x$ connects to 2. Take $x = 0.02$: then $1 - x = 0.98 = \frac{49}{50}$, so

$$\sqrt{0.98} = \frac{7}{\sqrt{50}} = \frac{7\sqrt{2}}{10} \implies \sqrt{2} = \frac{10}{7}\sqrt{0.98}.$$

The series gives $\sqrt{0.98} \approx 1 - 0.01 - 0.00005 = 0.98995$, hence

$$\sqrt{2} \approx \frac{10}{7}(0.98995) \approx 1.41421,$$

agreeing with the true value to five decimal places, from just three terms. The key is small x: the closer x is to 0, the faster the series converges.

CAUTION Always state the interval of validity alongside an HL binomial expansion; the series is meaningless outside it. And the expansion needs the form $(1+\bullet)^n$: factor first, then expand.

At HL these expansions return as Maclaurin series in Ch. 17, where the same coefficients arise from repeated differentiation.

YOUR TURN 2.10 The Binomial Series HL

47. Expand each up to and including the term in x^3, stating the validity.

(a) $(1+x)^{-1}$ [2] (b) $(1-x)^{-2}$ [3]

48. Expand each up to and including the term in x^2, stating the validity.

(a) $(1+4x)^{1/2}$ [3] (b) $(1+x)^{1/3}$ [3]

49. Expand $\dfrac{1}{3-x}$ in ascending powers of x up to x^2, stating the validity. [3]

50. Expand $(8+x)^{1/3}$ in ascending powers of x up to x^2, stating the validity. [3]

51. Use the expansion of $(1+x)^{1/2}$ up to the term in x^2, with $x = 0.1$, to estimate $\sqrt{1.1}$ to four decimal places. [3]

Chapter 2 · Sequences & Series

Reflections

TOK Fold a sheet of paper 42 times and the stack reaches the Moon. The arithmetic is exact, but no one has ever folded a sheet more than about eight times, so the object the calculation describes has never existed. Can a statement be mathematically true and physically impossible at the same time? What is the calculation actually telling us about?

TOK The sum $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots = 1$ is an equality, yet no finite number of steps ever reaches 1. We accept the answer because we define the sum to be the limit of the partial sums. Is that a discovery about infinity, or a decision we made so that the addition has an answer?

RW Around 450 BCE Zeno argued that a runner can never finish a race. First he must reach the halfway point, then half of what remains, and so on without end. Each step of the argument looks correct, and it troubled mathematicians for two thousand years. The answer is one line of this chapter: $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots = 1$. Infinitely many steps can have a finite total, provided the steps shrink fast enough.

IM The triangle of binomial coefficients was known long before Blaise Pascal wrote about it in 1654. Around the 2nd century BCE the Indian scholar Pingala counted the possible patterns of short and long syllables in poetry, a problem that produces exactly these numbers; a later Indian commentary set them out as a triangle and called it *Meru Prastara*, the staircase of Mount Meru. The Persian mathematicians al-Karaji and Omar Khayyam used it around 1000–1100 CE, and it is still called the Khayyam triangle in Iran. The Chinese mathematician Yang Hui published it in 1261, and it is called Yang Hui's triangle in China. Tartaglia used it in Italy in the 1500s, where it is called Tartaglia's triangle. The same pattern was therefore found independently in several places over eighteen centuries, and each culture named it after a different person. What does that tell you about who a mathematical result belongs to?

EXT A geometric series with $|r| < 1$ is the simplest example of a **power series**, a sum of increasing powers of x:

$$\sum_{k=0}^{\infty} x^k = \frac{1}{1-x}, \qquad |x| < 1.$$

This one identity produces many others. Differentiate each term, or integrate each term, and you obtain the series for $\ln(1 + x)$, for $\arctan x$, and for many further functions. In calculus, the topic of this chapter is *the seed from which the analysis of functions grows.*

End-of-Chapter Problems — Chapter 2: Sequences & Series

1. An arithmetic sequence has first term a and common difference d. The 5th term is 17 and the sum of the first 10 terms is 155.

(a) Write down two equations in a and d. [2]
(b) Find a and d. [2]
(c) Find the first negative term of the sequence. [2]

2. An arithmetic sequence has $u_3 = 7$ and $u_9 = 25$.

(a) Find u_1 and d. [3]
(b) Find S_{20}. [2]
(c) Find the least value of n such that $S_n > 600$. [3]

3. An arithmetic series has $u_1 = 2$ and $S_{10} = 155$.

(a) Find the common difference. [2]
(b) Find the sum $\sum_{k=11}^{20} u_k$. [4]

4. An arithmetic sequence has 20 terms. The sum of the odd-positioned terms $u_1 + u_3 + u_5 + \cdots + u_{19}$ is 160, and the sum of the even-positioned terms $u_2 + u_4 + u_6 + \cdots + u_{20}$ is 140. Find the first term u_1 and the common difference d. [6]

5. The sum of the first n terms of a series is given by $S_n = n^2 + 3n$.

(a) Find u_1 and u_2. [2]
(b) Show that the series is arithmetic and state the common difference. [3]
(c) Find the value of n such that $u_n = 43$. [2]

6. An AP has $u_1 = 120$ and $d = -3$.

(a) Find the first term that is negative. [3]
(b) Find the value of n that gives the maximum partial sum S_n, and find that maximum. [4]

7. A geometric sequence has first term a and common ratio r, where $r > 0$. Given $u_3 + u_4 = 12$ and $u_5 + u_6 = 3$, find a and r. [5]

8. The first three terms of a geometric sequence are k, 6, and $k + 5$.

(a) Show that $k^2 + 5k - 36 = 0$, and hence find the possible values of k. [4]
(b) For the value of k that gives a convergent series, find S_∞. [3]

9.

(a) Show that $\sum_{k=1}^{n} r^{k-1} = \frac{1-r^n}{1-r}$ for $r \neq 1$ by multiplying S by r and subtracting. [3]
(b) Hence find $\sum_{k=1}^{10} \left(\frac{2}{3}\right)^{k-1}$, giving your answer as an exact fraction. [3]

10. A geometric series has $u_1 = 3$ and $S_\infty = 12$.

(a) Find the common ratio. [2]
(b) Show that $S_n = 12\left(1 - \left(\frac{3}{4}\right)^n\right)$. [2]
(c) Find the least value of n such that $S_n > 11.9$. [3]

11.

(a) Find $\sum_{k=1}^{n}(4k-3)$ in terms of n. [3]
(b) Find the value of n such that $\sum_{k=1}^{n}(4k-3) = 780$. [3]

12. Consider the geometric series $1 + 2x + 4x^2 + 8x^3 + \cdots$

(a) Write down the common ratio. [1]
(b) Find the values of x for which the sum to infinity exists. [2]
(c) For these values, find S_∞ as a function of x. [2]

13. A geometric sequence has all positive terms. The sum of the first two terms is 10 and the sum to infinity is 25. Find u_1 and r. [5]

14. The sum to infinity of a geometric series is $\frac{3}{2}$ times the sum of the first 4 terms. Find the common ratio. [5]

15. The repeating decimal $0.\overline{142857} = 0.142857142857\ldots$ can be written as an infinite geometric series.

(a) Write down u_1 and the common ratio r. [2]
(b) Find the exact fractional value of $0.\overline{142857}$. [3]
(c) Verify your answer by converting the fraction from part (b) back to a decimal using long division. [1]

16. GDC A machine is bought for $60 000 and depreciates at 18% per year.

(a) Find its value after 4 years. [2]

(b) Find the least number of complete years until the value falls below one quarter of the purchase price. [3]

17. GDC \$8000 is invested for 6 years at 5% per year, compounded quarterly. Over the same period, inflation runs at 3% per year.

(a) Find the future value of the investment. [2]

(b) Find the real value of the investment after 6 years. [2]

(c) Comment on what your answers say about the investment's true growth. [1]

18. GDC An investment account pays 5% per year, compounded annually. A sum P is invested.

(a) Write an expression for the value of the account after n years. [1]

(b) Find the value of P if the account is worth \$20 000 after 10 years. [3]

(c) Find the number of complete years for the account to triple in value. [3]

19. GDC The value of a car decreases by 18% each year. A new car costs \$32 000.

(a) Find the value of the car after 3 years. [2]

(b) Find the year in which the value first falls below \$10 000. [3]

(c) Find the total loss in value over the first 5 years. [3]

20. A school committee of 6 is to be chosen from 9 teachers and 7 students.

(a) Find the number of committees containing exactly 4 students. [2]

(b) Find the number of committees containing at least 4 students. [3]

21.

(a) Show that $\binom{n}{2} = \dfrac{n(n-1)}{2}$, and explain the connection with the sum $1+2+\cdots+(n-1)$. [3]

(b) Hence find n such that $\binom{n}{2} = 190$. [2]

22. In the expansion of $(1+kx)^8$, the coefficient of x^2 is 112.

(a) Find the possible values of k. [3]

(b) For $k > 0$, find the coefficient of x^3. [2]

23. Consider the expansion of $\left(x + \dfrac{2}{x^2}\right)^{12}$.

(a) Find the term independent of x. [4]

(b) Find the coefficient of x^3. [2]

24.

(a) Expand $(1+2x)^{-2}$ in ascending powers of x up to and including the term in x^3, and state the values of x for which the expansion is valid. [4]

(b) Use your expansion with $x = 0.01$ to estimate $\dfrac{1}{1.02^2}$ to 5 decimal places. [2]

25.

(a) Show that $(4+x)^{1/2} = 2\left(1+\dfrac{x}{4}\right)^{1/2}$, and expand it in ascending powers of x up to the term in x^2, stating the validity. [4]

(b) Hence find an approximation to $\sqrt{4.2}$, giving five decimal places. [2]

26. A ball is dropped from a height of 10 m. After each bounce it reaches 70% of the height it fell from.

(a) Find the height reached after the 3rd bounce. [2]

(b) Find the total vertical distance travelled by the ball before it comes to rest. [4]

27. The first, second, and fifth terms of an arithmetic sequence with first term 3 are, in that order, the first, second, and third terms of a geometric sequence.

(a) Given the arithmetic sequence is non-constant, show that its common difference is $d = 6$ and the geometric sequence has common ratio $r = 3$. [4]

(b) Determine whether the geometric series $3 + 9 + 27 + \cdots$ has a sum to infinity, giving a reason. [2]

28. A sequence is defined by $u_1 = 3$ and $u_{n+1} = 2u_n + 1$ for $n \geq 1$.

(a) Write down u_2, u_3, and u_4. [2]

(b) Let $v_n = u_n + 1$. Show that v_n is a geometric sequence. [3]

(c) Find an explicit formula for u_n. [2]

29. Consider the two arithmetic sequences $4, 7, 10, \ldots$ and $3, 7, 11, \ldots$ Both sequences contain the value 7.

(a) Find the general term of each sequence. [2]

(b) Find the next three values that appear in both sequences. [4]

(c) Show that the common terms form an arithmetic sequence and state its common difference. [2]

30. A student can invest \$3000 in one of two accounts. Account A pays 5% per year *sim-*

ple interest; account B pays 4% per year, compounded annually.

(a) Write down an expression for the value of each account after n years. [3]
(b) Find the value of each account after 10 years. [2]
(c) Using a table or graph on your GDC, find the least whole number of years after which account B is worth more than account A. [3]

31. GDC A savings plan pays \$500 into an account at the start of each year. The account earns 4% per year, compounded annually.

(a) Show that after 3 years the total value is $500(1.04)^3 + 500(1.04)^2 + 500(1.04)$. [2]
(b) Write this as a geometric series and find the total after 10 years. [4]
(c) Find the number of complete years until the total exceeds \$8 000. [3]

32. GDC A sum of L is borrowed. Interest is added at a rate i per month, and a fixed repayment M is made at the end of each month, so the debt after one month is $L(1+i) - M$.

(a) Explain why the debt after one month is $L(1+i) - M$. [2]
(b) Show that the debt after two months is $L(1+i)^2 - M\big[(1+i)+1\big]$. [3]
(c) Hence show that the debt after n months is $L(1+i)^n - M\left(\dfrac{(1+i)^n - 1}{i}\right)$. [4]
(d) A loan of \$$20\,000$ is taken at 6% per year compounded monthly, and is to be cleared in exactly 5 years. Find the monthly repayment. [4]
(e) Find the total amount repaid, and hence the total interest paid. [3]

33. Consider the expansion of $\left(2x - \dfrac{3}{x}\right)^{10}$.

(a) Write down an expression for the general term of the expansion. [2]
(b) Find the term that is independent of x. [4]
(c) Find the coefficient of x^4. [3]
(d) Explain why the expansion contains no term in x^3. [2]
(e) By substituting a suitable value of x, find the sum of all the coefficients. [2]
(f) State how many terms the expansion has, and which powers of x occur. [2]

34. An arithmetic sequence has first term a and common difference d, where $a > 0$ and $d \neq 0$. Its 1st, 4th and 13th terms form a geometric sequence.

(a) Show that $3d = 2a$. [4]
(b) Given that $a = 3$, find d and write down the first four terms of the arithmetic sequence. [3]
(c) Find the common ratio of the geometric sequence. [2]
(d) Find the sum of the first 20 terms of the arithmetic sequence. [3]
(e) Find the sum of the first 8 terms of the geometric sequence. [3]

Challenge Questions

35. Turning products into sums. This investigation shows why a logarithm converts a geometric sequence into an arithmetic one.

(a) Let $u_n = \log_2(3^n)$. Show that $\{u_n\}$ is arithmetic and state its common difference. [3]

(b) Find the exact value of $\sum_{k=1}^{20} u_k$. [3]

(c) Find the smallest value of n such that $u_n > 30$. [3]

(d) Now let $v_n = \log_b(a^n)$, where $a > 0$ and $b > 1$ are fixed. Show that $\{v_n\}$ is arithmetic for every such a and b, and state its common difference. [3]

(e) Let $\{w_n\}$ be any geometric sequence with all terms positive and common ratio r. Show that $\{\log_b w_n\}$ is arithmetic, and state its common difference. [4]

(f) Part (e) is the reason logarithmic scales are used for quantities that grow by multiplication. Explain the connection in your own words, and give one example of such a quantity. [3]

36. Counting multiples. Each set of multiples below is an arithmetic sequence, so its total is an arithmetic series. Throughout, work with the integers from 1 to 1000.

(a) Find the sum of those divisible by 3, and the sum of those divisible by 5. [3]

(b) Explain why adding your two answers counts some integers twice, and find the sum of those divisible by both. [3]

(c) Hence find the sum of those divisible by 3 or by 5. [2]

(d) Find the sum of those divisible by 3 or by 5, but not by both. [3]

(e) Let p and q be distinct primes and let N be a positive integer. Show that the sum of the integers from 1 to N divisible by p or q but not both is

$$S_p + S_q - 2S_{pq}, \qquad \text{where } S_m = m\,\frac{t(t+1)}{2}, \quad t = \left\lfloor \frac{N}{m} \right\rfloor.$$

[4]

(f) Use the result of part (e) with $p = 7$, $q = 11$ and $N = 1000$. Verify your answer by a second method. [3]

37. An infinite geometric series has $S_\infty = 4$ and $u_2 = \frac{3}{4}$.

(a) Find two possible pairs of values for a (first term) and r. [4]

(b) For each solution, a second series is formed by squaring every term. Show each squared series is geometric and find its sum to infinity. [4]

(c) For the solution with $r < \frac{1}{2}$, find the sum to infinity of the series formed by taking the positive square root of every term. [3]

NUMBER & ALGEBRA

CHAPTER 3

Proofs

3

SYLLABUS 1.6 1.15

Mark some points on a circle. Join every pair of points with a straight line. Then count the regions inside the circle. Place the points carefully, so that no three lines cross at the same spot.

Count the regions for the first few cases.

Points n	**1**	**2**	**3**	**4**	**5**
Regions	1	2	4	8	16

The pattern seems clear: each new point doubles the number of regions. So six points must give 32.

Six points give **31**.

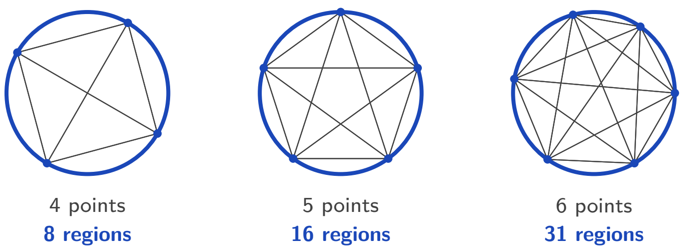


Figure 3.1 Four points give 8 regions and five give 16, but six give 31, not 32.

Five cases followed the doubling rule and the sixth did not. The first five gave no sign of it, and checking further cases would not have helped.

This is the difference between seeing a pattern and knowing that it always holds. A pattern is a guess. The word for such a guess is a **conjecture**: a statement believed true but not yet proved. The doubling rule was a conjecture, and the sixth case disproved it.

A **proof** is an argument in which each step follows from the one before, so the conclusion cannot fail, however many points you take.

By the end of this chapter you will be able to prove a statement directly from definitions, disprove one with a single counterexample, argue by contradiction, and prove a result about every positive integer by induction. The course asks for this because a formula is only trustworthy once it has been proved. Checking a rule on a few cases shows that it has not failed yet. A proof shows that it cannot fail.

3.1 Deductive Proof

A **proof** is a chain of logical steps. It starts from things already accepted, such as definitions, results proved earlier, and the rules of algebra, and it ends at the statement you want to establish. Each step must follow with certainty from the steps before it. If the chain is correct, the conclusion is not merely likely. It cannot be false.

This is **deductive** reasoning: you apply general principles to reach a specific, guaranteed result. The oldest example is not mathematical at all:

- Every human being is mortal. (a general principle)
- Socrates is a human being. (a specific case)
- Therefore Socrates is mortal. (the conclusion)

The conclusion is not a guess, and no amount of further evidence could improve it. If the first two statements are true, the third cannot be false. Mathematics uses arguments of exactly this shape:

- Every multiple of 4 is even.
- The number 132 is a multiple of 4.
- Therefore 132 is even.

Notice that you did not divide 132 by 2 to check. The conclusion was forced by the two statements above it.

The circle problem went the other way. It used a few particular cases to guess a general rule. Reasoning in that direction, from examples to a general claim, is called induction in the everyday sense of the word, and it can lead to a false conclusion. Deduction cannot.

Equality versus identity

Before you prove a statement about an equation, you must know which kind of statement it is. The equals sign is used for two very different claims.

DEF An **equation** is a statement that two expressions are equal for *some particular* values of the variable. An **identity** is a statement that they are equal for *every* value for which both sides are defined. An identity is written with the symbol $\equiv$:

$$(x+1)^2 \equiv x^2 + 2x + 1.$$

The equation $x^2 = 4$ is true only when $x = 2$ or $x = -2$. The identity $(x+1)^2 \equiv x^2 + 2x + 1$ is true for every real x. Solving an equation means finding the values that make it true. Proving an identity means showing that no value makes it false. The three-bar symbol $\equiv$ tells you which of the two you are dealing with: when you see it, the claim is "for all x", not "find x".

You already know one important identity from trigonometry:

$$\sin^2 x + \cos^2 x \equiv 1.$$

This is true for every angle x, not for some particular angles, which is why it is written with $\equiv$. It is proved and used in Ch. 9.

Proving an identity

The two sides of an identity have names. The expression to the left of the $\equiv$ sign is the **left-hand side**, written **LHS**. The expression to the right is the **right-hand side**, written **RHS**.

To prove an identity you take one side and transform it, step by step, into the other, using only valid algebra. Start from the more complicated side and simplify it: write down the LHS, change it one line at a time, and finish with the RHS. Write the labels LHS and RHS in your working, so the direction of the argument is clear to the marker.

CAUTION Do not start by assuming the identity is true and then manipulate both sides until you reach something obvious like $0 = 0$. That assumes what you are trying to prove. Work *one* side into the other instead.

EXAMPLE 3.1

The difficulty here is avoiding the common error of doing the same thing to both sides. Prove that $(a+b)^2 - (a-b)^2 \equiv 4ab$.
Start from the LHS and expand each square:

$$\text{LHS} = (a+b)^2 - (a-b)^2 = (a^2 + 2ab + b^2) - (a^2 - 2ab + b^2).$$

Now remove the brackets. Take care with the signs in the second bracket:

$$= a^2 + 2ab + b^2 - a^2 + 2ab - b^2 = 4ab = \text{RHS}. \quad \blacksquare$$

The LHS became the RHS using nothing but expansion. At no point did we assume the two sides were equal; we discovered it.

A quick **check** takes five seconds and finds most errors: at $a = 3$, $b = 2$ the left side is $25 - 1 = 24$ and the right side is $4(3)(2) = 24$. A check is not a proof. But if a check fails, you know for certain that something is wrong.

EXAMPLE 3.2

Show that $(x-3)^2 + 5 \equiv x^2 - 6x + 14$.

The LHS is the more complicated side, so start there and expand the square:

$$\text{LHS} = (x-3)^2 + 5 = (x^2 - 6x + 9) + 5.$$

Collect the constant terms:

$$= x^2 - 6x + 14 = \text{RHS}. \quad \blacksquare$$

Check at $x = 1$: the LHS is $(-2)^2 + 5 = 9$, and the RHS is $1 - 6 + 14 = 9$.

TIP Always check your own results by substituting a number, in proofs and everywhere else. Examiners expect you to be able to verify a result you have just derived, and one well-chosen value will reveal an algebra error immediately.

EXAMPLE 3.3

Show that $\dfrac{1}{4} + \dfrac{1}{12} = \dfrac{1}{3}$, and prove the algebraic generalisation

$$\frac{1}{m+1} + \frac{1}{m^2+m} \equiv \frac{1}{m}.$$

The numerical case is direct: $\frac{1}{4} + \frac{1}{12} = \frac{3}{12} + \frac{1}{12} = \frac{4}{12} = \frac{1}{3}$. Notice it is the claimed identity at $m = 3$: $m + 1 = 4$ and $m^2 + m = 12$.

For the general statement, start from the left side and factor the second denominator, $m^2 + m = m(m+1)$:

$$\frac{1}{m+1} + \frac{1}{m(m+1)} = \frac{m}{m(m+1)} + \frac{1}{m(m+1)} = \frac{m+1}{m(m+1)} = \frac{1}{m}. \quad \blacksquare$$

One calculation confirmed a single case. The algebra proved every case at once, for every $m \neq 0, -1$. This “verify, then generalise” structure is exactly how such questions are phrased in examinations.

Proving a general statement

Most proofs are not about identities. They are about claims such as “the sum of two odd numbers is even”. The first step is always the same: write the general object in algebra, so that one expression stands for every case at once.

The table below lists the standard forms. In every row, k and n are integers.

In words	In algebra
an even number	$2k$
an odd number	$2k+1$, or $2k-1$
two consecutive integers	n and $n+1$
three consecutive integers	$n-1$, n, $n+1$
two consecutive even numbers	$2k$ and $2k+2$
two consecutive odd numbers	$2k+1$ and $2k+3$
a multiple of 3	$3k$
a number that leaves remainder 1 when divided by 3	$3k+1$
a square number	k^2
a rational number	p/q, with p, q integers and $q \neq 0$

Choosing the right form matters. To prove a statement about *any* two odd numbers, use $2a+1$ and $2b+1$ with two different letters. Writing $2a+1$ twice would only prove the statement for two *equal* odd numbers.

Once the general case is written in symbols, ordinary algebra does the rest.

EXAMPLE 3.4

A single example such as $3+5=8$ confirms nothing about *all* odd numbers. Prove that the sum of any two odd numbers is even.

Let the two odd numbers be $2a+1$ and $2b+1$, where a and b are integers. Their sum is

$$(2a+1)+(2b+1) = 2a+2b+2 = 2(a+b+1).$$

Since $a+b+1$ is an integer, the sum is $2 \times (\text{integer})$, which is by definition even. ■

EXAMPLE 3.5

Prove that the product of two consecutive integers is always even.

Take the consecutive integers n and $n+1$. One of any two consecutive integers must be even: if n is even the product $n(n+1)$ has an even factor, and if n is odd then $n+1$ is even and again the product has an even factor. Either way $n(n+1)$ is a multiple of 2, so it is even. ■

CAUTION One example can never prove a statement that begins ``for all.'' It can only *disprove* one, as the next section shows. The proofs above work because the algebra stands for every case at once.

YOUR TURN 3.1 Deductive Proof

1. State whether each of the following is an equation or an identity.

(a) $3x = 12$ [1]
(b) $(x+2)^2 = x^2 + 4x + 4$ [1]

2. Prove each identity.

(a) $(a+b)^2 = a^2 + 2ab + b^2$ [2]
(b) $(a-b)^2 = a^2 - 2ab + b^2$ [2]
(c) $\dfrac{1}{x} - \dfrac{1}{x+1} = \dfrac{1}{x(x+1)}$ [3]

3. Prove each statement about odd and even integers.

(a) The sum of any two even integers is even. [2]
(b) The sum of an even integer and an odd integer is odd. [2]
(c) The product of any two odd integers is odd. [2]

4. Prove each statement for every integer n.

(a) $(2n+1)^2$ is odd. [2]
(b) $n^2 - n$ is even. [2]

3.2 Disproof by Counterexample HL

A universal statement claims something about every member of a collection. To show that such a claim is false, you do not need to discuss the whole collection. You need one member for which the claim fails.

DEF A **counterexample** is one specific case for which a general statement fails. A single counterexample is enough to prove that the statement is false.

The logic is exact. "Every n has property P" is false as soon as one value of n does not have property P. This is what happened with the circle problem. The claim "the number of regions doubles each time" was a universal statement, and $n = 6$ was a counterexample to it.

EXAMPLE 3.6

A famous formula, $n^2 + n + 41$, produces a prime number for $n = 0, 1, 2, \ldots, 39$, forty primes in a row. Decide whether it produces a prime for *every* non-negative integer n.

Test the claim at the first value the pattern does not cover. At $n = 40$:

$$40^2 + 40 + 41 = 1600 + 40 + 41 = 1681 = 41^2.$$

This is 41×41, not prime. So $n = 40$ is a counterexample, and the statement "$n^2 + n + 41$ is always prime" is false. ■

Forty cases supported the claim, and one case ended it. Examples can make a claim look true. They can never make it certain.

EXAMPLE 3.7

Disprove the claim: "every prime number is odd."

The number 2 is prime, and 2 is even. That single case is a counterexample, so the claim is false. ■

EXAMPLE 3.8

Show that this statement is not always true: "there are no positive integer solutions to the equation $x^2 + y^2 = 10$."

Take $x = 1$ and $y = 3$. Both are positive integers, and

$$1^2 + 3^2 = 1 + 9 = 10.$$

So a positive integer solution does exist, and the statement is false. ■

Notice what the answer contains: the values, the substitution, and the conclusion. Giving only "$x = 1$, $y = 3$" would not earn full marks.

TIP A counterexample must be *explained*, not just named. State the case, substitute it into the claim, show the calculation, and say in words why this makes the claim false. The IB is explicit that stating the counterexample alone is not sufficient.

Two more claims are worth testing before you read on. "If $a^2 > b^2$ then $a > b$" sounds obvious, but $a = -3$ and $b = 1$ give $9 > 1$ with $-3 < 1$, because squaring removes the sign. "The product of two irrational numbers is irrational" sounds reasonable, but $\sqrt{2} \times \sqrt{2} = 2$, since the two irrational parts cancel exactly.

RW In 1640 Fermat studied the numbers $F_n = 2^{2^n} + 1$ and found that $F_0 = 3$, $F_1 = 5$, $F_2 = 17$, $F_3 = 257$ and $F_4 = 65\,537$ are all prime. He conjectured that every F_n is prime. Almost a century later, in 1732, Euler tested the next one:

$$F_5 = 2^{32} + 1 = 4\,294\,967\,297 = 641 \times 6\,700\,417.$$

One counterexample ended the conjecture. No larger Fermat prime has ever been found.

CAUTION A counterexample only *disproves* a claim. Finding no counterexample after many attempts is not a proof. The circle problem is the warning: five cases agreed, and the sixth did not.

YOUR TURN 3.2 Disproof by Counterexample HL

5. Disprove each claim by giving a counterexample.

(a) Every integer is positive. [1]

(b) Every prime number is odd. [1]

(c) If n is divisible by 4 then n is divisible by 8. [1]

(d) The sum of two prime numbers is always even. [2]

6. Disprove each claim by giving a counterexample.

(a) $n^2 \geq 2n$ for all integers n. [2]

(b) $x^2 > x$ for all real x. [2]

(c) If n^2 is even then n is odd. [1]

3.3 Proof by Contradiction HL

Suppose a friend tells you they have written down the largest whole number that exists. You do not need to see it to know they are wrong. Whatever the number is, add 1 to it. The result is a whole number, and it is larger. The claim disproves itself.

You have just proved something without ever finding the answer. You did not search for the largest whole number. You assumed it existed and showed that the assumption leads to an impossible situation.

This is the idea of **proof by contradiction**. Assume the statement you want to prove is *false*. Reason correctly from that assumption until you reach something impossible. Correct reasoning cannot turn a true assumption into an impossible result, so the assumption must have been false. The original statement is therefore true.

> DEF In **proof by contradiction**, you assume the negation of what you want to prove, then derive a logical contradiction. The contradiction forces the assumption to be false, which establishes the original claim.

The method has a Latin name, *reductio ad absurdum*, meaning "reduction to the absurd". It is powerful because it gives you something to work with. Proving that no fraction equals $\sqrt{2}$ directly would mean checking infinitely many fractions. Assuming one exists gives you an equation to manipulate. The two proofs below are among the most famous in mathematics, and both work this way.

EXAMPLE 3.9

Prove that $\sqrt{2}$ is irrational, that is, it cannot be written as a fraction of two integers.

Assume the opposite: that $\sqrt{2}$ *is* rational. Then we can write

$$\sqrt{2} = \frac{p}{q},$$

where p and q are integers **sharing no common factor**, so the fraction is fully cancelled. The contradiction at the end of the proof depends on this condition, so state it explicitly. Squaring both sides gives $2 = \frac{p^2}{q^2}$, hence

$$p^2 = 2q^2.$$

So p^2 is even, which forces p itself to be even (the square of an odd number is odd). Write $p = 2m$. Then $p^2 = 4m^2$, and substituting,

$$4m^2 = 2q^2 \implies q^2 = 2m^2,$$

so q^2 is even, and therefore q is even too. But now p and q are both even, sharing the factor 2, contradicting our choice of a fully cancelled fraction. The assumption that $\sqrt{2}$ is rational is impossible, so $\sqrt{2}$ is irrational. ■

EXAMPLE 3.10

Prove that there are infinitely many prime numbers. This argument is due to Euclid, around 300 BCE.

Assume the opposite: that there are only finitely many primes, and list them all as $p_1, p_2, \dots, p_n$. Build the number

$$N = p_1 \times p_2 \times \cdots \times p_n + 1,$$

the product of every prime, plus one. Dividing N by any prime on the list leaves remainder 1, so

N is divisible by none of them. Yet every integer greater than 1 has at least one prime factor. That factor cannot be on our list, so our list was not complete, a contradiction. No finite list can hold all the primes, so there are infinitely many. ■

CAUTION State your assumption clearly at the start (``assume ...is rational'') and name the contradiction when you reach it. A proof by contradiction with no clearly identified contradiction proves nothing.

YOUR TURN 3.3 Proof by Contradiction HL

7. Prove each statement by contradiction.

(a) There is no largest integer. [2]

(b) There is no smallest positive real number. [3]

8. Let n be an integer. Prove each statement by contradiction.

(a) If n^2 is odd then n is odd. [3]

(b) If n^2 is even then n is even. [3]

9. Prove each statement by contradiction.

(a) There are no integers x and y with $2x + 4y = 5$. [2]

(b) The sum of a rational number and an irrational number is irrational. [3]

3.4 Proof by Induction HL

Picture an endless line of dominoes. You cannot knock over infinitely many by hand. But if you can show two things, that the first domino falls, and that *whenever* a domino falls it topples the next, then you know with certainty that every domino in the line will fall, however long the line.

Mathematical induction is exactly this idea, used to prove a statement P_n for every integer n from some starting value onward. We write the statement as P_n, with a subscript, because it is the n-th claim in an infinite list of claims $P_1, P_2, P_3, \ldots$, not a function being evaluated. You verify the first claim, then prove that each claim forces the one after it.

KEY **Principle of mathematical induction.** To prove that P_n holds for all integers $n \geq 1$:

1. Show that P_1 is true.
2. Assume that P_k is true for **some** $k \geq 1$ (the *inductive hypothesis*).
3. Show that P_k implies P_{k+1}.
4. Conclude: *since P_1 is true, and P_k true implies P_{k+1} true, by the principle of mathematical induction P_n is true for all integers $n \geq 1$.*

(If the statement starts later, at n_0, begin at P_{n_0} instead of P_1.)

Every induction proof follows these four steps, and the sentence in step 4 should be written out in full. It is not decoration. It is the step that turns the two facts you proved into a conclusion about every n, and it carries its own mark in examinations.

CAUTION In step 2, assume P_k for **some** k, never for *all* k. Assuming it for all k assumes the very thing you are proving, and examiners withhold the reasoning marks for it. The hypothesis is one domino falling, not the whole line.

Why induction works

The domino image can be made precise. Suppose steps 1 and 3 are done, and pick any n you like, say $n = 1000$. Then P_1 is true; P_1 forces P_2; P_2 forces P_3; and after 999 applications of the inductive step, P_{1000} is true. The same finite chain reaches *any* n, so no case escapes, even though you only ever proved two things.

There is a second way to see it, and it connects induction to the previous section. Suppose the two steps are done, yet P_n somehow failed for some values of n. Among the failures there would be a *smallest* one, say P_m. Now $m \neq 1$, because step 1 checked P_1. So $m-1 \geq 1$, and P_{m-1} is true, since m was the smallest failure. But step 3 says that P_{m-1} forces P_m, which contradicts the failure of P_m. So induction is itself a proof by contradiction, and the two techniques of this chapter are closely connected.

Induction is not only for sums. In this course it also proves divisibility statements and inequalities below, De Moivre's theorem for complex numbers in Ch. 18, and formulas for n-th derivatives in Ch. 14. Whenever a claim has the form "for every integer n", induction is the first method to consider.

Induction for a sum

How to approach it. Here P_n says that a sum equals a formula. Add the next term of the sum to both sides of the assumption P_k. Then rearrange the right-hand side until it becomes the original formula with $k + 1$ written wherever n appeared. Write that target expression down before you begin, so you know what you are aiming at.

EXAMPLE 3.11

Prove that $1+2+3+\cdots+n=\dfrac{n(n+1)}{2}$ for all integers $n\geq 1$.

Let P_n be the statement $1+2+\cdots+n=\dfrac{n(n+1)}{2}$.

Step 1 (P_1): the left side is 1; the right side is $\dfrac{1\cdot 2}{2}=1$. They agree, so P_1 is true.

Step 2: assume P_k is true for some $k\geq 1$, that is,

$$1+2+\cdots+k=\frac{k(k+1)}{2}.$$

Step 3: add the next term, $k+1$, to both sides:

$$1+2+\cdots+k+(k+1)=\frac{k(k+1)}{2}+(k+1).$$

Factor $(k+1)$ out of the right side:

$$=(k+1)\left(\frac{k}{2}+1\right)=(k+1)\cdot\frac{k+2}{2}=\frac{(k+1)(k+2)}{2}.$$

This is exactly P_{k+1}, the original formula with n replaced by $k+1$. So $P_k\Rightarrow P_{k+1}$.

Step 4: since P_1 is true, and P_k true implies P_{k+1} true, by the principle of mathematical induction P_n is true for all integers $n\geq 1$. ■

Induction for divisibility

The same four steps prove divisibility statements.

How to approach it. Here P_n says that an expression E_n is divisible by a fixed number d. Turn the assumption into an equation: divisible by d means $E_k = dm$ for some integer m. Then rewrite E_{k+1} so that E_k appears inside it, replace that part by dm, and take d outside as a common factor. The aim is to reach the form $d\times$ (integer).

EXAMPLE 3.12

Prove that 8^n-1 is divisible by 7 for all integers $n\geq 1$.

Let P_n be the statement "8^n-1 is divisible by 7."

Step 1 (P_1): $8^1-1=7$, which is divisible by 7. So P_1 is true.

Step 2: assume P_k is true for some $k\geq 1$, so $8^k-1=7m$ for some integer m.

Step 3: consider the next case:

$$8^{k+1}-1=8\cdot 8^k-1=8(8^k-1)+8-1=8(7m)+7=7(8m+1).$$

Since $8m+1$ is an integer, $8^{k+1}-1$ is divisible by 7, which is P_{k+1}. So $P_k\Rightarrow P_{k+1}$.

Step 4: since P_1 is true, and P_k true implies P_{k+1} true, by the principle of mathematical induction P_n is true for all integers $n\geq 1$. ■

Notice where the hypothesis was used: rewriting $8^k - 1$ as $7m$ is the only reason the final line factors cleanly.

Induction for an inequality

How to approach it. Here P_n says that $A_n > B_n$. Inequalities are proved by chaining: if $a > b$ and $b \geq c$, then $a > c$. Start from A_{k+1} and rewrite it so that A_k appears, then use the assumption to replace A_k by something smaller. That gives the first link. Now prove separately that this new quantity is at least B_{k+1}, which gives the second link. The two links join into a single chain from A_{k+1} down to B_{k+1}.

EXAMPLE 3.13

Prove that $2^n > n^2$ for all integers $n \geq 5$. The base case is not $n = 1$ here, and that matters: the inequality is false for $n = 2, 3, 4$.

Let P_n be the statement $2^n > n^2$.

Step 1 (P_5): $2^5 = 32$ and $5^2 = 25$, and $32 > 25$. So P_5 is true.

Step 2: assume P_k is true for some $k \geq 5$, that is, $2^k > k^2$.

Step 3: then

$$2^{k+1} = 2 \cdot 2^k > 2k^2.$$

It is now enough to show $2k^2 \geq (k+1)^2$. Expanding, this asks whether $2k^2 \geq k^2 + 2k + 1$, that is, $k^2 - 2k - 1 \geq 0$. For $k \geq 5$ this certainly holds, since $k^2 - 2k - 1 = (k-1)^2 - 2 \geq 16 - 2 > 0$. Therefore

$$2^{k+1} > 2k^2 \geq (k+1)^2,$$

which is P_{k+1}. So $P_k \Rightarrow P_{k+1}$.

Step 4: since P_5 is true, and P_k true implies P_{k+1} true for $k \geq 5$, by the principle of mathematical induction $2^n > n^2$ for all integers $n \geq 5$. ■

CAUTION Two ways an induction proof fails: forgetting the base case (the dominoes can topple each other yet never start), or proving the inductive step *without using* the inductive hypothesis. If you never used the assumption P_k, you have not built a chain, only restated the problem.

YOUR TURN 3.4 Proof by Induction HL

10. Prove each result by induction, for all $n \geq 1$.

(a) $1 + 2 + 3 + \cdots + n = \dfrac{n(n+1)}{2}$ [4]

(b) $1 + 3 + 5 + \cdots + (2n-1) = n^2$ [4]

11. Prove each result by induction, for all $n \geq 1$.

(a) $2 + 4 + 6 + \cdots + 2n = n(n + 1)$ [4]

(b) $1 + 2 + 4 + \cdots + 2^{n-1} = 2^n - 1$ [4]

12. Prove each divisibility result by induction, for all $n \geq 1$.

(a) $3^n - 1$ is divisible by 2. [3]

(b) $4^n - 1$ is divisible by 3. [3]

(c) $5^n - 1$ is divisible by 4. [3]

13.

(a) Prove by induction that $2^n \geq n + 1$ for all $n \geq 0$. [3]

(b) Explain why the base case here is $n = 0$ rather than $n = 1$. [2]

Chapter 3 · Proofs

Reflections

TOK A proof reduces a claim to things already accepted, but those starting points, the axioms, are themselves assumed, not proved. In 1931 Kurt Gödel showed that any logical system rich enough to describe ordinary arithmetic must contain true statements it cannot prove. Mathematical certainty, then, is real but not absolute: it is certainty *relative* to a chosen foundation. Does this make mathematics more like a discovered landscape or a built structure?

IM Induction was not invented in one place or time. A recognisable form appears in the work of the Persian mathematician al-Karaji around 1000 CE, who used it to establish results about sums of powers. The method was developed further by Levi ben Gerson in the fourteenth century and by Pascal in the seventeenth, and the name "mathematical induction" was finally fixed by Augustus De Morgan in 1838. The proof that primes are infinite is older still, written down by Euclid in Alexandria around 300 BCE, and it is still taught in his original form more than two thousand years later.

RW Proof is not confined to pure mathematics. In software engineering, *formal verification* proves that a program meets its specification exactly, a guarantee no amount of testing can give, used where failure is catastrophic: aircraft control, medical devices, spacecraft. Modern cryptography depends on proved properties of prime numbers. The security of online banking is, in the end, a theorem.

EXT In 1976 the four-colour theorem, that any map can be coloured with four colours so no two neighbouring regions match, became the first major theorem whose proof required a computer. The machine checked nearly two thousand separate cases, far more than a person could check by hand. This raised a new question: if a proof is too long for any human to read, but every step has been verified by a machine, is it still a proof? What exactly is a proof *for*, certainty, or understanding?

The language of proof

Mathematics has a precise vocabulary for talking about statements and their truth. The terms below appear throughout this book and in any mathematical reading; the symbolic forms ($\neg$, $\Rightarrow$, $\iff$, contrapositive) are language for reading and for ToK discussion rather than examinable AA content.

Term	Meaning	Example
statement	a sentence that is definitely true or false	"7 is prime"
conjecture	believed true, not yet proved	every even $n > 2$ is a sum of two primes
theorem	a statement that has been proved	$\sqrt{2}$ is irrational
axiom	accepted without proof as a starting point	$a + b = b + a$ for real numbers
lemma	a small result proved on the way to a bigger one	"if n^2 is even then n is even," inside the $\sqrt{2}$ proof
corollary	an easy consequence of a theorem	$3 + \sqrt{2}$ is irrational
negation $\neg P$	the opposite statement	$\neg(n \text{ is even})$: "n is odd"
implication $P \Rightarrow Q$	if P holds, Q must hold	n divisible by 4 $\Rightarrow$ n even
converse $Q \Rightarrow P$	the reversed implication, **not** automatic	n even $\Rightarrow$ divisible by 4 is *false* $(n = 6)$
contrapositive $\neg Q \Rightarrow \neg P$	logically identical to $P \Rightarrow Q$	n odd $\Rightarrow$ n not divisible by 4
equivalence $P \iff Q$	each implies the other	n even $\iff$ n^2 even
counterexample	one case that disproves a universal claim	$n = 40$ for "$n^2 + n + 41$ is prime"
identity $\equiv$	equal for every value of the variable	$(x+1)^2 \equiv x^2 + 2x + 1$
■ (or QED)	marks the end of a proof	—

Note that the converse row contains its own counterexample. Reversing a true implication produces a new claim, which needs its own proof, and $n = 6$ shows that this particular one is false.

A gallery of conjectures

Every theorem started as a conjecture. Here are eight famous ones, each simple enough to state in a sentence, together with what has happened to them so far. None of them is examinable. Together they show what this chapter is about, in real mathematics.

Goldbach conjecture (1742) — open. Every even number greater than 2 is the sum of two primes: $28 = 5 + 23$, $100 = 3 + 97$. Verified for every even number up to 4×10^{18}, and still unproven. The most famous "obviously true, nobody can prove it" statement in mathematics.

Twin prime conjecture — open. There are infinitely many primes p for which $p + 2$ is also prime, like $11, 13$ and $101, 103$. In 2013 Yitang Zhang proved that there are infinitely many prime pairs at most 70 million apart. Later work reduced that distance to 246. This is much closer to the conjecture, but 246 is still not 2.

Collatz conjecture — open. Take any positive integer; halve it if even, triple it and add 1 if odd; repeat. Does every starting number eventually reach 1? Try 27 on your GDC: it takes 111 steps and climbs to 9232 on the way. Verified for astronomically many starting values, proved for none of them collectively. Erdős said mathematics is "not yet ripe" for it.

Odd perfect numbers — open for ∼2300 years. A number is perfect if it equals the sum of its proper divisors, like $6 = 1 + 2 + 3$ and $28 = 1 + 2 + 4 + 7 + 14$. Every perfect number ever found is even. Whether an odd one exists is arguably the oldest unsolved problem in mathematics; any example must exceed 10^{1500}.

Legendre's conjecture — open. Between any two consecutive squares n^2 and $(n + 1)^2$ there is always at least one prime. True in every case ever checked, and still unproved.

Fermat's Last Theorem (1637) — proved, 1995. No positive integers satisfy $x^n + y^n = z^n$ for $n \geq 3$. Fermat wrote in the margin of a book that he had a proof, but that the margin was too small to hold it. The first accepted proof came 358 years later, from Andrew Wiles, who worked on it for seven years in near secrecy, and it uses mathematics far beyond anything available to Fermat. Conjectures can be settled, and this one was.

Pólya's conjecture (1919) — disproved. At least half of the numbers up to any N have an odd number of prime factors. Verified for millions of cases and widely believed for forty years, until it was shown to fail, with smallest known counterexample near $N = 906{,}150{,}257$. This is the circle problem from the start of the chapter, on a far larger scale: millions of cases agreed with the claim, one case did not, and the claim is false.

TOK Which of these deserves more confidence: Goldbach, verified for 4×10^{18} cases but unproven, or a theorem proved once and checked by a handful of humans? Pólya's conjecture had millions of confirmations and was false. What, then, is evidence actually worth in mathematics, and is that different from every other field of knowledge?

End-of-Chapter Problems — Chapter 3: Proofs

1. Prove that the product of any three consecutive integers is divisible by 6. [5]

2. Consider the cube of a binomial.

(a) Prove the identity $(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$. [3]

(b) Hence prove that $(n+1)^3 - n^3 = 3n^2 + 3n + 1$ for every integer n. [3]

3. Disprove the claim "$n^2 + 1$ is never divisible by 5" by giving two counterexamples. [3]

4. Prove by contradiction that $\sqrt{3}$ is irrational. [6]

5. Let a be a non-zero rational number and b an irrational number.

(a) Prove by contradiction that ab is irrational. [4]

(b) Hence state, with a reason, whether $\dfrac{\sqrt{2}}{5}$ is rational or irrational. [2]

6.

(a) Prove by contradiction that $x^2 - 6x + 10 = 0$ has no real solution, without using the discriminant. [3]

(b) Prove by contradiction that if n is an integer and n^2 is a multiple of 3, then n is a multiple of 3. [3]

7. Prove that $\sqrt{2} + \sqrt{3}$ is irrational. You may assume that $\sqrt{6}$ is irrational. [6]

8. Prove by contradiction that $\sqrt[3]{5}$ is irrational. [6]

9. Prove by induction that $7^n - 1$ is divisible by 6 for all $n \geq 1$. [5]

10. Prove by induction that $6^n + 4$ is divisible by 5 for all $n \geq 1$. [5]

11. Prove by induction that $n! > 2^n$ for all $n \geq 4$, stating why the base case is $n = 4$. [5]

12. Prove by induction that $5^{2n} - 1$ is divisible by 24 for all $n \geq 1$. [5]

13. Prove by induction that $2^n > n$ for all $n \geq 1$. [5]

14. Prove by induction that $9^n - 1$ is divisible by 8 for all $n \geq 1$. [5]

15. Prove by induction that $n^3 + 2n$ is divisible by 3 for all $n \geq 1$. [5]

16. Prove by induction that $\sum_{r=1}^{n} r^2 = \dfrac{n(n+1)(2n+1)}{6}$ for all $n \geq 1$. [6]

17. Prove by induction that $\sum_{r=1}^{n} r^3 = \dfrac{n^2(n+1)^2}{4}$ for all $n \geq 1$. [6]

18. Prove by induction that $\sum_{r=1}^{n} r(r+1) = \dfrac{n(n+1)(n+2)}{3}$ for all $n \geq 1$. [6]

19. Prove by induction that $\sum_{r=1}^{n} (3r-2) = \dfrac{n(3n-1)}{2}$ for all $n \geq 1$. [6]

20. Prove by induction that $\sum_{r=1}^{n} \dfrac{1}{r(r+1)} = \dfrac{n}{n+1}$ for all $n \geq 1$. [6]

21. A student claims that $n^2 - n + 11$ is prime for every positive integer n.

(a) Verify the claim for $n = 1$ and $n = 2$. [2]

(b) Disprove the claim with a counterexample. [3]

22. Consider parity of squares.

(a) Prove that if n is odd then n^2 is odd. [2]

(b) Hence prove by contradiction that if n^2 is even then n is even. [3]

23. Prove that $1 + 3 + 5 + \cdots + (2n-1) = n^2$ in two ways:

(a) directly, using the sum of an arithmetic sequence; [3]

(b) by induction. [3]

24. Prove that $\log_2 3$ is irrational. [6]

25. Prove by induction that $\sum_{r=1}^{n} r \cdot r! = (n+1)! - 1$ for all $n \geq 1$. [6]

26.

(a) Prove by induction that $\sum_{r=1}^{n} r = \dfrac{n(n+1)}{2}$ for all $n \geq 1$. [5]

(b) You may assume that $\sum_{r=1}^{n} r^2 = \dfrac{n(n+1)(2n+1)}{6}$. Find $\sum_{r=1}^{n} r(r-1)$, giving your answer in fully factorised form. [4]

(c) Hence evaluate $\sum_{r=1}^{20} r(r-1)$. [3]

(d) Show that $\sum_{r=1}^{n} r(r-1) = 2\binom{n+1}{3}$, and explain why this means the sum is always even. [3]

27.

(a) Prove that if n^2 is even, then n is even. [3]

(b) Hence prove by contradiction that $\sqrt{2}$ is irrational. [5]

(c) Explain precisely where the argument in part (b) breaks down if you attempt to use it to prove that $\sqrt{4}$ is irrational. [3]

(d) A student writes: "*$\sqrt{2}$ is irrational, so $\sqrt{2}\times\sqrt{2}$ must be irrational too.*" Explain the error, and state what it shows about products of irrational numbers. [3]

28. A sequence is defined by $u_1 = 1$ and $u_{n+1} = \dfrac{u_n}{1+u_n}$ for $n \geq 1$.

(a) Find u_2, u_3 and u_4. [3]

(b) Write down a conjecture for u_n in terms of n. [1]

(c) Prove your conjecture by induction. [5]

(d) A student's whole proof reads: "*Assume the statement is true for $n = k$. Then it is true for $n = k + 1$. Hence it is true for all n.*" State what is missing, and explain why the argument establishes nothing without it. [3]

(e) Give a statement $P(n)$ for which $P(k) \Rightarrow P(k+1)$ is true for every k, but $P(n)$ is false for every n. Justify both claims. [3]

Challenge Questions

29. **Counting the regions of a circle.** This investigation completes the problem from the start of the chapter. Throughout, n points are placed on a circle so that no three chords meet at a single interior point.

(a) By drawing, verify that $n = 1, 2, 3, 4, 5$ give $1, 2, 4, 8, 16$ regions. [2]

(b) Explain why each interior intersection point corresponds to exactly one choice of 4 of the n points, and deduce that there are $\binom{n}{4}$ interior points. [3]

(c) The chords and arcs form a network. Show that it has $V = n + \binom{n}{4}$ vertices and $E = n + \binom{n}{2} + 2\binom{n}{4}$ edges. *(Hint: each interior point adds one extra segment to each of the two chords through it)* [4]

(d) Euler's formula for a connected planar network states $V - E + F = 2$, where F counts all faces including the region outside the circle. Use it to prove that the number of regions inside the circle is $\binom{n}{4} + \binom{n}{2} + 1$. [4]

(e) Hence find the number of regions for $n = 6$ and $n = 10$. [2]

(f) If the n points are equally spaced, the count for $n = 6$ is 30 rather than 31. Explain what changes. [3]

30. Fermat's conjecture. In 1640 Fermat studied $F_n = 2^{2^n} + 1$.

(a) Show that F_0, F_1, F_2, F_3 and F_4 are prime. [3]

(b) Suppose m has an odd factor $d > 1$, so $m = de$. Using the factorisation

$$x^d + 1 = (x+1)\left(x^{d-1} - x^{d-2} + \cdots - x + 1\right) \quad (d \text{ odd}),$$

with $x = 2^e$, prove that $2^m + 1$ is not prime. [5]

(c) Deduce that if $2^m + 1$ is prime, then m must be a power of 2. [2]

(d) Verify that 641 divides $F_5 = 4\,294\,967\,297$, and state what this shows about Fermat's conjecture. [3]

31. A proof that names no example. Consider the number $\sqrt{2}^{\sqrt{2}}$.

(a) Suppose $\sqrt{2}^{\sqrt{2}}$ is rational. Explain why this immediately produces an irrational number raised to an irrational power that is rational. [2]

(b) Suppose instead $\sqrt{2}^{\sqrt{2}}$ is irrational. Show that $\left(\sqrt{2}^{\sqrt{2}}\right)^{\sqrt{2}} = 2$, and explain why this also produces such a pair. [4]

(c) Conclude that there exist irrational a and b with a^b rational. [2]

(d) This argument proves that such a pair exists without telling you which case is the true one. Compare it with the proof that $\sqrt{2}$ is irrational, and comment on what each proof gives you. [4]

32. Fibonacci identities. The Fibonacci numbers are defined by $F_1 = F_2 = 1$ and $F_{n+2} = F_{n+1} + F_n$.

(a) Prove by induction that $F_1 + F_2 + \cdots + F_n = F_{n+2} - 1$ for all $n \geq 1$. [5]

(b) Prove by induction that $F_1^2 + F_2^2 + \cdots + F_n^2 = F_n F_{n+1}$ for all $n \geq 1$. [5]

(c) Part (b) has a picture proof: squares of sides $F_1, F_2, \ldots, F_n$ tile a rectangle. State the dimensions of that rectangle and explain how it proves the identity. [3]

33. Divisibility by consecutive integers.

(a) Prove that $n^3 - n$ is divisible by 6 for every integer n. [3]

(b) Prove that $n^5 - n$ is divisible by 30 for every integer n. *(Hint: factor $n^5 - n$ and consider divisibility by 2, 3 and 5 separately)* [6]

(c) Investigate whether $n^7 - n$ is divisible by 42 for every integer n. State the general pattern you notice about $n^p - n$ when p is prime. [5]

FUNCTIONS

Chapter 4

Lines & Quadratics

4

SYLLABUS 1.16 2.1–2.4 2.6 2.7

On 17 January 1995 an earthquake struck near Kobe, in western Japan. The Akashi Kaikyō Bridge was under construction: its two towers already stood in the strait, but the roadway between them did not yet exist. The earthquake moved the towers apart.

The engineers did not start again. They measured the new gap and adjusted the design to fit it. The bridge that opened in 1998 has a main span of 1991 metres instead of the planned 1990.

They could do this because the design was written as equations, not as fixed drawings. The main cables of a suspension bridge carry a level roadway of even weight, and the curve they hang in is a *parabola*, the same curve as a thrown ball. Change one value in the equations and every other measurement follows from it.

That is the power of a **function**. It takes an input and returns exactly one output, so a whole structure becomes a single rule you can graph, study and recalculate. This chapter builds the functions the rest of the course rests on: lines, which change at a steady rate, and quadratics, which give the cable its curve.

By the end of this chapter you will be able to give a function's domain, range and inverse, and to find the equation of a line from whatever you are told. You will solve systems of linear equations, move a quadratic between its factorised, vertex and expanded forms, and use the discriminant to count the roots of an equation.

4.1 Functions

A function is a rule you can rely on. Give it an input and it returns one definite output, and it returns the same output every time. That is what makes it possible to reason about.

> DEF A **function** f assigns exactly one output $f(x)$ to each input x. The set of inputs it accepts is the **domain**; the set of outputs that actually occur is the **range**.

The phrase "exactly one" is the whole idea. A rule that could return two different outputs for the same input is not a function. On a graph this becomes the **vertical line test**: a curve represents a function only if no vertical line crosses it more than once.

Reading the domain

A function need not accept every number. Three operations, and only three, put limits on what you may substitute. Learn the three and you can read the domain of almost any function on the syllabus.

Where x appears	Condition	Reason
In a denominator	$\neq 0$	you cannot divide by zero
Inside an even root	≥ 0	a negative number has no real square root
Inside a logarithm	> 0	the logarithm of zero, or of a negative number, does not exist

Note the difference between the last two. A square root allows zero, because $\sqrt{0} = 0$. A logarithm does not, because no power of the base gives zero. That single distinction, $\geq$ against $>$, is the most common slip in this topic.

Function	Condition	Domain
$\sqrt{x+3}$	$x+3 \geq 0$	$x \geq -3$
$\sqrt{5-x}$	$5-x \geq 0$	$x \leq 5$
$\dfrac{1}{x-2}$	$x-2 \neq 0$	$x \neq 2$
$\ln(x-4)$	$x-4>0$	$x>4$
$\dfrac{1}{\sqrt{x-1}}$	$x-1>0$, since it is both under a root and in a denominator	$x>1$

When a function contains more than one of these, every condition must hold at once. Write them all down, then take the values that satisfy all of them.

EXAMPLE 4.1

State the domain of $f(x) = \dfrac{\sqrt{x+2}}{x-1}$.

There are two restrictions here. The square root needs $x+2 \geq 0$, so $x \geq -2$. The denominator needs $x-1 \neq 0$, so $x \neq 1$.

Both must hold, so the domain is

$$x \geq -2, \quad x \neq 1.$$

Notice that $x=1$ satisfies the first condition but fails the second, so it must still be removed.

EXAMPLE 4.2

State the domain of $g(x) = \ln(9-x^2)$.

A logarithm needs a positive argument, so $9 - x^2 > 0$, that is $x^2 < 9$. This is a quadratic inequality, solved in §4.7: the parabola x^2-9 lies below the axis between its roots. So

$$-3 < x < 3.$$

Both ends are excluded, because at $x = \pm 3$ the argument is zero and $\ln 0$ does not exist.

CAUTION Solve the condition, do not guess it. For $\sqrt{5-x}$ the domain is $x \leq 5$, not $x \geq 5$. Write the inequality down and rearrange it every time, however obvious it looks.

Reading the range

The range is the set of outputs that actually occur. There is no single rule, but four facts cover most functions you will meet, and each says that some expression can never go below zero.

Almost every range argument rests on one fact: some part of the function can never be negative. A square is never negative, so $x^2 \geq 0$; a modulus is never negative, so $|x| \geq 0$; the square root symbol means the non-negative root, so $\sqrt{x} \geq 0$; and a positive base raised to any power stays positive, so $a^x > 0$.

To use them, build the function up from the part you know. Ask what that part can be, then follow the operations one at a time.

Function	**Reasoning**	**Range**
$x^2 + 1$	$x^2 \geq 0$, then add 1	$y \geq 1$
$\|x\| - 3$	$\|x\| \geq 0$, then subtract 3	$y \geq -3$
$\sqrt{x} + 2$	$\sqrt{x} \geq 0$, then add 2	$y \geq 2$
$3 - \|x + 1\|$	$\|x + 1\| \geq 0$, so subtracting it takes 3 downward	$y \leq 3$
$-(x - 2)^2 + 5$	$(x - 2)^2 \geq 0$, and the minus sign turns the inequality round	$y \leq 5$

CAUTION A minus sign in front reverses the direction. Since $(x-2)^2 \geq 0$, it follows that $-(x-2)^2 \leq 0$, so $-(x-2)^2 + 5 \leq 5$. Multiplying an inequality by a negative number always flips the sign.

For a quadratic, complete the square first. The vertex form shows the smallest or largest output immediately.

EXAMPLE 4.3

State the range of $f(x) = x^2 - 6x + 11$.

Complete the square:

$$x^2 - 6x + 11 = (x-3)^2 - 9 + 11 = (x-3)^2 + 2.$$

Now use the first fact. Since $(x-3)^2 \geq 0$, adding 2 gives $f(x) \geq 2$. The range is $y \geq 2$. The smallest value, 2, occurs at $x = 3$, which is the vertex.

EXAMPLE 4.4

State the domain and range of $g(x) = \sqrt{x-2}$.

Domain. The square root needs $x - 2 \geq 0$, so $x \geq 2$.

Range. As x runs from 2 upward, $x - 2$ runs from 0 upward, and so does its square root. So $g(x) \geq 0$.

The domain comes from the inputs the rule permits. The range comes from the outputs those inputs give. Always ask both questions.

EXAMPLE 4.5

State the domain and range of $f(x) = \sqrt{2 - x}$.

The square root needs $2 - x \geq 0$, that is $x \leq 2$. So the domain is $x \leq 2$.

As x decreases from 2, the quantity $2 - x$ increases from 0 upward, and so does its square root. The range is $f(x) \geq 0$.

Compare this with $\sqrt{x - 2}$ above. The two look similar but their domains point in opposite directions, so read the inequality from the expression each time rather than from memory.

TIP On Paper 2, check a range on your GDC. Graph the function and look at the lowest and highest points the curve reaches. The algebra is still expected in the working, but the graph catches a sign error in seconds.

Reading a graph

Most of what you need from a function can be read straight from its graph. Five features matter, and each has its own name.

Feature	What it is	How to find it
x-intercepts (also called zeros, or roots)	the points where the curve meets the x-axis	set $y = 0$ and solve
y-intercept	the point where the curve meets the y-axis	set $x = 0$
Turning points	the local maximum and minimum points, where the curve changes direction	GDC, or the vertex form for a quadratic
Asymptote	a line that the curve gets closer and closer to, without ever reaching it	look for values the function cannot take
Intersection	a point that lies on both of two graphs	solve the two equations together

Your GDC finds all five directly. The graph menu has zero, min/max and intersect tools. On Paper 2 these are the intended method. Graph the function, read the feature, and write down what you asked the calculator to find. Many equations have no algebraic solution at all. For those the graph is not a faster method. It is the only method.

TIP Know the difference between the command terms: **sketch** means show the shape with key features labelled, not to scale; **draw** means an accurate, to-scale plot. A sketch of a parabola needs the vertex, the intercepts, and the correct opening direction, and nothing else.

Inverse functions

An **inverse function** reverses the rule. If f sends 3 to 7, then f^{-1} sends 7 back to 3. An inverse exists only when no two inputs give the same output. If two did, the way back would have two answers, and a function may return only one.

> DEF The **inverse** f^{-1} of a function f undoes it: $f^{-1}(f(x)) = x$. It exists when f is **one-to-one**, meaning no two inputs share an output (the *horizontal* line test).

The inverse swaps input and output. Two things follow. Its graph is the mirror image of f in the line $y = x$, and its domain and range are those of f exchanged. To find a formula, write $y = f(x)$, swap x and y, then solve for y.

EXAMPLE 4.6

Find the inverse of $f(x) = 2x + 3$.
Write $y = 2x + 3$, then swap x and y:

$$x = 2y + 3.$$

Solving for y,

$$y = \frac{x-3}{2}, \qquad \text{so} \qquad f^{-1}(x) = \frac{x-3}{2}.$$

Check: f doubles and adds three; f^{-1} subtracts three and halves. Each step of f is undone, in reverse order.

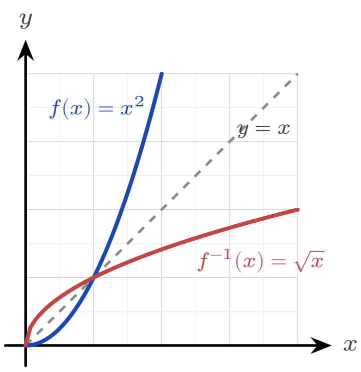


Figure 4.1 A function and its inverse are mirror images in the line $y = x$.

The inverse also gives another way to read an equation. Asking "for which x does $f(x) = 10$?" is the same as asking "what does f^{-1} send 10 to?"

KEY Solving $f(x) = 10$ is equivalent to evaluating $f^{-1}(10)$.

EXAMPLE 4.7

For $f(x) = 2x + 3$, solve $f(x) = 10$ in two ways.

Directly: $2x + 3 = 10$ gives $2x = 7$, so $x = \frac{7}{2}$.

Using the inverse found above, $f^{-1}(x) = \dfrac{x-3}{2}$, so

$$f^{-1}(10) = \frac{10-3}{2} = \frac{7}{2}.$$

The two routes must agree, because they ask the same question. When a question gives you f^{-1} already, the second route is the shorter one.

CAUTION The notation f^{-1} means the inverse, *not* the reciprocal. $f^{-1}(x)$ is not $\dfrac{1}{f(x)}$.

YOUR TURN 4.1 Functions

1. State the domain of each function.

(a) $\sqrt{x+3}$ [1] (b) $\sqrt{7-x}$ [1]

(c) $\dfrac{1}{x-2}$ [1] (d) $\dfrac{1}{x+5}$ [1]

(e) $\ln(x-4)$ [2] (f) $\ln(2x+1)$ [2]

(g) $\dfrac{1}{\sqrt{x-1}}$ [2] (h) $\dfrac{1}{x^2-9}$ [2]

2. State the domain of each function. Each has more than one restriction.

(a) $\dfrac{\sqrt{x+2}}{x-1}$ [3] (b) $\dfrac{\sqrt{x}}{x-9}$ [3]

3. State the range of each function on the domain given.

(a) $x^2+1, \quad x \in \mathbb{R}$ [1] (b) $|x|-3, \quad x \in \mathbb{R}$ [1]

(c) $\sqrt{x}+2, \quad x \geq 0$ [2] (d) $3-|x+1|, \quad x \in \mathbb{R}$ [2]

4. Find the range of each quadratic by completing the square. Each is defined for all real x. If you have not met this method yet, it is set out in §4.5.

(a) $x^2 - 6x + 11$ [3] (b) $x^2 - 4x$ [2]
(c) $-(x-2)^2 + 5$ [2] (d) $-x^2 + 6x - 4$ [3]

5. For $f(x) = x^2 - 3x$, find each value.

(a) $f(2)$ [1] (b) $f(-1)$ [1]
(c) $f(0)$ [1] (d) the values of x with $f(x) = 0$ [2]

6. Find the inverse of each function.

(a) $f(x) = 3x - 6$ [2] (b) $f(x) = \dfrac{x+1}{2}$ [2]
(c) $f(x) = 2x + 7$ [2] (d) $f(x) = \dfrac{1}{x+2}$ [3]
(e) $f(x) = \dfrac{2x+1}{x-3}$ [4] (f) $f(x) = \dfrac{x-4}{x+2}$ [4]

7. Consider $f(x) = \dfrac{3x+2}{x-1}$.

(a) State the domain of f. [1] (b) Find $f^{-1}(x)$. [3]
(c) State the domain of f^{-1}. [1] (d) Verify that $f^{-1}(f(2)) = 2$. [2]

8. Decide whether each rule defines y as a function of x, and justify your answer.

(a) $y^2 = x$ [2] (b) $y = x^2$ [1]

9. The function $f(x) = x^2$ has no inverse over all real numbers. State a restricted domain on which an inverse does exist, and give that inverse. [3]

4.2 Linear Functions

The simplest function that actually changes is the line: its output rises at a constant rate. That rate is the **gradient**, the rise divided by the run.

KEY · IB The gradient of the line through (x_1, y_1) and (x_2, y_2) is

$$m = \frac{y_2 - y_1}{x_2 - x_1}.$$

A gradient and one point on the line are enough to fix the line completely. From that fact come three ways to write its equation. All three describe the same line.

KEY · IB A straight line can be written

$$y = mx + c, \qquad y - y_1 = m(x - x_1), \qquad ax + by + d = 0.$$

Here m is the gradient and c the y-intercept.

Use the first form when you know the gradient and the y-intercept. Use the second when you know the gradient and any point. The third is the tidy general form, with whole-number coefficients.

Two lines are **parallel** when they rise at the same rate. They are **perpendicular** when you get one gradient from the other by turning the fraction upside down and changing the sign.

KEY For two lines with gradients m_1 and m_2:

$$m_1 = m_2 \text{ (parallel)}, \qquad m_1 m_2 = -1 \text{ (perpendicular)}.$$

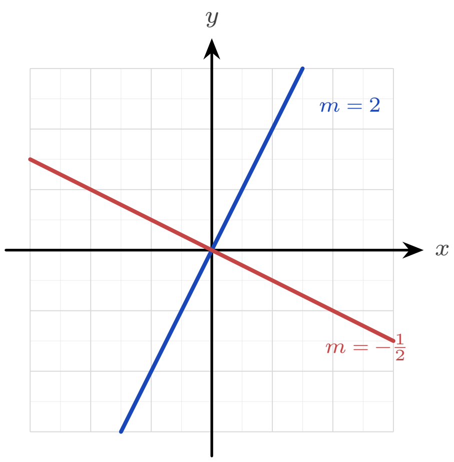


Figure 4.2 Perpendicular gradients: turn the fraction upside down and change the sign.

EXAMPLE 4.8

Find the equation of the line through $(1, 2)$ and $(4, 8)$.

First the gradient:

$$m = \frac{8-2}{4-1} = \frac{6}{3} = 2.$$

Then use the point $(1, 2)$ in point–gradient form:

$$y - 2 = 2(x - 1) \implies y = 2x.$$

EXAMPLE 4.9

Find the equation of the line through $(0, 4)$ perpendicular to $y = 3x - 1$.

The given line has gradient 3, so the perpendicular gradient is $-\frac{1}{3}$. Through $(0, 4)$,

$$y = -\tfrac{1}{3}x + 4.$$

For a perpendicular gradient, turn the fraction upside down and change the sign.

YOUR TURN 4.2 Linear Functions

10. Find the gradient of the line through each pair of points.

(a) $(1, 3)$ and $(5, 11)$ [1]

(b) $(2, 1)$ and $(4, 7)$ [1]

(c) $(-1, 4)$ and $(3, -4)$ [2]

(d) $(0, 5)$ and $(2, 5)$ [2]

11. Find the equation of each line, giving your answer as $y = mx + c$.

(a) gradient 2, through $(0, -3)$ [1]

(b) through $(2, 1)$ and $(4, 7)$ [2]

(c) gradient $-\frac{1}{2}$, through $(4, 0)$ [2]

(d) through $(0, 5)$, perpendicular to $y = -2x$ [3]

12. Write each equation as $y = mx + c$ and state the gradient.

(a) $2x + 3y = 6$ [2]

(b) $4x - 2y = 8$ [2]

13. Answer each question about parallel and perpendicular lines.

(a) Write down the gradient of a line perpendicular to $y = 4x + 1$. [1]

(b) Write down the gradient of a line parallel to $3x + y = 7$. [2]

(c) Are $y = 2x + 1$ and $y = 2x - 3$ parallel? Give a reason. [2]

(d) Are $y = 3x - 1$ and $y = -\frac{1}{3}x + 2$ perpendicular? Give a reason. [2]

4.3 Systems of Linear Equations HL

Two lines in a plane usually cross at one point. Finding that point means finding the values of x and y that satisfy *both* equations at once, a **system** of equations. The standard tool is **elimination**: combine the equations to cancel one variable, solve for the other, then substitute back.

EXAMPLE 4.10

Solve $2x + y = 7$ and $x - y = 2$.

Adding the two equations cancels y:

$$(2x + y) + (x - y) = 7 + 2 \implies 3x = 9 \implies x = 3.$$

Substituting into $x - y = 2$ gives $3 - y = 2$, so $y = 1$. The lines cross at $(3, 1)$.

Not every system has a single answer, and the picture shows why. Two lines can do three things. They cross once, which gives a **unique solution**. They stay parallel and never meet, so there is **no solution**. Or they are the same line drawn twice, which gives **infinitely many solutions**.

CAUTION If two equations have the same gradient but different intercepts, such as $2x + y = 3$ and $4x + 2y = 10$, the lines are parallel and the system has *no* solution. Elimination then produces a false statement such as $0 = 4$.

The same idea extends to three equations in three unknowns, where each equation is now a plane in space. You eliminate one variable at a time to reduce a 3×3 system to a 2×2 one, or read the solution directly from a GDC.

EXAMPLE 4.11

Solve the system

$$x + y + z = 6, \qquad 2x - y + z = 3, \qquad x + 2y - z = 2.$$

Eliminate z in pairs. Subtracting the first equation from the second:

$$(2x - y + z) - (x + y + z) = 3 - 6 \implies x - 2y = -3.$$

Adding the first and third equations:

$$(x + 2y - z) + (x + y + z) = 2 + 6 \implies 2x + 3y = 8.$$

Now solve this 2×2 system. From $x - 2y = -3$ we get $x = 2y - 3$; substituting,

$$2(2y - 3) + 3y = 8 \implies 7y = 14 \implies y = 2,$$

so $x = 1$. Finally the first equation gives $1 + 2 + z = 6$, hence $z = 3$. The solution is $(x, y, z) = (1, 2, 3)$.

Eliminating one variable at a time, working down the system, is called **row reduction**. It is what your GDC's simultaneous-equation solver does inside. Use the solver to check your work. You still need the algebra in an examination, and you need it when the solver returns only an error message.

The augmented matrix

Look again at the elimination above and notice what you were really changing: never the letters x, y, z, only the numbers in front of them. The letters only marked the columns, so you can stop writing them. Reduce the system to its coefficients and constants, written in a grid called the **augmented matrix**:

$$\begin{cases} x+y+z=6 \\ 2x-y+z=7 \\ x+2y-z=3 \end{cases} \quad\longrightarrow\quad \left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & -1 & 1 & 7 \\ 1 & 2 & -1 & 3 \end{array}\right)$$

The vertical bar shows where the equals signs were. Nothing new is happening: the matrix records the same elimination in a shorter form, with fewer symbols to copy incorrectly.

Elimination becomes three **row operations**. None of them changes the solution set. Each one is reversible, and each does to a whole equation only what you were always allowed to do:

> KEY The three elementary row operations:
>
> $$R_i \leftrightarrow R_j \ \text{(swap two rows)}, \qquad R_i \to kR_i \ \text{(multiply a row by } k \neq 0), \qquad R_i \to R_i + kR_j.$$
>
> *Swapping reorders the equations, scaling multiplies one through by a constant, and the third adds a multiple of one equation to another: the same three moves as elimination, written compactly.*

The goal is a step shape, with zeros below the first entry of each row. The last row then contains one unknown, the row above it two, and so on. This shape is called **row echelon form**. Once you reach it, work upwards from the last row, finding one unknown at a time. That is **back-substitution**.

EXAMPLE 4.12

Solve, by row reduction,

$$x+y+z=6, \qquad 2x-y+z=7, \qquad x+2y-z=3.$$

Work on the augmented matrix, clearing the first column below the pivot. $R_2 \to R_2 - 2R_1$ and $R_3 \to R_3 - R_1$:

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & -1 & 1 & 7 \\ 1 & 2 & -1 & 3 \end{array}\right) \to \left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -3 & -1 & -5 \\ 0 & 1 & -2 & -3 \end{array}\right)$$

A leading entry of 1 gives simpler arithmetic than -3, so swap $R_2 \leftrightarrow R_3$. Then clear below it with $R_3 \to R_3 + 3R_2$:

$$\to \left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & -2 & -3 \\ 0 & -3 & -1 & -5 \end{array}\right) \to \left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & -2 & -3 \\ 0 & 0 & -7 & -14 \end{array}\right)$$

The step shape is complete. Now read it from the bottom up. The last row says $-7z = -14$, so $z = 2$. The middle row says $y - 2z = -3$, so $y = 4 - 3 = 1$. The top row gives $x = 6 - y - z = 3$. The solution is $(x, y, z) = (3, 1, 2)$.
Every step here is the elimination you already know. The matrix keeps the working short and the columns lined up, and misaligned columns are the usual cause of mistakes.

Once a system is in row echelon form, the last row determines the outcome. There are only three possible outcomes, and every 3×3 system gives exactly one of them.

Last row reduces to	What it means	The three planes
$(0\ 0\ a \mid b)$, $a \neq 0$	one unique solution	meet at a single point
$(0\ 0\ 0 \mid c)$, $c \neq 0$	no solution: the row says $0 = c$, which is false	have no common point; two are parallel, or the three meet in a triangular prism
$(0\ 0\ 0 \mid 0)$	infinitely many solutions: the row says $0 = 0$, which is true but tells us nothing	share a whole line, or a whole plane if two rows vanish

Count the rows that vanish. No vanishing row gives one solution. One vanishing row gives a line of solutions, described by one free variable. Two vanishing rows mean all three equations describe the same plane, and the solutions form that whole plane, described by two free variables.

CAUTION Read the constant on the right before deciding. The rows $(0\ 0\ 0 \mid 0)$ and $(0\ 0\ 0 \mid 5)$ look almost identical, and they give opposite answers: infinitely many solutions, and none at all.

When a system has no unique solution

Each equation in three unknowns is a plane in space. Three planes usually meet at exactly one point. That point is the unique solution found in the example above.

Two other arrangements are possible. If two planes are parallel, or if the three meet in three parallel lines like the faces of a triangular prism, no point lies on all three. The system is **inconsistent**, and there is no solution. If instead the three planes share a whole line, like pages sharing a spine, every point of that line is a solution, so there are infinitely many.

You can separate the two cases by algebra alone, without a picture. If elimination produces a false statement such as $0 = 4$, the planes have no common point, and there is no solution. If it produces $0 = 0$, one equation repeats information already contained in the other two. There are infinitely many solutions, and one variable may take any value.

EXAMPLE 4.13

Consider the system

$$x + y + 2z = 2, \qquad 2x - y + z = 1, \qquad x + 4y + 5z = k.$$

Find the value of k for which the system has infinitely many solutions, and give the general solution. Show that for every other value of k there is no solution.

The question asks for infinitely many solutions, so expect a row that reduces to zeros. Eliminate x using the first equation. Second minus twice the first:

$$(2x - y + z) - 2(x + y + 2z) = 1 - 4 \implies -3y - 3z = -3 \implies y + z = 1.$$

Third minus the first:

$$(x + 4y + 5z) - (x + y + 2z) = k - 2 \implies 3y + 3z = k - 2 \implies y + z = \frac{k-2}{3}.$$

The same left side, $y+z$, has appeared twice. The two remaining equations are consistent only if

$$1 = \frac{k-2}{3} \implies k = 5.$$

If $k \neq 5$, subtracting them gives $0 = \frac{k-5}{3} \neq 0$, a false statement: the system is inconsistent, and no solution exists for any such k.

If $k = 5$, the third equation gives no new information, and the system reduces to two equations. Let the free variable be $z = t$. Then $y = 1 - t$, and the first equation gives

$$x = 2 - y - 2z = 2 - (1 - t) - 2t = 1 - t.$$

The general solution is

$$(x, y, z) = (1 - t,\ 1 - t,\ t), \qquad t \in \mathbb{R},$$

a whole line of solutions: the line where the three planes meet. Check with $t = 0$: the point $(1, 1, 0)$ satisfies all three original equations. So does $(0, 0, 1)$, from $t = 1$. One free parameter gives one line, so the geometry and the algebra give the same result.

CAUTION ``No unique solution'' covers two different cases, so always separate them. A row that reduces to $0 = 0$ gives infinitely many solutions. A row that reduces to $0 = c$, with $c \neq 0$, gives none. Examiners ask for both, and for the general solution in terms of a parameter.

YOUR TURN 4.3 Systems of Linear Equations HL

14. Solve each system by elimination.

(a) $x + y = 5, \quad x - y = 1$ [2]

(b) $3x - y = 5, \quad x + y = 3$ [2]

(c) $x + y = 4, \quad 2x + y = 7$ [2]

(d) $2x + 3y = 12, \quad x - y = 1$ [3]

15. Solve each system by substitution.

(a) $y = 2x, \quad x + y = 6$ [2]

(b) $y = x - 1, \quad 2x + y = 8$ [2]

16. State how many solutions each system has, and give a reason.

(a) $2x + y = 4, \quad 4x + 2y = 9$ [2]

(b) $y = x + 1, \quad y = x + 3$ [2]

(c) $x + y = 3, \quad 2x + 2y = 6$ [2]

(d) $x + y = 3, \quad x - y = 1$ [2]

17. Solve each system of three equations.

(a) $x + y + z = 6, \quad x - y + z = 2, \quad x + y - z = 0$ [3]

(b) $x + y + z = 4, \quad 2x - y + z = 8, \quad x + 2y - z = -3$ [4]

(c) $x + y + z = 2, \quad 2x - y + 3z = -1, \quad x - 2y + z = -1$ [4]

(d) $2x + y - z = -4, \quad x + 3y + 2z = 3, \quad 3x - y + z = 9$ [4]

18. For parts (a) to (c), write the augmented matrix, reduce it to row echelon form, and state the number of solutions.

(a) $x + y + z = 6, \quad x + 2y - z = 2, \quad 2x + 3y + z = 11$ [4]

(b) $x + y + z = 3, \quad 2x + 2y + 2z = 7, \quad x - y + z = 1$ [3]

(c) $x + y + z = 6, \quad x - y + z = 2, \quad 2x + 2z = 8$ [4]

(d) For $x + y + z = 3, \quad 2x + y + 3z = 7, \quad 3x + 2y + 4z = k$, find the value of k for which the system has infinitely many solutions, and state what happens for every other value of k. [4]

4.4 Quadratic Functions

A **quadratic** is a function whose highest power is x^2. Its graph is the **parabola**, the curve of Galileo's cannonball: a single smooth arch, symmetric about a vertical line through its turning point.

The same quadratic can be written in three forms, and each form shows a different feature immediately.

KEY A quadratic can be written in three forms:

$$y = ax^2 + bx + c, \qquad y = a(x-p)(x-q), \qquad y = a(x-h)^2 + k.$$

Standard form shows the y-intercept c; **factored** form shows the roots p and q; **vertex** form shows the turning point (h, k).

The sign of a sets the direction. If $a > 0$ the parabola opens upward and the turning point is a minimum. If $a < 0$ it opens downward and the turning point is a maximum. The curve is symmetric, and its mirror line passes through the vertex.

KEY · IB The graph of $f(x) = ax^2 + bx + c$ has axis of symmetry

$$x = -\frac{b}{2a}.$$

To move from standard form to vertex form, **complete the square**. This means rewriting $x^2 + \frac{b}{a}x$ as a perfect square, then subtracting whatever that added. It is the most useful step in the whole topic.

EXAMPLE 4.14

Write $y = x^2 - 6x + 5$ in vertex form, and state its vertex and axis of symmetry.

Take the $x^2 - 6x$ part and complete the square. Half of -6 is -3, and $(-3)^2 = 9$:

$$x^2 - 6x = (x-3)^2 - 9.$$

So

$$y = (x-3)^2 - 9 + 5 = (x-3)^2 - 4.$$

The vertex is $(3, -4)$ and the axis of symmetry is $x = 3$. Because $a = 1 > 0$, this vertex is the minimum point.

EXAMPLE 4.15

Sketch $y = x^2 - 2x - 3$, showing the vertex, axis of symmetry, and all intercepts.

Factor for the roots: $x^2 - 2x - 3 = (x-3)(x+1)$, so the x-intercepts are $x = -1$ and $x = 3$. The y-intercept is $c = -3$. Completing the square, $y = (x-1)^2 - 4$, so the vertex is $(1, -4)$ and the axis of symmetry is $x = 1$, which is exactly halfway between the two roots.

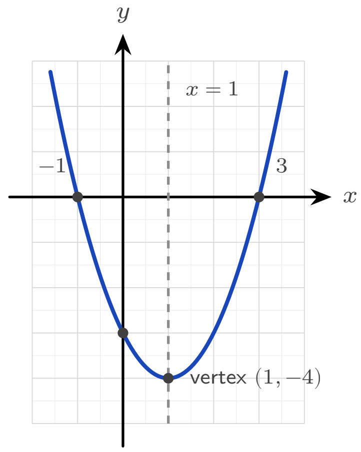


Figure 4.3 $y = x^2 - 2x - 3$, with its vertex, axis of symmetry and intercepts.

YOUR TURN 4.4 Quadratic Functions

19. Write each quadratic in vertex form and state its vertex.

(a) $y = x^2 + 4x + 1$ [2] (b) $y = x^2 - 4x$ [2]

(c) $y = x^2 - 6x + 11$ [2] (d) $y = x^2 + 2x + 5$ [2]

20. State the vertex and the axis of symmetry of each quadratic.

(a) $y = (x-2)^2 + 5$ [2] (b) $y = x^2 - 8x + 3$ [3]

(c) $y = -(x+1)^2 + 2$ [2] (d) $y = 2(x-3)^2 - 1$ [2]

21. Consider $y = (x-1)(x+5)$.

(a) Write down its roots. [1] (b) Find its y-intercept. [1]

(c) Find its axis of symmetry. [2] (d) Find its vertex. [2]

22. State whether each quadratic has a maximum or a minimum, and give a reason.

(a) $y = -x^2 + 2x$ [1] (b) $y = 3x^2 - x + 1$ [1]

4.5 Quadratic Equations

Setting a quadratic equal to zero is the same as asking where its parabola crosses the x-axis. There are three ways to find those crossings, and all three give the same answers. Choosing the right method saves time in an examination.

Method	Use it when	Be careful with
Factorisation	the quadratic factorises with whole numbers	many quadratics do not factorise, and searching for factors takes time
Completing the square	you also want the vertex, or an exact surd answer	the arithmetic is harder when $a \neq 1$
The formula	always. It works for every quadratic	slower than factorising when factors are obvious

Solving by factorisation

If the quadratic factorises, you can read the roots immediately. The reason is the zero product rule: a product is zero only when one of its factors is zero.

EXAMPLE 4.16

Solve $x^2 - 5x + 6 = 0$.

Find two numbers that multiply to 6 and add to -5. They are -2 and -3:

$$(x-2)(x-3) = 0.$$

A product is zero only when a factor is zero, so $x - 2 = 0$ or $x - 3 = 0$, giving $x = 2$ or $x = 3$.

CAUTION The zero product rule needs a zero on the right. From $(x-2)(x-3) = 2$ you may *not* conclude $x - 2 = 2$. Expand, collect everything on one side, and factorise again.

Solving by completing the square

Completing the square works whether or not the quadratic factorises, and it gives exact answers in surd form. Rewrite the quadratic as a perfect square plus a constant, then take square roots of both sides.

EXAMPLE 4.17

Solve $x^2 + 8x + 7 = 0$ by completing the square.

Half of 8 is 4, and $4^2 = 16$, so $x^2 + 8x = (x+4)^2 - 16$. The equation becomes

$$(x+4)^2 - 16 + 7 = 0 \implies (x+4)^2 = 9.$$

Take the square root of both sides, remembering both signs:

$$x + 4 = \pm 3 \implies x = -1 \text{ or } x = -7.$$

EXAMPLE 4.18

Solve $x^2 - 4x - 1 = 0$, giving exact answers.

Half of -4 is -2, and $(-2)^2 = 4$:

$$(x-2)^2 - 4 - 1 = 0 \implies (x-2)^2 = 5 \implies x - 2 = \pm\sqrt{5}.$$

So $x = 2 \pm \sqrt{5}$. This quadratic does not factorise with whole numbers, which is why the answer contains a surd.

TIP When $a \neq 1$, divide the whole equation by a first. For $2x^2 - 8x + 5 = 0$, divide by 2 to get $x^2 - 4x + 2.5 = 0$, then complete the square as usual.

Solving by formula

The formula works for every quadratic, and it is in the formula booklet.

KEY · IB The solutions of $ax^2 + bx + c = 0$, with $a \neq 0$, are

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

Substitute the three coefficients and the formula returns both roots at once.

The formula is not a rule to memorise without understanding. It is what you get by completing the square once, using letters instead of numbers. Start from $ax^2 + bx + c = 0$, divide by a, complete the square on $x^2 + \frac{b}{a}x$, then solve. The $\pm$ appears because the parabola is symmetric, with one root on each side of the axis.

EXAMPLE 4.19

Solve $2x^2 - 3x - 2 = 0$.

Here $a = 2$, $b = -3$, $c = -2$. The formula gives

$$x = \frac{3 \pm \sqrt{9+16}}{4} = \frac{3 \pm 5}{4},$$

so $x = 2$ or $x = -\frac{1}{2}$.

YOUR TURN 4.5 Quadratic Equations

23. Solve each equation by factorisation.

(a) $x^2 - 5x + 6 = 0$ [2] (b) $x^2 - x - 12 = 0$ [2]
(c) $x^2 + x - 20 = 0$ [2] (d) $2x^2 - 5x - 3 = 0$ [3]

24. Solve each equation by completing the square, giving exact answers.

(a) $x^2 + 8x + 7 = 0$ [3] (b) $x^2 - 2x - 5 = 0$ [3]
(c) $x^2 + 6x + 2 = 0$ [3] (d) $2x^2 - 8x + 5 = 0$ *(Hint: divide by 2 first)* [4]

25. Solve each equation with the quadratic formula, giving exact answers.

(a) $x^2 + 2x - 1 = 0$ [2] (b) $x^2 + 6x + 4 = 0$ [2]
(c) $2x^2 - 3x - 2 = 0$ [3] (d) $3x^2 + 2x - 1 = 0$ [3]

26. For each equation, state which of the three methods you would choose, and give a reason. You do not need to solve the equations.

(a) $x^2 - 7x + 12 = 0$ [1] (b) $x^2 + 4x - 2 = 0$ [1]
(c) $5x^2 - 3x - 1 = 0$ [1] (d) $x^2 - 9 = 0$ [1]

4.6 The Discriminant

The quadratic formula answers more than it is asked. Before you calculate a single root, the quantity under the square root sign already tells you how many real roots there are. That quantity has its own name.

KEY · IB The **discriminant** of $ax^2 + bx + c$ is

$$\Delta = b^2 - 4ac.$$

Its sign tells you how many real roots the equation has, without solving it.

The sign of Δ determines how many times the parabola meets the x-axis. The reason is that a negative number has no real square root, so $\pm\sqrt{\Delta}$ has no real value.

Discriminant	Real roots	Parabola and x-axis
$\Delta > 0$	two distinct	crosses twice
$\Delta = 0$	one repeated	touches once
$\Delta < 0$	none	does not meet it

EXAMPLE 4.20

Find the values of k for which $x^2 + kx + 9 = 0$ has exactly one (repeated) root.

One repeated root means $\Delta = 0$:

$$k^2 - 4(1)(9) = 0 \implies k^2 = 36 \implies k = \pm 6.$$

At these two values the parabola touches the x-axis at one point. Notice that the discriminant answers a question about the roots without finding them.

EXAMPLE 4.21

For the equation $3kx^2 + 2x + k = 0$, find the values of k that give two distinct real roots, two equal real roots, and no real roots.

Here the unknown constant appears in two places: $a = 3k$ and $c = k$, with $b = 2$. So

$$\Delta = b^2 - 4ac = 4 - 4(3k)(k) = 4 - 12k^2.$$

Now consider the three cases separately.

$$\Delta > 0: \ 4 - 12k^2 > 0 \implies k^2 < \tfrac{1}{3} \implies -\tfrac{1}{\sqrt{3}} < k < \tfrac{1}{\sqrt{3}}.$$

$$\Delta = 0: \ k^2 = \tfrac{1}{3} \implies k = \pm\tfrac{1}{\sqrt{3}}.$$

$$\Delta < 0: \ k^2 > \tfrac{1}{3} \implies k < -\tfrac{1}{\sqrt{3}} \ \text{ or } \ k > \tfrac{1}{\sqrt{3}}.$$

One value of k must be checked separately. If $k = 0$ the equation becomes $2x = 0$. That is linear, not quadratic, and it has the single root $x = 0$. So the answer for two distinct real roots is $-\frac{1}{\sqrt{3}} < k < \frac{1}{\sqrt{3}}$, *with* $k \neq 0$.

CAUTION Before using the discriminant, check that the equation really is quadratic. If the coefficient of x^2 contains the unknown constant, the value that makes it zero must be excluded and treated separately.

TIP A question that says ``two distinct roots'', ``equal roots'', ``a repeated root'', ``touches the x-axis'', or ``no real solutions'' is asking about the discriminant. Recognising the wording tells you which method to use.

YOUR TURN 4.6 The Discriminant

27. Evaluate the discriminant and state the nature of the roots.

(a) $x^2 - 4x + 4$ [2]
(b) $2x^2 + 3x + 5$ [2]
(c) $3x^2 - 2x - 1$ [2]
(d) $x^2 + 5x + 7$ [2]

28. Find the value or values of k in each case.

(a) $x^2 + 6x + k = 0$ has equal roots [2]
(b) $x^2 + kx + 16 = 0$ has equal roots [3]
(c) $x^2 - 3x + k = 0$ has no real roots [3]
(d) $kx^2 + 4x + k = 0$ has equal roots. State any value of k that must be excluded. [4]

29. The equation $x^2 + (k+1)x + 9 = 0$ has two distinct real roots. Find the set of values of k. [4]

4.7 Quadratic Inequalities

A quadratic **inequality** is a different question. It does not ask where the parabola meets the axis. It asks where the parabola lies above the axis, or below it. The method is always the same three steps.

KEY To solve a quadratic inequality:

1. Collect every term on one side, so the other side is 0.
2. Find the roots of the matching equation. These are the only places where the sign can change.
3. Sketch the parabola and read the intervals where it lies on the side you want.

EXAMPLE 4.22

Solve $x^2 - x - 6 > 0$.

Factor: $x^2 - x - 6 = (x-3)(x+2)$, so the roots are $x = -2$ and $x = 3$. The parabola opens upward. It is therefore *above* the axis outside the roots, and below the axis between them. So

$$x^2 - x - 6 > 0 \quad \text{when} \quad x < -2 \text{ or } x > 3.$$

A rough sketch of the parabola is faster and safer than memorising sign rules.

EXAMPLE 4.23

Solve $2x^2 \leq x + 6$.

First collect everything on one side:

$$2x^2 - x - 6 \leq 0.$$

Factorise: $(2x+3)(x-2) = 0$ gives roots $x = -\frac{3}{2}$ and $x = 2$. The parabola opens upward, so it lies *below* the axis between the roots. The sign $\leq$ includes the roots themselves:

$$-\tfrac{3}{2} \leq x \leq 2.$$

CAUTION Never divide an inequality by a term containing x. Its sign is unknown, and dividing by a negative quantity reverses the inequality. Collect on one side and factorise instead.

EXAMPLE 4.24

Solve $x^2 + x + 1 > 0$.

The discriminant is $1 - 4 = -3 < 0$, so the parabola never meets the x-axis. Since $a = 1 > 0$ it opens upward, so it lies entirely above the axis. The inequality holds for *every* real x.

When the discriminant is negative, a quadratic inequality has either every real number as its answer or none at all. The sign of a decides which.

YOUR TURN 4.7 Quadratic Inequalities

30. Solve each inequality.

(a) $x^2 - 9 < 0$ [2]

(b) $x^2 + 2x - 15 < 0$ [2]

(c) $x^2 - 4x + 3 \geq 0$ [3]

(d) $-x^2 + 4x > 0$ [3]

31. Solve each inequality. Two of them are special cases, so justify your answers.

(a) $2x^2 \leq x + 6$ [3]

(b) $x^2 \geq 16$ [2]

(c) $x^2 + x + 1 > 0$ [3]

(d) $x^2 + 4 < 0$ [2]

32. A rectangle has width x and length $x + 2$, both in centimetres.

(a) Show that its area is $x^2 + 2x$. [1]

(b) Find the values of x for which the area is less than 15. [4]

Chapter 4 · Lines & Quadratics

Reflections

TOK The same parabola appears in pure algebra, as the graph of x^2, and in the physical world, as the path of every thrown object, the cross-section of a satellite dish, and the cable of certain bridges. Why should a curve defined by an abstract equation also describe falling stones? This is what the physicist Eugene Wigner called the "unreasonable effectiveness of mathematics." Is mathematics invented to fit the world, or discovered as something already present in it?

IM Quadratics are among the oldest mathematics we have. Babylonian clay tablets from around 2000 BCE solve them to divide land and dig canals. The systematic method is younger: around 820 CE in Baghdad, al-Khwārizmī wrote *al-Kitāb al-mukhtaṣar fī ḥisāb al-jabr*, completing the square with geometric squares and rectangles. From its title, *al-jabr* ("restoring"), comes the word *algebra*, and from his name comes *algorithm*.

RW A parabola reflects every incoming parallel ray to a single point, called the focus. Satellite dishes and radio telescopes use this to collect weak signals at one point. Car headlights and torches use it in reverse: a bulb at the focus produces a parallel beam. Solar concentrators use it to focus sunlight and boil water. Quadratics also model optimisation everywhere: the maximum area for a fixed fence, the price that maximises profit, the angle at which a ball travels farthest.

EXT The discriminant divides quadratics into three cases. When $\Delta < 0$ the parabola does not meet the x-axis and there are no *real* roots, but the quadratic formula still returns answers, now containing $\sqrt{-1}$. Keeping these "impossible" numbers instead of rejecting them leads to the **complex numbers** of Chapter 18, where every quadratic has exactly two roots and the discriminant only tells you whether those roots are real.

End-of-Chapter Problems — Chapter 4: Lines & Quadratics

1. Consider $f(x) = 2x - 1$.

(a) Find $f^{-1}(x)$. [2]

(b) Verify that $f(f^{-1}(x)) = x$. [2]

2. Consider $f(x) = 3 - 2x$.

(a) State the x- and y-intercepts. [2]

(b) Find $f^{-1}(x)$. [2]

3. Consider $f(x) = \dfrac{2x+1}{x-3}$, $x \neq 3$.

(a) State the domain. [1]

(b) Find $f^{-1}(x)$. [4]

4. Find the equation of the line through $(3, -1)$ parallel to $2x - y = 4$. [3]

5. Consider the points $A(-1, 2)$ and $B(3, 10)$.

(a) Find the midpoint of AB. [1]

(b) Find the gradient of AB. [1]

(c) Find the length of AB. [2]

6. The line $2x + 3y = 12$ is given.

(a) Find its x- and y-intercepts. [2]

(b) State its gradient. [2]

7. A line passes through $A(1, 2)$ and $B(5, 4)$.

(a) Find the gradient of AB. [1]

(b) Find the equation of AB. [2]

(c) Find the equation of the perpendicular bisector of AB. [4]

8. Solve the system $3x + 2y = 7, \quad x - 2y = -3$. [3]

9. Solve the system

$$x + 2y + z = 4, \qquad 2x + y - z = 2, \qquad x - y + 2z = 2.$$

[6]

10. Consider the system

$$x + y + z = 3, \qquad x + 2y + 3z = 4, \qquad 2x + 3y + 4z = k.$$

(a) Find the value of k for which the system has solutions. [3]
(b) For this value of k, find the general solution. [3]
(c) Describe the configuration of the three planes when $k = 10$. [1]

11. A quadratic has roots 2 and -5.

(a) Write it in factored form (with leading coefficient 1). [1]
(b) Expand it into standard form. [2]

12. A parabola passes through $(0, 3)$, $(1, 0)$ and $(3, 0)$. Find its equation. [5]

13. Consider $f(x) = x^2 + 2x - 3$.

(a) Factorise $f(x)$ and state its roots. [2]
(b) Find the vertex. [2]
(c) Sketch the graph, showing all intercepts. [2]

14. Consider $f(x) = x^2 - 6x + 5$.

(a) Write $f(x)$ in vertex form. [2]
(b) State the vertex and the axis of symmetry. [2]
(c) Find the x- and y-intercepts. [3]

15. Solve the inequality $x^2 - 2x - 8 \leq 0$. [3]

16. The equation $x^2 - 4x + c = 0$ has no real roots. Find the set of values of c. [3]

17. Solve the inequality $x^2 + x - 6 > 0$. [3]

18. The equation $x^2 + kx + 4 = 0$ has two distinct real roots. Find the set of values of k. [4]

19. Find the values of k for which $x^2 + kx + (k + 3) = 0$ has equal roots. [4]

20. Two numbers have a sum of 10 and a product of 21. Find the numbers. [4]

21. The roots of $x^2 + bx + c = 0$ sum to 6 and differ by 4. Find b and c. [5]

22. Show that $x^2 + (k+1)x + k = 0$ has real roots for every real k, and find the value of k for which the roots are equal. [5]

23. The roots of $x^2 - px + q = 0$ are α and 2α. Show that $2p^2 = 9q$. [5]

24. Find the points of intersection of $y = x^2$ and $y = 2x + 3$. [4]

25. Consider the line $y = 2x - 5$ and the parabola $y = x^2 - 2x + 2$.

(a) Show that they do not intersect. [3]
(b) Explain what this means geometrically. [1]

26. A ball is thrown so that its height in metres after t seconds is $h(t) = 20t - 5t^2$.

(a) Find the time at which the ball lands. [2]
(b) Find the maximum height reached. [3]

27. Find the value of k for which the line $y = x + k$ is a tangent to the parabola $y = x^2$. [5]

28. A rectangle has length 3 more than its width w.

(a) Write its area A as a function of w. [2]
(b) Given the area is 40, find w. [3]

29. A rectangle has perimeter 20. Show that its area is $A = x(10-x)$, where x is one side, and find the maximum possible area. [5]

30. Prove that the line $y = mx + c$ is a tangent to the parabola $y = x^2$ if and only if $c = -\dfrac{m^2}{4}$. [6]

31. GDC Consider $f(x) = x^3 - 4x + 1$.

(a) Using your GDC, sketch the graph of f for $-3 \leq x \leq 3$, labelling all key features. [2]
(b) Find the zeros of f, giving your answers to 3 significant figures. [2]
(c) Find the coordinates of the local maximum and local minimum points. [2]
(d) Solve $f(x) = 2x - 1$, giving your answers to 3 significant figures. [2]

32. A bridge has a parabolic arch. The arch is 40 m wide at road level and 10 m high at its centre. Take the road as the x-axis, with the centre of the arch at $x = 0$.

(a) Show that the arch is modelled by $y = 10 - \frac{x^2}{40}$. [4]
(b) Find the height of the arch 12 m from the centre. [2]
(c) A lorry 8 m wide and 9.5 m high drives along the centre of the road. Determine whether it passes under the arch. [4]
(d) Find the greatest width of a lorry 9 m high that can pass under the arch, giving your answer to three significant figures. [4]
(e) The model gives a negative value of y when $|x| > 20$. Explain what this means for the model. [2]

33. Consider the line $y = mx + 3$ and the parabola $y = x^2 + 5x + 7$.

(a) Show that at any point of intersection, $x^2 + (5 - m)x + 4 = 0$. [3]
(b) Find the values of m for which the line is a tangent to the parabola. [4]
(c) Find the values of m for which the line does not meet the parabola. [3]
(d) For the smaller value of m found in part (b), find the coordinates of the point of tangency. [3]
(e) State the values of m for which the line meets the parabola twice. [2]

34. HL Consider the system of equations

$$x + y + z = 2, \qquad x + 2y + 3z = 5, \qquad 2x + 3y + az = b,$$

where a and b are real constants.

(a) Write the system as an augmented matrix and row reduce it. [4]
(b) Find the value of a for which the system does not have a unique solution. [3]
(c) For that value of a, find the value of b for which the system is consistent. [2]
(d) For those values of a and b, find the general solution of the system. [4]
(e) Describe how the three planes are arranged in each of the three cases: a not equal to the value in part (b); a equal to it with b consistent; and a equal to it with b not consistent. [3]

Challenge Questions

35. **Tangents to a parabola.** A line is a tangent to a curve when it meets the curve exactly once. This investigation turns that idea into a condition on the discriminant, then uses it to count the tangents from a point.

(a) Show that the line $y = mx + c$ meets $y = x^2$ where $x^2 - mx - c = 0$, and hence that the line is a tangent exactly when $c = -\frac{m^2}{4}$. [4]

(b) Find the two tangents to $y = x^2$ that pass through the point $(0, -4)$, and state where each one touches the parabola. [4]

(c) Repeat part (a) for the parabola $y = ax^2$, with $a \neq 0$, and show that the tangency condition becomes $c = -\dfrac{m^2}{4a}$. [3]

(d) Using part (c), find how many tangents to $y = x^2$ pass through the point $(0, k)$, for each value of k. State the three cases and explain each one from the picture. [5]

(e) A student claims that every point in the plane has either two tangents to $y = x^2$ or none. Decide whether the claim is true, and justify your answer. [4]

36. A family of planes. Two planes that meet do so in a line. This investigation shows that infinitely many planes contain that same line, and uses the idea to decide when a third equation is consistent.

(a) The planes $P_1 : x + y + z = 1$ and $P_2 : x - y + 2z = 2$ meet in a line. Find that line in the form (x, y, z) in terms of one parameter. [4]

(b) For any constant λ, the equation $P_1 + \lambda P_2 = 0$ describes a plane. Show that every point of the line from part (a) lies on it, whatever λ is. [3]

(c) Find the value of λ that produces the plane $2x + 3z = 3$, and hence explain why the system P_1, P_2, $2x + 3z = 3$ has infinitely many solutions. [4]

(d) Explain why the system P_1, P_2, $2x + 3z = k$ has no solution for every $k \neq 3$. Reduce the augmented matrix to support your answer. [4]

(e) A third plane is chosen at random. Which of the three outcomes in §4.3 is it most likely to produce, and why are the other two special? [3]

37. Sum and product of roots. A quadratic can be read from its roots without ever solving it. This investigation builds the relationships and then uses them to answer questions that would be slow by the formula.

(a) Let the roots of $x^2 - Sx + P = 0$ be α and β. By expanding $(x - \alpha)(x - \beta)$, show that $\alpha + \beta = S$ and $\alpha\beta = P$. [3]

(b) Show that $(\alpha - \beta)^2 = S^2 - 4P$, and hence write down the condition on S and P for the roots to be real. [3]

(c) The roots of a quadratic sum to 6 and differ by 4. Use part (b) to find the quadratic, without finding the roots first. [3]

(d) Show that $\alpha^2 + \beta^2 = S^2 - 2P$ and that $\dfrac{1}{\alpha} + \dfrac{1}{\beta} = \dfrac{S}{P}$. Verify both for $x^2 - 6x + 5 = 0$. [4]

(e) Find the condition on S and P for the two roots to be reciprocals of each other, and give one quadratic with that property and irrational roots. [3]

CHAPTER 5

Function Operations

5

SYLLABUS 2.5 2.10 2.11 2.14–2.16

Hold a card in front of a lamp and look at its shadow on the wall. Slide the card and the shadow slides. Move it closer to the lamp and the shadow grows. Turn it over and the shadow flips. The card itself never changes. Only what you do to it changes.

Graphs work the same way, and that is what makes this part of the course manageable. There are only a few shapes worth learning by heart, a line, a parabola, a square root, a reciprocal, together with a short list of moves that slide them, stretch them and reflect them. Every other curve you will meet is one of those few shapes with some of those moves applied to it.

So this chapter has two halves. The first is the list of moves, and what each one does to an equation. The second is **composition**: the rule for feeding the output of one function straight into the next, which is how those moves are chained together in the first place.

By the end of this chapter you will be able to compose two functions and find the domain of the result, and to describe and apply translations, stretches and reflections. You will solve equations by reading intersections off a graph, test a function for symmetry, and sketch the related graphs $|f(x)|$, $f(|x|)$ and $\dfrac{1}{f(x)}$. What ties these together is that every number in a formula corresponds to something you can see. Change the number and the graph moves in a way you can predict. A formula and a picture are two views of one function, and this chapter is about reading each from the other.

5.1 Composite Functions

Two functions can be joined end to end. Send a number into the first function. Take the number that comes out. Send that number into the second function. The single rule that carries out both steps at once is called a **composite** function.

$$x \longrightarrow \boxed{g} \longrightarrow g(x) \longrightarrow \boxed{f} \longrightarrow f(g(x))$$

Figure 5.1 Composition sends x through g first, then feeds the result into f. The inner function acts first, which is why $f \circ g$ is read from the inside out.

> DEF The **composite** $f \circ g$ is the function $(f \circ g)(x) = f(g(x))$: apply g first, then f. The **identity function** $I(x) = x$ returns every input unchanged.

The symbol $\circ$ is read "of", so $f \circ g$ is "f of g". It is also read "f after g", which says the same thing in the order the functions act. It is not multiplication. Read the notation from right to left: the function written closest to x is the one that acts on x first.

The key to composition is that the x in $f(x)$ is only a **placeholder**. It marks the position where the input goes. It does not have to be a single letter, and it does not have to be a number.

Read f as a rule with an empty slot. If $f(x) = x^2 + 1$, the rule says *square the input, then add one*. So $f(3) = 3^2 + 1$, and $f(2x) = (2x)^2 + 1$, and $f(\text{anything}) = (\text{anything})^2 + 1$. The input can be an expression, and it can be another function:

$$f(g(x)) = \big(g(x)\big)^2 + 1.$$

Nothing new is happening here. You are substituting a whole function exactly as you would substitute a number.

To build $(f \circ g)(x)$, apply g to x, then replace every x in the formula for $f(x)$ by the whole expression $g(x)$, then simplify.

The order changes the answer. Putting on socks then shoes is not the same as shoes then socks, and composing two functions behaves the same way.

EXAMPLE 5.1

For $f(x) = x^2 + 1$ and $g(x) = 2x$, find $(f \circ g)(x)$ and $(g \circ f)(x)$.

For $f \circ g$, apply g first. Its output is $2x$, so substitute $2x$ for every x in $f(x) = x^2 + 1$, then simplify:

$$(f \circ g)(x) = f(2x) = (2x)^2 + 1 = 4x^2 + 1.$$

For $g \circ f$, apply f first. Its output is $x^2 + 1$, so substitute $x^2 + 1$ for every x in $g(x) = 2x$, then simplify:

$$(g \circ f)(x) = g(x^2 + 1) = 2(x^2 + 1) = 2x^2 + 2.$$

The two answers are different. In general $f \circ g \neq g \circ f$, so always check which function acts first.

Composition also explains what an inverse does. The **inverse** of f, written f^{-1}, is the function that reverses f: if f sends 3 to 10, then f^{-1} sends 10 back to 3. Apply f and then apply f^{-1}, and you arrive back at the number you started with. The composite is therefore the identity function.

KEY For a function f with inverse f^{-1},

$$(f^{-1} \circ f)(x) = x \qquad \text{and} \qquad (f \circ f^{-1})(x) = x.$$

An inverse exists exactly when f is **one-to-one**: no two inputs share an output.

YOUR TURN 5.1 Composite Functions

1. For $f(x) = x + 3$ and $g(x) = 2x$, find each composite.

(a) $(f \circ g)(x)$ [2] (b) $(g \circ f)(x)$ [2]
(c) $(f \circ f)(x)$ [2] (d) $(g \circ g)(x)$ [2]

2. For $f(x) = x^2$ and $g(x) = x - 1$, find each of the following.

(a) $(f \circ g)(x)$ [2] (b) $(g \circ f)(x)$ [2]
(c) $(f \circ g)(3)$ [1] (d) $(g \circ f)(3)$ [1]

3. For $f(x) = x + 5$ and $g(x) = x^2$, evaluate each of the following.

(a) $(g \circ f)(2)$ [2] (b) $(f \circ g)(2)$ [2]
(c) $(g \circ f)(-3)$ [2] (d) $(f \circ g)(-3)$ [2]

4. For $f(x) = \dfrac{1}{x}$ and $g(x) = x + 1$, find each composite and state its domain.

(a) $(f \circ g)(x)$ [2] (b) the domain of $f \circ g$ [2]
(c) $(g \circ f)(x)$ [2] (d) the domain of $g \circ f$ [2]

5. Find the inverse of each function.

(a) $f(x) = 3x - 2$ [2] (b) $f(x) = \dfrac{x}{4} + 1$ [3]
(c) $f(x) = \dfrac{x - 5}{2}$ [2] (d) $f(x) = \dfrac{1}{x - 1}$ [3]

6. Let $f(x) = 2x + 1$.

(a) Find $f^{-1}(x)$. [2]
(b) Show that $(f \circ f^{-1})(x) = x$. [2]
(c) Show that $(f^{-1} \circ f)(x) = x$. [2]
(d) State what these two results tell you about f^{-1}. [1]

7. Let $f(x) = x^2$ and $g(x) = x + 2$.

(a) Find $(f \circ g)(x)$ and $(g \circ f)(x)$, and show that they are not the same. [3]
(b) Find the value of x for which the two composites are equal. [3]

5.2 Transformations of Graphs

A graph is an object you can move. There are three kinds of move. A **translation** slides it, a **reflection** flips it, and a **stretch** scales it.

Each move matches one small change in the equation. Learn which change produces which move, and you can read any transformed equation directly.

Move	Equation	Acts on	Effect on the graph
Translation	$y = f(x) + k$	output	up by k, or down if $k < 0$
Translation	$y = f(x - h)$	input	right by h, or left if $h < 0$
Reflection	$y = -f(x)$	output	in the x-axis
Reflection	$y = f(-x)$	input	in the y-axis
Stretch	$y = a\,f(x)$	output	vertically, factor a
Stretch	$y = f(bx)$	input	horizontally, factor $\frac{1}{b}$

Read the third column and the pattern appears. Everything depends on whether the change sits *outside* the bracket or *inside* it: a change to the output is always vertical, and a change to the input is always horizontal.

A change outside the bracket acts on the output, after f has been applied. It behaves as you would expect. Adding k raises every value by k, so the graph moves up by k. Multiplying by a multiplies every value by a, so the graph stretches vertically by a.

A change inside the bracket acts on the input, before f is applied, so its effect is reversed.

Take $y = f(x-h)$. At $x = p+h$ this returns $f(p)$, which is the value the original graph had at $x = p$. Every point therefore moves h units to the *right*, not left. In the same way, $y = f(bx)$ returns $f(p)$ at $x = \frac{p}{b}$, so every point moves from p to $\frac{p}{b}$. That is a horizontal stretch of factor $\frac{1}{b}$, which compresses the graph towards the y-axis whenever $b > 1$.

Translations

A translation slides the whole graph. Its shape and its size do not change, only its position.

Vertically, $y = f(x) + k$ adds k to every output, so every point moves up by k. A negative k moves it down.

Horizontally, $y = f(x-h)$ moves the graph right by h. This is the sign students most often reverse, so it is worth holding on to the reason above: the change is inside the bracket, so it acts on the input, and its effect is the opposite of its sign.

In examination questions a translation is usually written as a vector. The graph of $y = f(x-h)+k$ is the graph of $y = f(x)$ translated by $\binom{h}{k}$.

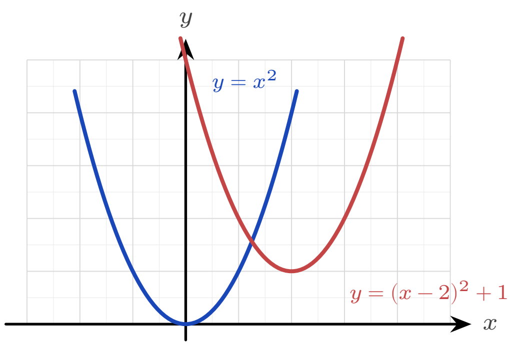


Figure 5.2 The graph of $y = (x-2)^2 + 1$ is the graph of $y = x^2$ translated 2 units right and 1 unit up. The vertex moves from $(0, 0)$ to $(2, 1)$, and the shape is unchanged.

EXAMPLE 5.2

Describe how the graph of $y = (x-2)^2 + 1$ is obtained from $y = x^2$.

Start with the bracket. The -2 sits inside it, so it acts on the input and moves the parabola 2 units *right*.

Now take the rest. The $+1$ sits outside the bracket, so it acts on the output and moves the parabola 1 unit *up*.

Together these send the vertex from $(0, 0)$ to $(2, 1)$, which is exactly what the completed-square form predicts. Transformations and the completed-square form are two descriptions of the same thing.

Reflections

A reflection flips the graph across an axis. Every point keeps its distance from that axis and moves to the other side of it.

For $y = -f(x)$ the sign of every output changes, so the point (p, q) moves to $(p, -q)$. That is a reflection in the x-axis. A maximum becomes a minimum, and any point already on the x-axis stays where it is.

For $y = f(-x)$ the sign of every input changes, so the point (p, q) moves to $(-p, q)$. That is a reflection in the y-axis. The y-intercept does not move, because changing the sign of 0 leaves it as 0.

EXAMPLE 5.3

Describe how the graph of $y = -(x-1)^2$ is obtained from $y = x^2$.

There are two changes. The -1 is inside the bracket, so it is horizontal: translate 1 unit right.

The minus sign is outside the bracket, so it is vertical: reflect in the x-axis.

One change is horizontal and the other is vertical, so they do not affect each other and may be done in either order.

The vertex moves from $(0, 0)$ to $(1, 0)$, and the parabola now opens downward.

Stretches

A stretch changes the size of the graph in one direction only. One line stays fixed while everything else moves away from it or towards it.

For $y = a\,f(x)$ every output is multiplied by a, so (p, q) moves to (p, aq). This is a vertical stretch of factor a. Points on the x-axis do not move, because $a \times 0 = 0$.

For $y = f(bx)$ every input is divided by b, as shown above, so (p, q) moves to $(\frac{p}{b}, q)$. This is a horizontal stretch of factor $\frac{1}{b}$. Points on the y-axis do not move.

CAUTION A factor smaller than 1 is still called a stretch. The graph of $y = \frac{1}{2}f(x)$ is a vertical stretch of factor $\frac{1}{2}$, even though it makes the graph shorter. Write ``stretch, factor $\frac{1}{2}$'', not ``squash''.

EXAMPLE 5.4

The graph of $y = f(x)$ has a maximum point at $(6, 4)$. State the coordinates of that maximum point on the graph of (a) $y = 3f(x)$, (b) $y = f(3x)$.

(a) The 3 is outside the bracket, so it is a vertical stretch of factor 3. It multiplies the y-coordinate and leaves the x-coordinate alone: $(6, 12)$.

(b) The 3 is inside the bracket, so it is a horizontal stretch of factor $\frac{1}{3}$. It divides the x-coordinate and leaves the y-coordinate alone: $(2, 4)$.

Each stretch changes one coordinate and leaves the other unchanged. Checking which coordinate moved is a quick way to confirm you have used the right one.

Composite transformations

Most graphs are several moves away from their parent function, applied one after another. This is a **composite transformation**. Describing one is a matter of taking the changes in a reliable order.

KEY To describe a composite transformation:

1. Split the changes into two groups: those **inside** the bracket, which are horizontal, and those **outside** it, which are vertical.
2. Do the two groups in either order. A horizontal move and a vertical move do not affect each other.
3. Inside each group the order matters. Apply the **stretch before the translation**.

Step 3 is the one to understand rather than memorise. A stretch multiplies everything it is given, including a translation that has already been carried out. Do the stretch first and it has only the original graph to multiply. The second example below shows exactly what goes wrong in the other order.

It also helps to write the target equation in one standard form before you start:

$$y = a\,f(b(x-h)) + k.$$

Then a is the vertical stretch factor, b gives the horizontal stretch factor $\frac{1}{b}$, h is the horizontal translation and k is the vertical translation, and every value can be read directly.

CAUTION Factorise inside the bracket first. In $y = f(2x-6)$ the translation is *not* 6. Written as $y = f(2(x-3))$, it is a horizontal stretch of factor $\frac{1}{2}$ followed by a translation of 3 to the right. Forgetting to factorise is the commonest error in this topic.

EXAMPLE 5.5

Describe a sequence of transformations that maps $y = \sqrt{x}$ onto $y = 2\sqrt{x+3} - 1$.

Take the change inside the bracket first. The $+3$ acts on the input, so it translates the graph 3 units *left* and gives $\sqrt{x+3}$.

Now take the vertical changes, stretch before translation. The factor 2 stretches the graph vertically by scale factor 2 and gives $2\sqrt{x+3}$. The -1 then translates it 1 unit *down*.

So the sequence is: translate 3 left, stretch vertically by factor 2, translate 1 down.

Check it by tracking the endpoint. The graph of $y = \sqrt{x}$ begins at $(0,0)$. The translation carries it to $(-3,0)$. The stretch leaves it there, because $2 \times 0 = 0$. The final translation carries it to $(-3,-1)$, which is exactly where $y = 2\sqrt{x+3} - 1$ begins.

The same care is needed inside the horizontal group, where a stretch and a translation also fail to commute. Within either group the order really does matter. Some pairs of moves give the same result in either order, and such a pair is said to **commute**. Other pairs do not, and swapping those produces a different curve.

EXAMPLE 5.6

Starting from $y = x^2$, apply a vertical stretch by factor 2 and a translation 1 unit up, in both orders. Are the results the same?

Stretch first, then translate. The stretch doubles every value, giving $2x^2$; the translation then adds 1:

$$x^2 \longrightarrow 2x^2 \longrightarrow 2x^2 + 1.$$

Translate first, then stretch. The translation adds 1, giving $x^2 + 1$; the stretch then doubles everything, *including* the 1:

$$x^2 \longrightarrow x^2 + 1 \longrightarrow 2(x^2 + 1) = 2x^2 + 2.$$

The two curves are different. A vertical stretch multiplies whatever is already there, so a translation applied first is multiplied as well. A vertical stretch and a vertical translation commute only when one of them does nothing, that is when the stretch factor is 1 or the translation is 0. In every other case you must state them in the order that actually builds the target curve.

TIP In examinations, ``describe the sequence of transformations'' expects named moves with their values: translation by $\binom{-3}{0}$, vertical stretch scale factor $2$, translation by $\binom{0}{-1}$. Vague words like ``moved'' or ``squashed'' lose the marks.

YOUR TURN 5.2 Transformations of Graphs

8. Describe the single transformation that takes $y = x^2$ to each graph.

(a) $y = x^2 + 3$ [1] (b) $y = (x + 4)^2$ [1]
(c) $y = -x^2$ [1] (d) $y = (x - 1)^2$ [1]

9. Describe the single transformation that takes $y = f(x)$ to each graph.

(a) $y = f(-x)$ [1] (b) $y = -f(x)$ [1]
(c) $y = 3f(x)$ [1] (d) $y = f(2x)$ [2]

10. The point $(2, 5)$ lies on $y = f(x)$. Find the coordinates of its image on each graph.

(a) $y = f(x - 3) + 1$ [2] (b) $y = 3f(x)$ [1]
(c) $y = f(2x)$ [2] (d) $y = -f(x) + 2$ [2]

11. The point $(-2, 3)$ lies on $y = f(x)$. Find the coordinates of its image on each graph.

(a) $y = f(2x)$ [2]
(b) $y = f(x) - 4$ [1]
(c) $y = f(-x)$ [1]
(d) $y = \frac{1}{2}f(x)$ [2]

12. Describe the sequence of transformations that maps $y = \sqrt{x}$ onto each graph.

(a) $y = \sqrt{x+3}$ [2]
(b) $y = 2\sqrt{x}$ [2]
(c) $y = 2\sqrt{x+3} - 1$ [3]
(d) $y = \sqrt{2x-6}$ *(Hint: factorise inside the root first)* [4]

13. Start from $y = x^2$.

(a) Apply a vertical stretch of factor 3, then a translation 2 units up. Write the resulting equation. [2]
(b) Apply the same two moves in the opposite order, and write the resulting equation. [2]

14. Consider $y = f(4x - 8)$.

(a) Write it in the form $y = f(b(x - h))$. [2]
(b) Describe the sequence of transformations from $y = f(x)$. [3]

5.3 Solving Equations Graphically

Every equation can be read as a question about two graphs. Solving $f(x) = g(x)$ means finding the values of x at which f and g return the same output. Those are exactly the points where the two graphs cross, because a crossing point lies on both curves at once.

So: sketch both graphs, find the crossings, and read their x-coordinates. Those x-coordinates are the solutions.

This matters most when no algebraic method exists. The equation $e^x = 3 - x$ has no solution in **closed form** (written as an exact formula using standard functions). But $y = e^x$ and $y = 3 - x$ clearly cross exactly once, and a calculator locates that crossing to as many decimal places as you need. Here **technology** is not a way of avoiding the mathematics. It is the method.

EXAMPLE 5.7

Solve $x^2 = x + 2$, and then solve $x^2 \leq x + 2$.

Start with the equation. Collect everything on one side so it can be factorised, then solve:

$$x^2 - x - 2 = 0 \implies (x-2)(x+1) = 0 \implies x = -1 \text{ or } x = 2.$$

These two values are where the parabola $y = x^2$ meets the line $y = x + 2$.
Now the inequality. It asks for the values of x where the parabola is *below* the line, or level with it. The crossings at $x = -1$ and $x = 2$ divide the x-axis into three parts, and the parabola is below the line only in the middle part:

$$x^2 \le x + 2 \quad \text{for} \quad -1 \le x \le 2.$$

This is the pattern for every inequality of this kind. Solving the equation finds the boundary points. The graph then shows which side of them to keep.

EXAMPLE 5.8

After a Finnish sauna is switched off, its temperature in °C after t minutes is modelled by $T(t) = 20 + 70e^{-0.12t}$. Find how long it takes the sauna to cool to $40\,°\text{C}$.
Read the question as two graphs: the cooling curve $y = T(t)$, and the horizontal line $y = 40$. The answer is the t-coordinate where they cross. Graph both on a graphic display calculator (GDC) and use intersect, which gives $t \approx 10.4$.
The algebra confirms it. Substitute $T(t) = 40$ and solve for t:

$$70e^{-0.12t} = 20 \implies e^{-0.12t} = \tfrac{2}{7} \implies t = \frac{\ln(7/2)}{0.12} \approx 10.4 \text{ minutes.}$$

The model also tells you something without any solving at all. As t grows, $e^{-0.12t}$ approaches 0, so $T(t)$ approaches 20. That 20 is the room temperature: the line $y = 20$ is a horizontal asymptote, and the sauna cools towards it without ever reaching it.

Inequalities between functions HL

The solution of $g(x) \ge f(x)$ is the set of x values for which the graph of g lies on or above the graph of f.

Work in two stages. First find the **boundary points**: the values of x where the two graphs meet, and also the values where either function is undefined. Then test each interval between them to see whether the inequality holds there.

Both kinds of boundary point matter. A vertical asymptote separates one interval from the next just as effectively as a crossing does, even though the graphs never meet there.

EXAMPLE 5.9

Solve $\dfrac{x+2}{x-1} \ge 3$.

Do not multiply both sides by $x - 1$. Its sign depends on x: it is negative when $x < 1$ and positive when $x > 1$. Multiplying an inequality by a negative number reverses the sign, so this method splits into cases and is easy to get wrong.
Instead, collect everything on one side and combine it into a single fraction:

$$\frac{x+2}{x-1} - 3 \geq 0 \implies \frac{(x+2)-3(x-1)}{x-1} \geq 0 \implies \frac{5-2x}{x-1} \geq 0.$$

A fraction changes sign only where it is zero or where it is undefined, so these are the boundary points. It is zero when $5 - 2x = 0$, that is $x = \frac{5}{2}$, and undefined when $x - 1 = 0$, that is $x = 1$. These two values split the x-axis into three intervals. Test the sign of $\dfrac{5-2x}{x-1}$ on each:

$$x < 1 : \ \tfrac{+}{-} < 0, \qquad 1 < x < \tfrac{5}{2} : \ \tfrac{+}{+} > 0, \qquad x > \tfrac{5}{2} : \ \tfrac{-}{+} < 0.$$

Only the middle interval satisfies the inequality, so the solution is

$$1 < x \leq \tfrac{5}{2}.$$

Check the two endpoints separately. Exclude $x = 1$, because the fraction is undefined there. Include $x = \frac{5}{2}$, because the inequality allows equality and the fraction is zero there.
A GDC graph of $y = \frac{x+2}{x-1}$ against $y = 3$ confirms it: the curve lies above the line only between the asymptote and the crossing.

CAUTION Never multiply an inequality by an expression whose sign you do not know. Move everything to one side, combine into a single fraction, and read the sign from the critical values.

YOUR TURN 5.3 Solving Equations Graphically

15. Solve each equation.

(a) $x^2 = 2x + 3$ [2] (b) $x^2 = 4 - 3x$ [2]
(c) $x^2 = x + 6$ [2] (d) $x^2 = 9$ [1]

16. Solve each inequality.

(a) $x^2 < 4$ [2] (b) $x^2 \geq 9$ [2]
(c) $x^2 \leq x + 6$ [3] (d) $x^2 > 2x + 3$ [3]

17. Find the coordinates of every intersection point of each pair of graphs.

(a) $y = x^2$ and $y = x + 6$ [3] (b) $y = x^2$ and $y = 2x + 8$ [3]
(c) $y = x^2 - 1$ and the x-axis [2] (d) $y = x^2$ and $y = -x$ [3]

18. Consider the equation $e^x = 3 - x$.

(a) Explain why it has exactly one real solution. [2]
(b) Find that solution, correct to three significant figures. [2]
(c) State why a graph is the appropriate method for this equation. [2]

19. Solve each inequality. HL

(a) $\dfrac{x+2}{x-1} \geq 3$ [4] (b) $\dfrac{2x-1}{x+2} \leq 1$ [4]

20. After a sauna is switched off, its temperature in °C after t minutes is $T(t) = 20 + 70e^{-0.12t}$.

(a) State the temperature at the moment it is switched off. [1]
(b) Find the temperature after 5 minutes. [2]
(c) Find how long it takes to cool to $40\,^\circ$C. [3]
(d) State the temperature the sauna approaches, and explain what it represents. [2]

5.4 Symmetry, Modulus and Related Graphs HL

Some graphs repeat themselves. If you know one half of such a graph, the symmetry determines the other half, so you only need to sketch half of it. It also reveals structure that the formula alone does not show.

Type	**Test**	**Symmetry of the graph**	**Examples**
Even	$f(-x) = f(x)$	in the y-axis	x^2, x^4, $\lvert x\rvert$, $\cos x$
Odd	$f(-x) = -f(x)$	rotational, order two, about the origin	x, x^3, $\dfrac{1}{x}$, $\sin x$
Neither	neither test holds	no symmetry of either kind	$x^2 + x$, $x + 1$, e^x

The names come from the powers. The functions $x^2, x^4, \ldots$ have even exponents and satisfy $f(-x) = f(x)$; the functions $x, x^3, \ldots$ have odd exponents and satisfy $f(-x) = -f(x)$. Most functions are neither, so "neither" is a complete and correct answer when the tests fail.

To test a function, substitute $-x$ for x, simplify, and compare the result with the original.

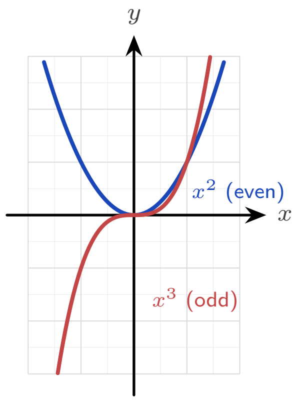


Figure 5.3 $y = x^2$ is even: reflecting it in the y-axis leaves it unchanged. $y = x^3$ is odd: reflecting it in the y-axis and then in the x-axis leaves it unchanged, which is a rotation of 180° about the origin.

A third kind of symmetry belongs to functions that are their own inverse. Recall that f^{-1} reverses f. If applying f twice already returns the input, so that $f(f(x)) = x$, then f reverses itself and $f = f^{-1}$. Such a function is called **self-inverse**, and its graph is symmetric in the line $y = x$.

EXAMPLE 5.10

Show that $f(x) = \dfrac{1}{x}$ is self-inverse, and sketch its graph.

Apply the function twice. Substitute the whole expression $\frac{1}{x}$ in place of x:

$$f(f(x)) = \frac{1}{\frac{1}{x}} = x.$$

Applying f twice returns the input, so f reverses itself and $f = f^{-1}$.

The sketch shows what this means geometrically. Reflecting any graph in the line $y = x$ produces the graph of its inverse. Since f *is* its own inverse, the reflection must leave the curve unchanged, so the curve is symmetric in $y = x$.

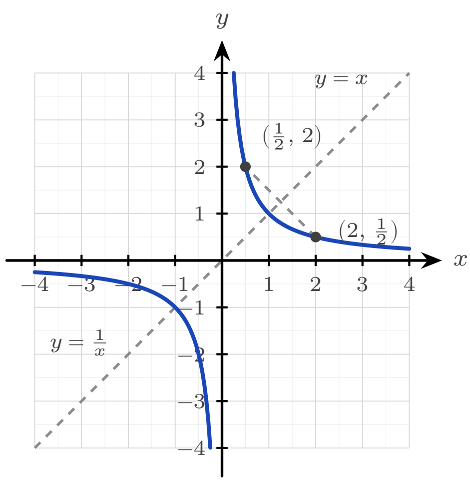


Figure 5.4 The reciprocal is symmetric in the line $y = x$. Each point on the curve has a partner with its coordinates swapped, so reflecting in $y = x$ returns the same curve.

Self-inverse functions are not rare, and they are not all this simple. The reflection $f(x) = 6 - x$ is self-inverse, and so is $f(x) = \dfrac{2x+1}{x-2}$, which you can check by the same substitution.

EXAMPLE 5.11

Determine whether $f(x) = x^3 - x$ is odd, even, or neither.
Substitute $-x$ for x, simplify, then look for a factor of -1:

$$f(-x) = (-x)^3 - (-x) = -x^3 + x = -(x^3 - x) = -f(x).$$

The result is $-f(x)$, so by the definition the function is **odd**. Its graph therefore has rotational symmetry of order two about the origin.

CAUTION A function like $f(x) = x^2$ is not one-to-one on all of $\mathbb{R}$, since $f(2) = f(-2)$, so it has no inverse there. Restricting the domain to $x \geq 0$ makes it one-to-one, and the inverse is $f^{-1}(x) = \sqrt{x}$.

Restricting the domain to make an inverse exist is a standard examination technique. Note what the caution above shows: the restriction you choose decides which branch of the inverse you get.

EXAMPLE 5.12

Let $f(x) = (x-3)^2$ for $x \geq 3$. Find $f^{-1}(x)$, stating its domain.
First check that an inverse exists. On $x \geq 3$ the parabola only rises, so no two inputs share an output and f is one-to-one.
Now find the inverse. Write $y = f(x)$ and solve for x:

$$x - 3 = \pm\sqrt{y} \implies x = 3 \pm \sqrt{y}.$$

Two branches appear, so the restriction must decide between them. Since $x \geq 3$, the expression $x - 3$ cannot be negative, so only the $+$ branch is possible. Hence

$$f^{-1}(x) = 3 + \sqrt{x}, \qquad x \geq 0,$$

where the domain of f^{-1} is the range of f. Had the restriction been $x \leq 3$ instead, the same algebra would give $f^{-1}(x) = 3 - \sqrt{x}$. The algebra is the same until that final choice of sign, but the inverse is different.

One last kind of repetition. A function is **periodic** if its graph repeats at regular intervals, that is, if $f(x + p) = f(x)$ for some fixed positive number p. The smallest such p is called the **period**. The trigonometric functions are the most important example: their graphs repeat forever along the x-axis.

The modulus $|f(x)|$

The remaining graphs in this section are also produced by symmetry. Each is built from a function f by one operation, and each has a single visual rule, so you are never expected to plot points.

The modulus of a number is its size without its sign, so $|f(x)|$ is never negative.

That gives the rule at once. Any part of the graph of f already above the x-axis is unchanged. Any part below the axis has its sign flipped, so it is reflected upward in the x-axis. The x-intercepts do not move, because $|0| = 0$.

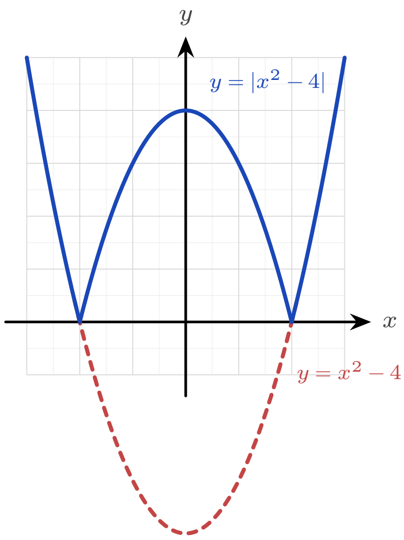


Figure 5.5 $y = |x^2 - 4|$ against $y = x^2 - 4$. The part of the parabola below the x-axis is reflected upward; everything already above the axis is untouched, so the two roots become corners.

Related graphs

Three more operations appear on the syllabus, and each follows one idea.

For $y = f(|x|)$ the modulus is applied to the *input*, so f receives the same number whether x is positive or negative. Keep the right-hand half of the graph and reflect it in the y-axis to make the left-hand half. The result is always an even function.

For $y = [f(x)]^2$ every output is squared, so no output is negative. Each root of f becomes a point where the new curve touches the x-axis from above rather than crossing it.

For $y = f(ax + b)$ the change is inside the bracket, so it is a combination of the horizontal transformations of §5.2.

YOUR TURN 5.4 Symmetry, Modulus and Related Graphs HL

21. Determine whether each function is odd, even, or neither.

(a) $f(x) = x^4 - 3x^2$ [2]
(b) $f(x) = x^3 + x$ [2]
(c) $f(x) = x^2 + x$ [1]
(d) $f(x) = x^5$ [1]
(e) $f(x) = 5$ [1]
(f) $f(x) = \dfrac{1}{x}$ [2]

22. Self-inverse functions.

(a) Show that $f(x) = \dfrac{1}{x}$ is self-inverse. [2]
(b) Show that $f(x) = 6 - x$ is self-inverse. [2]
(c) Show that $f(x) = \dfrac{2x+1}{x-2}$ is self-inverse. [3]
(d) Explain why $f(x) = 2x$ is not self-inverse. [2]

23. Restricting a domain so that an inverse exists.

(a) State a domain on which $f(x) = x^2$ has an inverse, and give that inverse. [2]
(b) Do the same for $f(x) = (x-1)^2$. [3]
(c) State the domain of the inverse you found in part (b). [1]
(d) Explain why $f(x) = x^3$ needs no restriction. [2]

24. Solve each equation.

(a) $|x - 3| = 2$ [2]
(b) $|2x + 1| = 7$ [2]
(c) $|3 - x| = 1$ [2]
(d) $|x| = 5$ [1]

25. The graph of $y = f(x)$ crosses the x-axis at $x = 2$ and has a minimum point at $(0, -4)$.

(a) State where $y = |f(x)|$ meets the x-axis. [1]
(b) State where $y = [f(x)]^2$ meets the x-axis. [1]
(c) State the point that $(0, -4)$ becomes on $y = |f(x)|$. [2]
(d) Explain why $y = f(|x|)$ is always an even function. [2]

5.5 The Reciprocal Function HL

The last of these operations deserves a section of its own. Given any function f, its **reciprocal** is

$$y = \frac{1}{f(x)},$$

defined at every x where f is defined and $f(x) \neq 0$.

You never need to calculate points to sketch it. Every feature of the reciprocal can be read from the graph of f, one feature at a time.

Where f does this	The reciprocal does this	Reason
$f(x) = 0$	vertical asymptote	you cannot divide by zero
$\lvert f(x)\rvert$ is large	the curve approaches the x-axis	a large number has a small reciprocal
$f(x) = 1$ or -1	the two graphs cross	1 and -1 are their own reciprocals
$f(x) > 0$	$\frac{1}{f(x)} > 0$	the sign is preserved
a local maximum	a local minimum	and a minimum becomes a maximum

The last row is worth a moment. Where f is large the reciprocal is small, so a high point of f becomes a low point of $\frac{1}{f}$. This holds wherever f is not zero, whatever the sign of f.

CAUTION $\dfrac{1}{f(x)}$ is *not* $f^{-1}(x)$. The reciprocal divides 1 by the output; the inverse undoes the function. For $f(x) = x+3$ the reciprocal is $\dfrac{1}{x+3}$ and the inverse is $x-3$, which are not the same. The notation f^{-1} never means ``one over f''.

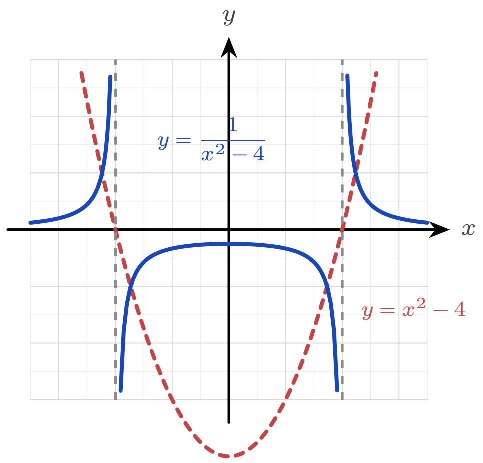


Figure 5.6 $y = \dfrac{1}{x^2-4}$ against $y = x^2 - 4$. Zeros of the original become vertical asymptotes at $x = \pm 2$, the minimum $(0, -4)$ becomes the maximum $(0, -\frac{1}{4})$ of the middle branch, and large values of f become values close to zero.

Check each rule against the figure. The dashed parabola is f and the solid curve is its reciprocal. The roots of f at $x = \pm 2$ have become vertical asymptotes. The minimum of f at $(0, -4)$ has become a maximum of $1/f$ at $(0, -\frac{1}{4})$. And where f takes large values, the reciprocal approaches the x-axis.

EXAMPLE 5.13

The graph of $y = f(x)$ has a single root at $x = 3$ and a local minimum at $(1, 2)$, and $f(x) \to \infty$ as $x \to \pm\infty$. Describe the corresponding features of $y = \dfrac{1}{f(x)}$.

Take the features one at a time.

The root at $x = 3$ makes the denominator zero, so $y = \dfrac{1}{f(x)}$ has a **vertical asymptote at** $x = 3$.

The minimum at $(1, 2)$ has $f(1) = 2$, so the reciprocal passes through $(1, \frac{1}{2})$. A minimum of f becomes a **maximum** of the reciprocal, so this is a local maximum at $(1, \frac{1}{2})$.

Since $f(x) \to \infty$ at both ends, $\dfrac{1}{f(x)} \to 0$, so $y = 0$ is a **horizontal asymptote**.

No algebra was needed. Every answer came from one feature of f.

TIP Sketching a reciprocal in an examination: mark the roots of f as vertical asymptotes first, then the points where $f = 1$ and $f = -1$, then the turning points with their y-coordinates inverted. Draw the curve through those, one branch at a time.

YOUR TURN 5.5 The Reciprocal Function HL

26. State the equations of the vertical asymptotes of $y = \dfrac{1}{f(x)}$ for each f.

(a) $f(x) = x - 4$ [1]
(b) $f(x) = x^2 - 9$ [2]
(c) $f(x) = (x - 1)(x + 5)$ [2]
(d) $f(x) = x^2 + 1$ [2]

27. Consider $y = \dfrac{1}{x^2 - 4}$.

(a) State the equations of the vertical asymptotes. [2]
(b) State the equation of the horizontal asymptote. [1]
(c) Find the value of y when $x = 0$. [1]
(d) State whether that point is a maximum or a minimum, and give a reason. [2]

28. The graph of $y = f(x)$ has a single root at $x = 3$ and a local minimum at $(1, 2)$, and $f(x) \to \infty$ as $x \to \pm\infty$.

(a) State the equation of the vertical asymptote of $y = \dfrac{1}{f(x)}$. [1]
(b) Find the coordinates of the turning point of $y = \dfrac{1}{f(x)}$, and state its type. [3]
(c) Describe the behaviour of $y = \dfrac{1}{f(x)}$ as $x \to \pm\infty$. [2]

(d) State the condition on $f(x)$ at a point where $y = f(x)$ and $y = \dfrac{1}{f(x)}$ cross. [2]

29. Sketch $y = \dfrac{1}{f(x)}$ for each function below, on separate axes. Show every asymptote and the coordinates of any turning point.

(a) $f(x) = x - 2$ [3]

(b) $f(x) = x^2 - 1$ [4]

(c) $f(x) = x^2 + 1$ [4]

(d) $f(x) = (x-1)(x-3)$ [4]

30. Let $f(x) = x - 2$.

(a) On one set of axes, sketch $y = f(x)$ and $y = \dfrac{1}{f(x)}$. [3]

(b) Find the coordinates of the two points where the graphs cross. *(Hint: they cross where $f(x) = \pm 1$)* [3]

(c) State the interval on which $\dfrac{1}{f(x)} > f(x)$ for $x > 2$. [3]

Chapter 5 · Function Operations

Reflections

TOK Slide the graph of $y = x^2$ two units right and you get $y = (x-2)^2$. Is this the same function or a different one? It has the same shape, the same name ("a parabola"), yet a different rule and different values everywhere. When we call two things "the same" in mathematics, we are always choosing what to ignore. The transformation makes that choice visible.

IM In 1872 Felix Klein, aged just twenty-three, proposed in his *Erlangen Programme* a radical reorganisation of geometry: a geometry *is* a set of transformations together with the properties they leave unchanged. Euclidean geometry studies what survives rotation and reflection; other geometries allow other moves. The transformations of this chapter are the same idea in miniature, and the question "what stays the same when I move this graph?" is one of the deepest in mathematics.

RW Transformations are the working language of digital media. Audio software shifts pitch (a horizontal stretch of a waveform) and adjusts volume (a vertical stretch); image editors scale, reflect, and translate pictures pixel by pixel; every animation moves one stored model through a stream of transformations. Engineers even split signals into an even part and an odd part, because each behaves differently in a circuit.

EXT Every function whatsoever splits, uniquely, into an even part and an odd part:

$$f(x) = \underbrace{\tfrac{1}{2}\big(f(x) + f(-x)\big)}_{\text{even}} + \underbrace{\tfrac{1}{2}\big(f(x) - f(-x)\big)}_{\text{odd}}.$$

Check it: the first bracket is unchanged when $x \to -x$, the second changes sign, and together they add back to $f(x)$. A single line of algebra, and every function is revealed as a sum of an even part and an odd part.

End-of-Chapter Problems — Chapter 5: Function Operations

1. Let $f(x) = 2x - 3$ and $g(x) = x^2$.

(a) Find $(f \circ g)(x)$. [2]
(b) Find $(g \circ f)(x)$. [2]
(c) Solve $(f \circ g)(x) = 5$. [2]

2. The graph of $y = f(x)$ has a maximum point at $(1, 4)$. State the coordinates of the maximum of:

(a) $y = f(x) + 2$; [1]
(b) $y = f(x - 3)$; [1]
(c) $y = 2f(x)$; [1]
(d) $y = -f(x)$ (state the resulting minimum). [1]

3. Let $f(x) = x^2 - 4x + 5$.

(a) Write $f(x)$ in the form $(x - h)^2 + k$. [2]
(b) Describe the two transformations mapping $y = x^2$ to $y = f(x)$. [2]
(c) State the range of f. [2]

4. Let $f(x) = \dfrac{x+1}{x-2}$, $x \neq 2$. Find $f^{-1}(x)$. [4]

5. Solve the equation $x^2 + 2 = 3x$, and hence solve the inequality $x^2 + 2 \leq 3x$. [5]

6. Determine whether each function is odd, even, or neither:

(a) $f(x) = x^3 - 2x$; [1]
(b) $f(x) = |x|$; [1]
(c) $f(x) = x^2 + x^3$. [1]

7. Consider $f(x) = x^2 + 2$ with domain $x \geq 0$.

(a) Find $f^{-1}(x)$. [3]
(b) State the domain of f^{-1}. [1]

8. Sketch the graph of $y = |x^2 - 1|$, clearly showing the x- and y-intercepts and the shape. [4]

9. Let $f(x) = x + 2$ and $g(x) = x^2$. Solve $(f \circ g)(x) = (g \circ f)(x)$. [4]

10. The graph of $y = f(x)$ passes through $(0, 1)$, $(2, 0)$ and $(3, 4)$. Find the images of these three points on $y = f(x + 1) - 2$. [3]

11. Let $f(x) = 3x + 2$.

(a) Find $f^{-1}(x)$. [2]
(b) Solve $f(x) = f^{-1}(x)$. [3]

12. Let $g(x) = \sqrt{x - 1}$.

(a) State the domain and range of g. [2]
(b) Find $g^{-1}(x)$ and state its domain. [3]

13. Let $f(x) = \dfrac{2}{x - 3}$.

(a) State the equations of the vertical and horizontal asymptotes. [2]
(b) Find $f^{-1}(x)$. [3]

14. The graph of $y = x^2$ is transformed to $y = -(x - 1)^2 + 2$. Describe the sequence of transformations and state the coordinates of the vertex. [4]

15. Solve the equation $|x - 2| = |x + 4|$. [4]

16. Solve the inequality $|2x - 1| < 5$. [3]

17. Determine whether each function is odd, even, or neither:

(a) $f(x) = \dfrac{x}{x^2 + 1}$; [1]
(b) $f(x) = \dfrac{1}{x^2 + 1}$; [1]
(c) $f(x) = x + 1$. [1]

18. The function f is even with $f(3) = 7$, and g is odd with $g(2) = -5$. Write down the values of $f(-3)$ and $g(-2)$. [2]

19. Let $f(x) = x^2 - 6x$.

(a) Write $f(x)$ in completed-square form. [2]
(b) Hence describe the transformations from $y = x^2$ and state the vertex. [3]

20. Find the coordinates of the point(s) where $y = |x|$ meets $y = 2 - x$. [3]

21. Let $f(x) = \dfrac{1}{x+2}$.

(a) State the domain and range. [2]

(b) State the equations of the two asymptotes. [2]

22. The function $f(x) = ax + b$ satisfies $f(1) = 5$ and $f(3) = 11$.

(a) Find a and b. [3]

(b) Find $f^{-1}(x)$. [2]

23. The quadratic $f(x) = x^2 + bx + c$ has vertex $(2, -1)$.

(a) Find b and c. [3]

(b) Describe the transformation of $y = x^2$ that gives $y = f(x)$. [2]

24. Let $f(x) = x^2 - 2x$.

(a) Find the roots and the vertex of f. [3]

(b) Sketch $y = |f(x)|$, showing the intercepts and the turning point. [3]

25. Solve the inequality $\dfrac{2x-1}{x+2} \leq 1$. [5]

26. Let $f(x) = (x+1)^2 - 4$ for $x \geq -1$.

(a) Explain why f has an inverse on this domain. [1]

(b) Find $f^{-1}(x)$, stating its domain. [4]

27.

(a) Solve $|x-2| = |3x+1|$. [4]

(b) Hence, or otherwise, solve $|x-2| \geq |3x+1|$. [3]

28. For $f(x) = \dfrac{x}{1+x}$, find $(f \circ f)(x)$ and simplify fully. [5]

29. Prove that if f and g are both odd functions, then $f \circ g$ is also odd. [5]

30. The function $f(x) = ax + b$ satisfies $f(f(x)) = x$ for all real x. Find all possible values of a and b. [6]

31. Prove that every function $f : \mathbb{R} \to \mathbb{R}$ can be written as the sum of an even function and an odd function. [6]

32. Show that $f(x) = \dfrac{2x+3}{x-2}$ is self-inverse, that is, $(f \circ f)(x) = x$. [5]

33. Let $f(x) = x^2 - 4x + 3$.

(a) Write $f(x)$ in completed-square form, and state the coordinates of the vertex. [3]
(b) Find the roots of f and the y-intercept. [3]
(c) Sketch $y = f(x)$, showing the vertex and both intercepts. [2]
(d) On separate axes, sketch $y = |f(x)|$ and $y = f(|x|)$, showing the key points of each. [4]
(e) The graph of $y = f(x)$ is translated 2 units to the right and 3 units up to give $y = g(x)$. Find $g(x)$ in the form $ax^2 + bx + c$. [3]

34. Let $f(x) = \dfrac{3x-1}{x+2}$ for $x \neq -2$, and let $g(x) = x + 2$.

(a) State the range of f, justifying your answer. [2]
(b) Find $f^{-1}(x)$, and state its domain. [4]
(c) Find $(f \circ g)(x)$, giving your answer as a single fraction. [3]
(d) State the domain of $f \circ g$. [2]
(e) Solve $(f \circ g)(x) = 2$. [4]

35. HL Let $f(x) = x^2 - 4$.

(a) Sketch $y = f(x)$, showing the intercepts. [2]
(b) Write down the equations of the vertical asymptotes of $y = \dfrac{1}{f(x)}$, and explain how you found them. [3]
(c) Find the coordinates of the maximum point of $y = \dfrac{1}{f(x)}$ on the interval $-2 < x < 2$, and explain why it is a maximum even though the value is negative. [3]
(d) Describe the behaviour of $y = \dfrac{1}{f(x)}$ as $x \to \pm\infty$, and state the horizontal asymptote. [2]
(e) Sketch $y = \dfrac{1}{f(x)}$, showing all asymptotes and the point found in part (c). [4]

Challenge Questions

36. **Self-inverse rational functions.** A function of the form

$$f(x) = \frac{ax+b}{cx+d}, \qquad c \neq 0,$$

is called *self-inverse* when $(f \circ f)(x) = x$ for every x in its domain. This investigation finds exactly which of them are.

(a) Show that $f(x) = \dfrac{x+1}{x-1}$ is self-inverse. [3]

(b) Show that for the general function above,

$$(f \circ f)(x) = \frac{(a^2 + bc)\,x + b(a+d)}{c(a+d)\,x + (bc + d^2)}.$$

[4]

(c) Hence prove that, when $c \neq 0$, the function f is self-inverse if and only if $a + d = 0$. *(Hint: two expressions in x are equal for all x only when their coefficients match)* [5]

(d) Show that if $ad - bc = 0$ then f is constant, and explain why the condition $ad - bc \neq 0$ must therefore be assumed throughout. [3]

(e) Write down two further self-inverse examples with $c \neq 0$, one of which has $b = 0$. Verify one of them. [3]

37. When does the order of two transformations matter? Section 5.2 showed that a vertical stretch and a vertical translation give different curves in different orders. This investigation settles the general question.

Throughout, start from $y = f(x)$.

(a) Apply a vertical stretch of factor a and then a translation of k units up, and write down the result. Do the same in the opposite order. Deduce the exact condition on a and k under which the two agree for every function f. [4]

(b) The horizontal moves are substitutions: move A replaces x by $x - h$, and move C replaces x by bx. Show that A followed by C gives $y = f(bx - h)$, while C followed by A gives $y = f(bx - bh)$. Deduce the condition under which these agree for every f. [5]

(c) Show that any vertical move and any horizontal move commute, whatever their parameters. *(Hint: consider which variable each move acts on)* [3]

(d) Using parts (a) to (c), state a single rule that predicts when two transformations may be applied in either order. Explain why your rule is consistent with all three cases above. [4]

38. Counting the solutions of $|f(x)| = k$. Let $f(x) = x^2 - 4$ and let $k \geq 0$ be a constant.

(a) Sketch $y = |f(x)|$, showing the x-intercepts and the coordinates of the local maximum. [3]

(b) Find the number of real solutions of $|f(x)| = k$ for each of $k = 0$, $k = 3$, $k = 4$ and $k = 6$. [4]

(c) Explain, using your sketch, why the number of solutions changes at $k = 4$ and not at any other value. [3]

(d) State the number of real solutions of $|f(x)| = k$ for every $k \geq 0$, giving your answer as a list of cases. [3]

(e) Now let g be any quadratic with a single minimum point of value m, where $m < 0$. State the corresponding list of cases for $|g(x)| = k$, in terms of m. [3]

Rational and Exponential Functions

6

SYLLABUS 1.11 2.8 2.9 2.13

Pour a cup of coffee at $90\,^{\circ}$C and set it down in a room at $20\,^{\circ}$C. It cools quickly at first, then more slowly, then so slowly that you can barely measure the change. Leave it long enough and it reaches room temperature. That is what we usually say. Now look at the numbers instead: the difference between the coffee and the room keeps halving. It becomes smaller and smaller, but it never becomes zero.

Isaac Newton measured this and found the rule. In each fixed interval of time, the temperature difference falls by the same *fraction*, not by the same amount. That property is what defines an **exponential**. Plot the temperature against time and the curve falls steeply, then bends and flattens, settling towards the horizontal line at $20\,^{\circ}$C without ever reaching it.

That line is an **asymptote**, and asymptotes appear in every section of this chapter. A curve can approach a value it never reaches. It can also increase without bound as its input approaches a value where the function is undefined. Reading that behaviour from the equation alone is the skill this chapter builds.

Two families of function behave this way. **Rational** functions are ratios of polynomials. The **transcendental** pair are the exponential and the logarithm, each the mirror image of the other in the line $y = x$. Between them they model cooling coffee, growing populations, decaying atoms and compound interest. By the end of the chapter you will be able to sketch any of them from its formula.

More precisely, you will find the asymptotes of a rational function and sketch it, handle an oblique asymptote by division, treat the exponential and the logarithm as inverses, and split a rational expression into partial fractions. These functions earn their place because they model what lines and parabolas cannot. One grows in proportion to its own size; another approaches a limit it never reaches.

6.1 Reciprocal and Rational Functions

The simplest function with an asymptote is the **reciprocal** $y = \dfrac{1}{x}$.

It is undefined at $x = 0$. As x approaches 0, the output grows without bound, towards $+\infty$ on one side and $-\infty$ on the other. For large $|x|$ the opposite happens: the output becomes very small, and the curve approaches zero.

So the reciprocal has two asymptotes: the vertical line $x = 0$ and the horizontal line $y = 0$. It is also **self-inverse**, meaning that applying it twice returns the original input, so its graph is symmetric in the line $y = x$. See Ch. 5.

It is defined for every input except $x = 0$, so its **domain** is all real numbers with $x \neq 0$. It produces every output except 0, so its **range** is all real numbers with $y \neq 0$.

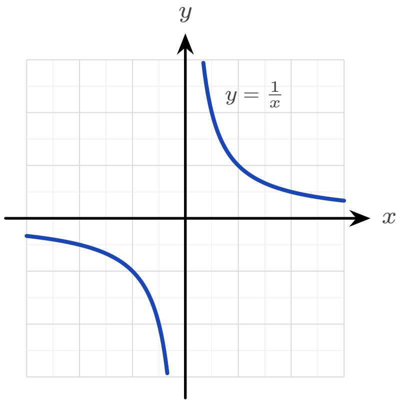


Figure 6.1 The reciprocal $y = \frac{1}{x}$. Its asymptotes are the axes themselves: the curve approaches $x = 0$ and $y = 0$ without meeting either. The domain is every real number except 0; the range is every real number except 0.

> DEF An **asymptote** is a line the graph of a function approaches arbitrarily closely. A **vertical asymptote** occurs at a value of x where the function is undefined and grows without bound. A **horizontal asymptote** is a value the function approaches as $x \to \pm\infty$.

Every function of the form $\dfrac{ax+b}{cx+d}$, with $c \neq 0$ and $ad - bc \neq 0$, is a reciprocal curve that has been shifted and stretched. It has one vertical asymptote, where the denominator is zero, and one horizontal asymptote, at the ratio of the leading coefficients.

Both conditions matter. If $c = 0$ the function is linear and has no asymptote at all; if $ad - bc = 0$ the numerator is a multiple of the denominator and the function is constant.

KEY For $f(x) = \dfrac{ax+b}{cx+d}$, with $c \neq 0$ and $ad - bc \neq 0$:

$$\text{vertical asymptote at } x = -\frac{d}{c}, \qquad \text{horizontal asymptote at } y = \frac{a}{c}.$$

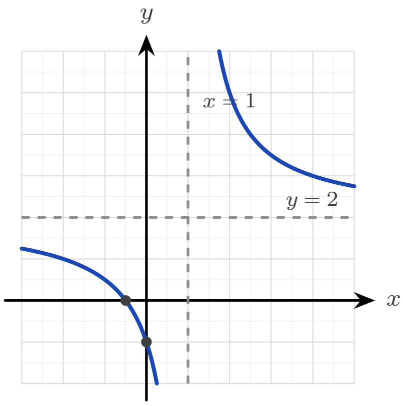


Figure 6.2 The curve $y = \frac{2x+1}{x-1}$, with its two asymptotes dashed and its two intercepts marked. These four features are what a sketch must show.

EXAMPLE 6.1

Sketch $f(x) = \dfrac{2x+1}{x-1}$, showing all asymptotes and intercepts.

Find the vertical asymptote first. The denominator is zero at $x = 1$, so the line $x = 1$ is the vertical asymptote.

Next the horizontal asymptote. The ratio of the leading coefficients gives $y = \dfrac{2}{1} = 2$.

Rewriting the function confirms the shape. Divide the numerator by the denominator:

$$\frac{2x+1}{x-1} = 2 + \frac{3}{x-1},$$

a reciprocal curve centred on the crossing of the two asymptotes, $(1, 2)$.

Now find the intercepts. Substitute $x = 0$ for the y-intercept: $f(0) = \dfrac{1}{-1} = -1$. Set the numerator to zero for the x-intercept: $2x + 1 = 0$, so $x = -\frac{1}{2}$. Both are marked on the figure above.

Two asymptotes and two intercepts fix the whole sketch.

TIP A sketch question is marked on exactly four features: both asymptotes and both intercepts. Label each one on the diagram; an unlabelled curve of the right shape scores very little.

YOUR TURN 6.1 Reciprocal and Rational Functions

1. State the vertical asymptote of each function.

(a) $f(x) = \dfrac{1}{x-3}$ [1] (b) $f(x) = \dfrac{1}{x-5}$ [1]

(c) $f(x) = \dfrac{3}{x+2}$ [1] (d) $f(x) = \dfrac{2x}{x+1}$ [1]

2. State the horizontal asymptote of each function.

(a) $f(x) = \dfrac{2x}{x+1}$ [1] (b) $f(x) = \dfrac{x+1}{x-2}$ [1]

(c) $f(x) = \dfrac{3x-6}{x+1}$ [2] (d) $f(x) = \dfrac{3}{x+2}$ [2]

3. State both asymptotes and both intercepts of each function.

(a) $f(x) = \dfrac{x-1}{x+2}$ [4]

(b) $f(x) = \dfrac{2x+3}{x-1}$ [4]

4. State the domain and range of each function.

(a) $f(x) = \dfrac{1}{x-5}$ [2] (b) $f(x) = \dfrac{1}{x} + 4$ [3]

5. Consider $f(x) = \dfrac{3x-6}{x+1}$.

(a) Write $f(x)$ in the form $A + \dfrac{B}{x+1}$. [3]

(b) Hence state the point on which the two asymptotes cross. [2]

(c) Sketch the curve, showing both asymptotes and both intercepts. [3]

6.2 Rational Functions and Oblique Asymptotes HL

When the numerator has a higher degree than the denominator, there is no horizontal asymptote. The curve approaches a slanted line instead.

The useful case is when the numerator's degree is exactly one more than the denominator's. Polynomial division then splits the function into a linear part plus a remainder, and that remainder approaches zero for large $|x|$. The linear part is the **oblique** (slant) asymptote.

EXAMPLE 6.2

Find the asymptotes of $f(x) = x + \dfrac{1}{x}$.

The function is undefined at $x = 0$, so the line $x = 0$ is a vertical asymptote.

For large $|x|$ the term $\dfrac{1}{x}$ approaches zero, so $f(x)$ approaches x. The line $y = x$ is therefore the oblique asymptote.
The graph has two separate branches. Each one lies between the y-axis and the line $y = x$.

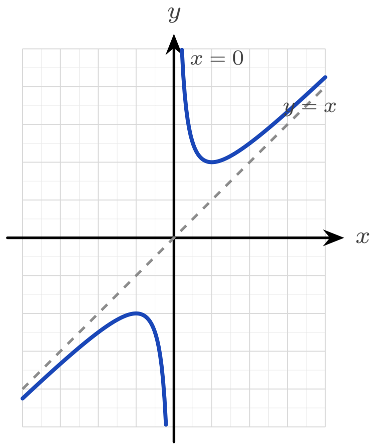


Figure 6.3 The two branches of $y = x + \frac{1}{x}$, each approaching the vertical asymptote $x = 0$ on one side and the oblique asymptote $y = x$ on the other.

In that example the split into a line plus a remainder was already written out for you. In general you must produce it yourself by **polynomial division**.

EXAMPLE 6.3

Find all asymptotes of $f(x) = \dfrac{x^2 - 2x + 3}{x - 1}$.
The denominator is zero at $x = 1$, so there is a vertical asymptote there. The numerator's degree exceeds the denominator's by one, so there is an oblique asymptote, found by dividing:

$$\frac{x^2 - 2x + 3}{x - 1} = x - 1 + \frac{2}{x - 1},$$

since $(x-1)(x-1) + 2 = x^2 - 2x + 3$. As $|x|$ grows the remainder term $\frac{2}{x-1}$ approaches zero, so the curve approaches the line

$$y = x - 1,$$

which is the oblique asymptote.

TIP Show the division, not only the answer. The quotient *is* the asymptote, and the division itself carries most of the marks.

The syllabus also names the reverse shape: a linear numerator over a quadratic denominator, $\dfrac{ax + b}{cx^2 + dx + e}$. Here the denominator has the higher degree, so it grows faster than the numerator and the curve approaches the horizontal asymptote $y = 0$. Each real root of the denominator gives its own vertical asymptote, so this form can have two.

EXAMPLE 6.4

Sketch the key features of $f(x) = \dfrac{x+1}{x^2-4}$.

Factorise the denominator to find the vertical asymptotes: $x^2 - 4 = (x-2)(x+2)$, so there are *two*, at $x = 2$ and $x = -2$.

The numerator has lower degree than the denominator, so the horizontal asymptote is $y = 0$.

For the intercepts, set the numerator to zero to get $x = -1$, and substitute $x = 0$ to get $f(0) = -\frac{1}{4}$.

The sign of f on each interval completes the sketch. The three values $x = -2$, $x = -1$ and $x = 2$ divide the x-axis into four intervals. Test the sign of $\dfrac{x+1}{(x-2)(x+2)}$ on each:

$$x < -2: \ \tfrac{-}{(-)(-)} < 0, \qquad -2 < x < -1: \ \tfrac{-}{(-)(+)} > 0,$$

$$-1 < x < 2: \ \tfrac{+}{(-)(+)} < 0, \qquad x > 2: \ \tfrac{+}{(+)(+)} > 0.$$

Now read the branches from the signs. Where f is positive next to an asymptote the curve rises to $+\infty$; where it is negative it falls to $-\infty$. The middle branch passes through both intercepts.

The rule for reading asymptotes off a rational function is worth stating once for all cases.

Degree of numerator vs denominator	Asymptote as $x \to \pm\infty$
numerator $<$ denominator	horizontal, $y = 0$
numerator $=$ denominator	horizontal, at the ratio of leading coefficients
numerator $=$ denominator $+1$	oblique (a slanted line)

In every case, vertical asymptotes sit at the zeros of the denominator (once common factors are cancelled).

YOUR TURN 6.2 Rational Functions and Oblique Asymptotes HL

6. State the vertical asymptotes of each function.

(a) $f(x) = \dfrac{x^2+1}{x-2}$ [1] (b) $f(x) = \dfrac{1}{(x-1)(x+3)}$ [2]

(c) $f(x) = \dfrac{x}{x^2-9}$ [2] (d) $f(x) = \dfrac{x^2+1}{x^2-4}$ [2]

7. Write each function as a polynomial plus a remainder, and state the oblique asymptote.

(a) $f(x) = \dfrac{x^2+3}{x}$ [2] (b) $f(x) = \dfrac{x^2}{x-1}$ [3]

(c) $f(x) = \dfrac{x^2+1}{x-2}$ [3]

(d) $f(x) = \dfrac{x^2+4x+1}{x+1}$ [3]

8. For each function, state which kind of asymptote it has as $x \to \pm\infty$, and give its equation.

(a) $f(x) = \dfrac{x}{x^2-9}$ [2]

(b) $f(x) = \dfrac{x^2+1}{x^2-4}$ [2]

(c) $f(x) = \dfrac{x^2-9}{x+2}$ [3]

9. Consider $f(x) = x + \dfrac{2}{x}$.

(a) State the vertical asymptote. [1]

(b) State the oblique asymptote. [1]

(c) Explain why the curve never crosses the oblique asymptote. [2]

(d) Sketch the two branches. [2]

6.3 Exponential Functions

An **exponential function** $y = a^x$ places the variable in the exponent. Compare it with a power such as x^3, where the variable is the base and the exponent is fixed. In an exponential the two roles are swapped.

That swap has one important consequence: an exponential changes at a rate proportional to its own size. A quantity that is twice as large grows twice as fast. This is why the same function describes cooling coffee, growing populations and decaying atoms.

For $a > 1$ the graph rises, and rises ever more steeply: this is **growth**. For $0 < a < 1$ it falls towards zero: this is **decay**.

In both cases the graph passes through $(0, 1)$, stays positive everywhere, and approaches the horizontal asymptote $y = 0$ on one side. So the domain of an exponential function is all real numbers, and its range is the positive real numbers, $y > 0$.

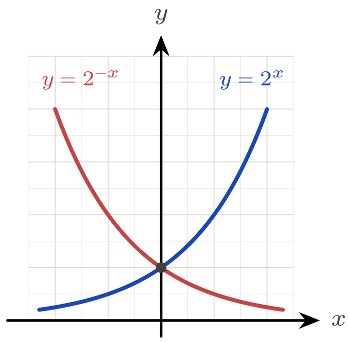


Figure 6.4 Growth and decay are reflections of each other in the y-axis. Both pass through $(0, 1)$ and approach $y = 0$, but on opposite sides.

One base is used more than any other: $e \approx 2.718$. It is the base for which the steepness of the curve equals its height at every point. That property makes the calculus of e^x the simplest of any exponential: its rate of change is again e^x. So every other exponential is usually rewritten in terms of e. See Ch. 14.

KEY · IB Any exponential can be written with base e, and exponential and logarithm undo each other:

$$a^x = e^{x \ln a}, \qquad \log_a(a^x) = x, \qquad a^{\log_a x} = x \quad (a, x > 0,\ a \neq 1).$$

EXAMPLE 6.5

For $f(x) = 5(0.4)^x$, state whether the function grows or decays, and give its domain, range and asymptote.

Look at the base first. Here $a = 0.4$, and $0 < 0.4 < 1$, so the function **decays**.

The factor 5 multiplies every output, so it stretches the graph vertically but changes nothing else. The outputs of $(0.4)^x$ are always positive, so multiplying by 5 keeps them positive.

Hence the domain is all real numbers, the range is $y > 0$, and the horizontal asymptote is $y = 0$.

Transforming an exponential graph

An exponential is rarely met on its own. In a model it usually appears as

$$y = k\,a^x + c,$$

and each constant has a separate effect. The factor k stretches the graph vertically. The constant c moves it up or down.

The vertical shift is the one that changes the asymptote. The graph of a^x approaches $y = 0$, so adding c makes it approach $y = c$ instead. **The horizontal asymptote moves with the graph.**

EXAMPLE 6.6

Sketch $y = 3(2)^x - 4$, stating the asymptote and both intercepts.

Start with the asymptote. The -4 shifts the whole graph down by 4, so the horizontal asymptote is $y = -4$, not $y = 0$.

For the y-intercept, substitute $x = 0$:

$$y = 3(2)^0 - 4 = 3 - 4 = -1.$$

For the x-intercept, set $y = 0$ and solve:

$$3(2)^x = 4 \implies 2^x = \tfrac{4}{3} \implies x = \log_2 \tfrac{4}{3} \approx 0.415.$$

The curve therefore rises from just above $y = -4$ on the left, crosses the axes at $(0, -1)$ and $(0.415, 0)$, and then rises steeply.

CAUTION A shifted exponential does *not* have the asymptote $y = 0$. In $y = k\,a^x + c$ the asymptote is $y = c$. Reading it as $y = 0$ without checking is the most common error in this topic.

Exponential models

Most real quantities are modelled with base e and time as the variable:

$$A(t) = A_0\,e^{kt},$$

where A_0 is the amount at $t = 0$. The sign of k decides the behaviour: $k > 0$ gives growth, and $k < 0$ gives decay. Its size decides the speed.

EXAMPLE 6.7

Japan's population was about 128.1 million in 2010 and about 126.1 million in 2020. Model the population as $P(t) = P_0 e^{kt}$, with t in years after 2010, and use it to estimate the population in 2030.

Take $P_0 = 128.1$, the value at $t = 0$. Now use the 2020 figure to find k. Substitute $t = 10$ and $P = 126.1$:

$$126.1 = 128.1\,e^{10k} \implies e^{10k} = \frac{126.1}{128.1} \implies k = \frac{1}{10}\ln\left(\frac{126.1}{128.1}\right) \approx -0.00157.$$

The value of k is negative, which confirms decay: the population is falling by about 0.16% each year.

For 2030, substitute $t = 20$:

$$P(20) = 128.1\,e^{-0.00157\times 20} \approx 124.1 \text{ million.}$$

Treat that figure with care. The model assumes the *rate* stays fixed, and Japan's rate of decline is itself increasing, so official projections for 2030 are lower than this. A two-point exponential fit tells you what would happen if nothing changed, which is rarely what happens.

TIP In a modelling question, read A_0 from the value at $t = 0$ before doing anything else. It is usually given directly, and one mark depends on stating it.

YOUR TURN 6.3 Exponential Functions

10. Evaluate each expression.

(a) 2^3 [1] (b) 2^{-2} [1]
(c) e^0 [1] (d) $9^{1/2}$ [1]

11. State the horizontal asymptote and the y-intercept of each graph.

(a) $y = 3^x$ [2] (b) $y = 5 \cdot 2^x$ [2]
(c) $y = 2^x + 5$ [2] (d) $y = 4(3)^x - 1$ [2]

12. State whether each function grows or decays, and give a reason.

(a) $y = (0.5)^x$ [1] (b) $y = 3^x$ [1]
(c) $y = 5e^{-0.2x}$ [2] (d) $y = 2^{-x}$ [2]

13. Solve each equation exactly.

(a) $2^x = 8$ [1] (b) $e^x = 1$ [1]
(c) $5 \cdot 3^x = 45$ [2] (d) $3(2)^x = 4$ [3]

14. Write each in the required form.

(a) 2^x in terms of base e [2] (b) 5^x in terms of base e [2]

15. GDC A population is modelled by $P(t) = 800e^{0.03t}$, with t in years.

(a) State P_0, and say whether the population grows or decays. [2]
(b) Find the population after 10 years. [2]
(c) Find how long it takes the population to reach 1200. [3]

16. GDC A radioactive sample decays according to $A(t) = 50e^{-0.2t}$, with t in days.

(a) Find $A(6)$, correct to three significant figures. [2]
(b) Find the half-life of the sample. [3]

6.4 Logarithmic Functions

Since $y = a^x$ is one-to-one, it has an inverse, and that inverse is the **logarithm** $y = \log_a x$. It answers the reversed question: not "what is a to the x?" but "to what power must I raise a to get x?" Everything about its graph follows from being the mirror image of the exponential in the line $y = x$.

Reflecting swaps the axes' roles. The exponential's horizontal asymptote $y = 0$ becomes the logarithm's *vertical* asymptote $x = 0$; its domain of all reals becomes the logarithm's range, and its range of positive numbers becomes the logarithm's domain. So $\log_a x$ exists only for $x > 0$, passes through $(1, 0)$, and rises without bound, ever more slowly. Its domain is therefore the positive real numbers, $x > 0$, and its range is all real numbers.

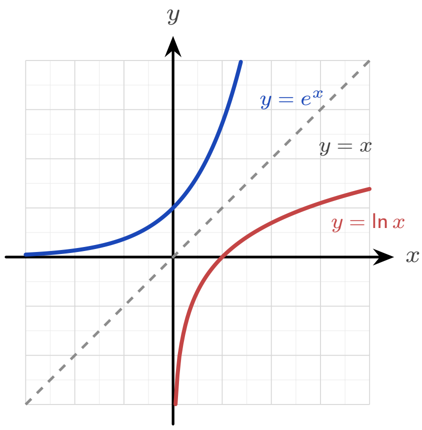


Figure 6.5 The logarithm is the exponential reflected in the line $y = x$. Reflecting exchanges the two axes, which is why one asymptote is horizontal and the other vertical.

CAUTION The logarithm is only defined for positive inputs. Writing $\ln(x-3)$ silently assumes $x > 3$; forgetting this domain restriction is the most common error with logarithmic functions.

Finding the domain

Because a logarithm accepts only positive inputs, every logarithmic function carries a hidden condition. To find the domain, set whatever sits inside the logarithm greater than zero, and solve.

EXAMPLE 6.8

State the domain of $f(x) = \ln(x-3)$ and of $g(x) = \log(5-2x)$.

For f, the input to the logarithm is $x - 3$. Require it to be positive:

$$x - 3 > 0 \implies x > 3.$$

For g, the input is $5 - 2x$. Require that to be positive:

$$5 - 2x > 0 \implies 5 > 2x \implies x < \tfrac{5}{2}.$$

Note the direction. Dividing by -2 would have reversed the inequality, so it is safer to move the term across as above.

Transforming a logarithmic graph

The same shifts apply as for any function, but they act on the *vertical* asymptote rather than a horizontal one.

In $y = \ln(x - h)$ the change is inside the bracket, so it moves the graph h units right. The asymptote travels with it, from $x = 0$ to $x = h$.

EXAMPLE 6.9

Sketch $y = \ln(x - 2)$, stating its asymptote, its domain and its x-intercept.
The input must be positive, so $x - 2 > 0$ and the domain is $x > 2$.
That same condition gives the asymptote. As x approaches 2 from above, $x - 2$ approaches 0 and $\ln(x - 2)$ decreases without bound, so $x = 2$ is a vertical asymptote.
For the x-intercept, set $y = 0$. A logarithm is zero when its input is 1:

$$x - 2 = 1 \implies x = 3.$$

So the curve rises from the asymptote at $x = 2$, crosses the axis at $(3, 0)$, and continues upward ever more slowly.

Using the inverse relationship

Because the exponential and the logarithm undo each other, you can use a logarithm to solve for a variable in an exponent, and an exponential to solve for a variable inside a logarithm. That is how the inverse of a function built from either one is found.

EXAMPLE 6.10

Find the inverse of $f(x) = 3e^{2x} + 1$, and state its domain.
Write $y = f(x)$, then undo each operation one at a time until x stands alone:

$$y = 3e^{2x} + 1 \implies \frac{y - 1}{3} = e^{2x}.$$

The variable is now in an exponent, so apply the natural logarithm to both sides:

$$2x = \ln\left(\frac{y - 1}{3}\right) \implies x = \tfrac{1}{2}\ln\left(\frac{y - 1}{3}\right).$$

Exchange the letters to write the inverse as a function of x:

$$f^{-1}(x) = \tfrac{1}{2}\ln\left(\frac{x-1}{3}\right).$$

Now the domain. The domain of f^{-1} is the range of f. Since $e^{2x} > 0$ for every x, we have $3e^{2x} > 0$ and so $f(x) > 1$. The domain of f^{-1} is therefore $x > 1$, which agrees with requiring $\frac{x-1}{3} > 0$ inside the logarithm.

TIP State the domain whenever a question asks for an inverse involving a logarithm. It is a separate mark, and the quickest check is that the domain of f^{-1} equals the range of f.

YOUR TURN 6.4 Logarithmic Functions

17. Evaluate each expression.

(a) $\log_2 8$ [1] (b) $\ln 1$ [1]
(c) $\ln e$ [1] (d) $\log_5 125$ [1]

18. Solve each equation.

(a) $\ln x = 0$ [1] (b) $\log_3 x = 2$ [1]
(c) $\ln x = 3$ [2] (d) $\log_2(x-1) = 4$ [2]

19. State the domain of each function.

(a) $y = \ln(x-2)$ [1] (b) $y = \log(x+5)$ [1]
(c) $y = \ln(2x-6)$ [2] (d) $y = \log(4-x)$ [2]

20. State the vertical asymptote and the x-intercept of each graph.

(a) $y = \ln x$ [2] (b) $y = \ln(x+1)$ [2]
(c) $y = \ln(x-2)$ [2] (d) $y = \log(x+5)$ [3]

21. Find the inverse of each function, and state its domain.

(a) $f(x) = e^x + 2$ [3]
(b) $f(x) = \ln(x-1)$ [3]
(c) $f(x) = 3e^{2x} + 1$ [4]

22. The domain of $y = \ln(9 - x^2)$ is not a single inequality.

(a) Write down the condition that the input must satisfy. [1]

(b) Solve it to find the domain. [3]

6.5 Partial Fractions HL

Adding two simple fractions produces one complicated fraction. **Partial fractions** reverse that process: they split a single fraction back into a sum of simpler ones, one for each factor of the denominator.

The split form is often much simpler to work with than the original. It is also what makes a rational function possible to integrate, so this section is the direct preparation for the integration of rational functions. See Ch. 16.

The method has one precondition, and then three cases decided by the factors in the denominator. Take the precondition first.

The fraction must be proper

A fraction is **proper** when the degree of its numerator is lower than the degree of its denominator. Only a proper fraction can be split into partial fractions.

If the numerator has degree equal to or higher than the denominator, the fraction is **improper**. Divide first, by polynomial division, to write it as a polynomial plus a proper remainder, then split the remainder. For example,

$$\frac{x^2}{x^2-1} = 1 + \frac{1}{x^2-1},$$

and only the proper part $\dfrac{1}{x^2-1}$ is decomposed further.

Case 1: distinct linear factors

This is the only case the AA HL syllabus examines. The denominator is a product of different linear factors, and each factor contributes one term with a constant on top.

KEY For a proper fraction over two distinct linear factors,

$$\frac{px+q}{(x-\alpha)(x-\beta)} = \frac{A}{x-\alpha} + \frac{B}{x-\beta}.$$

To find A and B, multiply through by the denominator, then substitute the values of x that make one factor zero. Each substitution isolates one unknown.

EXAMPLE 6.11

Express $\dfrac{3x+5}{(x-1)(x+2)}$ in partial fractions.

Write

$$\frac{3x+5}{(x-1)(x+2)} = \frac{A}{x-1} + \frac{B}{x+2},$$

and multiply through by $(x-1)(x+2)$:

$$3x+5 = A(x+2) + B(x-1).$$

Setting $x = 1$ isolates A: $8 = 3A$, so $A = \frac{8}{3}$. Setting $x = -2$ isolates B: $-1 = -3B$, so $B = \frac{1}{3}$. Therefore

$$\frac{3x+5}{(x-1)(x+2)} = \frac{8}{3(x-1)} + \frac{1}{3(x+2)}.$$

Choosing the two values that each make one factor zero is what makes the method quick. It turns a pair of simultaneous equations into two one-line calculations.

EXAMPLE 6.12

Express $\dfrac{x+7}{x^2+x-2}$ in partial fractions.

This denominator is not factorised, so factorise it first:

$$x^2+x-2 = (x+2)(x-1).$$

Now proceed as before:

$$\frac{x+7}{(x+2)(x-1)} = \frac{A}{x+2} + \frac{B}{x-1} \implies x+7 = A(x-1) + B(x+2).$$

Setting $x = 1$: $8 = 3B$, so $B = \frac{8}{3}$. Setting $x = -2$: $5 = -3A$, so $A = -\frac{5}{3}$. Hence

$$\frac{x+7}{x^2+x-2} = \frac{8}{3(x-1)} - \frac{5}{3(x+2)}.$$

TIP Examination questions usually give the denominator in expanded form. The factorising step is part of the answer and earns a mark, so write it out.

Case 2: a repeated linear factor HL

When a linear factor is squared, one term is not enough. A factor $(x-\alpha)^2$ needs a separate term for each power up to the square: one over $(x-\alpha)$ and one over $(x-\alpha)^2$.

KEY For a proper fraction over a repeated linear factor,

$$\frac{px+q}{(x-\alpha)^2} = \frac{A}{x-\alpha} + \frac{B}{(x-\alpha)^2}.$$

The reason for the second term is short. Placed over the common denominator $(x-\alpha)^2$, the single term $\dfrac{A}{x-\alpha}$ becomes $\dfrac{A(x-\alpha)}{(x-\alpha)^2}$. Its numerator is a multiple of $(x-\alpha)$, so it is zero at $x = \alpha$ and cannot match a general numerator there. The term $\dfrac{B}{(x-\alpha)^2}$ supplies the missing value.

Substituting $x = \alpha$ still isolates the top term, here B. To find A, compare the coefficients of a chosen power of x, since no second root is available.

EXAMPLE 6.13

Express $\dfrac{3x+1}{(x-1)^2}$ in partial fractions.

Write

$$\frac{3x+1}{(x-1)^2} = \frac{A}{x-1} + \frac{B}{(x-1)^2},$$

and multiply through by $(x-1)^2$:

$$3x + 1 = A(x-1) + B.$$

Setting $x = 1$ isolates B: $4 = B$. Comparing the coefficients of x gives $A = 3$. Therefore

$$\frac{3x+1}{(x-1)^2} = \frac{3}{x-1} + \frac{4}{(x-1)^2}.$$

A repeated factor uses substitution for the top term and coefficient comparison for the rest.

Case 3: an irreducible quadratic factor HL

A quadratic factor that does not factorise over the real numbers, meaning its discriminant is negative, is **irreducible**. Over such a factor the numerator is not a constant but a full linear expression $Bx + C$.

KEY For a proper fraction over a linear factor and an irreducible quadratic,

$$\frac{\text{proper}}{(x-\alpha)(x^2+bx+c)} = \frac{A}{x-\alpha} + \frac{Bx+C}{x^2+bx+c}.$$

The numerator over each factor always has degree one less than that factor: a constant over a linear factor, a linear expression over a quadratic. Substituting $x = \alpha$ isolates A; the remaining unknowns come from comparing coefficients.

EXAMPLE 6.14

Express $\dfrac{3x^2-2x+3}{(x-1)(x^2+1)}$ in partial fractions.

The factor x^2+1 has no real root, so it is irreducible. Write

$$\frac{3x^2-2x+3}{(x-1)(x^2+1)} = \frac{A}{x-1} + \frac{Bx+C}{x^2+1},$$

and multiply through by $(x-1)(x^2+1)$:

$$3x^2-2x+3 = A(x^2+1) + (Bx+C)(x-1).$$

Setting $x=1$ isolates A: $4=2A$, so $A=2$. Comparing the coefficients of x^2 gives $3=A+B$, so $B=1$. Comparing the constant terms gives $3=A-C$, so $C=-1$. Therefore

$$\frac{3x^2-2x+3}{(x-1)(x^2+1)} = \frac{2}{x-1} + \frac{x-1}{x^2+1}.$$

The three cases follow one rule: give the denominator its factors, then give each factor a numerator of degree one less than itself, adding an extra term for every power of a repeated factor.

Factor in the denominator	Term(s) it contributes
linear $(x-\alpha)$	$\dfrac{A}{x-\alpha}$
repeated linear $(x-\alpha)^2$	$\dfrac{A}{x-\alpha}+\dfrac{B}{(x-\alpha)^2}$
irreducible quadratic (x^2+bx+c)	$\dfrac{Bx+C}{x^2+bx+c}$

Cases 2 and 3 are beyond the AA HL syllabus, which restricts partial fractions to two distinct linear factors.

YOUR TURN 6.5 Partial Fractions HL

23. Express each fraction in partial fractions.

(a) $\dfrac{1}{(x-1)(x-2)}$ [3] (b) $\dfrac{5}{(x-2)(x+3)}$ [3]

(c) $\dfrac{3}{(x+1)(x+4)}$ [3] (d) $\dfrac{x}{(x-1)(x+1)}$ [3]

24. Express each fraction in partial fractions.

(a) $\dfrac{2x+1}{x(x+1)}$ [3] (b) $\dfrac{x+4}{x(x-2)}$ [3]

(c) $\dfrac{2x}{(x-3)(x+1)}$ [3] (d) $\dfrac{5x-1}{(x-1)(x+2)}$ [4]

25. Factorise the denominator first, then express each fraction in partial fractions.

(a) $\dfrac{4x}{x^2-4}$ [4] (b) $\dfrac{x+7}{x^2+x-2}$ [4]

26. Consider $\dfrac{x^2}{x^2-1}$.

(a) Explain why this fraction is improper. [1]

(b) Divide to write it as $1+\dfrac{1}{x^2-1}$. [2]

(c) Hence express the whole fraction in partial fractions. [4]

Chapter 6 · Rational and Exponential Functions

Reflections

TOK An asymptote describes a curve approaching a line it never meets, across infinitely many ever-smaller steps. This is the shape of Zeno's ancient paradox: to cross a room you must first cross half, then half of what remains, forever. Motion, Zeno argued, is therefore impossible. Calculus answers him: infinitely many shrinking steps can sum to a finite whole. Does mathematics dissolve the paradox, or merely rename it?

TOK The model in §6.3 fits two population figures and predicts 124.1 million people in Japan in 2030. Official projections are lower, because the rate of decline is itself changing. The model's arithmetic is not wrong. Its assumption is. Every model works by choosing which features of the world to leave out. So when a prediction fails, we have learned something either about the world or about our own choice of what to leave out. How would you tell which?

IM The word *transcendental* was Leibniz's, for quantities that "transcend" algebra, that no polynomial equation can capture. It took until 1873 for Charles Hermite to prove that e is transcendental, and 1882 for Ferdinand von Lindemann to prove the same of π, settling a two-thousand-year-old question: because π is transcendental, the circle cannot be squared with compass and straightedge. Some numbers lie forever beyond the reach of algebra.

RW Exponential decay is the clock of the physical world. Every radioactive isotope falls by half over a fixed *half-life*, from carbon-14's $5{,}730$ years, used to date ancient wood and bone, to isotopes that decay in seconds. The same exponential describes unrestricted population growth and compound interest. When growth is limited by a maximum, it becomes the logistic curve of a spreading rumour or an epidemic. Rational functions model saturation: reaction rates, drug concentrations, and average costs that flatten toward a limit.

EXT The exponential e^x has a *secret identity*. Give it an imaginary exponent and it becomes a rotation: Euler's formula $e^{i\theta} = \cos\theta + i\sin\theta$ ties the transcendental e to the trigonometry of Chapter 9 and the complex numbers of Chapter 18. Setting $\theta = \pi$ gives $e^{i\pi} + 1 = 0$, a single line linking e, π, i, 1, and 0, often called the most beautiful equation in mathematics.

End-of-Chapter Problems — Chapter 6: Rational and Exponential Functions

1. Consider $f(x) = \dfrac{2x+3}{x-1}$.

(a) State the vertical and horizontal asymptotes. [2]
(b) Find the x- and y-intercepts. [2]
(c) Sketch the graph. [2]

2. Consider $f(x) = \dfrac{x-2}{x+3}$.

(a) State the domain. [1]
(b) State the asymptotes. [2]
(c) Find $f^{-1}(x)$. [3]

3. Consider $f(x) = \dfrac{x^2-1}{x}$.

(a) State the vertical asymptote. [1]
(b) Find the oblique asymptote. [2]
(c) Find the x-intercepts. [2]

4. Solve $3^{x+1} = 27$. [2]

5. Solve $2^{2x} = 32$. [2]

6. Let $f(x) = e^x$.

(a) Find $f^{-1}(x)$. [2]
(b) State the domain of f^{-1}. [1]

7. Solve $\ln(2x-1) = 0$. [2]

8. Solve $e^{2x} = 5$, giving your answer in exact form. [3]

9. Express $\dfrac{7x-1}{(x-1)(x+2)}$ in partial fractions. [4]

10. Consider $f(x) = \dfrac{3x}{x-2}$.

(a) State the asymptotes. [2]
(b) Show that $f^{-1}(x) = \dfrac{2x}{x-3}$. [4]

11. A bacterial population is modelled by $P(t) = 500 \cdot 2^t$, where t is in hours.

(a) State the initial population. [1]
(b) Find the population after 3 hours. [2]
(c) Find when the population reaches 8000. [3]

12. A radioactive sample has mass $M(t) = 80 \cdot (0.5)^{t/5}$ grams, t in days.

(a) State the initial mass. [1]
(b) State the half-life. [2]
(c) Find the mass after 10 days. [2]

13. Solve $\log_2 x + \log_2(x-2) = 3$. [5]

14. Express $\dfrac{1}{x^2-4}$ in partial fractions. [4]

15. Solve $5 \cdot 3^x = 45$. [2]

16. The value of a car after t years is $V(t) = 20000(0.85)^t$.

(a) State the initial value. [1]
(b) Find the value after 4 years. [2]
(c) State the long-term behaviour of $V(t)$. [1]

17. Consider $f(x) = \dfrac{x^2+x-2}{x-1}$.

(a) Simplify $f(x)$ by factoring. [2]
(b) State the domain. [1]
(c) Describe the graph. [2]

18. Solve $2^x = 3$, giving your answer to 3 significant figures. [3]

19. Given $\log 2 = 0.301$ and $\log 3 = 0.477$, find $\log 6$ and $\log 1.5$. [3]

20. Express $\dfrac{x+5}{x^2+x}$ in partial fractions. [4]

21. Consider $f(x) = \dfrac{4x-1}{2x+1}$.

(a) State the horizontal asymptote. [1]
(b) State the vertical asymptote. [1]
(c) State the y-intercept. [1]

22. Solve $e^{2x} - 3e^x + 2 = 0$. [5]

23. Sketch, on the same axes, $y = 2^x$ and $y = 2^{-x}$, stating the horizontal asymptote and the common point. [4]

24. Solve $\log_3(x+6) = 2$. [2]

25. Show that $f(x) = \dfrac{ax+b}{cx+d}$ is self-inverse (that is, $f(f(x)) = x$) if and only if $a + d = 0$. [6]

26. Solve $4^x - 2^{x+1} = 8$. [5]

27. Express $\dfrac{x^2+1}{x(x-1)(x+1)}$ in partial fractions. [6]

28. Simplify $\log_2 3 \cdot \log_3 8$. [4]

29. The curve $y = \dfrac{x^2+ax+b}{x-1}$ has oblique asymptote $y = x + 3$.

(a) Find a. [3]

(b) Find the value of b for which the curve has a hole at $x = 1$ instead of a vertical asymptote. [3]

30. HL Consider $f(x) = \dfrac{x^2-4x+5}{x-2}$.

(a) Show that $f(x) = (x-2) + \dfrac{1}{x-2}$. [3]

(b) Write down the equations of the vertical and oblique asymptotes. [3]

(c) Find the y-intercept, and show that the curve has no x-intercept. [3]

(d) Show that the curve never meets its oblique asymptote. [2]

(e) Sketch the curve, showing both asymptotes and the y-intercept. [4]

31. GDC The number of people who have heard a rumour t days after it starts is modelled by

$$N(t) = \frac{5000}{1 + 49e^{-0.8t}}.$$

(a) Find the number of people who had heard the rumour at $t = 0$. [2]

(b) Find $N(5)$, giving your answer to the nearest whole number. [2]

(c) Find the time at which 2500 people have heard the rumour, giving an exact answer and a value to three significant figures. [4]

(d) State the value N approaches as t becomes large, and interpret it in context. [3]

(e) Show that $\dfrac{5000-N}{N} = 49e^{-0.8t}$, and hence explain why plotting $\ln\left(\dfrac{5000-N}{N}\right)$ against t gives a straight line. State its gradient and its vertical intercept. [5]

32. Let $f(x) = \ln(2x - 4)$.

(a) State the domain of f. [2]

(b) Find the exact x-intercept of the graph of f. [3]

(c) State the equation of the vertical asymptote, and describe the behaviour of $f(x)$ near it. [3]

(d) Find $f^{-1}(x)$, and state its domain and range. [4]

(e) Describe a sequence of transformations that maps $y = \ln x$ onto $y = \ln(2x - 4)$. [3]

Challenge Questions

33. **Splitting a sum with partial fractions.** Partial fractions are usually a step towards integration. This investigation uses them for something else: adding infinitely many terms exactly.

(a) Show that $\dfrac{1}{n(n+1)} = \dfrac{1}{n} - \dfrac{1}{n+1}$ for every positive integer n. [2]

(b) Hence write out the sum $\displaystyle\sum_{n=1}^{N} \frac{1}{n(n+1)}$ term by term, and show that almost every term cancels. Deduce that the sum equals $\dfrac{N}{N+1}$. *(Hint: write the first three and the last two terms)* [4]

(c) State the value the sum approaches as $N \to \infty$, and explain what that value means for the graph of the partial sums. [2]

(d) Show that $\dfrac{1}{n(n+2)} = \dfrac{1}{2}\left(\dfrac{1}{n} - \dfrac{1}{n+2}\right)$, and hence find $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n(n+2)}$. [4]

(e) Investigate $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n(n+k)}$ for a positive integer k. Find its value for $k = 3$ and $k = 4$, then state a general formula and justify it. [6]

34. **Newton's law of cooling.** A body at temperature T in a room at temperature R cools so that the difference $T - R$ decays exponentially:

$$T(t) = R + (T_0 - R)\, e^{-kt}, \qquad k > 0,$$

where T_0 is the temperature at $t = 0$ and t is measured in minutes.

(a) Show that the temperature difference $T(t) - R$ falls by the same factor over any interval of fixed length, and that this factor does not depend on T_0. [4]

(b) Deduce that the difference halves every $\dfrac{\ln 2}{k}$ minutes. [3]

(c) A cup of coffee at $90\,^\circ$C is left in a room at $20\,^\circ$C. After 10 minutes it has cooled to $60\,^\circ$C. Find k in exact form, and hence find the half-life of the temperature difference.

[4]

(d) Find, to the nearest minute, how long the coffee takes to reach $40\,^\circ$C. [3]

(e) Explain why the model predicts that the coffee never reaches $20\,^\circ$C, and state what feature of the graph carries that fact. [3]

35. **When does an oblique asymptote exist?** Consider

$$f(x) = \frac{ax^2 + bx + c}{dx + e}, \qquad a \neq 0,\ d \neq 0.$$

(a) By polynomial division, show that

$$f(x) = \frac{a}{d}\,x + \frac{bd - ae}{d^2} + \frac{r}{dx + e}$$

and find r in terms of a, b, c, d and e. [5]

(b) Hence write down the equation of the oblique asymptote, and explain why the remainder term does not affect it. [3]

(c) Find the condition on a, b, d and e for the oblique asymptote to pass through the origin. [3]

(d) Show that f has no vertical asymptote exactly when $ae^2 - bde + cd^2 = 0$, and explain what the graph does at $x = -\frac{e}{d}$ in that case. [4]

(e) Verify your condition on $f(x) = \dfrac{x^2 + 3x - 4}{x - 1}$, and state what this function really is. [3]

FUNCTIONS

Chapter 7

Polynomials

7

SYLLABUS 2.12 HL only

Nature rarely changes in sudden jumps. Temperature, water level and the position of a moving object all change smoothly. Polynomials are the simplest functions that behave the same way, so they are used to model a great deal of the physical world.

The degree matches the situation. Distance travelled at a constant speed is linear. A thrown ball follows a parabola, the graph of a quadratic, and so do the surface of spun water and the cross-section of a satellite dish. The volume of a sphere grows with the cube of its radius, $V = \frac{4}{3}\pi r^3$, and the power a ship needs grows roughly with the cube of its speed. An evenly loaded beam bends into a quartic curve. Over a short enough interval, almost any smooth curve can be matched closely by a polynomial, so measurements are often fitted with one when no exact formula is known. See Ch. 17.

That smoothness comes from the way a polynomial is built. A polynomial uses one variable and a few constants, combined by addition and multiplication only. There is no division by the variable, no square root, and no trigonometry. A polynomial is therefore defined for every real number, so its graph has no gaps and no asymptotes, and it has no corners anywhere. The sum of two polynomials is a polynomial, and so is their product.

Describing a polynomial is easy. Solving one is not. To solve $f(x) = 0$ is to find every value of x that makes the polynomial equal to zero. An exact formula for those values exists for degree 2, degree 3 and degree 4. For degree 5 and above, no such formula exists. Niels Abel proved this in 1824. This chapter gives the methods that work without a formula: testing whether a value is a root, factorising a polynomial once one root is known, and finding the sum and product of the roots from the coefficients alone.

Before any of that you will learn to read a polynomial's graph from its factorised form: where it crosses the axis, where it only touches, and what it does at either end. That is the thread running through the chapter. A polynomial's coefficients and its roots describe the same object, and nearly every technique here moves between the two.

7.1 Polynomial Functions and their Graphs

Much of the graph of a polynomial can be described before a single point is plotted. The degree and the coefficients carry that information, so begin with the formal definition.

DEF A **polynomial function** of **degree** n has the form

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0, \qquad a_n \neq 0,$$

where the coefficients are constants and n is a non-negative integer.

Polynomials are named by their degree. Degree 1 is **linear**, degree 2 is **quadratic**, degree 3 is **cubic**, degree 4 is **quartic**, and degree 5 is **quintic**. The degree also limits the shape: it limits how many times the curve can meet the x-axis, and how often it can change direction.

The constants $a_n, a_{n-1}, \dots, a_1, a_0$ are the **coefficients** of the powers of x. The coefficient of the highest power is a_n, called the **lead coefficient** (also written leading coefficient), and the term $a_n x^n$ that contains it is the **leading order term**.

A coefficient may take any value, with one restriction: the lead coefficient cannot be zero. If a_n were zero, the term $a_n x^n$ would be zero, and what remained would be a polynomial of degree $n - 1$ instead.

The sign of the lead coefficient determines which way the ends of the curve point. A polynomial with $a_n > 0$ is sometimes called a **positive polynomial**, and one with $a_n < 0$ a **negative polynomial**. These names describe the lead coefficient only. They do not mean that the function is positive or negative everywhere: $y = x^2 - 4$ has a positive lead coefficient, yet y is negative for $-2 < x < 2$.

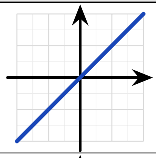
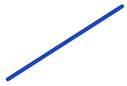
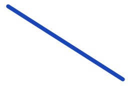
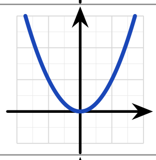
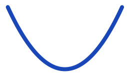
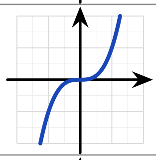
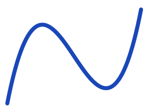
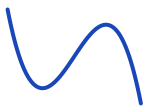
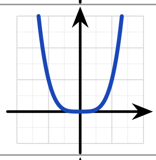
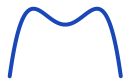
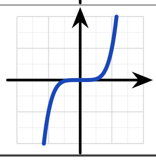
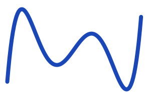
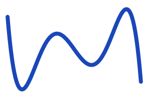
Three features are visible before you plot a single point.

The **end behaviour**, meaning the direction of the curve as $x \to \pm\infty$, is determined entirely by the leading term. If n is odd, the two ends point in opposite directions. If n is even, they point the same way, and the sign of a_n determines whether that direction is upward or downward.

The number of **roots**, the places where the graph meets the x-axis, can never exceed the degree. A cubic meets the axis at most three times, a quartic at most four.

A **turning point** is a place where the curve stops rising and begins to fall, or stops falling and begins to rise. A polynomial of degree n has at most $n - 1$ turning points. The possible numbers then decrease in steps of two: $n-1$, $n-3$, and so on. The number changes by two rather than by one because a maximum and a minimum are always lost together, never one alone.

The table collects all three features for the degrees you meet most often.

n	$y = x^n$	**Shape when** $a_n > 0$	**Shape when** $a_n < 0$	**Times it meets the** x**-axis**	**Turning points**
1				1	0
2				0, 1 or 2	1
3				1, 2 or 3	0 or 2
4				0, 1, 2, 3 or 4	1 or 3
5				1, 2, 3, 4 or 5	0, 2 or 4

Where a factor is repeated, the graph behaves differently. The **multiplicity** of a root is the number of times its factor appears in the factorised polynomial. In $(x-1)(x-2)^2$ the root $x = 1$ has multiplicity one, and the root $x = 2$ has multiplicity two.

Multiplicity decides what the curve does at the axis. A root of multiplicity one, from a single factor $(x-a)$, crosses the axis. A root of even multiplicity, such as $(x-a)^2$, only *touches* the axis and turns back. Practise both directions: reading the multiplicities from a sketch, and predicting the sketch from the factors.

EXAMPLE 7.1

Sketch $y = (x+1)(x-2)^2$, showing the intercepts and the behaviour at each root.

Read the roots and their multiplicities from the factors. The factor $(x+1)$ is single, so the graph crosses the axis at $x = -1$. The factor $(x-2)^2$ is squared, so the graph touches the axis at $x = 2$ and turns back.

Find the y-intercept. Substitute $x = 0$: $y = (1)(-2)^2 = 4$, the point $(0, 4)$.

Fix the ends from the leading term. Multiplying out leads with x^3: an odd degree with a positive coefficient, so the curve runs from $-\infty$ at the left to $+\infty$ at the right.

These three facts fix the shape.

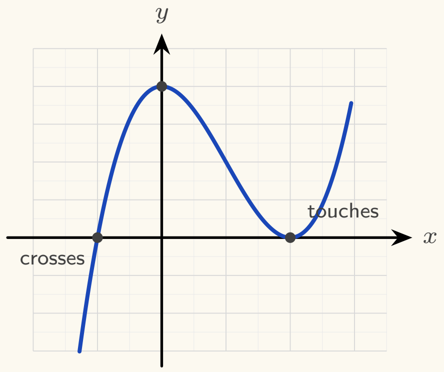


Figure 7.1 A sketch of $y = (x+1)(x-2)^2$. The simple root at $x = -1$ crosses the axis; the double root at $x = 2$ touches it and turns back. The y-intercept is $(0, 4)$, and the ends follow x^3.

Every factorised polynomial is sketched the same way: the roots and their multiplicities give the behaviour at the x-axis, the value at $x = 0$ gives the y-intercept, and the leading term gives the two ends.

When the polynomial is written out in full rather than factorised, the y-intercept is even quicker to read. Every term except the constant contains x, so substituting $x = 0$ makes all of them zero. The constant term a_0 is what remains.

KEY The graph of $y = a_n x^n + \cdots + a_1 x + a_0$ crosses the y-axis at $(0, a_0)$. *The constant term is the y-intercept.*

For $y = 2x^3 - 3x + 5$, the constant term is 5, so the graph crosses the y-axis at $(0, 5)$.

From the graph back to the equation

The reverse problem is also examined. Given a sketch, you are asked for the polynomial that produced it. The intercepts give you almost everything you need.

Each x-intercept contributes a factor, and the behaviour there gives its multiplicity: a crossing gives a single factor, a touch gives a squared factor. That fixes the polynomial up to one unknown number, the leading coefficient, and any other point on the curve determines it.

KEY **Finding a polynomial from its graph.** Write one factor for each x-intercept, with the multiplicity the graph shows at that point. Then substitute the coordinates of any other known point to find the leading coefficient a_n.

EXAMPLE 7.2

The graph below shows a cubic. It touches the x-axis at $x = -1$, crosses it at $x = 2$, and passes through $(0, -1)$. Find its equation.

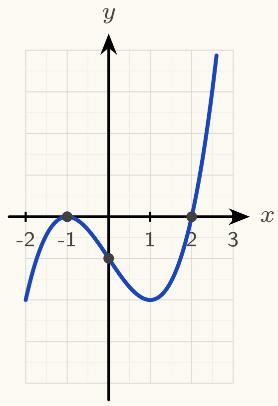


Figure 7.2 The curve touches the x-axis at $x = -1$, crosses it at $x = 2$, and passes through $(0, -1)$. These three facts determine the equation.

Write the factors from the x-intercepts. The curve touches at $x = -1$, so that root is repeated and contributes $(x + 1)^2$. It crosses at $x = 2$, so that root is single and contributes $(x - 2)$. Those factors already account for degree three, so

$$y = a(x + 1)^2(x - 2)$$

for some constant a.

Find a from the remaining point. Substitute $x = 0$ and $y = -1$:

$$-1 = a(1)^2(-2) = -2a,$$

so $a = \frac{1}{2}$ and

$$y = \tfrac{1}{2}(x + 1)^2(x - 2).$$

Check the answer against the graph. The leading term is $\frac{1}{2}x^3$, which is positive, so the curve rises to the right, and the sketch agrees.

The intercepts fix the factors; one further point fixes the leading coefficient.

YOUR TURN 7.1 Polynomial Functions and their Graphs

1. State the degree and the lead coefficient of each polynomial.

(a) $f(x) = 3x^4 - 2x + 7$ [2] (b) $f(x) = 5 - x^3$ [2]

(c) $f(x) = (x - 1)(x + 2)(x - 3)$ [2] (d) $f(x) = -2x^5 + x^2$ [2]

2. Describe the end behaviour of each graph as $x \to +\infty$ and as $x \to -\infty$.

(a) $y = x^3$ [2] (b) $y = -x^4$ [2]

(c) $y = 2x^5 - x$ [2] (d) $y = 3 - x^2$ [2]

3. Answer each question about degree and shape.

(a) State the maximum number of real roots of a degree-5 polynomial. [1]

(b) State the maximum number of turning points of a quartic. [1]

(c) List every possible number of turning points a cubic can have. [2]

(d) Explain why a cubic must have at least one real root. [2]

4. State the roots of each polynomial, and say at each one whether the graph crosses or touches the x-axis.

(a) $y = (x-2)(x+3)(x-5)$ [2]

(b) $y = (x-1)^2(x+4)$ [2]

(c) $y = x(x-3)^2$ [2]

(d) $y = (x+1)^2(x-2)^2$ [3]

5. Consider $y = (x+1)(x-3)^2$.

(a) State the roots and their multiplicities. [2]

(b) Find the y-intercept. [2]

(c) State the end behaviour. [2]

(d) Sketch the curve. [3]

6. A cubic touches the x-axis at $x = 2$, crosses it at $x = -1$, and passes through $(0, 4)$.

(a) Write down the factors, with the correct multiplicities. [2]

(b) Find the lead coefficient. [3]

(c) Write the equation of the cubic. [1]

7.2 Polynomial Division

You can already add, subtract and multiply polynomials. Addition and subtraction collect like terms, and multiplication expands every product before collecting. Division is the operation that still needs a method, and it is the one the rest of this chapter depends on.

Factorising a polynomial means writing it as a product. Once you know one factor, the other is found by dividing. So a method for dividing one polynomial by another is needed before any factorising can be done.

Polynomial division follows the pattern of whole-number division. When you divide 47 by 5 you write

$$47 = 5 \times 9 + 2,$$

which records a divisor, a quotient, and a remainder smaller than the divisor. Polynomials behave the same way, with "smaller" measured by degree instead of size.

KEY **The division algorithm.** For polynomials $f(x)$ and $d(x)$, with $d(x)$ not the zero polynomial, there are unique polynomials $q(x)$ and $r(x)$ such that

$$f(x) = d(x)\,q(x) + r(x),$$

where the degree of r is less than the degree of d. *Dividend equals divisor times quotient, plus remainder.*

Fix the four names now, because every method below uses them. The polynomial being divided, f, is the **dividend**. The polynomial you divide by, d, is the **divisor**. The result q is the **quotient**, and the part that remains, r, is the **remainder**. When $r = 0$ the division is exact, and d is a factor of f.

Long division works for a divisor of any degree. Three steps repeat until the remainder has lower degree than the divisor.

Divide the leading term of the dividend by the leading term of the divisor. That result is the next term of the quotient. Multiply the whole divisor by that term, then subtract the product from the dividend. Repeat on what is left.

EXAMPLE 7.3

Divide $2x^3 - 3x^2 + x - 5$ by $(x - 2)$.
Set the work out in columns, one column for each power of x.

$$\begin{array}{rrrrrl}
 & 2x^2 & +x & +3 & & \text{quotient} \\ \cline{2-5}
x-2\,) & 2x^3 & -3x^2 & +x & -5 & \text{dividend} \\
 & 2x^3 & -4x^2 & & & \\ \cline{2-3}
 & & x^2 & +x & & \\
 & & x^2 & -2x & & \\ \cline{3-4}
 & & & 3x & -5 & \\
 & & & 3x & -6 & \\ \cline{4-5}
 & & & & 1 & \text{remainder}
\end{array}$$

The divisor $x - 2$ is written to the left of the bracket, and the dividend $2x^3 - 3x^2 + x - 5$ inside it. The quotient is written on the top line, one term at each stage, and the remainder appears on the bottom line.
Each stage repeats the same three steps: divide, multiply, subtract.
Stage 1. Divide the leading term of the dividend by the leading term of the divisor: $2x^3 \div x = 2x^2$. Write $2x^2$ in the quotient. Multiply the divisor by it: $2x^2(x-2) = 2x^3-4x^2$. Subtract that from the dividend, which leaves $x^2 + x$.
Stage 2. Repeat on x^2+x. Divide: $x^2 \div x = x$. Multiply: $x(x-2) = x^2 - 2x$. Subtract, which leaves $3x - 5$.

Stage 3. Repeat on $3x-5$. Divide: $3x \div x = 3$. Multiply: $3(x-2) = 3x-6$. Subtract, which leaves 1.
The degree of 1 is lower than the degree of $x-2$, so the division stops.
The quotient is $2x^2+x+3$ and the remainder is 1:

$$2x^3-3x^2+x-5 = (x-2)(2x^2+x+3)+1.$$

Always write the answer in this form. Multiplying out the right side is a complete check of the division.

CAUTION Write the dividend in descending powers and insert a zero term for every missing power before dividing. Divide x^3-7x+6 as x^3+0x^2-7x+6. Without that placeholder the columns shift, and every subtraction after it is wrong.

EXAMPLE 7.4

Divide x^3-7x+6 by $(x-1)$.
There is no x^2 term, so write it as x^3+0x^2-7x+6 and keep a column for it.

$$\begin{array}{r|rrrr}
 & & x^2 & +x & -6 \\ \hline
x-1\,) & x^3 & +0x^2 & -7x & +6 \\
 & x^3 & -x^2 & & \\ \hline
 & & x^2 & -7x & \\
 & & x^2 & -x & \\ \hline
 & & & -6x & +6 \\
 & & & -6x & +6 \\ \hline
 & & & & 0
\end{array}$$

The remainder is 0, so the division is exact and $(x-1)$ is a factor. The quotient x^2+x-6 factorises further:

$$x^3-7x+6 = (x-1)(x^2+x-6) = (x-1)(x+3)(x-2).$$

A remainder of zero shows that the divisor is a factor. The next two sections are built on that fact.

YOUR TURN 7.2 Polynomial Division

7. Simplify each expression.

(a) $(x^3+2x-5)+(3x^3-x^2+1)$ [2]   (b) $(x^4+3x^2-1)-(2x^3-x^2+2)$ [2]

(c) $(x^2+2x-1)(x^2-x+3)$ [3] (d) $(x-2)(x^2+2x+4)$ [2]

8. Divide, and state the quotient and the remainder.

(a) x^2+5x+6 by $(x+2)$ [2] (b) x^3-3x^2+2x by $(x-1)$ [3]
(c) x^3+2x^2-x+1 by $(x-1)$ [3] (d) x^3-8 by $(x-2)$ [3]

9. Divide, and state the quotient and the remainder. Insert a zero term for each missing power first.

(a) $2x^3-x^2-3$ by $(x+1)$ [3] (b) x^3+3x^2-4 by $(x+2)$ [3]
(c) $2x^4-3x^3+x-6$ by $(x-2)$ [4]

10. Divide by the quadratic divisor.

(a) x^4-1 by (x^2-1) [3] (b) x^4-1 by (x^2+1) [3]

7.3 The Remainder Theorem

Here are two operations that appear to have nothing to do with each other. *Evaluating* f at 2 gives the height of the graph above one point. *Dividing* f by $(x-2)$ is the long calculation of the previous section. The remainder theorem states that the two produce the same number, and the proof is one line long.

Take the division algorithm with a linear divisor $(x-a)$:

$$f(x)=(x-a)\,q(x)+r,$$

where the remainder r must be a *constant*, because its degree is less than the degree of $x-a$, and less than one means zero.

Now substitute $x=a$ into that identity. The factor $(x-a)$ becomes zero, so the whole first term is zero, leaving $f(a)=r$. The height of the graph at a *is* the remainder, because everything else in the polynomial is divisible by $(x-a)$ and so contributes nothing at that point.

KEY **Remainder theorem.** When a polynomial $f(x)$ is divided by $(x-a)$, the remainder is $f(a)$. *One substitution replaces the entire division.*

EXAMPLE 7.5

Find the remainder when $f(x) = x^3 - 2x + 5$ is divided by $(x-2)$.

By the remainder theorem, the remainder is

$$f(2) = 2^3 - 2(2) + 5 = 8 - 4 + 5 = 9.$$

No division needed; the answer is a single evaluation.

EXAMPLE 7.6

Find the remainder when $f(x) = 4x^3 - 4x^2 + 5x - 1$ is divided by $(2x-1)$.

The divisor need not have the form $(x - a)$. The theorem works for any linear divisor, provided you evaluate at the **zero of the divisor**, meaning the value of x that makes the divisor equal to zero. Here $2x - 1 = 0$ when $x = \frac{1}{2}$, so the remainder is

$$f\left(\tfrac{1}{2}\right) = 4 \cdot \tfrac{1}{8} - 4 \cdot \tfrac{1}{4} + \tfrac{5}{2} - 1 = \tfrac{1}{2} - 1 + \tfrac{5}{2} - 1 = 1.$$

Evaluate at the zero of the divisor, not at one of its coefficients. Dividing by $(2x-1)$ means substituting $\frac{1}{2}$, never 1 or 2.

EXAMPLE 7.7

The polynomial $f(x) = x^3 + ax + b$ leaves remainder 2 when divided by $(x-1)$ and remainder -13 when divided by $(x+2)$. Find a and b.

Each remainder condition is one evaluation, so two conditions give two equations:

$$f(1) = 1 + a + b = 2 \implies a + b = 1,$$

$$f(-2) = -8 - 2a + b = -13 \implies -2a + b = -5.$$

Subtracting, $3a = 6$, so $a = 2$ and $b = -1$. The remainder theorem has turned two divisions into a pair of linear equations; this reversal, from remainders back to coefficients, is the standard examination form of the theorem.

YOUR TURN 7.3 The Remainder Theorem

11. Find the remainder in each case.

(a) $x^3 + 2x - 1$ divided by $(x-1)$ [2]

(b) $x^3 - 4x^2 + x + 6$ divided by $(x-2)$ [2]

(c) $2x^3 - 3x + 1$ divided by $(x+1)$ [2]

(d) $x^3 + x^2 + x + 1$ divided by $(x+2)$ [2]

12. Find the remainder in each case. The divisor is not of the form $(x-a)$.

(a) $4x^3-4x^2+5x-1$ divided by $(2x-1)$ [3]

(b) $2x^3+x^2-2$ divided by $(2x+1)$ [3]

13. Interpret each statement.

(a) The remainder when $f(x)$ is divided by $(x-3)$ is 0. What does this tell you? [2]

(b) The remainder when $f(x)$ is divided by $(x-2)$ is 7. State $f(2)$. [1]

(c) $f(-1)=0$. Write down a factor of $f(x)$. [1]

(d) x^4-1 is divided by $(x-1)$. Find the remainder without dividing. [2]

14. The polynomial $f(x)=x^3+ax+b$ leaves remainder 2 when divided by $(x-1)$ and remainder -13 when divided by $(x+2)$.

(a) Write down two equations in a and b. [3]

(b) Solve them to find a and b. [2]

7.4 The Factor Theorem

The most useful case of the remainder theorem is when the remainder is zero. If dividing by $(x-a)$ leaves no remainder, then $(x-a)$ divides $f(x)$ exactly: it is a factor. This is what makes factorising possible.

KEY **Factor theorem.** $(x-a)$ is a factor of $f(x)$ if and only if $f(a)=0$.

To factorise an unfamiliar cubic, search for a root by testing small values. The factors of the constant term are the natural candidates. The reason is proved in §7.6: the product of the roots is the constant term divided by the leading coefficient, apart from a sign. So any integer root must divide the constant term. Once one root a is found, $(x-a)$ is a factor, and dividing it out leaves a quadratic you can finish by hand.

EXAMPLE 7.8

Show that $(x-1)$ is a factor of $f(x)=x^3-6x^2+11x-6$, and factorise f completely.

Test $x=1$:

$$f(1)=1-6+11-6=0,$$

so by the factor theorem $(x-1)$ is a factor.

Now divide it out. Instead of long division, *compare coefficients*. The quotient must be a quadratic, so write

$$x^3 - 6x^2 + 11x - 6 = (x-1)(x^2 + bx + c),$$

then compare the coefficients of matching powers on the two sides. Compare the constant terms: $-c = -6$, so $c = 6$. Compare the coefficients of x^2: $b - 1 = -6$, so $b = -5$. The coefficient of x has not been used, so use it as a check: on the right it is $c - b = 6 + 5 = 11$, which matches the left. ✓ Hence

$$f(x) = (x-1)(x^2 - 5x + 6) = (x-1)(x-2)(x-3).$$

The constant term -6 gives the candidates $\pm1, \pm2, \pm3, \pm6$. Testing $x = 1$ succeeded on the first attempt, and comparing coefficients carried out the division with a check included.

TIP Comparing coefficients is usually faster than long division in an examination. Each unknown comes from one comparison, and any coefficient you did not use provides a check. Long division earns the same marks if you prefer it.

YOUR TURN 7.4 The Factor Theorem

15. Decide whether the given linear expression is a factor. Show your working.

(a) Is $(x-2)$ a factor of $x^3 - 3x^2 + 4$? [2]

(b) Is $(x+1)$ a factor of $x^3 + 2x^2 - x - 2$? [2]

(c) Is $(x-1)$ a factor of $x^3 + x^2 + x + 1$? [2]

(d) Is $(x+2)$ a factor of $x^3 + 8$? [2]

16. Factorise each polynomial completely.

(a) $x^3 - x$ [2]

(b) $x^3 - 4x$ [2]

(c) $x^3 + 2x^2 - x - 2$ [3]

(d) $2x^3 + x^2 - 5x + 2$ [4]

17. Answer each part.

(a) One factor of $x^2 - 5x + 6$ is $(x-2)$. State the other. [1]

(b) Show that $x^3 - 3x^2 + 4$ has a repeated factor. [3]

7.5 Solving Polynomial Equations

You already know how to solve linear and quadratic equations. A linear equation is rearranged directly. A quadratic is solved by factorising, by completing the square, or by the quadratic formula.

For a cubic or any higher degree, there is no formula of that kind for you to use. The factor theorem is used instead. Find one root by testing values, divide that factor out, and repeat until what remains is a quadratic. Then solve the quadratic in the usual way.

Every real root is an x-intercept of the graph, so the algebra and the graph always agree.

EXAMPLE 7.9

Solve $x^3 - 3x^2 - x + 3 = 0$.
Test the divisors of 3. At $x = 1$: $1 - 3 - 1 + 3 = 0$, so $(x - 1)$ is a factor. Dividing,

$$x^3 - 3x^2 - x + 3 = (x - 1)(x^2 - 2x - 3) = (x - 1)(x - 3)(x + 1).$$

The roots are $x = 1,\ 3,\ -1$.

CAUTION A cubic with real coefficients always has at least one real root, because one end goes to $+\infty$ and the other to $-\infty$, so the curve must cross the axis. If no rational root can be found by testing, one root is irrational and the equation must be solved by technology.

YOUR TURN 7.5 Solving Polynomial Equations

18. Solve each equation. Each is already factorised or factorises at sight.

(a) $(x - 1)(x + 2)(x - 4) = 0$ [1] (b) $(x - 3)^2(x + 1) = 0$ [2]
(c) $x^3 = 4x$ [2] (d) $x^3 + x^2 - 2x = 0$ [2]

19. Solve each equation, using the factor theorem to find the first root.

(a) $x^3 - 6x^2 + 11x - 6 = 0$ [3] (b) $x^3 - 2x^2 - x + 2 = 0$ [3]
(c) $2x^3 + x^2 - 5x + 2 = 0$ [4]

20. Solve each equation.

(a) $x^4 - 5x^2 + 4 = 0$ *(Hint: substitute $u = x^2$)* [3]
(b) $x^3 - 1 = 0$, giving only the real root, and explain why there is exactly one. [3]

7.6 Sum and Product of Roots

The coefficients of a polynomial determine the sum and the product of its roots. You can therefore find those two values *without solving the equation at all*. This matters whenever the roots are irrational, or when the equation is too hard to solve, because the sum and the product remain easy to obtain.

Take the quadratic case first. If $ax^2 + bx + c = 0$ has roots α and β, then the polynomial *is* $a(x-\alpha)(x-\beta)$, and expanding tells us what the coefficients must be:

$$a(x-\alpha)(x-\beta) = a(x^2 - (\alpha+\beta)x + \alpha\beta) = ax^2 - a(\alpha+\beta)\,x + a\,\alpha\beta.$$

Comparing with $ax^2 + bx + c$: the middle coefficient is $b = -a(\alpha+\beta)$ and the constant is $c = a\,\alpha\beta$. Solve each for the roots' totals:

$$\alpha + \beta = -\frac{b}{a}, \qquad \alpha\beta = \frac{c}{a}.$$

The minus sign on the sum is not a convention to memorise. It comes from the brackets, one $-\alpha$ and one $-\beta$.

The same expansion explains every degree at once, and the reason is the counting argument from Ch. 2. Expanding

$$a_n(x-r_1)(x-r_2)\cdots(x-r_n)$$

means choosing, from each bracket, either the x or the root, exactly as in the binomial theorem. To build the x^{n-1} term you take x from all brackets but one, so each root appears exactly once, each carrying a single minus sign: the coefficient collects $-(r_1 + r_2 + \cdots + r_n)$, the *sum*. To build the constant term you take the root from *every* bracket: all n roots multiplied together, carrying n minus signs, which is where the $(-1)^n$ comes from. So the outermost coefficients of a polynomial record the sum and the product of its roots.

KEY · IB For the equation $a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0 = 0$,

$$\text{sum of roots} = -\frac{a_{n-1}}{a_n}, \qquad \text{product of roots} = (-1)^n\,\frac{a_0}{a_n}.$$

The second coefficient collects the sum of the roots; the constant term collects their product.

These are the outermost of a whole family of relations (Vieta's formulas), but the sum and product are the two you will use most, for checking answers, for building equations with required roots, and for finding an unknown coefficient.

One more relation fills the gap between them. Building the x^{n-2} term means taking the root from exactly two brackets and x from the rest, so that coefficient collects the roots multiplied together in pairs. For a cubic $a_3x^3 + a_2x^2 + a_1x + a_0 = 0$ with roots α, β, γ, the three relations together are

$$\alpha+\beta+\gamma=-\frac{a_2}{a_3},\qquad \alpha\beta+\beta\gamma+\gamma\alpha=\frac{a_1}{a_3},\qquad \alpha\beta\gamma=-\frac{a_0}{a_3}.$$

The syllabus asks you to know only the first and last. The middle relation is a short extension, and it is the one to use whenever a question involves all three roots at once.

EXAMPLE 7.10

The equation $2x^2-5x+3=0$ has roots α and β. Find $\alpha+\beta$ and $\alpha\beta$ without solving, then verify. Here $a=2$, $b=-5$, $c=3$, so

$$\alpha+\beta=-\frac{-5}{2}=\frac{5}{2},\qquad \alpha\beta=\frac{3}{2}.$$

Solving confirms it: the roots are 1 and $\frac{3}{2}$, whose sum is $\frac{5}{2}$ and product is $\frac{3}{2}$. The sum and product were available from the coefficients before the roots were found.

EXAMPLE 7.11

The cubic $x^3-4x^2+x+6=0$ has roots α,β,γ. Write down $\alpha+\beta+\gamma$ and $\alpha\beta\gamma$. With $a_3=1$, $a_2=-4$, $a_0=6$ and degree $n=3$,

$$\alpha+\beta+\gamma=-\frac{-4}{1}=4,\qquad \alpha\beta\gamma=(-1)^3\,\frac{6}{1}=-6.$$

(The roots happen to be $-1,2,3$: sum 4, product -6, as promised.)

The method is most useful when the roots are irrational. The next three examples never find a root, yet answer precise questions about them. This is a common examination form.

EXAMPLE 7.12

The equation $x^2-3x+1=0$ has roots α and β. Without solving, find $\alpha^2+\beta^2$ and $\dfrac{1}{\alpha}+\dfrac{1}{\beta}$. The roots are irrational, but their sum and product are not: $\alpha+\beta=3$ and $\alpha\beta=1$. A **symmetric combination** of the roots is an expression whose value does not change when the two roots are exchanged. Both $\alpha^2+\beta^2$ and $\frac{1}{\alpha}+\frac{1}{\beta}$ are symmetric. Every symmetric combination can be rebuilt from the sum and the product. For the sum of squares, start from the square of the sum:

$$(\alpha+\beta)^2=\alpha^2+2\alpha\beta+\beta^2 \implies \alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta=9-2=7.$$

For the sum of reciprocals, combine over a common denominator:

$$\frac{1}{\alpha}+\frac{1}{\beta}=\frac{\alpha+\beta}{\alpha\beta}=\frac{3}{1}=3.$$

The identity $\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta$ is used often in this topic. Learn it as a rearrangement of $(\alpha+\beta)^2$, not as a separate formula.

EXAMPLE 7.13

With α, β as above, find a quadratic equation with integer coefficients whose roots are 2α and 2β.

A quadratic is determined by the sum and product of its roots: $x^2-(\text{sum})x+(\text{product})=0$. The new totals follow from the old ones:

$$2\alpha+2\beta=2(\alpha+\beta)=6, \qquad (2\alpha)(2\beta)=4\alpha\beta=4.$$

So the required equation is

$$x^2-6x+4=0.$$

No root was ever computed: transforming an equation's roots means transforming their sum and product, nothing more.

EXAMPLE 7.14

The roots of $x^2+kx+12=0$ differ by 1. Find the possible values of k.

Let the roots be α and β with $\alpha-\beta=1$. The coefficients give $\alpha+\beta=-k$ and $\alpha\beta=12$. Connect the difference to these totals by squaring it:

$$(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta \implies 1=k^2-48,$$

so $k^2=49$ and $k=\pm7$. Check with $k=-7$: the equation $x^2-7x+12=0$ has roots 3 and 4, which indeed differ by 1. A condition on the roots became an equation in the coefficient, using the identity $(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta$.

YOUR TURN 7.6 Sum and Product of Roots

21. State the sum and the product of the roots of each equation.

(a) $x^2-7x+10=0$ [2]
(b) $2x^2+3x-5=0$ [2]
(c) $x^2+4x+4=0$ [2]
(d) $3x^2-2x-8=0$ [2]

22. For each cubic or quartic, state the sum and the product of the roots.

(a) $x^3-6x^2+11x-6=0$ [2]
(b) $2x^3-4x^2+x-3=0$ [3]
(c) $x^4-3x^3+2x-5=0$ [3]

23. Build the required equation.

(a) Write a quadratic with lead coefficient 1 whose roots sum to 5 and multiply to 6. [2]

(b) The roots of $x^2 + bx + 12 = 0$ are 3 and one other. Find the other root and b. [3]

(c) Find the value of k for which $x^2 - kx + 9 = 0$ has equal roots. [3]

(d) Write a cubic with lead coefficient 1 whose roots are -1, 2 and 4. [3]

24. The roots of $2x^2-7x+6 = 0$ are α and β. Without solving the equation, find each value.

(a) $\alpha + \beta$ and $\alpha\beta$ [2]

(b) $\dfrac{1}{\alpha} + \dfrac{1}{\beta}$ [2]

(c) $\alpha^2 + \beta^2$ [2]

(d) $(\alpha - \beta)^2$ [3]

25. The roots of $x^2 + 5x + 3 = 0$ are α and β.

(a) Write down $\alpha + \beta$ and $\alpha\beta$. [2]

(b) Find the sum and product of $\alpha + 2$ and $\beta + 2$. [3]

(c) Hence find a quadratic equation with integer coefficients whose roots are $\alpha + 2$ and $\beta + 2$. [2]

26. The roots of $x^3 - 4x^2 + x + 6 = 0$ are α, β and γ.

(a) Write down $\alpha + \beta + \gamma$ and $\alpha\beta\gamma$. [2]

(b) Write down $\alpha\beta + \beta\gamma + \gamma\alpha$. [2]

(c) Hence find $\alpha^2 + \beta^2 + \gamma^2$. [3]

Chapter 7 · Polynomials

Reflections

IM The history behind this chapter is unusual. In Italy in the 1530s, mathematicians competed in public problem-solving contests, and a loser could lose his university post. Methods were therefore kept secret. Niccolò Tartaglia found a way to solve the cubic and told no one. Gerolamo Cardano persuaded him to share it under an oath of silence, then published it in 1545, together with his student Ferrari's solution of the quartic. Almost three hundred years passed before Niels Abel, who died of tuberculosis at twenty-six, proved that no such formula exists for the quintic. Évariste Galois, killed in a duel at twenty, explained exactly which equations can be solved by a formula and which cannot. To do this he created *group theory*, the mathematical study of symmetry, which is now used in particle physics and cryptography.

TOK Abel and Galois proved something surprising. The roots of a general quintic *exist*, but they cannot be written using only addition, subtraction, multiplication, division and roots. The equation $x^5 - x - 1 = 0$ has a real solution. That number is completely determined, and a computer can calculate it to a thousand decimal places. Even so, no formula of that kind can express it exactly. What does it mean, then, to "know" a number? Is such a root any less real than one you can write down?

RW Polynomials are used constantly in computing, because they need only multiplication and addition. Those are the two operations a processor performs fastest. The curved shapes in fonts and animation are drawn with *Bézier curves*, which are polynomial curves whose shape is controlled by a few chosen points. *Error-correcting codes* are built from polynomials as well. These are methods of adding extra information to a message so that it can still be read after part of it is damaged, which is how a scratched disc still plays and how a faint signal from a spacecraft is recovered. Polynomials are also used to fit smooth models to scattered measurements in science and finance.

EXT The factor theorem raises a question it cannot answer. How many roots does a polynomial really have? Over the real numbers, $x^2 + 1$ has none. The **Fundamental Theorem of Algebra** gives the answer. Over the *complex* numbers of Chapter 18, every polynomial of degree n has exactly n roots, counted with multiplicity. The sum and product formulas then hold for every polynomial, because the roots they describe always exist.

End-of-Chapter Problems

1. Consider $f(x) = x^3 - 2x^2 - 5x + 6$.

(a) Show that $x = 1$ is a root. [2]
(b) Factorise $f(x)$ completely. [3]
(c) Solve $f(x) = 0$. [1]

2. Consider $f(x) = 2x^3 + 3x^2 - 11x - 6$.

(a) Show that $(x - 2)$ is a factor. [2]
(b) Factorise $f(x)$ completely. [3]

3. When $x^3 + ax + b$ is divided by $(x - 1)$ the remainder is 4, and $(x + 1)$ is a factor. Find a and b. [5]

4. The polynomial $x^3 - 3x^2 + kx - 8$ has $(x - 2)$ as a factor. Find k. [3]

5. Sketch $y = (x + 1)(x - 1)(x - 3)$, showing all intercepts and the shape. [4]

6. A quadratic $x^2 + px + q$ has roots 2 and -5. Find p and q. [3]

7. Consider $f(x) = x^3 + 2x^2 - x - 2$.

(a) Factorise $f(x)$ completely. [3]
(b) Solve $f(x) = 0$. [1]

8. Solve $x^3 - 7x + 6 = 0$. [4]

9. The roots of $x^2 - 5x + 3 = 0$ are α and β. Find $\alpha^2 + \beta^2$. [4]

10. The polynomial $x^3 + ax^2 + bx - 2$ has $(x - 1)$ and $(x - 2)$ as factors. Find a and b. [5]

11. A cubic has roots 1, 2 and 3.

(a) Write it in factored and expanded form (leading coefficient 1). [3]
(b) State the sum and product of the roots and verify them from the expanded form. [2]

12. Consider $f(x) = x^4 - 5x^2 + 4$.

(a) Solve $f(x) = 0$. [3]

(b) State how many x-intercepts the graph has. [1]

13. Show that $x = 1$ is a root of $x^3 + 4x^2 + x - 6 = 0$, and find the other roots. [4]

14. The equation $x^2 - kx + 9 = 0$ has equal roots. Use the sum and product of roots to find k. [4]

15. The roots of $3x^2 - x - 2 = 0$ are α and β. Find $\dfrac{1}{\alpha} + \dfrac{1}{\beta}$. [3]

16. Consider $f(x) = 2x^3 - x^2 - 13x - 6$.

(a) Show that $(x + 2)$ is a factor. [2]

(b) Factorise $f(x)$ completely and solve $f(x) = 0$. [4]

17. The remainder when $x^3 + cx + d$ is divided by $(x - 1)$ is 6, and $(x - 2)$ is a factor. Find c and d. [5]

18. Solve $2x^3 + 3x^2 - 3x - 2 = 0$. [5]

19. The cubic $x^3 + px^2 + qx + r$ has roots -1, 2 and 4. Find p, q and r. [5]

20. The roots of $x^2 - 6x + 7 = 0$ are α and β. Find $\alpha^2 + \beta^2$ and $(\alpha - \beta)^2$. [5]

21. The polynomial $x^3 - kx + 2$ has $(x - 2)$ as a factor.

(a) Find k. [2]

(b) Find the other two roots, in exact form. [4]

22. Sketch $y = x^2(x - 3)$, describing the behaviour at each root. [4]

23. The quadratic with roots α and β is $x^2 - 4x + 1$. Find a quadratic (leading coefficient 1) whose roots are $\alpha + 1$ and $\beta + 1$. [5]

24. The polynomial $x^3 - 2x^2 + x - 2$ factorises as $(x - 2)(x^2 + 1)$.

(a) Verify this by expansion. [2]

(b) State the only real root, and explain why there are no others. [2]

25. When a polynomial $p(x)$ is divided by $(x - 1)$ the remainder is 3, and by $(x - 2)$ the remainder is 5. Find the remainder when $p(x)$ is divided by $(x - 1)(x - 2)$. [6]

26. The roots of $x^3 - 6x^2 + 11x - 6 = 0$ are α, β, γ. Find $\alpha^2 + \beta^2 + \gamma^2$ without solving the equation. [5]

27. Prove that if a polynomial with integer coefficients has a rational root $\frac{p}{q}$ in lowest terms, then p divides the constant term and q divides the leading coefficient. [6]

28. The cubic $x^3 + px + q = 0$ has a repeated root. Show that $4p^3 + 27q^2 = 0$. [6]

29. Find all real values of k for which $x^3 - 3x + k = 0$ has three distinct real roots. [6]

30. Consider $f(x) = 2x^3 - 3x^2 - 11x + 6$.

(a) Show that $(x - 3)$ is a factor of $f(x)$. [2]
(b) Factorise $f(x)$ completely. [4]
(c) Solve $f(x) = 0$. [2]
(d) State the y-intercept, and describe the behaviour of $f(x)$ as $x \to \pm\infty$. [3]
(e) Sketch $y = f(x)$, showing all intercepts. [2]
(f) Hence solve the inequality $f(x) \leq 0$. [3]

31. The polynomial $p(x) = x^3 + ax^2 + bx + 6$ leaves a remainder of 12 when divided by $(x - 2)$, and $(x + 1)$ is a factor.

(a) Show that $4a + 2b = -2$ and $a - b = -5$. [4]
(b) Hence find a and b. [2]
(c) Factorise $p(x)$ as far as possible over the real numbers. [4]
(d) State how many real roots $p(x) = 0$ has, giving a reason. [2]
(e) Find the remainder when $p(x)$ is divided by $(x - 1)(x + 1)$. *(Hint: the remainder has the form $rx + s$)* [4]

32. The cubic equation $x^3 + px^2 + qx + r = 0$ has roots α, β and γ.

(a) Write down $\alpha + \beta + \gamma$, $\alpha\beta + \beta\gamma + \gamma\alpha$ and $\alpha\beta\gamma$ in terms of p, q and r. [3]
(b) The equation $x^3 - 6x^2 + 3x + 10 = 0$ has roots α, β, γ. Find $\alpha^2 + \beta^2 + \gamma^2$ without solving the equation. [4]
(c) Find $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$. [3]
(d) Given that one root is -1, find the other two. [3]
(e) Write down a cubic equation with integer coefficients whose roots are 2α, 2β and 2γ. [3]

Challenge Questions

33. Power sums of the roots. The equation $x^2 - px + q = 0$ has roots α and β. For each integer $k \geq 0$ define

$$S_k = \alpha^k + \beta^k.$$

(a) Write down the values of S_0 and S_1. [2]
(b) Show that $S_2 = p^2 - 2q$. [2]
(c) Find S_3 in terms of p and q. [3]
(d) Explain why $\alpha^k = p\,\alpha^{k-1} - q\,\alpha^{k-2}$ for every $k \geq 2$, and deduce that $S_k = p\,S_{k-1} - q\,S_{k-2}$. [4]
(e) The equation $x^2 - 3x + 1 = 0$ has roots α and β. Use the recurrence to find S_2, S_3 and S_4. [4]
(f) The roots of $x^2 - 3x + 1 = 0$ are irrational. Explain why S_k is nevertheless an integer for every k. [3]

34. Palindromic polynomials. A polynomial is *palindromic* if its list of coefficients reads the same forwards as backwards. For example $x^4 + x^3 - 4x^2 + x + 1$ has coefficients $1, 1, -4, 1, 1$.

(a) Explain why $x = 0$ is never a root of a palindromic polynomial. [2]
(b) Let $P(x)$ be palindromic of degree n. Show that $x^n P\left(\frac{1}{x}\right) = P(x)$, and deduce that if α is a root of P then so is $\frac{1}{\alpha}$. [4]
(c) Divide $x^4 + x^3 - 4x^2 + x + 1 = 0$ through by x^2 and substitute $u = x + \frac{1}{x}$. Show that the equation becomes $u^2 + u - 6 = 0$. [4]
(d) Hence find all four roots of $x^4 + x^3 - 4x^2 + x + 1 = 0$ in exact form. [5]
(e) Show that every palindromic polynomial of odd degree has $x = -1$ as a root. [3]

GEOMETRY & TRIGONOMETRY

Chapter 8

Solids, Circles & Triangles

8

SYLLABUS 3.1 3.2 3.3 3.4

About a thousand years ago, a scholar climbed a hill in what is now Pakistan and measured the size of the Earth.

Al-Bīrūnī knew the older method. Eratosthenes had compared shadows in two Egyptian towns, but that required a team to pace out the long distance between them. Al-Bīrūnī wanted a measurement that one person could make from a single place.

He used a fact you can check yourself. From the top of a hill the horizon is not level with your eye. It lies a little below it. At the fort of Nandana he measured that small angle and found it was about 34 minutes of arc, a little over half a degree. He already knew the height of the hill.

That is enough, because your line of sight to the horizon only touches the Earth. Where it touches, it meets the radius at a right angle. So the hill, the horizon and the centre of the Earth form one right-angled triangle. If the hill has height h and the angle of dip is θ, then $\cos\theta = \dfrac{R}{R+h}$, and rearranging gives the radius:

$$R = \frac{h\cos\theta}{1-\cos\theta}.$$

His answer was close to the modern value of 6371 km.

The same idea reaches things you can never travel to. Look at the Moon and it covers about half a degree of your view. That half degree is an angle at your eye. The Moon is about $384\,000$ km away, and multiplying that distance by the angle in radians gives the Moon's width: about 3500 km. The measured value is 3475 km.

One angle and one distance, and you have the size of a world. That is what this chapter builds. The triangle gives the sine and cosine rules. The solid gives volume and surface area. The circle gives the radian, the unit that links an arc to a full turn.

8.1 Three-Dimensional Geometry

Everything you know about coordinates in the plane extends to space by adding a third axis, z. The distance formula is the clearest example. In two dimensions it is Pythagoras' theorem; in three, it is Pythagoras applied *twice*, once across the base and once up.

KEY · IB The distance between (x_1, y_1, z_1) and (x_2, y_2, z_2) is

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}.$$

The midpoint is the average of the coordinates, $\left(\dfrac{x_1 + x_2}{2}, \dfrac{y_1 + y_2}{2}, \dfrac{z_1 + z_2}{2}\right)$.

To see why, find the distance across the base first: $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$. Treat that as one leg of a new right triangle, whose other leg is the height $(z_2 - z_1)$. A second use of Pythagoras gives the formula. A sphere is then just the set of points a fixed distance r from a centre (a, b, c): $(x - a)^2 + (y - b)^2 + (z - c)^2 = r^2$.

The angle between a line and a plane

A line that is not parallel to a plane meets it at an angle, but which angle? The honest answer is the smallest one: the angle between the line and its own **shadow** on the plane, the projection you get by dropping perpendiculars from the line onto the surface.

DEF The **angle between a line and a plane** is the angle between the line and its perpendicular projection onto the plane. You find it by dropping a perpendicular from a point on the line to the plane, forming a right triangle, and using right-angled trigonometry.

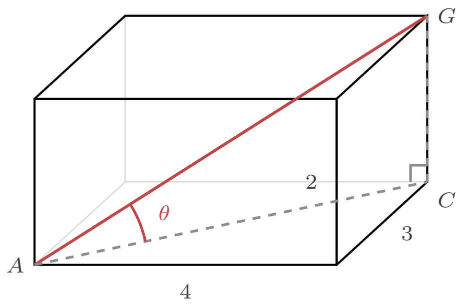


Figure 8.1 The space diagonal AG in scarlet, and its projection AC on the base, dashed. The vertical CG meets the base at a right angle, so ACG is a right triangle and θ is the angle between the diagonal and the base.

EXAMPLE 8.1

A cuboid has a 4×3 base and height 2. Find the angle between the space diagonal AG and the base.

The diagonal of the base is $AC = \sqrt{4^2 + 3^2} = 5$. This is the projection of AG onto the base, and $CG = 2$ is vertical, meeting the base at a right angle. In the right triangle ACG,

$$\tan\theta = \frac{CG}{AC} = \frac{2}{5} \implies \theta = \arctan(0.4) \approx 21.8^\circ.$$

The key move is always the same: find the projection on the plane, then the perpendicular height, then use tan.

The syllabus also asks for the angle between two intersecting *lines*, and in a solid the two lines are usually edges meeting at a vertex. Build the triangle they span, then solve it. The distance and midpoint formulas supply every length you need.

EXAMPLE 8.2

A square-based pyramid has base corners $A(0,0,0)$, $B(6,0,0)$, $C(6,6,0)$, $D(0,6,0)$ and apex $V(3,3,4)$. Find the angle between the slant edges VA and VB.

The two edges meet at V and span the triangle VAB, so the angle we want is $\angle AVB$. First measure the triangle. By the distance formula,

$$VA = \sqrt{(3-0)^2 + (3-0)^2 + (4-0)^2} = \sqrt{34},$$

and $VB = \sqrt{34}$ by the same calculation, while $AB = 6$ along the base.

The triangle is isosceles, so split it down the middle. The midpoint of AB is $M(3,0,0)$, and VM meets AB at right angles. In the right triangle VMA, the side $AM = 3$ is opposite the half-angle at V, and the hypotenuse is $VA = \sqrt{34}$:

$$\sin\frac{\theta}{2} = \frac{3}{\sqrt{34}} \implies \frac{\theta}{2} \approx 31.0^\circ \implies \theta \approx 61.9^\circ.$$

Everything came from the two formulas of this section: distances measured the triangle, the midpoint split it, and right-angled trigonometry finished it.

YOUR TURN 8.1 Three-Dimensional Geometry

1. Find the distance between each pair of points.

(a) $(1,2,3)$ and $(4,6,3)$ [2]

(b) $(0,0,0)$ and $(2,3,6)$ [2]

(c) $(-1,0,2)$ and $(2,-4,2)$ [2]

(d) $(2,-1,4)$ and $(5,3,4)$ [2]

2. Answer each part.

(a) Find the midpoint of $(1,2,3)$ and $(5,4,1)$. [1]

(b) Find the midpoint of $(2,-1,4)$ and $(6,3,0)$. [2]

3. Write the equation of each sphere.

(a) centre $(0,0,0)$, radius 5 [1]

(b) centre $(1,2,3)$, radius 4 [2]

(c) centre $(2,-1,3)$, radius 6 [2]

(d) centre $(0,0,2)$, radius $\sqrt{7}$ [2]

4. A cuboid has base 3×4 and height 12.

(a) Find the length of the diagonal of the base. [2]

(b) Find the length of the space diagonal. [2]

(c) Find the angle between the space diagonal and the base. [3]

8.2 Volumes and Surface Areas of Solids

Beyond the cuboids, cylinders, and prisms you already know, three curved solids appear again and again. Their formulas are given in the exam. Knowing where each piece comes from is what prevents you from misusing them.

KEY · IB

Sphere, radius r	$V = \frac{4}{3}\pi r^3$, $A = 4\pi r^2$
Cone, radius r, height h	$V = \frac{1}{3}\pi r^2 h$, curved $A = \pi r l$
Pyramid, base area B, height h	$V = \frac{1}{3}Bh$
Cylinder, radius r, height h	$V = \pi r^2 h$, curved $A = 2\pi r h$

Two cautions save most errors. The cone's curved surface uses the **slant height** l, not the vertical height h; the two are linked by Pythagoras, $l = \sqrt{r^2 + h^2}$. And a "total surface area" means every face. A solid hemisphere has a curved part $2\pi r^2$ *and* a flat circular base πr^2, so its total is $3\pi r^2$. The same hemisphere as an open bowl has only $2\pi r^2$.

EXAMPLE 8.3

A solid is a cone of base radius 3 and height 4 sitting on top of a hemisphere of radius 3. Find its total surface area.

The exposed surface is the cone's curved part plus the hemisphere's curved part; the two flat circles where they join are hidden inside. The cone's slant height is $l = \sqrt{3^2 + 4^2} = 5$, so its curved area is $\pi r l = 15\pi$. The hemisphere's curved area is $2\pi r^2 = 18\pi$. The total is

$$15\pi + 18\pi = 33\pi \approx 104.$$

Deciding which faces are exposed is the harder step. Once you know that, the formulas are routine.

YOUR TURN 8.2 Volumes and Surface Areas of Solids

5. Find the volume of each solid. Leave your answers in terms of π where appropriate.

(a) sphere of radius 3 [1]
(b) cone of radius 3, height 4 [2]
(c) pyramid with base area 20, height 9 [1]
(d) cylinder of radius 4, height 10 [2]

6. Find the surface area asked for in each case.

(a) the surface area of a sphere of radius 5 [1]
(b) the curved surface area of a cone of radius 5, slant height 13 [1]
(c) the total surface area of a solid hemisphere of radius 4 [3]
(d) the curved surface area of a cylinder of radius 4, height 10 [2]

7. A cone has radius 6 and height 8.

(a) Find its slant height. [1]
(b) Hence find its curved surface area. [2]

8. A solid is a cone of radius 3 and height 4 standing on a hemisphere of radius 3.

(a) Explain which surfaces are exposed. [2]
(b) Find the total surface area, in terms of π. [3]
(c) Find the total volume, in terms of π. [3]

8.3 The Sine and Cosine Rules

Right-angled triangles are solved with sin, cos, and tan: with sides labelled opposite, adjacent, and hypotenuse relative to an angle, $\sin\theta = \frac{o}{h}$, $\cos\theta = \frac{a}{h}$, $\tan\theta = \frac{o}{a}$. But most triangles have no right angle, and two rules extend trigonometry to all of them. Throughout, we label a side by the lower-case letter of the angle *opposite* it, so side a faces angle A.

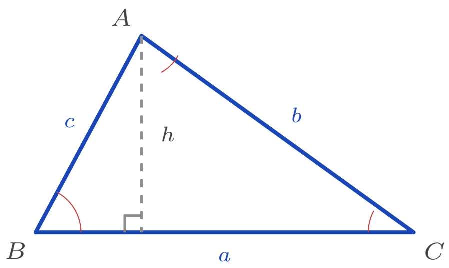


Figure 8.2 Each side is named after the angle opposite it: side a faces angle A, and so on. The dashed altitude h splits the triangle into two right triangles, which is where the sine rule comes from.

The sine rule

Drop the altitude h from A onto side a. It splits the triangle into two right triangles, and in each one h is the opposite side: $h = c \sin B$ and $h = b \sin C$. Setting these equal, $c \sin B = b \sin C$, which rearranges to $\frac{b}{\sin B} = \frac{c}{\sin C}$. Dropping a different altitude gives the third ratio.

KEY · IB **Sine rule.** In any triangle,

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}.$$

The sine rule pairs a side with the angle facing it, so use it when you know such a pair, plus one more side or angle (cases AAS, ASA, or SSA).

The ambiguous case

The sine rule needs one extra check when you use it to find an *angle*.

Suppose you know two sides and an angle that is *not* between them. This is the SSA case. The sine rule then gives an equation of the form $\sin B = k$, and that equation has two solutions between $0°$ and $180°$: the angle B, and its supplement $180° - B$. A sine and its supplement are equal, so the equation cannot tell them apart.

Your calculator returns only the first one. You must test the second yourself. Add it to the angle you were given: if the total is still below $180°$, a second triangle exists and both answers are correct.

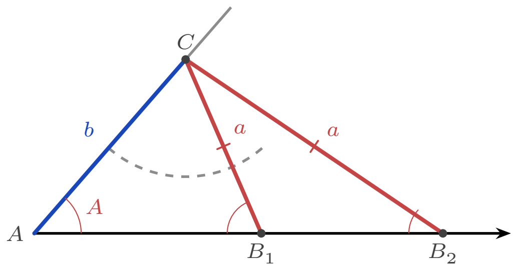


Figure 8.3 Two sides and a non-included angle. Swinging the fixed length a from C meets the base at B_1

and at B_2, so two different triangles have the same a, b and A. The angles at B_1 and B_2 add to $180°$, which is why the sine rule returns both.

EXAMPLE 8.4

In triangle ABC, $a = 8$, $b = 10$, and $A = 40°$. Find angle B.
By the sine rule, $\dfrac{\sin B}{10} = \dfrac{\sin 40°}{8}$, so

$$\sin B = \frac{10 \sin 40°}{8} \approx 0.803.$$

The calculator gives $B \approx 53.5°$. The supplement is $180° - 53.5° = 126.5°$, so test it: $A + B = 40° + 126.5° = 166.5°$, which is below $180°$. A third angle of $13.5°$ remains, so that triangle exists too.
Both values of B are correct. There are two triangles, and a complete answer reports both.

CAUTION This check is needed only when the sine rule gives an *angle*. If you are finding a *side*, or using the cosine rule, no supplement arises. The cosine rule returns an angle between $0°$ and $180°$ directly, because cos is negative for obtuse angles and positive for acute ones, so it distinguishes the two.

The cosine rule

When you know two sides and the angle *between* them, no side-and-opposite-angle pair is available, and the sine rule cannot start. The cosine rule is used instead. It is Pythagoras' theorem with a correction term for the angle not being $90°$. Coordinates prove it in three lines. Place the angle C at the origin with side a along the axis, so $B = (a, 0)$. Then A lies at $(b \cos C,\ b \sin C)$, by the definitions of cosine and sine. The distance formula measures the third side:

$$c^2 = (a - b\cos C)^2 + (b \sin C)^2 = a^2 - 2ab\cos C + b^2\cos^2 C + b^2 \sin^2 C,$$

and since $\cos^2 C + \sin^2 C = 1$, the last two terms collapse to b^2.

KEY · IB **Cosine rule.** In any triangle,

$$c^2 = a^2 + b^2 - 2ab\cos C, \qquad \text{equivalently} \qquad \cos C = \frac{a^2 + b^2 - c^2}{2ab}.$$

When $C = 90°$, $\cos C = 0$ and the rule reduces to $c^2 = a^2 + b^2$, which is Pythagoras' theorem. The first form finds the third side from two sides and the included angle (SAS); the rearranged form finds an angle from all three sides (SSS).

The area of a triangle

The familiar area $\frac{1}{2}$ base $\times$ height becomes a trigonometric formula the moment you write the height in terms of a side. With base b and height $h = a \sin C$ (from the right triangle at C),

$$\text{Area} = \tfrac{1}{2}\, b\, (a \sin C).$$

KEY · IB **Area of a triangle.** Area $= \dfrac{1}{2} ab \sin C$, using two sides and the angle between them.

YOUR TURN 8.3 The Sine and Cosine Rules

9. Use the sine rule to find the required side or angle.

(a) $a = 7$, $A = 40^\circ$, $B = 60^\circ$: find b [2]

(b) $a = 9$, $A = 35^\circ$, $B = 55^\circ$: find b [2]

(c) $A = 45^\circ$, $B = 60^\circ$, $a = 10$: find C and b [3]

(d) $a = 10$, $b = 12$, $A = 35^\circ$: find the acute angle B [3]

10. Use the cosine rule.

(a) $b = 5$, $c = 8$, $A = 60^\circ$: find a [2]

(b) $a = 6$, $b = 9$, $C = 110^\circ$: find c [3]

(c) $a = 6$, $b = 7$, $c = 8$: find angle A [3]

(d) sides 5, 6, 7: find the largest angle [3]

11. Find the area of each triangle.

(a) sides 5 and 8, included angle 30° [2]

(b) $a = 9$, $b = 6$, $C = 90^\circ$ [1]

(c) sides 7 and 9, included angle 40° [2]

12. In triangle ABC, $a = 7$, $b = 9$ and $A = 45^\circ$.

(a) Find the two possible values of angle B. [3]

(b) Show that both give a valid triangle. [3]

(c) Explain why this situation cannot arise when the given angle is included between the two given sides. [2]

8.4 Applications: Elevation, Depression, and Bearings

These rules are used most often in problems about the physical world, where the information is given as angles and directions rather than as a labelled triangle. Two conventions cover almost every case.

An **angle of elevation** is measured upward from the horizontal to a line of sight; an **angle of depression** is measured downward. The two horizontals are parallel, so the angle of depression from a cliff-top down to a boat equals the angle of elevation from the boat up to the cliff-top. They are alternate angles.

A **bearing** gives a direction as an angle measured clockwise from north. It runs from $000°$ to $360°$, and it is always written with three figures, so due east is $090°$ and not $90°$. The three figures make the reading unambiguous on a chart. Due south is $180°$, due west is $270°$, and north-west is $315°$.

The **back bearing** is the direction of the return journey. It differs from the outward bearing by $180°$. Add $180°$ if the bearing is less than $180°$, and subtract $180°$ if it is $180°$ or more, so that the answer stays between $000°$ and $360°$.

EXAMPLE 8.5

A ship sails 60 km from port P on a bearing of $050°$ to a point Q, then 80 km on a bearing of $140°$ to R. Find the distance PR.

The two bearings differ by $140° - 50° = 90°$, so the ship turns through a right angle at Q: the angle PQR is $90°$. Then PR is a hypotenuse, and Pythagoras gives

$$PR = \sqrt{60^2 + 80^2} = \sqrt{10000} = 100 \text{ km}.$$

Sketching the bearings from north lines at P and Q is what reveals the right angle; the arithmetic follows from the picture.

Most journeys do not turn so conveniently. When the angle at the corner is not $90°$, the same sketch supplies the data for the cosine rule, and the sine rule then gives the direction back. Nearly every examination bearings question has this shape, so it is worth fixing the order of steps.

KEY **Solving a bearings problem.**

1. Sketch the journey, and draw a north line at **every** point where a bearing is given.
2. Mark each bearing as an angle clockwise from its own north line.
3. Find the interior angle of the triangle at the corner. Take the back bearing of the first leg, then subtract the second bearing.
4. Use the cosine rule for the missing distance, then the sine rule for the missing angle.
5. Turn that angle back into a bearing by adding it to the bearing of the first leg.

Step 1 is the one students leave out, and without it step 3 cannot be done.

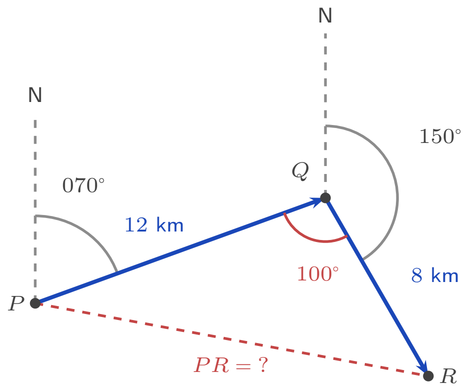


Figure 8.4 The two bearings are measured clockwise from a north line at P and at Q. Together they fix the interior angle $\angle PQR = 100^\circ$, which is what the cosine rule needs.

EXAMPLE 8.6

A hiker walks 12 km from P on a bearing of 070° to Q, then 8 km on a bearing of 150° to R. Find the distance PR and the bearing of R from P.

First construct the angle at the corner. The bearing back from Q to P is $070^\circ + 180^\circ = 250^\circ$, and the onward bearing is 150°, so the interior angle is $\angle PQR = 250^\circ - 150^\circ = 100^\circ$. Subtracting the back bearing is the standard method here, and the sketch confirms the result: the angle at the corner is clearly obtuse.

No right angle, so the cosine rule finds the closing side:

$$PR^2 = 12^2 + 8^2 - 2(12)(8)\cos 100^\circ \approx 241.3 \implies PR \approx 15.5 \text{ km}.$$

For the bearing of R from P, find the angle $\angle QPR$ by the sine rule:

$$\frac{\sin \angle QPR}{8} = \frac{\sin 100^\circ}{15.54} \implies \sin \angle QPR \approx 0.507 \implies \angle QPR \approx 30.5^\circ.$$

The bearing of R from P is therefore $070^\circ + 30.5^\circ \approx 100.5^\circ$, reported as 100° to the nearest degree. The labelled diagram carries the whole solution. The bearings give the corner angle, the cosine rule gives the closing side, and the sine rule gives the final direction.

YOUR TURN 8.4 Applications: Elevation, Depression, and Bearings

13. Find the angle of elevation in each case.

(a) a 50 m tower, from 120 m away on level ground [2]

(b) a 45 m tower, from 90 m away on level ground [2]

14. In each case, find the horizontal distance from the base of the tall object to the object below.

(a) From the top of an 80 m cliff, the angle of depression to a boat is $25°$. [2]

(b) From the top of a 30 m building, the angle of depression to a car is $40°$. [2]

(c) From the top of a 60 m cliff, the angle of depression to a buoy is $20°$. [3]

15. Answer each question about bearings.

(a) Write the bearing of due west as a three-figure bearing. [1]

(b) A ship travels on a bearing of $120°$. Find its back bearing. [1]

(c) B is on a bearing of $065°$ from A. Find the bearing of A from B. [2]

(d) Find the back bearing of $200°$, and explain why you subtract here rather than add. [2]

16. A ladder 6 m long rests against a vertical wall at $70°$ to the ground.

(a) Find the height it reaches on the wall. [2]

(b) Find the distance from the foot of the ladder to the wall. [2]

17. GDC A ship sails 40 km from P on a bearing of $030°$ to Q, then 50 km on a bearing of $120°$ to R.

(a) Show that angle $PQR = 90°$. [2]

(b) Find the distance PR. [2]

(c) Find the bearing of R from P. [3]

8.5 Radians, Arc Length, and Sector Area

The degree is a human invention: the Babylonians chose to cut the circle into 360 parts, and we inherited it. The circle itself suggests a more natural unit. Wrap the radius around the circumference; the angle it subtends at the centre is one **radian**. Since the circumference is $2\pi r$, exactly 2π radii fit around, so a full turn is 2π radians.

KEY 2π radians $= 360^\circ$, so π radians $= 180^\circ$. Convert by multiplying: degrees to radians $\times \dfrac{\pi}{180}$; radians to degrees $\times \dfrac{180}{\pi}$.

The advantage appears immediately, because arc length and sector area become simple fractions of the whole circle. An arc is the same fraction of the circumference as its angle θ is of a full turn: $\dfrac{l}{2\pi r} = \dfrac{\theta}{2\pi}$, and the 2π's cancel.

KEY · IB For a sector of radius r and angle θ **in radians**,

$$\text{arc length } l = r\theta, \qquad \text{sector area } A = \tfrac{1}{2}r^2\theta.$$

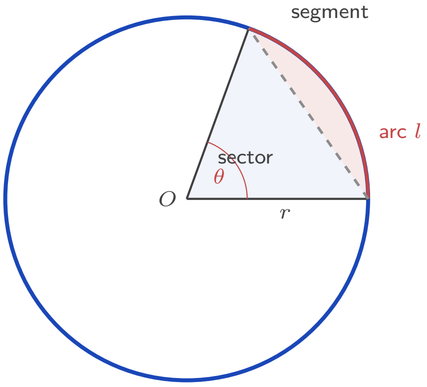


Figure 8.5 The shaded pie slice is the *sector*, bounded by two radii and the arc l. The smaller region cut off by the chord is the *segment*.

The area follows the same way: the sector is the fraction $\dfrac{\theta}{2\pi}$ of the whole disc πr^2, giving $A = \dfrac{\theta}{2\pi} \cdot \pi r^2 = \tfrac{1}{2}r^2\theta$. A **segment**, the region between a chord and its arc, is then a sector with the triangle cut away:

$$\text{segment} = \tfrac{1}{2}r^2\theta - \tfrac{1}{2}r^2 \sin\theta = \tfrac{1}{2}r^2(\theta - \sin\theta).$$

EXAMPLE 8.7

A sector has radius 10 cm and angle $\dfrac{2\pi}{5}$ radians. Find its arc length and area.

Directly from the formulas,

$$l = r\theta = 10 \cdot \frac{2\pi}{5} = 4\pi \approx 12.6 \text{ cm}, \qquad A = \tfrac{1}{2}r^2\theta = \tfrac{1}{2}(100)\frac{2\pi}{5} = 20\pi \approx 62.8 \text{ cm}^2.$$

For the segment cut off by the chord, subtract the triangle from the sector:

$$\tfrac{1}{2}r^2(\theta - \sin\theta) = 50\left(\frac{2\pi}{5} - \sin\frac{2\pi}{5}\right) \approx 50(1.2566 - 0.9511) \approx 15.3 \text{ cm}^2.$$

All three formulas require θ in radians. Using degrees here is a common mistake, and it makes every answer wrong.

EXAMPLE 8.8

The opener mentioned the older method. Here it is in full. Eratosthenes measured the sun's rays at 7.2° to the vertical in Alexandria, while they were vertical in Syene, and took the arc between the towns to be 5000 stadia. Find the radius and circumference of the Earth in stadia.

Convert the angle first:

$$7.2^\circ \times \frac{\pi}{180} = \frac{\pi}{25} \text{ radians.}$$

The two towns and the Earth's centre make a sector whose arc is the distance between them, so $l = r\theta$ can be used in reverse:

$$5000 = r \cdot \frac{\pi}{25} \implies r = \frac{125\,000}{\pi} \approx 39\,800 \text{ stadia,}$$

and the circumference is $2\pi r = 250\,000$ stadia, which is just the one-fiftieth argument in formula form. Taking a stadion as roughly 158 m gives a circumference near $39\,500$ km, within a couple of percent of the true $40\,075$ km. One angle, one arc, one formula, and the planet is measured.

YOUR TURN 8.5 Radians, Arc Length, and Sector Area

18. Convert to radians, in terms of π.

(a) 60° [1]
(b) 135° [1]
(c) 210° [2]
(d) 90° [1]

19. Convert to degrees.

(a) $\dfrac{3\pi}{4}$ [1]
(b) $\dfrac{2\pi}{3}$ [1]
(c) 1 radian, to three significant figures [2]

20. Find the arc length and the area of each sector.

(a) radius 8, angle 1.5 radians [2]
(b) radius 6, angle $\dfrac{\pi}{3}$ radians [2]
(c) radius 9, angle 2.4 radians [3]
(d) radius 4, angle 90° [3]

21. In each case the angle is unknown. Find it from the information given.

(a) A sector has radius 10 and arc length 25. Find its angle in radians. [1]

(b) A sector has radius 6 and area 54. Find its angle in radians, then find its arc length. [3]

22. A chord subtends an angle of 2 radians at the centre of a circle of radius 8.

(a) Find the area of the sector. [2]

(b) Find the area of the triangle formed by the chord and the two radii. [2]

(c) Find the area of the minor segment. [2]

(d) Find the perimeter of the sector. [2]

Chapter 8 · Solids, Circles & Triangles

Reflections

IM Trigonometry was built by many people across many centuries. The Greek astronomer Hipparchus, around 150 BCE, made the first table of chords. Ptolemy extended it in the *Almagest*. In India, Aryabhata and those after him replaced the chord with the half-chord, which they called the *jyā*. That is our sine. The word travelled into Arabic as *jiba*, was misread as *jaib*, meaning "bay", and was then translated into Latin as *sinus*. Islamic astronomers such as al-Bīrūnī developed the sine and cosine rules further, for their astronomy and for finding the direction of Mecca. The word "sine" still carries that chain of translations inside it.

TOK The formulas $l = r\theta$ and $A = \frac{1}{2}r^2\theta$ are only this simple when θ is in radians. Later you will meet another example: the derivative of $\sin x$ is exactly $\cos x$, again only in radians. The degree is a human choice. The Babylonians picked 360 because it divides so many ways. The radian is not a choice: it is the angle for which the arc equals the radius, and the circle itself fixes it. Is the radian discovered, then, where the degree is only invented?

RW Fixing an unknown position from measured angles is called **triangulation**, and it is still how we map the world. Surveyors once carried chains of triangles across whole countries. GPS finds your position from measured distances to satellites. Astronomers find the distance to nearby stars by *parallax*: as the Earth moves around the Sun, a near star seems to shift slightly against the far ones, and that tiny angle gives the distance. The unit it produces is called the parsec. Each of these is a triangle solved for a side nobody can reach.

EXT The ambiguous case is really a statement about congruence. Give a triangle by two angles and a side (ASA or AAS), or by three sides (SSS), or by two sides and the angle between them (SAS), and only one triangle fits. These are the congruence rules. But two sides and an angle that is *not* between them (SSA) is not a congruence rule. It can leave the triangle undecided, and the sine rule's two answers are that gap showing through. The missing congruence rule and the doubled answer are the same fact.

End-of-Chapter Problems — Chapter 8: Solids, Circles & Triangles

1. A cuboid has dimensions $6 \times 8 \times 10$.

(a) Find the length of the space diagonal. [2]

(b) Find the angle between the space diagonal and the base. [3]

2. Consider the points $A(2, 0, 0)$, $B(0, 4, 0)$ and $C(0, 0, 4)$.

(a) Find the lengths AB, AC and BC. [3]

(b) Find angle BAC. [3]

3. A cone has radius 5 and height 12.

(a) Find the slant height. [1]

(b) Find the volume. [2]

(c) Find the total surface area. [2]

4. A solid consists of a cylinder of radius 3 and height 10 with a hemisphere of radius 3 on top. Find its total surface area. [5]

5. In triangle ABC, $a = 8$, $b = 11$, $c = 15$.

(a) Find the largest angle. [3]

(b) Find the area of the triangle. [2]

6. In triangle PQR, $p = 10$, $q = 7$ and $R = 95^\circ$.

(a) Find r. [2]

(b) Find the area of the triangle. [2]

7. A triangle has two sides of length 9 and 12 with an included angle of 40°.

(a) Find the length of the third side. [2]

(b) Find the area of the triangle. [2]

8. From port P, a ship sails 40 km on a bearing of 030° to Q, then 50 km on a bearing of 120° to R.

(a) Show that angle $PQR = 90^\circ$. [2]

(b) Find the distance PR. [2]

(c) Find the bearing of R from P. [3]

9. A sector has radius 12 cm and angle 1.2 radians.

(a) Find the arc length. [1]
(b) Find the area of the sector. [2]
(c) Find the perimeter of the sector. [2]

10. A sector has area 30 cm^2 and radius 6 cm.

(a) Find the angle of the sector in radians. [2]
(b) Find the arc length. [2]

11. A chord subtends an angle of 1.5 radians at the centre of a circle of radius 10 cm.

(a) Find the area of the sector. [2]
(b) Find the area of the triangle formed by the chord and the two radii. [2]
(c) Find the area of the minor segment. [2]

12. In triangle ABC, $A = 50^\circ$, $a = 6$ and $b = 7$.

(a) Find the two possible values of angle B. [3]
(b) Find the two possible areas of the triangle. [3]

13. A regular hexagon is inscribed in a circle of radius 8. Find its area. [4]

14. A cone has volume 100π and radius 5.

(a) Find the height. [2]
(b) Find the slant height. [2]
(c) Find the curved surface area. [1]

15. From a point on level ground, the angle of elevation of the top of a tower is 30°. Moving 40 m closer, the angle of elevation becomes 45°. Find the height of the tower. [5]

16. A square-based pyramid has base side 10 and height 12.

(a) Find the volume. [2]
(b) Find the slant height of a triangular face. [2]
(c) Find the total surface area. [2]

17. The area of a triangle is 24, with $a = 8$ and $b = 10$. Find the two possible values of the included angle C. [3]

18. Two ships leave a port at the same time. One sails on a bearing of $040°$ at 12 km/h, the other on a bearing of $100°$ at 16 km/h. Find the distance between them after 2 hours. [5]

19. Points $A(0,0,0)$ and $B(4,4,7)$ are given.

(a) Find $|AB|$. [2]
(b) Find the midpoint of AB. [1]
(c) Show that B lies on the sphere $x^2+y^2+z^2=81$. [1]

20. A pendulum of length 0.8 m swings through an angle of 0.5 radians. Find the length of the arc traced by the bob. [2]

21. Convert to radians, in terms of π: (a) $30°$; (b) $225°$; (c) $300°$. [3]

22. Two observers stand 200 m apart on level ground, with a balloon in the vertical plane between them. From one the angle of elevation to the balloon is $40°$; from the other it is $55°$. Find the height of the balloon. [6]

23. In triangle ABC, $AB=12$, $BC=9$ and angle $BAC=35°$. Find the two possible values of angle BCA. [4]

24. Convert to degrees: (a) $\dfrac{5\pi}{6}$; (b) $\dfrac{7\pi}{4}$; and find (c) the arc length of a sector of radius 9 and angle $\dfrac{5\pi}{6}$. [4]

25. A sector of a circle of radius 12 and angle 1.5 radians is rolled into a cone, so that the radius becomes the slant height and the arc becomes the base circumference. Find the radius and height of the cone. [6]

26. A sector of a circle has perimeter 20 and area 16. Find the radius r and angle θ. [6]

27. A chord of a circle of radius 10 has length 12. Find the area of the minor segment it cuts off. [6]

28. The area of a triangle is 84, and two of its sides have lengths 13 and 14. Find both possible lengths of the third side. [6]

29. GDC A yacht leaves harbour H and sails 25 km on a bearing of $048°$ to a buoy A. It then sails 18 km on a bearing of $130°$ to a second buoy B.

(a) Show that angle $HAB=98°$. [3]
(b) Find the distance HB. [4]
(c) Find the bearing of B from H. [4]

(d) Find the area of triangle HAB. [3]

(e) The yacht returns directly from B to H. Find the total distance it has sailed. [2]

30. GDC A rectangular box $ABCDEFGH$ has a horizontal base $ABCD$ with $AB = 8$ cm and $BC = 6$ cm, and vertical edges of length 5 cm, so that G is the vertex above C.

(a) Find the length of the base diagonal AC. [2]

(b) Find the exact length of the space diagonal AG. [3]

(c) Find the angle that AG makes with the base $ABCD$. [3]

(d) Find the angle $G\hat{A}B$ between the diagonal AG and the edge AB. [4]

(e) Find the total surface area of the box. [3]

31. GDC A container is a cone of base radius 6 cm and height 15 cm, held with its point downwards.

(a) Find the volume of the cone when full, in terms of π. [2]

(b) Water is poured in to a depth of 10 cm. Using similar triangles, find the radius of the circular water surface. [3]

(c) Find the volume of water, in terms of π. [3]

(d) Find the fraction of the cone's volume that is filled, and explain why it is not $\frac{2}{3}$ even though the water reaches $\frac{2}{3}$ of the height. [4]

(e) Find the depth of water when the cone is exactly half full, giving your answer to three significant figures. [3]

Challenge Questions

32. **How many triangles fit the data?** In triangle ABC you are given the side b, the angle A, and the side a opposite A. This is the SSA case. This investigation finds exactly how many triangles fit, for every possible value of a.

Throughout, take A to be acute, and let h be the distance from C to the line AB.

(a) Show that $h = b \sin A$. [2]

(b) Take $b = 10$ and $A = 40^\circ$. Find the number of triangles when $a = 5$, when $a = 8$, and when $a = 12$. [4]

(c) Explain why no triangle exists when $a < b \sin A$. *(Hint: what is the shortest distance from C to the line AB?)* [3]

(d) Explain why exactly one triangle exists when $a = b \sin A$, and state what is special about it. [3]

(e) Show that two triangles exist exactly when $b \sin A < a < b$, and explain what goes wrong at $a = b$. [4]

(f) Now let A be obtuse. State how many triangles exist for $a \le b$ and for $a > b$, and justify each answer. [4]

33. The best cone from a sector. A sector of radius R and angle θ radians is cut from a sheet of paper. The two straight edges are joined to make a cone. This investigation finds the angle that gives the largest cone.

Write $t = \dfrac{\theta}{2\pi}$, so that $0 < t < 1$.

(a) The arc of the sector becomes the circumference of the base. Show that the base radius is $r = Rt$. [3]

(b) Show that the height of the cone is $h = R\sqrt{1-t^2}$. [3]

(c) Hence show that the volume is $V = \frac{1}{3}\pi R^3 t^2 \sqrt{1-t^2}$. [3]

(d) Taking $R = 1$, use technology to find the value of t that makes V largest, and hence find θ to the nearest degree. [4]

(e) Show that at that maximum $\dfrac{r}{R} = \sqrt{\dfrac{2}{3}}$, and find the exact value of $\dfrac{h}{r}$. [5]

34. How reliable was al-Bīrūnī's measurement? The opener described his method. From a hill of height h he measured the angle of dip θ to the horizon, and the radius of the Earth follows from one right-angled triangle. This investigation asks how much trust the answer deserves.

(a) The line of sight touches the Earth, so it meets the radius there at a right angle. Show that $\cos\theta = \dfrac{R}{R+h}$, and hence that $R = \dfrac{h\cos\theta}{1-\cos\theta}$. [3]

(b) Al-Bīrūnī recorded $h = 652$ cubits and $\theta = 34'$. Find R in cubits. *(Hint: $34' = \frac{34}{60}$ of a degree)* [3]

(c) Repeat the calculation for $\theta = 33'$ and for $\theta = 35'$. Find the percentage change in R in each case. [4]

(d) For small θ the approximation $\cos\theta \approx 1 - \frac{1}{2}\theta^2$ holds, with θ in radians. Use it to show that $R \approx \dfrac{2h}{\theta^2}$. [3]

(e) Hence explain why an error of about 3% in θ produces an error of about 6% in R. Comment on what this means for how confidently his result can be quoted. [4]

CHAPTER 9

Trigonometric Functions

9

SYLLABUS 3.5–3.11

Your voice is a wave. So is the light from your screen, the signal carrying this page to your device, the current in the wall socket, the tide in Tokyo Bay, and the beating of your own heart.

They do not look alike. A heartbeat is a sharp spike, a spoken vowel is jagged, a tide is a slow rise and fall over twelve hours. In 1822 Joseph Fourier made a startling claim: a repeating signal, however jagged, can be built up from pure sine waves. Choose enough of them, with the right heights and the right timings, and the shape appears. Your phone uses this idea many times a second to compress your voice.

So the same curve describes all of them. Where does it come from?

Watch one seat on a turning Ferris wheel. As the wheel turns steadily, the seat rises, slows near the top, falls, slows near the bottom, and rises again. Plot its height against time and the sine curve appears. The seat is a point moving round a circle, and its height is the vertical coordinate of that point. The circle and the wave are the same motion seen two ways: one going round, one going along.

That is why this chapter starts with a circle. In Ch. 8 sine and cosine were ratios inside a right-angled triangle, so the angle had to be acute. On the circle they become functions of *any* angle, and they repeat forever.

By the end of this chapter you will be able to read exact values of sine, cosine and tangent from the unit circle. You will use identities to rewrite one expression as another, sketch and transform the graphs of these functions, and work with the reciprocal and inverse functions. You will also solve trigonometric equations, where the answers come in families: infinitely many solutions described by a single formula.

9.1 The Unit Circle

You already have a definition. In a right triangle,

$$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}}, \qquad \cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}}, \qquad \tan\theta = \frac{\text{opposite}}{\text{adjacent}}.$$

This works, but only for acute angles. The three angles of a triangle add to 180°, and one of them is already 90°, so θ must be less than 90°. There is no right triangle containing an angle of 120°, or 400°, or -50°. Under this definition $\sin 120^\circ$ has no meaning at all.

Yet a rotation of 120° is perfectly ordinary. We need a wider definition, one that still agrees with the old one wherever the old one works.

The unit circle gives it. Take the circle of radius 1 centred at the origin, and a point P on it at angle θ, measured anticlockwise from the positive x-axis.

For an acute θ, drop a perpendicular from P to the x-axis. This makes a right triangle whose hypotenuse is the radius, so the hypotenuse is 1. The old ratios then read

$$\sin\theta = \frac{\text{opposite}}{1} = \text{opposite}, \qquad \cos\theta = \frac{\text{adjacent}}{1} = \text{adjacent},$$

and those two sides are exactly the coordinates of P. So for acute angles the new definition agrees with the old one.

The new form never mentions a triangle. A point can sit anywhere on the circle, at any angle, and it always has coordinates. That is what extends the definition to every real θ.

DEF On the unit circle, for a point P at angle θ:

$$\cos\theta = x\text{-coordinate of } P, \qquad \sin\theta = y\text{-coordinate of } P.$$

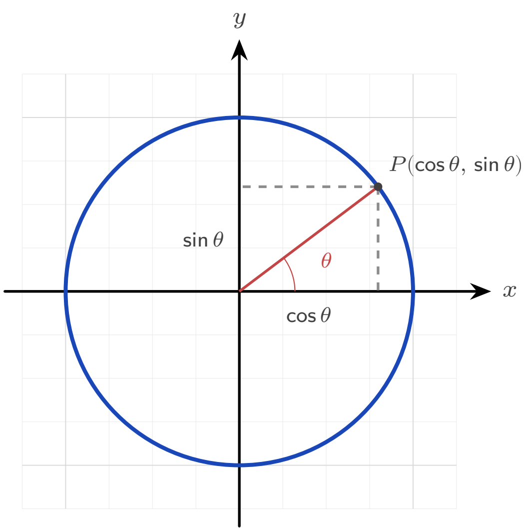


Figure 9.1 On a circle of radius 1 the two shorter sides of the right triangle *are* the coordinates of P, so $\cos\theta$ is read across and $\sin\theta$ up.

The tangent is the gradient of the line OP, so $\tan\theta = \dfrac{\text{rise}}{\text{run}} = \dfrac{\sin\theta}{\cos\theta}$. It is undefined wherever $\cos\theta = 0$.

KEY · IB $\tan\theta = \dfrac{\sin\theta}{\cos\theta}$.

Exact values

A few angles give exact coordinates, and they come from the 30–60–90 and 45–45–90 triangles. These are worth learning by heart.

θ	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
$\sin\theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos\theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\tan\theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	—

The first-quadrant values repeat into the other three quadrants with signs changed. Collecting them all onto one circle gives the diagram to learn.

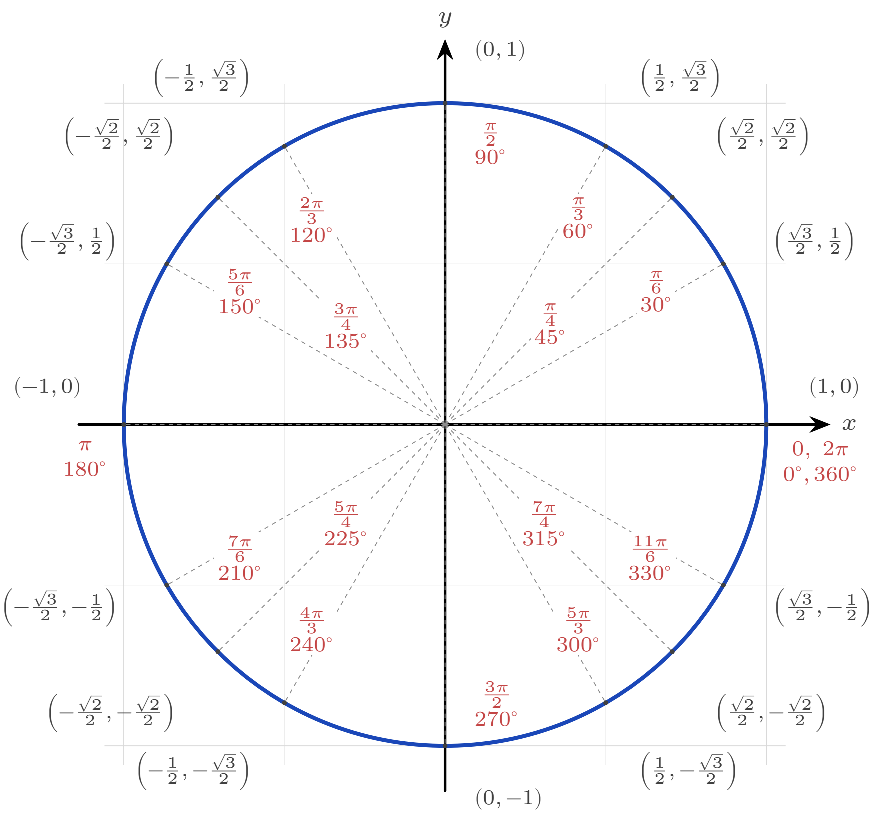


Figure 9.2 The unit circle with every special angle. The scarlet label inside the circle is the angle in radians; the dark label outside is the point $(\cos\theta,\ \sin\theta)$. To read a value, find the angle and take the matching coordinate.

Notice how little there is to learn. Only three pairs of numbers appear, $\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$, $\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$ and $\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$, together with the four points on the axes. Everything else is one of those with a sign changed.

Symmetry and the sign of each function

Since $\cos\theta$ and $\sin\theta$ are the coordinates of P, their signs are just the signs of x and y in each quadrant. Nothing needs to be memorised separately: in the second quadrant x is negative and y is positive, so $\cos\theta < 0$ and $\sin\theta > 0$, and $\tan\theta = \frac{\sin}{\cos}$ is negative.

Quadrant	Angles	(x, y)	Positive there	Mnemonic
First	0 to $\frac{\pi}{2}$	$(+,+)$	all three	**A**ll
Second	$\frac{\pi}{2}$ to π	$(-,+)$	sin only	**S**tudents
Third	π to $\frac{3\pi}{2}$	$(-,-)$	tan only	**T**ake
Fourth	$\frac{3\pi}{2}$ to 2π	$(+,-)$	cos only	**C**alculus

Reading the last column anticlockwise from the first quadrant gives **All Students Take Calculus**, which is the usual way to remember the pattern. Use it as a check, not as the reason: the reason is always the sign of the coordinates.

Each symmetry of the circle now gives an identity, and each one is a statement about where a reflected point lands.

Reflect in the y-axis. The point at angle $\pi - \theta$ is the mirror image, across the y-axis, of the point at angle θ. Reflecting in the y-axis negates the x-coordinate and leaves y alone, so the two points are $(-\cos\theta,\ \sin\theta)$ and $(\cos\theta,\ \sin\theta)$. Comparing coordinates:

$$\sin(\pi - \theta) = \sin\theta, \qquad \cos(\pi - \theta) = -\cos\theta.$$

Reflect in the x-axis. Measuring $-\theta$ means turning clockwise instead of anticlockwise, which lands on the mirror image of P across the x-axis. That reflection negates y and leaves x alone, so

$$\cos(-\theta) = \cos\theta, \qquad \sin(-\theta) = -\sin\theta.$$

Turn a full circle. Adding 2π to the angle sends P once round and back to exactly the same point, so both coordinates are unchanged:

$$\sin(\theta + 2\pi) = \sin\theta, \qquad \cos(\theta + 2\pi) = \cos\theta.$$

This is why both functions are **periodic** with period 2π. Every identity of this kind says the same thing: move the point, see which coordinate changes sign.

You have met the first of these relations before. Two angles, θ and $\pi - \theta$, share the same sine. That shared value is what produced the ambiguous case of the sine rule in Ch. 8. A triangle built from a sine value can sometimes close in two ways, one acute and one obtuse.

YOUR TURN 9.1 The Unit Circle

1. Write down the exact value of each.

(a) $\sin\dfrac{\pi}{6}$ [1]　(b) $\cos\dfrac{\pi}{4}$ [1]

(c) $\tan\dfrac{\pi}{3}$ [1]　(d) $\tan\dfrac{\pi}{4}$ [1]

2. Use the symmetry of the circle to find each exact value.

(a) $\sin\dfrac{5\pi}{6}$ [2]　(b) $\cos\dfrac{2\pi}{3}$ [2]

(c) $\sin\dfrac{4\pi}{3}$ [2]　(d) $\cos\dfrac{7\pi}{4}$ [2]

3. State the quadrant or quadrants in which each condition holds.

(a) $\sin\theta > 0$ [1]　(b) $\cos\theta < 0$ [1]

(c) $\tan\theta > 0$ [2]　(d) $\sin\theta < 0$ and $\cos\theta > 0$ [2]

4. The point P on the unit circle has angle θ.

(a) State the coordinates of P in terms of θ. [1]

(b) Explain why $\cos^2\theta + \sin^2\theta = 1$ for every θ. [2]

(c) Explain why $\tan\theta$ is undefined at $\theta = \dfrac{\pi}{2}$. [2]

9.2 Trigonometric Identities

An **identity** is an equation true for every angle. The most fundamental one is Pythagoras' theorem read off the unit circle: the point $(\cos\theta, \sin\theta)$ is distance 1 from the origin, so $\cos^2\theta + \sin^2\theta = 1$.

KEY · IB $\cos^2\theta + \sin^2\theta = 1$.

Double and compound angles

The functions of a sum of two angles can be expanded, and from the sum formulas everything else follows. These **compound-angle** identities are usually just quoted, but the proof is short and uses only the unit circle and the cosine rule from Ch. 8.

Put two points on the unit circle, P at angle A and Q at angle B. Their coordinates are $P(\cos A,\ \sin A)$ and $Q(\cos B,\ \sin B)$, and the angle $\angle POQ$ at the centre is $A - B$.

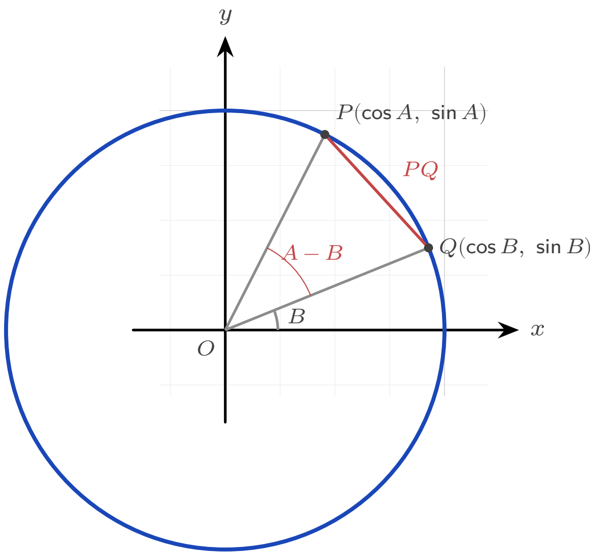


Figure 9.3 Two points on the unit circle, at angles A and B. The angle between the radii is $A - B$, and the chord PQ can be measured either by the distance formula or by the cosine rule.

Now find the distance PQ in two different ways.

By the distance formula:

$$PQ^2 = (\cos A - \cos B)^2 + (\sin A - \sin B)^2.$$

Expanding gives $\cos^2 A - 2\cos A\cos B + \cos^2 B + \sin^2 A - 2\sin A\sin B + \sin^2 B$. Both Pythagorean pairs collapse to 1, so

$$PQ^2 = 2 - 2(\cos A\cos B + \sin A\sin B).$$

By the cosine rule in triangle OPQ, where both sides from the centre have length 1 and the angle between them is $A - B$:

$$PQ^2 = 1^2 + 1^2 - 2(1)(1)\cos(A - B) = 2 - 2\cos(A - B).$$

The two expressions are the same length, so

$$\cos(A - B) = \cos A \cos B + \sin A \sin B.$$

Every other compound-angle formula follows from this one. Replacing B by $-B$, and using $\cos(-B) = \cos B$ with $\sin(-B) = -\sin B$, gives the formula for $\cos(A + B)$. The sine versions come from $\sin\theta = \cos(\frac{\pi}{2} - \theta)$, and the tangent versions from dividing the sine formula by the cosine formula.

KEY · IB HL **Compound angles:**

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B, \qquad \cos(A \pm B) = \cos A \cos B \mp \sin A \sin B,$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}.$$

Setting $B = A$ reduces each of these to a **double-angle** identity, the most useful family in the topic. The derivation is worth doing once, because it shows there is nothing new to learn here.

Put $B = A$ in the sine formula:

$$\sin 2A = \sin(A + A) = \sin A \cos A + \cos A \sin A = 2 \sin A \cos A.$$

Do the same in the cosine formula:

$$\cos 2A = \cos(A + A) = \cos A \cos A - \sin A \sin A = \cos^2 A - \sin^2 A.$$

That is the first form. The other two come from $\sin^2 A + \cos^2 A = 1$, used to remove one function or the other. Replacing $\sin^2 A$ by $1 - \cos^2 A$ gives

$$\cos 2A = \cos^2 A - (1 - \cos^2 A) = 2\cos^2 A - 1,$$

and replacing $\cos^2 A$ by $1 - \sin^2 A$ instead gives

$$\cos 2A = (1 - \sin^2 A) - \sin^2 A = 1 - 2\sin^2 A.$$

The tangent version follows in the same way, by putting $B = A$ in the tangent formula.

The sine and cosine versions sit in the SL half of the booklet; the tangent version is HL.

KEY · IB **Double angles:**

$$\sin 2\theta = 2\sin\theta\cos\theta, \qquad \cos 2\theta = \cos^2\theta - \sin^2\theta = 2\cos^2\theta - 1 = 1 - 2\sin^2\theta.$$

KEY · IB HL $\tan 2\theta = \dfrac{2\tan\theta}{1-\tan^2\theta}$.

The three forms of $\cos 2\theta$ are one identity rewritten by $\sin^2 + \cos^2 = 1$. Choose whichever leaves you with only sines or only cosines. That choice is what makes the identity so useful when solving equations.

EXAMPLE 9.1

Find the exact value of $\sin 75°$.

Write $75° = 45° + 30°$ and use the compound-angle formula:

$$\sin 75° = \sin 45° \cos 30° + \cos 45° \sin 30° = \frac{\sqrt{2}}{2}\cdot\frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{2}\cdot\frac{1}{2} = \frac{\sqrt{6}+\sqrt{2}}{4}.$$

Splitting an awkward angle into a sum of two known ones is the standard method for exact values beyond the basic table.

Identities also let you move between ratios *without ever finding the angle*. Examiners set this often: the angle stays unknown from start to finish, and the Pythagorean identity supplies everything needed.

EXAMPLE 9.2

Given that $\cos x = \dfrac{3}{4}$ and x is acute, find the exact value of $\sin 2x$ without finding x.

From $\cos^2 x + \sin^2 x = 1$, $\sin^2 x = 1 - \frac{9}{16} = \frac{7}{16}$, and since x is acute, $\sin x = \frac{\sqrt{7}}{4}$ (positive). Then

$$\sin 2x = 2\sin x\cos x = 2\cdot\frac{\sqrt{7}}{4}\cdot\frac{3}{4} = \frac{3\sqrt{7}}{8}.$$

The word "acute" fixed the sign of $\sin x$. If x had instead been in the fourth quadrant, the same working would give $\sin 2x = -\frac{3\sqrt{7}}{8}$: always let the quadrant choose the sign before doubling.

Proving an identity is the same game played in reverse: start from one side, substitute known identities, and simplify until the other side appears.

EXAMPLE 9.3

Prove that $\dfrac{\sin 2\theta}{1+\cos 2\theta} = \tan\theta$.

Work on the left side, choosing the form $\cos 2\theta = 2\cos^2\theta - 1$ because it cancels the 1:

$$\frac{\sin 2\theta}{1+\cos 2\theta} = \frac{2\sin\theta\cos\theta}{1+(2\cos^2\theta-1)} = \frac{2\sin\theta\cos\theta}{2\cos^2\theta} = \frac{\sin\theta}{\cos\theta} = \tan\theta. \quad \blacksquare$$

Choosing the form of $\cos 2\theta$ that cancels the constant is the same idea as in equation solving. Pick the version that matches the terms beside it.

YOUR TURN 9.2 Trigonometric Identities

5. Simplify each expression.

(a) $\sin^2\theta + \cos^2\theta$ [1] (b) $2\sin\theta\cos\theta$ [1]
(c) $\cos^2\theta - \sin^2\theta$ [1] (d) $1 - 2\sin^2\theta$ [1]

6. Given $\sin\theta = \frac{3}{5}$ with θ acute, find each exact value.

(a) $\cos\theta$ [2] (b) $\tan\theta$ [1]
(c) $\sin 2\theta$ [2] (d) $\cos 2\theta$ [2]

7. Given $\cos\theta = -\frac{3}{5}$ with θ in the second quadrant, find each exact value.

(a) $\sin\theta$ [2] (b) $\tan\theta$ [2]
(c) $\sin 2\theta$ [2] (d) $\cos 2\theta$ [2]

8. Use a compound-angle formula to find each exact value.

(a) $\cos 15^\circ$, from $45^\circ - 30^\circ$ [3] (b) $\sin 75^\circ$, from $45^\circ + 30^\circ$ [3]

9. Prove each identity.

(a) $\dfrac{\sin 2\theta}{1+\cos 2\theta} = \tan\theta$ [3]
(b) $\dfrac{1-\cos 2\theta}{\sin 2\theta} = \tan\theta$ [3]

9.3 Graphs of Trigonometric Functions

The graphs are the circle unrolled. Send the point P round the circle and plot its y-coordinate against the angle: that plot *is* $y = \sin x$. Plot its x-coordinate instead and you get $y = \cos x$.

Every feature of the graph can be read off the circle before you plot anything.

On the circle	On the graph of $y = \sin x$
the y-coordinate runs from -1 to 1	the curve stays between -1 and 1: amplitude 1
P returns to its start after one turn	the curve repeats every 2π: period 2π
$y = 0$ at $\theta = 0,\ \pi,\ 2\pi$	the curve crosses the axis at $0,\ \pi,\ 2\pi$
y is largest at the top, $\theta = \frac{\pi}{2}$	maximum 1 at $x = \frac{\pi}{2}$
$y > 0$ in the upper half of the circle	the curve is above the axis for $0 < x < \pi$

Read the x-coordinate instead and the cosine graph appears the same way. It starts at 1, because P starts on the positive x-axis, and falls to 0 at $\frac{\pi}{2}$, where P is at the top and its x-coordinate is zero.

That also explains why the two curves have the same shape. The x-coordinate at angle θ is the y-coordinate a quarter turn later, so $\cos\theta = \sin(\theta + \frac{\pi}{2})$, and the cosine curve is the sine curve shifted left by $\frac{\pi}{2}$.

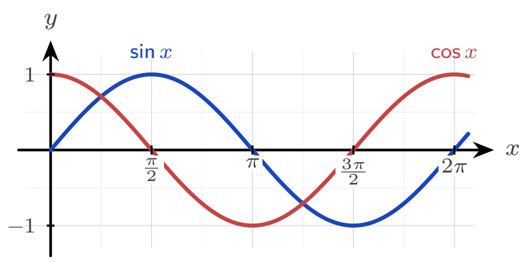


Figure 9.4 One turn of the circle unrolled. Both waves have amplitude 1 and period 2π; the cosine is the sine shifted left by $\frac{\pi}{2}$.

The tangent behaves differently, and the circle explains why. Since $\tan\theta = \frac{y}{x}$, it is undefined wherever $x = 0$, which is at the top and the bottom of the circle. That gives vertical asymptotes at $x = \frac{\pi}{2} + n\pi$.

Its period is π, not 2π. Half a turn sends P to the point diametrically opposite, whose coordinates are $(-x, -y)$. The ratio $\frac{-y}{-x}$ equals $\frac{y}{x}$, so the tangent repeats after only half a revolution.

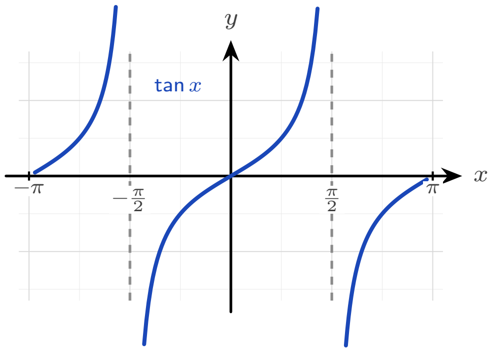


Figure 9.5 The tangent repeats every π, not every 2π, with a vertical asymptote wherever $\cos x = 0$.

Transforming the wave

The general sinusoid stretches and slides the basic sine, and each parameter has a plain meaning.

KEY For $y = a\sin(b(x+c)) + d$:

$$\text{amplitude} = |a|, \qquad \text{period} = \frac{2\pi}{|b|}, \qquad \text{horizontal shift} = -c, \qquad \text{vertical shift (midline)} = d.$$

Questions often run the other way, giving you the graph's largest and smallest values and asking for a and d. Since the maximum is $d + |a|$ and the minimum is $d - |a|$, adding and subtracting recovers both constants.

Quantity	**From the equation**	**From max and min**
Amplitude	$\|a\|$	$\dfrac{\text{max} - \text{min}}{2}$
Midline	$y = d$	$y = \dfrac{\text{max} + \text{min}}{2}$
Maximum	$d + \|a\|$	—
Minimum	$d - \|a\|$	—
Period	$\dfrac{2\pi}{\|b\|}$	the distance between repeats

The two right-hand formulas are worth reading as sentences. The amplitude is half the total rise and fall, and the midline sits halfway between the top and the bottom. For $y = 4\sin(3x) - 2$ the maximum is 2 and the minimum is -6, so the amplitude is $\frac{2-(-6)}{2} = 4$ and the midline is $\frac{2+(-6)}{2} = -2$, which matches the equation.

EXAMPLE 9.4

Describe $y = 3\sin(2x) - 1$ and state its range.

The amplitude is 3, so the wave rises and falls 3 either side of its midline. The period is $\frac{2\pi}{2} = \pi$, so it completes a full cycle twice as fast as $\sin x$. The midline is $y = -1$. The graph therefore runs from $-1 - 3 = -4$ up to $-1 + 3 = 2$, giving range $-4 \le y \le 2$.

The four parameters are not just labels on a graph; they are how a sinusoid is fitted to the real world. The opening story can now be settled with actual numbers.

EXAMPLE 9.5

When it opened in 1999, the Daikanransha in Odaiba, Tokyo, was briefly the world's tallest Ferris wheel: 115 m tall, its wheel 100 m across, turning once every 16 minutes. It lost the record to the London Eye within the year, and it was taken down in 2022, but its numbers make a clean model. A seat starts at the bottom of the wheel. Write the seat's height $h(t)$ in metres after t minutes, and find its height after 5 minutes.

Read each parameter off the wheel. The axle sits at $115 - 50 = 65$ m, so the midline is $d = 65$. The radius gives the amplitude, $a = 50$. One revolution takes 16 minutes, so $b = \frac{2\pi}{16} = \frac{\pi}{8}$. The seat starts at the *bottom*, 15 m up, which is what an upside-down cosine does at $t = 0$:

$$h(t) = 65 - 50\cos\left(\frac{\pi t}{8}\right).$$

Check the skeleton before trusting it: $h(0) = 65 - 50 = 15$ (bottom) and $h(8) = 65 + 50 = 115$ (top, half a turn later). Then

$$h(5) = 65 - 50\cos\frac{5\pi}{8} \approx 65 + 19.1 \approx 84.1 \text{ m}.$$

Choosing $-\cos$ to start at the minimum, rather than forcing a phase shift into a sine, is the modelling move examiners reward: match the *shape* of the motion first, then the numbers.

Symmetry of the graphs HL

Each reflection symmetry of the circle from Section 9.1 becomes a visible symmetry of the graphs. The cosine curve is mirror-symmetric in the y-axis, so $\cos(-x) = \cos x$: cosine is an **even** function. The sine and tangent curves map onto themselves under a half-turn about the origin, so $\sin(-x) = -\sin x$ and $\tan(-x) = -\tan x$: both are **odd**, in the sense of Ch. 5. The same circle symmetries give the supplementary-angle relations

$$\sin(\pi - \theta) = \sin\theta, \qquad \cos(\pi - \theta) = -\cos\theta, \qquad \tan(\pi - \theta) = -\tan\theta,$$

which you can see on the graphs directly: the sine curve is symmetric about the vertical line $x = \frac{\pi}{2}$, while cosine and tangent flip sign across it. In an exam these relations save you from ever reaching for a calculator: $\cos\frac{3\pi}{4} = -\cos\frac{\pi}{4} = -\frac{\sqrt{2}}{2}$ in one step.

YOUR TURN 9.3 Graphs of Trigonometric Functions

10. State the amplitude, the period and the midline of each graph.

(a) $y = 3\sin x$ [2]
(b) $y = \sin 2x$ [2]
(c) $y = 4\sin(3x) - 2$ [3]
(d) $y = 5\sin(4x) + 3$ [3]

11. State the range of each function.

(a) $y = 2\sin x + 1$ [2]
(b) $y = 3\sin(2x) - 1$ [2]
(c) $y = 2\cos\left(\frac{x}{2}\right) - 1$ [3]

12. Answer each question about $y = \tan x$.

(a) State its period. [1]
(b) State the equations of two of its asymptotes. [2]
(c) Explain why the asymptotes occur where they do. [2]

13. A wheel of diameter 40 m has its lowest point 2 m above the ground and turns once every 20 seconds. A seat starts at the bottom.

(a) State the amplitude and the midline of the seat's height. [2]
(b) Find the value of b in $h(t) = d - a\cos(bt)$. [2]
(c) Write down $h(t)$. [2]
(d) Find the height after 5 seconds. [2]

9.4 Reciprocal Trigonometric Functions HL

At Higher Level three more functions are added, the reciprocals of the first three. They introduce no new mathematics: each is one divided by a function you already know. What they do introduce is a set of graphs with asymptotes, and two identities that make certain equations solvable.

KEY · IB $\sec\theta = \dfrac{1}{\cos\theta}, \qquad \operatorname{cosec}\theta = \dfrac{1}{\sin\theta}.$

The third, the **cotangent**, is $\cot\theta = \dfrac{1}{\tan\theta} = \dfrac{\cos\theta}{\sin\theta}$. Its definition is not printed in the booklet, so learn it.

Two facts follow from the reciprocal alone, and between them they fix every graph.

First, a reciprocal has an asymptote wherever the original function is zero, since you cannot divide by zero. So sec has asymptotes where $\cos\theta = 0$, at $\theta = \frac{\pi}{2} + n\pi$, and cosec and cot have them where $\sin\theta = 0$, at $\theta = n\pi$.

Second, sin and cos never exceed 1 in size, so their reciprocals are never smaller than 1 in size. The range of $\sec\theta$ and of $\operatorname{cosec}\theta$ is therefore $y \leq -1$ or $y \geq 1$: neither ever takes a value strictly between -1 and 1. This is the fact examiners test most often about these functions.

Function	**Definition**	**Asymptotes at (n any integer)**	**Range**
$\sec\theta$	$\dfrac{1}{\cos\theta}$	$\theta = \frac{\pi}{2} + n\pi$	$y \leq -1$ or $y \geq 1$
$\operatorname{cosec}\theta$	$\dfrac{1}{\sin\theta}$	$\theta = n\pi$	$y \leq -1$ or $y \geq 1$
$\cot\theta$	$\dfrac{\cos\theta}{\sin\theta}$	$\theta = n\pi$	all real y

The two Pythagorean variants

The identity $\cos^2\theta + \sin^2\theta = 1$ has two companions, one for each reciprocal pair, and both come from dividing it through.

Dividing $\cos^2\theta + \sin^2\theta = 1$ by $\cos^2\theta$ gives

$$1 + \tan^2\theta = \sec^2\theta,$$

since $\frac{\sin\theta}{\cos\theta} = \tan\theta$ and $\frac{1}{\cos\theta} = \sec\theta$. Dividing the same identity by $\sin^2\theta$ instead gives the companion result. Deriving them takes one line each, which is quicker than trusting your memory of which is which.

KEY · IB $1 + \tan^2\theta = \sec^2\theta, \qquad 1 + \cot^2\theta = \operatorname{cosec}^2\theta.$

These identities exist to turn mixed equations into single-function ones, exactly as the double angles did.

EXAMPLE 9.6

Solve $\tan^2 x + \sec x = 1$ for $0 \le x \le 2\pi$.

The equation mixes tan and sec; the identity $\tan^2 x = \sec^2 x - 1$ rewrites it in $\sec x$ alone:

$$\sec^2 x - 1 + \sec x = 1 \implies \sec^2 x + \sec x - 2 = 0 \implies (\sec x + 2)(\sec x - 1) = 0.$$

So $\sec x = -2$ or $\sec x = 1$, which is $\cos x = -\frac{1}{2}$ or $\cos x = 1$. From the circle,

$$x = \frac{2\pi}{3},\ \frac{4\pi}{3} \quad \text{or} \quad x = 0,\ 2\pi.$$

Converting back to cos at the end is the safest final step, because the unit circle is read in sine and cosine, not in secant.

YOUR TURN 9.4 Reciprocal Trigonometric Functions HL

14. Evaluate each exactly.

(a) $\sec\frac{\pi}{3}$ [1] (b) $\operatorname{cosec}\frac{\pi}{2}$ [1]

(c) $\cot\frac{\pi}{4}$ [1] (d) $\sec\frac{2\pi}{3}$ [2]

15. Simplify each expression.

(a) $\sec^2\theta - \tan^2\theta$ [1] (b) $\operatorname{cosec}^2\theta - \cot^2\theta$ [2]

(c) $\dfrac{1}{\cot\theta}$ [1] (d) $\tan\theta\cot\theta$ [2]

9.5 Inverse Trigonometric Functions HL

Everything so far has gone one way: give an angle, get a ratio. The inverse functions run the other way, recovering the angle from the ratio.

You have already used them without naming them. When you solve $\sin x = \frac{1}{2}$ and press the $\sin^{-1}$ key, that key is the inverse function.

Two notations, one function. The inverse of sine is written either $\arcsin x$ or $\sin^{-1} x$. They mean exactly the same thing. This book uses arcsin, because the other form invites a serious confusion:

$$\sin^{-1} x \ \text{ means the } \textit{inverse}, \qquad \text{but} \qquad (\sin x)^{-1} = \frac{1}{\sin x} = \operatorname{cosec} x.$$

The -1 is not an exponent here. Writing arcsin, arccos and arctan removes the ambiguity, and your calculator's $\sin^{-1}$ key is the same function under the other name.

Why a restriction is needed

There is an immediate difficulty. A function has an inverse only if it is one-to-one (Ch. 5), and sin, cos and tan are as far from one-to-one as a function can be. Each takes every value in its range infinitely often, so the horizontal line $y = \frac{1}{2}$ meets $y = \sin x$ infinitely many times.

That is a real problem, not a technicality. If you ask "which angle has sine $\frac{1}{2}$?", there is no single answer: $\frac{\pi}{6}$, $\frac{5\pi}{6}$, $\frac{13\pi}{6}$ and infinitely many others all qualify. A function must return one output, so it has to choose.

The fix is the one from Ch. 5: cut the domain down until the function is one-to-one, then invert what is left. Which piece to keep is a choice, and mathematicians have agreed on one for each function. Those agreed pieces are shaded in the diagrams below, and they are what fix the ranges in the table.

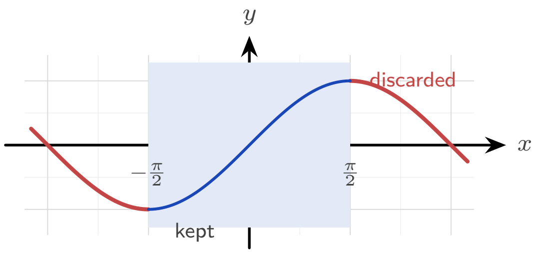


Figure 9.6 To invert $y = \sin x$, keep only the shaded piece from $-\frac{\pi}{2}$ to $\frac{\pi}{2}$. On that piece the curve rises steadily, so it is one-to-one and each output comes from exactly one input.

The same is done for the other two: cosine keeps $0 \le x \le \pi$, where it falls steadily, and tangent keeps $-\frac{\pi}{2} < x < \frac{\pi}{2}$, one branch between two asymptotes. Those kept pieces become the *ranges* of the inverses, because domain and range swap when a function is inverted.

Function	Domain	Range
$\arcsin x$	$-1 \le x \le 1$	$-\dfrac{\pi}{2} \le y \le \dfrac{\pi}{2}$
$\arccos x$	$-1 \le x \le 1$	$0 \le y \le \pi$
$\arctan x$	all real x	$-\dfrac{\pi}{2} < y < \dfrac{\pi}{2}$

Each inverse is a reflection of its restricted parent in the line $y = x$ (Ch. 5), so the graphs are the familiar curves turned on their sides. Arcsine climbs steeply from $(-1, -\frac{\pi}{2})$ to $(1, \frac{\pi}{2})$. Arccosine falls from $(-1, \pi)$ to $(1, 0)$. Arctangent flattens towards the horizontal asymptotes $y = \pm\frac{\pi}{2}$, which are the vertical asymptotes of tangent after the reflection.

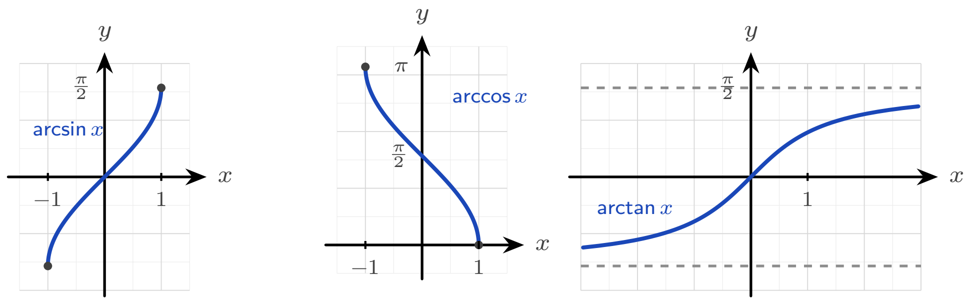


Figure 9.7 Each inverse is its restricted parent reflected in $y = x$. The marked endpoints show the range each one is confined to.

EXAMPLE 9.7

Find the exact values of (a) $\arcsin\left(-\frac{1}{2}\right)$, (b) $\arccos\left(-\frac{\sqrt{3}}{2}\right)$, (c) $\arcsin\left(\sin\frac{5\pi}{6}\right)$.

(a) The angle in $[-\frac{\pi}{2}, \frac{\pi}{2}]$ whose sine is $-\frac{1}{2}$ is $-\frac{\pi}{6}$. Not $\frac{11\pi}{6}$ and not $\frac{7\pi}{6}$: those solve the equation but lie outside arcsine's range.

(b) The angle in $[0, \pi]$ whose cosine is $-\frac{\sqrt{3}}{2}$ is $\pi - \frac{\pi}{6} = \frac{5\pi}{6}$. Arccosine handles negatives by reaching into the second quadrant.

(c) $\sin\frac{5\pi}{6} = \frac{1}{2}$, and $\arcsin\frac{1}{2} = \frac{\pi}{6}$. The composition returns $\frac{\pi}{6}$, not $\frac{5\pi}{6}$: arcsine can only answer with angles from its own range.

CAUTION $\arcsin(\sin x)$ does not always return x. Because arcsin only outputs angles in $[-\frac{\pi}{2}, \frac{\pi}{2}]$, feeding it $\sin(2)$ gives $\pi - 2$, not 2. The restricted range is doing exactly its job.

YOUR TURN 9.5 Inverse Trigonometric Functions HL

16. Evaluate each exactly.

(a) $\arcsin\dfrac{1}{2}$ [1]

(b) $\arccos 0$ [1]

(c) $\arctan(-1)$ [2]

(d) $\arccos\left(-\dfrac{1}{2}\right)$ [2]

17. These compositions do not simply cancel.

(a) State the range of $\arcsin x$ and the range of $\arccos x$. [2]

(b) Evaluate $\arcsin\left(\sin\dfrac{3\pi}{4}\right)$, and explain why the answer is not $\dfrac{3\pi}{4}$. [3]

(c) Evaluate $\arccos\left(\cos\dfrac{4\pi}{3}\right)$. [3]

9.6 Solving Trigonometric Equations

An equation like $\sin x = \frac{1}{2}$ has not one solution but infinitely many, because the wave crosses any horizontal line over and over. Your calculator returns only one of them, the **principal value**. Everything else you must find yourself, and the circle is what tells you how.

Here is the reason there is always a second angle. Solving $\sin x = k$ means finding the points on the circle whose y-coordinate is k. A horizontal line at height k cuts the circle in *two* places, not one, so two angles share that sine. The same is true for $\cos x = k$, where a vertical line at $x = k$ cuts the circle twice.

Where the second angle sits depends on which function you have. The rule is short, and it is worth learning as one line.

Equation	Second angle
$\sin x = k$	$\pi - \alpha$
$\cos x = k$	$2\pi - \alpha$
$\tan x = k$	$\alpha + \pi$

Each line is a symmetry of the circle. For sine the two points sit at the same height, mirrored in the y-axis. For cosine they sit at the same horizontal position, mirrored in the x-axis. For tangent the second point is diametrically opposite, half a turn away.

Then check the signs. Your two angles should lie in the two quadrants where that function is positive, or in the other two if k is negative. If only one of them does, you have used the wrong row.

Once you have both angles in one turn, add or subtract whole turns of 2π to reach every solution the interval asks for.

KEY **Solving a trigonometric equation.**

1. Rearrange until a single trigonometric function is alone on one side.
2. If the argument is not plain x, substitute u for it and **transform the interval as well**.
3. Find the principal value α from your calculator.
4. Find the second angle from the table above.
5. Add or subtract 2π (or $2\pi b$, in terms of u) until you have every solution in the interval.
6. If you substituted, convert back to x at the very end.

Step 2 is where marks are most often lost, and step 5 is where solutions go missing.

EXAMPLE 9.8

Solve $\cos x = -\dfrac{1}{2}$ for $0 \le x \le 2\pi$.

The calculator gives the principal value $\alpha = \arccos\left(-\frac{1}{2}\right) = \frac{2\pi}{3}$.

This is a cosine equation, so the partner angle is $2\pi - \alpha$:

$$2\pi - \frac{2\pi}{3} = \frac{4\pi}{3}.$$

Both lie in the interval, so $x = \frac{2\pi}{3}$ or $x = \frac{4\pi}{3}$.

Check the signs against the circle. Cosine is negative in the second and third quadrants, and $\frac{2\pi}{3}$ and $\frac{4\pi}{3}$ are exactly one angle from each. That check catches a wrong partner immediately.

EXAMPLE 9.9

Solve $\sin x = \dfrac{1}{2}$ for $0 \le x \le 2\pi$.

The principal value is $x = \dfrac{\pi}{6}$. Sine is also positive in the second quadrant, where the partner angle is $\pi - \dfrac{\pi}{6} = \dfrac{5\pi}{6}$. Both lie in the interval, so

$$x = \frac{\pi}{6} \quad \text{or} \quad x = \frac{5\pi}{6}.$$

Always ask which *other* quadrant shares the sign; that second angle is the one candidates most often miss.

TIP On a calculator paper, always check your solution count graphically: plot both sides and count the intersections in the interval. If your algebra found three solutions and the screen shows four, the screen is right.

When the argument is not plain x but $2x$ or $3(x-4)$, substitute for the whole argument and, crucially, transform the interval along with it. A doubled argument sweeps the circle twice as fast, so it collects twice as many solutions.

EXAMPLE 9.10

Solve $\sin 2x = \dfrac{1}{2}$ for $0 \le x \le 2\pi$.

Let $u = 2x$. As x runs over $[0, 2\pi]$, u runs over $[0, 4\pi]$: two full turns. In the first turn, $\sin u = \frac{1}{2}$ at $u = \frac{\pi}{6}$ and $\frac{5\pi}{6}$; the second turn repeats them 2π later, at $\frac{13\pi}{6}$ and $\frac{17\pi}{6}$. Halving each u recovers x:

$$x = \frac{\pi}{12}, \frac{5\pi}{12}, \frac{13\pi}{12}, \frac{17\pi}{12}.$$

Four solutions, not two. Solving in u over the *enlarged* interval first, and only then dividing, is what protects you from losing the second pair.

Using identities

An equation mixing $\sin 2x$ with $\sin x$, or $\cos 2x$ with $\cos x$, cannot be solved as it stands. An identity rewrites everything in terms of a single function, turning the equation into an algebraic one, usually a quadratic.

EXAMPLE 9.11

Solve $\cos 2x + \cos x = 0$ for $0 \le x \le 2\pi$.
Choose the form of $\cos 2x$ that gives only cosines, $\cos 2x = 2\cos^2 x - 1$:

$$2\cos^2 x - 1 + \cos x = 0 \implies 2\cos^2 x + \cos x - 1 = 0.$$

This factors as $(2\cos x - 1)(\cos x + 1) = 0$, so $\cos x = \frac{1}{2}$ or $\cos x = -1$. From the circle,

$$x = \frac{\pi}{3}, \frac{5\pi}{3} \quad \text{or} \quad x = \pi.$$

Picking the form of the double-angle identity that matches the other terms is the key step in these problems.

Enrichment: the wave form $a\sin\theta + b\cos\theta$

This last idea sits just beyond the AA syllabus. It is included because it is both elegant and useful. A sum of a sine and a cosine of the same angle is itself a single shifted sine wave: the compound-angle identity, run backwards. Writing it as $R\sin(\theta + \alpha)$ makes both its maximum and its equation solvable at a glance.

KEY $a\sin\theta + b\cos\theta = R\sin(\theta + \alpha)$, where $R = \sqrt{a^2 + b^2}$ and $\tan\alpha = \dfrac{b}{a}$. The maximum value is R, the minimum $-R$.

A related idea appears when a problem asks for *all* solutions rather than those in a fixed interval. The answer is then a family generated by the period. For example, $\sin x = k$ has general solution $x = \arcsin k + 2n\pi$ or $x = (\pi - \arcsin k) + 2n\pi$, for every integer n. The AA syllabus never asks for general solutions in an exam, but seeing the infinite family explains why a question must always restrict you to an interval.

YOUR TURN 9.6 Solving Trigonometric Equations

18. Solve for $0 \le x \le 2\pi$.

(a) $\sin x = \dfrac{1}{2}$	[2]	(b) $\cos x = 0$	[2]
(c) $\tan x = 1$	[2]	(d) $\cos x = 1$	[1]

19. Solve for $0 \leq x \leq 2\pi$.

(a) $\sin x = -\frac{1}{2}$ [2]

(b) $2\cos x = \sqrt{3}$ [2]

(c) $\sqrt{2}\sin x = -1$ [3]

(d) $\tan x = -\sqrt{3}$ [3]

20. Solve for $0 \leq x \leq 2\pi$. Transform the interval before solving.

(a) $\sin 2x = 0$ [2]

(b) $\cos 2x = \frac{1}{2}$ [3]

(c) $\tan 2x = 1$ [4]

21. Solve for $0 \leq x \leq 2\pi$, using an identity first.

(a) $2\sin^2 x - \sin x = 0$ [3]

(b) $2\cos^2 x + \cos x - 1 = 0$ [4]

(c) $\cos 2x + \sin x = 0$ [4]

Chapter 9 · Trigonometric Functions

Reflections

IM Trigonometry began as astronomy. Babylonian and Greek observers needed to predict the wandering of the planets, and the sine was born as a tool for the geometry of the celestial sphere. Its greatest early development came in India and in the Islamic world. Astronomers there tabulated sines to many decimal places, to fix the times of prayer and the direction of Mecca. The subject returned to Europe through Latin translations, which gave us the word "sine" itself. For two thousand years, the functions of this chapter were the mathematics of the heavens.

TOK A sine wave is periodic: it repeats forever, identical in every cycle. Yet it is built from the endlessly irrational π and takes irrational values at almost every point. How can something so regular be made of something so unruly? The physicist Joseph Fourier then proved that *any* repeating signal, however jagged, is a sum of pure sine waves. So the smooth circle is hidden inside every square wave and every spoken word. Is the sinusoid a human tool, or the alphabet in which nature writes all its rhythms?

RW Because a sine wave is what steady rotation looks like from the side, it models every oscillation in science and engineering. Alternating current rises and falls as a sine while the generator spins. Sound is pressure oscillating in the same way, and its pitch is the frequency of the wave. Light, radio, and every signal your phone sends and receives are sinusoids too. Fourier analysis, decomposing a signal into its sine components, is how music is compressed, images are stored, and earthquakes and heartbeats are read.

End-of-Chapter Problems — Chapter 9: Trigonometric Functions

1. Use the unit circle to write down the exact value of each.

(a) $\sin\dfrac{2\pi}{3}$ [1]
(b) $\cos\dfrac{2\pi}{3}$ [1]
(c) $\tan\dfrac{2\pi}{3}$ [2]
(d) $\cos\dfrac{5\pi}{4}$ [2]
(e) $\sin\dfrac{11\pi}{6}$ [2]
(f) $\tan\dfrac{3\pi}{4}$ [2]

2. The point P lies on the unit circle at angle θ.

(a) State the coordinates of P when $\theta = \dfrac{5\pi}{6}$. [2]

(b) P has coordinates $\left(-\dfrac{\sqrt{2}}{2}, -\dfrac{\sqrt{2}}{2}\right)$. Find θ for $0 \le \theta \le 2\pi$. [2]

(c) P lies in the third quadrant. State the sign of $\sin\theta$, of $\cos\theta$ and of $\tan\theta$. [3]

(d) Given $\sin\theta = \dfrac{\sqrt{3}}{2}$, write down the two possible values of θ in $0 \le \theta \le 2\pi$, and explain why there are exactly two. [3]

3. Prove each identity. They are arranged in increasing order of difficulty.

(a) $\cos^2\theta - \sin^2\theta = 1 - 2\sin^2\theta$ [2]

(b) $\dfrac{1-\cos 2\theta}{\sin 2\theta} = \tan\theta$ [3]

(c) $\dfrac{\sin 2\theta}{1+\cos 2\theta} = \tan\theta$ [3]

(d) $\cos^4\theta - \sin^4\theta = \cos 2\theta$ [3]

(e) $\dfrac{1-\cos 2\theta}{1+\cos 2\theta} = \tan^2\theta$ [4]

(f) $\dfrac{1}{\sec\theta - \tan\theta} = \sec\theta + \tan\theta$ [4]

(g) $\dfrac{\sin\theta}{1+\cos\theta} + \dfrac{1+\cos\theta}{\sin\theta} = 2\operatorname{cosec}\theta$ [5]

(h) $\dfrac{1+\sin 2\theta}{\cos 2\theta} = \dfrac{\cos\theta + \sin\theta}{\cos\theta - \sin\theta}$ [5]

4. Given $\cos\theta = -\dfrac{3}{5}$ with θ in the second quadrant, find the exact value of each.

(a) $\sin\theta$ [2]
(b) $\tan\theta$ [1]
(c) $\sin 2\theta$ [2]
(d) $\cos 2\theta$ [2]

5. Given $\tan\theta = 2$ with θ acute, find the exact value of each.

(a) $\sin\theta$ and $\cos\theta$ [2] (b) $\sin 2\theta$ [2]
(c) $\cos 2\theta$ [2]

6. Use a compound-angle formula to find each exact value.

(a) $\tan 75°$ [4] (b) $\cos 15°$ [3]

7. Prove each result.

(a) $\sin(A+B)+\sin(A-B)=2\sin A\cos B$ [4]
(b) $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$ [5]
(c) $\sin 3\theta = 3\sin\theta - 4\sin^3\theta$ [5]

8. Solve for $0 \le x \le 2\pi$.

(a) $\sin x = \dfrac{1}{2}$ [2] (b) $\sqrt{3}\tan x = 1$ [3]
(c) $2\cos^2 x = 1$ [4] (d) $\tan^2 x = 3$ [4]

9. Solve for $0 \le x \le 2\pi$. Each becomes a quadratic.

(a) $2\sin^2 x + \sin x - 1 = 0$ [5] (b) $4\sin^2 x - 3 = 0$ [4]
(c) $\cos 2x = \cos x$ [5] (d) $\sin 2x = \sin x$ [5]

10. Solve for $0 \le x \le 2\pi$.

(a) $2\sin x\cos x = \dfrac{1}{2}$ [4]
(b) $\cos 2x + 3\sin x = 2$ [6]
(c) $\sin x + \cos x = 1$ [6]

11. Given the identity $1+\tan^2\theta = \sec^2\theta$, solve $\sec^2 x = 4$ for $0 \le x \le \dfrac{\pi}{2}$. [4]

12. Evaluate, giving exact values.

(a) $\arctan 1$ [1] (b) $\arcsin\left(-\dfrac{1}{2}\right)$ [1]
(c) $\arccos(-1)$ [1] (d) $\arcsin\left(\sin\dfrac{5\pi}{6}\right)$ [3]

13. Consider $f(x) = 3\sin(2x)+1$.

(a) State the amplitude and the period. [2]
(b) State the maximum and minimum values. [2]

14. The graph of $y = a\sin(bx)+d$ has maximum 7, minimum 1, and period π. Find a, b

and d. [4]

15. Sketch $y = \cos x - 1$ for $0 \le x \le 2\pi$, stating the maximum, the minimum, and the intercepts. [4]

16. The depth of water in a harbour is $d(t) = 5 + 3\sin\left(\frac{\pi t}{6}\right)$ metres, t hours after midnight.

(a) State the maximum and minimum depths. [2]
(b) Find the period, and say what it represents. [2]

17. A seat on a Ferris wheel has height $h(t) = 10 - 8\cos\left(\frac{\pi t}{15}\right)$ metres, t seconds after the start.

(a) State the minimum and maximum heights. [2]
(b) Find the time for one full revolution. [2]

18. Express $3\sin x + 4\cos x$ in the form $R\sin(x + \alpha)$ with $R > 0$.

(a) Find R and α. [3]
(b) Hence state the maximum value. [1]

19. Find the maximum value M of $5\sin x + 12\cos x$, and the value of x in $[0, 2\pi]$ at which it occurs. [5]

20. Use the symmetry of the unit circle to answer each part.

(a) Given $\sin 0.6 = 0.565$, write down $\sin(\pi - 0.6)$. [2]
(b) Given $\cos 1.2 = 0.362$, write down $\cos(-1.2)$ and $\cos(2\pi - 1.2)$. [3]
(c) Explain why $\sin(\theta + 2\pi) = \sin\theta$ for every θ. [2]
(d) Explain why $\tan(\theta + \pi) = \tan\theta$, but $\sin(\theta + \pi) \neq \sin\theta$. [4]

Challenge Questions

21. **The wave form, and when an equation can be solved.** Section 9.5 states that $a\sin\theta + b\cos\theta$ can be written as $R\sin(\theta + \alpha)$. This investigation proves it and then uses it to decide, without solving, whether an equation has any solutions at all.

Throughout take $a > 0$ and $b > 0$.

(a) Expand $R\sin(\theta+\alpha)$ and compare it with $a\sin\theta + b\cos\theta$ to show that $R\cos\alpha = a$ and $R\sin\alpha = b$. [3]

(b) Hence show that $R = \sqrt{a^2+b^2}$ and $\tan\alpha = \dfrac{b}{a}$. *(Hint: square and add)* [3]

(c) Write $7\sin\theta + 24\cos\theta$ in the form $R\sin(\theta+\alpha)$, and state its maximum and minimum values. [3]

(d) Explain why the equation $a\sin\theta + b\cos\theta = k$ has a solution if and only if $|k| \le R$. [4]

(e) Solve $3\sin x + 4\cos x = 5$ for $0 \le x \le 2\pi$, giving your answer to three significant figures. Explain why this equation has exactly one solution in the interval. [5]

22. **$\cos n\theta$ as a polynomial in $\cos\theta$.** You have already met $\cos 2\theta = 2\cos^2\theta - 1$ and, in the end-of-chapter problems, $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$. Each is a polynomial in $\cos\theta$ alone. This investigation shows that this always happens, and finds the pattern.

Write $c = \cos\theta$ throughout.

(a) Use the compound-angle formulas to show that

$$\cos((n+1)\theta) = 2\cos\theta\,\cos(n\theta) - \cos((n-1)\theta).$$

(Hint: expand $\cos(n\theta+\theta)$ and $\cos(n\theta-\theta)$, then add) [5]

(b) Starting from $\cos 0 = 1$ and $\cos\theta = c$, use the recurrence to obtain $\cos 2\theta$ and $\cos 3\theta$ in terms of c. [3]

(c) Use it again to find $\cos 4\theta$ and $\cos 5\theta$ in terms of c. [4]

(d) State the degree of the polynomial for $\cos n\theta$, and the coefficient of its highest power. Justify both from the recurrence. [4]

(e) Explain why no such polynomial in $\cos\theta$ alone exists for $\sin n\theta$ when n is even. *(Hint: compare $\sin(-\theta)$ with $\sin\theta$)* [3]

23. **Counting solutions before finding them.** A common examination error is to lose solutions when the argument is not plain x. This investigation finds how many solutions there are, so that you can check your own answer.

Take b to be a positive integer and consider $\sin(bx) = k$ on the interval $0 \le x \le 2\pi$.

(a) Solve $\sin x = \frac{1}{2}$, then $\sin 2x = \frac{1}{2}$, then $\sin 3x = \frac{1}{2}$, on $0 \le x \le 2\pi$. State how many solutions each has. [4]

(b) Explain why substituting $u = bx$ turns the interval $0 \le x \le 2\pi$ into $0 \le u \le 2\pi b$. [2]

(c) Hence state, with reasons, the number of solutions of $\sin(bx) = k$ when $|k| < 1$. [3]

(d) State the number of solutions when $k = 1$, and when $|k| > 1$. Explain the difference. [4]

(e) How many solutions does $\cos(bx) = k$ have when $|k| < 1$? Justify your answer, and say whether the endpoints $x = 0$ and $x = 2\pi$ need special care. [4]

GEOMETRY & TRIGONOMETRY

Chapter 10

Vectors

10

SYLLABUS 3.12–3.18 HL only

A stool with three legs never wobbles. A stool with four legs, on the same uneven floor, almost always does.

This is not bad carpentry. It is geometry. Three points always lie in one flat surface, whatever their heights, so all three feet touch the floor at once. A fourth point does not have to lie in that surface, and when it does not, the stool rocks between two pairs of legs. Photographers and surveyors know this, which is why a tripod has three legs and not four.

That fact cannot be written down in ordinary arithmetic. It is not about how big anything is; it is about position, direction, and flatness. To state it, and to prove it, we need a language in which a direction is something you can calculate with.

A **vector** is a quantity with both a size and a direction. Speed is a number, but velocity is a vector. Distance is a number, but displacement is a vector. A pilot flying due north through a wind from the west travels neither due north nor at their airspeed, because their motion and the wind point different ways and both must be accounted for.

Once a quantity has a direction, addition works differently. Adding three and three no longer has to give six. If the two point the same way you get six, if they point opposite ways you get zero, and any value in between is possible. The answer depends on the angle.

By the end of this chapter you will be able to add and scale vectors, use the scalar product to find angles and test for perpendicularity, and use the vector product to find areas and normals. You will write the equation of a line and of a plane, decide how two lines or three planes are arranged, and find shortest distances. A vector carries a size and a direction in one object, which makes it the natural tool for describing position and movement in three dimensions. Physics, computer graphics and navigation all depend on it, from the pilot's crosswind to the satellites overhead.

10.1 Vectors: Magnitude and Direction

A vector has exactly two properties: a **magnitude**, which is its length, and a **direction**. Where it sits in space does not matter. Two arrows of the same length pointing the same way are the same vector.

An arrow drawn to represent a vector is called a **directed line segment**. “Directed” because it has an arrowhead, and “line segment” because it is a piece of a line with two ends. Examination questions use this phrase, so it is worth knowing that it means an arrow and nothing more.

Vectors must be kept apart from ordinary numbers, which in this context are called **scalars**. Three notations are in common use, and they all mean the same thing. Printed books, including this one, use a bold letter **a**. By hand you cannot write bold type, so you write either a letter with an arrow above it, $\vec{a}$, or an underlined letter, $\underline{a}$. An arrow is also used for a vector named by its two endpoints: $\overrightarrow{AB}$ means the vector from the point A to the point B.

The most useful representation is the **column vector**. It lists how far to travel along each axis to get from the tail to the head.

$$\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = a_1\hat{\mathbf{i}} + a_2\hat{\mathbf{j}} + a_3\hat{\mathbf{k}},$$

Here $\hat{\mathbf{i}}$, $\hat{\mathbf{j}}$ and $\hat{\mathbf{k}}$ are the **base vectors**: arrows of length 1 along the x-, y- and z-axes. The hat is there to record that each one has length 1, a point returned to in the next subsection. The formula booklet prints them without hats, as **i**, **j**, **k**; the two mean the same thing. The column form and the base-vector form also mean the same thing, so use whichever is convenient.

The algebra of arrows

Vectors are added **tip to tail**. Draw the second vector starting where the first one ends. The sum is the arrow from the start of the first to the end of the second. In components it is simple: everything happens coordinate by coordinate.

KEY $$\begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} + \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} = \begin{pmatrix} a_1 + b_1 \\ a_2 + b_2 \\ a_3 + b_3 \end{pmatrix}, \qquad k \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = \begin{pmatrix} ka_1 \\ ka_2 \\ ka_3 \end{pmatrix}.$$

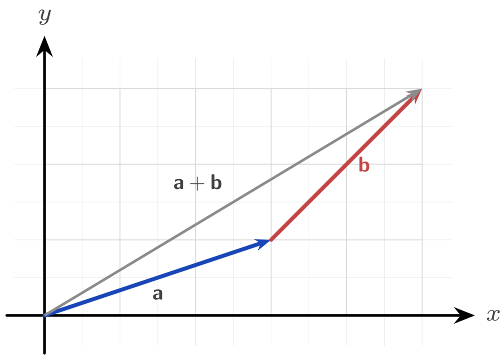


Figure 10.1 Adding tip to tail. Draw **b** starting where **a** ends; the sum runs from the start of **a** to the end of **b**.

Scalar multiplication by k scales the length by $|k|$ and reverses the direction when k is negative. This gives a clean test: two nonzero vectors are **parallel** exactly when one is a scalar multiple of the other, $\mathbf{a} = k\mathbf{b}$.

Two particular vectors are worth naming.

The **negative** of $\mathbf{v}$, written $-\mathbf{v}$, is the vector of the same length pointing the opposite way. It is the case $k = -1$ of scalar multiplication. Subtraction is then just addition of the negative:

$$\mathbf{a} - \mathbf{b} = \mathbf{a} + (-\mathbf{b}).$$

The **zero vector** $\mathbf{0}$ has magnitude 0 and no direction at all. It is what you get from $\mathbf{v}+(-\mathbf{v})$, or from $0\mathbf{v}$. It is the only vector whose direction is undefined, and it is written in bold to keep it apart from the number 0.

Magnitude and unit vectors

Three words are used for the same quantity. The **magnitude** of a vector, its **length**, and its **modulus** all mean the size of the arrow, and all are written $|\mathbf{v}|$. This book uses whichever reads more naturally in the sentence. The value comes straight from Pythagoras, extended into three dimensions.

KEY · IB The magnitude of $\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$ is $|\mathbf{v}| = \sqrt{v_1^2 + v_2^2 + v_3^2}$.

Dividing a vector by its own length leaves the direction untouched but sets the length to 1. The result is a **unit vector**, written $\hat{\mathbf{v}}$, the pure direction stripped of any magnitude, ready to be scaled to any length you like.

KEY $\hat{\mathbf{v}} = \dfrac{\mathbf{v}}{|\mathbf{v}|}$ is the unit vector in the direction of $\mathbf{v}$.

The base vectors are unit vectors, which is what their hats record: $|\hat{\mathbf{i}}| = |\hat{\mathbf{j}}| = |\hat{\mathbf{k}}| = 1$, one unit along each axis. So writing $\mathbf{a} = a_1\hat{\mathbf{i}} + a_2\hat{\mathbf{j}} + a_3\hat{\mathbf{k}}$ says that $\mathbf{a}$ is built by taking a_1 steps of length 1 along x, then a_2 along y, then a_3 along z.

EXAMPLE 10.1

Find a vector of magnitude 6 in the direction of $\mathbf{v} = \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$.

First the length: $|\mathbf{v}| = \sqrt{2^2 + (-1)^2 + 2^2} = \sqrt{9} = 3$. The unit vector is $\hat{\mathbf{v}} = \frac{1}{3}\mathbf{v}$, and scaling it to length 6 means multiplying by 6:

$$6\hat{\mathbf{v}} = \frac{6}{3}\begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix} = \begin{pmatrix} 4 \\ -2 \\ 4 \end{pmatrix}.$$

Reduce to a unit vector first, then rescale. Those two steps answer every question of the form "find a vector of magnitude m in the direction of".

YOUR TURN 10.1 Vectors: Magnitude and Direction

1. Find the magnitude of each vector.

(a) $\begin{pmatrix} 3 \\ 4 \\ 0 \end{pmatrix}$ [1] (b) $\begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$ [1]

(c) $\begin{pmatrix} 0 \\ 3 \\ 4 \end{pmatrix}$ [1] (d) $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ [2]

2. Given $\mathbf{a} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 0 \\ -1 \\ 4 \end{pmatrix}$, evaluate each of the following.

(a) $\mathbf{a} + \mathbf{b}$ [1] (b) $\mathbf{a} - \mathbf{b}$ [1]

(c) $\mathbf{a} + 2\mathbf{b}$ [2] (d) $3\mathbf{a} - \mathbf{b}$ [2]

3. Unit vectors and scaling.

(a) Find the unit vector in the direction of $\begin{pmatrix} 0 \\ 3 \\ 4 \end{pmatrix}$. [2]

(b) Find the unit vector in the direction of $\begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$. [2]

(c) Find a vector of magnitude 6 in the direction of $\begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$. [3]

(d) Explain why every unit vector has magnitude exactly 1, whatever its direction. [2]

4. Decide whether each pair of vectors is parallel, and justify your answer.

(a) $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix}$ [1] (b) $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ 4 \\ 5 \end{pmatrix}$ [2]

10.2 Position Vectors and Geometry

Fix the origin O. The **position vector** of a point A is the arrow $\overrightarrow{OA}$ from the origin to A; its components are simply the coordinates of A. Position vectors are the bridge between the free-floating algebra of arrows and the fixed points of a geometry problem.

The arrow from one point to another is found by subtraction. Almost everything that follows rests on this one fact.

KEY $\overrightarrow{AB} = \mathbf{b} - \mathbf{a}$, the position vector of the head minus that of the tail.

The distance between A and B is then the magnitude of that displacement, $|\overrightarrow{AB}| = |\mathbf{b} - \mathbf{a}|$, and the midpoint of AB has position vector $\frac{1}{2}(\mathbf{a} + \mathbf{b})$.

EXAMPLE 10.2

Points $A(1, 2, -1)$ and $B(4, 0, 3)$ are given. Find $\overrightarrow{AB}$ and the distance AB.

$$\overrightarrow{AB} = \mathbf{b} - \mathbf{a} = \begin{pmatrix} 4 \\ 0 \\ 3 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 3 \\ -2 \\ 4 \end{pmatrix}, \qquad AB = \sqrt{3^2 + (-2)^2 + 4^2} = \sqrt{29}.$$

Head minus tail gives the arrow; the magnitude of that arrow is the distance. Every three-dimensional distance problem is this one line.

Three points A, B, C are **collinear** (on one straight line) when $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are parallel, that is, one is a scalar multiple of the other. This is the vector version of "same direction from a shared point", and the equation of a line in the next section is built from it.

Proving geometry with vectors

Vectors give a way of proving geometric facts by algebra, with no diagram-chasing and no coordinates. The method is always the same three steps.

KEY **Proving a geometric property with vectors.**

1. Choose one point as the origin, and name the position vectors of the others with single letters.
2. Write every point in the figure in terms of those letters.
3. Show the required relation by algebra.

Choosing the origin well is what keeps the algebra short. Put it at a vertex that many edges meet.

EXAMPLE 10.3

Prove that the diagonals of a parallelogram bisect each other.

Let the parallelogram be $OABC$, with the origin at O. Write $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OC} = \mathbf{c}$.

Because $OABC$ is a parallelogram, $\overrightarrow{CB} = \overrightarrow{OA} = \mathbf{a}$, so the fourth vertex is

$$\overrightarrow{OB} = \mathbf{a} + \mathbf{c}.$$

The two diagonals are OB and AC. Find the midpoint of each.

Midpoint of OB: $\frac{1}{2}(\mathbf{0} + \mathbf{a} + \mathbf{c}) = \frac{1}{2}(\mathbf{a} + \mathbf{c})$.

Midpoint of AC: $\frac{1}{2}(\mathbf{a} + \mathbf{c})$.

The two midpoints are the same point. A single point lying halfway along both diagonals is exactly what "bisect each other" means. ■

Notice that no coordinates were used, and the result holds for every parallelogram, in a plane or in space.

EXAMPLE 10.4

M and N are the midpoints of the sides OA and OB of a triangle. Prove that MN is parallel to AB and half its length.

Take the origin at O, with $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$. The midpoints are then

$$\overrightarrow{OM} = \tfrac{1}{2}\mathbf{a}, \qquad \overrightarrow{ON} = \tfrac{1}{2}\mathbf{b}.$$

Now find the two displacement vectors, each as head minus tail:

$$\overrightarrow{MN} = \tfrac{1}{2}\mathbf{b} - \tfrac{1}{2}\mathbf{a} = \tfrac{1}{2}(\mathbf{b} - \mathbf{a}), \qquad \overrightarrow{AB} = \mathbf{b} - \mathbf{a}.$$

So $\overrightarrow{MN} = \frac{1}{2}\overrightarrow{AB}$.

One vector is a scalar multiple of the other, so the two segments are parallel, and the scalar is $\frac{1}{2}$, so MN is half the length of AB. ■

The whole proof is one line of algebra. This result is known as the midpoint theorem.

YOUR TURN 10.2 Position Vectors and Geometry

5. Points $A(1, 2, -1)$ and $B(4, 0, 3)$ are given.

(a) Find $\overrightarrow{AB}$. [1]

(b) Find the distance AB. [2]

(c) Find the midpoint of AB. [2]

(d) Find $\overrightarrow{BA}$. [1]

6. Further position-vector work.

(a) A is $(1, 1, 1)$ and $\overrightarrow{AB} = \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}$. Find the coordinates of B. [2]

(b) B is the midpoint of AC, with $A(2, 1, 3)$ and $B(4, 5, 3)$. Find C. [3]

(c) Show that $A(1, 1, 1)$, $B(2, 3, 5)$ and $C(4, 7, 13)$ are collinear. [3]

7. Proving geometry with vectors. In each part, take the origin at O and write $\overrightarrow{OA} = \mathbf{a}$, $\overrightarrow{OB} = \mathbf{b}$.

(a) $OACB$ is a parallelogram. Write $\overrightarrow{OC}$ in terms of **a** and **b**. [2]

(b) Find the midpoint of the diagonal OC, and the midpoint of the diagonal AB. [3]

(c) Hence prove that the diagonals of a parallelogram bisect each other. [2]

(d) M and N are the midpoints of OA and OB. Prove that $\overrightarrow{MN} = \frac{1}{2}\overrightarrow{AB}$, and state what this shows about the two segments. [4]

10.3 The Scalar Product

There are two ways to multiply vectors, and they answer two different questions. The first, the **scalar product** (or dot product), takes two vectors and returns a number, and that number carries the angle between them. It has two forms, and they always agree.

KEY · IB **Scalar product.** For **a**, **b** with angle θ between them,

$$\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2 + a_3 b_3 \qquad \text{and} \qquad \mathbf{a} \cdot \mathbf{b} = |\mathbf{a}|\,|\mathbf{b}| \cos\theta.$$

The second form is worth pausing on, because the $\cos\theta$ is not decoration. Ordinary multiplication assumes both quantities are measured along the same line: 3×4 makes sense because both numbers sit on one number line. Two vectors usually point different ways, so multiplying their lengths would measure nothing in particular.

The repair is to bring them onto the same line first. Keep **a** as it is, and use only the part of **b** that lies along **a** — the shadow **b** casts on **a**, which has length $|\mathbf{b}|\cos\theta$. Multiplying those two gives

$$\mathbf{a}\cdot\mathbf{b} = |\mathbf{a}| \times |\mathbf{b}|\cos\theta,$$

the length of **a** times how much of **b** goes in **a**'s direction. Doing it the other way round, casting **a** onto **b**, gives $|\mathbf{b}| \times |\mathbf{a}|\cos\theta$, the same number. That is why the order does not matter.

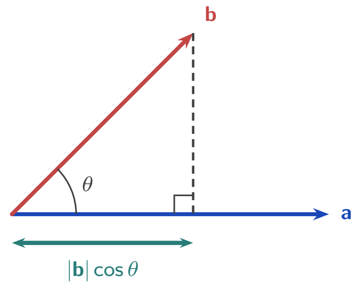


Figure 10.2 The shadow of **b** on **a**. Dropping the tip of **b** perpendicularly onto **a** leaves a length of $|\mathbf{b}|\cos\theta$, and the scalar product multiplies that by $|\mathbf{a}|$.

The scalar product then obeys the rules you would expect of a multiplication.

- **Commutative:** $\mathbf{a}\cdot\mathbf{b} = \mathbf{b}\cdot\mathbf{a}$. The order does not matter.
- **Distributive:** $\mathbf{a}\cdot(\mathbf{b}+\mathbf{c}) = \mathbf{a}\cdot\mathbf{b} + \mathbf{a}\cdot\mathbf{c}$. Brackets expand as usual.
- **Scalars move outside:** $(k\mathbf{a})\cdot\mathbf{b} = k(\mathbf{a}\cdot\mathbf{b})$. A constant factor can be taken out.
- **Dotted with itself:** $\mathbf{a}\cdot\mathbf{a} = |\mathbf{a}|^2$, since $\theta = 0$ and $\cos 0 = 1$. The result is the squared length, and this is the line used most often in proofs.

The base vectors show both extremes at once. Each has length 1 and lies along its own axis, so dotting one with itself gives $1\times1\times\cos 0 = 1$. Different axes are at right angles, so dotting two different base vectors gives 0.

$$\hat{\mathbf{i}}\cdot\hat{\mathbf{i}} = \hat{\mathbf{j}}\cdot\hat{\mathbf{j}} = \hat{\mathbf{k}}\cdot\hat{\mathbf{k}} = 1, \qquad \hat{\mathbf{i}}\cdot\hat{\mathbf{j}} = \hat{\mathbf{j}}\cdot\hat{\mathbf{k}} = \hat{\mathbf{k}}\cdot\hat{\mathbf{i}} = 0.$$

This is exactly why the component form works. Expanding $(a_1\hat{\mathbf{i}}+a_2\hat{\mathbf{j}}+a_3\hat{\mathbf{k}})\cdot(b_1\hat{\mathbf{i}}+b_2\hat{\mathbf{j}}+b_3\hat{\mathbf{k}})$ produces nine terms, and every term pairing two *different* base vectors is wiped out by that zero. Only the three matching terms survive, leaving $a_1b_1 + a_2b_2 + a_3b_3$.

The component form is easy to compute; the cosine form carries the geometry. Setting the two equal is what lets you find the angle between any two directions, and it is used throughout the chapter.

> KEY · IB $\cos\theta = \dfrac{a_1b_1 + a_2b_2 + a_3b_3}{|\mathbf{a}|\,|\mathbf{b}|}.$

EXAMPLE 10.5

Find the angle between $\mathbf{a} = \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$.

Compute the three ingredients: $\mathbf{a} \cdot \mathbf{b} = 2 + 2 + 4 = 8$, $|\mathbf{a}| = \sqrt{4+1+4} = 3$, $|\mathbf{b}| = 3$. Then

$$\cos\theta = \frac{8}{9} \implies \theta = \arccos\frac{8}{9} \approx 27.3^\circ.$$

Dot product over product of lengths: the recipe never changes, whether the vectors come from a triangle, a pair of lines, or a pair of planes.

The most important special case is a right angle. Since $\cos 90^\circ = 0$, the scalar product of perpendicular vectors is zero, and the converse holds too. This gives a test for perpendicularity that needs no angles at all, just a multiply-and-add.

KEY $\mathbf{a} \perp \mathbf{b} \iff \mathbf{a} \cdot \mathbf{b} = 0$ (for nonzero $\mathbf{a}, \mathbf{b}$).

EXAMPLE 10.6

Find the angle between $\mathbf{a} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 2 \\ 0 \\ -1 \end{pmatrix}$.

The dot product is $\mathbf{a} \cdot \mathbf{b} = (1)(2)+(2)(0)+(2)(-1) = 0$. Because it vanishes, the vectors are perpendicular: $\theta = 90^\circ$. Notice there was no need to find either magnitude, a zero dot product settles the angle instantly.

YOUR TURN 10.3 The Scalar Product

8. Evaluate each scalar product.

(a) $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 4 \\ -1 \\ 2 \end{pmatrix}$ [1]

(b) $\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ [1]

(c) $\begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$ [1]

(d) $\begin{pmatrix} 3 \\ 4 \end{pmatrix} \cdot \begin{pmatrix} 4 \\ -3 \end{pmatrix}$ [1]

9. Find the angle between each pair of vectors.

(a) $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$ [2]

(b) $\begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$ [3]

(c) $\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$ [3]

10. Perpendicularity.

(a) Given $|\mathbf{a}| = 3$, $|\mathbf{b}| = 4$ and an angle of 60° between them, find $\mathbf{a} \cdot \mathbf{b}$. [2]

(b) Find k so that $\begin{pmatrix} 2 \\ k \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} = 0$. [2]

(c) Find k so that $\begin{pmatrix} 1 \\ k \\ 2 \end{pmatrix}$ is perpendicular to $\begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix}$. [3]

(d) Explain why $\mathbf{a} \cdot \mathbf{a} = |\mathbf{a}|^2$ for every vector $\mathbf{a}$. [2]

11. A rhombus has sides $\mathbf{a}$ and $\mathbf{b}$ drawn from one vertex, so $|\mathbf{a}| = |\mathbf{b}|$.

(a) Write the two diagonals in terms of $\mathbf{a}$ and $\mathbf{b}$. [2]

(b) Expand $(\mathbf{a} + \mathbf{b}) \cdot (\mathbf{a} - \mathbf{b})$. [3]

(c) Hence prove that the diagonals of a rhombus are perpendicular. [3]

(d) Verify your result for $\mathbf{a} = \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$. [3]

10.4 The Vector Equation of a Line

A line is fixed by two pieces of information: one point it passes through, and a direction it runs along. Start at a known point $\mathbf{a}$ and move any distance along a direction $\mathbf{b}$. As the **parameter** λ takes every real value, this reaches every point on the line.

KEY · IB **Vector equation of a line:** $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b}$, where $\mathbf{a}$ is the position vector of a point on the line and $\mathbf{b}$ is a direction vector.

Writing $\mathbf{r} = (x, y, z)$ and reading off each coordinate gives the **parametric form**; eliminating λ between them gives the **Cartesian form**.

KEY · IB **Parametric:** $x = x_0 + \lambda l, \quad y = y_0 + \lambda m, \quad z = z_0 + \lambda n.$

Cartesian: $\dfrac{x - x_0}{l} = \dfrac{y - y_0}{m} = \dfrac{z - z_0}{n}$, where $\mathbf{b} = \begin{pmatrix} l \\ m \\ n \end{pmatrix}$.

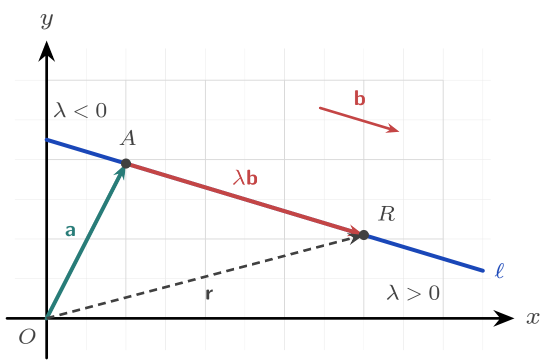


Figure 10.3 A line built from a point and a direction. The vector $\mathbf{a}$ reaches the known point A; travelling $\lambda\mathbf{b}$ from there lands on a general point R, whose position vector $\mathbf{r}$ closes the triangle. Negative λ runs back

the other way, so the whole line is covered. The direction **b** has no fixed home, so the free copy drawn above is the same vector.

The direction vector carries the geometry of how two lines relate. The **angle between two lines** is the angle between their direction vectors, found by the scalar product (take the acute value). Two lines are **parallel** when their directions are scalar multiples. And in three dimensions there is a possibility with no two-dimensional analogue: lines that are neither parallel nor crossing.

Two lines in space fall into exactly one of four cases. Two questions settle which: are the directions parallel, and do the lines share a point?

Case	Directions	Points shared	Description
Coincident	parallel	all of them	the same line written two ways
Parallel	parallel	none	different lines, same direction
Intersecting	not parallel	exactly one	they cross at a single point
Skew	not parallel	none	they pass without meeting, like an overpass above a road

Only the last case is new. Two lines in a plane must either be parallel or cross, but in three dimensions they can miss each other entirely.

To classify a pair of lines, first compare directions. If they are scalar multiples, check whether a point of one lies on the other: yes means coincident, no means parallel. If the directions differ, set the two equations equal and try to solve for the parameters: a consistent solution means they intersect (and gives the point); an inconsistent system means they are skew.

EXAMPLE 10.7

Do the lines $\mathbf{r}_1 = \begin{pmatrix}1\\0\\2\end{pmatrix} + \lambda\begin{pmatrix}1\\1\\1\end{pmatrix}$ and $\mathbf{r}_2 = \begin{pmatrix}2\\3\\1\end{pmatrix} + \mu\begin{pmatrix}1\\-1\\2\end{pmatrix}$ intersect?

The directions are not multiples of each other, so the lines are not parallel. Setting components equal:

$$1+\lambda = 2+\mu, \qquad \lambda = 3-\mu, \qquad 2+\lambda = 1+2\mu.$$

From the first two, $\lambda - \mu = 1$ and $\lambda + \mu = 3$, so $\lambda = 2$, $\mu = 1$. Check in the third: $2+2=4$

and $1 + 2(1) = 3$. Since $4 \neq 3$, the system is inconsistent, so the lines are **skew**. Two equations fix the parameters; the third decides whether the lines really meet.

Lines in motion: kinematics

So far the parameter has had no meaning attached to it. Give it one and the line equation describes a moving object. Read λ as *time* t, and $\mathbf{r} = \mathbf{a} + t\mathbf{b}$ says: start at $\mathbf{a}$, and move with constant **velocity** $\mathbf{b}$. The direction of $\mathbf{b}$ is the heading; its magnitude $|\mathbf{b}|$ is the **speed**. This is precisely the pilot from the opening page, with velocity the tip-to-tail sum of airspeed and wind.

EXAMPLE 10.8

An aircraft passes the point $(10, 5)$ (coordinates in km) at $t = 0$ and flies with constant velocity $\begin{pmatrix} 150 \\ 200 \end{pmatrix}$ km h^{-1}. Find its speed, its position at $t = 2$ hours, and whether its path passes through the airport at $(310, 405)$.

The speed is the magnitude of the velocity: $\sqrt{150^2 + 200^2} = 250$ km h^{-1}. The position at time t is

$$\mathbf{r} = \begin{pmatrix} 10 \\ 5 \end{pmatrix} + t\begin{pmatrix} 150 \\ 200 \end{pmatrix}, \qquad \text{so at } t = 2: \quad \mathbf{r} = \begin{pmatrix} 310 \\ 405 \end{pmatrix}.$$

That *is* the airport, reached at exactly $t = 2$. To test any point, solve the first coordinate for t, then check the others. One value of t must satisfy every equation at once. This is the same test used for intersecting lines.

YOUR TURN 10.4 The Vector Equation of a Line

12. Writing the equation of a line.

(a) Write a vector equation of the line through $A(1, 2, 3)$ with direction $\begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}$. [1]

(b) Find a direction vector of the line through $A(1, 0, 2)$ and $B(4, 3, 5)$. [2]

(c) Write a vector equation of that line. [2]

13. Forms of the equation.

(a) Write $\mathbf{r} = \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix} + \lambda\begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}$ in parametric form. [2]

(b) Write $\mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \lambda\begin{pmatrix} 2 \\ 1 \\ 4 \end{pmatrix}$ in Cartesian form. [2]

(c) Does $(3, 3, 2)$ lie on $\mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \lambda\begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}$? Justify your answer. [3]

14. An aircraft passes $(10, 5)$ at $t = 0$, with velocity $\begin{pmatrix} 150 \\ 200 \end{pmatrix}$ km h^{-1}.

(a) Find its speed. [2] (b) Write its position vector at time t. [2]
(c) Find its position after 2 hours. [2]

10.5 The Vector Product

The second way to multiply vectors returns not a number but a *vector*. The **vector product** (or cross product) $\mathbf{a} \times \mathbf{b}$ points perpendicular to *both* of the vectors it is built from, and its length measures the area they span. Where the dot product was about alignment, the cross product is about the plane the two vectors sweep out.

KEY · IB **Vector product.** For $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$,

$$\mathbf{a} \times \mathbf{b} = \begin{pmatrix} a_2b_3 - a_3b_2 \\ a_3b_1 - a_1b_3 \\ a_1b_2 - a_2b_1 \end{pmatrix}, \qquad \mathbf{a} \times \mathbf{b} = |\mathbf{a}|\,|\mathbf{b}| \sin\theta\, \hat{\mathbf{n}},$$

where θ is the angle between $\mathbf{a}$ and $\mathbf{b}$, and $\hat{\mathbf{n}}$ is the unit vector perpendicular to both, with direction given by the right-hand rule. Taking magnitudes of the second form gives $|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}|\,|\mathbf{b}| \sin\theta$, since $|\hat{\mathbf{n}}| = 1$.

The direction is fixed by the **right-hand rule**. Four properties are worth keeping in mind.

- **Perpendicular to both.** The result is at right angles to the plane of $\mathbf{a}$ and $\mathbf{b}$. This is what makes it the tool for building normal vectors in the next section.
- **Not commutative.** It is **anti-commutative**: $\mathbf{b} \times \mathbf{a} = -(\mathbf{a} \times \mathbf{b})$. The order matters, and reversing it reverses the answer.
- **Distributive, and scalars come out.** $\mathbf{u} \times (\mathbf{v} + \mathbf{w}) = \mathbf{u} \times \mathbf{v} + \mathbf{u} \times \mathbf{w}$ and $(k\mathbf{v}) \times \mathbf{w} = k(\mathbf{v} \times \mathbf{w})$.
- **Zero for parallel vectors.** If $\mathbf{a}$ and $\mathbf{b}$ are parallel then $\sin\theta = 0$, so $\mathbf{a} \times \mathbf{b} = \mathbf{0}$. In particular $\mathbf{v} \times \mathbf{v} = \mathbf{0}$, the mirror image of $\mathbf{v} \cdot \mathbf{v} = |\mathbf{v}|^2$.

The base vectors again show the pattern most clearly. Each is perpendicular to the other two, so crossing two different ones gives a unit vector along the third axis, and the right-hand rule fixes which way it points. Crossing one with itself gives $\mathbf{0}$, since the angle is 0 and $\sin 0 = 0$.

$$\hat{\mathbf{i}} \times \hat{\mathbf{j}} = \hat{\mathbf{k}}, \qquad \hat{\mathbf{j}} \times \hat{\mathbf{k}} = \hat{\mathbf{i}}, \qquad \hat{\mathbf{k}} \times \hat{\mathbf{i}} = \hat{\mathbf{j}}, \qquad \hat{\mathbf{i}} \times \hat{\mathbf{i}} = \hat{\mathbf{j}} \times \hat{\mathbf{j}} = \hat{\mathbf{k}} \times \hat{\mathbf{k}} = \mathbf{0}.$$

Read the first three in the cyclic order $\hat{\mathbf{i}} \to \hat{\mathbf{j}} \to \hat{\mathbf{k}} \to \hat{\mathbf{i}}$: going forwards gives a plus, and going backwards gives a minus, so $\hat{\mathbf{j}} \times \hat{\mathbf{i}} = -\hat{\mathbf{k}}$.

One array, two products

Both products come out of the same nine numbers. Multiply every component of **a** by every component of **b** and lay the results in a square.

	b_1	b_2	b_3
a_1	a_1b_1	a_1b_2	a_1b_3
a_2	a_2b_1	a_2b_2	a_2b_3
a_3	a_3b_1	a_3b_2	a_3b_3

The **diagonal** is the scalar product. Those are the three terms where a base vector meets itself, and $\hat{\mathbf{i}} \cdot \hat{\mathbf{i}} = 1$ keeps them; everything off the diagonal is killed by $\hat{\mathbf{i}} \cdot \hat{\mathbf{j}} = 0$. Adding the shaded entries gives $a_1b_1 + a_2b_2 + a_3b_3$.

The **off-diagonal** entries are the vector product, taken in opposite pairs. Here the diagonal is what dies, since $\hat{\mathbf{i}} \times \hat{\mathbf{i}} = \mathbf{0}$, and each pair of mirror-image entries subtracts to give one component:

$$\underbrace{a_2b_3 - a_3b_2}_{\text{the pair missing 1}}, \qquad \underbrace{a_3b_1 - a_1b_3}_{\text{the pair missing 2}}, \qquad \underbrace{a_1b_2 - a_2b_1}_{\text{the pair missing 3}}.$$

So the component of $\mathbf{a} \times \mathbf{b}$ in a given direction is built from the two rows and columns that leave that direction out — which is exactly why the answer is perpendicular to both. The two products are the same array read two ways: down the diagonal, or across it.

Area of a parallelogram and a triangle

The magnitude has a clean geometric meaning: it is the area of the parallelogram with **a** and **b** as sides. To see why, read the two factors of $|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}|\,|\mathbf{b}|\sin\theta$ separately. Take $|\mathbf{a}|$ as the base. Then $|\mathbf{b}|\sin\theta$ is the perpendicular height, because **b** is the slanted side and $\sin\theta$ keeps only the part of it at right angles to the base. So the magnitude is base $\times$ height, which is the area.

> KEY · IB Area of a parallelogram $= |\mathbf{a} \times \mathbf{b}|$, where **a** and **b** are two adjacent sides.

A triangle is half of such a parallelogram, so its area is $\frac{1}{2}|\mathbf{a} \times \mathbf{b}|$; remember the half, because the booklet prints only the parallelogram.

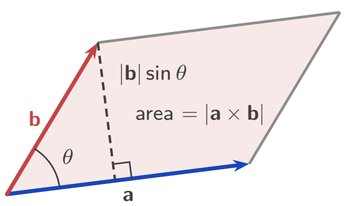


Figure 10.4 The parallelogram spanned by **a** and **b**. Its area is $|\mathbf{a} \times \mathbf{b}|$, and a triangle on the same two sides

has half that area.

EXAMPLE 10.9

Find the area of the triangle with vertices $A(0,0,0)$, $B(2,1,0)$, $C(1,0,2)$.

Take $\overrightarrow{AB} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$ and $\overrightarrow{AC} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}$. Their cross product is

$$\overrightarrow{AB} \times \overrightarrow{AC} = \begin{pmatrix} (1)(2) - (0)(0) \\ (0)(1) - (2)(2) \\ (2)(0) - (1)(1) \end{pmatrix} = \begin{pmatrix} 2 \\ -4 \\ -1 \end{pmatrix}.$$

Its magnitude is $\sqrt{4+16+1} = \sqrt{21}$, so the triangle's area is $\frac{1}{2}\sqrt{21}$. The cross product does the work here: one calculation and one square root replace any trigonometry.

YOUR TURN 10.5 The Vector Product

15. Evaluate each vector product.

(a) $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ [1]

(b) $\begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ 3 \\ 0 \end{pmatrix}$ [1]

(c) $\begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}$ [2]

(d) $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \times \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}$ [3]

16. Areas and magnitudes.

(a) Given $|\mathbf{a}| = 3$, $|\mathbf{b}| = 5$ and an angle of 30°, find $|\mathbf{a} \times \mathbf{b}|$. [2]

(b) Find the area of the parallelogram with sides $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix}$. [2]

(c) Find the area of the triangle with vertices $A(0,0,0)$, $B(2,1,0)$, $C(1,0,2)$. [3]

(d) Explain why $\mathbf{v} \times \mathbf{v} = \mathbf{0}$ for every vector $\mathbf{v}$. [2]

10.6 Planes

A plane is a flat, endless sheet, and like a line it can be built from a point and some directions. A line needed only one direction, because a line is one-dimensional: one number, the parameter, is enough to say where you are on it. A plane is two-dimensional, so it needs two directions and two numbers. Anchor at $\mathbf{a}$, then travel λ lots of $\mathbf{b}$ and μ lots of $\mathbf{c}$. The pair (λ, μ) acts as a coordinate system inside the sheet, and every point of the plane has exactly one such pair. The two directions must not be parallel: if they were, every combination of them would still point along a single line, and the sheet would collapse to that line.

KEY · IB **Vector equation of a plane:** $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b} + \mu\mathbf{c}$.

Two parameters make this form awkward to use. It is hard to tell whether a point lies on the plane, or whether two such equations describe the same plane. A better description uses not the directions the plane contains but the single direction it does *not*: the **normal vector** $\mathbf{n}$, perpendicular to the whole sheet. Every displacement within the plane is perpendicular to $\mathbf{n}$, and the scalar product states exactly that.

KEY · IB **Normal (scalar-product) form:** $\mathbf{r} \cdot \mathbf{n} = \mathbf{a} \cdot \mathbf{n}$,

Cartesian form: $n_1x + n_2y + n_3z = d$, where $\mathbf{n} = \begin{pmatrix} n_1 \\ n_2 \\ n_3 \end{pmatrix}$ and $d = \mathbf{a} \cdot \mathbf{n}$.

The picture behind this is worth holding on to. Take any point R on the plane. The displacement $\mathbf{r} - \mathbf{a}$ from the known point A to R lies flat in the sheet, so it is perpendicular to $\mathbf{n}$ and its scalar product with $\mathbf{n}$ is zero. Expanding $(\mathbf{r} - \mathbf{a}) \cdot \mathbf{n} = 0$ gives the form above.

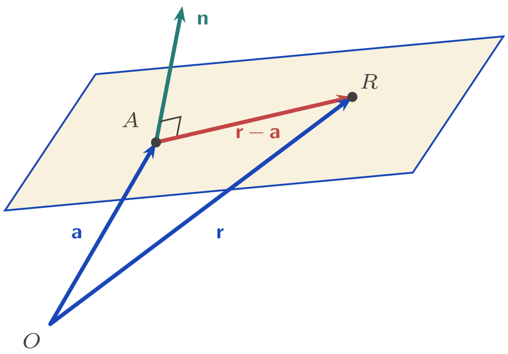


Figure 10.5 A plane held by one point and one perpendicular direction. Every displacement inside the sheet, such as $\mathbf{r} - \mathbf{a}$, is at right angles to the normal $\mathbf{n}$, so $(\mathbf{r} - \mathbf{a}) \cdot \mathbf{n} = 0$ for every point R of the plane, and for no other point.

A Cartesian plane equation can now be read directly: the coefficients of x, y, z *are* the normal vector. To find a plane through three points, use the cross product of two edge vectors to build $\mathbf{n}$, then fix d from any one point.

EXAMPLE 10.10

Find the Cartesian equation of the plane with normal $\mathbf{n} = \begin{pmatrix} 3 \\ -2 \\ 2 \end{pmatrix}$ through $A(4, 8, -1)$.

With $\mathbf{n} = (3, -2, 2)$ the equation is $3x - 2y + 2z = d$. Substitute A to find d:

$$d = 3(4) - 2(8) + 2(-1) = 12 - 16 - 2 = -6, \quad \text{so} \quad 3x - 2y + 2z = -6.$$

The normal supplies the coefficients and a single point supplies the constant. That is the whole method.

When no normal is given, the cross product produces one.

EXAMPLE 10.11

Find the Cartesian equation of the plane through $A(2,0,0)$, $B(0,3,0)$, $C(0,0,6)$.

Two edge vectors span the plane: $\overrightarrow{AB} = \begin{pmatrix} -2 \\ 3 \\ 0 \end{pmatrix}$ and $\overrightarrow{AC} = \begin{pmatrix} -2 \\ 0 \\ 6 \end{pmatrix}$. Their cross product is normal to both, hence to the plane:

$$\mathbf{n} = \overrightarrow{AB} \times \overrightarrow{AC} = \begin{pmatrix} (3)(6) - (0)(0) \\ (0)(-2) - (-2)(6) \\ (-2)(0) - (3)(-2) \end{pmatrix} = \begin{pmatrix} 18 \\ 12 \\ 6 \end{pmatrix} = 6 \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}.$$

Any scalar multiple of a normal is a normal, so use the cleaner $(3,2,1)$. Through A: $d = 3(2) = 6$, giving $3x + 2y + z = 6$. Check with B: $2(3) = 6$; and C: $1(6) = 6$. Both lie on it. Scaling the normal down before computing d keeps every later number small.

That example settles the stool from the opening page. Any three points not in a straight line give two edge vectors; their cross product is a normal, and a normal with a point fixes a plane. So three points always have exactly one plane through them, and three feet always reach the floor. A fourth point comes with no such guarantee, and the fourth leg is the one left hanging.

Angles with a plane

Two planes meet at an angle, and that angle is the angle between their normals, computed by the ordinary scalar product. To see why, look along the line where the planes meet, so that each plane is seen edge-on and appears as a line. Turning one arm of an angle through 90°, and then the other arm through 90°, leaves the angle unchanged. The normals are exactly the two arms turned in this way, so they carry the same angle.

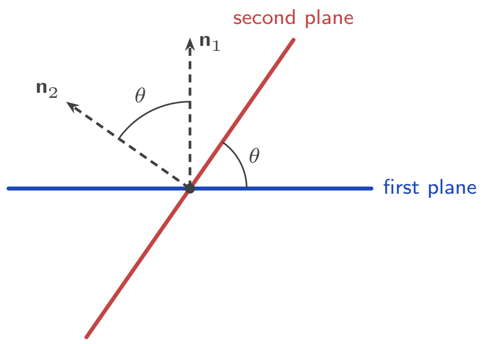


Figure 10.6 Two planes seen end-on, looking straight along the line where they meet. Each plane shows as a line. The angle between the planes and the angle between their normals are the same angle.

Taking the absolute value of the dot product guarantees the acute answer.

KEY Angle between two planes (normals $\mathbf{n}_1, \mathbf{n}_2$): $\cos\theta = \dfrac{|\mathbf{n}_1 \cdot \mathbf{n}_2|}{|\mathbf{n}_1|\,|\mathbf{n}_2|}$.

EXAMPLE 10.12

Find the acute angle between the planes $x + 2y - z = 3$ and $2x - y + z = 1$.

Read the normals straight off the coefficients: $\mathbf{n}_1 = (1, 2, -1)$ and $\mathbf{n}_2 = (2, -1, 1)$. Then $\mathbf{n}_1 \cdot \mathbf{n}_2 = 2 - 2 - 1 = -1$, and both magnitudes are $\sqrt{6}$, so

$$\cos\theta = \frac{|-1|}{\sqrt{6}\cdot\sqrt{6}} = \frac{1}{6} \implies \theta \approx 80.4^\circ.$$

The dot product came out negative, which means the angle between the normals is obtuse. Taking the absolute value gives the acute angle instead, which is what the question asks for.

A line and a plane need one extra step. The angle we want is between the line and its projection *in* the plane: the shadow the line would cast if light fell straight down onto the sheet. But the scalar product gives the angle φ between the line and the *normal*. These two angles add to 90°, so subtract from 90° to get the one we want.

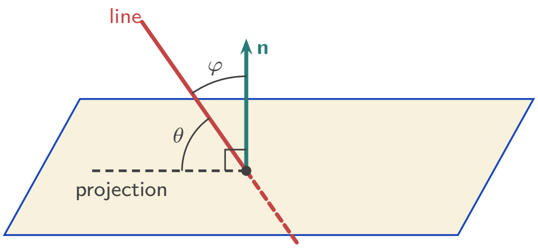


Figure 10.7 The angle between a line and a plane is measured to the line's projection in the plane, marked θ, not to the normal. Since the normal stands at 90° to the projection, the two angles at the point of entry satisfy $\theta + \varphi = 90^\circ$.

KEY Angle between a line (direction $\mathbf{d}$) and a plane (normal $\mathbf{n}$): $\sin\theta = \dfrac{|\mathbf{d}\cdot\mathbf{n}|}{|\mathbf{d}|\,|\mathbf{n}|}$.

Using sin instead of cos is what builds the "90° minus" into the formula. The line's angle to the plane is the complement of its angle to the normal.

Intersections: line and plane, plane and plane

To find where a line meets a plane, put the line's parametric coordinates into the plane's Cartesian equation. That gives one equation in λ. Solve it, then substitute back to get the point.

EXAMPLE 10.13

Find where the line $\mathbf{r} = \begin{pmatrix}1\\0\\2\end{pmatrix} + \lambda\begin{pmatrix}2\\1\\-1\end{pmatrix}$ meets the plane $3x - 2y + 2z = 11$.

Parametrically, $x = 1 + 2\lambda$, $y = \lambda$, $z = 2 - \lambda$. Substituting into the plane,

$$3(1+2\lambda) - 2\lambda + 2(2-\lambda) = 11 \implies 2\lambda + 7 = 11 \implies \lambda = 2,$$

so the point is $(5, 2, 0)$. Check: $3(5) - 2(2) + 2(0) = 11$. One substitution, one linear equation, one point. If instead the λ terms had cancelled, the line would be parallel to the plane. There is then no solution if the constants disagree, and the whole line lies in the plane if they agree.

Two non-parallel planes do not meet at a point; they meet along a whole **line**, like the spine of an open book. That line runs within both planes, so its direction is perpendicular to both normals, and the cross product builds it directly.

EXAMPLE 10.14

Find a vector equation of the line of intersection of the planes $x + y + z = 6$ and $2x - y + z = 3$.

The direction is perpendicular to both normals:

$$\mathbf{d} = \mathbf{n}_1 \times \mathbf{n}_2 = \begin{pmatrix}1\\1\\1\end{pmatrix} \times \begin{pmatrix}2\\-1\\1\end{pmatrix} = \begin{pmatrix}2\\1\\-3\end{pmatrix}.$$

For a point on the line, pick a coordinate and set it to zero. With $z = 0$: $x + y = 6$ and $2x - y = 3$, so $x = 3$, $y = 3$. Hence

$$\mathbf{r} = \begin{pmatrix}3\\3\\0\end{pmatrix} + t\begin{pmatrix}2\\1\\-3\end{pmatrix}.$$

Check the direction: $(2, 1, -3) \cdot (1, 1, 1) = 0$ and $(2, 1, -3) \cdot (2, -1, 1) = 0$. Perpendicular to both normals, as the geometry demands. Setting $z = 0$ fails only if the line never crosses the plane $z = 0$. In that case set $x = 0$ or $y = 0$ instead.

Three planes

Three planes form a system of three linear equations, the same systems solved by row reduction in Ch. 4. Nothing new has to be calculated here. What is new is that every algebraic outcome now has a shape, and the shape is often easier to remember than the algebra.

There are five arrangements worth knowing. The first two have solutions; the last three do not, and they are told apart by how many of the normals are parallel.

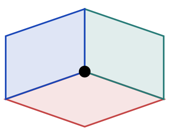
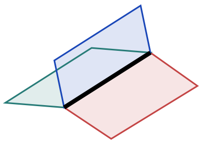
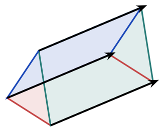
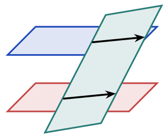
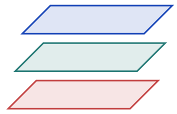
Unique solution	Infinitely many solutions	No solutions (inconsistent system)		
		No normals parallel	Two normals parallel	Three normals parallel
				
Three planes meet at a point	Three planes meet along a line	Three planes form a prism	One plane cutting two parallel planes	Three parallel planes

Figure 10.8 The five arrangements of three planes. Only the first has a single solution; the second has a whole line of them. The last three have none, and the number of parallel normals says which one you are looking at.

One further case is possible but rarely asked: if the three equations are multiples of one another, they all describe the same plane, and every point of that plane is a solution.

Row reduction tells you which case you are in without any picture. A row of zeros with a zero on the right means a dependent system, so the planes share a line. A row of zeros with a non-zero on the right means an inconsistent one, so there is no common point. The pictures only tell you what that answer looks like.

EXAMPLE 10.15

The planes $x + y + z = 6$ and $2x - y + z = 3$ from the previous example are joined by a third, $3x + 2z = c$. Describe the configuration of the three planes for $c = 9$ and for $c \neq 9$.

Adding the first two equations gives $3x + 2z = 9$, so every point on their line of intersection satisfies $3x + 2z = 9$. Indeed, check the line $\mathbf{r} = (3, 3, 0) + t(2, 1, -3)$ in the third plane: $3(3 + 2t) + 2(-3t) = 9 + 6t - 6t = 9$, for every t.

So when $c = 9$ the third plane contains the entire line, and all three planes share it, like an open book with three pages on one spine. When $c \neq 9$ the third plane is parallel to that line but misses it. The three planes then bound a triangular prism: each pair meets in a line, the three lines are parallel, and no point lies on all three. In the language of Ch. 4, the system is dependent when $c = 9$ and inconsistent otherwise; the geometry is what those words look like.

YOUR TURN 10.6 Planes

17. Equations of planes.

(a) Write the Cartesian equation of the plane with normal $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ through the origin. [1]

(b) State a normal vector to the plane $2x - y + 3z = 5$. [1]

(c) Find the Cartesian equation of the plane with normal $\begin{pmatrix}1\\1\\1\end{pmatrix}$ through $(2, 0, 1)$. [2]

(d) Does $(1, 1, 1)$ lie on the plane $x + y + z = 3$? [1]

18. Planes from points, and angles.

(a) Find the Cartesian equation of the plane through $A(2, 0, 0)$, $B(0, 3, 0)$ and $C(0, 0, 6)$. [4]

(b) Find the acute angle between the planes $x + 2y - z = 3$ and $2x - y + z = 1$. [4]

19. Intersections.

(a) Find where the line $\mathbf{r} = \begin{pmatrix}1\\0\\2\end{pmatrix} + \lambda\begin{pmatrix}2\\1\\-1\end{pmatrix}$ meets the plane $3x - 2y + 2z = 11$. [4]

(b) Find a direction vector of the line of intersection of $x+y+z = 6$ and $2x-y+z = 3$. [3]

(c) Find the angle between the line in part (a) and the plane $3x - 2y + 2z = 11$. [4]

10.7 Distances

Two objects that never meet are still some distance apart, and asking for that distance always means the same thing: the *shortest* one. There is a single idea behind every case in this section, and it needs no new formula.

> KEY **The shortest link is perpendicular.** Of all the segments joining two objects, the shortest is the one at right angles to what it joins. Perpendicular means a scalar product of zero, so the geometry turns straight into an equation.

That gives a method rather than a formula to memorise. Nothing here is printed in the formula booklet, so working it out is the point.

KEY **Finding a shortest distance.**

1. Write the unknown point on each object in terms of its parameter.
2. Set the scalar product of the joining vector with each direction it must be perpendicular to equal to zero.
3. Solve for the parameter or parameters, which locates the feet of the perpendicular.
4. Find the length of the joining vector.

From a point to a line

A general point on the line is $F = \mathbf{a} + \lambda\mathbf{b}$. As λ varies, F slides along the line and the length PF changes. It is shortest exactly when $\overrightarrow{PF}$ is perpendicular to the line, so $\overrightarrow{PF} \cdot \mathbf{b} = 0$. That is one equation in one unknown.

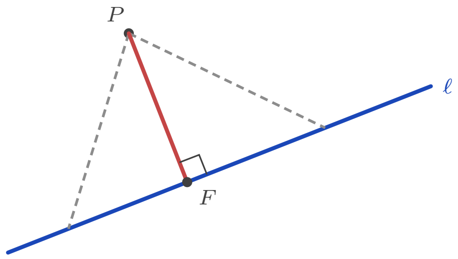


Figure 10.9 Every dashed segment joins P to the line. The shortest of them meets the line at right angles, at the foot of the perpendicular F.

EXAMPLE 10.16

Find the distance from $P(0, 1, 3)$ to the line $\mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$.

A general point on the line is $F = (1 + \lambda,\ 2\lambda,\ -1 + 2\lambda)$, so

$$\overrightarrow{PF} = \begin{pmatrix} 1 + \lambda \\ 2\lambda - 1 \\ 2\lambda - 4 \end{pmatrix}.$$

Now make it perpendicular to the direction:

$$\overrightarrow{PF} \cdot \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} = (1 + \lambda) + 2(2\lambda - 1) + 2(2\lambda - 4) = 9\lambda - 9 = 0,$$

so $\lambda = 1$ and the foot is $F(2, 2, 1)$. Then $\overrightarrow{PF} = (2, 1, -2)$ and the distance is $\sqrt{4 + 1 + 4} = 3$. Notice what the method gives you for free: not just the distance but the point on the line that is closest to P, which many questions ask for as well.

From a point to a plane

For a plane the perpendicular direction is already known: it is the normal $\mathbf{n}$. So no scalar product needs to be set to zero. Instead, walk from P along $\mathbf{n}$ until you land on the plane. The general point on that walk is $\mathbf{p} + t\mathbf{n}$, and putting it into $\mathbf{r} \cdot \mathbf{n} = d$ gives one equation in t. The distance travelled is then $|t|\,|\mathbf{n}|$.

EXAMPLE 10.17

Find the distance from $P(1,1,1)$ to the plane $2x - y + 2z = 12$.

The normal is $\mathbf{n} = (2, -1, 2)$, so a point on the walk from P is

$$\mathbf{p} + t\mathbf{n} = (1 + 2t,\ 1 - t,\ 1 + 2t).$$

This lies on the plane when

$$2(1 + 2t) - (1 - t) + 2(1 + 2t) = 12 \implies 9t + 3 = 12 \implies t = 1.$$

The foot is $(3, 0, 3)$, and the distance is $|t|\,|\mathbf{n}| = 1 \times 3 = 3$. Check the foot on the plane: $6 - 0 + 6 = 12$. ✓

Between two skew lines

Skew lines never meet, yet there is still a shortest gap between them, and the same idea finds it. Take a general point on each, P_1 on the first and P_2 on the second. The joining vector $\overrightarrow{P_1P_2}$ is shortest when it is perpendicular to *both* directions at once. That is two scalar products set to zero, so two equations in the two unknowns λ and μ.

EXAMPLE 10.18

Find the shortest distance between $l_1 : \mathbf{r} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ and $l_2 : \mathbf{r} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$.

General points are $P_1 = (\lambda,\ \lambda,\ 1 + \lambda)$ and $P_2 = (1 + \mu,\ 1,\ \mu)$, so

$$\overrightarrow{P_1P_2} = \begin{pmatrix} 1 + \mu - \lambda \\ 1 - \lambda \\ \mu - 1 - \lambda \end{pmatrix}.$$

Perpendicular to both directions gives two equations:

$$\overrightarrow{P_1P_2} \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = 1 - 3\lambda + 2\mu = 0, \qquad \overrightarrow{P_1P_2} \cdot \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = 2\mu - 2\lambda = 0.$$

The second gives $\mu = \lambda$; the first then gives $1 - \lambda = 0$, so $\lambda = \mu = 1$. The feet are $P_1(1, 1, 2)$

and $P_2(2,1,1)$, and

$$\overrightarrow{P_1P_2} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \qquad \text{distance} = \sqrt{2}.$$

Two ordinary simultaneous equations, and the answer comes with both endpoints of the shortest segment attached.

The remaining cases

Nothing else needs a new method. Every other pairing either meets, and so has distance zero, or is parallel, and then any convenient point reduces it to one of the three cases above.

The two objects	What to do
Two parallel lines	Take any point on one line, then use point to line
Two intersecting lines	They meet, so the distance is 0
Two skew lines	Two scalar products set to zero, as above
A line parallel to a plane	Take any point on the line, then use point to plane
A line meeting a plane	They meet, so the distance is 0
Two parallel planes	Take any point on one plane, then use point to plane
Two non-parallel planes	They meet along a line, so the distance is 0

The first step in any of these questions is therefore not calculation but classification. Decide whether the objects meet, and if they do not, decide which of the three methods the pair reduces to.

EXAMPLE 10.19

Find the distance between the parallel planes $2x - y + 2z = 12$ and $2x - y + 2z = 3$.
Both have normal $(2, -1, 2)$, so they are parallel and never meet. Take any point on the second plane; setting $y = z = 0$ gives $x = \frac{3}{2}$, so $A(\frac{3}{2}, 0, 0)$ lies on it. Now walk from A along the normal to the first plane:

$$2\left(\tfrac{3}{2} + 2t\right) - (-t) + 2(2t) = 12 \implies 9t + 3 = 12 \implies t = 1,$$

so the distance is $|t|\,|\mathbf{n}| = 3$. Any other starting point on the second plane gives the same answer, which is what parallel means.

YOUR TURN 10.7 Distances

20. Distance from a point to a line. Throughout, $l : \mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$.

(a) Find the foot of the perpendicular from $P(4, 0, 2)$ to l, and hence the distance from P to l. [4]

(b) Find the distance from $Q(1, 4, 4)$ to l. [3]

(c) The line m has equation $\mathbf{r} = \begin{pmatrix} 4 \\ 0 \\ 2 \end{pmatrix} + t \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$. Explain why m is parallel to l, and write down the distance between them. [2]

(d) Check your answer to part (c) by repeating the method with the point on m where $t = 1$. [3]

21. Distance from a point to a plane. Throughout, $\Pi : x + 2y + 2z = 14$.

(a) Find the foot of the perpendicular from $P(1, 1, 1)$ to Π, and hence the distance from P to Π. [4]

(b) Find the distance from the origin to Π. [3]

(c) Find the distance between Π and the parallel plane $x + 2y + 2z = 5$. [3]

(d) Show that the line $\mathbf{r} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix}$ is parallel to Π, and write down the distance between the line and the plane. [3]

22. Distance between two lines. Throughout, $l_1 : \mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ and $l_2 : \mathbf{r} = \begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$.

(a) Show that l_1 and l_2 are skew. [4]

(b) Write down general points P_1 on l_1 and P_2 on l_2, and find the values of λ and μ for which $\overrightarrow{P_1P_2}$ is perpendicular to both direction vectors. [5]

(c) Hence state the two feet of the common perpendicular and the shortest distance between l_1 and l_2. [3]

(d) The lines $n_1 : \mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$ and $n_2 : \mathbf{r} = \begin{pmatrix} 2 \\ 3 \\ 3 \end{pmatrix} + \mu \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$ are not skew. Apply the same method to them and explain what your answer means. [4]

Chapter 10 · Vectors

Reflections

TOK Both operations are called products, yet neither behaves like multiplication. The dot product takes two vectors and gives back a number. The cross product changes sign when you swap the order, and returns zero for two vectors that are not themselves zero.

So is either one really multiplication, or have we borrowed a comfortable word for something new? And why these definitions in particular? Nothing forced $|\mathbf{a}||\mathbf{b}|\cos\theta$ on us. It was chosen because it answered a question people wanted answered. How much of mathematics is discovered, and how much is chosen and then found to be useful?

TOK The same plane can be a drawing, a vector equation, or the single line $2x - y + 3z = 7$. The symbols are easier to calculate with. The picture is easier to believe.

If the two disagree in your head, which one do you trust, and what happens in four dimensions, where the picture is no longer available to you?

RW Every three-dimensional game and animated film runs on this chapter. The dot product decides how brightly a surface is lit, by comparing the direction of the light with the direction the surface faces: a value near 1 points at the light and is bright, a value near 0 faces edge-on and stays dark. The cross product builds the normals that say which way each surface points in the first place. At the same time the physics engine adds force and velocity vectors tip to tail, thousands of times a second, which is why a thrown object curves the way your eye expects.

EXT The cross product only exists because we live in three dimensions.

In two dimensions there is no room for a vector perpendicular to two others. In four dimensions or more there is too much room: no single perpendicular direction is picked out, because a whole family of them works. Three dimensions is the only case where two vectors determine exactly one perpendicular line. That is why $\mathbf{a} \times \mathbf{b}$ can be a vector at all.

End-of-Chapter Problems

1. Given $\mathbf{a} = \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 1 \\ 3 \\ -1 \end{pmatrix}$:

(a) find $\mathbf{a} + 2\mathbf{b}$; [2]

(b) find $|\mathbf{a}|$; [1]

(c) find the unit vector in the direction of $\mathbf{a}$. [2]

2. For $A(1, -1, 2)$ and $B(3, 2, 4)$:

(a) find $\overrightarrow{AB}$; [1]

(b) find $|\overrightarrow{AB}|$; [2]

(c) find the midpoint of AB. [2]

3. The vectors $\mathbf{a} = \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 1 \\ 2 \\ k \end{pmatrix}$ are perpendicular. Find k. [3]

4. Find the angle between $\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$. [4]

5. Consider the line $l : \mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}$.

(a) Find the point on l when $\lambda = 2$. [2]

(b) Determine whether $P(5, 2, 3)$ lies on l. [2]

6. Find, in degrees, the acute angle between two lines with direction vectors $\begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$ and $\begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix}$. [4]

7. Given $\mathbf{a} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 2 \\ 0 \\ -1 \end{pmatrix}$, find $\mathbf{a} \times \mathbf{b}$ and $|\mathbf{a} \times \mathbf{b}|$. [4]

8. Find the area of the triangle with vertices $A(1, 0, 0)$, $B(0, 1, 0)$, $C(0, 0, 1)$. [4]

9. A plane Π has normal $\begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$ and passes through $A(1, 2, -1)$.

(a) Find the Cartesian equation of Π. [3]

(b) Determine whether $B(2, 1, 0)$ lies on Π. [1]

10. Find the Cartesian equation of the plane through $A(1, 0, 0)$, $B(0, 1, 0)$ and $C(0, 0, 1)$. [4]

11. Given $\mathbf{a} = \begin{pmatrix} 4 \\ -3 \\ 0 \end{pmatrix}$:

(a) find $|\mathbf{a}|$; [1]
(b) find the unit vector $\hat{\mathbf{a}}$; [2]
(c) find the vector of magnitude 10 in the direction of $\mathbf{a}$. [2]

12. Points $A(2,1,3)$ and $B(4,5,3)$ are given. Find the point C such that B is the mid-point of AC. [3]

13. The line $l : \mathbf{r} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$ meets the plane $z = 4$. Find the point of intersection. [4]

14. Find the angle between the vectors $\begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$. [4]

15. Show that the points $A(1,2,3)$, $B(2,4,5)$ and $C(4,8,9)$ are collinear. [4]

16. Given $|\mathbf{a}| = 5$, $|\mathbf{b}| = 3$ and $\mathbf{a} \cdot \mathbf{b} = 9$, find the angle between $\mathbf{a}$ and $\mathbf{b}$. [4]

17. The lines $l_1 : \mathbf{r} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$ and $l_2 : \mathbf{r} = \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ intersect. Find the point of intersection. [5]

18. The plane Π has equation $2x + y - 2z = 6$.

(a) State a normal vector to Π. [1]
(b) Find where the line $\mathbf{r} = \lambda \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}$ meets Π. [4]

19. Given $\mathbf{a} = \begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}$:

(a) find $\mathbf{a} \cdot \mathbf{b}$; [2]
(b) hence state the angle between $\mathbf{a}$ and $\mathbf{b}$. [2]

20. Find the area of the triangle with vertices $A(0,0,0)$, $B(3,0,0)$, $C(0,4,0)$ using the vector product. [4]

21. Find a vector equation of the line through $A(2,-1,3)$ and $B(4,1,7)$. [4]

22. Find the acute angle between the planes $x + y + z = 1$ and $x - y + z = 3$. [5]

23. Given $\mathbf{a} = \begin{pmatrix} 3 \\ 4 \\ 0 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 0 \\ 0 \\ 5 \end{pmatrix}$, find $\mathbf{a} \times \mathbf{b}$ and $|\mathbf{a} \times \mathbf{b}|$. [4]

24. Find the value of t for which $\begin{pmatrix} 1 \\ t \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ -1 \\ t \end{pmatrix}$ are perpendicular. [4]

25. At time $t = 0$ seconds a drone is at the point $(2,-1,10)$ (coordinates in metres) and moves with constant velocity $\begin{pmatrix} 3 \\ 4 \\ 0 \end{pmatrix}$ m s^{-1}.

(a) Write down a vector equation for the drone's position at time t. [2]

(b) Find the drone's speed. [2]

(c) Determine whether the drone passes through the point $(17, 19, 10)$, and if so, at what time. [3]

26. Find a vector equation of the line of intersection of the planes $x + y - z = 2$ and $2x - y + z = 1$. [6]

27. Consider the planes $\Pi_1 : x + y + z = 3$, $\Pi_2 : x - y + z = 1$ and $\Pi_3 : x + z = c$, where $c \in \mathbb{R}$.

(a) Find a vector equation of the line of intersection of Π_1 and Π_2. [4]

(b) Show that when $c = 2$ the three planes intersect in this line. [2]

(c) Describe geometrically the configuration of the three planes when $c \neq 2$. [2]

28. Show that the lines $l_1 : \mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$ and $l_2 : \mathbf{r} = \begin{pmatrix} 2 \\ 3 \\ 3 \end{pmatrix} + \mu \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$ intersect, and find the point of intersection. [6]

29. Find the shortest distance from the point $P(1, 2, 3)$ to the plane $2x - y + 2z = 0$, by finding the foot of the perpendicular from P. [6]

30. Find a unit vector perpendicular to both $\mathbf{a} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 2 \\ 0 \\ -1 \end{pmatrix}$. [5]

31. The points $A(1, 0, 0)$, $B(0, 2, 0)$ and $C(0, 0, 3)$ are given.

(a) Find the Cartesian equation of the plane ABC. [3]

(b) Find the area of triangle ABC. [3]

32. Find the angle that the line $l : \mathbf{r} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$ makes with the plane $x + 2y + 2z = 5$. [6]

33. The points $A(2, 0, 1)$, $B(1, 3, 2)$ and $C(4, 1, 0)$ are given, and O is the origin.

(a) Find $\overrightarrow{AB}$ and $\overrightarrow{AC}$. [2]

(b) Find $\overrightarrow{AB} \times \overrightarrow{AC}$. [3]

(c) Find the exact area of triangle ABC. [2]

(d) Find the Cartesian equation of the plane through A, B and C. [4]

(e) Find the exact shortest distance from O to that plane. [3]

(f) Hence find the volume of the tetrahedron $OABC$. [2]

34. Consider the lines

$$l_1 : \mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}, \qquad l_2 : \mathbf{r} = \begin{pmatrix} 4 \\ 1 \\ 4 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}.$$

(a) Show that l_1 and l_2 are not parallel. [2]
(b) Determine whether the lines intersect, and if so find the point of intersection. [5]
(c) Find the acute angle between the lines. [3]
(d) Find the Cartesian equation of the plane containing both lines. [4]
(e) Explain why such a plane exists only because of your answer to part (b). [2]

35. GDC Two aircraft fly at the same altitude. At time t hours their position vectors, in km, are

$$\mathbf{r}_A = \begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix} + t\begin{pmatrix} 300 \\ 400 \\ 0 \end{pmatrix}, \qquad \mathbf{r}_B = \begin{pmatrix} 600 \\ 0 \\ 2 \end{pmatrix} + t\begin{pmatrix} 0 \\ 500 \\ 0 \end{pmatrix}.$$

(a) Find the speed of each aircraft. [3]
(b) Find $\overrightarrow{AB}$ at time t. [3]
(c) Show that the distance d between them satisfies $d^2 = 100\,000t^2 - 360\,000t + 360\,000$. [4]
(d) Find the time at which the aircraft are closest, and the least distance between them. [4]

Challenge Questions

36. **The shortest distance from a point to a line.** Section 10.7 works this out one case at a time. Here you build a formula that does every case, then a second formula, and show the two agree.

Let l be the line $\mathbf{r} = \mathbf{a} + \lambda\mathbf{d}$, and P a point not on it.

(a) Explain why the shortest distance from P to l is measured along the perpendicular. [2]
(b) Let $F = \mathbf{a} + \lambda\mathbf{d}$ be the foot of that perpendicular. Explain why $\overrightarrow{PF} \cdot \mathbf{d} = 0$, and use this to show that

$$\lambda = \frac{(\mathbf{p} - \mathbf{a}) \cdot \mathbf{d}}{|\mathbf{d}|^2}.$$

[5]
(c) For $l : \mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + \lambda\begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}$ and $P(4, 3, 1)$, find λ, the foot F, and the distance. [5]
(d) A second formula is $\text{distance} = \dfrac{|\overrightarrow{AP} \times \mathbf{d}|}{|\mathbf{d}|}$, where A is any point on l. Check that it agrees with part (c). [4]
(e) Explain why the second formula works, using $|\mathbf{u} \times \mathbf{v}| = |\mathbf{u}||\mathbf{v}| \sin\theta$. *(Hint: what is $|\overrightarrow{AP}| \sin\theta$ in the diagram?)* [4]

37. **How far apart are two skew lines?** Two skew lines never meet, yet there is still a shortest distance between them. This investigation finds it.

Let $l_1 : \mathbf{r} = \mathbf{a}_1 + \lambda\mathbf{d}_1$ and $l_2 : \mathbf{r} = \mathbf{a}_2 + \mu\mathbf{d}_2$ be skew.

(a) Show that $l_1 : \mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + \lambda\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ and $l_2 : \mathbf{r} = \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix} + \mu\begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$ are skew. [4]

(b) The shortest join between the two lines is perpendicular to both. Explain why its direction must be parallel to $\mathbf{n} = \mathbf{d}_1 \times \mathbf{d}_2$. [3]

(c) Hence explain why the shortest distance is the length of the projection of $\mathbf{a}_2 - \mathbf{a}_1$ onto $\mathbf{n}$, that is

$$\text{distance} = \frac{|(\mathbf{a}_2 - \mathbf{a}_1) \cdot \mathbf{n}|}{|\mathbf{n}|}.$$

[4]

(d) Use the formula on the two lines in part (a), giving an exact answer. [5]

(e) What does the formula give when the two lines intersect? Explain why your answer is what it should be. [3]

38. The triple product and volume. Combining both products gives $\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})$, a single number called the *scalar triple product*. It measures volume.

Throughout, take $\mathbf{u} = \begin{pmatrix}1\\2\\0\end{pmatrix}$, $\mathbf{v} = \begin{pmatrix}0\\1\\2\end{pmatrix}$ and $\mathbf{w} = \begin{pmatrix}2\\0\\1\end{pmatrix}$.

(a) Find $\mathbf{v} \times \mathbf{w}$, then $\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})$. [4]

(b) The parallelepiped with edges $\mathbf{u}$, $\mathbf{v}$, $\mathbf{w}$ has base area $|\mathbf{v} \times \mathbf{w}|$. Explain why its volume is $|\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})|$. *(Hint: volume = base area × perpendicular height)* [4]

(c) Evaluate $\mathbf{v} \cdot (\mathbf{w} \times \mathbf{u})$ and $\mathbf{u} \cdot (\mathbf{w} \times \mathbf{v})$. State what each result shows about the triple product. [4]

(d) Three vectors are *coplanar* when they all lie in one plane. Explain why this happens exactly when the triple product is zero, and test $\begin{pmatrix}1\\2\\0\end{pmatrix}$, $\begin{pmatrix}0\\1\\2\end{pmatrix}$, $\begin{pmatrix}2\\5\\2\end{pmatrix}$. [4]

(e) A tetrahedron on the same three edges has one sixth of the parallelepiped's volume. Find the volume of the tetrahedron with vertices at the origin and at the three points $\mathbf{u}$, $\mathbf{v}$, $\mathbf{w}$ above. [3]

Chapter 11

Statistics

11

SYLLABUS 4.1 4.2 4.3 4.4 4.10

In the 1880s Francis Galton kept a notebook of heights. He measured 928 grown children and the 205 sets of parents who raised them. He wanted to answer one question: how well does a parent's height predict a child's?

Tall parents did tend to have tall children. But Galton noticed something stranger. The children of very tall parents were, on average, a little shorter than their parents. The children of very short parents were a little taller. Each generation drifted back toward the middle. He called this "regression toward mediocrity", and although the insult has since been dropped, we still call the technique *regression* today.

To see the pattern at all, Galton first had to do something far more ordinary. He had 928 numbers. One number on its own says very little, and a thousand numbers say too much at once. He had to summarise them, measure how much they varied, and then compare one list against another.

That is what this chapter builds. You will learn to collect data honestly, and to describe a single list of numbers: where its centre lies and how widely it spreads. You will display a distribution as a histogram, a cumulative frequency curve or a box plot. Then you will compare two lists, measure whether they move together, and find the line that best predicts one from the other.

Every one of these tools has a range where it can be trusted and a range where it cannot, and knowing the difference is part of the subject. A correlation does not establish a cause. A line fitted between two values says nothing reliable beyond them. Galton's drift is worth holding on to as an example: it is not a fact about families, and no biology is needed to explain it. You will meet the reason at the end of the chapter.

11.1 Sampling and Data

You almost never have all the data. Galton could not measure every family in England, a factory cannot test every light bulb it makes, and a biologist cannot weigh every fish in a lake. So you measure a part and reason about the whole. The honesty of that reasoning depends entirely on how the part was chosen.

The complete group you want to know about is the **population**. The smaller group you actually measure is the **sample**. A number describing the population is a **parameter**; the matching number computed from a sample is a **statistic**. You use the statistic to estimate the parameter you cannot see.

> DEF A **population** is the entire set of individuals under study. A **sample** is a subset of the population from which data are actually collected.
> A **parameter** describes a population; a **statistic** describes a sample.

Types of data

Data come in two broad kinds. **Qualitative** data describe a quality or category, such as eye colour or country of birth. **Quantitative** data are numerical. Quantitative data split again. **Discrete** data take separate values you can count, such as the number of goals in a match. **Continuous** data can take any value in a range, such as height or time, limited only by the precision of your instrument.

The distinction matters because it decides how you may display and summarise the data. You can find the mean number of goals, but "mean eye colour" is meaningless.

Sampling techniques

A good sample resembles the population on a smaller scale. The techniques below are different ways of achieving that. Each one balances fairness against the effort required.

DEF **Simple random:** every member of the population has an equal chance of selection.
Systematic: order the population and select every k-th member from a random start.
Stratified: divide the population into groups (strata), then sample from each group in proportion to its size.
Quota: like stratified, but members within each group are chosen non-randomly until each quota is filled.
Convenience: select whoever is easiest to reach.

Stratified sampling is the most useful technique for populations with clear subgroups. If a school is 60% junior and 40% senior students, a stratified sample of 50 takes 30 juniors and 20 seniors, preserving the structure.

EXAMPLE 11.1

A school has 800 juniors and 400 seniors. A stratified sample of 60 students is taken. How many should be seniors?

Seniors are $\frac{400}{1200} = \frac{1}{3}$ of the school. The sample must keep that proportion:

$$\frac{1}{3} \times 60 = 20 \text{ seniors} \quad (\text{and } 40 \text{ juniors}).$$

Stratified sampling guarantees both groups are represented in their true ratio, which a simple random sample only achieves on average.

Bias and reliability

A sample is **biased** when its method systematically favours some outcomes over others. The statistic then misses the parameter, no matter how large the sample grows. A survey about reading habits conducted in a library will overstate how much people read. Bias is a fault in the method, not in the sample size; collecting more biased data only makes you more confident in the wrong answer.

Reliability also depends on the records themselves. Real data sets contain missing entries and recording errors. Before any analysis, ask how the data were collected, what might be missing, and whether any value could have been recorded wrongly.

CAUTION A large sample is not automatically a good one. The 1936 *Literary Digest* poll surveyed 2.4 million people and still predicted the wrong US election winner. Its list of names came from telephone and car owners, who were wealthier than the average voter. Size cannot repair a biased method.

Outliers

An **outlier** is a value far from the rest of the data. It may be a genuine extreme value, or it may be a recording error. Either way it can distort a summary badly. We will give it a precise definition once we have quartiles, in §11.4. For now, treat any suspiciously distant value as something to investigate, not something to delete immediately.

YOUR TURN 11.1 Sampling and Data

1. State whether each variable is discrete or continuous.

(a) the number of goals scored in a match [1]
(b) the mass of a parcel [1]
(c) the number of students in a class [1]
(d) the time taken to run 100 m [1]

2. A college has 1500 students, 600 of them male. A stratified sample of 50 is taken.

(a) Find the number of males in the sample. [2]
(b) Find the number of females in the sample. [1]
(c) Explain one advantage of stratified sampling over simple random sampling here. [2]

3. Sampling methods.

(a) A sample of 40 is taken from 2000 people by choosing every kth person. Find k. [1]
(b) Name the sampling method in part (a). [1]
(c) A researcher surveys people at a gym about how much they exercise. Explain why this is biased. [2]
(d) A town of $15\,000$ has 4000 residents over 65. In a stratified sample of 150, find the number over 65. [2]

11.2 Measures of Central Tendency

Once you have a list of numbers, the first question is always the same: what is a typical value? A single number that stands in for the whole list is a measure of **central tendency**. There are three, and they answer slightly different questions.

The **mean** is the balance point: add everything and share it equally. The **median** is the middle value when the data are ordered. The **mode** is the most frequent value.

The mean

For n raw values, the mean is the total divided by the count. When values repeat, you weight each value by how often it occurs, which is just a shortcut for the same sum.

KEY · IB

$$\bar{x} = \frac{\sum_{i=1}^{k} f_i x_i}{n}, \qquad \text{where } n = \sum_{i=1}^{k} f_i$$

Here x_i are the distinct values, f_i their frequencies. A table listing each value alongside its frequency is called a **frequency distribution**, and the next example shows one. It is the standard compact way to present discrete data. With no repeats, every $f_i = 1$ and this collapses to $\bar{x} = \frac{\sum x_i}{n}$, the familiar form.

EXAMPLE 11.2

In 20 football matches a team scored the following numbers of goals. Find the mean.

Goals x	**0**	**1**	**2**	**3**	**4**
Frequency f	4	7	5	3	1

Multiply each value by its frequency, then divide by the total count:

$$\bar{x} = \frac{(0)(4)+(1)(7)+(2)(5)+(3)(3)+(4)(1)}{4+7+5+3+1} = \frac{30}{20} = 1.5.$$

The mean need not be a value the data can actually take. No team scores 1.5 goals, but 1.5 is the fair share.

Median and mode

The median splits ordered data into two equal halves. For n values, it sits at position $\frac{n+1}{2}$. If n is odd this is a single middle value; if n is even it is the average of the two middle values.

EXAMPLE 11.3

Find the median and mode of $4, 7, 2, 9, 4, 11, 6$.

Order first: $2, 4, 4, 6, 7, 9, 11$. With $n = 7$, the median is at position $\frac{7+1}{2} = 4$, the fourth value: 6.

The mode is the most frequent value: 4 appears twice, everything else once, so the mode is 4.

Grouped data

Continuous data are often recorded in classes, such as "time between 10 and 20 minutes." You no longer know the individual values, only the class each fell into. To estimate the mean, assume every value sits at its class **midpoint**, then apply the frequency formula. The result is an estimate, because the assumption is rarely exact.

EXAMPLE 11.4

Fifty phone calls to a help line had the durations below. Estimate the mean duration.

Duration (min)	$0 \le t < 4$	$4 \le t < 8$	$8 \le t < 12$	$12 \le t < 16$
Frequency	8	18	16	8

Use the midpoint of each class: $2, 6, 10, 14$.

$$\bar{x} = \frac{(2)(8) + (6)(18) + (10)(16) + (14)(8)}{50} = \frac{16 + 108 + 160 + 112}{50} = \frac{396}{50} = 7.92 \text{ min.}$$

The **modal class** is $4 \le t < 8$, the class with the highest frequency. With grouped data you report a modal class, not a single modal value.

Which measure to use

The three measures disagree when data are skewed, and that disagreement tells you something. The mean uses every value, so a single very large value raises it. The median depends only on position, so it changes very little. This is why incomes are reported by median, not mean: a few billionaires would make the mean meaningless as a "typical" wage.

CAUTION The mean is sensitive to outliers; the median is resistant to them. When a distribution is strongly skewed or contains a suspect extreme value, the median usually describes the centre better.

YOUR TURN 11.2 Measures of Central Tendency

4. For the data $5,\ 8,\ 8,\ 10,\ 14,\ 15$, find each of the following.

(a) the mean [1]
(b) the median [2]
(c) the mode [1]
(d) the range [1]

5. Find the required measure in each case.

(a) the median of $3, 7, 9, 12, 15, 20$ [2]
(b) the mode of $2, 4, 4, 5, 7, 7, 7, 9$ [1]

(c) the mean of $3, 7, 7, 8, 10, 11, 12, 14$ [2]

(d) the median of $12, 15, 15, 18, 20, 22$ [2]

6. Work backwards from a given mean.

(a) The numbers $4, 7, x, 12, 15$ have a mean of 10. Find x. [2]

(b) The mean of six numbers is 14. A seventh number is added and the mean becomes 15. Find the seventh number. [3]

7. The table shows a frequency distribution.

Value x	1	2	3	4	5
Frequency f	3	5	8	4	2

(a) Find n. [1]

(b) Find $\sum fx$. [2]

(c) Find the mean. [2]

(d) State the mode. [1]

8. The grouped data below record a measurement x.

Class	$0 \le x < 10$	$10 \le x < 20$	$20 \le x < 30$	$30 \le x < 40$
Frequency	6	10	9	5

(a) Write down the four class midpoints. [1]

(b) Estimate the mean. [3]

(c) State the modal class. [1]

(d) Explain why the mean is only an estimate. [2]

11.3 Measures of Dispersion

The centre alone does not describe a data set. Consider two classes that both average 60% on a test. In the first, every student scored between 58 and 62. In the second, half scored 30 and half scored 90. The mean is the same; the two classes are not. To capture the difference you need a measure of **spread**.

Range and interquartile range

The simplest measure is the **range**: the largest value minus the smallest. It is easy to find but unreliable. It depends only on the two most extreme values, which are exactly the ones most likely to be outliers.

A better approach is to ignore the extremes and measure the spread of the middle. Just as the median cuts the data in half, the **quartiles** cut it into quarters. Q_1 is the value one quarter of the way through the ordered data, Q_2 is the median, and Q_3 is three quarters of the way. The spread of the central half is the **interquartile range**.

KEY · IB

$$\text{IQR} = Q_3 - Q_1$$

Because it discards the top and bottom quarters, the IQR is unaffected by outliers, which is why it is normally reported alongside the median.

EXAMPLE 11.5

Find the range and interquartile range of

$$3,\ 5,\ 6,\ 8,\ 8,\ 9,\ 11,\ 13,\ 15.$$

The data are already ordered, with $n = 9$. The range is $15 - 3 = 12$.
The median (Q_2) is the 5th value, 8. The lower quartile is the median of the four values below it, $3, 5, 6, 8$:

$$Q_1 = \frac{5+6}{2} = 5.5.$$

The upper quartile is the median of the four values above, $9, 11, 13, 15$:

$$Q_3 = \frac{11+13}{2} = 12.$$

So IQR $= 12 - 5.5 = 6.5$. Methods for placing quartiles vary slightly between textbooks and calculators; in this course, report the value your GDC gives.

Variance and standard deviation

The IQR uses only the middle half of the data. A measure that uses every value would be better. The idea: measure how far each value sits from the mean, then average those distances.

You cannot simply average the deviations $x_i - \bar{x}$, because by the definition of the mean they cancel to zero. The solution is to square them first, which removes the signs. The average squared deviation is the **variance**; its square root, which returns to the original units, is the **standard deviation**.

KEY

$$\sigma^2 = \frac{\sum_{i=1}^{k} f_i (x_i - \bar{x})^2}{n}, \qquad \sigma = \sqrt{\sigma^2}$$

The standard deviation σ is, roughly, the typical distance of a value from the mean. A small σ means the values lie close to the centre. A large σ means they are widely spread.

CAUTION In this course you find σ with your **GDC** (graphic display calculator), not by hand. The formula is given so that you understand what the calculator computes, not so that you evaluate it by hand under exam conditions. Know what σ *means*.

EXAMPLE 11.6

Return to the two classes, both with mean 60. Class A scored $58, 59, 60, 61, 62$; Class B scored $30, 45, 60, 75, 90$. Compare their standard deviations.

Both means are 60. For Class A the deviations are $-2, -1, 0, 1, 2$:

$$\sigma_A = \sqrt{\frac{4+1+0+1+4}{5}} = \sqrt{2} \approx 1.41.$$

For Class B the deviations are $-30, -15, 0, 15, 30$:

$$\sigma_B = \sqrt{\frac{900+225+0+225+900}{5}} = \sqrt{450} \approx 21.2.$$

The means are identical, but the standard deviations differ by a factor of 15. The single number σ captures exactly the difference that the mean alone does not show.

The effect of changing the data

Suppose every value in a data set is changed in the same way, either by adding a constant or by multiplying by a constant. Temperatures convert from Celsius to Fahrenheit; prices rise by a fixed tax. You can predict the new mean and standard deviation without recomputing anything.

Adding a constant c to every value shifts the whole distribution sideways. The centre moves by c, but the spread does not change. Multiplying every value by k stretches the distribution away from zero, scaling both the centre and the spread.

KEY If $y_i = kx_i + c$ for every value, then

$$\bar{y} = k\bar{x} + c, \qquad \sigma_y = |k|\,\sigma_x, \qquad \sigma_y^2 = k^2\sigma_x^2.$$

The constant c does not affect the spread. Moving every value by the same amount changes no distance between them. Only the multiplier k rescales σ.

EXAMPLE 11.7

A set of test marks has mean 54 and standard deviation 8. Every mark is scaled by multiplying by 1.5 and then adding 5. Find the new mean and standard deviation.
Here $k = 1.5$ and $c = 5$:

$$\bar{y} = 1.5(54) + 5 = 86, \qquad \sigma_y = 1.5 \times 8 = 12.$$

The $+5$ moves the mean but does not change the spread. Only the $\times 1.5$ affects σ.

YOUR TURN 11.3 Measures of Dispersion

9. For the data $2,\ 4,\ 5,\ 7,\ 8,\ 10,\ 12,\ 14,\ 16,\ 20$, find each of the following.

(a) the range [1]
(b) Q_1 [2]
(c) Q_3 [2]
(d) the interquartile range [1]

10. GDC Use your GDC for each part.

(a) Find the mean and standard deviation of $5,\ 8,\ 9,\ 11,\ 13,\ 14$. [2]
(b) Find the mean and standard deviation of the values $10, 20, 30, 40$ with frequencies $2, 5, 8, 3$. [3]
(c) Find the standard deviation of $12,\ 15,\ 15,\ 18,\ 20,\ 22$. [2]

11. Each data set is transformed. Find the new mean and the new standard deviation.

(a) Mean 50, standard deviation 6; every value is multiplied by 3. [2]
(b) Mean 30, standard deviation 5; the value 10 is added to every data point. [2]
(c) Mean 60, standard deviation 12; each value is transformed by $y = 0.5x + 20$. [2]
(d) Mean 40, standard deviation 9; each value is transformed by $y = 2x - 5$. [3]

12. Explain each of the following.

(a) Why adding a constant to every value leaves the standard deviation unchanged. [2]
(b) Why the interquartile range is preferred to the range when the data may contain outliers. [2]

11.4 Visualising Distributions

Numbers summarise a data set; a picture shows its shape. A mean and standard deviation tell you where the data sit and how far they spread. They do not tell you whether the distribution is symmetric, or leans to one side, or is concentrated at one end. Three kinds of display are used in this course.

Histograms

A **histogram** groups continuous data into classes and draws a bar for each, with height equal to the frequency. Unlike a bar chart for categories, the bars touch, because the horizontal axis is a continuous number line. In this course every class has the same width.

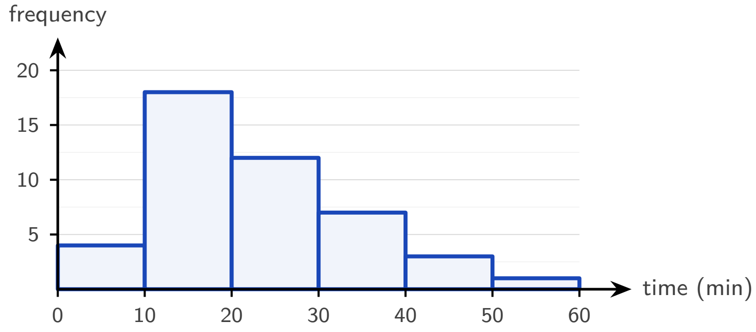


Figure 11.1 A histogram of 45 waiting times in classes of equal width. The bars touch because the horizontal axis is a continuous number line. The long tail to the right is positive skew.

The shape is what a histogram is for. It may be a symmetric peak, a long tail to the right (**positive skew**), as above, or a long tail to the left (**negative skew**).

In a positively skewed distribution the few large values raise the mean above the median. In a negatively skewed one the order reverses. Comparing mean and median is a quick numerical test for skew.

Cumulative frequency

A **cumulative frequency** curve answers "how many values fall below this point?" You plot the running total of frequencies against the *upper* boundary of each class, then join the points with a smooth curve. The result is an S-shape, and the median and quartiles can be read from it directly. Enter from the vertical axis at one quarter, one half and three quarters of the total. Move across to the curve, then drop to the horizontal axis to read Q_1, Q_2 and Q_3.

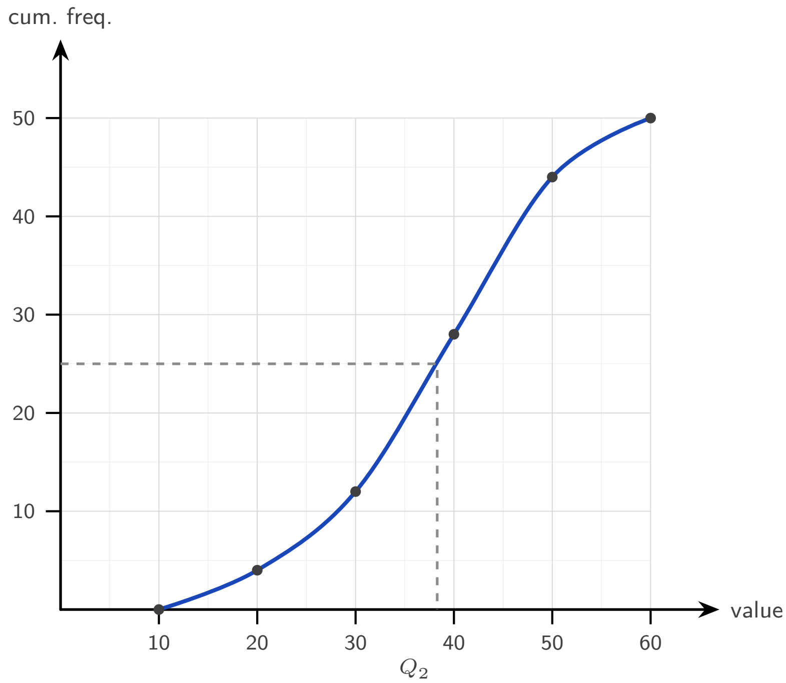


Figure 11.2 A cumulative frequency curve for 50 values. Each point is plotted at the *upper* boundary of its class. The dashed line reads the median: enter at half the total frequency, cross to the curve, drop to the value axis.

The dashed line shows the median. Enter at half the total frequency, here 25 of 50, cross to the curve, then drop to the value axis. Note that each point is plotted at the *upper* boundary of its class, and the curve is drawn through the points.

The same read-off gives any **percentile**: the p-th percentile is the value below which $p\%$ of the data lie. Enter the vertical axis at $\frac{p}{100} \times n$ and read across and down exactly as for the median. The median is the 50th percentile, and the quartiles Q_1 and Q_3 are the 25th and 75th. For the curve above, the 90th percentile is read off at a cumulative frequency of $0.9 \times 50 = 45$, giving a value of roughly 51.

Box-and-whisker plots

A **box-and-whisker plot** draws the **five-number summary**: minimum, Q_1, median, Q_3, maximum. The box spans the interquartile range, and a line inside it marks the median. The whiskers reach to the smallest and largest values that are *not* outliers. Any outlier is marked separately with a cross. In one small picture you see the centre, the spread, and the skew.

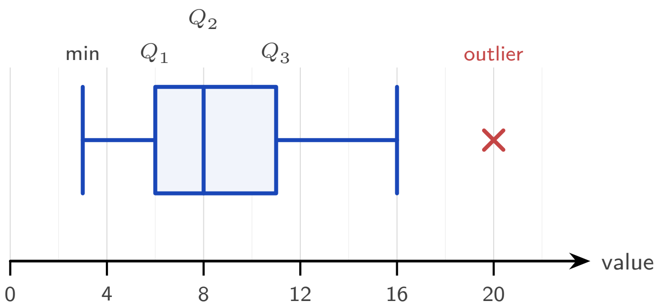


Figure 11.3 A box-and-whisker plot of the five-number summary. The box spans the interquartile range and the inner line marks the median. The right whisker stops at 16, the largest value that is not an outlier; the value 20 is marked with a cross.

A box positioned to the left, with a long right whisker, as above, indicates positive skew. Here the value 20 is an outlier, by the rule below. The whisker therefore stops at 16, the largest value that is not an outlier, and 20 is marked with a cross. Comparing two box plots side by side is the fastest way to contrast two distributions.

A box plot also gives a first hint about the normal distribution of Ch. 13. If the box and whiskers are roughly symmetric about the median, the data *may* be normally distributed. Clear skew shows that they are not.

The outlier rule

Now "outlier" can be defined precisely. A value is treated as an outlier if it lies more than 1.5 interquartile ranges beyond the nearer quartile.

KEY A value x is an **outlier** if

$$x < Q_1 - 1.5\,(\text{IQR}) \qquad \text{or} \qquad x > Q_3 + 1.5\,(\text{IQR}).$$

EXAMPLE 11.8

A data set has $Q_1 = 24$, $Q_3 = 40$. Is a value of 70 an outlier?

First IQR $= 40 - 24 = 16$. The upper boundary is

$$Q_3 + 1.5(\text{IQR}) = 40 + 1.5(16) = 40 + 24 = 64.$$

Since $70 > 64$, the value is an outlier. On a box plot it would be marked with a cross, with the whisker stopping at the largest value that is not an outlier.

YOUR TURN 11.4 Visualising Distributions

13. Consider the data $4,\ 6,\ 7,\ 9,\ 10,\ 12,\ 13,\ 15,\ 18.$

(a) Find the median. [1]

(b) Find Q_1 and Q_3. [2]

(c) Write down the five-number summary. [2]

(d) State the interquartile range. [1]

14. Apply the $1.5 \times$ IQR outlier rule.

(a) A data set has $Q_1 = 30$ and $Q_3 = 50$. Determine whether 95 is an outlier. [2]

(b) A data set has $Q_1 = 12$ and $Q_3 = 20$. Determine whether 2 is an outlier. [2]

(c) A data set has $Q_1 = 9$ and $Q_3 = 19$. Find both outlier boundaries. [3]

15. Reading displays.

(a) A box plot has $Q_1 = 12$ and $Q_3 = 24$. Find the interquartile range. [1]

(b) A cumulative frequency curve is drawn for 80 values. State the cumulative frequencies at which you read off Q_1, the median and Q_3. [2]

(c) Explain why the bars of a histogram touch, while those of a bar chart do not. [2]

11.5 Correlation

Everything so far has described one variable. Galton's question was different: he had *two* measurements on each family, parent height and child height, and wanted to know whether they were linked. This is the study of **bivariate** data, meaning data in which two variables are recorded for each individual. The first step is always to look.

Scatter diagrams

A **scatter diagram** plots each pair (x, y) as a point. The cloud of points has a shape, and the shape is the relationship. If y tends to rise as x rises, the correlation is **positive**. If y falls as x rises, it is **negative**. If there is no tilt, there is no linear correlation. The tighter the points cluster around a straight line, the **stronger** the correlation.

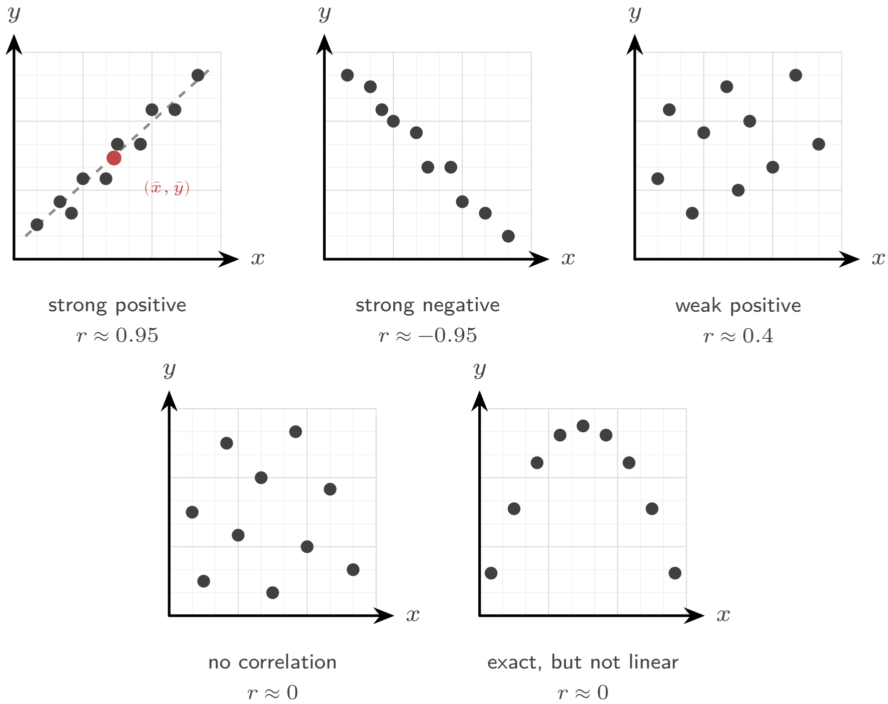


Figure 11.4 Five clouds and the value of r each gives. The sign of r is the direction of the tilt and the size of r is how tightly the points hug a line. The last panel is the warning: the points lie exactly on a curve, so the relationship could not be stronger, yet r is near zero because the curve is not straight.

Read the panels from left to right. The first two tilt clearly, one up and one down, and the points sit close to a line: those are strong correlations, positive and negative. The third tilts upward but the points scatter widely, so the correlation is weak. The fourth has no tilt at all.

The last panel is the one to remember. Its points lie exactly on a curve, so the two variables are perfectly related, yet r is near zero. That is because r only measures how close the points are to a *straight* line.

The first panel also shows a **line of best fit** drawn by eye: a straight line balancing the points above and below it. To position it, first compute the **mean point** $(\bar{x}, \bar{y})$, the mean of the x-values paired with the mean of the y-values, marked in scarlet. Draw your line through it. Every correctly drawn line of best fit passes through the mean point; in §11.6 the calculator makes this exact.

Pearson's correlation coefficient

You can judge correlation well by eye, but not precisely. **Pearson's product-moment correlation coefficient**, written r, turns the strength and direction into a single number between -1 and 1.

> DEF Pearson's correlation coefficient r measures the strength and direction of the **linear** relationship between two variables. It satisfies $-1 \leq r \leq 1$.

The sign of r gives the direction, and its size gives the strength. The extremes $r = \pm 1$ mean the points lie exactly on a straight line; $r = 0$ means no linear relationship. A rough reading:

$\lvert r \rvert$ **near**	**Interpretation**
1	strong linear correlation
0.5	moderate linear correlation
0	weak or no linear correlation

You compute r with your GDC by entering the two lists. Reading and interpreting r matters far more than the formula behind it.

CAUTION r measures *linear* association only. Data lying perfectly on a parabola can have r near 0, because the relationship, though exact, is not straight. Always look at the scatter diagram before trusting r.

Correlation is not causation

A strong r says two variables move together. It does not say one causes the other. There are at least three quite different reasons a correlation can appear without any causal link, and it is worth being able to tell them apart.

A third variable drives both. Ice-cream sales and drowning deaths rise and fall together across the year. Neither causes the other. Summer heat raises both, because hot weather sends people to the ice-cream shop and to the water. A variable like this, sitting behind both and explaining the link, is called a **confounding variable**.

The arrow points the other way. Hospitals with the most advanced equipment can show the highest death rates. The equipment is not killing patients. The most seriously ill patients are sent to the best-equipped hospitals, so the severity of illness is causing the choice of hospital, not the reverse.

Coincidence, found by searching. Between 2000 and 2009, cheese consumption per person in the United States tracked the number of people who died by becoming entangled in their bedsheets, with r above 0.94. Nobody thinks these are connected. If you compare thousands of pairs of unrelated variables, some pairs will match closely by chance alone, and a determined search will always find them.

The third case is the one to keep in mind whenever a surprising correlation is reported. A high r is easy to find if you go looking through enough data.

CAUTION A high correlation never establishes cause. There may be a hidden third variable, or the link may be coincidence. Causation must be argued from the context, never from r alone.

YOUR TURN 11.5 Correlation

16. Describe the correlation indicated by each value of r.

(a) $r = -0.85$ [1] (b) $r = 0.12$ [1]
(c) $r = 0.97$ [1] (d) $r = -0.40$ [2]

17. GDC Find Pearson's correlation coefficient for each data set.

(a) $x = 1, 2, 3, 4, 5$ and $y = 2, 5, 4, 7, 9$ [2]
(b) $x = 2, 4, 6, 8, 10$ and $y = 5, 9, 10, 14, 17$ [2]

18. Interpreting r.

(a) A set of points lies exactly on the parabola $y = x^2$ for $-3 \leq x \leq 3$. Explain why r is close to 0 even though the relationship is exact. [3]
(b) A study finds a strong positive correlation between shoe size and reading ability in children. Explain why shoe size does not cause reading ability. [3]

11.6 Linear Regression

If the scatter diagram shows a clear linear trend, you can do more than measure it. You can draw the line that best fits the data and use it to predict. The question is what "best" should mean.

The least-squares line

For any candidate line, each data point sits some vertical distance above or below it. That distance is the **residual**, the error of the line at that point. The **least-squares regression line** is the one line that makes the total of the squared residuals as small as possible. Squaring, again, stops positive and negative errors from cancelling, and it counts large errors much more heavily than small ones.

KEY The **regression line of y on x** has the form

$$y = ax + b,$$

chosen to minimise the sum of squared vertical residuals. It always passes through the mean point $(\bar{x}, \bar{y})$.

You find a and b from your GDC. You must also be able to say what they *mean* in context. The gradient a is the change in y for each one-unit increase in x. The intercept b is the predicted y when $x = 0$. The line always passes through $(\bar{x}, \bar{y})$, which is a useful check and an occasional exam shortcut.

EXAMPLE 11.9

A GDC gives the regression line of exam score y on hours studied x as $y = 6.2x + 41$, valid for students who studied between 2 and 9 hours. Predict the score of a student who studied 5 hours, interpret the values 6.2 and 41 in context, and comment on predicting the score for 20 hours.

For $x = 5$, substitute into the line:

$$y = 6.2(5) + 41 = 31 + 41 = 72.$$

In context, the gradient says each additional hour of study raises the predicted score by 6.2 marks. The intercept 41 is the predicted score of a student who did not study at all. That reading is itself a mild extrapolation, because $x = 0$ lies below the data range.

Predicting at $x = 20$ is unsafe. The data only cover 2 to 9 hours, so 20 lies far outside the known range, and the linear trend may not continue there.

Interpolation and extrapolation

Predicting *inside* the range of the data is **interpolation**, and it is reliable when the correlation is strong. Predicting *outside* the range is **extrapolation**, and it is unreliable. You are assuming the pattern continues where you have no evidence. The further beyond the data you go, the less trustworthy the prediction.

CAUTION A regression line is only trustworthy for prediction within the range of the original data, and only when $|r|$ is reasonably large. Extrapolating far beyond the data, or fitting a line to weakly correlated points, produces predictions worth little.

The two regression lines

There is an asymmetry hidden in "least squares." The line of y on x minimises *vertical* residuals, because it treats y as the quantity to predict from x. If instead you want to predict x from y, you minimise *horizontal* residuals, giving a different line: the **regression line of x on y**.

> KEY Use the regression line of y **on** x to predict y from a given x. Use the regression line of x **on** y to predict x from a given y. Both lines pass through $(\bar{x}, \bar{y})$, and they coincide only when $r = \pm 1$.

The rule is simple. Choose the line whose name begins with the quantity you want to predict. Choosing the wrong line is a common and avoidable error.

EXAMPLE 11.10

For a set of bivariate data, $\bar{x} = 10$ and $\bar{y} = 25$. The regression line of y on x is $y = 2x + 5$. Verify that the mean point lies on this line.
Substitute $x = \bar{x} = 10$:

$$y = 2(10) + 5 = 25 = \bar{y}. \checkmark$$

Every least-squares line of y on x passes through $(\bar{x}, \bar{y})$, so this is a quick check on a computed line.

YOUR TURN 11.6 Linear Regression

19. GDC Find the regression line of y on x for each data set.

(a) $x = 1, 2, 3, 4, 5$ and $y = 3, 5, 4, 8, 10$ [3]
(b) $x = 2, 4, 6, 8, 10$ and $y = 5, 9, 10, 14, 17$ [3]

20. Using a regression line.

(a) A regression line is $y = 2.5x + 4$. Predict y when $x = 6$. [1]
(b) A regression line $y = -1.5x + 30$ was fitted using data with $2 \leq x \leq 15$. Predict y when $x = 8$. [2]
(c) Explain why using that same line to predict y at $x = 40$ is unreliable. [2]

21. Two further results.

(a) The mean point of a data set is $(8, 20)$ and the regression line of y on x has gradient 1.5. Find the equation of the line. [3]

(b) You are given a value of y and want to predict the corresponding x. State which regression line you should use, and why. [3]

Chapter 11 · Statistics

Reflections

TOK Almost everything you are told about the world arrives as a number. Your grade. The mass of an atom. The temperature tomorrow. How happy a country is. How intelligent you are. Each of these is reported as a figure, and the figure is then treated as the thing itself.

But the number and the thing are not the same. Mass really is a quantity, and 12.011 is a fair description of carbon. Happiness is not a quantity in that sense; a score of 7.2 is the output of a questionnaire somebody designed, with questions somebody chose. Between those two extremes sit most of the numbers in your life. A grade of 6 compresses two years of work into one digit. An IQ of 115 compresses a mind.

So there are two different claims hidden in any statistic. The first is that the thing being measured has a size at all. The second is that the particular number we assign captures it. Both can fail, and they fail independently.

The Pythagoreans held that all things *are* number. Modern physics is written almost entirely in quantities, which suggests they had a point. Yet the moment we count something that was never a quantity to begin with, the mathematics still works perfectly and the conclusion can still be nonsense.

Which of the numbers used to describe you measure something that was already there, and which create the thing they claim to measure?

RW Galton's "regression toward mediocrity" is now called regression to the mean, and it affects decision-making everywhere. A pilot who flies badly and is scolded usually does better next time. The reason is not the scolding. An extreme performance is, on average, followed by a less extreme one. Mistaking this statistical drift for cause and effect has misled trainers, investors, and doctors for over a century.

IM The word "statistics" shares a root with "state." It began in the 18th century as the numerical bookkeeping of nations: population, taxes, soldiers, harvests. The mathematics of describing variation grew out of the practical needs of governments counting their citizens, long before it became a tool of science.

EXT Your calculator returns r without showing its working. The formula behind it is

$$r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \ \sum(y_i - \bar{y})^2}},$$

and it is worth reading rather than memorising, because every part of it does something recognisable.

Look at one term of the numerator, $(x_i - \bar{x})(y_i - \bar{y})$. It asks where a single point sits relative to *both* means. If the point is above average in x and above average in y, both brackets are positive and the product is positive. If it is below average in both, both brackets are negative and the product is positive again. But if the point is above average in one and below average in the other, the product is negative.

So each point votes. Points that fit a rising pattern vote positive, points that fit a falling pattern vote negative, and points near a mean contribute almost nothing. The numerator is the total vote.

That total still depends on the units. Measure heights in millimetres instead of metres and every deviation grows by a thousand. The denominator fixes this: dividing by $\sqrt{\sum(x_i - \bar{x})^2}$ and $\sqrt{\sum(y_i - \bar{y})^2}$ cancels the units of each variable, so r is a pure number.

It also explains the bound $-1 \leq r \leq 1$. The numerator can never exceed the denominator, a fact known as the Cauchy–Schwarz inequality, and equality holds exactly when every point lies on one straight line. That is why $r = \pm 1$ means perfect linearity, and nothing else does.

End-of-Chapter Problems — Chapter 11: Statistics

1. The times, in minutes, taken by 40 students to complete a puzzle are summarised below.

Time (min)	$0 \le t < 10$	$10 \le t < 20$	$20 \le t < 30$	$30 \le t < 40$	$40 \le t < 50$
Frequency	4	11	15	8	2

(a) Estimate the mean time. [3]
(b) Write down the modal class. [1]
(c) State the class containing the median. [2]

2. A data set is $12, 15, 18, 19, 21, 22, 24, 25, 27, 28, 30, 46$.

(a) Find the five-number summary. [3]
(b) Determine whether 46 is an outlier, showing your working. [3]
(c) Describe the shape of the distribution. [1]

3. GDC The table shows the number of hours x eight students revised and their test score y.

x	2	3	5	6	8	9	11	12
y	10	14	15	20	24	27	30	34

(a) Find Pearson's correlation coefficient and interpret it. [3]
(b) Find the equation of the regression line of y on x. [2]
(c) Interpret, in context, the gradient and the y-intercept of your line. [2]
(d) Use your line to estimate the score of a student who revised for 7 hours. [2]
(e) Explain whether your line should be used to estimate the score for 25 hours. [2]

4. A set of data has mean 45 and standard deviation 8.

(a) Each value is increased by 12. Write down the new mean and standard deviation. [2]
(b) Instead, each value is multiplied by 1.5. Write down the new mean and standard deviation. [2]
(c) Explain why adding a constant changes the mean but not the standard deviation. [2]

5. A region has populations of 4000, 10 000, and 6000 in three towns A, B, and C. A stratified sample of 100 households is required.

(a) Find the number of households to sample from each town. [3]

(b) State one advantage of stratified sampling over simple random sampling here. [2]

6. The cumulative frequencies of a data set of 60 values are given.

Value $\leq$	10	20	30	40	50
Cumulative frequency	4	16	34	52	60

(a) Draw a cumulative frequency curve. [3]
(b) Use it to estimate the median and the interquartile range. [4]
(c) Estimate the 70th percentile. [2]

7. In a test, 20 students in class A had a mean score of 72, and 30 students in class B had a mean score of 65. Find the mean score of all 50 students combined. [3]

8. A frequency distribution of the values $1, 2, 3, 4$ has frequencies $5, 8, k, 4$. The mean is 2.5. Find the value of k. [4]

9. Two machines fill bottles. A sample from each gives the same mean volume of 500 mL, but machine X has standard deviation 2 mL and machine Y has standard deviation 9 mL.

(a) Explain what this tells you about the two machines. [2]
(b) State which machine is more reliable, with a reason. [2]

10. GDC The table gives the area x (in m^2) and monthly rent y (in \$) of five apartments.

x	10	20	30	40	50
y	14	20	25	33	38

(a) Find the regression line of y on x. [2]
(b) Use it to estimate the rent of a 35 m^2 apartment. [2]
(c) Find the regression line of x on y, and use it to estimate the area of an apartment renting for \$30. [3]

11. A data set is $4, 6, 7, 8, 9, 10, 55$.

(a) Find the mean and the median. [3]
(b) The value 55 is removed. Find the new mean and median. [2]
(c) Comment on which measure of centre is more affected by the outlier, and why. [2]

12. A student's final grade is 40% coursework and 60% final exam. The student scores 70 on coursework and 85 on the exam. Find the final grade. [2]

13. The masses, in grams, of seven eggs are $52, 55, 58, 60, 61, 63, 90$.

(a) Find Q_1, Q_3, and the interquartile range. [3]
(b) Show that 90 is an outlier. [2]
(c) State, with a reason, whether the mean or the median better represents a typical egg. [2]

14. A regression line of y on x passes through the mean point $(6, 19)$ and has gradient -2.

(a) Find the equation of the regression line. [2]
(b) Use it to predict y when $x = 4$. [2]

15. The five-number summaries of test scores in two classes are given below.

	Min	Q_1	Median	Q_3	Max
Class A	32	45	55	62	90
Class B	42	50	58	66	74

(a) Write down the interquartile range for Class A. [1]
(b) Show that the score of 90 in Class A is an outlier. [2]
(c) Compare the two distributions, referring to centre and spread. [2]
(d) State, with a reason, whether the scores in Class B may be normally distributed. [2]

16. GDC The waiting times, in minutes, of a group of customers are summarised below. The estimated mean waiting time is 20 minutes.

Time (min)	$0 \le t < 10$	$10 \le t < 20$	$20 \le t < 30$	$30 \le t < 40$
Frequency	6	k	12	8

(a) Show that $k = 18$. [3]
(b) Write down the modal class. [1]
(c) State the class containing the median. [2]
(d) Use your GDC to estimate the standard deviation. [2]

17. A set of examination marks x has mean 62 and standard deviation 10. The marks are scaled linearly to $y = ax + b$, $a > 0$, so that the scaled marks have mean 70 and standard deviation 8.

(a) Find the values of a and b. [4]
(b) A student's original mark was 75. Find their scaled mark. [1]
(c) The median of the original marks was 61. Write down the median of the scaled marks. [1]

18. GDC The table shows the maximum temperature x (°C) and the number of ice creams sold y on six days.

x (°C)	18	20	23	25	27	30
y	110	125	150	160	175	195

(a) Find Pearson's correlation coefficient r and comment on it. [2]
(b) Find the equation of the regression line of y on x. [2]
(c) Interpret, in context, the gradient of your line. [1]
(d) Estimate the number of ice creams sold on a day with maximum temperature $24\,^\circ$C. [2]
(e) Explain why the line should not be used when the maximum temperature is $40\,^\circ$C. [1]

19. For a bivariate data set, the regression line of y on x is $y = 2x + 1$ and the regression line of x on y is $x = 0.4y + 1$.

(a) Find $\bar{x}$ and $\bar{y}$. [4]
(b) Estimate the value of x when $y = 20$. [2]
(c) Explain why the regression line of y on x should not be used in part (b). [1]

20. A supermarket wants to survey its customers.

(a) State the sampling method used when every 20th customer entering the store is selected. [1]
(b) State the sampling method used when customers are grouped by age band and sampled in proportion to the size of each band. [1]
(c) State the sampling method used when the first 30 customers on Monday morning are selected. [1]
(d) Explain why the method in (c) is likely to produce a biased sample. [2]
(e) A school of 320 juniors and 480 seniors is to be surveyed with a stratified sample of 50. Find the number of juniors and seniors in the sample. [2]

21. A group of n students has a mean mark of 68. When 5 students with mean mark 60 join the group, the overall mean falls to 66. Find n. [4]

22.

(a) Match each value of r to one description: $r = -0.92$, $r = -0.35$, $r = 0.08$, $r = 0.85$. Descriptions: strong positive, weak negative, strong negative, no linear correlation. [2]
(b) A study finds a strong positive correlation between the hours children spend watching television and their body mass. Explain why this does not show that television causes weight gain, and suggest a possible third variable that could explain both. [2]
(c) Heights are recorded in centimetres and r is calculated. The heights are converted to metres. State, with a reason, the effect on the value of r. [2]

23. The table gives the monthly mean air temperature in Tokyo, in °C, taken from the Japan Meteorological Agency climatological normals for 1991–2020.

Month	J	F	M	A	M	J	J	A	S	O	N	D
Temp (°C)	5.4	6.1	9.4	14.3	18.8	21.9	25.7	26.9	23.3	18.0	12.5	7.7

(a) Find the five-number summary of the twelve values. [4]
(b) Show that the data contain no outliers. [3]
(c) The JMA publishes an annual normal of $15.8\,^{\circ}$C. Calculate the mean of the twelve monthly values and compare it with the published figure. [3]
(d) Describe the shape of the distribution, and explain why this box plot tells you very little about the temperature on any particular day in Tokyo. [3]

24. A data set of n values has mean $\bar{x}$. A new value equal to $\bar{x}$ is added to the set.

(a) Show that the mean is unchanged. [3]
(b) Show that the standard deviation decreases (or stays the same). State the condition for equality. [4]

25. Five positive integers have a mean of 8, a median of 7, and a mode of 4 (occurring exactly twice). Find all possible sets of five integers. [6]

26. The regression line of y on x for a data set is $y = 0.8x + 3$, and the regression line of x on y is $x = 1.05y - 2$.

(a) Both lines pass through $(\bar{x}, \bar{y})$. Find $\bar{x}$ and $\bar{y}$. [4]
(b) It can be shown that the product of the two gradients equals r^2. Find r. [3]

27. A set of n data values has mean $\bar{x}$ and standard deviation σ. Prove algebraically that adding a constant c to every value leaves σ unchanged, starting from the definition $\sigma^2 = \frac{1}{n}\sum(x_i - \bar{x})^2$. [5]

Challenge Questions

28. How much can one value move a summary? Consider the data set

$$3,\ 5,\ 7,\ 9,\ 11.$$

(a) Write down the mean and the median. [2]
(b) The largest value is replaced by k, where $k \geq 9$. Find the mean in terms of k, and write down the median. [4]
(c) Find the value of k for which the mean is 20. [2]
(d) Explain why the median is 7 for *every* $k \geq 9$. [3]
(e) A data set has n values. Explain why changing a single value can move the mean by any amount, however large, but can move the median by at most one place in the ordered list. Hence explain what is meant by calling the median *resistant* and the mean *not resis-*

tant. [5]

(f) State, in terms of n, the smallest number of values that must be changed to make the median as large as you wish. Justify your answer for $n = 5$. [3]

29. Combining two groups. Group A has n_1 values with mean $\bar{x}_1$ and variance σ_1^2. Group B has n_2 values with mean $\bar{x}_2$ and variance σ_2^2. The two groups are combined into one set of $n_1 + n_2$ values.

(a) Show that the combined mean is $\bar{x} = \dfrac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2}$. [3]

(b) Group A has 12 values with mean 50, and Group B has 8 values with mean 60. Find the combined mean. [2]

(c) Using $\sum x^2 = n(\sigma^2 + \bar{x}^2)$ for each group, show that the combined variance is

$$\sigma^2 = \frac{n_1(\sigma_1^2 + \bar{x}_1^2) + n_2(\sigma_2^2 + \bar{x}_2^2)}{n_1 + n_2} - \bar{x}^2.$$

[5]

(d) Group A has standard deviation 4 and Group B has standard deviation 3. Find the standard deviation of the combined set. [4]

(e) Show that $\sigma^2 \geq \dfrac{n_1\sigma_1^2 + n_2\sigma_2^2}{n_1 + n_2}$, and state the condition for equality. Interpret this result. [4]

30. Putting data on a common scale. A data set $x_1, x_2, \ldots, x_n$ has mean $\bar{x}$ and standard deviation σ, with $\sigma \neq 0$. Each value is replaced by its *standardised value*

$$z_i = \frac{x_i - \bar{x}}{\sigma}.$$

(a) Show that the mean of the z_i is 0. [4]

(b) Show that the standard deviation of the z_i is 1. [4]

(c) The data set $4, 8, 10, 12, 16$ has mean 10 and standard deviation 4. Write down the five standardised values, and verify your answers to parts (a) and (b). [4]

(d) Student P scores 78 on a test with mean 70 and standard deviation 5. Student Q scores 84 on a different test with mean 75 and standard deviation 8. Determine which student performed better relative to their own group. [4]

(e) Explain why standardised values allow marks from two different tests to be compared, when the raw marks cannot be. [2]

STATISTICS AND PROBABILITY

Chapter 12

12

Probability

SYLLABUS 4.5 4.6 4.11 4.13

Probability puts a number on how likely an event is. That single step is what turns risk into something you can measure, compare and act on, so that a decision can rest on more than a feeling about what might happen.

Here is a decision that turns on such a number. A health service screens everyone in an age group for a rare disease. The test is a good one. Of every 100 people who have the disease, it correctly identifies 99. Of every 100 people who do not have it, it correctly clears 95. About one person in a hundred has the disease.

Your result comes back positive. How likely is it that you have the disease?

Most people answer about 99%, and doctors asked the same question often answer the same way. The correct answer is about 17%. The test is not faulty and none of the numbers are unusual. The error is in the reasoning. That 99% says how the test behaves *when the disease is present*, and the quantity you actually want is a different one: how likely the disease is *now that the result is positive*. Those two numbers are not the same, and this chapter builds the rule that connects them.

So probability gives clear answers, and it is easy to apply a clear answer to the wrong question. Two habits guard against that, and both are practised throughout the chapter. The first is to state precisely which event you are asking about. The second is to keep track of where a probability came from: some are calculated from the symmetry of a fair die, others only measured by counting what happened in 200 trials, and the two are not equally trustworthy.

We begin with the simplest experiment there is, a single roll of a die. When we return to the test in the last section, the answer will take three lines.

12.1 The Language of Probability

Probability measures how likely something is, on a scale from 0 (impossible) to 1 (certain). To compute it we need to be precise about what "something" means.

Five words are used precisely from here on. The die is used to illustrate each one.

- **Experiment** — any process with an uncertain result. Rolling a die is an experiment.
- **Trial** — one repetition of the experiment. One roll is one trial.
- **Outcome** — the result of a single trial, such as a 3.
- **Sample space** U — the set of all possible outcomes. For one die, $U = \{1, 2, 3, 4, 5, 6\}$.
- **Event** — any subset of the sample space, that is, a collection of outcomes you are interested in. "An even number" is the event $\{2, 4, 6\}$.

When every outcome is equally likely, probability is just counting: the fraction of outcomes that belong to the event.

KEY · IB

$$P(A) = \frac{n(A)}{n(U)}$$

Here $n(A)$ is the number of outcomes in event A, and $n(U)$ the number in the whole sample space. Every probability obeys $0 \le P(A) \le 1$.

EXAMPLE 12.1

A fair die is rolled once. Find the probability of (a) an even number, (b) a number greater than 4.

The sample space has $n(U) = 6$ equally likely outcomes.

$$P(\text{even}) = \frac{n(\{2,4,6\})}{6} = \frac{3}{6} = \frac{1}{2}, \qquad P(>4) = \frac{n(\{5,6\})}{6} = \frac{2}{6} = \frac{1}{3}.$$

Counting works only because the die is fair. If the outcomes were not equally likely, this formula would not apply.

Relative frequency

When the outcomes are not equally likely, counting fails, and the probability has to be estimated by experiment. Repeat the trial many times and record how often the event occurs. The fraction of trials in which it happens is the **relative frequency** of the event, and it serves as an experimental estimate of the probability.

KEY After n trials,

$$P(A) \approx \frac{\text{number of trials in which } A \text{ occurred}}{n}$$

The estimate is rough when n is small, and it becomes more stable as n grows. This is the difference between the two kinds of probability you will meet: a *theoretical* probability comes from the structure of the experiment (a fair die, by symmetry), while an *experimental* one comes from data, and is what you fall back on when there is no symmetry to use.

EXAMPLE 12.2

A drawing pin is dropped 200 times and lands point-up 78 times. Estimate the probability that a dropped pin lands point-up.

There is no symmetry argument for a drawing pin, so use the relative frequency:

$$P(\text{point-up}) \approx \frac{78}{200} = 0.39.$$

A further 200 drops would almost certainly not give exactly 78 again, but the estimate becomes more accurate as the number of trials increases.

The complement

Sometimes the event you want is awkward to count, but its opposite is easy. The **complement** of A, written A', is the event that A does *not* happen. Since A either happens or it does not, their probabilities must fill the whole scale.

KEY · IB

$$P(A) + P(A') = 1 \qquad \Longrightarrow \qquad P(A') = 1 - P(A)$$

The complement is worth trying whenever the event itself is hard to count. In particular, whenever a question contains the phrase “at least one,” work out the complement first.

EXAMPLE 12.3

A fair coin is tossed three times. Find the probability of getting at least one head.

Counting every way to get one, two, or three heads is possible but slow. The complement of “at least one head” is “no heads at all,” a single outcome: TTT.

$$P(\text{at least one head}) = 1 - P(\text{TTT}) = 1 - \frac{1}{8} = \frac{7}{8}.$$

One subtraction replaces a count of seven outcomes. The saving becomes far greater as the number of tosses increases.

Expected number of occurrences

If an event has probability p and you repeat the experiment n times, you expect it to happen about np times.

KEY If an event has probability p and the experiment is repeated n times, the **expected number of occurrences** is

$$\text{expected number} = np.$$

This is not a guarantee. It is a long-run average. If you repeated the whole set of n trials many times and averaged the counts, that average would be close to np.

EXAMPLE 12.4

A fair die is rolled 60 times. How many sixes would you expect?

Each roll gives a six with probability $\frac{1}{6}$, so over 60 rolls:

$$\text{expected sixes} = 60 \times \frac{1}{6} = 10.$$

You will rarely get exactly 10, but the average over many sets of 60 rolls will be close to 10.

YOUR TURN 12.1 The Language of Probability

1.

(a) A fair die is rolled. Find the probability that the number is odd. [1]

(b) A card is drawn from a deck of 52. Find the probability that it is a heart. [1]

(c) A bag holds 4 red, 6 green and 5 blue counters. Find the probability that a counter drawn at random is green. [2]

(d) A letter is chosen at random from the word STATISTICS. Find the probability that it is an S. [2]

(e) A spinner numbered 1 to 8 is spun. Find the probability that it lands on a multiple of 3. [2]

2.

(a) Given $P(A) = 0.35$, find $P(A')$. [1]

(b) A fair die is rolled 90 times. Find the expected number of 5s. [2]

(c) Two fair coins are tossed. Find the probability of at least one tail. [2]

3.

(a) Two fair coins are tossed. List the sample space, and find the probability of exactly one head. [3]

(b) A spinner has five sectors numbered 1 to 5, but it is not fair. In 400 spins it lands on 3 exactly 96 times. Estimate the probability of landing on 3. [2]

(c) Using that estimate, find the expected number of 3s in 250 spins. [2]

(d) Explain why the probability in part (a) can be found without any experiment, while the one in part (b) cannot. [2]

4.

(a) A bag holds 12 counters, 5 of which are red. Find the probability that a counter drawn at random is not red. [2]

(b) The probability that a bus is late on a given day is 0.18. Find the probability that it is not late. [1]

(c) A fair coin is tossed four times. Find the probability of at least one tail. [3]

12.2 Combining Events

Real questions rarely involve a single event. You want the probability that a student plays football *or* tennis, or that a card is a king *and* a heart. Two operations cover every case: the **union** $A \cup B$ ("A or B or both") and the **intersection** $A \cap B$ ("A and B together").

The word *or* needs care, because everyday English and mathematics do not use it the same way. "Tea or coffee?" offers you one drink, not both. In mathematics $A \cup B$ always includes the case where both happen. This is called the **non-exclusivity of or**, and it is the reason the addition rule below needs a correction term. When a question really does mean one or the other but not both, it will say so.

A **Venn diagram** makes the structure visible. The rectangle is the sample space U, each circle is an event, and every point of the diagram lies in exactly one of four regions. Combining an event with a complement names each region precisely.

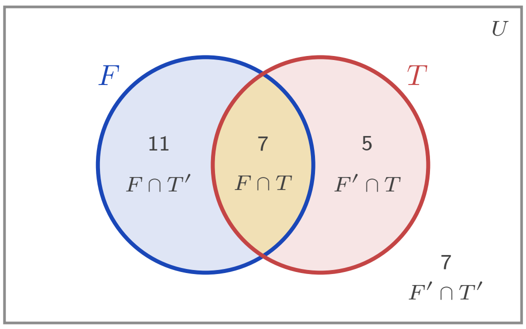


Figure 12.1 The four regions of a two-event Venn diagram, with the class of 30 filled in. Football only is $F \cap T'$, both is $F \cap T$, tennis only is $F' \cap T$, and neither is $F' \cap T'$, the region outside both circles. The four counts add to 30.

Fill a Venn diagram from the middle outwards. Put the overlap in first, then subtract it from each circle total to get the "only" regions, then use the grand total to find the outside. Working the other way round double counts the overlap, which is the same mistake the addition rule is built to correct.

The addition rule

If you add $P(A)$ and $P(B)$ to find $P(A \cup B)$, you count the overlap twice: once in A, once in B. Subtracting it once corrects the double count.

KEY · IB

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

When two events cannot happen together, they are **mutually exclusive**: their intersection is empty, $P(A \cap B) = 0$, and the rule simplifies.

KEY · IB For mutually exclusive events:

$$P(A \cup B) = P(A) + P(B)$$

EXAMPLE 12.5

In a class of 30, 18 students play football, 12 play tennis, and 7 play both. A student is chosen at random. Find the probability that the student plays football or tennis, and the probability that they play neither.

Apply the addition rule directly:

$$P(F \cup T) = \frac{18}{30} + \frac{12}{30} - \frac{7}{30} = \frac{23}{30}.$$

"Neither" is the complement of "football or tennis":

$$P(\text{neither}) = 1 - \frac{23}{30} = \frac{7}{30}.$$

The Venn diagram above shows the same thing: 11 play only football, 7 play both, 5 play only tennis, and 7 play neither, totalling 30.

EXAMPLE 12.6

A single card is drawn from a standard deck of 52. Find the probability that it is a king or a

heart.
There are 4 kings and 13 hearts, but the king of hearts is both, so it is counted in each:

$$P(\text{king} \cup \text{heart}) = \frac{4}{52} + \frac{13}{52} - \frac{1}{52} = \frac{16}{52} = \frac{4}{13}.$$

Forgetting to subtract the king of hearts gives $\frac{17}{52}$, which is the most common error in this topic.

Three events need three circles, and the same instruction applies: start in the middle. The centre region belongs to all three events, and every other region is found by subtracting what has already been placed.

EXAMPLE 12.7

In a year group of 40 students, 20 study French, 18 study German and 15 study Spanish. Of these, 8 study French and German, 6 study German and Spanish, 7 study French and Spanish, and 3 study all three. A student is chosen at random. Find the probability that the student studies (a) exactly one of the three languages, (b) none of them.
Start with the 3 in the centre. The 8 who study French and German include those 3, so $8 - 3 = 5$ study those two and no other. In the same way $6 - 3 = 3$ study German and Spanish only, and $7 - 3 = 4$ study French and Spanish only.
Now the "only" regions. French has 20 altogether, of which $5 + 3 + 4 = 12$ are already placed, leaving 8 for French only. German gives $18-(5+3+3) = 7$, and Spanish gives $15-(4+3+3) = 5$.

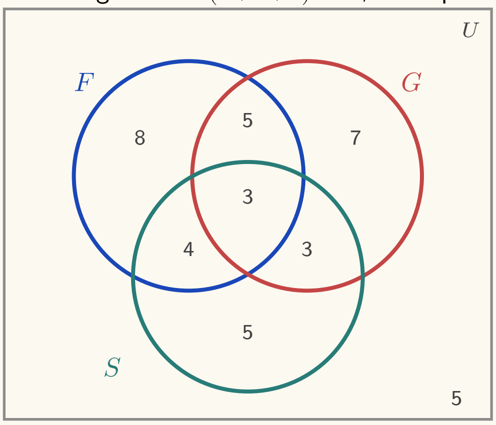


Figure 12.2 The three-circle diagram, filled from the centre outwards. The seven inner counts total 35, so 5 of the 40 students lie outside all three circles.

(a) Exactly one language means the three outer regions: $8 + 7 + 5 = 20$, so

$$P(\text{exactly one}) = \frac{20}{40} = \frac{1}{2}.$$

(b) The seven regions inside the circles total 35, so 5 students study none:

$$P(\text{none}) = \frac{5}{40} = \frac{1}{8}.$$

No formula was used. Once the diagram is filled, every probability is a count divided by 40.

Sample space diagrams and two-way tables

When an experiment has two parts, drawing the whole sample space is often faster than any formula. A **sample space diagram** plots one part on each axis. For two dice, the 36 equally likely outcomes form a grid, and any event is just a set of grid points to count.

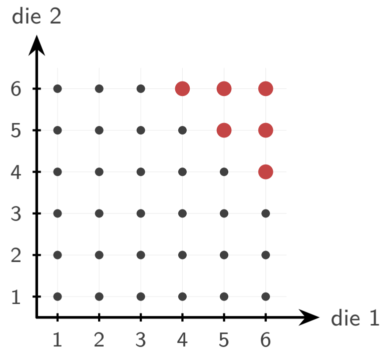


Figure 12.3 The 36 equally likely outcomes of rolling two dice. The six larger scarlet points are the outcomes whose total is at least 10.

The six larger scarlet points are the outcomes with a total of at least 10. There are six of them, so

$$P(\text{total} \geq 10) = \frac{6}{36} = \frac{1}{6}.$$

No rule was needed, only a picture and a count.

A **two-way table** (a table of outcomes) does the same for grouped data: it splits a population by two characteristics at once, and every probability is a cell, row, or column total divided by the grand total.

EXAMPLE 12.8

The table shows the 100 students of a year group by age group and club membership.

	Club	No club	Total
Junior	14	26	40
Senior	21	39	60
Total	35	65	100

A student is chosen at random. Find the probability that the student (a) is a junior in a club, (b) is in a club or is a junior.

(a) Read the single cell where the two conditions meet:

$$P(J \cap C) = \frac{14}{100} = 0.14.$$

(b) Apply the addition rule with the row and column totals:

$$P(J \cup C) = \frac{40}{100} + \frac{35}{100} - \frac{14}{100} = \frac{61}{100} = 0.61.$$

The table already contains every count used in both calculations. We will return to this table in the next section, where it answers conditional questions just as directly.

YOUR TURN 12.2 Combining Events

5.

(a) Given $P(A) = 0.4$, $P(B) = 0.5$ and $P(A \cap B) = 0.2$, find $P(A \cup B)$. [2]

(b) Events A and B are mutually exclusive, with $P(A) = 0.6$ and $P(B) = 0.3$. Find $P(A \cup B)$. [1]

(c) Given $P(A \cup B) = 0.8$, $P(A) = 0.5$ and $P(B) = 0.4$, find $P(A \cap B)$. [2]

(d) Given $P(A) = 0.3$, $P(B) = 0.4$ and $P(A \cup B) = 0.7$, determine whether A and B are mutually exclusive. [2]

6.

(a) A card is drawn from a deck of 52. Find the probability that it is a face card or a spade. [3]

(b) In a group of 40 people, 25 like coffee, 20 like tea and 10 like both. Find the probability that a person chosen at random likes coffee or tea. [2]

(c) Two events satisfy $n(A \text{ only}) = 8$, $n(B \text{ only}) = 5$, $n(A \cap B) = 4$ and $n(\text{neither}) = 3$. Find $P(A)$. [2]

(d) Given $n(U) = 50$, $n(A) = 22$, $n(B) = 18$ and $n(A \cap B) = 7$, find $n(A \cap B')$ and $n(A' \cap B')$. [3]

(e) In a group of 30 students, 15 play the piano, 12 the guitar and 9 the drums. Of these, 6 play piano and guitar, 4 play guitar and drums, 5 play piano and drums, and 2 play all three. Find the probability that a student chosen at random plays exactly one instrument, and the probability that they play none. [4]

7. Two fair dice are rolled.

(a) Find the probability that the total is 7. [2]

(b) Find the probability that the two numbers differ by exactly 1. [2]

(c) Find the probability that the two dice show the same number. [2]

8. The table shows 80 people classified by sex and by writing hand.

	Left	Right	Total
Male	7	33	40
Female	5	35	40
Total	12	68	80

One person is chosen at random.

(a) Find the probability that the person is left-handed. [1]
(b) Find the probability that the person is female and right-handed. [2]
(c) Find the probability that the person is female or left-handed. [2]

12.3 Conditional Probability and Independence

Information changes probability. The chance that a random person has a particular illness is small, but the chance *given* that they tested positive is different. **Conditional probability** is the probability of one event once you know another has happened.

The essential idea is that knowing B has occurred **reduces the sample space** to B alone. Within that smaller set, you ask what fraction also lies in A.

KEY · IB

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \qquad P(B) \neq 0$$

Read $P(A \mid B)$ as "the probability of A given B." The denominator $P(B)$ is the new, restricted sample space; the numerator is the part of it where A also holds.

EXAMPLE 12.9

Using the class of 30 from before, a student is chosen at random. Given that they play football, find the probability that they also play tennis.
Restrict to the 18 football players. Of those, 7 also play tennis:

$$P(T \mid F) = \frac{P(T \cap F)}{P(F)} = \frac{7/30}{18/30} = \frac{7}{18}.$$

Note this differs from $P(F \mid T) = \frac{7}{12}$. Conditioning is not symmetric: the order matters.

The multiplication rule

Rearranging the conditional formula gives a rule for the probability that two events both happen.

KEY

$$P(A \cap B) = P(A)\,P(B \mid A)$$

In words: the chance of both is the chance of the first times the chance of the second given the first. This is exactly how you move along the branches of a tree diagram, as the next section shows.

Independence

Two events are **independent** when knowing one tells you nothing about the other, so $P(A \mid B) = P(A)$. Substituting this into the multiplication rule gives a cleaner, symmetric test.

KEY · IB A and B are **independent** if and only if

$$P(A \cap B) = P(A)\,P(B),$$

or equivalently

$$P(A \mid B) = P(A) = P(A \mid B').$$

To test independence, compute both sides of either form and compare. The conditional form states directly what independence means: whether B happened or did not happen, the probability of A is unchanged.

EXAMPLE 12.10

A fair coin and a fair die are tossed together. Let H be "heads" and S be "rolling a six." Are H and S independent?

Here $P(H) = \frac{1}{2}$ and $P(S) = \frac{1}{6}$. The outcome "heads and a six" is one of the 12 equally likely combined outcomes:

$$P(H \cap S) = \frac{1}{12} = \frac{1}{2} \times \frac{1}{6} = P(H)\,P(S).$$

The two sides match, so the events are independent, which fits intuition: a coin cannot influence a die.

The two-way table of §12.2 passes the same test. There, $P(C \mid J) = \frac{14}{40} = 0.35$ and $P(C) = \frac{35}{100} = 0.35$: knowing a student is a junior does not change the chance they are in a club, so age group and club membership are independent.

CAUTION Mutually exclusive and independent are opposites, not synonyms. If two events with non-zero probability are mutually exclusive, then one happening prevents the other completely, which is the strongest possible dependence. Never assume an ``or'' situation is also an ``independent'' one.

YOUR TURN 12.3 Conditional Probability and Independence

9.

(a) Given $P(A \cap B) = 0.2$ and $P(B) = 0.5$, find $P(A \mid B)$. [2]
(b) Given $P(A) = 0.6$ and $P(B \mid A) = 0.3$, find $P(A \cap B)$. [2]
(c) Given $P(A \mid B) = 0.7$ and $P(B) = 0.4$, find $P(A \cap B)$. [2]
(d) Two fair dice are rolled. Given that the first die shows an even number, find the probability that the total is 8. [3]

10.

(a) Events A and B are independent, with $P(A) = 0.5$ and $P(B) = 0.4$. Find $P(A \cap B)$. [2]
(b) Given $P(A) = 0.3$, $P(B) = 0.5$ and $P(A \cap B) = 0.2$, determine whether A and B are independent. [2]
(c) Two cards are drawn from a deck of 52 without replacement. Find the probability that both are kings. [2]

11. In a class of 25 students, 15 study Spanish, 10 study art and 6 study both.

(a) Given that a student studies Spanish, find the probability that they also study art. [3]
(b) Given that a student studies art, find the probability that they also study Spanish. [2]
(c) Determine whether studying Spanish and studying art are independent. [3]
(d) A card is drawn from a deck of 52. Given that it is a face card, find the probability that it is a king. [2]

12.4 Tree Diagrams

When events happen in sequence, a **tree diagram** organises every path. Each branch carries the probability of that step, and two rules then do the whole calculation: **multiply along the branches** of a single path (the multiplication rule), and **add across paths** that satisfy the condition (the paths are mutually exclusive).

When each draw is made **with replacement**, the object goes back before the next draw, every stage repeats the same probabilities, and the stages are independent. Care is needed with sequences **without replacement**, where the first outcome changes the probabilities on the second branches.

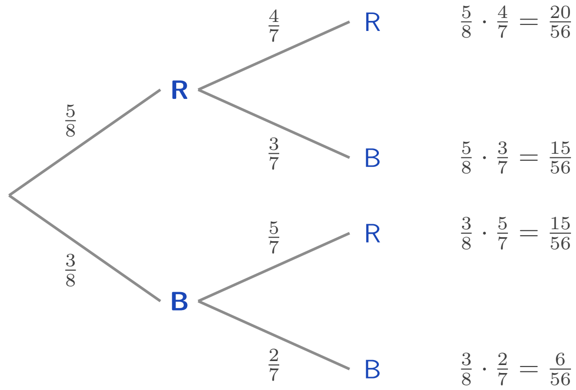


Figure 12.4 A tree diagram for drawing two marbles from 5 red and 3 blue without replacement. The probabilities on the second stage differ from the first, because one marble has already been removed. Multiply along a path; add across paths.

EXAMPLE 12.11

A bag holds 5 red and 3 blue marbles. Two are drawn without replacement. Find the probability that (a) both are red, (b) one of each colour is drawn, (c) at least one is blue.

After drawing one red, only 4 reds remain out of 7 marbles, which is why the second branch changes.

(a) Multiply along the top path:

$$P(\text{both red}) = \frac{5}{8} \times \frac{4}{7} = \frac{20}{56} = \frac{5}{14}.$$

(b) "One of each" covers two paths, red-then-blue and blue-then-red; add them:

$$P(\text{one of each}) = \frac{5}{8} \cdot \frac{3}{7} + \frac{3}{8} \cdot \frac{5}{7} = \frac{15+15}{56} = \frac{30}{56} = \frac{15}{28}.$$

(c) Use the complement of "at least one blue," which is "both red":

$$P(\text{at least one blue}) = 1 - \frac{5}{14} = \frac{9}{14}.$$

All three rules appear here: multiply along a path, add across alternatives, and take the complement for "at least one."

Putting the marble back changes every second-stage branch. The bag returns to 5 red and 3 blue, so both stages read $\frac{5}{8}$ and $\frac{3}{8}$, and the two draws are independent.

EXAMPLE 12.12

The same bag holds 5 red and 3 blue marbles, but now the first marble is replaced before the second is drawn. Find the probability that both are red, and compare with the answer without replacement.

Every branch keeps its original probability, so

$$P(\text{both red}) = \frac{5}{8} \times \frac{5}{8} = \frac{25}{64} \approx 0.391.$$

Without replacement the answer was $\frac{5}{14} \approx 0.357$, which is smaller. Removing a red marble leaves fewer reds for the second draw, so the second red becomes less likely. Replacing it removes that effect.

Replacement also makes the most useful pattern in this chapter available. When the same trial is repeated n times independently, "at least one success" has a complement that is a single product, because "no successes at all" means failure every time.

KEY For n independent trials in which the event has probability p each time,

$$P(\text{at least one}) = 1 - (1-p)^n.$$

EXAMPLE 12.13

A machine part fails on any given day with probability 0.02, independently of other days. Find the probability that it fails at least once during a 30-day month.

Counting the ways to fail once, twice, three times and so on would take all day. The complement is a single product: the part survives all 30 days, each day with probability 0.98.

$$P(\text{at least one failure}) = 1 - (0.98)^{30} = 1 - 0.545 = 0.455.$$

A part that is safe on 98 days out of 100 still has close to an even chance of failing within a month. A small probability repeated often stops being small, which is why maintenance is scheduled by the month and not by the day.

YOUR TURN 12.4 Tree Diagrams

12.

(a) A bag holds 3 white and 2 black balls. Two are drawn with replacement. Find the probability that both are white. [2]

(b) Repeat part (a) for two balls drawn without replacement. [2]

(c) A box holds 6 working and 4 defective bulbs. Two are drawn without replacement. Find the probability that exactly one is defective. [3]

13.

(a) A machine works on any given day with probability 0.9, independently of other days. Find the probability that it works on two consecutive days. [2]

(b) The probability of rain is 0.3. If it rains, the probability of being late is 0.6; if it does not, the probability is 0.1. Find the probability of being late. [3]

(c) A fair die is rolled three times. Find the probability of at least one six. [3]

(d) Using the machine in part (a), find the probability that it fails on at least one of five consecutive days. [3]

14.

(a) A bag holds 4 red and 6 blue counters. Three are drawn without replacement. Find the probability that all three are blue. [3]

(b) For the same three draws, find the probability that exactly two are red. [4]

(c) A component works with probability 0.95, independently of other components. Find the probability that all four components in a circuit work. [2]

(d) Find the probability that at least one of the four components fails. [2]

12.5 Bayes' Theorem HL

We can now return to the letter. The difficulty is that the test gives you $P(\text{positive} \mid \text{disease})$, but what you actually want is the reverse, $P(\text{disease} \mid \text{positive})$. Conditional probability is not symmetric, so these are different numbers. **Bayes' theorem** is the tool that reverses the condition.

It is built from two facts you already have. The conditional formula gives $P(A \mid B) = \frac{P(A\cap B)}{P(B)}$, and the denominator can be split according to whether A happened or not: $P(B) = P(A)P(B \mid A) + P(A')P(B \mid A')$. Putting these together:

KEY · IB HL

$$P(A \mid B) = \frac{P(A)\,P(B \mid A)}{P(A)\,P(B \mid A) + P(A')\,P(B \mid A')}$$

The numerator is the single path through A; the denominator is every path that leads to B. You are asking what fraction of all the ways to observe B passed through A.

That description is the key to using it. Written out in full the formula is long, but a tree diagram carries the same information in a form that is quicker to set up. Draw the first stage, A and A'. Draw the second stage, B given each of them. Multiply along every path. Then divide the path that passes through A by the total of all the paths that end in B.

TIP Do not memorise the Bayes formula as a string of symbols. Draw the tree, multiply along the paths, and divide the path you want by the total of the paths that match the evidence. It gives the same answer, it is far harder to get wrong under exam pressure, and it earns the same marks.

EXAMPLE 12.14

A disease affects 1% of a population. A test is positive for 99% of those who have it, and (falsely) positive for 5% of those who do not. A person tests positive. Find the probability they actually have the disease.

Let D be "has the disease" and $+$ be "tests positive." We know $P(D) = 0.01$, $P(+ \mid D) = 0.99$, and $P(+ \mid D') = 0.05$. Apply Bayes:

$$P(D \mid +) = \frac{(0.01)(0.99)}{(0.01)(0.99) + (0.99)(0.05)} = \frac{0.0099}{0.0099 + 0.0495} = \frac{0.0099}{0.0594} \approx 0.167.$$

The same calculation reads straight off a tree. The upper path gives $0.01 \times 0.99 = 0.0099$, and the lower path gives $0.99 \times 0.05 = 0.0495$. Divide the upper path by the total of both.

Only about 17%. The reason is the base rate. The disease is so rare that the few true positives are greatly outnumbered by the false positives, which come from the very large healthy majority. The test is good; the disease is simply very rare.

Three alternatives

When the first stage has three possibilities instead of two, the denominator simply sums over all three paths. The structure is identical.

EXAMPLE 12.15

Three machines produce all the parts in a factory: machine A makes 50%, B makes 30%, C makes 20%. Their defect rates are 1%, 2%, and 3% respectively. A part is found to be defective. Find the probability it came from machine C.

Weight each machine's defect rate by its share of production, then take C's share of the total:

$$P(C \mid \text{def}) = \frac{(0.20)(0.03)}{(0.50)(0.01) + (0.30)(0.02) + (0.20)(0.03)} = \frac{0.006}{0.005 + 0.006 + 0.006} = \frac{0.006}{0.017} \approx 0.353.$$

Machine C makes only a fifth of the parts but, because it is the least reliable, accounts for over a third of the defects.

YOUR TURN 12.5 Bayes' Theorem HL

15.

(a) Using the situation in question 13 part (b), a person is late. Find the probability that it rained. [2]

(b) A disease affects 2% of a population. A test is positive for 95% of those who have it and for 10% of those who do not. Given a positive result, find the probability of having the disease. [3]

(c) Factory A makes 60% of all items, of which 2% are defective; factory B makes the other 40%, of which 5% are defective. Given that an item is defective, find the probability it came from factory B. [3]

16.

(a) In a mailbox, 30% of messages are spam. The word "free" appears in 90% of spam messages and in 5% of the others. A message contains the word "free". Find the probability that it is spam. [3]

(b) A box contains two coins: one fair, and one with heads on both sides. A coin is chosen at random and tossed once, showing heads. Find the probability that it is the two-headed coin. [3]

(c) Of the students in a year group, 70% revise for a test. Of those who revise, 80% pass; of those who do not, 30% pass. A student passes. Find the probability that the student revised. [3]

17. Three suppliers provide the components used by a factory. Supplier A provides 50%, supplier B 30% and supplier C 20%. Of their components, 1%, 3% and 4% respectively are defective.

(a) Find the probability that a component chosen at random is defective. [3]

(b) Given that a component is defective, find the probability that it came from supplier B. [3]

(c) Given that a component is defective, find the probability that it came from supplier A. [2]

(d) Supplier A provides the most components but accounts for the fewest defective ones. Explain why. [2]

Chapter 12 · Probability

Reflections

IM Probability began at a gambling table. In 1654 the French writer Antoine Gombaud, the Chevalier de Méré, asked Blaise Pascal why a betting strategy that seemed sound was losing him money. Pascal's correspondence with Pierre de Fermat over that single question laid the foundations of probability theory. A subject now central to physics, finance, and medicine grew out of a dispute about dice.

TOK Bayes' theorem describes how a rational mind should update a belief when new evidence arrives: start from a *prior*, the probability believed before the evidence, weigh the evidence, and arrive at a *posterior*, the probability believed afterwards. Some philosophers argue this is not just a formula but a model of knowledge itself. If that is right, then the disease-test paradox is not a quirk of medicine. It is a warning that evidence means little until you account for how rare the thing you are testing for actually is.

TOK We say a fair die has probability $\frac{1}{6}$ because of its symmetry, and we say a drawing pin lands point-up with probability 0.39 because that is what 200 drops showed. The first number was never measured; the second was never proved.

Are these two the same kind of claim? And when a measured value disagrees with a calculated one, which of the two should be revised?

TOK Probability was built to answer questions about gambling, and casinos use it now to guarantee themselves a profit. The mathematics itself is neutral, but the uses of it are not.

Should a mathematician who works out the odds be held responsible for how those odds are used?

RW The base-rate paradox has real costs. Courtrooms have convicted people on "one-in-a-million" DNA matches without asking how many people in the population would also match by chance. Mass screening programmes for rare cancers can produce more false alarms than true detections. Understanding $P(D \mid +)$ versus $P(+ \mid D)$ is not only an examination topic. It changes verdicts and medical decisions.

End-of-Chapter Problems — Chapter 12: Probability

1. In a class of 50 students, 28 study Mathematics (M), 20 study Art (A), and 10 study neither.

(a) Find the number of students who study both Mathematics and Art. [3]
(b) Find the probability that a randomly chosen student studies only Art. [2]
(c) Find $P(A \mid M)$. [2]

2. The two-way table shows whether 100 people exercise regularly.

	Exercises	Does not	Total
Male	30	20	50
Female	24	26	50
Total	54	46	100

(a) Find the probability that a person is male and exercises. [2]
(b) Find the probability that a person exercises, given that they are female. [2]
(c) Determine whether the events "male" and "exercises" are independent. [3]

3. A bag contains 7 red and 3 green sweets. Two are drawn without replacement.

(a) Draw a tree diagram for the two draws. [3]
(b) Find the probability that both sweets are red. [2]
(c) Find the probability that both sweets are the same colour. [3]
(d) Find the probability that at least one sweet is green. [2]

4. A condition affects 5% of a population. A screening test is positive for 90% of those who have it, and for 15% of those who do not.

(a) Find the probability that a randomly chosen person tests positive. [3]
(b) Given a positive result, find the probability the person has the condition. [3]
(c) Comment on what your answer says about screening for rare conditions. [2]

5. Events A and B satisfy $P(A) = 0.5$ and $P(B) = 0.3$.

(a) Find $P(A \cup B)$ if A and B are independent. [3]
(b) Find $P(A \cup B)$ if A and B are mutually exclusive. [2]
(c) Explain why A and B cannot be both independent and mutually exclusive here. [2]

6. Two fair dice are rolled and the scores noted.

(a) Find the probability that the sum is 7. [2]
(b) Find the probability that at least one die shows a 6. [2]
(c) Given that at least one die shows a 4, find the probability that the sum is 7. [3]

7. Three suppliers provide components: X supplies 25%, Y supplies 45%, and Z supplies 30%. Their defect rates are 4%, 2%, and 5% respectively.

(a) Find the probability that a randomly chosen component is defective. [3]
(b) Given that a component is defective, find the probability it came from Z. [3]

8. Events A and B are independent. Given $P(A) = 0.4$ and $P(A \cup B) = 0.7$, find $P(B)$. [4]

9. A basketball player makes each free throw independently with probability 0.8. She takes 4 free throws.

(a) Find the probability she makes all four. [2]
(b) Find the probability she misses at least one. [2]

10. At a school, 60% of students study French and 25% study both French and Spanish. Find the probability that a student studies Spanish, given that they study French. [3]

11. In a class of 40 students, the numbers studying two activities A and B are: only A is $2x$, both is x, only B is $x + 4$, and neither is 8.

(a) Find the value of x. [3]
(b) Find $P(A \cap B)$. [1]
(c) Find $P(A \cup B)$. [2]

12. A biased four-sided spinner numbered 1 to 4 was spun 250 times. The results were:

Number	1	2	3	4
Frequency	95	60	55	40

(a) Estimate the probability that the spinner lands on 1. [2]
(b) Estimate the probability that the spinner lands on an odd number. [2]
(c) The spinner is spun a further 60 times. Find the expected number of times it lands on 4. [2]

13. Events A and B satisfy $P(A) = 0.45$, $P(B) = 0.4$, and $P((A \cup B)') = 0.25$.

(a) Find $P(A \cap B)$. [2]
(b) Find the probability that A occurs but B does not. [1]

(c) Find $P(A \mid B)$. [2]

(d) Determine whether A and B are independent. [2]

14. A fair four-sided die (numbered 1–4) and a fair six-sided die (numbered 1–6) are rolled together, and the product of the two scores is recorded.

(a) Represent the outcomes in a sample space diagram. [2]

(b) Find the probability that the product is 12. [2]

(c) Find the probability that the product is odd. [2]

(d) Given that the product is even, find the probability that the four-sided die shows a 4. [3]

15. A football team's key striker is fit for a match with probability 0.85. The team wins with probability 0.7 if she is fit, and with probability 0.4 if she is not.

(a) Draw a tree diagram to represent this information. [2]

(b) Find the probability that the team wins the match. [2]

(c) Given that the team won, find the probability that the striker was fit. [3]

16. A box contains 4 red and 2 green counters. Three counters are drawn at random without replacement.

(a) Find the probability that all three counters are red. [2]

(b) Find the probability that exactly one counter is green. [3]

(c) Find the probability that at least one counter is green. [2]

17. Events A and B satisfy $P(A) = 0.6$, $P(A \mid B) = 0.6$, and $P(A \cup B) = 0.84$.

(a) Explain why A and B are independent. [1]

(b) Find $P(B)$. [3]

(c) Write down $P(B \mid A')$, justifying your answer. [2]

18. The 80 students of a year group are classified by gender and handedness: 10 of the 40 boys and 6 of the 40 girls are left-handed.

(a) Display this information in a two-way table, including totals. [2]

(b) Find the probability that a randomly chosen student is a right-handed girl. [1]

(c) Given that a student is left-handed, find the probability that the student is a boy. [2]

(d) Determine, using conditional probability, whether the events "boy" and "left-handed" are independent. [3]

19. An airline routes 50% of its flights through airport A, 30% through airport B, and 20% through airport C. The probabilities that a flight is delayed are 0.06, 0.03, and 0.10 respectively.

(a) Find the probability that a randomly chosen flight is delayed. [3]
(b) Given that a flight was delayed, find the probability that it was routed through airport A. [3]

20. Events A, B, and C are mutually exclusive and exhaustive, with $P(A) = 2k$, $P(B) = 3k$, and $P(C) = k + 0.1$.

(a) Find the value of k. [2]
(b) Find $P(A \cup B)$. [2]
(c) Find $P(A \mid B')$. [3]

21. GDC An archer hits the target with probability 0.9 on each shot, independently of all other shots.

(a) Find the probability that her first miss is on her third shot. [2]
(b) Find the probability that she misses at least once in five shots. [2]
(c) Find the least number of shots for which the probability of at least one miss exceeds 0.95. [3]

22. A family has two children. Assume each child is equally likely to be a boy or a girl, independently.

(a) List the sample space. [1]
(b) Given that at least one of the children is a boy, find the probability that both are boys. [3]
(c) Given instead that the *elder* child is a boy, find the probability that both are boys. [2]
(d) Explain why the answers to (b) and (c) differ. [1]

23. In a survey of 100 people, 50 like tea, 60 like coffee, and 35 like juice. Also, 30 like tea and coffee, 20 like tea and juice, 25 like coffee and juice, and 10 like all three.

(a) Find the number who like at least one of the three drinks. [3]
(b) Find the number who like none. [1]
(c) Find the number who like exactly one drink. [3]

24. A disease affects 1% of a population. A test is positive for 99% of those with the disease and for 5% of those without. A person tests positive *twice*, the two tests being independent given disease status.

(a) Find the probability the person has the disease after one positive test. [3]
(b) Find the probability the person has the disease after two positive tests. [4]
(c) Comment on the effect of the second test. [1]

25. Events A and B are independent. Prove that A and B' are also independent, that is, show $P(A \cap B') = P(A)\,P(B')$. [5]

26. From a group of 5 men and 4 women, three people are chosen at random. Find the probability that exactly two are women. [5]

27. Of the 200 students at a school, 130 cycle to school (C), 90 travel by train (T), and 40 do both. The remaining students walk.

(a) Draw a Venn diagram showing the number of students in each of the four regions. [3]
(b) Find the probability that a student chosen at random walks to school. [2]
(c) Find $P(C \mid T)$. [3]
(d) Determine whether C and T are independent, justifying your answer. [3]
(e) Two students are chosen at random, without replacement. Find the probability that both of them walk. [3]
(f) State whether your answer to part (e) would be larger or smaller if the two students were chosen with replacement, and explain why. [2]

28. A machine produces components, of which 6% are defective. Each component is inspected. The inspection rejects a defective component with probability 0.95, and wrongly rejects a component that is not defective with probability 0.02.

(a) Draw a tree diagram to represent this information. [3]
(b) Find the probability that a randomly chosen component is rejected. [3]
(c) Find the probability that a rejected component was in fact not defective. [3]
(d) Find the probability that a component which passes inspection is defective. [3]
(e) In a batch of 5000 components, find the expected number that are defective but pass inspection. [2]
(f) Using your answers to parts (d) and (e), comment on how well the inspection protects a customer. [2]

29. A box contains 10 batteries, of which 3 are flat. Batteries are taken from the box one at a time, without replacement, and tested until a working one is found. Let N be the number of batteries tested.

(a) Find $P(N = 1)$. [2]
(b) Find $P(N = 2)$. [3]
(c) Find $P(N \leq 3)$. [4]
(d) Explain why N can never be greater than 4. [2]
(e) Find $P(N = 3)$ and $P(N = 4)$, and verify that the four probabilities sum to 1. [4]

Challenge Questions

30. The birthday problem. In a group of n people, assume every birthday is equally likely among 365 days and ignore leap years.

(a) For $n = 2$, show that the probability the two birthdays differ is $\dfrac{364}{365}$. [2]

(b) Explain why, for $n = 3$, the probability that all three differ is $\dfrac{364}{365} \times \dfrac{363}{365}$. [3]

(c) Write down an expression for the probability that all n birthdays differ. [3]

(d) Hence find, to three decimal places, the probability that at least two people share a birthday when $n = 10$, $n = 23$ and $n = 50$. [4]

(e) Find the smallest n for which the probability of a shared birthday exceeds $\frac{1}{2}$. [3]

(f) Most people guess that n must be close to 180. Explain why the true answer is so much smaller, by considering how many *pairs* of people a group of n contains. [3]

31. Three doors. A prize is behind one of three closed doors. You choose a door. The host knows where the prize is, and opens one of the other two doors, always an empty one. You may keep your door or switch to the last closed door.

(a) Write down the probability that your first choice is correct. [1]

(b) Find the probability of winning if you always switch. Treat the case where your first choice is correct separately from the case where it is not. [5]

(c) State the probability of winning if you always stay, and say which strategy is better. [2]

(d) The host always opens an empty door, so his action seems to tell you nothing. Explain why it does. [4]

(e) The game is now played with n doors and one prize, and the host opens every door except your choice and one other. Show that switching wins with probability $\dfrac{n-1}{n}$. [4]

32. The Chevalier's puzzle. In 1654 a French gambler won money on one of these two bets and lost money on the other. He could not see why.

(a) Bet 1: roll one die four times, and win if at least one six appears. Show that the probability of winning is $\dfrac{671}{1296}$. [4]

(b) Bet 2: roll a pair of dice 24 times, and win if at least one double six appears. Find this probability, to four decimal places. [4]

(c) State which bet the gambler should take, and explain why the two probabilities are so close. [3]

(d) He argued that 24 rolls of two dice must be as good as 4 rolls of one, because 24 is to 36 as 4 is to 6. Explain the error. [3]

(e) Find the smallest number of rolls of a pair of dice for which Bet 2 is favourable. [4]

■ ■ ■

STATISTICS AND PROBABILITY

Chapter 13

Probability Distributions

13

SYLLABUS 4.7 4.8 4.9 4.12 4.14

Measure the same kind of thing many times and you rarely get the same answer twice. Masses differ from packet to packet, heights differ from person to person, and a repeated experiment gives a different count each time. One number cannot describe a quantity that behaves like this. What describes it is the list of values it can take together with how likely each one is, and that list is called a **distribution**. This chapter is about the few distributions that cover a great many situations.

A factory fills bags of rice, each meant to hold 500 grams, and weighs 5000 of them. No two masses agree exactly. Grouping the readings and drawing them as a graph gives a shape: most bags near 500 g, with fewer and fewer further out on either side, falling away symmetrically into one smooth hump. A bell.

The same curve turns up in places that have nothing to do with rice. Astronomers found it in the *errors* of their telescope readings, and biologists in the heights of people. The reason is the same in every case. A bag's mass is not decided by one thing but by many: the size of the individual grains, a slight vibration in the machine, the exact instant the valve closes. Each effect is tiny and as likely to push the mass up as down, and when many such effects add together, the bell is what you get.

A distribution is worth having because it turns uncertainty into something you can calculate with. From one you can predict an average result, measure how far individual results are likely to stray from it, and put a number on the risk of an extreme value. That is what makes these few curves useful in medicine, manufacturing and finance. It is also where care is needed, because every answer depends on the distribution being the right description of the situation, and choosing the wrong one gives confident answers that are wrong.

We start with the simplest case, a single uncertain number taking a few values, and work up through the distribution used for coin flips and quality control, until the bell curve appears as their limit.

13.1 Discrete Random Variables

A **random variable** is a number whose value depends on chance. The number of heads in three coin tosses, the score on a die, the number of defective items in a box: each is a rule that assigns a number to a random outcome. We write random variables with capital letters, X, and particular values with lower case, x.

A random variable is **discrete** when it can only take separate, countable values. To describe it fully, you list every value it can take alongside the probability of each. This list is its **probability distribution**.

DEF A **discrete random variable** X takes countable values $x_1, x_2, \ldots$ Its **probability distribution** gives $P(X = x_i)$ for each value, subject to

$$P(X = x_i) \geq 0 \qquad \text{and} \qquad \sum_i P(X = x_i) = 1.$$

The second condition is the one you use most often. Since some value must occur, all the probabilities add to exactly 1. This single fact lets you find a missing probability.

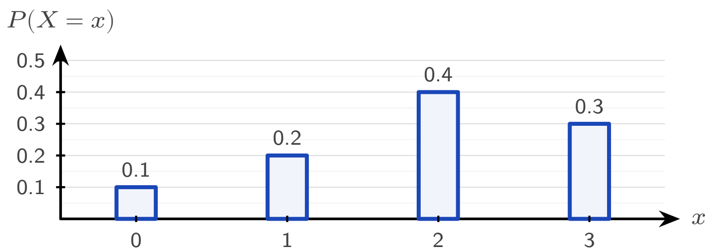


Figure 13.1 The probability distribution of a discrete random variable. Each bar is the probability of one value, and the four bars must total exactly 1.

EXAMPLE 13.1

A discrete random variable X has the distribution below. Find k, then $P(X \geq 3)$.

x	**1**	**2**	**3**	**4**
$P(X = x)$	0.1	0.3	k	0.2

The probabilities must sum to 1:

$$0.1 + 0.3 + k + 0.2 = 1 \implies k = 0.4.$$

Then $P(X \geq 3) = P(X = 3) + P(X = 4) = 0.4 + 0.2 = 0.6$.

EXAMPLE 13.2

A random variable X takes values $1, 2, 3, 4$ with $P(X = x) = kx$. Find k.
Sum the probabilities and set equal to 1:

$$k(1) + k(2) + k(3) + k(4) = 10k = 1 \implies k = \frac{1}{10}.$$

When the distribution is given by a formula, the total-probability condition still determines the constant.

YOUR TURN 13.1 Discrete Random Variables

1.

(a) For $x = 1, 2, 3$ the probabilities are 0.2, 0.5 and k. Find k. [1]

(b) A variable X has $P(X = x) = 0.15,\ 0.25,\ 0.35,\ k$ for $x = 0, 1, 2, 3$. Find k and $P(X \leq 1)$. [2]

(c) A variable takes values $x = 0, 1, 2, 3$ with $P(X = x) = k(x + 1)$. Find k. [2]

(d) A variable takes values $x = 1, 2, 3, 4$ with $P(X = x) = \dfrac{x^2}{30}$. Find $P(X = 3)$. [2]

(e) For the distribution $P(X = x) = 0.1,\ 0.4,\ 0.3,\ 0.2$ at $x = 0, 1, 2, 3$, find $P(X > 1)$ and $P(1 \leq X \leq 2)$. [2]

2.

(a) Show that $P(X = x) = \frac{1}{18}(4 + x)$ for $x = 1, 2, 3$ defines a probability distribution, and find $P(X \geq 2)$. [3]

(b) A variable takes values $x = 1, 2, 3, 4$ with probabilities 0.1, a, 0.3, b. Given that $P(X \leq 2) = 0.4$, find a and b. [3]

(c) Hence find $P(X > 2)$ for the distribution in part (b). [1]

(d) A variable takes values $x = 1, 2, 3$ with $P(X = x) = k \cdot 2^x$. Find k. [2]

13.2 Expected Value and Variance

If you played a game many times, what would you win on average per play? That long-run average is the **expected value** $E(X)$. It is not the most likely outcome, nor a value X need ever take. It is the balance point of the distribution: each value weighted by its probability.

KEY · IB

$$E(X) = \mu = \sum_i x_i \, P(X = x_i)$$

This is exactly the mean from Ch. 11, with probabilities playing the role of relative frequencies. Over many trials, the relative frequency of each value approaches its probability, so the average approaches $E(X)$.

EXAMPLE 13.3

A game costs \$2 to play. You roll a fair die and win, in dollars, the number shown. Is the game worth playing?

Let X be the amount won. Each face is equally likely:

$$E(X) = \tfrac{1}{6}(1+2+3+4+5+6) = \tfrac{21}{6} = 3.5.$$

You expect to win \$3.50 per play but pay \$2, so the expected net gain is \$1.50. Over many games you gain money on average.

That last calculation names an idea the syllabus uses directly. A game is **fair** when the expected net gain is zero, so that neither side profits in the long run. Since the net gain is the winnings minus the stake, a fair stake is exactly the expected winnings.

KEY · IB A game is **fair** when $E(\text{gain}) = 0$, that is, when the stake equals $E(\text{winnings})$.

EXAMPLE 13.4

A fair die is rolled. You win \$10 for a six, \$4 for a four or a five, and nothing otherwise. Find the stake that makes the game fair, and state whether a stake of \$4 favours the player or the organiser.

Let W be the amount won. The three cases have probabilities $\frac{1}{6}$, $\frac{2}{6}$ and $\frac{3}{6}$, so

$$E(W) = 10\left(\tfrac{1}{6}\right) + 4\left(\tfrac{2}{6}\right) + 0\left(\tfrac{3}{6}\right) = \tfrac{10+8}{6} = 3.$$

A stake of \$3 makes the game fair: the expected gain is $3 - 3 = 0$. At a stake of \$4 the expected gain is $3 - 4 = -1$, a loss of one dollar per play on average, so the game favours the organiser. Note that no single play ever produces a loss of exactly \$1; the figure is the average over many plays.

Variance and standard deviation HL

Expected value locates the centre; **variance** measures the spread, just as in descriptive statistics. The variance is the expected squared distance from the mean. A short calculation turns it into a more practical form.

KEY · IB HL

$$\text{Var}(X) = E((X-\mu)^2) = E(X^2) - \mu^2, \qquad \text{where } E(X^2) = \sum_i x_i^2\, P(X = x_i)$$

The form $E(X^2) - \mu^2$ is mostly easier to use: find the mean of the squares, then subtract the square of the mean. The standard deviation is $\sigma = \sqrt{\text{Var}(X)}$.

EXAMPLE 13.5

Find the variance and standard deviation of X:

x	**0**	**1**	**2**	**3**
$P(X = x)$	0.1	0.2	0.4	0.3

First the mean:

$$\mu = 0(0.1) + 1(0.2) + 2(0.4) + 3(0.3) = 1.9.$$

Then $E(X^2)$:

$$E(X^2) = 0^2(0.1) + 1^2(0.2) + 2^2(0.4) + 3^2(0.3) = 0.2 + 1.6 + 2.7 = 4.5.$$

So $\text{Var}(X) = 4.5 - 1.9^2 = 4.5 - 3.61 = 0.89$, and $\sigma = \sqrt{0.89} \approx 0.943$.

Linear transformations HL

If you scale and shift a random variable, its mean and variance respond predictably, exactly as a data set does. For constants a and b:

KEY · IB HL

$$E(aX + b) = aE(X) + b, \qquad \text{Var}(aX + b) = a^2\, \text{Var}(X)$$

The shift b moves the mean but not the spread. The scale a stretches the mean and stretches the variance by a^2, since variance is built from squared distances.

EXAMPLE 13.6

For the variable above, $E(X) = 1.9$ and $\text{Var}(X) = 0.89$. Let $Y = 2X + 3$. Find $E(Y)$ and $\text{Var}(Y)$.

$$E(Y) = 2(1.9) + 3 = 6.8, \qquad \text{Var}(Y) = 2^2(0.89) = 3.56.$$

The $+3$ leaves the variance untouched; only the factor of 2, squared, scales it.

YOUR TURN 13.2 Expected Value and Variance

3.

(a) Find $E(X)$ for $x = 1, 2, 3$ with $P(X = x) = 0.2,\ 0.3,\ 0.5.$ [2]

(b) Find $E(X)$ for $x = 0, 1, 2, 3$ with $P(X = x) = 0.1,\ 0.4,\ 0.3,\ 0.2.$ [2]

(c) A fair coin pays \$3 for heads and loses \$2 for tails. Find the expected gain per toss. [2]

(d) A game costs \$1. You win \$5 with probability $\frac{1}{8}$, and otherwise lose your stake. Find the expected gain, and state whether the game is fair. [3]

4.

(a) For the distribution in question 3 part (b), find $E(X^2)$ and Var(X). [3]

(b) Given $E(X) = 4$ and Var$(X) = 2$, find $E(3X - 1)$ and Var$(3X - 1)$. [2]

5.

(a) A wheel is spun for a prize. It pays \$8 with probability 0.1, \$2 with probability 0.3, and nothing otherwise. Find the expected amount won, and the stake that would make the game fair. [3]

(b) The stall charges \$2 to spin. Find the expected gain per spin for a player, and the expected loss over 50 spins. [3]

(c) A variable has $E(X) = 3$ and Var$(X) = 5$. Find $E(2X + 1)$, Var$(2X + 1)$ and $E(X^2)$. [4]

13.3 The Binomial Distribution

Many situations are the same experiment repeated: ten coin flips, twenty items checked for defects, a quiz of multiple-choice guesses. When each repetition is independent and has just two outcomes, the count of "successes" follows one specific distribution, the **binomial**.

Four conditions define it. Check each one before reaching for the distribution.

- **Fixed number of trials.** The value of n is decided in advance, not by the results.
- **Two outcomes only.** Every trial ends as a success or a failure, with nothing in between.
- **Constant probability.** Every trial has the same probability of success p.

- **Independence.** The result of one trial does not change the probabilities for any other.

When all four hold, write $X \sim B(n, p)$.

DEF If X counts the successes in n independent trials each with success probability p, then $X \sim B(n, p)$, and

$$P(X = r) = \binom{n}{r} p^r (1-p)^{n-r}, \qquad r = 0, 1, 2, \dots, n.$$

Each factor has a meaning: p^r for r successes, $(1-p)^{n-r}$ for the rest failing, and $\binom{n}{r}$ for the number of different orders in which those successes can fall. In practice you evaluate it with a GDC, using the binomial pdf for $P(X = r)$ and the binomial cdf for $P(X \leq r)$.

EXAMPLE 13.7

A student guesses on a 10-question multiple-choice test, each question having 4 options. Find the probability of (a) exactly 3 correct, (b) at most 2 correct.
Here $X \sim B(10, 0.25)$, since each guess is correct with probability $\frac{1}{4}$.
(a) Using the binomial pdf:

$$P(X = 3) = \binom{10}{3}(0.25)^3(0.75)^7 \approx 0.250.$$

(b) Using the binomial cdf:

$$P(X \leq 2) = P(0) + P(1) + P(2) \approx 0.526.$$

Despite 10 questions, scoring 3 or fewer is more likely than not. Guessing rarely produces a good score.

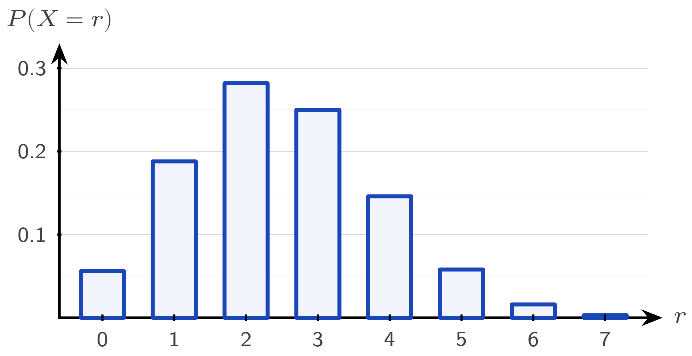


Figure 13.2 The distribution $B(10,\ 0.25)$, the guessing student of the example above. The bars are tallest at $r = 2$ and $r = 3$, and the long tail to the right shows why a high score by guessing is so unlikely.

Mean and variance of the binomial

You could find the mean by summing $r\,P(X = r)$, but there is no need. For n trials each succeeding with probability p, the results are clean.

KEY · IB For $X \sim B(n, p)$:

$$E(X) = np, \qquad \text{Var}(X) = np(1 - p)$$

The mean is intuitive: 10 trials at probability 0.25 give 2.5 expected successes. The variance formula is largest when $p = 0.5$, when each trial is most unpredictable.

EXAMPLE 13.8

A factory's items are defective with probability 0.05, independently. A sample of 20 is taken. Find the expected number of defectives and the probability that at least one is defective.

Let $X \sim B(20, 0.05)$. The expected number is

$$E(X) = 20 \times 0.05 = 1.$$

For "at least one," use the complement:

$$P(X \geq 1) = 1 - P(X = 0) = 1 - (0.95)^{20} \approx 0.642.$$

Even with a low defect rate, a sample of 20 is more likely than not to contain a defective item. The probability of "at least one" rises quickly as the sample size increases.

CAUTION Check independence and a fixed n before using the binomial. Drawing items *without replacement* breaks independence, since each draw changes the next probability, so that count is not binomial.

YOUR TURN 13.3 The Binomial Distribution

6. GDC

(a) $X \sim B(8, 0.5)$. Find $P(X = 4)$. [2]
(b) $X \sim B(10, 0.3)$. Find $P(X = 2)$. [2]
(c) $X \sim B(6, 0.4)$. Find $P(X \leq 2)$. [2]
(d) $X \sim B(12, 0.25)$. Find $P(X \geq 1)$. [2]

7.

(a) $X \sim B(20, 0.1)$. Find the mean and the variance. [2]
(b) $X \sim B(15, 0.6)$. Find $E(X)$ and $\text{Var}(X)$. [2]

8. GDC In a factory, 8% of items are defective, independently of one another. A sample of 25 items is taken.

(a) Find the probability that exactly 2 items are defective. [2]
(b) Find the probability that at least 1 item is defective. [2]
(c) Find the expected number of defective items in the sample. [1]
(d) A quiz has 12 questions, each with 5 options, and a student guesses every answer. Find the probability of at least 4 correct answers. [3]
(e) A box holds 10 items, 3 of them defective, and 4 items are drawn without replacement. Explain why the number of defective items drawn is not binomial. [2]

13.4 Continuous Random Variables HL

A discrete variable jumps between separate values. A continuous one, such as a height, a mass or a waiting time, can take any value in a range. You cannot list a probability for each value, because there are infinitely many of them. Something has to replace the list.

The bags of rice show what. Group the 5000 masses into bars one gram wide, and the probability of landing in a group is the *area* of its bar. Now halve the bar width. There are twice as many bars, each about half as tall, and the total area is unchanged at 1. Halve it again, and again. The tops of the bars settle into a smooth curve, and the area under that curve is still 1.

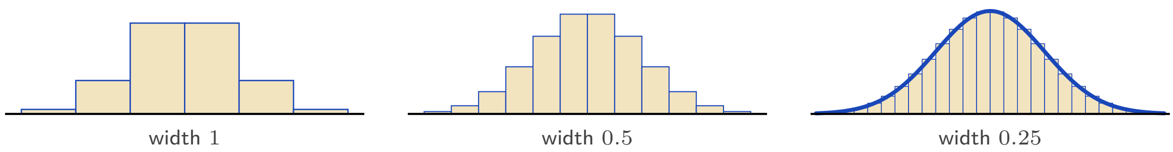


Figure 13.3 The same 5000 masses grouped into bars of decreasing width. Every version encloses total area 1; as the bars narrow, their tops approach a smooth curve. That curve is the density.

So probability for a continuous variable is **area**, and the curve whose area gives it is called a **probability density function**, written $f(x)$. Two conditions make f a valid density: it is never negative, and the total area under it is 1.

DEF A **probability density function** $f(x)$ satisfies

$$f(x) \geq 0 \quad \text{for all } x, \qquad \int_{-\infty}^{\infty} f(x)\,dx = 1,$$

and probabilities are areas: $P(a \leq X \leq b) = \displaystyle\int_a^b f(x)\,dx.$

One consequence looks strange at first. A bar of zero width has zero area, so $P(X = c) = 0$ for every single value c. This does not mean that value is impossible. It means no exact value carries a share of the total: a person's height is never exactly 170 cm to unlimited precision, only somewhere in a range near it. Since the endpoints contribute nothing, $P(a \leq X \leq b)$ and $P(a < X < b)$ are the same number, and you may be careless about $<$ and $\leq$ here in a way you cannot be with a discrete variable.

The expected value and variance carry over from the discrete case, with integrals replacing sums.

KEY

$$E(X) = \int_{-\infty}^{\infty} x\,f(x)\,dx, \qquad \text{Var}(X) = \int_{-\infty}^{\infty} x^2 f(x)\,dx - \mu^2$$

Three numbers describe where a density sits, and each is read off the curve in its own way. The **mode** is the value where f is highest. The **median** is the value that cuts the area in half. The **mean** is the balance point, $\int x f(x)\,dx$. For a symmetric curve all three coincide; for a lopsided one they do not, and the next example shows all three at work.

EXAMPLE 13.9

A continuous random variable has density $f(x) = kx$ for $0 \leq x \leq 2$, and $f(x) = 0$ elsewhere. Find k, then $P(X < 1)$, the mean, and the median.

Total area must be 1:

$$\int_0^2 kx\,dx = k\left[\tfrac{x^2}{2}\right]_0^2 = 2k = 1 \implies k = \tfrac{1}{2}.$$

Then

$$P(X<1)=\int_0^1 \tfrac{x}{2}\,dx=\left[\tfrac{x^2}{4}\right]_0^1=\tfrac{1}{4}.$$

The mean:

$$E(X)=\int_0^2 x\cdot\tfrac{x}{2}\,dx=\int_0^2 \tfrac{x^2}{2}\,dx=\left[\tfrac{x^3}{6}\right]_0^2=\tfrac{8}{6}=\tfrac{4}{3}.$$

The **median** m splits the area in half:

$$\int_0^m \tfrac{x}{2}\,dx=\tfrac{1}{2}\implies\tfrac{m^2}{4}=\tfrac{1}{2}\implies m=\sqrt{2}.$$

The **mode** is where f is greatest. Since $f(x) = \frac{x}{2}$ increases across $[0,2]$, the mode is at the right end, $x=2$.

CAUTION The mean, median, and mode of a continuous variable usually differ. The mode maximises $f(x)$; the median halves the area; the mean is the balance point $\int x f(x)\,dx$. Only for a symmetric density do all three coincide.

Piecewise densities

A density need not be a single formula. Any function that is non-negative and encloses total area 1 qualifies, including one built from pieces. The only new requirement is care with the arithmetic: integrals must be split at the joins, and the median may lie in either piece.

EXAMPLE 13.10

A continuous random variable has density

$$f(x)=\begin{cases}\dfrac{2x}{3} & 0\le x\le 1,\\ \dfrac{3-x}{3} & 1<x\le 3,\\ 0 & \text{otherwise.}\end{cases}$$

Verify that f is a valid density, find $P(X>2)$, and find the median.
Both pieces are non-negative, and the total area splits at $x=1$:

$$\int_0^1 \tfrac{2x}{3}\,dx+\int_1^3 \tfrac{3-x}{3}\,dx=\left[\tfrac{x^2}{3}\right]_0^1+\tfrac{1}{3}\left[3x-\tfrac{x^2}{2}\right]_1^3=\tfrac{1}{3}+\tfrac{2}{3}=1.\checkmark$$

For $P(X>2)$ only the second piece matters:

$$P(X>2)=\int_2^3 \tfrac{3-x}{3}\,dx=\tfrac{1}{3}\left[3x-\tfrac{x^2}{2}\right]_2^3=\tfrac{1}{6}.$$

For the median, first locate it: the area up to $x = 1$ is $\frac{1}{3} < \frac{1}{2}$, so the median lies in the second piece. Collect the remaining area there:

$$\tfrac{1}{3} + \int_{1}^{m} \tfrac{3-x}{3}\,dx = \tfrac{1}{2} \implies 3m - \tfrac{m^2}{2} - \tfrac{5}{2} = \tfrac{1}{2} \implies m^2 - 6m + 6 = 0,$$

giving $m = 3 - \sqrt{3} \approx 1.27$ (taking the root inside the interval). The mode is $x = 1$, where the two pieces meet at the peak of the triangle.

Modes by calculus

When a density rises and then falls, its mode is an interior maximum, and you find it by the standard method: differentiate and set $f'(x) = 0$. The mode can also sit at an endpoint or at a corner, as in the examples above, so always check where the maximum actually is.

EXAMPLE 13.11

A continuous random variable has density $f(x) = kx^2(3-x)$ for $0 \le x \le 3$, and zero elsewhere. Find k, the mode, and $E(X)$.

Total area 1:

$$\int_{0}^{3} k(3x^2 - x^3)\,dx = k\Big[x^3 - \tfrac{x^4}{4}\Big]_0^3 = k\left(27 - \tfrac{81}{4}\right) = \tfrac{27k}{4} = 1 \implies k = \tfrac{4}{27}.$$

For the mode, maximise f:

$$f'(x) = k(6x - 3x^2) = 3kx(2-x) = 0 \implies x = 0 \text{ or } x = 2.$$

Since f rises on $(0,2)$ and falls on $(2,3)$, the mode is $x = 2$. Finally,

$$E(X) = \int_{0}^{3} k(3x^3 - x^4)\,dx = \tfrac{4}{27}\left[\tfrac{3x^4}{4} - \tfrac{x^5}{5}\right]_0^3 = \tfrac{4}{27}\cdot\tfrac{243}{20} = \tfrac{9}{5}.$$

Mean 1.8, mode 2: the density is not symmetric, so the balance point sits below the peak.

YOUR TURN 13.4 Continuous Random Variables HL

9.

(a) A density is $f(x) = kx^2$ for $0 \le x \le 3$, and zero elsewhere. Find k. [2]

(b) A density is $f(x) = k$ for $0 \le x \le 4$, and zero elsewhere. Find k and $P(X < 1)$. [2]

(c) A density is $f(x) = \dfrac{x}{8}$ for $0 \le x \le 4$, and zero elsewhere. Find $P(X > 2)$. [2]

(d) For the density in part (c), find $E(X)$. [3]

(e) A density is $f(x) = 3x^2$ for $0 \le x \le 1$, and zero elsewhere. Find the median. [3]

10.

(a) A continuous random variable has density $f(x) = kx(4-x)$ for $0 \le x \le 4$, and zero elsewhere. Find k. [3]

(b) Write down the mode of this density, and justify your answer. [2]

(c) Find $E(X)$. [2]

(d) A second variable has density $f(x) = x$ for $0 \le x \le 1$, $f(x) = \frac{1}{2}$ for $1 < x \le 2$, and zero elsewhere. Verify that the total area is 1, and find $P(X < 1.5)$. [4]

(e) Find the median of the density in part (d). [2]

13.5 The Normal Distribution

Refine the histogram of bag masses far enough and the bars settle into one particular density. It is the **normal distribution**, and everything from the last section applies to it: probability is area, and the total area is 1. What makes it worth a section of its own is how often it turns up — heights, masses, measurement errors, examination marks — so much of the natural and measured world that it appears in almost every area of statistics.

Two numbers fix a normal distribution completely, and it is worth being clear about what each one does.

- The **mean** μ locates the centre. Changing μ slides the whole curve left or right without altering its shape at all.
- The **standard deviation** σ sets the width. Increasing σ stretches the curve sideways, and because the area underneath must stay equal to 1, stretching it sideways also flattens it. A large σ therefore gives a wide, low bell and a small σ a narrow, tall one.

We write $X \sim N(\mu, \sigma^2)$, with the *variance* in the second slot, so remember to take a square root before using σ. The curve is symmetric about μ, so the mean, median and mode all sit together at the peak.

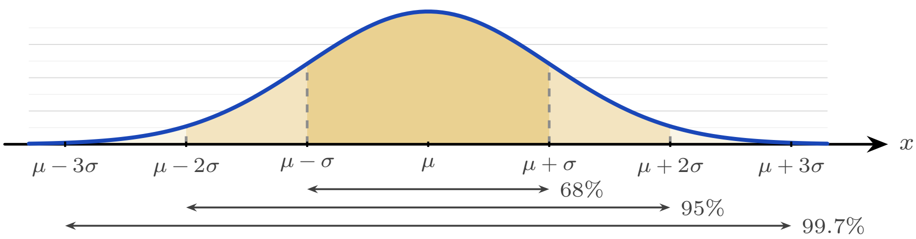


Figure 13.4 The normal curve, symmetric about μ. The three shaded bands reach one, two and three standard deviations either side of the mean, and hold about 68%, 95% and 99.7% of the values. Almost nothing lies beyond three standard deviations.

This symmetry gives a quick approximation, the **68–95–99.7 rule**: about 68% of values lie within one standard deviation of the mean, about 95% within two, and about 99.7% within three. It lets you judge, from μ and σ alone, where most of the data lie, without a calculator.

Read from the other end, the same rule measures how surprising a value is. Falling outside two standard deviations happens to about 5% of values, roughly one in twenty. Falling outside three happens to about 0.3%, roughly one in 370. That is why a reading three standard deviations from the mean is treated as a signal that something has changed, rather than as ordinary variation.

The curve has an equation, and although you will not have to use it, seeing it explains where the shape comes from.

KEY **The normal density** (not assessed):

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}}\, e^{-\frac{(x-\mu)^2}{2\sigma^2}}.$$

Read the exponent first, because it carries the whole shape. The quantity $(x-\mu)^2$ is a parabola: it is zero at $x = \mu$ and grows as x moves away from the mean, and it depends only on the *distance* from μ, not on the direction. The minus sign turns that growth into decay, since e^{-u} is largest when $u = 0$ and shrinks towards zero as u grows. So the curve is tallest at the mean, falls away at the same rate on both sides, and never quite reaches the axis. The $2\sigma^2$ underneath sets the speed of the fall: a large σ makes the exponent grow slowly, giving a wide, low bell. The factor $\dfrac{1}{\sigma\sqrt{2\pi}}$ in front changes no part of the shape; it is the number that makes the total area exactly 1.

This formula is not in the formula booklet, and no examination question will ask you to use it. Every normal probability in this course is found with a GDC. It is here so that the bell is a consequence of something rather than a shape you are asked to accept.

For exact probabilities you use a GDC, which computes the area under the curve between any two values directly from μ and σ.

EXAMPLE 13.12

The heights of adult women are modelled by $X \sim N(170, 8^2)$ (in cm). Find (a) $P(X < 180)$, (b) $P(160 < X < 180)$.

Both are areas under the normal curve, read from a GDC with $\mu = 170$, $\sigma = 8$.

(a) $P(X < 180) \approx 0.894$.

(b) $P(160 < X < 180) \approx 0.789$.

The second answer is close to the 68% from the rule of thumb, because 160 and 180 are roughly one standard deviation either side of the mean.

YOUR TURN 13.5 The Normal Distribution

11. GDC

(a) $X \sim N(50, 10^2)$. Find $P(X < 60)$. [2]
(b) $X \sim N(100, 15^2)$. Find $P(X > 120)$. [2]
(c) $X \sim N(20, 4^2)$. Find $P(16 < X < 24)$. [2]
(d) $Z \sim N(0, 1)$. Find $P(Z < 1.5)$. [1]
(e) $X \sim N(500, 50^2)$. Find $P(X < 400)$. [2]

12. GDC The masses of packets of rice are modelled by $X \sim N(500, 25^2)$, in grams.

(a) Find the probability that a packet weighs more than 540 g. [2]
(b) Find $P(475 < X < 525)$, and compare your answer with the 68% from the rule of thumb. [3]
(c) In a batch of 800 packets, find the expected number weighing more than 540 g. [2]
(d) Write down the range of masses within which about 95% of packets lie. [2]

13.6 Standardization and the Inverse Normal

Every normal distribution is the same bell, just shifted and scaled. **Standardizing** expresses any value as the number of standard deviations it sits from its mean. That number is the **z-value**, and it converts any normal variable to the **standard normal** $Z \sim N(0, 1)$.

KEY · IB

$$z = \frac{x - \mu}{\sigma}$$

A z-value of 1.5 means "one and a half standard deviations above the mean," regardless of the original units. Standardizing lets you compare values from different normal distributions, and it is the key to working backwards.

EXAMPLE 13.13

Exam scores are $N(60, 12^2)$. A student scores 78. How many standard deviations above the mean is this?

$$z = \frac{78 - 60}{12} = \frac{18}{12} = 1.5.$$

The score sits 1.5 standard deviations above the mean. A student scoring 78 on a different exam with $N(70, 4^2)$ would have $z = 2$, and so be higher *relative to their cohort*, despite the equal raw mark.

The inverse normal

Often the question runs the other way: you know a probability and want the value that produces it. "What score is the 90th percentile?" The **inverse normal** on a GDC takes a probability and returns the corresponding value x.

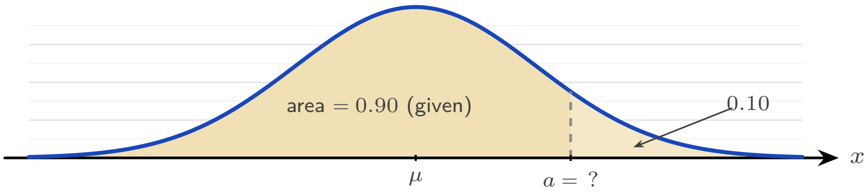


Figure 13.5 The inverse normal reverses the usual question. Instead of being given a and finding the area, you are given the area and must find a. A calculator expects the area to the *left* of a, so a question phrased as a right-hand tail, here 0.10, must be turned into its left-hand partner 0.90 first.

EXAMPLE 13.14

For heights $X \sim N(170, 8^2)$, find the height a below which 90% of women fall.
You need a with $P(X < a) = 0.90$. The inverse normal gives

$$a \approx 180.3 \text{ cm.}$$

Equivalently, the 90th percentile sits at $z \approx 1.282$, so $a = 170 + 1.282(8) \approx 180.3$, matching.

Finding an unknown μ or σ

The inverse normal also lets you recover a missing parameter. If you know a probability and a value, standardize to get a z, then solve the equation $z = \frac{x-\mu}{\sigma}$ for whichever of μ or σ is unknown.

EXAMPLE 13.15

A machine fills bottles so that volumes are normal with $\sigma = 5$ mL. Given that 80% of bottles contain less than 20 mL, find the mean μ.
Here $P(X < 20) = 0.80$. The inverse normal for the standard curve gives the matching z:

$$z = 0.8416, \qquad \text{so} \qquad 0.8416 = \frac{20-\mu}{5}.$$

Solving, $20 - \mu = 4.21$, hence $\mu \approx 15.8$ mL.

EXAMPLE 13.16

Scores are normal with mean 50. Given that 10% of scores exceed 60, find σ.
$P(X > 60) = 0.10$ means $P(X < 60) = 0.90$, so $z = 1.282$:

$$1.282 = \frac{60 - 50}{\sigma} \implies \sigma = \frac{10}{1.282} \approx 7.80.$$

Both parameters unknown

When neither μ nor σ is known, one probability statement is not enough: you need two. Each statement standardizes to an equation of the form $x = \mu + z\sigma$, and the pair solves simultaneously, exactly like any two linear equations.

EXAMPLE 13.17

The masses of a species of fish are normally distributed. It is known that 5% of the fish have mass below 50 g and 10% have mass above 80 g. Find μ and σ.
Translate each statement into a z-value with the inverse normal:

$$P(X < 50) = 0.05 \implies z_1 = -1.645, \qquad P(X < 80) = 0.90 \implies z_2 = 1.282.$$

Each gives an equation $x = \mu + z\sigma$:

$$50 = \mu - 1.645\sigma, \qquad 80 = \mu + 1.282\sigma.$$

Subtracting, $30 = 2.926\sigma$, so $\sigma \approx 10.3$ g. Substituting back, $\mu = 50 + 1.645(10.25) \approx 66.9$ g.
Note the signs carefully: a value *below* the mean has a negative z. One wrong sign here changes both answers, so check each z against the picture before solving.

YOUR TURN 13.6 Standardization and the Inverse Normal

13. GDC

(a) $X \sim N(60, 12^2)$. Find the z-value of $x = 84$. [2]
(b) $X \sim N(70, 5^2)$. Find the value x for which $P(X < x) = 0.95$. [2]
(c) $X \sim N(80, 10^2)$. Find the value x for which $P(X > x) = 0.15$. [3]

14. GDC

(a) $X \sim N(\mu, 8^2)$ with $P(X < 50) = 0.30$. Find μ. [3]
(b) $X \sim N(\mu, 6^2)$ with $P(X > 40) = 0.20$. Find μ. [3]
(c) $X \sim N(100, \sigma^2)$ with $P(X < 90) = 0.25$. Find σ. [3]

15. GDC

(a) Two examinations have marks $A \sim N(65, 10^2)$ and $B \sim N(72, 6^2)$. A student scores 75 in A and 80 in B. Find both z-values, and state which is the better performance relative to the cohort. [4]

(b) $X \sim N(\mu, \sigma^2)$ with $P(X < 20) = 0.10$ and $P(X < 35) = 0.90$. Find μ and σ. [5]

(c) For $X \sim N(50, 8^2)$, find the lower quartile, the upper quartile, and the interquartile range. [3]

Chapter 13 · Probability Distributions

Reflections

RW The bell curve's universality has a name: the central limit theorem. It says that the sum of many independent random effects tends toward a normal distribution, whatever the shape of the individual effects. This is why measurement errors, heights, and the means of large samples are all approximately normal. The pattern was noticed in 1846; the theorem that explains it was not fully proven until the 1930s.

TOK In 1846 a Belgian astronomer found this curve in the chest measurements of 5738 soldiers, and drew from it a conclusion the data did not support. He invented *l'homme moyen*, the "average man," and treated this statistical fiction as an ideal that real people deviate from. No actual person has all the average values at once. When we summarise a population by its mean, are we describing a real thing, or creating one? The normal curve is a fact about variation; the "average man" was an interpretation added to it.

TOK A game is called fair when the expected gain is zero. Every casino game fails that test by design, and the house edge is published.

If both sides know the odds, is a game with a built-in edge unfair, or merely unequal? And does the answer change when the player cannot do the arithmetic?

TOK This chapter offers several models for the same kind of situation: a binomial count can be approximated by a normal curve, and a real data set is never exactly either.

What criteria should decide between two models that both fit? Simplicity, accuracy, and agreement with existing theory can point different ways, and the choice is made by people.

EXT The binomial becoming normal is a theorem, not a loose comparison. Plot $B(n, 0.5)$ for increasing n and the discrete bars take the shape of the continuous bell. Galton built a machine, the quincunx, to show it physically: balls dropped through rows of pegs, each bounce an independent left-or-right trial, pile up at the bottom in a binomial that looks exactly normal. The distributions of this chapter are not separate facts but one story told at increasing scale.

End-of-Chapter Problems — Chapter 13: Probability Distributions

1. A discrete random variable X has the distribution below.

x	1	2	3	4
$P(X=x)$	0.1	k	0.3	0.2

(a) Find k. [2]
(b) Find $E(X)$. [2]
(c) Find Var(X). [3]

2. A charity raffle sells 500 tickets at $2 each. There is one prize of $300 and five prizes of $50.

(a) Find the expected winnings (before cost) for a single ticket. [3]
(b) Hence find the expected gain or loss per ticket, and comment. [2]

3. GDC In a production line, 8% of items are faulty, independently. A random sample of 15 items is taken. Let X be the number of faulty items.

(a) State the distribution of X. [1]
(b) Find $P(X=0)$. [2]
(c) Find $P(X\geq 2)$. [2]
(d) Find the expected number of faulty items. [1]

4. GDC A fair six-sided die is rolled 30 times. Let X be the number of sixes.

(a) State the distribution of X and find its mean. [2]
(b) Find $P(X=5)$. [2]
(c) Find $P(X\geq 7)$. [2]

5. A continuous random variable has density $f(x)=kx$ for $0\leq x\leq 3$, and zero elsewhere.

(a) Find k. [2]
(b) Find $P(X<2)$. [2]
(c) Find $E(X)$. [3]
(d) Find the median and the mode. [3]

6. GDC The heights of a species of plant are $X\sim N(170,8^2)$ in cm.

(a) Find $P(X<160)$. [2]

(b) Find $P(160 < X < 185)$. [2]

(c) Find the proportion of plants taller than 180 cm. [2]

7. GDC Test scores are normally distributed with mean 500 and standard deviation 80.

(a) Find the score at the 90th percentile. [2]

(b) The top 5% of candidates receive a distinction. Find the minimum score for a distinction. [3]

8. GDC A variable X is normally distributed. It is known that $P(X < 40) = 0.10$ and $P(X < 70) = 0.75$.

(a) Write down two equations involving μ and σ using standardized values. [3]

(b) Find μ and σ. [4]

9. The random variable X has $E(X) = 5$ and $\text{Var}(X) = 4$. Let $Y = 2X - 1$.

(a) Find $E(Y)$. [1]

(b) Find $\text{Var}(Y)$ and the standard deviation of Y. [3]

10. GDC Reaction times are $X \sim N(60, 12^2)$ in milliseconds. A reaction is "slow" if it exceeds 75 ms.

(a) Find the probability that a single reaction is slow. [2]

(b) Thirty reactions are recorded independently. Find the expected number of slow reactions. [2]

(c) Find the probability that at least one of the thirty is slow. [3]

11. A continuous random variable has density $f(x) = \frac{1}{8}(4 - x)$ for $0 \leq x \leq 4$, and zero elsewhere.

(a) Verify that f is a valid probability density function. [3]

(b) Find $P(X < 2)$. [2]

(c) Find the mode. [1]

12. A discrete random variable X takes the values $1, 2, 3, 4$ with $P(X = 1) = 0.2$, $P(X = 2) = a$, $P(X = 3) = b$, and $P(X = 4) = 0.1$. It is known that $E(X) = 2.4$.

(a) Find the values of a and b. [4]

(b) Find $P(X > 2 \mid X \geq 2)$. [3]

13. A stall game costs $\$c$ to play. A player wins \$10 with probability 0.2, wins \$2 with probability 0.3, and otherwise wins nothing.

(a) Find the expected winnings from a single play. [2]

(b) Find the value of c for which the game is fair. [1]

(c) The stall sets $c = 3$. Find the player's expected loss over 100 plays. [2]

14. GDC A gardener plants 12 seeds. Each seed germinates with probability 0.85, independently of the others. Let X be the number that germinate.

(a) State two conditions that make a binomial model appropriate here. [2]

(b) Find $P(X = 10)$. [2]

(c) Find $P(X \geq 10)$. [2]

(d) Find the mean and standard deviation of X. [2]

15. GDC A darts player hits the bullseye with probability 0.3 on each throw, independently.

(a) In 8 throws, find the probability of exactly 3 bullseyes. [2]

(b) Find the least number of throws needed for the probability of at least one bullseye to exceed 0.98. [3]

16. GDC The random variable $X \sim B(10, p)$ satisfies $P(X = 0) = 0.0282$.

(a) Find the value of p. [3]

(b) Find $P(X = 3)$. [2]

(c) Write down $E(X)$ and Var(X). [2]

17. GDC The lifetime of a battery is $X \sim N(120, 15^2)$, in hours.

(a) Find the probability that a battery lasts more than 140 hours. [2]

(b) Find $P(100 < X < 130)$. [2]

(c) A company ships 10 000 batteries. Find the expected number that last more than 140 hours. [2]

18. GDC The masses of apples are $X \sim N(160, 20^2)$, in grams. The heaviest 8% are sold as premium, and the lightest 5% are rejected.

(a) Find the minimum mass of a premium apple. [2]

(b) Find the maximum mass of a rejected apple. [2]

(c) Write down the proportion of apples that are neither premium nor rejected. [1]

19. GDC Journey times are normally distributed with mean 25 minutes and standard deviation σ. It is known that $P(X < 30) = 0.85$.

(a) Find σ. [3]

(b) Hence write down $P(X < 20)$, justifying your answer. [2]

20. GDC The masses of loaves of bread are normally distributed with unknown mean μ and standard deviation σ. It is known that 2% of loaves have mass below 300 g and 5% have mass above 340 g.

(a) Write down two equations for μ and σ using standardized values. [3]
(b) Find μ and σ. [3]

21. A continuous random variable has density

$$f(x) = \begin{cases} k & 0 \le x \le 1, \\ k(2-x) & 1 < x \le 2, \\ 0 & \text{otherwise.} \end{cases}$$

(a) Show that $k = \dfrac{2}{3}$. [3]
(b) Sketch the graph of f. [2]
(c) Find $P(X > 1.5)$. [2]
(d) Find the median of X. [2]
(e) Find $E(X)$. [3]

22. A continuous random variable has density $f(x) = kx(1-x)$ for $0 \le x \le 1$, and zero elsewhere.

(a) Show that $k = 6$. [2]
(b) Use calculus to find the mode of X. [2]
(c) Find $E(X)$. [2]
(d) Find $\text{Var}(X)$. [3]

23. GDC Scores on an examination are $X \sim N(55, 10^2)$. The pass mark is 45.

(a) Find the probability that a randomly chosen candidate passes. [2]
(b) Six candidates are chosen at random. Find the probability that exactly five of them pass. [3]
(c) Find the probability that at least five of them pass. [2]

24. GDC In 1846 a Belgian astronomer published the chest measurements, in inches, of 5738 Scottish militiamen, and used them as evidence that human measurements follow a normal distribution. His table is reproduced below.

Chest (in)	33	34	35	36	37	38	39	40
Frequency	3	18	81	185	420	749	1073	1079
Chest (in)	41	42	43	44	45	46	47	48
Frequency	934	658	370	92	50	21	4	1

(a) Show that the mean chest measurement is 39.8 inches, to 3 significant figures. [3]

(b) The standard deviation of the measurements is 2.05 inches. Assuming $X \sim N(39.8,\, 2.05^2)$, find $P(X < 38.5)$. [2]

(c) Find the proportion of the 5738 soldiers who actually measured 38 inches or less, and compare it with your answer to part (b). [3]

(d) Find the proportion of soldiers whose measurement lies within one standard deviation of the mean, and compare it with the value predicted by the normal model. [3]

(e) Comment on how well the normal distribution describes this real data set. [2]

Challenge Questions

25. What makes a game fair? At a fair, a stall offers this game. You pay a stake, roll two fair dice, and are paid \$20 if the total is 12, \$5 if the total is 10 or 11, and nothing otherwise. Let W be the amount paid out.

(a) Write down the probability distribution of W. [3]

(b) Find $E(W)$. [3]

(c) State the stake that would make the game fair, and explain what "fair" means here. [3]

(d) The stall charges \$2. Find the expected profit per game for the stall, and the expected profit over 500 games. [3]

(e) Find Var(W) and the standard deviation of W. [3]

(f) A player who plays ten times often finishes ahead, even though the game favours the stall. Use your answer to part (e) to explain why, and explain why the stall is not worried. [4]

26. GDC **From the binomial to the bell.** The Reflections claim that a binomial distribution comes to look normal as n grows. This question tests that claim, and finds the adjustment that makes it work.

Throughout, let $X \sim B(50, 0.5)$ and let $Y \sim N(\mu, \sigma^2)$ have the same mean and variance as X.

(a) Write down $E(X)$ and Var(X), and hence state μ and σ for Y. [2]

(b) Find $P(20 \leq X \leq 30)$ using the binomial. [2]

(c) Find $P(20 < Y < 30)$ using the normal. Comment on how well it matches part (b). [3]

(d) X takes whole-number values, while Y takes every value in between. The bar of the histogram for $X = 20$ covers the interval from 19.5 to 20.5. Using this idea, find $P(19.5 < Y < 30.5)$ and compare it with part (b). [4]

(e) Repeat parts (b), (c) and (d) for $X \sim B(10, 0.5)$ and the range $4 \leq X \leq 6$. [4]

(f) State which of the two normal calculations should be used, and describe how the quality of the approximation changes as n increases. [3]

27. The most likely value. Let $X \sim B(n,p)$ with $0 < p < 1$. This question finds the mode of X, the value it takes most often.

(a) Show that for $r = 1, 2, \dots, n$,

$$\frac{P(X=r)}{P(X=r-1)} = \frac{(n-r+1)\,p}{r\,(1-p)}.$$

[3]

(b) Hence show that $P(X=r) \geq P(X=r-1)$ if and only if $r \leq (n+1)p$. [3]

(c) Deduce that the mode of X is $\lfloor (n+1)p \rfloor$ when $(n+1)p$ is not an integer, and use this to find the most likely number of successes when $X \sim B(20, 0.3)$. [3]

(d) Explain what happens when $(n+1)p$ is an integer, and illustrate your answer with $X \sim B(9, 0.5)$. [3]

CALCULUS

Chapter 14

Limits & Derivatives

14

SYLLABUS 5.1–5.4 5.6 5.7 5.12 5.13 5.15

A car's speedometer reads 80 km/h. Ask what that number means, and you meet a problem that mathematicians could not solve for two thousand years.

Speed is distance divided by time. But the speedometer shows your speed *right now*, at a single instant. In one instant no time passes. No distance is covered either. So speed at an instant appears to be $0 \div 0$, which has no value. How can the needle point to 80?

The way out is to stop asking about the instant. Ask about short intervals around it instead.

Measure your average speed over the next hour. Then over the next minute. Then the next second, then the next hundredth of a second. Each interval is shorter than the one before, and each average is closer to one particular number. That number is the speed at the instant. The averages reach it, even though the instant alone cannot give it.

This idea is called the **limit**. You cannot reach a value directly, so you find it by seeing what nearby values approach. All of calculus is built on it.

From the limit comes the **derivative**. It measures the exact rate at which something changes: how steep a curve is at one point, or how fast an apple is falling at the moment it lands. This chapter builds the derivative from the limit, and then develops methods for calculating it quickly.

By the end of this chapter you will be able to evaluate limits, say when a function is continuous and differentiable, and differentiate from first principles. You will apply the product, quotient and chain rules, differentiate the standard functions of the course, and find the equation of a tangent and a normal. Calculus begins here because a derivative answers two questions at once. Geometrically it is the gradient of a curve at a point; physically it is a rate of change. Every later application is one of those two readings.

14.1 Limits

A **limit** is the value a function approaches as its input approaches some point. It does not matter what the function does at that point itself. We write

$$\lim_{x \to a} f(x) = L$$

to mean: as x gets **arbitrarily close** (as close as you like) to a, $f(x)$ gets arbitrarily close to L. The words "whatever happens at a" matter. A limit can exist at a point where the function itself is undefined.

For most functions at most points, the limit is found by direct substitution. If f is defined at a and its graph passes through that point unbroken, then the value it approaches is simply the value it takes:

$$\lim_{x \to 2} (x^2 + 3x) = 2^2 + 3(2) = 10.$$

That case is straightforward. The useful cases are the ones where substitution fails.

EXAMPLE 14.1

Find $\lim_{x \to 3} \dfrac{x^2 - 9}{x - 3}$.

Substituting $x = 3$ gives $\frac{0}{0}$, which is undefined: the function has a hole at $x = 3$. But for every $x \neq 3$ we can factor and cancel:

$$\frac{x^2 - 9}{x - 3} = \frac{(x-3)(x+3)}{x - 3} = x + 3.$$

Away from the hole the function is simply $x + 3$, so as $x \to 3$ it approaches 6:

$$\lim_{x \to 3} \frac{x^2 - 9}{x - 3} = 6.$$

At $x = 3$ the function has no value at all, yet 6 is the value it approaches. The difference between "approaches" and "equals" is what a limit measures.

The form $\frac{0}{0}$ is called **indeterminate**. It does not mean the limit fails to exist. It means more work is needed.

One limit of this kind is worth memorising, because the derivatives of the trigonometric functions rest on it:

$$\lim_{x \to 0} \frac{\sin x}{x} = 1.$$

When there is no algebraic method, you can *estimate* a limit. Calculate the function at values closer and closer to the target, from both sides. Here is the result for this one, with x in radians:

x	-0.1	-0.01	0.01	0.1
$\dfrac{\sin x}{x}$	0.99833	0.99998	0.99998	0.99833

Both sides approach 1. A table or a zoomed-in graph is a fair way to **conjecture** (guess, before proving) the value of a limit, and on a calculator paper it is often the fastest way to check an algebraic answer.

Reading a limit off the graph

A graph often shows why a limit takes the value it does, and the two fractions $\dfrac{\sin x}{x}$ and $\dfrac{\cos x}{x}$ make the contrast clearly. Both have denominator $x \to 0$, yet they behave completely differently, and the reason is what the *numerator* does at the same moment.

Sketch $y = \sin x$ and $y = x$ on the same axes. Near the origin the two graphs are almost indistinguishable: at $x = 0.1$, $\sin x = 0.0998$. Numerator and denominator shrink to zero together, at the same rate, so their ratio approaches 1.

Now sketch $y = \cos x$ against $y = x$. As $x \to 0$ the numerator approaches 1 while the denominator approaches 0. A fixed quantity divided by something ever smaller grows without bound, so the fraction does not approach any finite value.

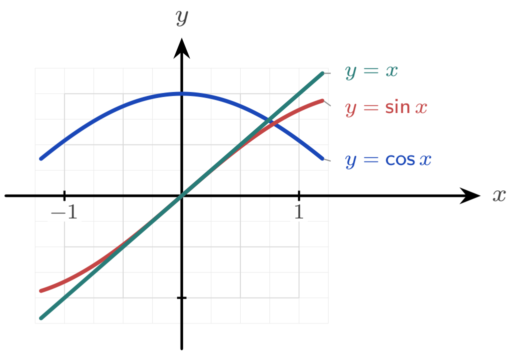


Figure 14.1 Near the origin the graphs of $\sin x$ and x are almost the same, so their ratio tends to 1. But $\cos x$ tends to 1 while x tends to 0, so $\frac{\cos x}{x}$ has no finite limit.

CAUTION Be precise about the second case. As $x \to 0^+$ the fraction $\frac{\cos x}{x}$ increases without bound, but as $x \to 0^-$ it decreases without bound, because x is negative there. The two sides disagree, so $\lim_{x\to 0} \frac{\cos x}{x}$ does not exist. Writing that it ``equals infinity'' conceals that disagreement, so it is not an acceptable answer.

Not every limit exists. A limit **converges** when the values approach one finite number, and **diverges** when they do not: $\frac{1}{x^2}$ grows without bound as $x \to 0$, and $\sin x$ oscillates between -1 and 1 as $x \to \infty$ without approaching any value. You have met this distinction before: the infinite geometric series of Ch. 2 converges exactly when $|r| < 1$, because its partial sums approach a single value, and diverges otherwise.

CAUTION A limit can fail to exist. If $f(x)$ approaches different values from the left and the right of a, as at a jump, there is no single limit. ``Approaches $\frac{0}{0}$'' is fixable; ``approaches two different things'' is not.

YOUR TURN 14.1 Limits

1. Find each limit by direct substitution.

(a) $\lim_{x\to 2}(3x-1)$ [1] (b) $\lim_{x\to 0}(x^2+2x+5)$ [1]

(c) $\lim_{x\to 3}(x^2-9)$ [1] (d) $\lim_{x\to 1}\dfrac{x+1}{x+3}$ [2]

2. Each of these gives $\frac{0}{0}$. Factorise, cancel, then take the limit.

(a) $\lim_{x\to 4}\dfrac{x^2-16}{x-4}$ [2] (b) $\lim_{x\to 1}\dfrac{x^2-1}{x-1}$ [2]

(c) $\lim_{x\to 2}\dfrac{x^2-x-2}{x-2}$ [2] (d) $\lim_{x\to -3}\dfrac{x^2-9}{x+3}$ [2]

3. Write down the value of $\lim_{x\to 0}\dfrac{\sin x}{x}$. [1]

4. Explain why $\lim_{x\to 0}\dfrac{|x|}{x}$ does not exist. [2]

5. State whether each of the following converges or diverges as $x \to \infty$, and give the limit where one exists.

(a) $\dfrac{1}{x}$ [1] (b) $\sin x$ [1]

(c) x^2 [1] (d) $\dfrac{3x+1}{x}$ [2]

6. GDC Estimate $\lim_{x\to 0}\dfrac{1-\cos x}{x^2}$ by evaluating the expression at $x = 0.1$ and $x = 0.01$ radians. State the exact value you think the limit takes. [3]

14.2 The Derivative

Calculus is the mathematics of change. You already know how to measure change: it is the gradient.

Rise over run gives the change in y for each unit of change in x. If the gradient doubles, the quantity changes twice as fast.

There is one problem. A gradient is defined for a *straight line* only. One number describes the whole line, because a line changes at the same rate everywhere.

Very few real quantities behave like that. A cooling cup, a thrown ball, a growing population: in each case the rate of change is itself changing. So the question “what is the gradient of this curve?” has no single answer. The answer is different at every point.

So ask a smaller question. Do not ask for the gradient of the curve. Ask for the gradient of the curve *at one point*. Once you can answer that for one point, you can answer it for every point, and the rate of change becomes a function of its own.

Here is the method. On the graph of $y = f(x)$, choose your point x and a second point $x + h$ close to it. Join the two points with a straight line, called the **secant**, and find its gradient. You can already calculate that:

$$\frac{f(x+h) - f(x)}{h},$$

the rise over the run. This is the average rate of change over the interval of width h.

The secant gradient is still not the answer. It is an average over an interval of width h, and the question was about a single point.

Now make h smaller and smaller. The second point moves towards the first, and the secant rotates towards a limiting position, meeting the curve at one point only. That line is the **tangent**. Its gradient is the rate of change at a single point, and it is called the **derivative**.

That is the complete idea. A gradient is defined only for a line, so we find the one line that matches the curve at that point, and use its gradient instead.

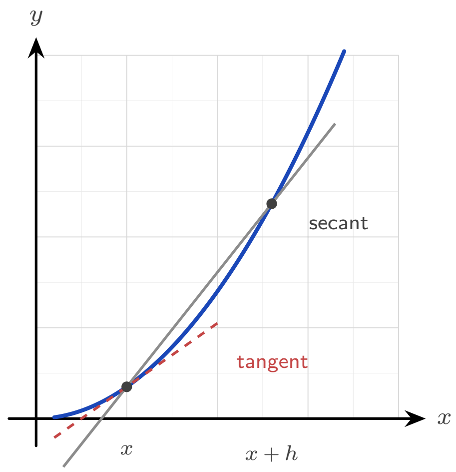


Figure 14.2 The secant joins two points on the curve and measures an average rate of change. As $h \to 0$ the second point moves to the first and the secant approaches the tangent.

DEF · IB The **derivative** of f at x, written $f'(x)$, is

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h},$$

when this limit exists. Computing a derivative directly from this definition is called differentiation **from first principles**.

The derivative is itself a function. Give it a value of x and it gives back the gradient of the curve at that value. Three notations mean the same thing: $f'(x)$, $\dfrac{dy}{dx}$ and $\dfrac{d}{dx}f(x)$.

EXAMPLE 14.2

Differentiate $f(x) = x^2$ from first principles.

Form the difference quotient and expand:

$$\frac{f(x+h) - f(x)}{h} = \frac{(x+h)^2 - x^2}{h} = \frac{x^2 + 2xh + h^2 - x^2}{h} = \frac{2xh + h^2}{h} = 2x + h.$$

Now let $h \to 0$:

$$f'(x) = \lim_{h\to 0}(2x + h) = 2x.$$

You must cancel the h before taking the limit. While $h \neq 0$ the division is allowed. Only after cancelling can h be set to zero safely.

EXAMPLE 14.3

Differentiate $f(x) = \dfrac{1}{x}$ from first principles.

$$\frac{f(x+h) - f(x)}{h} = \frac{1}{h}\left(\frac{1}{x+h} - \frac{1}{x}\right) = \frac{1}{h} \cdot \frac{x - (x+h)}{x(x+h)} = \frac{-h}{h\,x(x+h)} = \frac{-1}{x(x+h)}.$$

As $h \to 0$:

$$f'(x) = \lim_{h\to 0} \frac{-1}{x(x+h)} = -\frac{1}{x^2}.$$

TIP In examinations, first-principles differentiation is asked only for polynomials. The $\frac{1}{x}$ example above shows how far the definition extends; it is not itself an examination task.

Continuity and differentiability HL

The derivative is a limit, so it can fail to exist. There are two ways this matters.

A function is **continuous** at a point when its limit there exists and equals its value. Informally: the graph passes through the point with no hole and no jump, and you could draw it without lifting the pen.

A function is **differentiable** at a point when the first-principles limit exists. This is stronger.

The graph must be unbroken and also *smooth*, with one clear tangent at that point.

Every differentiable function is continuous, but the reverse is not true. The standard example is the absolute value.

EXAMPLE 14.4

Show that $f(x) = |x|$ is continuous at $x = 0$ but not differentiable there.
Continuity is clear: as $x \to 0$ from either side, $|x| \to 0 = f(0)$. The graph is an unbroken V.
For differentiability, form the difference quotient at 0:

$$\frac{f(0+h) - f(0)}{h} = \frac{|h|}{h} = \begin{cases} 1 & h > 0, \\ -1 & h < 0. \end{cases}$$

The quotient approaches 1 from the right and -1 from the left. The two sides disagree, so the limit does not exist and f has no derivative at 0. On the graph, the V has a *corner*, and a corner has no single tangent. A graph can be unbroken and still fail to be smooth.

TIP Examinations will not ask you to *test* a function for continuity or differentiability. What they expect is the idea itself: you may differentiate only where the curve is smooth.

YOUR TURN 14.2 The Derivative

7. Differentiate each from first principles, showing the limit.

(a) $f(x) = x^2$ [3] (b) $f(x) = 3x^2$ [3]
(c) $f(x) = x^2 + x$ [3] (d) $f(x) = x^2 - 3x$ [3]

8. The definition of the derivative can also be used numerically. For $f(x) = \sin x$, evaluate $\dfrac{f(0+h) - f(0)}{h}$ for $h = 0.1$, $h = 0.01$ and $h = 0.001$, working in radians. State the value the quotient is approaching, and name the standard derivative it confirms. [4]

9. GDC Repeat the same calculation for $f(x) = e^x$ at $x = 0$, then at $x = 1$. Explain what the two answers suggest about the derivative of e^x. [4]

10. Write down the three notations used in this book for the derivative of $y = f(x)$. [2]

11. Explain why $f(x) = |x|$ has no derivative at $x = 0$, even though its graph is unbroken there. [3]

14.3 Differentiating Powers

First principles works, but it is slow. The two worked examples above show a pattern: x^2 gave $2x$, and $\frac{1}{x} = x^{-1}$ gave $-x^{-2}$.

In both cases two things happened. The power decreased by one, and the original power became a multiplier in front. This is the **power rule**.

KEY · IB

$$\frac{d}{dx}x^n = nx^{n-1}$$

At Standard Level this is used with integer powers. It holds just as well for every rational power, which is the form needed from §14.5 onwards and the reason the next example can differentiate $\sqrt{x}$.

Two more rules let you differentiate any polynomial. A constant multiplier is unchanged, and a sum is differentiated one term at a time:

$$\frac{d}{dx}(c\,f(x)) = c\,f'(x), \qquad \frac{d}{dx}(f(x)+g(x)) = f'(x)+g'(x).$$

A constant on its own has derivative zero, because its graph is a horizontal line with gradient zero.

EXAMPLE 14.5

Differentiate $y = 3x^4 - 2x^2 + 7$.

Differentiate each term with the power rule:

$$\frac{dy}{dx} = 3(4x^3) - 2(2x) + 0 = 12x^3 - 4x.$$

The constant 7 has derivative zero, as every constant does.

EXAMPLE 14.6

Differentiate $y = \sqrt{x} + \dfrac{1}{x^2}$.

Rewrite both terms as powers before differentiating:

$$y = x^{1/2} + x^{-2} \implies \frac{dy}{dx} = \tfrac{1}{2}x^{-1/2} - 2x^{-3} = \frac{1}{2\sqrt{x}} - \frac{2}{x^3}.$$

The power rule requires the function in the form x^n. Convert every root and every reciprocal to a power before differentiating, and the rest of the question becomes routine.

EXAMPLE 14.7

Find the gradient of $y = x^3 - 4x$ at the point where $x = 2$.
Differentiate, then substitute:

$$\frac{dy}{dx} = 3x^2 - 4, \qquad \left.\frac{dy}{dx}\right|_{x=2} = 3(4) - 4 = 8.$$

The derivative is a function. A gradient *at a point* is that function evaluated at the point.

YOUR TURN 14.3 Differentiating Powers

12. Differentiate each of the following.

(a) $y = x^7$ [1]
(b) $y = 5x^3 - 2x$ [2]
(c) $y = x^4 - 3x^2 + 6$ [2]
(d) $y = 4x^5 + x$ [2]
(e) $y = 2x^3 - 7x^2 + x - 9$ [2]
(f) $y = 6$ [1]

13. Rewrite each function as a power of x, then differentiate. In part (f), divide through first.

(a) $y = 2/x$ [2]
(b) $y = \sqrt{x}$ [2]
(c) $y = x^3 + 1/x$ [2]
(d) $y = 1/x^3$ [2]
(e) $y = \sqrt[3]{x}$ [2]
(f) $y = (x^2 + 1)/x$ [3]

14. Find the gradient of each curve at the point given.

(a) $y = x^2 - 5x$ at $x = 3$ [2]
(b) $y = x^3 - 4x$ at $x = -1$ [2]
(c) $y = \sqrt{x}$ at $x = 4$ [2]
(d) $y = 2x^3 + x$ at $x = 1$ [2]

15. Find the value or values of x at which each curve has the stated gradient.

(a) $y = x^2 - 6x$, gradient 0 [2]
(b) $y = x^3$, gradient 12 [3]

16. Explain why $y = \sqrt[4]{x}$ must be rewritten before the power rule can be applied, and state the rewritten form. [2]

14.4 Tangents and Normals

The derivative gives the gradient of the tangent at any point, and a gradient plus a point determines a line. So you can now write the equation of the **tangent** to a curve at a given point. The **normal** is the line perpendicular to the tangent there, with gradient the negative reciprocal.

The tangent and the normal are both straight lines, so both are written with the same equation, the point-gradient form from Ch. 4:

$$y - y_1 = m(x - x_1),$$

where (x_1, y_1) is a known point on the line and m is its gradient. Only two things ever change between the two lines. The point (x_1, y_1) is the same for both, because the tangent and the normal meet the curve at the same place, so all that differs is the gradient m.

KEY · IB At the point (x_1, y_1) on the curve $y = f(x)$, both lines have the form $y - y_1 = m(x - x_1)$, with

$$m_{\text{tangent}} = f'(x_1), \qquad m_{\text{normal}} = -\frac{1}{f'(x_1)}.$$

So every question of this type follows three steps: find y_1 by substituting x_1 into f, find $f'(x_1)$ for the gradient, then put both into the point-gradient form.

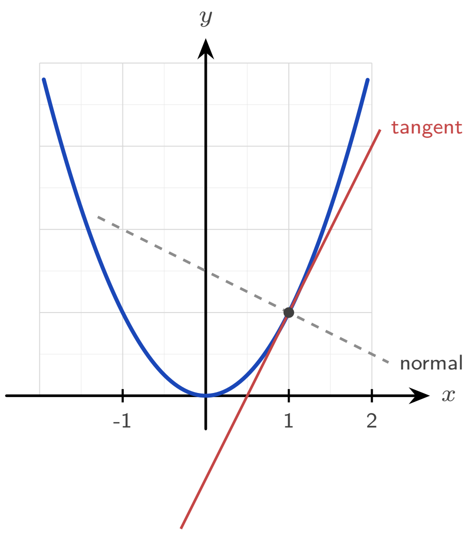


Figure 14.3 At a point on a curve the tangent has gradient $f'(x)$, and the normal is perpendicular to it, with gradient $-1/f'(x)$.

EXAMPLE 14.8

Find the equations of the tangent and normal to $y = x^2 - 3x + 2$ at the point where $x = 2$.

First the point: $y(2) = 4 - 6 + 2 = 0$, so the point is $(2, 0)$.
Then the gradient: $\frac{dy}{dx} = 2x - 3$, so at $x = 2$ the tangent gradient is $2(2) - 3 = 1$.
Tangent, through $(2, 0)$ with gradient 1:

$$y - 0 = 1(x - 2) \implies y = x - 2.$$

The normal is perpendicular, so its gradient is -1:

$$y - 0 = -1(x - 2) \implies y = -x + 2.$$

Always find the y-coordinate first. A common error is to write the equation of the line using the x-value alone.

YOUR TURN 14.4 Tangents and Normals

17. Find the equation of the tangent to each curve at the point given.

(a) $y = x^2$ at $(2, 4)$ [2]
(b) $y = x^2 - 4x + 3$ at $x = 1$ [3]
(c) $y = 1/x$ at $x = 1$ [3]
(d) $y = e^x$ at $x = 0$ [3]

18. Find the equation of the normal to each curve at the point given.

(a) $y = x^3$ at $(1, 1)$ [3]
(b) $y = \sqrt{x}$ at $x = 4$ [3]

19. A tangent and a normal are drawn at the same point of a curve. Explain the relationship between their gradients, and state what happens when the tangent is horizontal. [3]

20. The tangent to $y = x^2$ at the point where $x = a$ passes through $(0, -4)$. Find the two possible values of a. [4]

21. Find the coordinates of the point on $y = x^2 - 6x$ at which the tangent is horizontal. [3]

14.5 The Standard Derivatives

The power rule covers algebraic functions. The trigonometric, exponential and logarithmic functions have derivatives of their own. Each can be proved from first principles. You can also find each one by reading gradients from a graph, and it is worth doing that once before you memorise anything.

The method is always the same. Move along the graph of f from left to right. At each point, ask how steep the graph is, and plot that steepness as a height. The curve you plot is f'.

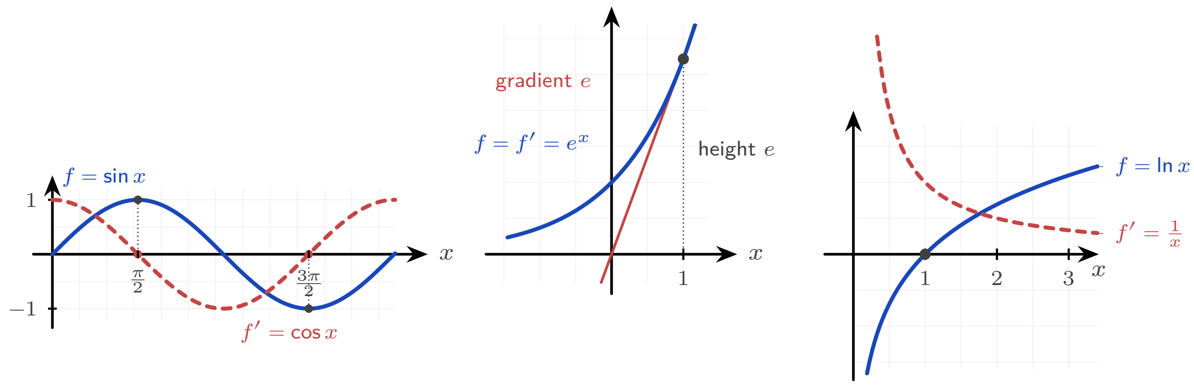


Figure 14.4 Each derivative read from the gradient of its own graph. **Left:** where $\sin x$ is flat, at $\frac{\pi}{2}$ and $\frac{3\pi}{2}$, the curve $\cos x$ crosses zero. **Centre:** the tangent to e^x at $x = 1$ has gradient e, which is also the height of the curve there. **Right:** $\ln x$ is steep near 0 and flattens as x grows, and $\frac{1}{x}$ does exactly that.

Read each panel separately.

Sine. At $x = 0$ the sine graph is rising most steeply, so f' starts at its maximum. At the top of the curve, $x = \frac{\pi}{2}$, the graph is flat for an instant, so f' is zero. Past the crest it falls, so f' goes negative. A curve that starts at 1, drops through zero at $\frac{\pi}{2}$ and reaches -1 at π is exactly $\cos x$.

The exponential. Measure the gradient of $y = e^x$ at any point and you get the height of the curve at that point. That is the defining property of e, so here $f' = f$.

The logarithm. Just to the right of 0 the graph of $\ln x$ is almost vertical, so f' is very large. As x increases the curve becomes flatter, so f' decreases towards zero. It stays positive everywhere. That describes $\dfrac{1}{x}$.

Those are the four standard derivatives to learn.

KEY · IB

$$\frac{d}{dx}\sin x = \cos x, \quad \frac{d}{dx}\cos x = -\sin x, \quad \frac{d}{dx}e^x = e^x, \quad \frac{d}{dx}\ln x = \frac{1}{x}$$

The exponential is the one to look at twice: e^x is its own derivative. More precisely, the functions equal to their own derivative are exactly those of the form $y = Ce^x$, and e^x is the one passing through $(0, 1)$. That property is what makes e important. Note also that the results for sin and cos require x in *radians*.

At Higher Level a wider set of standard derivatives is provided. HL They are listed here for reference and used freely in the rules that follow. The reciprocal and inverse trigonometric functions in the table are defined in Ch. 9; only their derivatives are needed here.

$$\frac{d}{dx}\tan x = \sec^2 x$$
$$\frac{d}{dx}\sec x = \sec x \tan x$$
$$\frac{d}{dx}\operatorname{cosec} x = -\operatorname{cosec} x \cot x$$
$$\frac{d}{dx}\cot x = -\operatorname{cosec}^2 x$$

KEY · IB

$$\frac{d}{dx}a^x = a^x \ln a$$
$$\frac{d}{dx}\log_a x = \frac{1}{x\ln a}$$
$$\frac{d}{dx}\arcsin x = \frac{1}{\sqrt{1-x^2}}$$
$$\frac{d}{dx}\arccos x = -\frac{1}{\sqrt{1-x^2}}$$
$$\frac{d}{dx}\arctan x = \frac{1}{1+x^2}$$

Notice one pair. The derivative of arccos is the negative of the derivative of arcsin, because the two graphs are reflections of each other.

EXAMPLE 14.9

Differentiate $y = 3\sin x + 2e^x - 5\ln x$.

Apply the standard derivatives term by term:

$$\frac{dy}{dx} = 3\cos x + 2e^x - \frac{5}{x}.$$

Constant multipliers are unchanged, exactly as with powers.

YOUR TURN 14.5 The Standard Derivatives

22. Differentiate each of the following.

(a) $y = \sin x + \cos x$ [1] (b) $y = e^x - 3\ln x$ [2]
(c) $y = 4\cos x$ [1] (d) $y = \tan x$ [1]
(e) $y = \arctan x$ [1] (f) $y = 3^x$ [2]

23. Differentiate each of the following. All are in the Higher Level list.

(a) $y = 2\sin x - 5e^x$ [2] (b) $y = \ln x + \tan x$ [2]
(c) $y = \sec x$ [1] (d) $y = \arcsin x$ [1]

24. Find the gradient of $y = \sin x$ at $x = \dfrac{\pi}{3}$, giving an exact answer. [2]

25. Find the value of x at which the curve $y = e^x$ has gradient 1, and state the significance of that point. [3]

26. The results $\dfrac{d}{dx}\sin x = \cos x$ and $\dfrac{d}{dx}\cos x = -\sin x$ hold only when x is measured in radians. Explain why the choice of unit matters. [2]

14.6 The Rules of Differentiation

Most functions are built from simpler ones, by composing, multiplying or dividing them. Three rules cover these three cases. With the standard derivatives, they let you differentiate almost any function in the course.

The chain rule

A **composite** function is a function inside a function, such as $(3x^2+1)^5$ or $\sin(2x)$. The **chain rule** differentiates it by working from the outside in: differentiate the outer function, then multiply by the derivative of the inner.

KEY · IB If $y = g(u)$ where $u = f(x)$, then

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

The chain rule appears in most differentiation questions once the functions stop being simple powers. Two habits make it reliable:

1. **Identify the inner function first**, before differentiating anything. Ask which part is wrapped inside the other.
2. **Multiply by the derivative of the inner function.** Differentiating the outer function alone and stopping there is a common error, and it is worth checking for at the end of every chain rule question.

EXAMPLE 14.10

Differentiate (a) $y = (3x^2 + 1)^5$ and (b) $y = e^{x^2}$.

(a) Outer is u^5, inner is $u = 3x^2 + 1$:

$$\frac{dy}{dx} = 5(3x^2 + 1)^4 \cdot 6x = 30x(3x^2 + 1)^4.$$

(b) Outer is e^u, inner is $u = x^2$:

$$\frac{dy}{dx} = e^{x^2} \cdot 2x = 2x\, e^{x^2}.$$

In each case the second factor, $6x$ and $2x$, is the derivative of the inner function.

The product rule

To differentiate a product of two functions, you cannot simply multiply their derivatives. It is worth seeing why, because the mistake is a natural one to make.

Take $u = x$ and $v = x$, so that $y = uv = x^2$. The derivative of x^2 is $2x$. But $u' = 1$ and $v' = 1$, and $u'v' = 1$. The two answers do not agree, so multiplying derivatives is simply wrong.

The reason is that a product changes for two separate reasons at the same time.

Suppose both u and v increase. Part of the increase in uv comes from the change in u, multiplied by the whole of v. The rest comes from the change in v, multiplied by the whole of u. The correct rule adds these two parts, which is why each term keeps one factor undifferentiated.

KEY · IB If $y = uv$, then

$$\frac{dy}{dx} = u\frac{dv}{dx} + v\frac{du}{dx}$$

EXAMPLE 14.11

Differentiate $y = x^2e^x$.

Let $u = x^2$ and $v = e^x$, so $u' = 2x$ and $v' = e^x$:

$$\frac{dy}{dx} = x^2e^x + e^x(2x) = e^x(x^2 + 2x).$$

Factorise the answer where you can. A factorised form makes later work much easier, for example finding where $\frac{dy}{dx} = 0$.

The quotient rule

For a quotient of two functions, the rule is similar but the order of terms matters, since subtraction is involved.

KEY · IB If $y = \dfrac{u}{v}$, then

$$\frac{dy}{dx} = \frac{v\dfrac{du}{dx} - u\dfrac{dv}{dx}}{v^2}$$

EXAMPLE 14.12

Differentiate $y = \dfrac{x}{x+1}$.

Let $u = x$ and $v = x + 1$, so $u' = 1$ and $v' = 1$:

$$\frac{dy}{dx} = \frac{(x+1)(1) - x(1)}{(x+1)^2} = \frac{1}{(x+1)^2}.$$

The order in the numerator is: bottom times derivative of top, minus top times derivative of bottom. Reversing the two terms changes the sign of the answer, and it is a common error.

YOUR TURN 14.6 The Rules of Differentiation

27. Use the chain rule to differentiate each of the following.

(a) $y = (2x+1)^4$ [2]

(b) $y = \sin(x^2)$ [2]

(c) $y = e^{3x}$ [2]

(d) $y = \sqrt{3x+1}$ [3]

28. Use the product rule to differentiate each of the following.

(a) $y = x\cos x$ [2] (b) $y = x^2e^x$ [2]
(c) $y = x^3\ln x$ [3] (d) $y = x\sin x$ [2]

29. Use the quotient rule to differentiate each of the following.

(a) $y = \dfrac{x}{x+1}$ [2] (b) $y = \dfrac{x+1}{x-1}$ [3]
(c) $y = \dfrac{\sin x}{x}$ [3] (d) $y = \dfrac{e^x}{x}$ [3]

30. Explain, with a counterexample, why the derivative of a product is not the product of the derivatives. [3]

31. Find the gradient of $y = (x^2 - 1)^3$ at the point where $x = 2$. [3]

32. A student writes $\dfrac{d}{dx}\sin(3x) = \cos(3x)$. Identify the error and give the correct derivative. [2]

14.7 Higher Derivatives and Concavity

The derivative $f'(x)$ is itself a function, so it can be differentiated again. The result is the **second derivative** $f''(x)$, also written $\dfrac{d^2y}{dx^2}$.

If f' measures how fast f changes, then f'' measures how fast that *rate* changes. For motion this is easy to name: f is position, f' is velocity and f'' is acceleration.

You can continue. Differentiating n times gives the nth derivative, written $\dfrac{d^ny}{dx^n}$ or $f^{(n)}(x)$. You will need that notation for Maclaurin series. See Ch. 17.

Together, the signs of f' and f'' describe the shape of a graph completely. They answer two different questions, so keep them separate.

Sign	f': going up or down?	f'': bending which way?
positive	**increasing**	**concave up** (bends upward)
negative	**decreasing**	**concave down** (bends downward)
zero	**stationary point**: the curve is flat there	**point of inflexion**, *if* the concavity also changes

The two are independent. A curve can be increasing and concave down at the same time: it is rising, but by less and less. It can also be decreasing and concave up: it is falling, but by less and less.

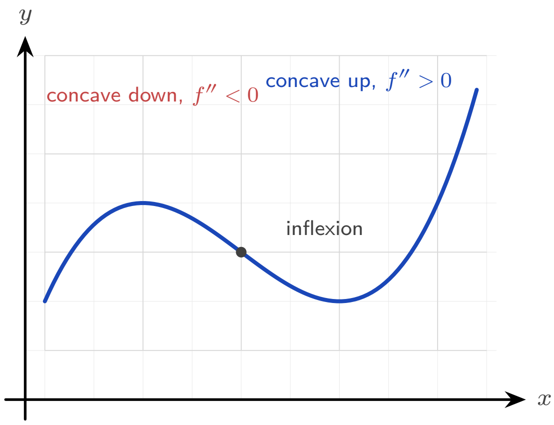


Figure 14.5 The second derivative decides which way the curve bends. Where the bending changes direction the curve has a point of inflexion.

EXAMPLE 14.13

For $f(x) = x^3 - 3x^2$, find $f'(x)$ and $f''(x)$, then state where f is increasing and where it is concave up.

Differentiate twice:

$$f'(x) = 3x^2 - 6x = 3x(x-2), \qquad f''(x) = 6x - 6 = 6(x-1).$$

f is increasing where $f'(x) > 0$, that is where $3x(x-2) > 0$, giving $x < 0$ or $x > 2$.
f is concave up where $f''(x) > 0$, that is $x > 1$.
At $x = 1$ the concavity changes from down to up: this is a **point of inflexion**, which you will study in Ch. 15.

Reading f, f' and f'' together

A common examination question shows one of these three graphs and asks about another. It helps to see all three at the same time.

Below are $f(x) = x^3 - 3x$ and its two derivatives, drawn one above the other on the same x-axis. The vertical scales are not the same. Only the zeros and the signs matter here.

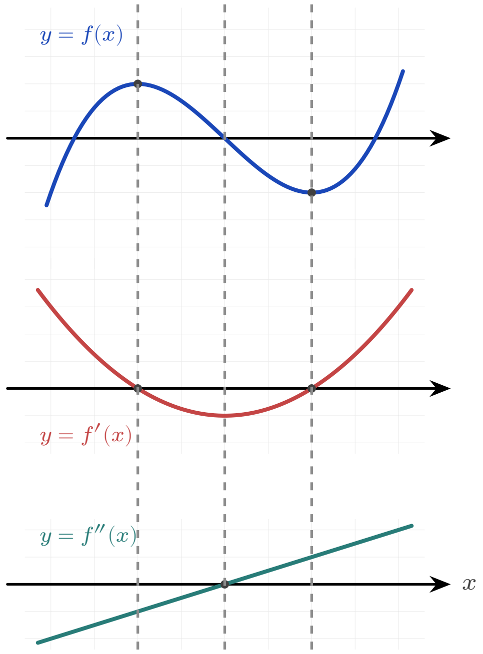


Figure 14.6 f, f' and f'' for $f(x) = x^3 - 3x$, stacked over a shared x-axis. Read down each dashed line: a turning point of f sits above a zero of f', and the inflexion of f sits above a turning point of f' and a zero of f''.

Read down each dashed line. At every dashed line, something happens in all three graphs at once, and the three results below are worth learning. Examinations test them in both directions: from f to f', and from f' back to f.

- A **turning point of** f sits above a **zero of** f'. At $x = -1$ the curve f has a maximum, and there f' crosses zero from positive to negative. At $x = 1$ the curve has a minimum, and there f' crosses zero from negative to positive.
- An **interval where** f **rises** sits above an **interval where** f' **is positive**, that is, where the graph of f' is above the axis.
- An **inflexion of** f sits above a **turning point of** f', and above a **zero of** f''. In this example all three occur at $x = 0$.

Use the table in either direction. Given any one of the three graphs, you can sketch the other two.

YOUR TURN 14.7 Higher Derivatives and Concavity

33. Find the second derivative of each of the following.

(a) $f(x) = x^3 - 6x^2$ [2] (b) $y = x^4$ [2]
(c) $y = \sin x$ [2] (d) $y = e^{2x}$ [2]

34. Find the interval or intervals on which each function is increasing.

(a) $f(x) = x^2 - 4x$ [2] (b) $f(x) = x^4 - 2x^2$ [4]

35. Find the interval on which each curve is concave up.

(a) $y = x^3$ [2] (b) $y = x^3 - 3x^2$ [3]

36. For $f(x) = x^3 - 3x^2$, find the coordinates of the point of inflexion and confirm that the concavity changes there. [4]

37. The graph of $y = f'(x)$ is a parabola with roots at $x = -2$ and $x = 3$, opening upwards. State the x-coordinates of the stationary points of f, and identify the nature of each. [4]

38. A particle moves so that its displacement after t seconds is $s = 2t^3 - 3t^2$ metres. Find expressions for the velocity and the acceleration, and find the time at which the acceleration is zero. [4]

14.8 L'Hôpital's Rule HL

This chapter began with the indeterminate form $\frac{0}{0}$, and solved it by factorising. Factorising fails as soon as the functions are trigonometric, exponential or logarithmic.

There is a second method for those cases. If a limit gives $\frac{0}{0}$ or $\frac{\infty}{\infty}$, the derivative gives the answer. The rule is named after the French mathematician Guillaume de l'Hôpital, and the name is said *lo - pee - TAL*: the letter s is silent, and so is the h.

KEY **L'Hôpital's rule.** If $\lim_{x\to a} \frac{f(x)}{g(x)}$ gives $\frac{0}{0}$ or $\frac{\infty}{\infty}$, then

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)},$$

provided the right-hand limit exists.

Differentiate the top and the bottom *separately*. This is not the quotient rule. Then try the limit again, and if it is still indeterminate, apply the rule a second time.

Why the rule works HL

The reason is short, and it explains why the rule needs the $\frac{0}{0}$ form.

Suppose $f(a) = 0$ and $g(a) = 0$. Then near $x = a$ each function is close to its own tangent line at a, so

$$f(x) \approx f'(a)(x-a), \qquad g(x) \approx g'(a)(x-a).$$

Divide one by the other. The factor $(x-a)$ appears in both, so it cancels:

$$\frac{f(x)}{g(x)} \approx \frac{f'(a)(x-a)}{g'(a)(x-a)} = \frac{f'(a)}{g'(a)}.$$

So the quotient of the functions approaches the quotient of their gradients. This is why both functions must vanish at a: only then does each one reduce to a multiple of $(x-a)$, and only then does the cancelling work.

EXAMPLE 14.14

Evaluate (a) $\lim_{x\to 0} \frac{\sin x}{x}$ and (b) $\lim_{x\to 0} \frac{1-\cos x}{x^2}$.

(a) Substituting gives $\frac{0}{0}$. Differentiate top and bottom:

$$\lim_{x\to 0} \frac{\sin x}{x} = \lim_{x\to 0} \frac{\cos x}{1} = \cos 0 = 1.$$

This confirms in one line the limit stated in §14.1.

(b) Substituting gives $\frac{0}{0}$. One application gives $\frac{\sin x}{2x}$, still $\frac{0}{0}$, so apply the rule again:

$$\lim_{x\to 0} \frac{1-\cos x}{x^2} = \lim_{x\to 0} \frac{\sin x}{2x} = \lim_{x\to 0} \frac{\cos x}{2} = \frac{1}{2}.$$

CAUTION L'Hôpital's rule applies *only* to the indeterminate forms $\frac{0}{0}$ and $\frac{\infty}{\infty}$. Always check that the limit is indeterminate before differentiating. Applying it to a form like $\frac{2}{0}$ or $\frac{3}{5}$ gives a wrong answer.

EXAMPLE 14.15

Evaluate $\lim_{x\to\infty} \frac{\ln x}{x}$.
As $x \to \infty$ this is $\frac{\infty}{\infty}$. Differentiate top and bottom:

$$\lim_{x\to\infty} \frac{\ln x}{x} = \lim_{x\to\infty} \frac{1/x}{1} = \lim_{x\to\infty} \frac{1}{x} = 0.$$

The denominator increases without bound while the numerator increases very slowly, so the ratio approaches zero. L'Hôpital's rule makes that comparison exact.

YOUR TURN 14.8 L'Hôpital's Rule HL

39. Evaluate each limit, checking first that it is indeterminate.

(a) $\lim_{x\to 0} \frac{e^x - 1}{x}$ [2]

(b) $\lim_{x\to 0} \frac{\sin 2x}{x}$ [2]

(c) $\lim_{x\to 0} \frac{\tan x}{x}$ [2]

(d) $\lim_{x\to\infty} \frac{\ln x}{x^2}$ [2]

40. Each of these needs the rule applied twice. Evaluate them.

(a) $\lim_{x\to 0} \frac{1 - \cos x}{x^2}$ [3]

(b) $\lim_{x\to 0} \frac{e^x - 1 - x}{x^2}$ [3]

41. Evaluate $\lim_{x\to\infty} \frac{x^2+1}{2x^2-3}$, first by L'Hôpital's rule and then by dividing every term by x^2. Confirm that the two methods agree. [4]

42. A student evaluates $\lim_{x\to 0} \frac{x+2}{x+1}$ by differentiating top and bottom, obtaining 1. Explain the error and give the correct value. [3]

43. State the two indeterminate forms to which L'Hôpital's rule applies, and explain why the rule must not be used on a form such as $\frac{2}{0}$. [3]

Chapter 14 · Limits & Derivatives

Reflections

TOK The early calculus was built on "infinitesimals," quantities treated as nonzero when dividing and as zero when convenient. The philosopher George Berkeley mocked them in 1734 as "ghosts of departed quantities," and he had a point: the logic was shaky. The method gave correct answers for 150 years before the limit gave it a rigorous foundation. Is a mathematical technique justified by its results, or only by its proof? Calculus worked long before anyone could say exactly why.

TOK A derivative gives a rate of change at a single instant, yet no instant can be measured: every measurement takes time. Is an instantaneous rate something the world contains, or a device invented to describe it? Does it matter, if the device predicts correctly every time?

TOK The expression $\frac{0}{0}$ has no value, and yet a limit of that form often has one exactly. Can a symbol that means nothing still carry information? Where does the meaning sit: in the symbols, or in the process that produced them?

IM Calculus was created twice, independently. Isaac Newton developed his "method of fluxions" in England in the 1660s; Gottfried Leibniz developed his version, with the $\frac{dy}{dx}$ notation we still use, in Germany in the 1670s. A bitter priority dispute split European mathematics for a century. Newton's notation was used in Britain and Leibniz's in the rest of Europe. British mathematics fell behind, partly because it kept the more awkward symbols. Notation is not neutral: the right symbol can speed up a whole field.

RW The derivative is the language of change, so it appears wherever anything varies. Velocity and acceleration in physics, marginal cost in economics, reaction rates in chemistry, growth rates in biology: each is a derivative. When a news report says inflation is "rising but more slowly," it is making a claim about a second derivative, that the rate of change of prices is itself decreasing.

EXT L'Hôpital's rule has a deeper cousin. Near a point, many functions are well approximated by a polynomial, the Maclaurin series: $\sin x \approx x - \frac{x^3}{6}$, $e^x \approx 1 + x + \frac{x^2}{2}$. Substituting these into an indeterminate limit resolves it without differentiating repeatedly. The same problem that L'Hôpital's rule solves leads on to representing functions as infinite polynomials, one of the great ideas of analysis.

End-of-Chapter Problems — Chapter 14: Limits & Derivatives

1. Differentiate $y = e^x \sin x$, and hence find the gradient of the curve at $x = 0$. [4]

2. Find the equation of the tangent to $y = x^2$ that is parallel to the line $y = 6x - 1$. [4]

3. Evaluate $\lim_{x \to 0} \frac{e^{2x} - 1}{x}$. [4]

4. Differentiate $f(x) = x^3$ from first principles, showing every step of the limit. [4]

5. Using the quotient rule on $\tan x = \frac{\sin x}{\cos x}$, prove that $\frac{d}{dx} \tan x = \sec^2 x$. [4]

6. Consider $f(x) = x^3 - 3x$.

(a) Find $f'(x)$. [1]
(b) Find the coordinates of the points where the tangent is horizontal. [4]

7. The function $f(x) = \frac{e^x - 1}{x}$ is not defined at $x = 0$.

(a) Copy and complete the table of values of $f(x)$, giving 5 significant figures: $x = -0.1,\ -0.01,\ 0.01,\ 0.1$. [2]
(b) Suggest the value of $\lim_{x \to 0} f(x)$. [1]
(c) Verify your conjecture using l'Hôpital's rule. [2]

8. Differentiate $f(x) = 3x^2 - 2x$ from first principles. [5]

9. Consider $f(x) = |x - 2|$.

(a) Sketch the graph of f. [1]
(b) Explain briefly why f is continuous at $x = 2$. [2]
(c) Explain why f is not differentiable at $x = 2$. [2]

10. Evaluate $\lim_{x \to 0} \frac{1 - \cos 3x}{x^2}$. [5]

11. Find the coordinates of the points on the curve $y = x^3 - 3x$ where the tangent is parallel to the line $y = 9x + 1$. [5]

12. A curve has equation $y = \frac{e^{2x}}{1 + x^2}$.

(a) Find $\dfrac{dy}{dx}$. [4]

(b) Find the gradient of the curve at $x = 0$. [1]

13. Evaluate $\lim_{x \to 0} \dfrac{x - \sin x}{x^3}$, justifying each application of L'Hôpital's rule. [5]

14. Consider $f(x) = x^2 - 2x$.

(a) Find $f'(x)$ from first principles. [4]

(b) Hence find the gradient of the curve at $x = 3$. [2]

15. The curve $y = x^2 - 2x$ is considered at the point where $x = 3$.

(a) Find the coordinates of the point. [1]

(b) Find the equation of the tangent at this point. [3]

(c) Find the equation of the normal at this point. [2]

16. Evaluate the following limits.

(a) $\lim_{x \to 0} \dfrac{1 - \cos x}{x^2}$ [3]

(b) $\lim_{x \to 0} \dfrac{e^x - 1 - x}{x^2}$ [3]

17. A curve has equation $y = \dfrac{\ln x}{x}$ for $x > 0$.

(a) Find $\dfrac{dy}{dx}$. [3]

(b) Show that the curve has a horizontal tangent at $x = e$. [3]

18. Differentiate each of the following, simplifying your answers.

(a) $y = (2x^3 - 1)^4$ [2]

(b) $y = x^2 \ln x$ [2]

(c) $y = \dfrac{\sin x}{x}$ [2]

19. Consider the curve $y = e^x$ at the point where $x = 1$.

(a) Show that the tangent there passes through the origin. [4]

(b) Find the equation of the normal at the same point. [2]

20. Differentiate each of the following.

(a) $y = \arctan(3x)$ [2]

(b) $y = \sec(2x)$ [2]

(c) $y = \log_2 x$ [2]

21. Consider $f(x) = x^3 - 6x^2 + 9x + 1$.

(a) Find the intervals on which f is increasing. [3]
(b) Find the interval on which f is concave up. [2]
(c) State the x-coordinate of the point of inflexion. [1]

22. Find the equations of *both* tangents to the curve $y = x^2$ that pass through the point $(0, -1)$. [6]

23. Differentiate each of the following, simplifying where possible.

(a) $y = (3x^2 + 1)^5$ [2]
(b) $y = x^2 e^x$ [2]
(c) $y = \dfrac{x}{x+1}$ [3]

24. Differentiate each of the following.

(a) $y = \sec x$ [1]
(b) $y = \ln(\cos x)$ [3]
(c) $y = x \arctan x$ [3]

25. The displacement of a particle is $s(t) = t^3 - 6t^2 + 9t$ metres at time t seconds, for $t \geq 0$.

(a) Find the velocity $v(t) = s'(t)$. [2]
(b) Find the times at which the particle is momentarily at rest. [3]
(c) Find the acceleration at $t = 3$. [2]

26. The derivative of a function f is $f'(x) = (x+2)(x-4)$.

(a) Find the intervals on which f is increasing. [2]
(b) Find the x-coordinates of the turning points of f and determine their nature. [3]
(c) Find the x-coordinate of the point of inflexion of f. [2]

27. Evaluate $\lim\limits_{x \to 0} \dfrac{\tan x - x}{x^3}$. You may use $\lim\limits_{x \to 0} \dfrac{\tan x}{x} = 1$ after establishing it. [7]

28. Consider $f(x) = x^3 - 3x^2 - 9x$.

(a) Find $f'(x)$ and the values of x where $f'(x) = 0$. [3]
(b) State the intervals on which f is increasing. [2]
(c) Find $f''(x)$ and the x-coordinate of the point of inflexion. [3]

29. Consider $f(x) = x^3 - 3x^2 - 9x + 5$.

(a) Find $f'(x)$ and factorise it fully. [2]
(b) Find the coordinates of the two stationary points. [3]
(c) Find $f''(x)$, and use it to determine the nature of each stationary point. [3]
(d) Find the coordinates of the point of inflexion. [2]
(e) State the interval on which f is concave up. [2]

30. Let $g(x) = \dfrac{2x-1}{x+2}$, $x \neq -2$.

(a) Write down the equations of the vertical and horizontal asymptotes. [2]
(b) Use the quotient rule to show that $g'(x) = \dfrac{5}{(x+2)^2}$. [3]
(c) Hence explain why g is increasing on every interval of its domain. [2]
(d) Find the equation of the tangent to the curve at the point where $x = 0$. [2]
(e) Find the equation of the normal at the same point. [2]

31. Let $y = x^2 e^{-x}$.

(a) Use the product rule to show that $\dfrac{dy}{dx} = x(2-x)e^{-x}$. [3]
(b) Find the coordinates of the two stationary points. [3]
(c) Show that $\dfrac{d^2y}{dx^2} = (x^2 - 4x + 2)e^{-x}$. [3]
(d) Use the second derivative to determine the nature of each stationary point. [2]
(e) Describe the behaviour of y as $x \to \infty$, and state the equation of the asymptote. [2]

32. GDC A closed cylindrical can has volume 500 cm^3. Its radius is r cm and its height is h cm.

(a) Show that $h = \dfrac{500}{\pi r^2}$. [2]
(b) Hence show that the total surface area is $S = 2\pi r^2 + \dfrac{1000}{r}$. [3]
(c) Find $\dfrac{dS}{dr}$. [2]
(d) Find the value of r that minimises S, giving your answer to three significant figures. [3]
(e) Verify, using the second derivative, that this value gives a minimum. [2]
(f) Show that at this radius the height is exactly twice the radius. [2]

Challenge Questions

33. **The power rule, proved.** Section 14.3 stated the power rule after two examples suggested it. Here you prove it for every positive integer n, using the binomial theorem.

(a) Write down the first three terms of $(x+h)^n$ in ascending powers of h. [3]

(b) Hence show that

$$\frac{(x+h)^n - x^n}{h} = nx^{n-1} + \binom{n}{2}x^{n-2}h + \cdots + h^{n-1},$$

and explain why every term after the first still contains h. [5]

(c) Deduce that $\frac{d}{dx}x^n = nx^{n-1}$ for every positive integer n. [3]

(d) Check your working by expanding in full for $n = 4$. [3]

(e) The rule holds for negative and fractional n as well, but this argument does not reach them. State which step fails when $n = -1$, then confirm that case by differentiating $\frac{1}{x}$ from first principles. [4]

34. Where the derivative of sine comes from. The standard derivatives in §14.5 rest on one limit. Here you establish that limit, then use it to prove the first of them.

(a) Evaluate $\dfrac{\sin x}{x}$ at $x = 0.1$, 0.01 and 0.001 radians, and at the matching negative values. Use the results as evidence that $\lim\limits_{x\to 0} \dfrac{\sin x}{x} = 1$. [3]

(b) Show that $\dfrac{\cos x - 1}{x} = \dfrac{-\sin^2 x}{x(\cos x + 1)}$. Hence show that $\lim\limits_{x\to 0} \dfrac{\cos x - 1}{x} = 0$. *(Hint: multiply above and below by $\cos x + 1$)* [5]

(c) The compound-angle identity gives $\sin(x+h) = \sin x \cos h + \cos x \sin h$. Use it to show that

$$\frac{\sin(x+h) - \sin x}{h} = \sin x\left(\frac{\cos h - 1}{h}\right) + \cos x\left(\frac{\sin h}{h}\right).$$

[4]

(d) Hence prove from first principles that $\dfrac{d}{dx}\sin x = \cos x$. [3]

(e) Explain why the proof fails if x is measured in degrees. [2]

35. Two routes to the same limit. An indeterminate limit can be settled by L'Hôpital's rule, or by replacing each function with its Maclaurin series. Here you use both on $\lim\limits_{x\to 0} \dfrac{x - \sin x}{x^3}$ and compare them.

(a) Evaluate the limit by applying L'Hôpital's rule repeatedly, checking at each stage that the form is still indeterminate. [5]

(b) The Maclaurin series for $\sin x$ begins $x - \dfrac{x^3}{3!} + \dfrac{x^5}{5!} - \cdots$. Substitute it into the fraction and simplify, and the limit follows in one step. [4]

(c) Use the series method on $\lim\limits_{x\to 0} \dfrac{e^x - 1 - x}{x^2}$, given that $e^x = 1 + x + \dfrac{x^2}{2!} + \dfrac{x^3}{3!} + \cdots$. [3]

(d) Give one advantage of each method, and one limit for which the series method would be harder to use. [4]

CALCULUS

Chapter 15

Derivative Applications

15

SYLLABUS 5.8 5.9 5.14

Throw a ball straight up. It rises, it slows, and at the very top it stops for an instant before it falls back. At that instant its velocity is exactly zero. A moment earlier the ball was moving up; a moment later it is moving down. The top of the flight is the one place where it is doing neither.

That instant is the key to a large class of problems. The top of the flight is where the height is greatest, and it is also where the velocity is zero. The same pattern holds everywhere: when a quantity reaches its largest or smallest value, its rate of change is zero at that moment, and the curve flattens.

This is why the derivative is the right tool for finding a best value. The largest volume from a fixed sheet of metal, the least metal for a can of a given size, the quickest route, the cheapest design: in each case the answer sits where the derivative is zero. In this chapter the derivative stops being something you calculate and becomes something you use. You will find the peaks and valleys of a curve and decide which is which, solve optimisation problems, describe motion, and work with quantities that change together.

The ball at the top of its flight holds the whole idea: to find an extreme value, find where the rate of change is zero.

By the end of this chapter you will be able to find and classify the stationary points and points of inflexion of a curve, and to read its shape from the signs of f' and f''. You will use a derivative to find a largest or smallest value under a constraint, describe motion in a straight line by differentiating displacement, and differentiate implicitly when two rates are linked. The course spends this long on it because a derivative is not only a gradient. It is a rate of change, and many real questions ask how fast something is changing, or where a quantity is greatest or least.

15.1 Stationary Points and Their Nature

A **stationary point** is a point where the gradient is zero: the tangent is horizontal. Since the derivative gives the gradient, stationary points are exactly the solutions of $f'(x) = 0$. They are the points where a curve stops rising or stops falling.

For a smooth curve, every local maximum and minimum is a stationary point. The condition $f'(x) = 0$ finds them all. It does not find an extreme value at a corner, where the derivative does not exist: $f(x) = |x|$ has a minimum at $x = 0$ and no stationary point there. See Ch. 14.

DEF A **stationary point** of f occurs where $f'(x) = 0$. At a **local maximum** the curve changes from increasing to decreasing; at a **local minimum**, from decreasing to increasing.

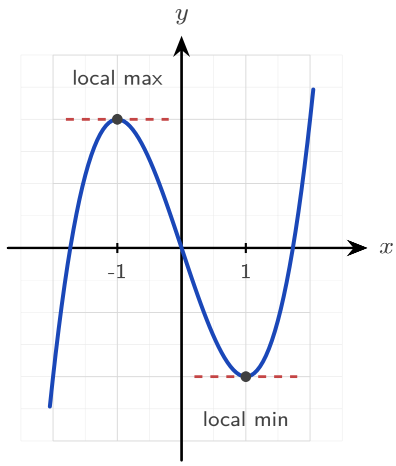


Figure 15.1 At each stationary point the tangent is horizontal. The second derivative decides which is the maximum and which the minimum.

Finding stationary points is only half the work. The other half is deciding what kind each one is: a maximum, a minimum, or neither. There are two tests.

The first derivative test

The sign of f' tells you whether the curve is rising or falling, so it can be read on either side of the stationary point. Just before a maximum the curve rises, $f' > 0$, and just after it falls, $f' < 0$. A minimum is the reverse. Record the three signs in order and the shape is decided.

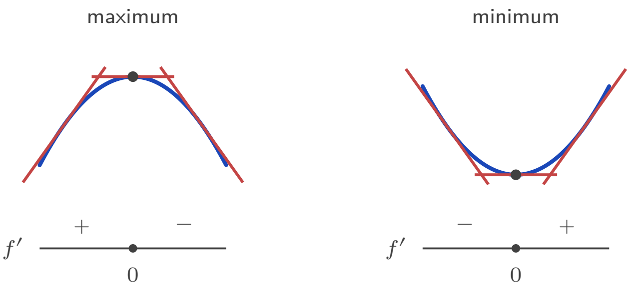


Figure 15.2 The first derivative test. The scarlet strokes are tangents: rising, flat, then falling at a maximum, and the reverse at a minimum. The strip below records the sign of f' on each side.

The second derivative test

This test is usually faster, and it uses the second derivative, which measures concavity. There is a picture behind it. At a minimum the curve bends upward, so it sits *above* its own horizontal tangent, and $f'' > 0$. At a maximum it bends downward, sits *below* the tangent, and $f'' < 0$.

Figure 15.3 The second derivative test. A curve that bends upward lies above its horizontal tangent, and one that bends downward lies below it. That is what the sign of f'' records.

KEY At a stationary point where $f'(x) = 0$:

$$f''(x) > 0 \Rightarrow \text{local minimum}, \qquad f''(x) < 0 \Rightarrow \text{local maximum}.$$

If $f''(x) = 0$ this test gives no answer. Use the sign of f' on either side instead.

EXAMPLE 15.1

Find and classify the stationary points of $f(x) = x^3 - 3x^2 - 9x + 5$.
Differentiate and solve $f'(x) = 0$:

$$f'(x) = 3x^2 - 6x - 9 = 3(x-3)(x+1) = 0 \implies x = -1 \text{ or } x = 3.$$

The second derivative is $f''(x) = 6x - 6$.
At $x = -1$: $f''(-1) = -12 < 0$, a local maximum. $f(-1) = 10$, so $(-1, 10)$.
At $x = 3$: $f''(3) = 12 > 0$, a local minimum. $f(3) = -22$, so $(3, -22)$.
Here the maximum is above the minimum, which is normal. The word "local" means highest or lowest *among nearby points*, not over the whole curve.

YOUR TURN 15.1 Stationary Points and Their Nature

1. Find the stationary points of each curve.

(a) $y = x^2 - 6x + 5$ [2]
(b) $y = x^3 - 12x$ [2]
(c) $y = 2x^3 - 3x^2 - 12x + 1$ [3]
(d) $y = x^4 - 2x^2$ [3]

2.

(a) Use the second derivative to classify the stationary point of $y = x^2 - 6x + 5$. [2]
(b) Use the second derivative to classify both stationary points of $y = x^3 - 12x$. [3]
(c) Find the stationary point of $y = x + \dfrac{1}{x}$ for $x > 0$, and state its nature. [3]
(d) A curve has $\dfrac{dy}{dx} = (x-1)(x-4)$. Write down the x-coordinates of its stationary points, and state the nature of each without differentiating again. [4]

3. The curve $y = x^3$ has a stationary point at the origin.

(a) Show that $x = 0$ is a stationary point. [2]
(b) Explain why the second derivative test cannot classify it. [2]
(c) Use the sign of $\dfrac{dy}{dx}$ on either side to classify it. [3]

15.2 Points of Inflexion and Concavity

A stationary point is not the only feature worth naming. A **point of inflexion** is where the curve changes the *direction of its bend*: from concave up to concave down, or the reverse. It is where f'' changes sign, which requires $f''(x) = 0$.

DEF A **point of inflexion** is a point where the concavity changes sign. A necessary condition is $f''(x) = 0$, but $f''(x) = 0$ alone is not sufficient: the concavity must actually change.

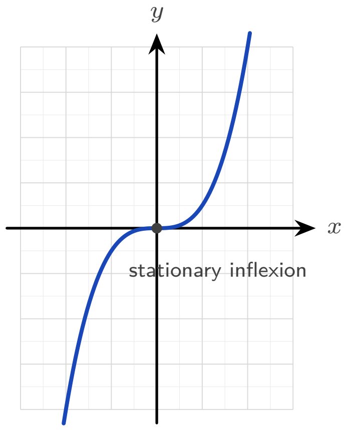


Figure 15.4 At the origin $y = x^3$ has both a horizontal tangent and a change of concavity, so this inflexion is also a stationary point.

An inflexion can happen at any gradient, and the syllabus names both kinds.

In $y = x^3$ above, the inflexion at the origin has **zero gradient**. The tangent is horizontal, so this point is also a stationary point, but it is neither a maximum nor a minimum.

In the next example the inflexion has a **non-zero gradient**. The curve is falling as it passes through.

Only one thing makes a point an inflexion: the concavity changes there. The gradient can take any value.

EXAMPLE 15.2

Find the point of inflexion of $f(x) = x^3 - 3x^2$.
Differentiate twice:

$$f'(x) = 3x^2 - 6x, \qquad f''(x) = 6x - 6.$$

Set $f''(x) = 0$: $6x - 6 = 0 \implies x = 1$. To confirm it is genuine, check that f'' changes sign: for $x < 1$, $f'' < 0$ (concave down); for $x > 1$, $f'' > 0$ (concave up). It changes, so $x = 1$ is a point of inflexion. Since $f(1) = -2$, the point is $(1, -2)$.
Note the gradient there: $f'(1) = 3 - 6 = -3 \neq 0$. This is a *non-stationary* inflexion, the curve changing its bend mid-descent.

CAUTION $f''(x) = 0$ does not by itself guarantee an inflexion. For $f(x) = x^4$, $f''(0) = 0$, yet the curve is concave up on both sides: $x = 0$ is a minimum, not an inflexion. Always confirm the concavity actually changes.

YOUR TURN 15.2 Points of Inflexion and Concavity

4. Find the point of inflexion of each curve.

(a) $y = x^3 - 6x^2 + 5$ [2]
(b) $y = x^3 + 3x^2$ [2]
(c) $y = 2x^3 - 9x^2 + 12x$ [2]

5.

(a) Explain why $y = x^4$ has no point of inflexion at $x = 0$. [2]
(b) Determine, for $y = x^5$, whether the point where $f'' = 0$ is a genuine inflexion. [3]
(c) State the interval on which $y = x^3 - 6x^2 + 5$ is concave down. [2]

15.3 Optimization

Optimisation means finding the largest or smallest value of a real quantity: the greatest volume, the least cost, the shortest time.

It is worth asking why the derivative helps at all. Without it, the only way to find a best value would be to try inputs one at a time and compare the results. That can suggest an answer but never settles it, because between any two inputs you tried there are infinitely many you did not. The derivative removes the search. A maximum or a minimum leaves a mark that algebra can detect: the curve is flat there, so $f'(x) = 0$. An impossible hunt through infinitely many values becomes one equation to solve, and the handful of solutions are the only candidates worth checking. That is the whole reason this chapter exists.

In practice the calculus is the easy part; the work is in setting up the function.

Almost every optimisation question follows the same five steps.

1. **Name the quantity to be optimised** and write an expression for it. At this stage it may contain more than one variable.
2. **Find the constraint**, the second equation linking those variables. It is always in the question, often as a fixed volume, a fixed length of material, or a fixed perimeter.
3. **Use the constraint to eliminate variables**, until the quantity is a function of one variable only. State the values that variable may take.
4. **Differentiate, set the derivative to zero, and solve.**
5. **Justify the nature of the point**, then answer the question that was actually asked.

Steps 3 and 5 are the ones to watch. Reducing to a single variable is where the algebra goes wrong, and skipping the justification is where marks are dropped at the end.

TIP An optimisation question is not finished when you solve $f'(x) = 0$. Two marks usually depend on what follows: justify the nature of the point, using f'' or the sign of f', and then answer the question that was asked. If the question asks for the maximum volume, give the volume, not the value of x that produces it.

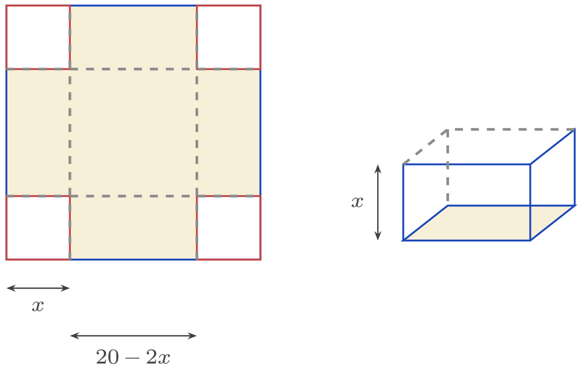


Figure 15.5 Cutting a square of side x from each corner of the card, then folding the flaps up. The base of the box measures $20 - 2x$ each way and its height is x, so $V = x(20 - 2x)^2$.

EXAMPLE 15.3

An open box is made from a 20×20 cm square of card by cutting a square of side x from each corner and folding up the sides. Find the value of x that maximises the volume.
After cutting, the base is $(20 - 2x)$ by $(20 - 2x)$ and the height is x, so

$$V(x) = x(20 - 2x)^2, \qquad 0 < x < 10.$$

Differentiate (expanding the bracket helps):

$$V'(x) = (20 - 2x)^2 + x \cdot 2(20 - 2x)(-2) = (20 - 2x)(20 - 6x).$$

Setting $V'(x) = 0$ gives $x = 10$ (rejected, no box) or $x = \frac{10}{3}$.
Since $V' > 0$ before $\frac{10}{3}$ and $V' < 0$ after, this is a maximum. The maximum volume is

$$V\left(\tfrac{10}{3}\right) = \frac{16000}{27} \approx 593 \text{ cm}^3.$$

The restriction $0 < x < 10$ comes from the geometry. If x is 0 the box has no height; if x is 10 it has no base.

EXAMPLE 15.4

A closed cylindrical can must hold $355\,\text{cm}^3$. Find the radius that minimises the surface area.
Volume fixes the height: $\pi r^2 h = 355$, so $h = \dfrac{355}{\pi r^2}$. The surface area is two circles plus the

curved side:

$$S = 2\pi r^2 + 2\pi r h = 2\pi r^2 + \frac{710}{r}.$$

Differentiate and set to zero:

$$\frac{dS}{dr} = 4\pi r - \frac{710}{r^2} = 0 \implies r^3 = \frac{710}{4\pi} \implies r \approx 3.84 \text{ cm}.$$

Since $\frac{d^2S}{dr^2} = 4\pi + \frac{1420}{r^3} > 0$, this is a minimum. Using the constraint to remove h is the step that reduces two variables to one.

CAUTION Setting $f' = 0$ finds only the best values *inside* an interval. If the variable is restricted, for example $0 < r \leq 3$, the best value may sit at an **endpoint** instead, where the derivative need not be zero. HL Always list the candidates: every stationary point in the interval, and both endpoints. Then compare their values and take the largest or the smallest.

EXAMPLE 15.5

For the can above, the manufacturer's machinery cannot roll a radius above 3 cm, so $0 < r \leq 3$. Find the radius that minimises the surface area now.

The unconstrained minimum at $r \approx 3.84$ is no longer allowed. Consider the derivative on $(0, 3]$:

$$\frac{dS}{dr} = 4\pi r - \frac{710}{r^2}.$$

Both terms increase as r increases, so $\frac{dS}{dr}$ is increasing on this interval. Testing the largest value is therefore enough: at $r = 3$, $12\pi - \frac{710}{9} \approx -41 < 0$. The derivative is negative at the right-hand end, and smaller still to the left, so it is negative throughout and S is decreasing across the whole interval. The minimum therefore sits at the endpoint $r = 3$, with $S = 18\pi + \frac{710}{3} \approx 293$ cm^2, even though $S'(3) \neq 0$. Whenever the domain is restricted, check its endpoints as well. The condition $f' = 0$ finds only the best values that lie *inside* the interval.

YOUR TURN 15.3 Optimization

6.

(a) Two numbers have a sum of 20. Find the maximum value of their product, and justify that it is a maximum. [4]

(b) A rectangle has a perimeter of 40 cm. Find the dimensions giving the maximum area. [4]

(c) Find the minimum value of $S = x^2 + \frac{16}{x}$ for $x > 0$, justifying that it is a minimum. [4]

7. An open box is made from a 12×12 cm square of card by cutting a square of side x cm from each corner and folding up the sides.

(a) Show that the volume is $V = x(12 - 2x)^2$, and state the possible values of x. [3]

(b) Find the value of x that maximises the volume. [4]

(c) Find the maximum volume. [2]

8.

(a) A rectangular field is fenced on three sides, the fourth being a wall, using 100 m of fencing. Find the maximum area. [4]

(b) An open-topped cylindrical tank must hold 1000 cm^3. Show that its surface area is $S = \pi r^2 + \dfrac{2000}{r}$, and find the radius that minimises it. [5]

(c) For the tank in part (b), explain why the answer would change if the manufacturer could not make a radius larger than 5 cm. [3]

15.4 Kinematics

Motion is change of position, so calculus describes it exactly. If a particle's **displacement** from a fixed origin is $s(t)$, then its **velocity** is the rate of change of displacement, $v = \dfrac{ds}{dt}$, and its **acceleration** is the rate of change of velocity.

KEY · IB

$$a = \frac{dv}{dt} = \frac{d^2s}{dt^2}$$

Velocity carries a sign, so it records direction as well as how fast the particle moves. **Speed** is the size of the velocity, written $|v|$, and it is never negative.

The particle is **momentarily at rest** (at rest for an instant) when $v = 0$. This is the stationary-point condition of §15.1, applied to displacement.

Speed follows a separate rule. The particle **speeds up when v and a have the same sign**, because the acceleration acts in the direction of motion. It slows down when the two signs are different.

One quantity needs integration rather than differentiation: the **total distance travelled**, which keeps counting when the particle turns and comes back. See Ch. 16.

EXAMPLE 15.6

A particle moves so that its displacement is $s(t) = t^3 - 9t^2 + 24t$ metres, for $t \geq 0$ seconds. Find when it is at rest, and its acceleration at those times.

Differentiate for velocity:

$$v(t) = 3t^2 - 18t + 24 = 3(t-2)(t-4).$$

The particle is at rest when $v = 0$: at $t = 2$ and $t = 4$ seconds.

Acceleration is $a(t) = 6t - 18$. At $t = 2$, $a = -6\,\mathrm{m\,s^{-2}}$; at $t = 4$, $a = 6\,\mathrm{m\,s^{-2}}$. The sign change of velocity at each rest point means the particle reverses direction there.

Now ask whether the particle is speeding up at $t = 5$. There $v(5) = 3(3)(1) = 9 > 0$ and $a(5) = 6(5) - 18 = 12 > 0$. The two signs are the same, so the particle is speeding up: it moves in the positive direction and the acceleration acts the same way.

TIP In kinematics questions, state the units with every answer: $\mathrm{m\,s^{-1}}$ for velocity, $\mathrm{m\,s^{-2}}$ for acceleration. A correct number without its unit loses the accuracy mark.

CAUTION Velocity and speed are not the same. A particle with $v = -5$ moves in the negative direction at a speed of 5. Maximum speed can therefore occur where the velocity is most negative, so compare $|v|$, not v.

YOUR TURN 15.4 Kinematics

9. A particle moves in a straight line.

(a) Its displacement is $s = t^2 - 4t + 3$. Find when it is at rest, and its acceleration then. [2]

(b) Its displacement is $s = 2t^3 - 9t^2 + 12t$. Find the times when it is at rest. [3]

(c) Its displacement is $s = t^3 - 3t^2$. Find its velocity and acceleration at $t = 2$. [2]

(d) Its velocity is $v = t^2 - 5t + 6$. Find when it is at rest, and its acceleration at those times. [3]

10. A particle has velocity $v = 3t^2 - 12t + 9$ for $t \geq 0$.

(a) Find the times when it is at rest. [2]

(b) Find its acceleration at those times. [2]

(c) Determine whether the particle is speeding up or slowing down at $t = 1.5$, giving a reason. [3]

(d) A second particle has $v = t - 4$ and $a = 1$ at $t = 2$. State, with a reason, whether it is speeding up or slowing down. [2]

15.5 Implicit Differentiation HL

So far every curve has been given as a function $y = f(x)$, with y written on its own. But an equation such as $x^2 + y^2 = 25$ also defines a curve, without y being isolated.

Implicit differentiation finds the gradient of such a curve directly, without first solving for y.

The method is the chain rule. Treat y as an unknown function of x.

A term in y then differentiates as usual, but it gains an extra factor $\frac{dy}{dx}$, because you are differentiating a function of y with respect to x. For example, $\frac{d}{dx}(y^2) = 2y\frac{dy}{dx}$.

EXAMPLE 15.7

Find $\frac{dy}{dx}$ for the circle $x^2 + y^2 = 25$, and the gradient at $(3, 4)$.
Differentiate both sides with respect to x:

$$2x + 2y\frac{dy}{dx} = 0 \implies \frac{dy}{dx} = -\frac{x}{y}.$$

At $(3, 4)$ the gradient is $-\frac{3}{4}$. The answer depends on both coordinates, and it must: a circle has a different gradient at every point.

EXAMPLE 15.8

Find $\frac{dy}{dx}$ for $x^2 + xy + y^2 = 7$.
The middle term needs the product rule. Differentiating term by term:

$$2x + \left(y + x\frac{dy}{dx}\right) + 2y\frac{dy}{dx} = 0.$$

Collect the $\frac{dy}{dx}$ terms:

$$\frac{dy}{dx}(x + 2y) = -(2x + y) \implies \frac{dy}{dx} = -\frac{2x + y}{x + 2y}.$$

The procedure is always the same: differentiate, collect the $\frac{dy}{dx}$ terms on one side, factorise, divide.

This means every tangent and normal method from Ch. 14 now works on curves that are not functions.

EXAMPLE 15.9

Find the equation of the tangent to the curve $x^2 + xy + y^2 = 7$ at the point $(1, 2)$.

First confirm the point lies on the curve: $1 + 2 + 4 = 7$. From the previous example,

$$\frac{dy}{dx} = -\frac{2x+y}{x+2y} = -\frac{2+2}{1+4} = -\frac{4}{5} \quad \text{at } (1,2).$$

The tangent is $y - 2 = -\frac{4}{5}(x-1)$, or $4x + 5y = 14$. Implicit differentiation supplies the gradient. The rest is the usual equation of a line through a given point.

YOUR TURN 15.5 Implicit Differentiation HL

11. Find $\dfrac{dy}{dx}$ for each curve.

(a) $x^2 + y^2 = 16$ [2]
(b) $xy = 12$ [2]
(c) $x^2 - y^2 = 9$ [2]
(d) $x^3 + y^3 = 6$ [2]

12. Each of these needs the product rule.

(a) Find $\dfrac{dy}{dx}$ for $x^2 + xy = 5$. [3]
(b) Find $\dfrac{dy}{dx}$ for $x^2y + y^2 = 4$. [3]

13.

(a) Find the gradient of $x^2 + y^2 = 25$ at the point $(4, -3)$. [3]
(b) Find the equation of the tangent to $xy = 4$ at the point $(2, 2)$. [4]
(c) On the circle $x^2 + y^2 = 25$, find the two points at which the tangent is horizontal. [3]
(d) Explain why implicit differentiation gives a gradient in terms of both x and y, while differentiating $y = f(x)$ gives one in terms of x alone. [2]

15.6 Related Rates HL

When two quantities are linked by an equation and both change over time, their rates of change are linked too. **Related rates** problems use the chain rule to connect those rates. You differentiate the equation that relates the quantities with respect to time.

The method has three steps. Write an equation relating the quantities. Differentiate every term with respect to t, so that each variable contributes its own rate. Then substitute the values known at the instant in question.

TIP Substitute the given values only *after* differentiating. Putting $r = 5$ into the volume formula first turns a variable into a constant, and a constant has rate of change zero.

EXAMPLE 15.10

Air is pumped into a spherical balloon at $100\,\text{cm}^3\text{s}^{-1}$. Find the rate at which the radius is increasing when $r = 5$ cm.
Volume and radius are related by $V = \frac{4}{3}\pi r^3$. Differentiate with respect to t:

$$\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}.$$

Substitute $\dfrac{dV}{dt} = 100$ and $r = 5$:

$$100 = 4\pi(25)\frac{dr}{dt} \implies \frac{dr}{dt} = \frac{1}{\pi} \approx 0.318 \text{ cm s}^{-1}.$$

The rate of change of volume is constant, but the radius increases more slowly as the balloon grows, because the same volume is spread over a larger surface.

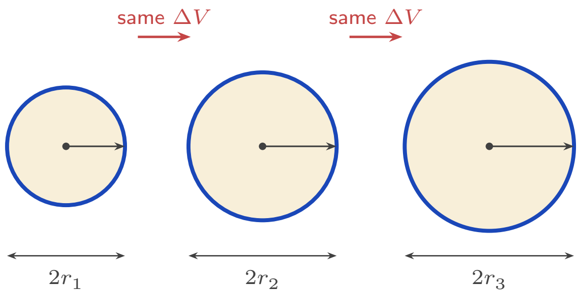


Figure 15.6 Equal volumes of air are added at each step, but the radius grows by less each time. The volume added is spread over the surface, and the surface area $4\pi r^2$ is itself growing, which is why $\frac{dr}{dt} = \frac{1}{4\pi r^2}\frac{dV}{dt}$ falls as r increases.

EXAMPLE 15.11

A 5 m ladder leans against a wall. The foot slides away from the wall at $0.5\,\text{m s}^{-1}$. How fast is the top sliding down when the foot is 3 m from the wall?
Let x be the distance of the foot from the wall and y the height of the top. Then $x^2 + y^2 = 25$. Differentiate with respect to t:

$$2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0 \implies \frac{dy}{dt} = -\frac{x}{y}\frac{dx}{dt}.$$

When $x = 3$, $y = \sqrt{25 - 9} = 4$. With $\dfrac{dx}{dt} = 0.5$:

$$\frac{dy}{dt} = -\frac{3}{4}(0.5) = -0.375 \text{ m s}^{-1}.$$

The negative sign means the top moves downward, as expected.

EXAMPLE 15.12

A water tank is an inverted cone with its vertex at the bottom, twice as tall as it is wide: at every level, the water surface has radius $r = \frac{h}{2}$, where h is the water depth. Water flows in at $2\,\text{m}^3\,\text{min}^{-1}$. How fast is the depth rising when $h = 3$ m?

Write the volume in terms of h alone, using the shape constraint $r = \frac{h}{2}$:

$$V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi\left(\frac{h}{2}\right)^2 h = \frac{\pi h^3}{12}.$$

Differentiate with respect to time:

$$\frac{dV}{dt} = \frac{\pi h^2}{4}\frac{dh}{dt} \implies 2 = \frac{9\pi}{4}\frac{dh}{dt} \implies \frac{dh}{dt} = \frac{8}{9\pi} \approx 0.283\ \text{m}\,\text{min}^{-1}.$$

Remove r *before* differentiating. Differentiating an expression that still contains two linked variables is where most errors happen.

The algebra also shows the physics: the same inflow raises the level more slowly as the cone widens.

YOUR TURN 15.6 Related Rates HL

14.

(a) A square's side grows at $2\,\text{cm}\,\text{s}^{-1}$. Find the rate of change of its area when the side is 5 cm. [3]

(b) A circle's radius grows at $3\,\text{cm}\,\text{s}^{-1}$. Find the rate of change of its area when $r = 4$ cm. [3]

(c) A cube's side grows at $1\,\text{cm}\,\text{s}^{-1}$. Find the rate of change of its volume when the side is 10 cm. [3]

(d) A circle's radius increases at $0.4\,\text{cm}\,\text{s}^{-1}$. Find the rate of increase of its circumference. [2]

15.

(a) A spherical balloon is inflated at $50\,\text{cm}^3\text{s}^{-1}$. Find the rate of change of the radius when $r = 10$ cm. [3]

(b) A ladder 5 m long leans against a vertical wall. Its foot slides away from the wall at $2\,\text{m}\,\text{s}^{-1}$. Find the rate at which the top slides down when the foot is 3 m from the wall. [5]

(c) Water is poured into a cone at $20\,\text{cm}^3\text{s}^{-1}$. The cone's height is always twice its surface radius, so $V = \frac{2}{3}\pi r^3$. Find the rate at which r increases when $r = 5$ cm. [4]

Chapter 15 · Derivative Applications

Reflections

TOK The ball is motionless at the top of its flight, but only for an instant, and an instant has no duration. Is being motionless for no time at all a physical state, or an artefact of the mathematics? Does calculus describe motion, or replace it with something we can calculate?

TOK This chapter is full of conditions that are necessary but not sufficient: $f'(x) = 0$ holds at every smooth maximum, and $f''(x) = 0$ at every point of inflexion, yet neither condition guarantees one. What kind of knowledge is a necessary condition? Does knowing where something must be count as knowing where it is?

TOK The can that uses least metal has its height equal to its diameter, but real cans are taller and narrower, because the model ignores printing, shipping and the size of a hand. The mathematics is exact and the conclusion is still wrong. Does mathematically optimal always mean practically best?

IM The idea that nature minimises something appears across physics under many names. In seventeenth-century France, Pierre de Fermat proposed that light travels the path taking the least time, which explains refraction. Euler and Lagrange later generalised this into the calculus of variations and the *principle of least action*, which restates the whole of classical mechanics as an optimisation problem. The belief that the world is in some sense optimal has produced a great deal of physics.

RW The method behind modern machine learning is *gradient descent*, which is this chapter's idea at scale. A model has millions of adjustable numbers, and how wrong its predictions are is a function of those numbers; training means making that function as small as possible. With millions of variables the minimum cannot be found directly, so the computer does what a walker in fog would do: measure the slope where it stands, step down it, and repeat. As the slope flattens the steps shrink and the model settles near a minimum. That is $f' = 0$, applied billions of times.

EXT Implicit differentiation and related rates are the same idea twice. Both are the chain rule applied when the variables depend on something else: on x in the first case, on time in the second.

Carried further it becomes multivariable calculus, where a point has a different rate of change in every direction. Choosing the steepest of them is exactly what gradient descent does.

End-of-Chapter Problems — Chapter 15: Derivative Applications

1. For $f(x) = x^4$, show that $f''(0) = 0$ but that $(0, 0)$ is *not* a point of inflexion. [4]

2. A stone dropped in a pond sends out a circular ripple whose radius grows at $0.5\,\mathrm{m\,s^{-1}}$. Find the rate at which the enclosed area is increasing when the radius is 8 m. [4]

3. A particle's velocity is $v(t) = 3t^2 - 12t + 9\,\mathrm{m\,s^{-1}}$, $t \geq 0$.

(a) Find the times when the particle is at rest. [2]

(b) Find the minimum velocity and the time at which it occurs. [3]

4. A company's profit from selling x units is $P(x) = -2x^2 + 120x - 1000$ dollars. Find the value of x that maximises the profit, justifying that it is a maximum, and state the maximum profit. [5]

5. The curve $\sin y + x^2 = 1$ passes through the point $(1, 0)$. Find the gradient of the curve at that point. [5]

6. A curve has equation $x^2 + xy + y^2 = 3$.

(a) Find $\dfrac{dy}{dx}$ in terms of x and y. [3]

(b) Find the equation of the tangent at the point $(1, 1)$. [3]

7. Water drains from an inverted cone whose radius always equals its height. The volume is $V = \frac{1}{3}\pi r^2 h$.

(a) Show that $V = \frac{1}{3}\pi h^3$. [2]

(b) Given the volume decreases at $2\,\mathrm{cm^3 s^{-1}}$, find the rate at which the depth h is falling when $h = 4$ cm. [4]

8. Consider $f(x) = x^3 - 3x$ on the closed interval $0 \leq x \leq 3$.

(a) Find the stationary point(s) in the interval and classify them. [3]

(b) Find the global maximum and minimum of f on the interval. [3]

9. Consider $g(x) = x + \dfrac{4}{x}$ on the interval $1 \leq x \leq 3$.

(a) Find the stationary point of g in the interval and classify it. [4]

(b) Find the maximum value of g on the interval, and state where it occurs. [2]

10. The curve C has equation $x^2 + y^2 - 4x + 2y = 15$.

(a) Show that the point $(0, 3)$ lies on C. [1]

(b) Use implicit differentiation to find $\frac{dy}{dx}$. [3]

(c) Find the equation of the tangent to C at $(0, 3)$. [2]

11. A street lamp is 5 m tall. A person of height 1.8 m walks away from it at $1.2\,\text{m s}^{-1}$. Find the rate at which the length of the person's shadow is increasing. [6]

12. A water tank is an inverted cone in which the water surface radius is always $\frac{3}{4}$ of the depth h. Water is pumped in at $1.5\,\text{m}^3\,\text{min}^{-1}$. Find the rate at which the depth is rising when $h = 2$ m. [6]

13. Find the point on the curve $y = \sqrt{x}$ closest to the point $(4, 0)$, and state the minimum distance. [6]

14. For the curve $x^2 + y^2 = 25$, use implicit differentiation twice to show that $\frac{d^2y}{dx^2} = -\frac{25}{y^3}$. [6]

15. A closed cylindrical can must hold $500\,\text{cm}^3$.

(a) Show that the surface area is $S = 2\pi r^2 + \frac{1000}{r}$. [3]

(b) Find the radius that minimises S, and justify it is a minimum. [4]

16. A rectangular pen of area $200\,\text{m}^2$ is built against a wall, needing fencing on three sides. Let the side perpendicular to the wall be x.

(a) Show that the length of fencing is $L = 2x + \frac{200}{x}$. [3]

(b) Find the dimensions that minimise the fencing used. [4]

17. The curve $y = x^3 + ax^2 + bx$ has stationary points at $x = 1$ and $x = -2$.

(a) Find the values of a and b. [4]

(b) Classify each stationary point. [3]

18. An open rectangular tank with a square base of side x must hold $32\,\text{m}^3$.

(a) Show that the surface area (base + four sides) is $A = x^2 + \frac{128}{x}$. [3]

(b) Find the value of x that minimises the material used. [4]

19. An open-topped tank with a square base of side x cm is made from exactly $1200\,\text{cm}^2$ of sheet metal.

(a) Show that the volume is $V = \dfrac{1200x - x^3}{4}$. [3]

(b) Find the dimensions that maximise the volume, and the maximum volume. [4]

20. A cone is inscribed in a sphere of radius R. Show that the cone of maximum volume has height $h = \frac{4R}{3}$. [7]

21. A 5 m ladder slides down a wall, its foot moving away at a constant $0.5\,\mathrm{m\,s^{-1}}$.

(a) Find the rate at which the top descends when the foot is 3 m from the wall. [3]

(b) Find the rate at which the area of the triangle formed by the ladder, wall, and ground is changing at that instant. [4]

22. A particle moves along the curve $y = x^2$ so that $\dfrac{dx}{dt} = 2$ at all times. Find the rate at which its distance from the origin is increasing as it passes through the point $(1, 1)$. [7]

23. Consider $f(x) = x^3 - 3x^2 - 9x + 5$.

(a) Find and classify the stationary points. [4]

(b) Find the point of inflexion. [2]

(c) State the intervals on which f is decreasing. [2]

24. A particle moves with displacement $s(t) = t^3 - 6t^2 + 9t$ metres, $t \geq 0$.

(a) Find the velocity and the times when the particle is at rest. [3]

(b) Find the acceleration when $t = 1$. [2]

(c) Determine whether the particle is speeding up or slowing down at $t = 1.5$. [3]

25. Consider $f(x) = x^3 - 6x^2 + 9x + 2$.

(a) Find and classify the stationary points. [5]

(b) Find the point of inflexion and show that the gradient there is not zero. [3]

26. A particle moves with displacement $s(t) = t^3 - 9t^2 + 24t - 10$ metres, for $0 \leq t \leq 6$ seconds.

(a) Find the times at which the particle is at rest. [3]

(b) Find its acceleration at each of those times. [2]

(c) Find the intervals of time during which the particle's *speed* is increasing. [3]

27. Consider $f(x) = xe^{-x}$.

(a) Find $f'(x)$ and the coordinates of the stationary point. [3]

(b) Determine its nature using the second derivative. [3]

(c) Find the x-coordinate of the point of inflexion. [2]

28. Consider $f(x) = x^4 - 4x^3$.

(a) Find $f'(x)$ and factorise it fully. [2]
(b) Find the coordinates of the stationary points. [3]
(c) Find $f''(x)$, and use it to classify the stationary point at $x = 3$. [3]
(d) Explain why the second derivative test fails at $x = 0$, and use the sign of f' to classify that point instead. [4]
(e) Find the coordinates of any points of inflexion. [3]

29. An open-topped box is made from a 12×12 cm square of card by cutting a square of side x cm from each corner and folding up the sides.

(a) Show that the volume is $V = x(12 - 2x)^2$, and state the possible values of x. [3]
(b) Find $\dfrac{dV}{dx}$ and show that it is zero at $x = 2$ and $x = 6$. [3]
(c) Explain why $x = 6$ must be rejected. [2]
(d) Find the maximum volume, justifying that it is a maximum. [4]

30. A particle moves in a straight line with displacement $s = t^3 - 6t^2 + 9t$ metres, for $t \geq 0$ seconds.

(a) Find expressions for the velocity and the acceleration. [3]
(b) Find the times at which the particle is momentarily at rest. [2]
(c) Find the displacement at each of those times. [2]
(d) Find the time at which the acceleration is zero, and state what is happening to the velocity at that instant. [3]
(e) Determine whether the particle is speeding up or slowing down at $t = 4$, giving a reason. [3]

31. The curve C has equation $x^3 + y^3 = 9xy$.

(a) Verify that the point $(2, 4)$ lies on C. [2]
(b) Use implicit differentiation to show that $\dfrac{dy}{dx} = \dfrac{x^2 - 3y}{3x - y^2}$. [4]
(c) Find the gradient of C at $(2, 4)$. [2]
(d) Find the equation of the tangent to C at $(2, 4)$. [3]

Challenge Questions

32. The shape of the best can. A closed cylindrical can must hold a fixed volume V. This investigation finds the shape that uses least metal, and shows that the answer does not depend on V.

(a) Show that the surface area is $S = 2\pi r^2 + \dfrac{2V}{r}$. [3]

(b) Find $\dfrac{dS}{dr}$, and show that S is stationary when $r^3 = \dfrac{V}{2\pi}$. [4]

(c) Verify, using the second derivative, that this gives a minimum. [3]

(d) Show that at this radius the height equals the diameter, whatever the value of V. [4]

(e) Real drink cans are taller and narrower than this. Suggest two costs the model leaves out. [2]

33. Closest approach. Finding the nearest point of a curve to a fixed point is an optimisation problem in disguise.

(a) Let $(x, \sqrt{x})$ be a point on $y = \sqrt{x}$ and let D be its distance from $(4, 0)$. Show that $D^2 = x^2 - 7x + 16$. [3]

(b) Explain why minimising D^2 gives the same answer as minimising D, and why using D^2 is easier. [3]

(c) Find the point on the curve closest to $(4, 0)$, and the least distance. [4]

(d) Show that at this closest point the line joining it to $(4, 0)$ is perpendicular to the tangent to the curve. [5]

(e) State the general result that part (d) illustrates. [2]

34. Quantifying curvature. A straight line does not bend; a small circle bends sharply. This investigation builds a number that measures how sharply a curve bends at a point. It is defined by

$$\kappa = \frac{|y''|}{\left(1 + (y')^2\right)^{3/2}},$$

and κ is called the *curvature*.

(a) Show that every straight line has $\kappa = 0$ at every point, and explain why the definition must give this. [3]

(b) For $y = x^2$, find κ as a function of x. Show that the curvature is greatest at the vertex, and find its value there. [4]

(c) Show that $\kappa \to 0$ as $x \to \pm\infty$ for $y = x^2$, and explain what this says about the shape of a parabola far from its vertex. [3]

(d) Use implicit differentiation on $x^2 + y^2 = R^2$ to show that a circle of radius R has $\kappa = \dfrac{1}{R}$ at every point. Explain why a constant curvature is exactly what you should expect. [5]

(e) The *radius of curvature* is defined as $\rho = \dfrac{1}{\kappa}$. Find ρ for $y = x^2$ at the origin, and state the circle that best matches the parabola there. [3]

CALCULUS

CHAPTER 16

Integral Calculus

16

SYLLABUS 5.5 5.9 5.10 5.11 5.15 5.16 5.17

Your car's speedometer tells you how fast you are going at every instant. Suppose you record it for an entire journey. Can you work out how far you travelled, using only the speed?

Over a tiny slice of time, your speed barely changes, so the distance covered in that slice is roughly speed times the slice of time. Add up all the slices and you have the total distance. As the slices shrink toward zero width, the rough sum becomes exact. What you are computing is the area under the speed-time graph.

This is the second great idea of calculus, and it arrives as two ideas rather than one. The first is **reversing a derivative**: given a rate of change, recover the quantity it came from. The second is **accumulating**: adding up a great many thin slices to get a total. The area under a graph is exactly this kind of total.

They have no obvious connection. Reversing a derivative is a problem in algebra. Adding slices is a problem in geometry. Yet the car journey links them, because the distance travelled is both the reverse of the velocity and the area under the velocity graph.

The chapter is built in that order. Section 16.1 does the reversal on its own, with no mention of area. Section 16.2 defines the area on its own, with no mention of derivatives, and then proves that the two are the same calculation. That result is the Fundamental Theorem of Calculus, and once it is in place the accumulated totals, areas between curves, volumes of revolution and distances travelled all become routine.

16.1 Antiderivatives

This chapter runs differentiation backwards. Given a function f, we look for a function whose derivative is f. Such a function is called an **antiderivative** of f.

There is never just one. The derivative of a constant is zero, so a constant can be added to any antiderivative without changing its derivative. Antiderivatives therefore come in families, all differing by a constant.

> DEF F is an **antiderivative** of f when $F'(x) = f(x)$. If F is one antiderivative, then $F(x) + C$ is an antiderivative for every constant C, and these are all of them.

Reversing the power rule means raising the power by one and dividing by the new power.

> KEY · IB An antiderivative of x^n is $\dfrac{x^{n+1}}{n+1} + C$, for $n \neq -1$.

The exception $n = -1$ is handled separately in the next section. Constant multiples and sums are reversed term by term, just as they are differentiated term by term.

EXAMPLE 16.1

Find the antiderivatives of $3x^2 - 4x + 1$.

Reverse each term, raising each power by one:

$$F(x) = x^3 - 2x^2 + x + C.$$

Check by differentiating: $\frac{d}{dx}(x^3 - 2x^2 + x + C) = 3x^2 - 4x + 1$. ✓ Differentiating your answer costs nothing and checks it completely.

Boundary conditions

The $+C$ means an antiderivative is not unique until you pin it down. One extra piece of information, a point the curve passes through, fixes C and selects a single function.

EXAMPLE 16.2

A curve has gradient $\dfrac{dy}{dx} = 6x^2 - 2$ and passes through $(1, 4)$. Find its equation.

Antidifferentiate to recover y:

$$y = 2x^3 - 2x + C.$$

Use the point $(1, 4)$: $4 = 2 - 2 + C \implies C = 4$. So $y = 2x^3 - 2x + 4$.

A symbol for the family

Writing "the antiderivatives of f" every time is slow, so there is a symbol for it.

KEY

$$\int f(x)\,dx = F(x) + C \qquad \text{means} \qquad F'(x) = f(x).$$

This is called the **indefinite integral** of f, and the C is the **constant of integration**. It is easy to leave out, and an answer written without it names one antiderivative instead of all of them.

For now the symbol is nothing more than shorthand: $\int f(x)\,dx$ means the family of antiderivatives, and no other meaning has been given to it. Its shape is worth noticing, though. The sign is an elongated letter S, and the S stands for *sum*. Nothing in this section has involved a sum of any kind, and that is deliberate. In §16.2 we shall define a completely separate quantity, the area under a curve, as a limit of sums, and only then prove that the two coincide. Until that proof, read the symbol as shorthand and nothing more.

YOUR TURN 16.1 Antiderivatives

1. Find the antiderivatives of each function.

(a) $4x^3 - 6x$ [1]
(b) $x^2 + 3x - 5$ [2]
(c) $\dfrac{6}{x^3}$ [2]
(d) $\sqrt{x}$ [2]
(e) $x^2 - \dfrac{1}{x^2}$ [2]
(f) $\dfrac{x^3 - 2x}{x}$ [3]
(g) $x^{-1/2}$ [2]
(h) $(2x)^3$ [2]

2.

(a) A curve has $\dfrac{dy}{dx} = 4x - 3$ and passes through $(2, 5)$. Find y. [3]

(b) A curve has $\dfrac{dy}{dx} = 3\sqrt{x}$ and passes through $(4, 20)$. Find y. [4]

(c) A curve has $\dfrac{d^2y}{dx^2} = 6x$, with $\dfrac{dy}{dx} = 1$ and $y = 2$ when $x = 0$. Find y. [4]

(d) Explain why a gradient function alone does not determine a curve, and what extra information does. [2]

(e) Explain why part (d) needed two pieces of information rather than one. [2]

16.2 Definite Integrals and Area

Set antiderivatives aside for the moment. This section starts a second, unrelated problem, and only later do the two meet.

The problem is area. For a region with straight edges there are formulas, but under a curve there are none, because the boundary is different at every point. What can be done is to approximate.

Cut the interval from a to b into n strips, each of width Δx. Replace each strip by a rectangle whose height is the value of the function somewhere in that strip. Every rectangle has area height $\times$ width, so the total is

$$\sum_{i=1}^{n} f(x_i)\,\Delta x.$$

This is not the area. It is a sum of rectangles, and it is wrong by however much the rectangles miss or overshoot. But narrower strips leave less room for error, and as n grows the sum settles on a single number.

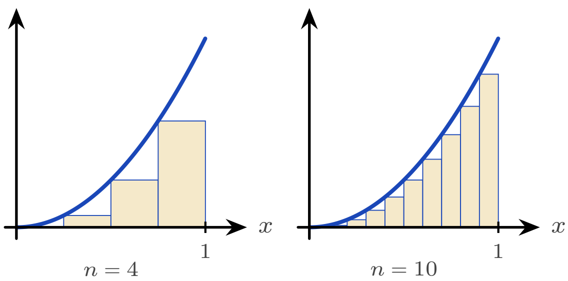


Figure 16.1 Rectangles under $y = x^2$ on $[0, 1]$, taking the height at the left of each strip. Four strips give 0.219 and ten give 0.285; the true area is $\frac{1}{3}$.

The numbers show it happening. Taking the height at the left of each strip underestimates the area, and taking it at the right overestimates it, so the true value is caught between them.

n	4	10	100	1000
low estimate	0.219	0.285	0.328	0.3328
high estimate	0.469	0.385	0.338	0.3338

The two columns close on each other, and the number they trap is $\frac{1}{3}$. That limit is what we mean by the area, and it is called the **definite integral**.

DEF The **definite integral** of f from a to b is the limit of the rectangle sums as the strips become arbitrarily narrow:

$$\int_a^b f(x)\,dx = \lim_{\Delta x\to 0}\sum_i f(x_i)\,\Delta x.$$

For $f(x)\geq 0$ it is the area between the curve and the x-axis from a to b.

Now the symbol makes sense. The elongated S is a sum, $f(x)\,dx$ is one rectangle of height $f(x)$ and vanishing width dx, and the integral adds them. That is what the notation was built to say.

It is also, so far, useless for calculation. Nothing here tells you how to find such a limit, and nobody wants to add a thousand rectangles by hand.

The Fundamental Theorem of Calculus (Extension)

The result is surprising. This limit of sums was defined without mentioning derivatives at all, yet it can be found by antidifferentiation. The argument below explains why. It is not examined, but it is the reason every method in the rest of the chapter works.

Fix the left end a and let the right end move. Write $A(x)$ for the area under the curve from a to x, so that A is a function of where you stop. Now ask how A changes when x increases by a small amount h. The extra area, $A(x+h)-A(x)$, is a single thin strip of width h.

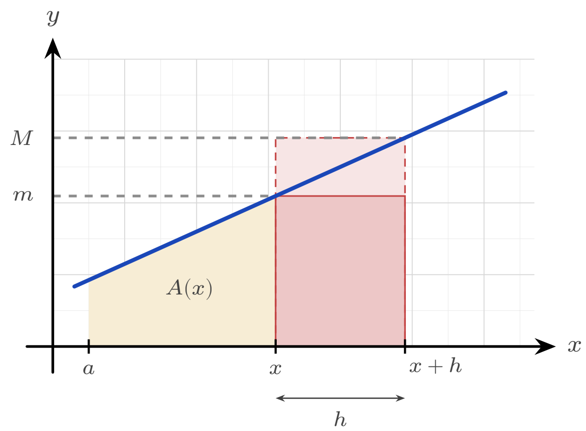


Figure 16.2 The extra area $A(x+h)-A(x)$ is one thin strip. It contains the short rectangle and is contained in the tall one, so its area lies between h times the smallest height of f on the strip and h times the largest.

That strip is trapped. It contains a rectangle of width h whose height is the *smallest* value of f on the strip, and it fits inside a rectangle of width h whose height is the *largest* value. Writing m and M for those two heights,

$$m\,h \;\leq\; A(x+h)-A(x) \;\leq\; M\,h.$$

Divide through by h:

$$m \le \frac{A(x+h) - A(x)}{h} \le M.$$

Now let $h \to 0$. The strip shrinks to the single point x, so its smallest and largest heights both close on $f(x)$. The quantity in the middle is squeezed between them, and it is exactly the definition of $A'(x)$. Therefore

$$A'(x) = f(x).$$

That is the whole argument, and it gives the result the chapter was built to reach: **the area function is an antiderivative of f.**

The computational rule follows in two lines. If F is any antiderivative of f, then A and F differ by a constant, so $A(x) = F(x) + C$. Putting $x = a$ gives $0 = F(a) + C$, since the area from a to a is nothing, so $C = -F(a)$. Putting $x = b$ then gives the area from a to b.

KEY **The Fundamental Theorem of Calculus.**

$$\int_a^b f(x)\,dx = \Big[F(x)\Big]_a^b = F(b) - F(a), \qquad \text{where } F' = f.$$

So the two ideas are one calculation. Reversing a derivative and adding up rectangles give the same answer. The constant C cancels in the subtraction, which is why a definite integral is a plain number while an indefinite one carries a $+C$.

KEY · IB

$$A = \int_a^b y\,dx \qquad (y \ge 0)$$

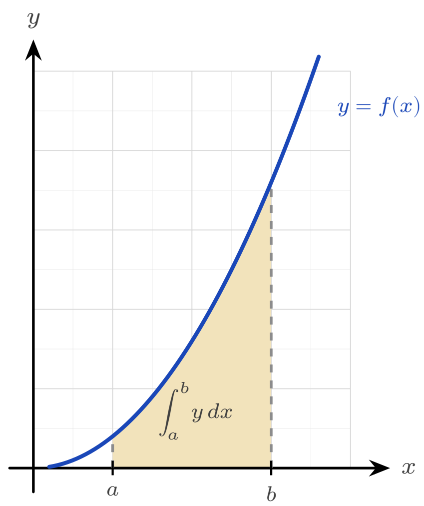


Figure 16.3 For $y \ge 0$ the definite integral measures the shaded area between the curve and the x-axis, from $x = a$ to $x = b$.

EXAMPLE 16.3

Evaluate $\int_0^2 3x^2\,dx$.

$$\int_0^2 3x^2\,dx = \left[x^3\right]_0^2 = 8 - 0 = 8.$$

TIP Your GDC evaluates definite integrals directly, and on a calculator paper that is often all a question requires. Even when analytic working is demanded, the GDC value is a free check of your antiderivative.

Areas below the axis

Where the curve dips below the x-axis, the integral is negative: it is a signed area. For a true geometric area you integrate the parts separately and take the absolute value of any negative piece.

EXAMPLE 16.4

Find the area enclosed between $y = x^2 - 4$ and the x-axis from $x = 0$ to $x = 2$.

On $[0, 2]$ the curve lies below the axis, so the integral is negative:

$$\int_0^2 (x^2 - 4)\,dx = \left[\tfrac{x^3}{3} - 4x\right]_0^2 = \tfrac{8}{3} - 8 = -\tfrac{16}{3}.$$

The area is the magnitude, $\dfrac{16}{3}$. An area is never negative, so $-\frac{16}{3}$ answers a different question from the one asked.

CAUTION Before finding an area, check where the curve crosses the x-axis. A curve that goes above and below will have positive and negative parts that partly cancel in a single integral, giving the wrong area. Split at the roots.

YOUR TURN 16.2 Definite Integrals and Area

3. Evaluate each definite integral.

(a) $\displaystyle\int_1^2 4x\,dx$ [2]

(b) $\displaystyle\int_0^3 (x^2+1)\,dx$ [2]

(c) $\displaystyle\int_1^e \frac{1}{x}\,dx$ [2]

(d) $\displaystyle\int_1^4 \frac{1}{\sqrt{x}}\,dx$ [2]

(e) $\displaystyle\int_0^{\pi/2} \cos x\,dx$ [2]

(f) $\displaystyle\int_{-1}^2 (3x^2-2x)\,dx$ [3]

4.

(a) Find the area under $y = x^2$ between $x = 1$ and $x = 3$. [2]

(b) Find the area enclosed between $y = x^2 - 1$ and the x-axis. [3]

(c) Find the area under $y = \sin x$ from $x = 0$ to $x = \pi$. [2]

(d) Evaluate $\int_0^{2\pi} \sin x \, dx$, and explain why the answer is not the area between the curve and the axis over that interval. [3]

(e) Find the area between $y = x^3$ and the x-axis from $x = -2$ to $x = 2$. [3]

5. The area under $y = x^2$ from $x = 0$ to $x = 2$ is to be estimated by four rectangles of equal width.

(a) Taking the height of each rectangle at the left of its strip, show that the estimate is 1.75. [3]

(b) Taking the height at the right instead, show that the estimate is 3.75. [2]

(c) Find the exact area, and confirm that it lies between the two estimates. [3]

(d) Repeat part (a) with eight rectangles, and show that the left-hand estimate rises to 2.1875. [3]

(e) State what happens to the two estimates as the number of rectangles increases. [2]

16.3 Standard Integrals

Every derivative you learned has a reverse. Reading the derivative table backwards gives the standard integrals.

KEY · IB

$$\int \frac{1}{x}\, dx = \ln|x| + C, \quad \int \sin x \, dx = -\cos x + C, \quad \int \cos x \, dx = \sin x + C, \quad \int e^x \, dx = e^x + C$$

The first fills the gap left by the power rule at $n = -1$: the antiderivative of x^{-1} is not a power but a logarithm. The absolute value lets it work for negative x too.

At HL, three more standard integrals are provided, the reverses of the exponential and inverse-trigonometric derivatives.

KEY · IB HL

$$\int a^x\,dx = \frac{a^x}{\ln a}+C, \quad \int \frac{1}{a^2+x^2}\,dx = \frac{1}{a}\arctan\frac{x}{a}+C, \quad \int \frac{1}{\sqrt{a^2-x^2}}\,dx = \arcsin\frac{x}{a}+C$$

Reading the rest of the HL derivative table backwards gives four more. They are worth learning, because the formula booklet gives only the derivatives:

KEY

$$\int \sec^2 x\,dx = \tan x + C, \qquad \int \sec x \tan x\,dx = \sec x + C,$$

$$\int \operatorname{cosec}^2 x\,dx = -\cot x + C, \qquad \int \operatorname{cosec} x \cot x\,dx = -\operatorname{cosec} x + C$$

Linear composites

Sometimes the function inside is linear, as in e^{2x} or $\cos 3x$. Integrate as usual, then divide by the coefficient of x inside. This reverses the chain rule, whose extra factor was that same coefficient.

KEY

$$\int f(ax+b)\,dx = \frac{1}{a}F(ax+b)+C, \qquad \text{where } F' = f$$

EXAMPLE 16.5

Find (a) $\int e^{2x}\,dx$, (b) $\int (2x+1)^5\,dx$, (c) $\int \cos 3x\,dx$.

Each has a linear function inside, so integrate and then divide by the coefficient of x:

$$\text{(a) } \tfrac{1}{2}e^{2x}+C, \qquad \text{(b) } \frac{(2x+1)^6}{12}+C, \qquad \text{(c) } \tfrac{1}{3}\sin 3x + C.$$

In (b), reversing the power rule gives $\frac{(2x+1)^6}{6}$, then dividing by the coefficient 2 gives the 12.

EXAMPLE 16.6

Find (a) $\displaystyle\int \frac{1}{x^2+4}\,dx$, (b) $\displaystyle\int \frac{1}{\sqrt{9-x^2}}\,dx$.
Match the standard forms with $a=2$ and $a=3$:

$$\text{(a) } \tfrac{1}{2}\arctan\frac{x}{2}+C, \qquad \text{(b) } \arcsin\frac{x}{3}+C.$$

The only work is recognising a^2 as 4 and as 9. After that the standard form gives the answer.

The linear-composite rule extends to the whole table. For instance, $\displaystyle\int \sec^2(2x+5)\,dx = \tfrac{1}{2}\tan(2x+5)+C$, the guide's own example: integrate as usual, divide by the inner coefficient.

YOUR TURN 16.3 Standard Integrals

6. Find each indefinite integral.

(a) $\displaystyle\int (2\cos x - \sin x)\,dx$ [2] (b) $\displaystyle\int \left(3e^x + \frac{1}{x}\right)dx$ [2]

(c) $\displaystyle\int e^{3x}\,dx$ [2] (d) $\displaystyle\int (3x-2)^4\,dx$ [2]

(e) $\displaystyle\int \sin 2x\,dx$ [2] (f) $\displaystyle\int \sec^2(4x)\,dx$ [2]

(g) $\displaystyle\int \cos(3x+1)\,dx$ [2] (h) $\displaystyle\int \frac{1}{2x+5}\,dx$ [2]

7. GDC Find each indefinite integral, matching it to a standard form.

(a) $\displaystyle\int \frac{1}{x^2+9}\,dx$ [2] (b) $\displaystyle\int \frac{1}{\sqrt{16-x^2}}\,dx$ [2]

(c) $\displaystyle\int 5^x\,dx$ [2] (d) $\displaystyle\int \frac{1}{x^2+4x+13}\,dx$ [4]

(e) $\displaystyle\int \frac{1}{\sqrt{9-4x^2}}\,dx$ [3] (f) $\displaystyle\int \frac{2}{4+x^2}\,dx$ [2]

16.4 Rewriting Before Integrating HL

Many integrands do not match a standard form as they stand, but can be rewritten until they do. Three rewritings cover almost every case: divide, complete the square, or split into partial fractions. Deciding between them takes one look at the denominator.

Divide first

When the numerator has degree at least that of the denominator, the fraction is **improper** and no standard form applies. Divide before anything else. For instance $\dfrac{x^2}{x+1} = x - 1 + \dfrac{1}{x+1}$, so the integral is $\frac{x^2}{2} - x + \ln|x+1| + C$. Only when the numerator has the lower degree are the next two methods available.

Completing the square HL

An examination question rarely presents a denominator already in the form $a^2 + x^2$. A quadratic such as $x^2 + 2x + 5$ has that shape, but only after you complete the square.

EXAMPLE 16.7

Find $\displaystyle\int \frac{1}{x^2+2x+5}\,dx$.

Complete the square: $x^2 + 2x + 5 = (x+1)^2 + 4$. This is the arctangent shape with $a = 2$ and the shifted variable $x + 1$:

$$\int \frac{dx}{(x+1)^2+4} = \frac{1}{2}\arctan\left(\frac{x+1}{2}\right) + C.$$

Any quadratic with no real roots can be written in the form $a^2 + u^2$ this way. The shift from x to $x+1$ does not change the integral, because the derivative of $x+1$ is 1.

Partial fractions HL

When the quadratic *does* factorise, use a different method. Split the fraction into partial fractions, as in Ch. 6. Each piece then has the form $\frac{1}{x+k}$, and integrates to a logarithm.

EXAMPLE 16.8

Find $\displaystyle\int \frac{1}{x^2+3x+2}\,dx$.

Factor and decompose: $\dfrac{1}{(x+1)(x+2)} = \dfrac{1}{x+1} - \dfrac{1}{x+2}$. Then

$$\int \left(\frac{1}{x+1} - \frac{1}{x+2}\right) dx = \ln|x+1| - \ln|x+2| + C = \ln\left|\frac{x+1}{x+2}\right| + C.$$

Factorisable denominator: partial fractions and logarithms. Irreducible denominator: complete the square and arctangent. Choosing between the two is the first decision, and factorising the denominator is what settles it.

YOUR TURN 16.4 Rewriting Before Integrating HL

8.

(a) Find $\displaystyle\int \frac{1}{x^2-9}\,dx$ using partial fractions. [4]

(b) Find $\displaystyle\int \frac{4x+5}{(x+1)(x+2)}\,dx$. [4]

(c) Explain how you decide, from the denominator alone, whether to use partial fractions or to complete the square. [2]

9. Rewrite each integrand before integrating.

(a) $\displaystyle\int \frac{x^2+3x}{x}\,dx$ [2]

(b) $\displaystyle\int \frac{2x+1}{x}\,dx$ [2]

(c) $\displaystyle\int \frac{x^3-1}{x^2}\,dx$ [3]

(d) $\displaystyle\int \frac{1}{x^2+6x+13}\,dx$ [4]

16.5 Integration by Substitution

Substitution reverses the chain rule. Look for a composite function whose inner function also appears, up to a constant multiple, as a factor outside. Put u equal to that inner function, and a hard integral in x becomes an easy one in u.

The method has four steps. Let u be the inner function. Replace dx using $du = \dfrac{du}{dx}\,dx$. Integrate in u. Then put x back.

EXAMPLE 16.9

Find $\displaystyle\int 2x(x^2+1)^3\,dx$.

Let $u = x^2+1$, so $\dfrac{du}{dx} = 2x$, giving $du = 2x\,dx$. The integral becomes

$$\int u^3\,du = \frac{u^4}{4} + C = \frac{(x^2+1)^4}{4} + C.$$

The factor $2x$ outside is exactly the $\frac{du}{dx}$ required. Without it the substitution would leave an x behind and fail.

EXAMPLE 16.10

Find $\int \frac{2x}{x^2+1}\,dx$.

With $u = x^2+1$, $du = 2x\,dx$:

$$\int \frac{1}{u}\,du = \ln|u| + C = \ln(x^2+1) + C.$$

An integral of the form $\frac{f'(x)}{f(x)}$ always integrates to $\ln|f(x)|$.

CAUTION For a definite integral by substitution, either convert the limits to u-values, or substitute back to x before applying the original limits. Applying the x-limits to a u-expression gives a wrong answer.

EXAMPLE 16.11

Evaluate $\int_0^{\sqrt{3}} x\sqrt{x^2+1}\,dx$.

Let $u = x^2+1$, so $du = 2x\,dx$. Convert the limits too: $x = 0 \Rightarrow u = 1$ and $x = \sqrt{3} \Rightarrow u = 4$. Then

$$\int_0^{\sqrt{3}} x\sqrt{x^2+1}\,dx = \frac{1}{2}\int_1^4 \sqrt{u}\,du = \frac{1}{2}\cdot\frac{2}{3}\left[u^{3/2}\right]_1^4 = \frac{1}{3}(8-1) = \frac{7}{3}.$$

Once the limits are converted, you never return to x at all; the whole calculation finishes in u.

YOUR TURN 16.5 Integration by Substitution

10. Find each integral by substitution.

(a) $\int 3x^2(x^3+1)^4\,dx$ [2]

(b) $\int xe^{x^2}\,dx$ [2]

(c) $\int 2x\sqrt{x^2+1}\,dx$ [3]

(d) $\int \frac{x}{(x^2+4)^2}\,dx$ [3]

(e) $\int \frac{\ln x}{x}\,dx$ [3]

(f) $\int \sin^3 x \cos x\,dx$ [3]

11.

(a) Evaluate $\int_0^1 3x^2(x^3+1)^4\,dx$ by converting the limits. [3]

(b) Evaluate the same integral by substituting back to x first, and confirm the answers agree. [3]

(c) Evaluate $\displaystyle\int_0^{\pi/2} \sin^2 x \cos x \, dx$ by converting the limits. [3]

16.6 Integration by Parts HL

Substitution reverses the chain rule; **integration by parts** reverses the product rule. It handles products that substitution cannot, such as xe^x or $\ln x$.

KEY · IB

$$\int u \frac{dv}{dx}\, dx = uv - \int v \frac{du}{dx}\, dx$$

You split the integrand into a part u to differentiate and a part $\dfrac{dv}{dx}$ to integrate. In practice it is easier to work with the same rule written in terms of the two functions you can see in front of you. Writing the integrand as a product uv, and letting $\displaystyle\int v\, dx$ stand for any antiderivative of v:

KEY

$$\int uv\, dx \;=\; u\int v\, dx \;-\; \int\left(\int v\, dx\right)\frac{du}{dx}\, dx$$

Read from left to right this says: integrate the second function, multiply by the first, then subtract the integral of that product with the first function differentiated. Nothing new has happened, but you never have to name v separately.

Everything now depends on which factor you call u. Choose the one that becomes simpler when differentiated, so that the second integral is easier than the one you started with. The order **ILATE** is a reliable guide: whichever type appears first in this list should be u.

Letter	Type of function	Example
I	Inverse trigonometric	$\arcsin x$, $\arctan x$
L	Logarithmic	$\ln x$
A	Algebraic	x, x^2, $3x-1$
T	Trigonometric	$\sin x$, $\cos x$
E	Exponential	e^x, 2^x

In $\int xe^x\,dx$ the factors are algebraic and exponential, so A comes before E and $u = x$. In $\int x\ln x\,dx$ they are logarithmic and algebraic, so L comes before A and $u = \ln x$. The order works because differentiating reduces the first three types to something simpler, while trigonometric and exponential functions repeat themselves however often you differentiate them.

CAUTION ILATE is a guide, not a theorem. It gives the right choice for the integrals you will meet in this course, but the test that always applies is the one behind it: does the second integral end up easier than the first?

EXAMPLE 16.12

Find $\displaystyle\int xe^x\,dx$.

Let $u = x$ (simplifies to 1) and $\dfrac{dv}{dx} = e^x$ (so $v = e^x$):

$$\int xe^x\,dx = xe^x - \int e^x\,dx = xe^x - e^x + C = e^x(x-1) + C.$$

Choosing $u = x$ is what makes the new integral, $\int e^x\,dx$, trivial. The other choice would make it worse.

EXAMPLE 16.13

Find $\displaystyle\int \ln x\,dx$.

There is only one function, but write $u = \ln x$ and $\dfrac{dv}{dx} = 1$ (so $v = x$):

$$\int \ln x\,dx = x\ln x - \int x\cdot\frac{1}{x}\,dx = x\ln x - x + C.$$

Taking $\dfrac{dv}{dx} = 1$ is worth remembering. It is how both $\ln x$ and $\arctan x$ are integrated.

Repeated integration by parts

If one application of parts leaves an integral that still needs parts, apply it again. Each round of differentiation reduces the power of x by one.

EXAMPLE 16.14

Find $\displaystyle\int x^2 e^x\,dx$.

First round, $u = x^2$:

$$\int x^2 e^x\,dx = x^2 e^x - 2\int x e^x\,dx.$$

The remaining integral is the earlier example, $\int x e^x\,dx = e^x(x-1)$:

$$\int x^2 e^x\,dx = x^2 e^x - 2e^x(x-1) + C = e^x(x^2 - 2x + 2) + C.$$

A power x^n takes n rounds of parts to clear, since only the power changes from round to round; the exponential returns unchanged each time.

Something cleverer happens when *neither* factor ever simplifies. For $e^x \sin x$, two rounds of parts bring the original integral back, and you solve for it like an equation.

EXAMPLE 16.15

Find $I = \displaystyle\int e^x \sin x\,dx$.

Parts with $u = \sin x$:

$$I = e^x \sin x - \int e^x \cos x\,dx.$$

Parts again on the new integral, $u = \cos x$:

$$\int e^x \cos x\,dx = e^x \cos x + \int e^x \sin x\,dx = e^x \cos x + I.$$

Substitute back:

$$I = e^x \sin x - e^x \cos x - I \implies 2I = e^x(\sin x - \cos x) \implies I = \tfrac{1}{2}e^x(\sin x - \cos x) + C.$$

The original integral reappeared on the right-hand side, which turned the problem into an equation for I. Watch for this whenever neither factor gets simpler: two rounds of parts bring I back, and algebra finishes the job.

Choosing a method

Five methods are now available, and most marks lost in this topic are lost by starting the wrong one. The choice is settled by looking at the integrand before writing anything.

What you see	Method	Example
A standard form exactly, or with a linear inside	read it off, dividing by the inner coefficient	$\cos 3x$, e^{2x}, $(2x+1)^5$
A function together with a multiple of its own derivative	substitution	$2x(x^2+1)^3$, $\dfrac{2x}{x^2+1}$
A product of two unlike functions, one of which simplifies when differentiated	parts	xe^x, $x\sin x$, $\ln x$
A fraction whose denominator factorises	partial fractions, then logarithms	$\dfrac{1}{x^2+3x+2}$
A fraction whose denominator is a quadratic with no real roots	complete the square, then arctangent	$\dfrac{1}{x^2+2x+5}$

Work down the list in that order, because the earlier tests are quicker to apply and the earlier methods are shorter. Two habits help further. Check first whether the fraction is improper, since dividing may remove the problem entirely. And if a substitution leaves an integral no easier than the one you started with, the substitution was the wrong one; go back rather than pressing on.

YOUR TURN 16.6 Integration by Parts HL

12. Find each integral by parts.

(a) $\displaystyle\int x\cos x\,dx$ [3]

(b) $\displaystyle\int xe^{2x}\,dx$ [3]

(c) $\displaystyle\int x\ln x\,dx$ [3]

(d) $\displaystyle\int \arctan x\,dx$ [4]

(e) $\displaystyle\int x^2\ln x\,dx$ [4]

(f) $\displaystyle\int x\sec^2 x\,dx$ [3]

13.

(a) Evaluate $\int_0^{\pi} x \sin x \, dx$. [3]

(b) Evaluate $\int_0^{1} x e^x \, dx$. [3]

(c) State which factor you chose as u in part (b), and why the other choice would not help. [2]

(d) Use ILATE to say which factor should be u in each of $\int x^2 e^x \, dx$, $\int x \arctan x \, dx$ and $\int e^x \sin x \, dx$. [3]

(e) Explain why the third integral in part (d) behaves differently from the first two. [2]

16.7 Areas Between Curves

To find the area between two curves, integrate the difference: the upper curve minus the lower. The signed-area problem disappears, because the difference is positive wherever the first curve is on top, regardless of the axis.

KEY

$$A = \int_a^b (y_{\text{top}} - y_{\text{bottom}}) \, dx$$

The limits a and b are usually the x-coordinates where the curves intersect, found by setting them equal.

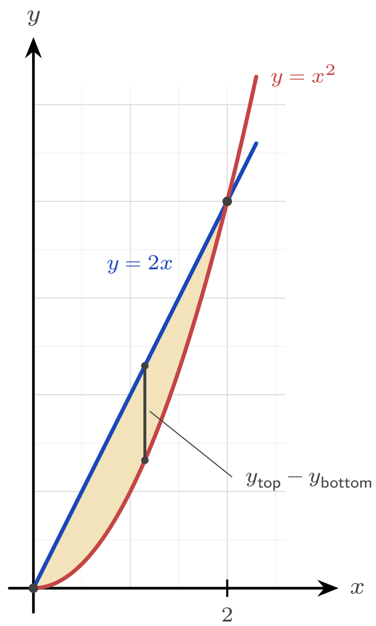


Figure 16.4 The region enclosed between $y = 2x$ and $y = x^2$. The curves meet where $2x = x^2$, at $x = 0$

and $x = 2$, and the line is the upper boundary in between, so the area is $\int_0^2 (2x - x^2)\,dx$.

EXAMPLE 16.16

Find the area enclosed between $y = 2x$ and $y = x^2$.

Intersections: $2x = x^2 \Rightarrow x = 0$ or $x = 2$. On $(0, 2)$ the line $y = 2x$ is above, so

$$A = \int_0^2 (2x - x^2)\,dx = \left[x^2 - \tfrac{x^3}{3}\right]_0^2 = 4 - \tfrac{8}{3} = \tfrac{4}{3}.$$

Area between a curve and the y-axis HL

Sometimes it is easier to integrate with respect to y: express x as a function of y and accumulate horizontal slices. This gives the area between the curve and the y-axis.

KEY · IB

$$A = \int_a^b x\,dy$$

EXAMPLE 16.17

Find the area between the curve $x = y^2$ and the y-axis from $y = 0$ to $y = 2$.

$$A = \int_0^2 y^2\,dy = \left[\tfrac{y^3}{3}\right]_0^2 = \tfrac{8}{3}.$$

Integrating in y avoids splitting the region, which would be awkward in x.

YOUR TURN 16.7 Areas Between Curves

14. Find each enclosed area.

(a) Between $y = x$ and $y = x^2$. [3]
(b) Between $y = 4 - x^2$ and the x-axis. [3]
(c) Between $y = 6 - x^2$ and $y = 2$. [3]
(d) Between $x = y^2$ and the y-axis, from $y = 0$ to $y = 3$. [2]
(e) Between $y = x^2$ and $y = 2 - x^2$. [3]

15. Consider the region enclosed by $y = x^2$ and $y = x + 2$.

(a) Find the x-coordinates where the two curves meet. [2]
(b) State which curve is the upper boundary between those values. [1]
(c) Find the area of the enclosed region. [3]

(d) State what answer you would get by integrating $x^2 - (x+2)$ instead, and explain what has gone wrong. [2]

16.8 Volumes of Revolution HL

Rotate a curve about an axis and it sweeps out a solid. To find the volume, slice the solid into thin disks perpendicular to the axis. Each disk is a cylinder of radius y (the curve's height) and thickness dx, with volume $\pi y^2\,dx$. Summing the disks and letting their thickness shrink turns the sum into an integral.

KEY · IB

$$V = \int_a^b \pi y^2\,dx \quad \text{(about the } x\text{-axis)}, \qquad V = \int_a^b \pi x^2\,dy \quad \text{(about the } y\text{-axis)}$$

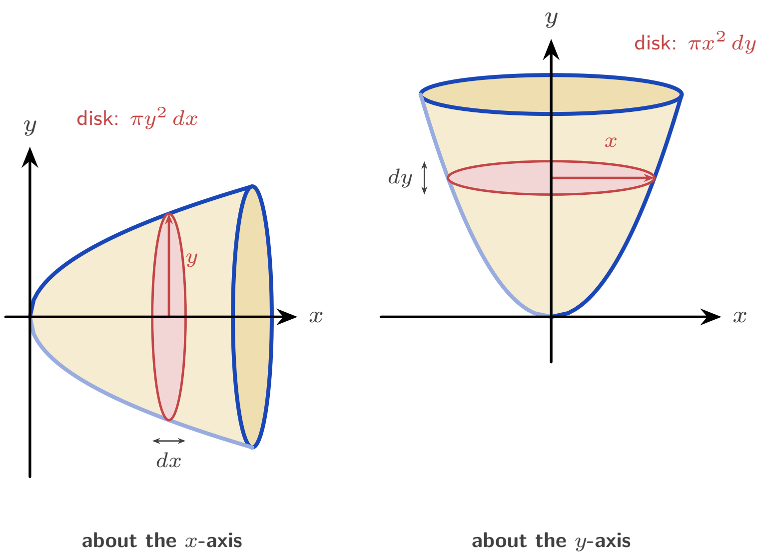


Figure 16.5 The same curve rotated about each axis. Turning it about the x-axis gives disks stacked along x: each has radius y and thickness dx, so its volume is $\pi y^2\,dx$. Turning it about the y-axis gives disks stacked along y: each has radius x and thickness dy, so its volume is $\pi x^2\,dy$. In both cases the integral adds the disks from one end of the solid to the other.

EXAMPLE 16.18

The curve $y = x^3$ for $0 \le x \le 2$ is rotated about the x-axis. Find the volume.

$$V = \int_0^2 \pi(x^3)^2\,dx = \pi\int_0^2 x^6\,dx = \pi\left[\tfrac{x^7}{7}\right]_0^2 = \frac{128\pi}{7}.$$

Square the function *before* integrating; squaring after is wrong.

EXAMPLE 16.19

The region under $y = x^2$ from $x = 0$ to $x = 1$ is rotated about the y-axis. Find the volume.
About the y-axis, slice into disks of radius x and thickness dy, with $x^2 = y$:

$$V = \int_0^1 \pi x^2 \, dy = \pi \int_0^1 y \, dy = \pi \left[\tfrac{y^2}{2} \right]_0^1 = \frac{\pi}{2}.$$

The limits are now y-values, and x^2 is rewritten in terms of y.

YOUR TURN 16.8 Volumes of Revolution HL

16. Find each volume of revolution about the x-axis.

(a) $y = x^2$, $0 \le x \le 2$. [3]
(b) $y = \sqrt{x}$, $0 \le x \le 4$. [2]
(c) $y = 2x$, $0 \le x \le 3$. [2]
(d) $y = \dfrac{1}{x}$, $1 \le x \le 3$. [3]
(e) $y = \sin x$, $0 \le x \le \pi$. [4]

17.

(a) The region under $y = x^2$, $0 \le x \le 2$, is rotated about the y-axis. Find the volume. [3]
(b) Explain why the answer differs from the earlier volume about the x-axis, even though the same curve and the same interval are used. [2]
(c) The region between $x = y^2$ and the y-axis, from $y = 0$ to $y = 2$, is rotated about the y-axis. Find the volume. [4]

16.9 Kinematics by Integration

In Ch. 15 you differentiated displacement to get velocity, and velocity to get acceleration. Integration reverses those two steps. Integrate acceleration to get velocity, and velocity to get displacement, using an initial condition each time to fix the constant.

KEY · IB

$$\text{displacement} = \int_{t_1}^{t_2} v\,dt, \qquad \text{distance travelled} = \int_{t_1}^{t_2} |v|\,dt$$

Displacement is the signed change in position; distance is the total path length. They differ whenever the particle reverses direction, because backward motion subtracts from displacement but adds to distance. On a velocity–time graph the difference is visible: area below the axis counts as negative for displacement, and as positive for distance.

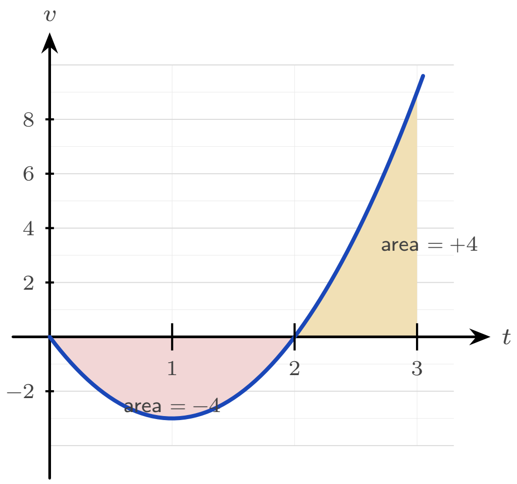


Figure 16.6 The velocity $v = 3t^2 - 6t$ over $0 \le t \le 3$. The particle moves backwards until $t = 2$ and forwards after it. Displacement adds the two signed areas, $-4 + 4 = 0$; distance adds their sizes, $4 + 4 = 8$ m.

EXAMPLE 16.20

A particle has velocity $v(t) = 3t^2 - 6t\,\mathrm{m\,s^{-1}}$. Find its displacement and the distance travelled over $0 \le t \le 3$.

Displacement integrates v directly:

$$\int_0^3 (3t^2 - 6t)\,dt = \left[t^3 - 3t^2\right]_0^3 = 27 - 27 = 0.$$

The particle ends where it began. For distance, note $v = 3t(t-2)$ is negative on $(0,2)$ and positive on $(2,3)$, so split:

$$\left|\int_0^2 v\,dt\right| + \int_2^3 v\,dt = |-4| + 4 = 8 \text{ m}.$$

Zero displacement but 8 m travelled: the particle went out 4 m and came back.

YOUR TURN 16.9 Kinematics by Integration

18. A particle moves in a straight line.

(a) Its velocity is $v = 2t + 1$. Find its displacement over $0 \leq t \leq 3$. [2]

(b) Its acceleration is $a = 6t$ and $v(0) = 1$. Find $v(t)$. [2]

(c) Its velocity is $v = t^2 - 4$. Find its displacement over $0 \leq t \leq 3$. [2]

(d) Its velocity is $v = 4 - 2t$. Find the distance travelled over $0 \leq t \leq 4$. [3]

(e) For the particle in part (d), find the displacement over the same interval, and explain why the two answers differ. [3]

(f) Its velocity is $v = 3t^2 - 12$. Find the time at which it changes direction, and the distance travelled over $0 \leq t \leq 3$. [4]

Chapter 16 · Integral Calculus

Reflections

IM Long before calculus, Archimedes (3rd century BCE, Syracuse) found the area of a parabolic segment by his "method of exhaustion": filling the region with ever-smaller triangles and summing them. He effectively computed an integral two thousand years before Newton and Leibniz gave the operation a name and a reverse-differentiation shortcut. The idea of accumulation by slicing is older than the algebra we use to carry it out, and his triangles are the ancestors of the rectangles in §16.2.

TOK The proof in §16.2 establishes *that* the area under a curve and the reverse of a derivative are the same calculation. It does not explain *why* the world should be arranged that way. Two ideas invented for different purposes, one geometric and one algebraic, turned out to be one idea.

Does that make the connection a discovery about quantities and their rates, or a consequence of how we chose to define both? Is mathematics invented or uncovered?

RW Integration is summation made continuous, so it appears wherever a total is built from a varying rate. The total charge from a varying current, the work done by a changing force, the area of an irregular plot of land, the probability accumulated under a distribution curve (Ch. 13): each is an integral. Whenever a quantity is the running total of a rate, integration computes it.

EXT Not every function has an antiderivative in terms of familiar functions: $\int e^{-x^2}\,dx$, central to statistics, cannot be written with powers, exponentials, and trigonometric functions. Yet the definite integral still exists as a perfectly real area, and is computed numerically. The reach of "find the area" is wider than the reach of "reverse the derivative."

End-of-Chapter Problems — Chapter 16: Integral Calculus

1. A curve has $\frac{dy}{dx} = 3x^2 - 4x + 1$ and passes through $(1, 2)$.

(a) Find the equation of the curve. [3]

(b) Find the value of y when $x = 2$. [2]

2. Consider $y = x^2 - 4$.

(a) Find the x-intercepts. [2]

(b) Find the area enclosed between the curve and the x-axis. [4]

3. The curves $y = x^2$ and $y = 8 - x^2$ intersect at two points.

(a) Find the coordinates of the intersection points. [3]

(b) Find the area enclosed between the curves. [4]

4.

(a) Using the substitution $u = x^2 + 4$, find $\int \frac{x}{x^2+4}\,dx$. [3]

(b) Hence evaluate $\int_0^2 \frac{x}{x^2+4}\,dx$, giving an exact answer. [3]

5.

(a) Find $\int xe^{2x}\,dx$ using integration by parts. [3]

(b) Evaluate $\int_0^1 xe^{2x}\,dx$, giving an exact answer. [2]

6. The region bounded by $y = \sqrt{x}$, the x-axis, and $x = 4$ is rotated 360° about the x-axis.

(a) Write down an integral for the volume. [2]

(b) Find the volume. [2]

7. A particle starts at the origin with velocity $5\,\mathrm{m\,s^{-1}}$ and has acceleration $a = 6t - 4\,\mathrm{m\,s^{-2}}$.

(a) Find the velocity $v(t)$. [2]

(b) Find the displacement $s(t)$. [2]

(c) Find the displacement when $t = 2$. [2]

8. Consider the region enclosed by $y = x^2$, the y-axis, and $y = 4$.

(a) Find the area of the region. [3]

(b) Find the volume when this region is rotated about the y-axis. [3]

9.

(a) Find $\displaystyle\int \frac{1}{x^2+4}\,dx$. [2]

(b) Evaluate $\displaystyle\int_0^2 \frac{1}{x^2+4}\,dx$, giving an exact answer. [3]

10. The velocity of a particle is $v(t) = 3t^2 - 12t + 9\,\mathrm{m\,s^{-1}}$, $t \geq 0$.

(a) Find the times when the particle is at rest. [2]

(b) Find the total distance travelled in the first 3 seconds. [5]

11. A curve has gradient function $\dfrac{dy}{dx} = 4x - 3$ and passes through the point $(2, 5)$.

(a) Find the equation of the curve. [3]

(b) Write down its y-intercept. [1]

12. Find the following integrals.

(a) $\displaystyle\int e^{3x}\,dx$ [2]

(b) $\displaystyle\int (4x-1)^6\,dx$ [2]

(c) $\displaystyle\int \sin 2x\,dx$ [2]

13. Evaluate $\displaystyle\int_1^4 \left(\frac{2}{\sqrt{x}} + 1\right) dx$. [4]

14. Consider the curve $y = x^2 - 1$ on the interval $0 \leq x \leq 2$.

(a) Find where the curve crosses the x-axis in this interval. [1]

(b) Find the total area enclosed between the curve and the x-axis on the interval. [5]

15. Find the area of the region enclosed by the curve $y = 4x - x^2$ and the line $y = x$. [5]

16. Using the substitution $u = x^2 + 1$, evaluate $\displaystyle\int_0^2 x(x^2+1)^3\,dx$. [5]

17. Evaluate $\displaystyle\int_0^2 \frac{x}{x^2+4}\,dx$, giving your answer in the form $a \ln b$. [5]

18. Find $\displaystyle\int \frac{x+7}{(x-1)(x+3)}\,dx$. [6]

19. Find $\displaystyle\int \frac{1}{x^2-4x+13}\,dx$. [5]

20. Find the following integrals.

(a) $\displaystyle\int x\cos x\,dx$ [3]

(b) $\displaystyle\int x^2 \ln x\,dx$ [4]

21. The region under $y=\sqrt{x}$ from $x=0$ to $x=4$ is rotated about the x-axis.

(a) Find the volume generated. [3]

(b) The same curve, written $x=y^2$ for $0\le y\le 2$, is rotated about the y-axis. Find the volume generated. [4]

22. A particle moves with velocity $v(t)=t^2-4t+3\,\mathrm{m\,s^{-1}}$ for $0\le t\le 4$.

(a) Find the displacement of the particle over the interval. [3]

(b) Find the total distance travelled. [5]

23. The area under $y=x^2+1$ from $x=0$ to $x=3$ is to be estimated by three rectangles of width 1.

(a) Find the estimate obtained by taking the height of each rectangle at the left of its strip. [3]

(b) Find the estimate obtained by taking the height at the right. [2]

(c) Evaluate $\displaystyle\int_0^3 (x^2+1)\,dx$ and verify that it lies between the two estimates. [3]

(d) State, with a reason, whether the left estimate would increase or decrease if six rectangles were used instead. [2]

24. The curve $y=x^2$ and the line $y=2-x$ enclose a region R.

(a) Find the x-coordinates of the two points of intersection. [3]

(b) Sketch the curve and the line, and shade R. [2]

(c) Show that the area of R is $\dfrac{9}{2}$. [5]

(d) Explain why integrating $y_{\text{curve}}-y_{\text{line}}$ instead would give the wrong sign. [2]

25. This question uses integration by parts throughout.

(a) Find $\displaystyle\int x\ln x\,dx$. [4]

(b) Hence evaluate $\displaystyle\int_1^e x \ln x \, dx$, giving your answer in terms of e. [3]

(c) Evaluate $\displaystyle\int_0^{\pi/2} x \cos x \, dx$. [5]

(d) In part (c), state which factor you took as u, and explain what would go wrong with the other choice. [2]

26. Consider the curve $y = x^3$ for $0 \le x \le 1$.

(a) The region between the curve and the x-axis is rotated 360° about the x-axis. Show that the volume is $\dfrac{\pi}{7}$. [4]

(b) The region between the curve and the y-axis is rotated 360° about the y-axis. Find this volume, giving an exact answer. [5]

(c) State which of the two solids has the greater volume, and explain the answer by describing the two regions. [3]

27. A particle moves in a straight line with acceleration $a = 6 - 2t \text{ m s}^{-2}$, and is at rest at $t = 0$.

(a) Show that $v = 6t - t^2$. [3]

(b) Find the time at which the particle is next at rest. [2]

(c) Find the maximum velocity, and the time at which it occurs. [3]

(d) Find the distance travelled from $t = 0$ until the particle is next at rest. [4]

(e) Explain why, for this motion, the distance travelled equals the displacement. [2]

28. Each of these integrals needs a different method. In each part, name the method before using it.

(a) $\displaystyle\int \frac{x^2+1}{x+1} \, dx$ [5]

(b) $\displaystyle\int_1^2 \frac{2x-1}{x^2-x+1} \, dx$ [4]

(c) $\displaystyle\int_0^{\pi/2} \sin x \cos^2 x \, dx$ [4]

(d) $\displaystyle\int \frac{1}{x^2-4x+13} \, dx$ [4]

Challenge Questions

29. **Solids you already know.** Every volume formula from Ch. 8 can be recovered by rotating a curve. This question derives three of them.

(a) The line $y = r$ for $0 \le x \le h$ is rotated about the x-axis. Show that the volume is $\pi r^2 h$, and name the solid. [3]

(b) The line $y = \dfrac{r}{h}x$ for $0 \le x \le h$ is rotated about the x-axis. Show that the volume is $\frac{1}{3}\pi r^2 h$. [4]

(c) The semicircle $y = \sqrt{r^2 - x^2}$ for $-r \le x \le r$ is rotated about the x-axis. Show that the volume is $\frac{4}{3}\pi r^3$. [5]

(d) The cone in part (b) has one third of the volume of the cylinder in part (a). Explain where the factor $\frac{1}{3}$ comes from in the integration. [3]

30. When parts goes in a circle. Some integrals return to where they started, and that is what solves them.

(a) Let $I = \displaystyle\int e^x \cos x \, dx$. Apply integration by parts twice, and show that $I = \frac{1}{2}e^x(\sin x + \cos x) + C$. [6]

(b) Explain why choosing $u = e^x$ at the first step and $u = \cos x$ at the second would not have worked. [3]

(c) Find $\displaystyle\int x^2 e^x \, dx$ by applying parts twice, and explain why this one terminates while part (a) does not. [5]

(d) Predict, without calculating, how many rounds of parts $\displaystyle\int x^4 e^x \, dx$ requires. Justify your answer. [2]

31. A reduction formula. Let $I_n = \displaystyle\int_0^{\pi/2} \sin^n x \, dx$ for $n \ge 0$.

(a) Evaluate I_0 and I_1 directly. [3]

(b) Writing $\sin^n x = \sin^{n-1} x \cdot \sin x$ and using integration by parts, show that

$$I_n = \frac{n-1}{n} I_{n-2}, \qquad n \ge 2.$$

[6]

(c) Use the formula to find I_2, I_3, I_4 and I_5 exactly. [4]

(d) Explain why I_n involves π when n is even but not when n is odd. [3]

CALCULUS

Chapter 17

Differential Equations

17

SYLLABUS 5.18 5.19 HL only

A cup of coffee cools fastest when it is hottest. Pull it from the machine at 80 degrees and it drops quickly at first; near room temperature it barely changes. The *rate* at which it cools depends on how far above the room its temperature currently is.

That sentence is a differential equation. Written in symbols, the rate of cooling $\frac{d\theta}{dt}$ is proportional to the gap $\theta - 20$ between the coffee and the 20-degree room:

$$\frac{d\theta}{dt} = -k(\theta - 20).$$

Notice what kind of equation this is. The unknown is not a number but a *function*, $\theta(t)$, and the equation links the function to its own derivative. Most laws of nature take exactly this form, because what we can observe directly is usually a rate: how fast something grows, cools, decays, or moves.

A **differential equation** relates a function to its derivatives. To **solve** one is to find the function itself, recovering $\theta(t)$ from a statement about $\frac{d\theta}{dt}$.

That is worth doing, because it turns a rule about change into a prediction. A law of nature tells you what is happening at this instant; solving its differential equation tells you the whole future.

By the end of this chapter you will be able to solve such an equation exactly by three methods: separating the variables, substituting to make them separate, and using an integrating factor. You will solve one approximately by Euler's method when no exact solution exists. And you will write a function as an infinite polynomial, a Maclaurin series, which approximates the function and gives a fifth route to a solution. The coffee equation is solved in the first section.

17.1 Separable Variables

Differential equations are classified by what they contain, and two words fix the ones in this chapter. The **order** is the highest derivative that appears, so every equation here, containing $\dfrac{dy}{dx}$ and no higher derivative, is **first order**. The equation is **linear** when y and its derivatives appear only to the first power and are never multiplied together; $\dfrac{dy}{dx} + xy = 1$ is linear, while $\dfrac{dy}{dx} = 2xy^2$ is not. The three exact methods below are chosen by which of these forms an equation takes.

The simplest case can be rearranged so that each variable sits on its own side. A first-order equation is **separable** if it can be written as a function of y times a function of x:

$$\frac{dy}{dx} = g(x)\,h(y).$$

Then you separate and integrate both sides. Treating $\dfrac{dy}{dx}$ as a ratio is informal but valid here: gather all the y terms with dy and all the x terms with dx.

KEY

$$\frac{dy}{dx} = g(x)\,h(y) \quad\Longrightarrow\quad \int \frac{1}{h(y)}\,dy = \int g(x)\,dx$$

One arbitrary constant appears, giving the **general solution**, a family of curves. An **initial condition** fixes the constant and selects the **particular solution**.

EXAMPLE 17.1

Solve $\dfrac{dy}{dx} = 2xy^2$ given $y(0) = 1$.

Separate the variables and integrate:

$$\int \frac{1}{y^2}\,dy = \int 2x\,dx \implies -\frac{1}{y} = x^2 + C.$$

Apply $y(0) = 1$: $-1 = 0 + C \implies C = -1$. So $-\dfrac{1}{y} = x^2 - 1$, giving

$$y = \frac{1}{1 - x^2}.$$

Always apply the initial condition as early as possible; carrying C further than necessary invites errors.

EXAMPLE 17.2

A cup of coffee obeys $\dfrac{d\theta}{dt} = -k(\theta - 20)$ with $\theta(0) = 80$. Solve for $\theta(t)$.
Separate and integrate:

$$\int \frac{1}{\theta - 20}\,d\theta = \int -k\,dt \implies \ln|\theta - 20| = -kt + C.$$

Exponentiating, $\theta - 20 = Ae^{-kt}$. The condition $\theta(0) = 80$ gives $A = 60$, so

$$\theta(t) = 20 + 60e^{-kt}.$$

The temperature decays exponentially toward the room's 20 degrees, never quite reaching it: exactly the behaviour we observed.

Separation also handles the most famous population model in ecology. In the 1960s, once the cattle disease rinderpest was eliminated around the Serengeti, the wildebeest herd began to grow, and its growth obeyed the **logistic equation**: growth proportional both to the herd itself and to the room left below the plains' carrying capacity.

EXAMPLE 17.3

The Serengeti wildebeest population n (in millions) satisfies $\dfrac{dn}{dt} = kn(1.4-n)$, with $n(0) = 0.25$ in 1963. Solve for $n(t)$.
Separate, and split the left side with the partial fractions of Ch. 6:

$$\int \frac{dn}{n(1.4-n)} = \int k\,dt, \qquad \frac{1}{n(1.4-n)} = \frac{1}{1.4}\left(\frac{1}{n} + \frac{1}{1.4-n}\right).$$

Integrating,

$$\frac{1}{1.4}\ln\left|\frac{n}{1.4-n}\right| = kt + C \implies \frac{n}{1.4-n} = Be^{1.4kt}.$$

The initial condition gives $B = \dfrac{0.25}{1.15} \approx 0.217$, and solving for n:

$$n(t) = \frac{1.4}{1 + 4.6\,e^{-1.4kt}}.$$

This is the logistic S-curve: growth that is nearly exponential at first, then levels off at the carrying capacity of 1.4 million. The bracket $(1.4 - n)$ is what slows it, since it shrinks towards zero as n approaches 1.4. The real herd behaved this way, climbing to about 1.3 million by 1977 and staying near that level since.

YOUR TURN 17.1 Separable Variables

1. Find the general solution of each equation.

(a) $\dfrac{dy}{dx} = \dfrac{x}{y}$ [2]

(b) $\dfrac{dy}{dx} = y$ [2]

(c) $\dfrac{dy}{dx} = \dfrac{1+y}{1+x}$ [3]

(d) $\dfrac{dy}{dx} = e^{x-y}$ [3]

2. Find the particular solution of each equation.

(a) $\dfrac{dy}{dx} = xy$ with $y(0) = 2$. [3]

(b) $\dfrac{dy}{dx} = 3x^2y$ with $y(0) = 1$. [3]

(c) $\dfrac{dy}{dx} = y \tan x$ with $y(0) = 4$, for $0 \le x < \frac{\pi}{2}$. [4]

(d) Explain why a general solution contains one arbitrary constant, and what the initial condition does to it. [2]

17.2 Homogeneous Differential Equations

Some equations are not separable as they stand, but become so after a substitution. A first-order equation is **homogeneous** if the right-hand side depends only on the ratio $\dfrac{y}{x}$. Such equations arise when a problem has no fixed scale, only directions. The standard example is a pursuit curve: a ferry that always aims at the dock while a current pushes it sideways. What matters at each moment is the direction from the ferry to the dock, and that direction is set by the ratio $\frac{y}{x}$ alone. You will solve the ferry in the challenge problems. The substitution $y = vx$ then separates the equation.

Since $y = vx$, the product rule gives $\dfrac{dy}{dx} = v + x\dfrac{dv}{dx}$. Substituting turns the equation into a separable one in v and x.

KEY For $\dfrac{dy}{dx} = f\left(\dfrac{y}{x}\right)$, substitute $y = vx$, so $\dfrac{dy}{dx} = v + x\dfrac{dv}{dx}$.

EXAMPLE 17.4

Solve $\dfrac{dy}{dx} = \dfrac{x+y}{x}$.

Rewrite the right side in terms of $\dfrac{y}{x}$: $\dfrac{x+y}{x} = 1 + \dfrac{y}{x}$. Substitute $y = vx$, $\dfrac{dy}{dx} = v + x\dfrac{dv}{dx}$:

$$v + x\frac{dv}{dx} = 1 + v \implies x\frac{dv}{dx} = 1.$$

This is separable:

$$\int dv = \int \frac{1}{x}\,dx \implies v = \ln|x| + C.$$

Finally replace $v = \dfrac{y}{x}$:

$$y = x(\ln|x| + C).$$

The single v on each side always cancels, and what is left is separable. That cancellation is what makes the substitution work.

YOUR TURN 17.2 Homogeneous Differential Equations

3. Use the substitution $y = vx$ to solve each equation.

(a) $\dfrac{dy}{dx} = \dfrac{2x+y}{x}$ [3]

(b) $\dfrac{dy}{dx} = \dfrac{y-x}{x}$ [3]

(c) $\dfrac{dy}{dx} = \dfrac{x+2y}{x}$ [4]

(d) $\dfrac{dy}{dx} = \dfrac{y}{x}$ for $x > 0$. [2]

4.

(a) Show that $\dfrac{dy}{dx} = \dfrac{x^2+y^2}{xy}$ is homogeneous, by writing the right-hand side in terms of $\dfrac{y}{x}$. [3]

(b) Hence find its general solution. [5]

17.3 The Integrating Factor

A **linear** first-order differential equation has the form

$$\frac{dy}{dx} + P(x)\,y = Q(x).$$

It is usually not separable. The method is to multiply through by a carefully chosen function, the **integrating factor**, which turns the left side into the derivative of a single product.

KEY · IB For $\dfrac{dy}{dx} + P(x)y = Q(x)$, the integrating factor is

$$I = e^{\int P(x)\,dx}.$$

After multiplying by I, the left side becomes exactly $\dfrac{d}{dx}(Iy)$, by the product rule run backwards. You then integrate both sides directly.

EXAMPLE 17.5

Solve $\dfrac{dy}{dx} + \dfrac{1}{x}y = x$ for $x > 0$.

Here $P(x) = \dfrac{1}{x}$, so the integrating factor is

$$I = e^{\int \frac{1}{x}\,dx} = e^{\ln x} = x.$$

Multiply through by x:

$$x\frac{dy}{dx} + y = x^2 \implies \frac{d}{dx}(xy) = x^2.$$

The left side has collapsed to a single derivative. Integrate both sides:

$$xy = \frac{x^3}{3} + C \implies y = \frac{x^2}{3} + \frac{C}{x}.$$

The whole method exists to create that $\dfrac{d}{dx}(xy)$ on the left; everything after is ordinary integration.

EXAMPLE 17.6

Solve $\dfrac{dy}{dx} + y = e^x$.

$P(x) = 1$, so $I = e^{\int 1\,dx} = e^x$. Multiply through:

$$\frac{d}{dx}(e^x y) = e^x \cdot e^x = e^{2x}.$$

Integrate: $e^x y = \frac{1}{2}e^{2x} + C$, hence $y = \frac{1}{2}e^x + Ce^{-x}$.

Where do such equations come from? The classic source is a **mixing problem**: a tank where liquid flows in and out, and the balance of inflow and outflow writes itself as a linear equation. When the volume itself changes, the equation is linear but not separable, so the integrating factor is the method to reach for.

EXAMPLE 17.7

A 100-litre vat in a brewery holds 5 kg of dissolved sugar. Syrup carrying 0.2 kg of sugar per litre flows in at 1 litre per minute, the vat is kept well stirred, and the mixture drains at 2 litres per minute. Find the sugar $S(t)$ in the vat at time t minutes.

The volume shrinks: $V(t) = 100 - t$. Sugar arrives at $0.2 \times 1 = 0.2$ kg min^{-1} and leaves at concentration times outflow, $\dfrac{S}{100-t} \times 2$. So

$$\frac{dS}{dt} = 0.2 - \frac{2S}{100-t} \implies \frac{dS}{dt} + \frac{2}{100-t}S = 0.2.$$

The integrating factor is

$$I = e^{\int \frac{2}{100-t}\,dt} = e^{-2\ln(100-t)} = (100-t)^{-2}.$$

Multiplying through collapses the left side:

$$\frac{d}{dt}\Big(S\,(100-t)^{-2}\Big) = 0.2\,(100-t)^{-2} \implies S\,(100-t)^{-2} = \frac{0.2}{100-t} + C.$$

So $S = 0.2(100-t) + C(100-t)^2$, and $S(0) = 5$ gives $C = -0.0015$:

$$S(t) = 0.2(100-t) - 0.0015(100-t)^2.$$

Sense-check the endpoints: $S(0) = 5$, and $S(100) = 0$, an empty vat holds no sugar. In between, the sugar first *rises*, to about 6.7 kg near $t = 33$, before the draining takes over. That rise is not obvious from the description, and it is the equation rather than intuition that reveals it.

Choosing a method

Three exact methods are now available, and a question rarely says which to use. The form of the equation decides it, and the test takes one line.

What the equation looks like	Method	Example
The variables can be split, $\dfrac{dy}{dx} = g(x)h(y)$	separate and integrate	$\dfrac{dy}{dx} = 2xy^2$
The right side depends only on $\dfrac{y}{x}$	substitute $y = vx$	$\dfrac{dy}{dx} = \dfrac{x+y}{x}$
Linear in y: $\dfrac{dy}{dx} + P(x)y = Q(x)$	integrating factor $e^{\int P\,dx}$	$\dfrac{dy}{dx} + \dfrac{1}{x}y = x$
None of the above	Euler's method, numerically	$\dfrac{dv}{dt} = 9.8 - 0.1v^2$

Try them in that order, because each test is quicker than the one after it. Note that the first two overlap: an equation can be both separable and homogeneous, in which case separating directly is the shorter route. Note also that a linear equation is separable only when $Q(x)$ is zero, which is why the integrating factor exists at all.

YOUR TURN 17.3 The Integrating Factor

5. Solve each linear equation.

(a) $\dfrac{dy}{dx} + y = 4$ [3]

(b) $\dfrac{dy}{dx} + 2y = 6$ [3]

(c) $\dfrac{dy}{dx} + 2y = e^x$ [3]

(d) $\dfrac{dy}{dx} - y = e^{2x}$ [3]

6.

(a) Write down the integrating factor for $\dfrac{dy}{dx} + \dfrac{3}{x}y = x^2$, for $x > 0$. [2]

(b) Solve $\dfrac{dy}{dx} + \dfrac{2}{x}y = x$ for $x > 0$. [3]

(c) Solve $\dfrac{dy}{dx} + \dfrac{1}{x}y = x$ with $y(1) = 1$, for $x > 0$. [4]

(d) Explain what the integrating factor is chosen to achieve on the left-hand side. [2]

17.4 Euler's Method

Not every differential equation can be solved exactly, and most of the equations that describe the real world cannot. When no exact solution exists, the solution can still be approximated numerically. **Euler's method** does this by walking along the curve in short straight steps, taking the gradient of each step from the differential equation itself.

The idea is the tangent line of Ch. 14. Near a known point the curve is close to its tangent, so moving a small distance h along the tangent lands close to the curve:

$$y \approx y_0 + \underbrace{f(x_0, y_0)}_{\text{gradient}} \times \underbrace{h}_{\text{step}} .$$

The differential equation supplies the gradient at whatever point you have reached, so the same move can be repeated from the new point, and again from the one after that.

KEY · IB

$$y_{n+1} = y_n + h\, f(x_n, y_n), \qquad x_{n+1} = x_n + h$$

Here h is the fixed **step length**. Each step is exact only at its starting point and drifts from the curve thereafter, so shorter steps stay closer, at the cost of needing more of them.

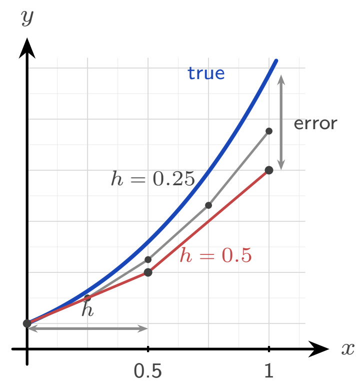


Figure 17.1 Euler's method on $\frac{dy}{dx} = x + y$ with $y(0) = 1$. Each straight step leaves the curve, which bends upward away from it, so the estimate falls short and the gap at $x = 1$ is the error. Halving the step length roughly halves that gap.

The figure shows two things at once. Each step is a straight line, so it leaves a curve that bends away from it, and the gap at the far end is the *error*. The gray path takes steps half as long and stays about twice as close to the curve. Shorter steps give more accuracy, at the cost of more of them.

EXAMPLE 17.8

Given $\dfrac{dy}{dx} = x + y$ with $y(0) = 1$, use Euler's method with $h = 0.1$ to estimate $y(0.3)$.

Apply $y_{n+1} = y_n + h(x_n + y_n)$ three times. A table is the clearest way to organise the work, and it is what an examiner expects to see.

n	x_n	y_n	$f = x_n + y_n$	$h\,f$
0	0	1	1	0.1
1	0.1	1.1	1.2	0.12
2	0.2	1.22	1.42	0.142
3	0.3	1.362		

Each y_n is the previous one plus the $h\,f$ on the line above. So $y(0.3) \approx 1.362$, against an exact value of about 1.3997.

Euler has undershot, and the reason is visible in the figure: this curve is concave up, so every tangent lies below it and each step falls short. That is the general rule. A concave-up solution is under-estimated and a concave-down one is over-estimated, so the sign of $\frac{d^2y}{dx^2}$ tells you which way your answer is wrong before you check it.

The method matters most when no exact solution is available. Add air resistance to a falling body and the equation becomes nonlinear, with no solution among the functions of this course, but Euler's method still produces a table of values. This is, in miniature, how weather forecasts are computed: equations too hard to solve exactly, stepped forward numerically, millions of times.

EXAMPLE 17.9

A skydiver falls from rest, and air resistance gives $\dfrac{dv}{dt} = 9.8 - 0.1v^2$ (in m s^{-1} and seconds). Use Euler's method with $h = 0.5$ to estimate the speed after 1.5 seconds, and compare with the terminal velocity.

Apply $v_{n+1} = v_n + 0.5(9.8 - 0.1v_n^2)$ three times from $v_0 = 0$:

$$\begin{aligned} v_1 &= 0 + 0.5(9.8) = 4.9 \\ v_2 &= 4.9 + 0.5(9.8 - 0.1(4.9)^2) = 4.9 + 0.5(7.399) = 8.60 \\ v_3 &= 8.60 + 0.5(9.8 - 0.1(8.60)^2) = 8.60 + 0.5(2.404) = 9.80 \end{aligned}$$

So $v(1.5) \approx 9.80$ m s^{-1}. The terminal velocity is where acceleration dies: $9.8 - 0.1v^2 = 0 \implies v = \sqrt{98} \approx 9.90$ m s^{-1}. After only three steps the estimate is already within 1% of the terminal speed. The physics is visible in the arithmetic: each bracket $9.8 - 0.1v_n^2$ is the net force per unit mass, and it shrinks towards zero as the speed builds.

YOUR TURN 17.4 Euler's Method

7. Use Euler's method for each of the following.

(a) $\dfrac{dy}{dx} = 2x$, $y(0) = 1$; use $h = 0.5$ to estimate $y(1)$. [3]

(b) $\dfrac{dy}{dx} = y$, $y(0) = 1$; use $h = 0.1$ to estimate $y(0.2)$. [3]

(c) $\dfrac{dy}{dx} = x + y$, $y(0) = 1$; use $h = 0.2$ to estimate $y(0.4)$. [3]

(d) $\dfrac{dy}{dx} = xy$, $y(1) = 1$; use $h = 0.1$ to estimate $y(1.2)$. [3]

8.

(a) For $\dfrac{dy}{dx} = xy$ with $y(1) = 2$, take one step with $h = 0.2$ to estimate $y(1.2)$. [2]

(b) For $\dfrac{dy}{dx} = x + y$ with $y(0) = 0$, take two steps with $h = 0.5$ to estimate $y(1)$. [3]

(c) Solve $\dfrac{dy}{dx} = y$ with $y(0) = 1$ exactly, and compare the exact $y(0.2)$ with your answer to question 7 part (b). [3]

(d) State whether Euler's method over- or under-estimates the solution of part (c), and explain why in terms of the shape of the curve. [3]

17.5 Maclaurin Series

Here is a different idea, and a powerful one: many functions can be written as an infinite polynomial.

It is worth knowing why anyone would want that. A polynomial is built from adding and multiplying, and those are the only two operations a machine can actually perform. Nothing in a calculator knows what $\sin x$ *is*. When you press the sine key it evaluates a polynomial, because a polynomial is something arithmetic can reach. Every function outside that small arithmetic world has to be turned into one before it can be computed, and a Maclaurin series is how that is done.

The idea is to match the function at a single point, as closely as possible. Near $x = 0$, look for a polynomial that shares the function's value, then its gradient, then its concavity, then every higher derivative at that point. The more derivatives agree, the longer the polynomial stays with the curve as you move away from 0. This is the **Maclaurin series**.

The coefficients are not chosen; they are forced. Suppose the series exists and write it as

$$f(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots$$

Putting $x = 0$ kills every term but the first, so $a_0 = f(0)$. Differentiating gives $f'(x) = a_1 + 2a_2 x + 3a_3 x^2 + \cdots$, and $x = 0$ now leaves $a_1 = f'(0)$. Differentiating again gives $f''(0) = 2a_2$, and once more $f'''(0) = 6a_3 = 3!\,a_3$. Each round of differentiating strips one term and multiplies the next by its own index, so in general $f^{(n)}(0) = n!\,a_n$.

KEY · IB

$$f(x) = f(0) + f'(0)\,x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \cdots$$

EXAMPLE 17.10

Find the Maclaurin series for $f(x) = e^x$.

Every derivative of e^x is e^x, and $e^0 = 1$, so $f^{(n)}(0) = 1$ for all n:

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

This series can be taken as the definition of e^x, and it computes the function to any accuracy you want. It also explains why e^x is its own derivative: differentiate the series term by term and you get the same series back.

The five standard series below are provided in the formula booklet. Each is exact, and truncating it gives a polynomial approximation that is excellent near $x = 0$.

KEY · IB

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots \qquad \sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots \qquad \ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots$$

$$\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots$$

The syllabus adds one more family: $(1+x)^p$ for any rational p, whose expansion $1 + px + \frac{p(p-1)}{2!}x^2 + \cdots$ is exactly the extended binomial theorem of Ch. 2 (booklet 1.10). With $p = \frac{1}{2}$, for instance, $\sqrt{1+x} \approx 1 + \frac{x}{2} - \frac{x^2}{8}$, which gives $\sqrt{1.1} \approx 1.04875$ without a calculator. The true value is 1.04881.

The figure below shows the first three odd-degree truncations of the sine series: the line $y = x$, the cubic, and the quintic. Each new term extends the interval over which the polynomial and the curve agree.

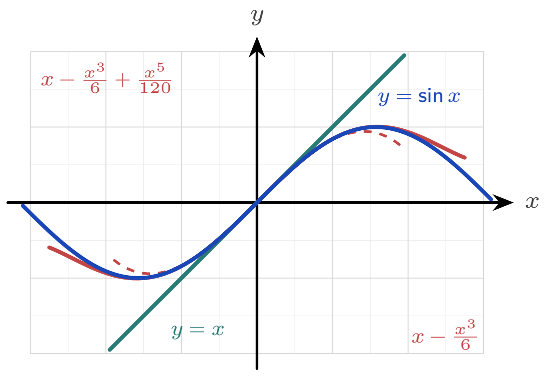


Figure 17.2 The first three odd-degree truncations of the sine series against $y = \sin x$. Each extra term widens the interval on which the polynomial and the curve agree, but every truncation eventually leaves the curve.

The same convergence can be watched numerically. Below are the successive truncations of the sine series at $x = 1$, against the true value $\sin 1 = 0.841471$.

terms up to	x	x^3	x^5	x^7
estimate	1	0.833333	0.841667	0.841468
error	0.159	0.0081	0.0002	0.0000027

Each new term cuts the error by a factor of tens, then hundreds, then thousands. That is the trade the series offers: a polynomial is simple enough to compute, and taking more terms buys as much accuracy as you are willing to pay for.

Building new series

New series are rarely built from the definition. They are made from the standard five by four operations, all of which the syllabus names.

- **Substitute.** Replacing x with $-x^2$ in the series for e^x gives

$$e^{-x^2} = 1 - x^2 + \frac{x^4}{2!} - \frac{x^6}{3!} + \cdots$$

- **Differentiate.** Differentiating the sine series term by term produces exactly the cosine series: the identity $\frac{d}{dx} \sin x = \cos x$ in polynomial form.
- **Integrate.** The geometric series of Ch. 2 gives $\dfrac{1}{1+x^2} = 1 - x^2 + x^4 - \cdots$ for $|x| < 1$, and integrating term by term yields $\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots$. The booklet's arctangent series is simply the geometric series integrated.
- **Multiply.** For a product, expand each factor and collect powers up to the degree you need.

EXAMPLE 17.11

Find the Maclaurin series of $e^x \sin x$ up to the term in x^3.

Multiply the two standard series, keeping only terms of degree ≤ 3:

$$\left(1 + x + \frac{x^2}{2} + \frac{x^3}{6}\right)\left(x - \frac{x^3}{6}\right) = x + x^2 + \left(\frac{1}{2} - \frac{1}{6}\right)x^3 + \cdots = x + x^2 + \frac{x^3}{3} + \cdots$$

The work is bookkeeping. Decide the target degree first, and discard anything above it before multiplying rather than after.

Small angles and series from differential equations

Truncating at the first term or two gives the **small-angle approximations** used throughout physics. For θ near 0, in radians,

$$\sin\theta \approx \theta, \qquad \cos\theta \approx 1 - \frac{\theta^2}{2}, \qquad \tan\theta \approx \theta.$$

These are not conveniences invented by physicists; they are the first one or two terms of the series, with the rest discarded. The pendulum formula depends on the first of them. For small swings the restoring force involves $\sin\theta$, and replacing $\sin\theta$ by θ is exactly what turns the equation of motion into simple harmonic motion, which can then be solved.

The series also prices the approximation. The first term discarded is $\frac{\theta^3}{6}$, so that is roughly the error, and the relative error stays under 1% for swings up to about 14°. Beyond that the pendulum's real period drifts away from the textbook formula, by an amount large enough to measure.

A Maclaurin series can also be built directly *from a differential equation*, by differentiating the equation itself to obtain the derivatives at 0. This joins the two halves of the chapter.

EXAMPLE 17.12

A function satisfies $\dfrac{dy}{dx} = 1 + y^2$ with $y(0) = 0$. Find the Maclaurin series of y up to the term in x^3.

Take the derivatives at 0 from the equation itself. First, $y'(0) = 1 + 0 = 1$. Differentiating the equation,

$$y'' = 2y\,y' \implies y''(0) = 0, \qquad y''' = 2(y')^2 + 2y\,y'' \implies y'''(0) = 2.$$

Feed these into the Maclaurin formula:

$$y = 0 + x + \frac{0}{2!}x^2 + \frac{2}{3!}x^3 + \cdots = x + \frac{x^3}{3} + \cdots$$

You may recognise the answer: $y = \tan x$ solves this equation, since $\frac{d}{dx}\tan x = \sec^2 x = 1 + \tan^2 x$, and these are indeed the first terms of its series. The equation was never solved. Each derivative at 0 was read off from the equation itself, one at a time, and that was enough to build the series.

TIP Maclaurin series suggest several good IA topics. Two worth exploring: Madhava's series $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \cdots$, which comes from setting $x = 1$ in the arctangent series, and why it converges so slowly that a better choice such as $\arctan\frac{1}{\sqrt{3}}$ reaches the same accuracy in a fraction of the terms; or how a calculator really evaluates $\sin x$, by reducing the angle to a small interval first and then using a short truncated series.

EXAMPLE 17.13

Use Maclaurin series to evaluate $\displaystyle\lim_{x\to 0} \frac{e^x - 1 - x}{x^2}$.

Expand the numerator with the series for e^x:

$$e^x - 1 - x = \left(1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \cdots\right) - 1 - x = \frac{x^2}{2} + \frac{x^3}{6} + \cdots$$

Divide by x^2:

$$\frac{e^x - 1 - x}{x^2} = \frac{1}{2} + \frac{x}{6} + \cdots \xrightarrow{x \to 0} \frac{1}{2}.$$

The series settles an indeterminate limit as neatly as L'Hôpital's rule, and it shows *why* the answer is $\frac{1}{2}$: that number is the leading coefficient.

YOUR TURN 17.5 Maclaurin Series

9. Find the first three nonzero terms of the Maclaurin series for each function.

(a) e^{2x} [2]
(b) $\cos x$ [2]
(c) $\ln(1 + 2x)$ [2]
(d) e^{-x^2} [3]

10. Find the first two nonzero terms of the Maclaurin series for each function.

(a) $\sin 2x$ [2]
(b) $\sin(x^2)$ [2]
(c) $\arctan 2x$ [2]
(d) $\sqrt{1 + x}$ [3]

11.

(a) Use the first three terms of the series for e^x to estimate $e^{0.1}$. [2]
(b) Use the first three terms of the cosine series to estimate $\cos 0.2$ to four decimal places. [3]
(c) Compare each estimate with the true value, and state which is the more accurate. Explain why. [3]
(d) Find the first three nonzero terms of the Maclaurin series for $e^x \cos x$. [4]

12. A function satisfies $\dfrac{dy}{dx} = x + y$ with $y(0) = 1$.

(a) Write down $y'(0)$. [1]
(b) By differentiating the equation, find $y''(0)$ and $y'''(0)$. [4]
(c) Hence find the Maclaurin series for y up to the term in x^3. [3]
(d) Use the series to estimate $y(0.3)$, and compare it with the Euler estimate 1.362 from the notes. [3]

Chapter 17 · Differential Equations

Reflections

RW Differential equations are the language in which physics is written. Newton's second law $F = ma$ is a differential equation for position; radioactive decay, population growth, electrical circuits, epidemics, and planetary orbits are all governed by them. To predict a system's future is, almost always, to solve a differential equation, exactly when you can and numerically (by methods like Euler's) when you cannot.

TOK Euler's method rests on a striking idea: knowing the rule for change at every point, plus a single starting state, determines the entire future of a system. This is mathematical determinism. Yet the same equations can be so sensitive to their starting values that long-term prediction becomes impossible in practice, the phenomenon of chaos. Determined but unpredictable: what does that tell us about the limits of knowledge?

IM Both halves of this chapter have a long international lineage.

The series came first from India. Mathematicians of the Kerala school, notably Madhava in the 14th century, found the expansions of $\sin x$, $\cos x$ and $\arctan x$ roughly three centuries before Newton, Leibniz and Maclaurin. The arctangent series, usually named after Gregory and Leibniz, was known in Kerala generations earlier.

Numerical methods have a similar story in the Islamic world. In the 15th century Jamshīd al-Kāshī, working at the observatory in Samarkand, needed $\sin 1^\circ$ to extraordinary precision for his astronomical tables. No formula gave it, so he wrote down an equation the value satisfied and solved it by repeated approximation, each round feeding its answer into the next, reaching sixteen correct decimal places. That is the same bargain Euler's method strikes: when no exact answer is available, iterate towards one and control the error. Two centuries earlier Sharaf al-Dīn al-Ṭūsī had studied when a cubic equation has solutions by examining where its maximum lies, an argument that reads today like the use of a derivative.

Mathematical discovery is rarely the work of one place or one name.

TOK A truncated series is deliberately wrong. It is chosen because it is simple enough to compute, and every extra term bought accuracy at the cost of that simplicity: four terms of the sine series give $\sin 1$ correct to five decimal places, and one term gives it to none.

Is there always a trade-off between accuracy and simplicity? And when a model is knowingly approximate, as every model in this chapter is, what makes it good enough to act on?

EXT The five standard series behave differently at the edges. The series for e^x, $\sin x$ and cos x converge for every x, but arctan x only for $|x| \leq 1$ and ln$(1 + x)$ only for $-1 < x \leq 1$, while the geometric series they are built from needs $|x| < 1$. Each series has a *radius of convergence*, and outside it the polynomial does not merely lose accuracy: it diverges completely. Finding that radius, and explaining why ln$(1 + x)$ has one at all when e^x does not, is the beginning of complex analysis, where the answer turns out to depend on where the function misbehaves in the complex plane. See Ch. 18.

EXT Maclaurin series turn calculus into algebra: integrals with no elementary antiderivative, like $\int e^{-x^2}\,dx$, can be integrated term by term once the function is a series. Series also solve differential equations that no other method in this course can reach, by assuming the answer is a power series and matching coefficients. The infinite polynomial is one of the most far-reaching ideas in all of analysis.

End-of-Chapter Problems

1. A population grows so that $\frac{dP}{dt} = 0.2P$, with $P = 500$ when $t = 0$.

(a) Solve the differential equation for $P(t)$. [4]

(b) Find the population when $t = 10$. [2]

2. A body cools according to $\frac{d\theta}{dt} = -0.1(\theta - 25)$, with $\theta(0) = 90$ (in $^\circ$C).

(a) Solve for $\theta(t)$. [4]

(b) Find the temperature after 10 minutes. [2]

(c) State the temperature the body approaches in the long run. [1]

3. Consider $\frac{dy}{dx} = \frac{x^2 + y^2}{xy}$ for $x > 0$.

(a) Use the substitution $y = vx$ to show that $xv\frac{dv}{dx} = 1$. [4]

(b) Hence find the general solution. [3]

4. Consider $\frac{dy}{dx} + \frac{1}{x}y = 3x$ for $x > 0$.

(a) Find the integrating factor. [2]

(b) Solve the equation given $y(1) = 2$. [4]

5. Consider $\frac{dy}{dx} = x + y$, with $y(0) = 1$.

(a) Use Euler's method with $h = 0.25$ to estimate $y(0.5)$, showing each step. [4]

(b) Solve the equation exactly (integrating factor) and find the true $y(0.5)$. [4]

(c) Comment on the size and direction of the error. [2]

6. Let $f(x) = \cos x$.

(a) Find $f(0), f'(0), f''(0), f'''(0), f^{(4)}(0)$. [3]

(b) Hence write the Maclaurin series of $\cos x$ up to the term in x^4. [2]

(c) Use it to estimate $\cos(0.2)$. [2]

7.

(a) Write down the Maclaurin series for $\sin x$ up to the term in x^5. [2]

(b) Hence evaluate $\lim_{x\to 0} \frac{x - \sin x}{x^3}$. [3]

8.

(a) Write down the Maclaurin series for e^x up to the term in x^3. [1]

(b) By substituting $-x^2$, find the series for e^{-x^2} up to x^4. [2]

(c) Hence estimate $\displaystyle\int_0^1 e^{-x^2}\,dx$ using terms up to x^4. [4]

9. A bowl of miso soup at 90°C is left in a 25°C room and obeys $\dfrac{d\theta}{dt} = -k(\theta - 25)$. After 10 minutes it has cooled to 65°C.

(a) Solve the differential equation, expressing θ in terms of k and t. [4]

(b) Find k. [2]

(c) Find how long, from the start, the soup takes to reach 40°C. [2]

10. Consider $\dfrac{dy}{dx} = \dfrac{y^2 + xy}{x^2}$ for $x > 0$, with $y(1) = 1$.

(a) Use the substitution $y = vx$ to show that $x\dfrac{dv}{dx} = v^2$. [3]

(b) Hence solve for y in terms of x. [4]

11. Solve each linear equation.

(a) $\dfrac{dy}{dx} - \dfrac{y}{x} = x^2$ for $x > 0$, given $y(1) = 1$. [6]

(b) $\dfrac{dy}{dx} + 2y = e^{-x}$ with $y(0) = 0$. [6]

12. Solve $\dfrac{dy}{dx} + (\tan x)\,y = \sec x$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$, given $y(0) = 2$. [7]

13. A fish population n (in thousands) in a lake satisfies the logistic equation $\dfrac{dn}{dt} = 0.4n(2 - n)$, with $n(0) = 0.5$.

(a) Using partial fractions, show that the solution is $n = \dfrac{2}{1 + 3e^{-0.8t}}$. [6]

(b) Find the time at which the population reaches 1500 fish. [3]

14. Consider $\dfrac{dy}{dx} = y - x^2$ with $y(0) = 1$.

(a) Use Euler's method with $h = 0.25$ to estimate $y(0.5)$. [5]

(b) The exact solution is concave up on $0 \le x \le 0.5$. State, with a reason, whether your estimate is above or below the true value. [2]

15. A skydiver falls from rest with $\dfrac{dv}{dt} = 9.8 - 0.2v^2$.

(a) Use Euler's method with $h = 0.5$ to estimate $v(1.5)$. [4]

(b) Find the terminal velocity, and comment on the behaviour of the Euler estimates around it. [3]

16. Find the Maclaurin series of $\ln(1+2x)$ up to the term in x^3

(a) by substitution into a standard series; [3]
(b) by computing derivatives from the definition. [3]

17.

(a) Find the Maclaurin series of $e^{-x}\sin x$ up to the term in x^3. [6]
(b) Use your series to estimate $e^{-0.1}\sin 0.1$, and compare it with the value from your calculator. [3]

18. By setting $x = 1$ in the Maclaurin series for $\arctan x$:

(a) write down a series for $\frac{\pi}{4}$; [2]
(b) estimate π using the first four terms; [2]
(c) comment on the rate of convergence. [1]

19. Use Maclaurin series to evaluate $\lim_{x\to 0} \frac{x - \sin x}{x^3}$. [5]

20. A function satisfies $\frac{dy}{dx} = x + y^2$ with $y(0) = 1$. Find the Maclaurin series of y up to the term in x^3. [7]

21. A drug is infused into the bloodstream at a constant rate r mg per hour, and the body removes it at a rate proportional to the concentration C, so that

$$\frac{dC}{dt} + kC = r,$$

where t is measured in hours and $C(0) = 0$.

(a) Explain why this equation is linear, and write down its integrating factor. [3]
(b) Solve the equation to show that $C = \frac{r}{k}(1 - e^{-kt})$. [4]
(c) A patient is given $r = 6$ and has $k = 0.3$. Find the concentration after 4 hours. [2]
(d) Find the concentration the patient approaches in the long run, and explain what this means for the treatment. [3]
(e) Find how long it takes to reach 90% of that level. [2]
(f) Once that level is reached the infusion is stopped, so that $\frac{dC}{dt} = -kC$. Solve this equation and find how long the concentration then takes to halve. [3]

22. In this question you will build a series for logarithms that converges quickly.

(a) Find the Maclaurin series of $\ln(1+x)$ up to the term in x^4. [4]

(b) Write down the Maclaurin series of $\ln(1-x)$ up to the term in x^4. [2]

(c) Hence show that $\ln\left(\dfrac{1+x}{1-x}\right) = 2\left(x + \dfrac{x^3}{3} + \dfrac{x^5}{5} + \cdots\right)$. [3]

(d) By choosing a suitable value of x, use the three terms shown to estimate $\ln 3$. [3]

(e) The series for $\ln(1+x)$ is valid only for $-1 < x \leq 1$. State the value of x that would be needed to reach $\ln 3$ from that series alone, and explain why the series in part (c) is the more useful one here. [2]

23. Water drains from a tank through a hole in its base. The depth h metres at time t minutes satisfies

$$\frac{dh}{dt} = -k\sqrt{h},$$

where k is a positive constant.

(a) Separate the variables to show that $2\sqrt{h} = c - kt$ for some constant c. [3]

(b) Given that the depth is h_0 when $t = 0$, show that $h = \left(\sqrt{h_0} - \frac{1}{2}kt\right)^2$. [2]

(c) A tank starts with $h_0 = 1.44$ and has $k = 0.2$. Find how long it takes to empty. [3]

(d) Find the depth after 6 minutes. [2]

(e) Use Euler's method with steps of 3 minutes to estimate the depth after 6 minutes, and find the error in this estimate. [4]

(f) Describe the shape of the graph of h against t, and use it to explain why the Euler estimate lies below the exact value. [2]

Challenge Questions

24. How fast does coffee cool? The chapter opened with $\dfrac{d\theta}{dt} = -k(\theta - 20)$, where θ is the temperature in degrees and t the time in minutes. A cup is poured at 80°, so $\theta(0) = 80$.

(a) Solve the equation to show that $\theta = 20 + 60e^{-kt}$. [3]

(b) The coffee has cooled to 60° after 10 minutes. Find k exactly, and to three significant figures. [3]

(c) Find the temperature after 20 minutes. [2]

(d) Show that the time taken to reach a temperature θ is $t = \dfrac{1}{k}\ln\left(\dfrac{60}{\theta - 20}\right)$, and explain why the coffee never reaches 20°. [4]

(e) A second cup is poured at 90° in the same room. Show that it takes exactly 10 minutes to cool from 80° to 60°, the same as the first cup. [3]

25. How wrong is Euler? Take $\dfrac{dy}{dx} = y$ with $y(0) = 1$, whose exact solution is $y = e^x$, so that $y(1) = e$.

(a) Show that one Euler step of length h gives $y_{n+1} = (1+h)y_n$, and hence that after n steps $y_n = (1+h)^n$. [3]

(b) Taking $h = \dfrac{1}{n}$, deduce that Euler's estimate of $y(1)$ is $\left(1+\dfrac{1}{n}\right)^n$. [2]

(c) Evaluate this estimate for $n = 2, 4, 8, 16$, and tabulate the error in each case. [4]

(d) Describe what happens to the error each time h is halved, and state what this says about the accuracy of the method. [3]

(e) Write down the limit of $\left(1+\dfrac{1}{n}\right)^n$ as $n \to \infty$, and explain why Euler's method had to give this answer. [3]

26. The ferry and the current. A ferry sets off from $(1, 0)$ toward a dock at the origin, always aiming straight at the dock, while a river current pushes it in the positive y-direction at a fraction k of the ferry's own speed. Its path satisfies

$$\frac{dy}{dx} = \frac{y - k\sqrt{x^2+y^2}}{x}.$$

(a) Show the equation is homogeneous, and that $y = vx$ leads to $x\dfrac{dv}{dx} = -k\sqrt{1+v^2}$. [3]

(b) Given that $\displaystyle\int \frac{dv}{\sqrt{1+v^2}} = \ln\left(v + \sqrt{1+v^2}\right) + C$, show that $v + \sqrt{1+v^2} = x^{-k}$. [3]

(c) Hence show that $y = \frac{1}{2}\left(x^{1-k} - x^{1+k}\right)$. [4]

(d) Taking $k = \frac{1}{2}$, find where the ferry lands. [2]

(e) Show that the ferry reaches the dock only when $k < 1$, and describe what happens when $k = 1$ and when $k > 1$. [4]

NUMBER & ALGEBRA

CHAPTER 18

Complex Numbers

18

SYLLABUS 1.12–1.14 HL ONLY

In the sixteenth century, Italian mathematicians learned to solve cubic equations. Their formula produced something unexpected. It sometimes asked for the square root of a negative number, even when the equation had three perfectly ordinary answers. No number they knew could give a negative square. Yet if they wrote the impossible quantity down and kept calculating, it disappeared before the end, and the answers were correct.

This chapter begins with that impossible quantity. We define a number i with $i^2 = -1$. No real number does this, so i is new. For two centuries mathematicians did not trust it, and the name "imaginary" was chosen as an insult. The name was unfair. Once i is allowed, every polynomial equation has a solution, a result now called the **Fundamental Theorem of Algebra**.

These new numbers are also geometric. Each one is a point in a plane, with the real numbers along one axis. Adding two of them moves a point across the plane. Multiplying two of them rotates and stretches. Multiplying by i rotates by a quarter-turn. In ordinary coordinates a rotation takes several steps; here it is one multiplication. This is why complex numbers are used throughout physics and engineering.

By the end of this chapter you will be able to write a complex number in three forms: $a + bi$, $r \operatorname{cis} \theta$ and $re^{i\theta}$. You will multiply, divide, and find powers and roots in each form. You will also use complex roots to factorise polynomials and to prove trigonometric identities.

Three forms may seem like more than you need. Each one makes a different calculation easy. Adding is simplest in $a + bi$ form. A power that would fill a page of working takes one line in polar form. Choosing the form before you start is the main skill in this chapter.

We start with that algebra: numbers of the form $a + bi$.

18.1 Cartesian Form

Everything starts from a single definition that no real number satisfies.

> DEF · IB The **imaginary unit** i satisfies $i^2 = -1$. A **complex number** is any $z = a + bi$ with $a, b \in \mathbb{R}$; $a = \text{Re}(z)$ is the **real part** and $b = \text{Im}(z)$ the **imaginary part**.

You add, subtract and multiply complex numbers just as you would binomials in i, with the single extra rule that i^2 is replaced by -1 wherever it appears.

That one rule is enough to find every power of i, because the powers repeat in a cycle of four:

$$i^1 = i, \qquad i^2 = -1, \qquad i^3 = i^2 \cdot i = -i, \qquad i^4 = (i^2)^2 = 1,$$

and $i^5 = i^4 \cdot i = i$ starts the cycle again. To simplify any power, divide the exponent by 4 and keep only the remainder: $i^{2026} = i^{4(506)+2} = i^2 = -1$.

EXAMPLE 18.1

Given $z = 2 + i$ and $w = 5 - 3i$, find $z + w$ and zw.

Adding is done part by part: $z + w = (2 + 5) + (1 - 3)i = 7 - 2i$. Multiplying, expand and then use $i^2 = -1$:

$$zw = (2 + i)(5 - 3i) = 10 - 6i + 5i - 3i^2 = 10 - i + 3 = 13 - i.$$

The whole of complex arithmetic is ordinary algebra plus the one substitution $i^2 = -1$.

The conjugate and division

Every complex number has a mirror image across the real axis, called its **conjugate**. The conjugate is what makes division possible.

> DEF The **complex conjugate** of $z = a + bi$ is $z^* = a - bi$.

The conjugate matters because a number times its own conjugate is always real: $zz^* = (a + bi)(a - bi) = a^2 + b^2$. This is exactly the tool for division, in the same way that a surd is rationalised. To divide, multiply top and bottom by the conjugate of the denominator, turning the denominator real.

EXAMPLE 18.2

Find $\dfrac{3 + 2i}{1 - i}$.

Multiply numerator and denominator by the conjugate $1+i$:

$$\frac{3+2i}{1-i}\cdot\frac{1+i}{1+i}=\frac{3+3i+2i+2i^2}{1^2+1^2}=\frac{3+5i-2}{2}=\frac{1+5i}{2}=\tfrac{1}{2}+\tfrac{5}{2}i.$$

Multiplying by the conjugate removes i from the denominator every time; the answer is then read off as real and imaginary parts.

One more Cartesian tool: two complex numbers are equal exactly when their real parts agree *and* their imaginary parts agree. A single complex equation therefore carries two real equations at once, and **equating parts** solves for unknowns.

EXAMPLE 18.3

Find the complex number z satisfying $z+2z^*=9+2i$.
Write $z=a+bi$, so $z^*=a-bi$:

$$(a+bi)+2(a-bi)=3a-bi=9+2i.$$

Equate real parts: $3a = 9 \implies a = 3$. Equate imaginary parts: $-b = 2 \implies b = -2$. So $z=3-2i$. One equation was enough for two unknowns, because the real and imaginary parts each give an equation.

The Argand diagram

A complex number carries two real numbers, so it is naturally a *point* in a plane: the horizontal axis for the real part, the vertical for the imaginary. This picture, the **Argand diagram**, is what makes the whole subject geometric.

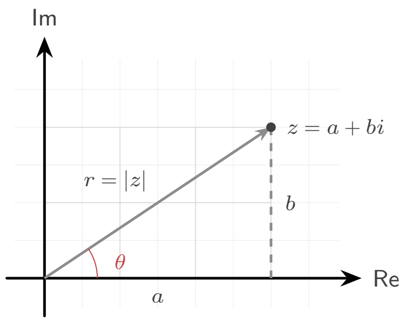


Figure 18.1 A complex number $z=a+bi$ drawn as the point (a,b) of the Argand diagram. Its distance from the origin is the modulus r, and the angle its arrow makes with the positive real axis is the argument θ.

The picture immediately explains the arithmetic. **Addition is vector addition**: to add w to z, place w's arrow tail-to-tip on z's, exactly as in Ch. 10, so $z+w$ is the far corner of the parallelogram the two arrows span. And **conjugation is reflection**: z^* is z flipped across the real axis, which is why zz^* lies on the real axis.

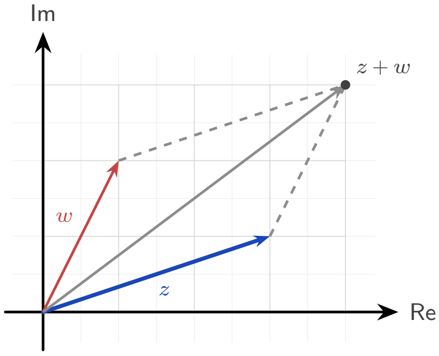


Figure 18.2 Addition is vector addition. Placing w's arrow tail-to-tip on z's puts $z+w$ at the far corner of the parallelogram the two arrows span.

There are two natural numbers to attach to a point in a plane: how far it is from the origin, and in what direction. These are the modulus and the argument, and they are the subject of the next section.

YOUR TURN 18.1 Cartesian Form

1. Evaluate each expression, giving your answer in the form $a+bi$.

(a) $(3+2i)+(1-4i)$ [1]
(b) $(5-i)-(2+3i)$ [1]
(c) $(2+3i)(1-i)$ [2]
(d) $(2+i)(2-i)$ [1]

2. Simplify each power of i.

(a) i^3 [1]
(b) i^4 [1]
(c) i^{15} [2]
(d) i^{100} [2]

3. Let $z=7-4i$.

(a) Write down Re(z) and Im(z). [1]
(b) Write down z^*. [1]
(c) Evaluate zz^*. [2]

4. Write each quotient in the form $a+bi$.

(a) $\dfrac{4+3i}{2-i}$ [3]
(b) $\dfrac{1}{1+i}$ [2]

5. Find the complex number z in each case.

(a) $z + 3z^* = 8 - 4i$ [3]

(b) $2z - z^* = 3 + 6i$ [3]

18.2 Modulus and Argument

The distance of z from the origin is its **modulus** $|z|$, straight from Pythagoras; the angle its arrow makes with the positive real axis is its **argument** $\arg(z)$, measured anticlockwise.

KEY For $z = a + bi$: $\quad |z| = \sqrt{a^2 + b^2}, \qquad \tan(\arg z) = \dfrac{b}{a}$.

CAUTION Finding the argument needs care: $\tan\theta = \frac{b}{a}$ alone cannot tell which quadrant z is in, because tan repeats every π. Always sketch z on the Argand diagram first, then choose the angle in the correct quadrant, conventionally in the range $-\pi < \theta \leq \pi$ (the **principal argument**).

Knowing the modulus r and argument θ is enough to rebuild the number, since $a = r\cos\theta$ and $b = r\sin\theta$. This gives the **modulus–argument** (or polar) form, and the equivalent exponential version, **Euler's form**.

KEY · IB **Polar and Euler form:** $\quad z = r(\cos\theta + i\sin\theta) = r\,\text{cis}\,\theta = re^{i\theta}$,

where $r = |z|$ and $\theta = \arg(z)$.

That $e^{i\theta} = \cos\theta + i\sin\theta$ is startling on first sight, and you already have the tools to see why it is true. Take the Maclaurin series for e^x from Ch. 17 and substitute $x = i\theta$, using the cycle of powers of i from §18.1:

$$e^{i\theta} = 1 + i\theta + \frac{(i\theta)^2}{2!} + \frac{(i\theta)^3}{3!} + \frac{(i\theta)^4}{4!} + \cdots = \left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \cdots\right) + i\left(\theta - \frac{\theta^3}{3!} + \cdots\right).$$

The even terms, where i has become ±1, form the cosine series; the odd terms, which still carry an i, form the sine series. The exponential function and the trigonometric functions are two views of one function on the complex plane. Setting $\theta = \pi$ gives the famous $e^{i\pi} + 1 = 0$, which ties e, i, π, 1 and 0 together in a single line, and now you know it is not a coincidence but a calculation.

EXAMPLE 18.4

Write $z = 1 - i$ in polar and Euler form, with $-\pi < \theta \leq \pi$.

The modulus is $r = \sqrt{1^2 + (-1)^2} = \sqrt{2}$. The point $(1, -1)$ lies in the fourth quadrant, so the argument is $\theta = -\frac{\pi}{4}$. Hence

$$z = \sqrt{2}\,(\cos\tfrac{-\pi}{4} + i\sin\tfrac{-\pi}{4}) = \sqrt{2}\,\text{cis}(-\tfrac{\pi}{4}) = \sqrt{2}\,e^{-i\pi/4}.$$

Sketch first, then read off the quadrant. A common error is to take arctan without checking which quadrant the point is in.

YOUR TURN 18.2 Modulus and Argument

6. Find the modulus of each number.

(a) $3+4i$ [1]
(b) $5-12i$ [1]
(c) $1+i$ [1]
(d) $-2i$ [1]

7. Find the principal argument of each number, with $-\pi < \theta \leq \pi$.

(a) $1+i$ [1]
(b) i [1]
(c) -3 [1]
(d) $-1-i$ [2]

8. Write each number in modulus–argument form and in Euler form.

(a) $2i$ [2]
(b) -3 [2]
(c) $1-i$ [3]

9. Write each number in Cartesian form.

(a) $4\,\text{cis}\,\frac{\pi}{3}$ [2]
(b) $2e^{-i\pi/4}$ [2]

18.3 Products, Quotients and De Moivre's Theorem

Polar form is worth the extra notation because it makes multiplication easy. When two complex numbers are multiplied, their moduli multiply and their arguments *add*, which is just the law of exponents applied to $re^{i\theta}$.

> KEY $|zw| = |z|\,|w|$ and $\arg(zw) = \arg z + \arg w$; $\left|\dfrac{z}{w}\right| = \dfrac{|z|}{|w|}$ and $\arg\!\left(\dfrac{z}{w}\right) = \arg z - \arg w$.

Geometrically this is striking: multiplying by a complex number *rotates* by its argument and *scales* by its modulus. So multiplying by $i = 1 \cdot e^{i\pi/2}$, whose modulus is 1, is a quarter-turn anticlockwise with no stretching at all.

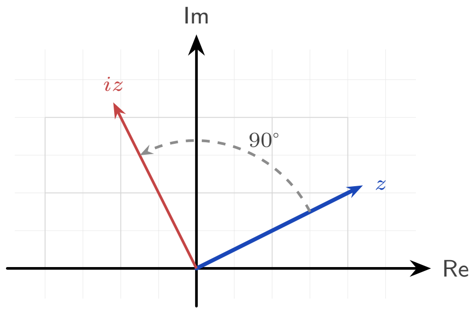


Figure 18.3 Multiplying by i turns a complex number through a quarter-turn anticlockwise about the origin, leaving its modulus unchanged.

If multiplying adds arguments, then raising to a power *multiplies* the argument. This is **De Moivre's theorem**, and everything in the rest of the chapter follows from it.

> KEY · IB **De Moivre's theorem:** $\big(r(\cos\theta + i\sin\theta)\big)^n = r^n(\cos n\theta + i\sin n\theta) = r^n e^{in\theta}$, for $n \in \mathbb{Z}$.

The syllabus extends the theorem to *rational* exponents (indeed it holds for all real n): a power like $z^{1/n}$ obeys the same rule, though it becomes multi-valued, which is exactly the story of the nth roots in the next section. For positive integers the theorem is proved by the induction of Ch. 3, using the compound-angle identities of Ch. 9.

EXAMPLE 18.5

Prove De Moivre's theorem for $n \in \mathbb{Z}^+$ by mathematical induction.

Let P_n be the statement $(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$.

Show P_1 is true: the left side is $(\cos\theta + i\sin\theta)^1$, the right is $\cos\theta + i\sin\theta$. Equal.

Assume P_k is true for some $k \in \mathbb{Z}^+$. Then

$$(\cos\theta + i\sin\theta)^{k+1} = (\cos k\theta + i\sin k\theta)(\cos\theta + i\sin\theta)$$

$$= (\cos k\theta\cos\theta - \sin k\theta\sin\theta) + i(\sin k\theta\cos\theta + \cos k\theta\sin\theta) = \cos(k+1)\theta + i\sin(k+1)\theta,$$

by the compound-angle identities. So P_k implies P_{k+1}.

Hence, since P_1 is true and $P_k \implies P_{k+1}$, the statement P_n is true for all $n \in \mathbb{Z}^+$ by mathematical induction. ■

Notice what the proof shows: each step of De Moivre is one use of the compound-angle identities. The theorem is those identities of Ch. 9 applied n times.

Raising a complex number to a high power means expanding a bracket again and again in Cartesian form. In polar form it takes one line: raise the modulus to the power, multiply the angle by the power, done.

EXAMPLE 18.6

Find $(1+i)^8$.

In polar form $1+i = \sqrt{2}\ \text{cis}\ \frac{\pi}{4}$. By De Moivre,

$$(1+i)^8 = (\sqrt{2})^8\ \text{cis}(8 \cdot \tfrac{\pi}{4}) = 16\ \text{cis}(2\pi) = 16.$$

Eight multiplications reduce to one power of the modulus and one multiple of the angle.

Draw the powers and you can watch the theorem work. Each multiplication by $1+i$ turns the arrow 45° anticlockwise and stretches it by $\sqrt{2}$, so the powers lie on a spiral that winds outward from the origin:

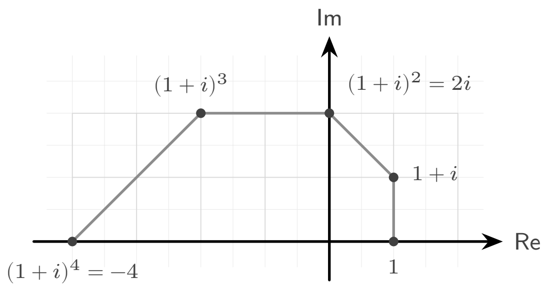


Figure 18.4 The first five powers of $1+i$. Each multiplication turns the arrow through 45° and stretches it by $\sqrt{2}$, so the powers lie on a spiral winding outward from the origin.

Four more steps continue the spiral through the lower half-plane and back to the positive real axis at $(1+i)^8 = 16$: a full turn of $8 \times 45^\circ = 360^\circ$, with the modulus grown to $(\sqrt{2})^8 = 16$. The worked example above is this picture written in algebra.

Numbers on the unit circle

One case of De Moivre is worth stating separately, because it is used often. If $|z| = 1$ then $z = \text{cis}\,\theta$, and dividing 1 by a number of modulus 1 simply reverses its angle, so $\dfrac{1}{z} = \text{cis}(-\theta) = z^*$. Add the two and the imaginary parts cancel while the real parts double; subtract them and the opposite happens.

KEY For $z = \operatorname{cis}\theta$ and any $n \in \mathbb{Z}$:

$$z^n + \frac{1}{z^n} = 2\cos n\theta, \qquad z^n - \frac{1}{z^n} = 2i\sin n\theta.$$

Read from left to right this converts powers into multiple angles, and that is what makes it useful: an expression like $\cos^4\theta$, awkward to integrate as it stands, becomes a sum of $\cos 2\theta$ and $\cos 4\theta$ terms that can be integrated directly. You will build exactly that in the end-of-chapter problems.

YOUR TURN 18.3 Products, Quotients and De Moivre's Theorem

10. Evaluate each expression, giving your answer in modulus–argument form.

(a) $\left(2\operatorname{cis}\frac{\pi}{6}\right)\left(3\operatorname{cis}\frac{\pi}{3}\right)$ [2]

(b) $\dfrac{6\operatorname{cis}\frac{2\pi}{3}}{2\operatorname{cis}\frac{\pi}{3}}$ [2]

(c) $\left(\operatorname{cis}\frac{\pi}{5}\right)^5$ [2]

(d) $\left(2\operatorname{cis}\frac{\pi}{6}\right)^3$ [2]

11. Given $|z| = 3$, $|w| = 4$, $\arg z = \frac{\pi}{4}$ and $\arg w = \frac{\pi}{6}$, write down

(a) $|zw|$ [1]

(b) $\arg(zw)$ [1]

(c) $\left|\dfrac{z}{w}\right|$ [1]

(d) $\arg\left(\dfrac{z}{w}\right)$ [1]

12.

(a) Expand $(\cos\theta + i\sin\theta)^4$ using De Moivre's theorem. [1]

(b) Use De Moivre's theorem to evaluate $(1+i)^6$. [3]

13. Let $z = \operatorname{cis}\theta$. Write each expression in terms of a single cosine or sine.

(a) $z + \dfrac{1}{z}$ [1]

(b) $z^3 + \dfrac{1}{z^3}$ [2]

(c) $z^2 - \dfrac{1}{z^2}$ [2]

18.4 Roots of Complex Numbers

De Moivre also works in reverse. Just as a positive real number has two square roots, a complex number has exactly n distinct nth roots, and they are evenly spaced around a circle. To find them, write the number in polar form, take the nth root of the modulus, divide the argument by n, and then step around by $\frac{2\pi}{n}$ to collect all n of them.

KEY The nth roots of $z = r \operatorname{cis}\theta$ are $\sqrt[n]{r} \operatorname{cis}\left(\dfrac{\theta + 2\pi k}{n}\right)$, for $k = 0, 1, \dots, n-1$.

Because the roots share a modulus $\sqrt[n]{r}$ and their arguments differ by equal steps of $\frac{2\pi}{n}$, they sit at the vertices of a **regular n-gon** centred at the origin. The special case $z^n = 1$ gives the **roots of unity**, a regular n-gon with one vertex at 1.

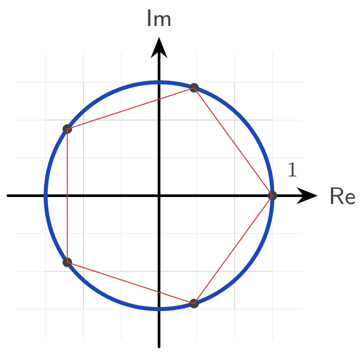


Figure 18.5 The five fifth roots of unity, at the vertices of a regular pentagon inscribed in the unit circle with one vertex at 1.

The roots of unity are worth a closer look, because the whole set is generated by just one of them. Write $\omega = \operatorname{cis}\dfrac{2\pi}{n}$ for the first root anticlockwise from 1. Then $\omega^k = \operatorname{cis}\dfrac{2\pi k}{n}$ is the kth root round the circle, and $\left(\omega^k\right)^n = \left(\omega^n\right)^k = 1^k = 1$, so every power of ω is again a root. The n roots are therefore

$$1,\ \omega,\ \omega^2,\ \dots,\ \omega^{n-1},$$

one number and its powers, stepping round the polygon and returning to the start.

They also **sum to zero**. Geometrically the n arrows point symmetrically in every direction and cancel, but the geometric series of Ch. 6 proves it outright: since $\omega \neq 1$,

$$1 + \omega + \omega^2 + \cdots + \omega^{n-1} = \frac{\omega^n - 1}{\omega - 1} = \frac{1-1}{\omega - 1} = 0.$$

The same conclusion follows from Ch. 7: $z^n - 1$ has no z^{n-1} term, so the sum of its roots vanishes. For $n = 3$ this gives the identity $1 + \omega + \omega^2 = 0$, which appears whenever cube roots of unity are used.

These evenly spaced points are also the basis of the *discrete Fourier transform*, which separates a signal into pure frequencies by sampling it against precisely these n points on the circle. The fast algorithm for computing it, the FFT, uses the polygon's symmetry, and it is what makes audio compression and digital signal processing fast enough to run in real time on a phone.

EXAMPLE 18.7

Find the three cube roots of 8.

Write $8 = 8\,\mathrm{cis}\,0$. The modulus of each root is $\sqrt[3]{8} = 2$, and the arguments are $\frac{0+2\pi k}{3}$ for $k = 0, 1, 2$:

$$2\,\mathrm{cis}\,0 = 2, \qquad 2\,\mathrm{cis}\,\tfrac{2\pi}{3} = -1 + \sqrt{3}\,i, \qquad 2\,\mathrm{cis}\,\tfrac{4\pi}{3} = -1 - \sqrt{3}\,i.$$

The three roots form an equilateral triangle on the circle of radius 2, one real and a conjugate pair.

Square roots without polar form

Polar form gives roots of every order, but it has one drawback: it needs the argument, and unless the angle is one of the standard ones, arctan returns a decimal and the answer is no longer exact. For *square* roots there is a second route that stays in Cartesian form and keeps the surds intact. Assume the root is $a + bi$, square it, and equate real and imaginary parts as in §18.1.

EXAMPLE 18.8

Find the square roots of $3 + 4i$ in the form $a + bi$.

Let $(a + bi)^2 = 3 + 4i$. Expanding the left side gives $a^2 - b^2 + 2abi$, so equating parts,

$$a^2 - b^2 = 3 \qquad \text{and} \qquad 2ab = 4.$$

The second gives $b = \dfrac{2}{a}$. Substituting into the first,

$$a^2 - \frac{4}{a^2} = 3 \implies a^4 - 3a^2 - 4 = 0 \implies (a^2 - 4)(a^2 + 1) = 0.$$

Since a is real, $a^2 = 4$, so $a = \pm 2$ and correspondingly $b = \pm 1$. The square roots are $2 + i$ and $-2 - i$, that is $\pm(2 + i)$.

Two features of that calculation are worth noting. The substitution always produces a **quadratic in a^2**, and the negative solution for a^2 is always rejected, because a is real. The two answers always differ by a sign, since square roots lie at opposite ends of a diameter, half a turn apart.

CAUTION Use Cartesian equating for square roots when the answer should be exact. For cube roots and higher, go back to polar form: equating parts would give a cubic or an equation of even higher degree, while De Moivre gives roots of any order in one line.

YOUR TURN 18.4 Roots of Complex Numbers

14.

(a) Write down a square root of -9. [1]

(b) State the modulus of each cube root of 27. [1]

(c) State how many distinct fifth roots a nonzero complex number has. [1]

(d) State the shape formed by the nth roots of a complex number on the Argand diagram. [1]

15. State

(a) the three cube roots of unity; [2]

(b) the four fourth roots of unity. [2]

16. Find the square roots of each number in the form $a+bi$ by equating real and imaginary parts.

(a) $5+12i$ [3]

(b) $-8+6i$ [3]

17. Find, in modulus–argument form,

(a) the three cube roots of 8; [3]

(b) the four fourth roots of 16. [3]

18.5 Complex Roots of Polynomials

Complex numbers were invented to solve equations, and they solve *every* polynomial equation. This is the **Fundamental Theorem of Algebra**: every polynomial of degree $n \geq 1$ has exactly n roots in the complex numbers, counted with multiplicity. The roots of a polynomial can all lie off the real line, but none of them can lie outside the complex plane.

The simplest case is one you have met before in a different form. In Ch. 4, a quadratic with negative discriminant had "no solutions." It always had two roots; they were simply not on the real line.

EXAMPLE 18.9

Solve $z^2 - 4z + 13 = 0$.

The quadratic formula works unchanged:

$$z = \frac{4 \pm \sqrt{16-52}}{2} = \frac{4 \pm \sqrt{-36}}{2} = \frac{4 \pm 6i}{2} = 2 \pm 3i.$$

The negative discriminant, which once ended the question, now simply produces an i: the two roots are $2+3i$ and $2-3i$, a conjugate pair placed symmetrically above and below the real axis.

That symmetry is no accident. When a polynomial has *real* coefficients, its non-real roots always come in conjugate pairs.

KEY **Conjugate root theorem.** If a polynomial has real coefficients and $z = a+bi$ is a root, then $z^* = a - bi$ is also a root.

This is why a cubic with real coefficients always has at least one real root: its three roots are either all real, or one real root plus a conjugate pair. A non-real root never appears alone in a polynomial with real coefficients.

EXAMPLE 18.10

Given that $2 + i$ is a root of $z^3 - 3z^2 + \cdots$ with real coefficients, state another root and find a quadratic factor.

By the conjugate root theorem, $2 - i$ is also a root. The two together give a quadratic factor with real coefficients:

$$(z-(2+i))(z-(2-i)) = (z-2)^2 - i^2 = z^2 - 4z + 5.$$

A conjugate pair of roots always multiplies to a real quadratic, here $z^2 - 4z + 5$, which can then be divided out to find the remaining real root.

Trigonometric identities from De Moivre

De Moivre's theorem also produces trigonometric identities. Expand $(\cos\theta + i\sin\theta)^n$ by the binomial theorem of Ch. 2, then compare real and imaginary parts with $\cos n\theta + i \sin n\theta$, and a multiple-angle formula follows for any n.

EXAMPLE 18.11

Use De Moivre's theorem to show that $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$, and find a matching identity for $\sin 3\theta$.

Write $c = \cos\theta$, $s = \sin\theta$, and expand by the binomial theorem:

$$(c+is)^3 = c^3 + 3c^2(is) + 3c(is)^2 + (is)^3 = (c^3 - 3cs^2) + i(3c^2 s - s^3).$$

By De Moivre this equals $\cos 3\theta + i \sin 3\theta$, so the parts must match separately. Real parts, using

$s^2 = 1 - c^2$:

$$\cos 3\theta = c^3 - 3c(1 - c^2) = 4\cos^3\theta - 3\cos\theta.$$

Imaginary parts, using $c^2 = 1 - s^2$:

$$\sin 3\theta = 3(1 - s^2)s - s^3 = 3\sin\theta - 4\sin^3\theta.$$

One expansion gives two identities, and the same method works for any n. Results that took several steps with compound angles in Ch. 9 now follow from a single binomial expansion.

TIP Complex numbers offer several good IA topics. Three directions worth exploring: iterate $z \mapsto z^2 + c$ for different c and investigate which starting points escape to infinity (this is how the Mandelbrot set is defined); investigate which regular polygons can be constructed with ruler and compass, a question Gauss answered at nineteen using exactly the roots of unity above; or measure phase shift in a simple AC circuit and model it with rotating complex numbers.

YOUR TURN 18.5 Complex Roots of Polynomials

18. Solve each equation.

(a) $z^2 + 9 = 0$ [1]
(b) $z^2 + 4 = 0$ [1]
(c) $z^2 - 2z + 5 = 0$ [2]
(d) $z^2 + 2z + 2 = 0$ [2]

19.

(a) Given that $3 + i$ is a root of a polynomial with real coefficients, state another root. [1]
(b) Write down a quadratic with real coefficients whose roots are $2i$ and $-2i$. [2]
(c) Write down a quadratic with real coefficients whose roots are $1 + i$ and $1 - i$. [2]

20.

(a) The equation $z^2 + bz + c = 0$ has real coefficients and $1 + 2i$ as a root. Find b and c. [3]
(b) Explain why a cubic with real coefficients must have at least one real root. [2]

Chapter 18 · Complex Numbers

Reflections

IM The name "imaginary" was an insult. Descartes coined it in the seventeenth century to dismiss these numbers as fictions, and Leibniz called them "an amphibian between being and non-being." Their rehabilitation was slow and international. The Norwegian surveyor Caspar Wessel and the Swiss bookkeeper Jean-Robert Argand independently drew the plane picture that made them concrete. It was the German mathematician Carl Friedrich Gauss who finally made them respectable: he gave the first accepted proof of the Fundamental Theorem of Algebra, and he popularised the term "complex" itself. A number system that began in suspicion became, within two centuries, essential.

TOK Does the naming shape the difficulty? "Imaginary" was coined as a dismissal and "real" as its opposite, and both names outlived the objection by centuries. Would these numbers have been accepted sooner if they had been called something neutral, such as "lateral"? If a change of vocabulary can change how readily an idea is believed, how much of what we call mathematical knowledge is carried by its language rather than its content?

TOK Euler's identity $e^{i\pi} + 1 = 0$ is routinely called the most beautiful equation in mathematics. What is being judged when a mathematician calls a result beautiful, and is that judgement evidence of anything? Beauty in mathematics is often described as economy, surprise, or unexpected connection, all of which this identity has. Yet if beauty guides which results we pursue and teach, it is shaping the subject, not merely decorating it.

TOK Are complex numbers discovered or invented? They began as a deliberate fiction, a symbol i defined into existence to make an equation solvable, which argues for invention. Yet once admitted, they turned out to describe electrical circuits, quantum states, and fluid flow with uncanny precision, as though they had been waiting in nature all along. When a tool built for pure convenience turns out to model the physical world so faithfully, what does that tell us about the relationship between mathematics and reality? Is the "unreasonable effectiveness" of complex numbers a coincidence, or a clue?

RW Complex numbers are the natural language of anything that oscillates. An alternating current, a radio wave, or a vibration is described by a rotating complex number whose real part is the physical signal; engineers add and scale these "phasors" to design circuits and filters that would otherwise need long trigonometric calculations. Quantum mechanics goes further still: the state of a particle is a complex-valued wave, and i sits at the centre of the equation that governs how it evolves. Signal processing in phones, impedance in power grids, and the mathematics of quantum computing all use the numbers of this chapter.

EXT De Moivre's theorem hints at a whole hidden unity. The identity $e^{i\theta} = \cos\theta + i\sin\theta$ says that the exponential function and the trigonometric functions, which look nothing alike on the real line, are the same function seen along different directions in the complex plane. Growth and oscillation, e^x and $\sin x$, are one thing viewed two ways. Push this further and you reach complex analysis, the study of functions of a complex variable. Such functions turn out to be far more tightly constrained than real ones, and problems about ordinary integers, such as how the prime numbers are distributed, are answered by studying functions over the complex plane.

End-of-Chapter Problems

1. Given $z = 3 + 4i$ and $w = 1 - 2i$:

(a) find $z + w$; [1]
(b) find zw; [2]
(c) find $\frac{z}{w}$; [2]
(d) find $|z|$. [1]

2. Express $\dfrac{2+i}{3-i}$ in the form $a + bi$. [4]

3. Find the modulus and the principal argument of

(a) $z = -1 + i$; [3]
(b) $z = \sqrt{3} - i$. [3]

4. Write $z = 1 + \sqrt{3}\,i$ in modulus–argument form and in Euler form. [4]

5. Use De Moivre's theorem to evaluate

(a) $(1 + \sqrt{3}\,i)^6$; [3]
(b) $(1 + i)^{10}$; [3]
(c) $\left(\dfrac{1+i}{\sqrt{2}}\right)^{12}$. [3]

6. Find the square roots of each number in the form $a + bi$.

(a) $3 + 4i$ [4]
(b) $-5 + 12i$ [4]

7. Given $z = 2\,\mathrm{cis}\,\frac{\pi}{3}$:

(a) find z^2; [2]
(b) find z^3. [2]

8. The equation $z^2 + bz + c = 0$ has real coefficients and $1 - 2i$ as a root. Find b and c. [4]

9. Find the three cube roots of $8i$, giving each

(a) in modulus–argument form; [3]
(b) in exact Cartesian form. [3]

10. Simplify

(a) i^{2026}; [2]

(b) i^{-7}. [2]

11. Find all solutions of $z^3 = 27$, giving each in modulus–argument form. [4]

12. Find real numbers x and y such that $(x + yi)(2 - i) = 3 + i$. [4]

13.

(a) Express -8 in the form $re^{i\theta}$, with $-\pi < \theta \leq \pi$. [3]

(b) The complex number z has $|z| = 2$ and $\arg z = \frac{\pi}{6}$. Write z in Cartesian form. [3]

14.

(a) Find the four fourth roots of -16, giving each in the form $a + bi$. [5]

(b) Find all solutions of $z^4 = 1$, and show that they sum to zero. [4]

15. Solve $2z^2 + 2z + 5 = 0$. [4]

16. Given $z = \cos\theta + i\sin\theta$, show that $z + \dfrac{1}{z} = 2\cos\theta$. [4]

17. The points representing 1, i, -1 and $-i$ are plotted on an Argand diagram. State the shape they form and find its area. [3]

18. Given that $2 + 3i$ is a root of a polynomial with real coefficients, find a real quadratic factor of that polynomial. [4]

19. Solve $z^2 = 2i$, giving the roots in the form $a + bi$. [4]

20. Given $z = 2 + 2i$, find $|z|$ and $\arg z$, and hence write z in Euler form. [4]

21. Find the complex number z satisfying $2z - z^* = 3 + 12i$. [4]

22. Prove De Moivre's theorem, $(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$, for all $n \in \mathbb{Z}^+$ by mathematical induction. [7]

23. Use De Moivre's theorem to show that $\cos 4\theta = 8\cos^4\theta - 8\cos^2\theta + 1$. [6]

24. Use De Moivre's theorem to show that $\tan 3\theta = \dfrac{3t - t^3}{1 - 3t^2}$, where $t = \tan\theta$. [7]

25. One cube root of a complex number w is $1+\sqrt{3}\,i$. Find w and the other two cube roots. [6]

26. In this question you will investigate the fifth roots of unity, the solutions of $z^5 = 1$.

(a) Write down the five roots in the form $\operatorname{cis}\theta$. [3]
(b) Show that the five roots sum to zero, using the sum of a geometric series. [3]
(c) Let $\omega = \operatorname{cis}\frac{2\pi}{5}$. Show that every fifth root of unity is a power of ω. [2]
(d) Deduce that $\omega+\omega^2+\omega^3+\omega^4 = -1$. [2]
(e) The five roots are plotted on an Argand diagram and joined to form a regular pentagon. Show that its area is $\frac{5}{2}\sin\frac{2\pi}{5}$, and evaluate this to three significant figures. [5]

27. Let $z = \operatorname{cis}\theta$.

(a) Show that $z^n + \dfrac{1}{z^n} = 2\cos n\theta$ for any $n \in \mathbb{Z}^+$. [3]
(b) Expand $\left(z+\dfrac{1}{z}\right)^4$ using the binomial theorem, and hence show that $16\cos^4\theta = 2\cos 4\theta + 8\cos 2\theta + 6$. [5]
(c) Deduce that $\cos^4\theta = \frac{1}{8}(\cos 4\theta + 4\cos 2\theta + 3)$. [2]
(d) Hence evaluate $\displaystyle\int_0^{\pi/2} \cos^4\theta\, d\theta$ exactly. [4]
(e) Explain what this method achieved that integrating $\cos^4\theta$ directly would not. [2]

28. The polynomial $P(z) = z^4 - 2z^3 + 14z^2 - 18z + 45$ has real coefficients, and $1+2i$ is a root.

(a) Write down another root of $P(z)$, and give a reason. [2]
(b) Find a real quadratic factor of $P(z)$. [3]
(c) Find the other quadratic factor. [3]
(d) Hence write down all four roots of $P(z)$. [3]
(e) The four roots are plotted on an Argand diagram. Describe the symmetry of the picture, and state the feature of $P(z)$ that guarantees it. [3]

Challenge Questions

29. **A sum of cosines from a seven-sided polygon.** Let $\omega = \operatorname{cis}\frac{2\pi}{7}$.

(a) Show that every solution of $z^7 = 1$ other than $z=1$ satisfies $z^6+z^5+z^4+z^3+z^2+z+1 = 0$. [3]
(b) Show that $\omega^k + \omega^{7-k} = 2\cos\dfrac{2\pi k}{7}$ for $k = 1, 2, 3$. [3]

(c) Hence show that $\cos\dfrac{2\pi}{7} + \cos\dfrac{4\pi}{7} + \cos\dfrac{6\pi}{7} = -\dfrac{1}{2}$. [4]

(d) Verify this on your calculator. [2]

(e) Explain how the same argument applies to $z^{2m+1} = 1$, and state the general result. [3]

30. Rotation is multiplication. Throughout, points of the plane are identified with complex numbers.

(a) Explain why multiplying by $\operatorname{cis}\alpha$ rotates a point about the origin through α. [2]

(b) Show that rotating z about the point w through α gives $z' = w + (z - w)\operatorname{cis}\alpha$. [3]

(c) The triangle with vertices 0, 1 and z is equilateral. Find the two possible values of z. [4]

(d) Show that $\zeta = \operatorname{cis}\frac{\pi}{3}$ satisfies $\zeta^2 = \zeta - 1$. [2]

(e) A triangle has vertices a, b, c, labelled so that $c - a = (b - a)\zeta$. Show that $a^2 + b^2 + c^2 = ab + bc + ca$. [4]

31. How far does De Moivre reach? The syllabus states De Moivre's theorem for integer n and notes that it extends to rational exponents. This question examines what that extension has to say.

(a) Using the formula for nth roots, write down the three values of $(\operatorname{cis}\theta)^{1/3}$. [3]

(b) Explain why the statement $(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$ needs restating when n is not an integer. [3]

(c) Take $\theta = \pi$ and $n = \frac{1}{2}$. Evaluate $\cos n\theta + i\sin n\theta$, find both square roots of $\operatorname{cis}\pi$, and comment on what you find. [4]

(d) State a version of the theorem valid for rational $n = \frac{p}{q}$ in lowest terms, saying how many values the left-hand side takes. [3]

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